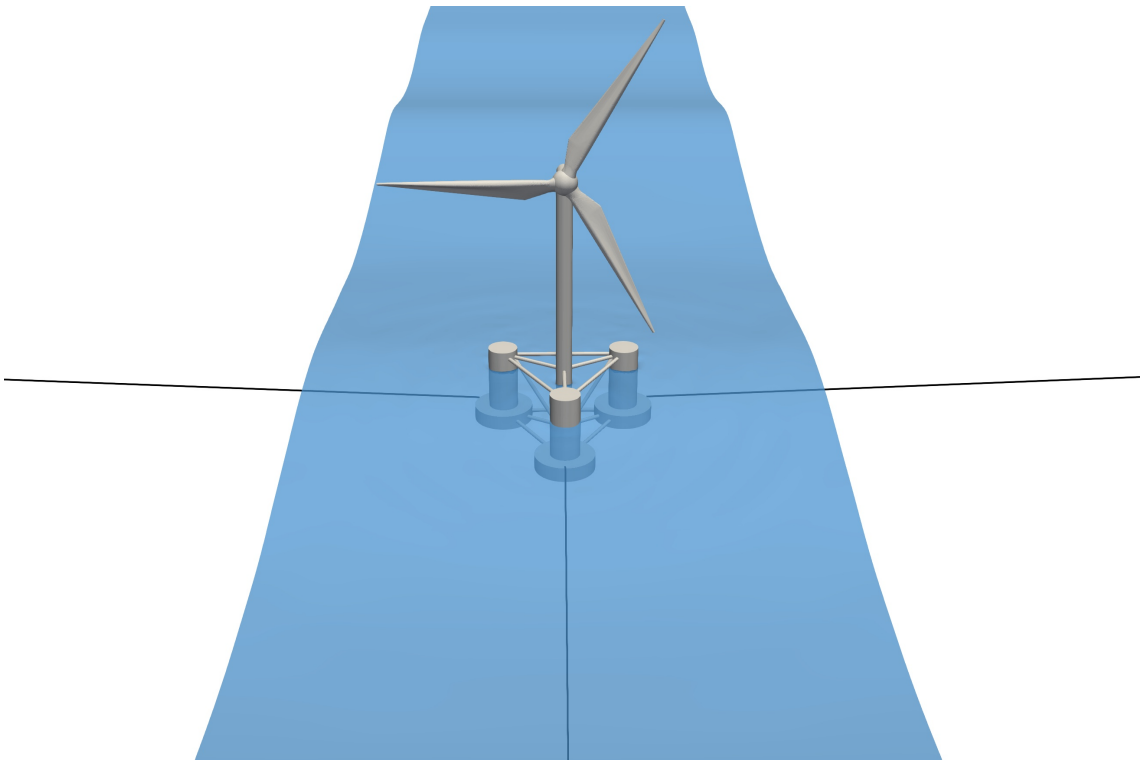
REEF3D :: User Guide



REEF3D 26.09

REEF3D
open-source
hydrodynamics

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AI declaration

This User Guide has been partially created with the assistance of Claude (Anthropic, model claude-opus-5-5).

Parts of the REEF3D open-source software have been created with the assistance of Claude (Anthropic, model claude-opus-5-5).

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1. Introduction

REEF3D is an open-source hydrodynamics framework for coastal, marine and hydraulic engineering. It combines several flow solvers that cover the relevant length scales, from ocean waves travelling over tens of kilometres to the detailed flow around a single structure:

- **REEF3D::CFD**: three-dimensional two-phase Navier-Stokes solver with interface capturing,
- **REEF3D::NHFLOW**: three-dimensional non-hydrostatic wave and flow model on a σ-grid,
- **REEF3D::FNPF**: fully nonlinear potential flow wave model,
- **REEF3D::SFLOW**: depth-averaged non-hydrostatic shallow water model,
- **REEF3D::SEASTATE**: phase-averaged spectral wave model for sea states over large coastal areas, stand-alone or as wave forcing for SFLOW and NHFLOW.

All modules share the same input file, wave generation library, parallelization and output routines. The mesh and the domain decomposition are generated with the accompanying open-source pre-processor DIVEMesh, which also handles solid geometries from STL files and bathymetry from measured data points. Multiphysics solvers for floating bodies (6DOF) with mooring lines and nets, the structural solver REEF3D::CORAL, sediment transport, porous media and turbulence are coupled to the flow solvers. Typical applications are wave propagation and transformation in coastal areas, wave loads on offshore and coastal structures including offshore wind foundations and floating structures, open channel and river flow, scour and sediment transport, and aquaculture structures.

REEF3D is released under the GNU General Public License, version 3 or later (GPL v3). The source code, the release notes and this guide are available from `www.reef3d.com` and from the GitHub repository `https://github.com/REEF3D/REEF3D`; DIVEMesh is available from `https://github.com/REEF3D/DIVEMesh`. Open source means that the code can be used at no cost, that users can see exactly which equations and algorithms are solved, and that developments benefit students, researchers and engineers working on academic and commercial problems alike.

REEF3D was designed under the following premises:

1. Hydraulic, coastal, marine and environmental engineering applications
    - robust modeling of complex free-surface flows and breaking waves
    - numerical wave tank and open channel boundary conditions
    - sediment transport and fluid-structure interaction
2. Ease of use: mesh generation should not limit user productivity
    - rectilinear grids with immersed boundaries instead of body-fitted meshes

- STL file input for solid geometries
- straightforward handling of natural bathymetries

3. Large computations can only be exploited in parallel
    - full parallelization based on domain decomposition and MPI
4. Stable, accurate and fast numerical simulations
    - high-order WENO discretization in space and TVD Runge-Kutta schemes in time
    - staggered grid with tight velocity-pressure coupling (CFD)
    - adaptive time stepping for stability
5. Maintainability and extensibility
    - modular, object-oriented C++
    - MPI as industry-standard library for high-performance computing, an in-house multigrid solver (REEFMG) and optionally HYPRE

How this guide is organised

- Chapter 2 (Model Overview) describes the hydrodynamic modules and helps to choose the right one for a given problem.
- Chapter 3 (Getting Started) walks through a first simulation.
- Chapter 4 (Theory and Numerical Methods) summarizes the governing equations and numerical methods of each module, including the spectral wave model SEASTATE.
- Chapter 5 (Multiphysics) summarizes the theory of the multiphysics models: floating bodies, moorings and nets, flexible structures, the structural solver CORAL, sediment transport, porous media and vegetation, heat and concentration, rheology, Lagrangian particles, sea ice and wind forcing.
- Chapter 6 (Functions) is the reference for all commands of the input file `ctrl.txt`, sorted by letter.
- Chapter 7 (Compiling the Code) explains how to install the dependencies and build REEF3D and DIVEMesh on Linux, macOS, Windows (WSL) and with Docker.
- Chapter 8 (Running the Code) explains the simulation folder, how to run DIVEMesh and REEF3D, how to restart from checkpoints, and how to run on HPC clusters.
- Chapter 9 (Post-Processing) describes the output files and their visualization in ParaView.
- The remaining chapters contain tutorials for SFLOW, FNPF, NHFLOW and CFD, which are also provided as input files in the `Tutorials` folder of the repository.

2. Model Overview

The hydrodynamic module is selected with **A 10** in `ctrl.txt`. Table 2.1 summarises the main differences between the modules, Section 2.1 gives guidance for the choice, and Chapter 4 describes the governing equations and numerical methods of each module. As a rule of thumb, the cheapest module that still resolves the relevant physics should be chosen: SEASTATE when only the sea state (wave height, period and direction) is needed over large areas, SFLOW or FNPF for large areas and long durations, NHFLOW for nearshore wave processes with breaking and run-up, and CFD where the three-dimensional flow, the two-phase interface or the detailed flow around structures matters.

2.1 Choosing a Module

The module is selected with A 10: 2 = SFLOW, 3 = FNPF, 5 = NHFLOW, 6 = CFD (default), 7 = SEASTATE. As a first orientation:

REEF3D::FNPF for wave propagation from deep to shallow water over large areas and complex coastlines, irregular and multi-directional sea states and wave kinematics for design, as long as the free surface is single-valued and viscous effects are not important. It is the most efficient module for waves; the Laplace equation is solved on a σ-grid with only a few cells in the vertical, and its results can be used as boundary conditions for NHFLOW or CFD, from state files (hydrodynamic coupling, HDC, B 92 61) or at runtime with both models in one MPI job (B 92 62, one- or two-way). Theory: Section 4.4.

REEF3D::NHFLOW for waves and currents in coastal areas where wave breaking, run-up with wetting and drying, the vertical flow structure or floating bodies matter, with a single-valued free surface. The shock-capturing finite volume scheme handles breaking waves and bores without an extra breaking model. Theory: Section 4.3.

REEF3D::SFLOW for depth-averaged problems such as open channel and river flow, long waves and bedload sediment transport, where the vertical flow structure is not important. The non-hydrostatic pressure extends the range to shallow and intermediate water waves. Theory: Section 4.5.

REEF3D::SEASTATE when the sea state rather than the individual waves is needed: wave height, period and direction over large coastal areas, for wind seas and swell from the boundary of the model to the coast. It solves the wave action balance for the directional spectrum (phase-averaged) on cells that need not resolve the wave length. Coupled with SFLOW or NHFLOW (A 750), it provides the wave forcing for wave-driven currents, setup and, with its surfbeat model, infragravity waves; it can also hand over 2D spectra at given points as the irregular wave input of FNPF or NHFLOW (A 760, B 85 11). Theory: Section 4.6, commands: Section 6.1.4.

REEF3D::CFD when the free surface can overturn or break violently, for wave impact and slamming, the detailed flow around structures and the forces on them, scour and two-phase flow in general. It is the most expensive module: restrict its domain to the region of interest and use FNPF to bring the waves there. Theory: Section 4.2.

All modules share the grid generation with DIVEMesh and the MPI parallelization, and the phase-resolving modules also share the high-order discretization schemes and the wave generation library (Section 4.1). A first simulation is described in Chapter 3, and each module has its own tutorial chapter (Chapters 10 to 13).

Table 2.1: Overview of the hydrodynamic modules.

	SEASTATE	**SFLOW**	**FNPF**	**NHFLOW**	**CFD**
A 10	7	2	3	5	6
Governing equations	wave action balance for the directional wave spectrum (phase-averaged)	depth-averaged non-hydrostatic shallow water equations	Laplace equation for the velocity potential, fully nonlinear free-surface boundary conditions	non-hydrostatic Navier-Stokes equations, Poisson equation for the dynamic pressure	incompressible RANS/Navier-Stokes equations, two-phase
Vertical resolution	none (2D horizontal grid, spectral grid of frequencies and directions)	none (2D horizontal grid)	σ-coordinates	σ-coordinates	full 3D rectilinear grid
Free surface	not resolved: H_s, periods and directions from the spectrum	water level η	single-valued η	single-valued η	level set (or VOF), overturning possible
Wave breaking	depth-induced breaking and whitecapping as source terms	breaking detection	breaking model (detection and damping)	shock-capturing (HLL-type Riemann solver)	resolved by the two-phase flow
Wetting and drying	cells shallower than A 705 are land	yes	coastline algorithm, optional wetting/drying (A 343)	yes	implicit through interface capturing
Floating bodies (6DOF)	no	one-way body motion (e.g. ship waves)	yes, resolved bodies with mooring	yes, with mooring and nets	yes, with mooring and nets
Sediment transport	no	bedload (with a suspended part, S 64)	no	bedload and suspended load	bedload and suspended load
Further physics	wind input, bottom friction, triads, quadruplets, vegetation, diffraction, obstacles, surfbeat; wave forcing for SFLOW and NHFLOW (A 750)	turbulence and diffusion, bed friction, meteorological forcing (wind, air pressure) and Coriolis force for storm surge	coupling to NHFLOW and CFD (HDC, or at runtime in one MPI job, one- or two-way)	turbulence (RANS/LES), VRANS porous media, wind and meteorological forcing, Coriolis force, FEM solids, coupling from FNPF (HDC, runtime)	turbulence (RANS/LES), VRANS porous media, heat and concentration transport, non-Newtonian rheology, FSI, coupling from FNPF (HDC, runtime, one- or two-way)
Relative cost	lowest per area (the cells need not resolve the wave length)	lowest of the phase-resolving modules	low	medium	highest
Typical use	sea states over large coastal areas, wave forcing for circulation and surfbeat, boundary spectra for FNPF and NHFLOW	open channel and river flow, long waves, tides and storm surge over large areas	wave propagation from deep to shallow water, design sea states, wave input for NHFLOW and CFD	nearshore waves with breaking and run-up, wave-structure interaction	detailed flow around structures, breaking wave loads, scour, dam break

3. Getting Started

This chapter takes you from an installed code to a first simulation and explains the steps you will repeat for every new case. It assumes that DIVEMesh and REEF3D have been compiled (Chapter 7) and that the executables `DiveMESH` and `REEF3D` are on your PATH (Section 8.1). The example is the smallest wave case of the tutorials: regular waves in a 2D numerical wave tank computed with REEF3D::FNPF. It runs in a few minutes on a laptop.

3.1 The Workflow at a Glance

Every REEF3D simulation follows the same five steps (Fig. 3.1):

1. Create a simulation folder for the case.
2. Describe the domain, the grid, solid geometries and the domain decomposition in `control.txt` and run **DiveMESH**. It writes one grid file per subdomain into `DIVEMesh_Grid`.
3. Describe the physics, numerics and output in `ctrl.txt`: which module (A 10), boundary conditions and waves (B), water level (F 60), simulation time (N 41), output (P) and number of processes (M 10).
4. Run REEF3D in parallel: **mpirun -n** *M 10* **REEF3D**.
5. Look at the results: 3D fields in ParaView, time series from gauges and probes with Python, Matlab or similar (Chapter 9).

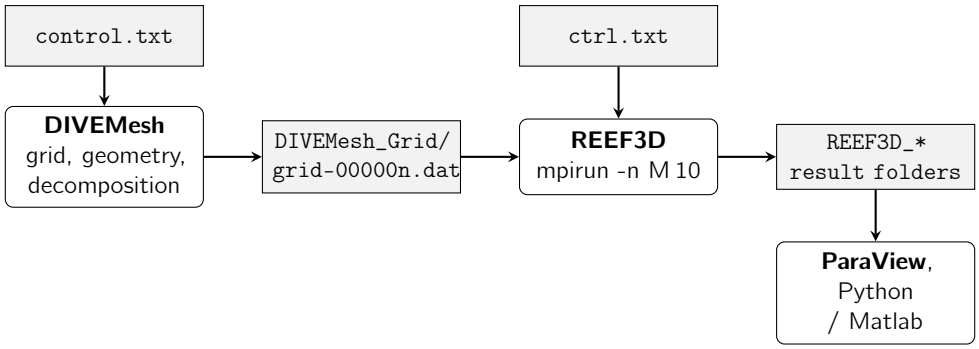

Figure 3.1: Workflow of a REEF3D simulation. The number of subdomains M 10 must be the same in `control.txt`, in `ctrl.txt` and in the mpirun command.

Both input files are plain text files with one command per line: a capital letter for the group, a number for the option and the values, e.g. `F 60 2.0`. Everything after a / is a comment. The syntax and its pitfalls are explained in Chapter 6; the most important points are summarized in Section 3.5.

3.2 Your First Simulation: Regular Waves in a 2D Wave Tank

The case is tutorial `Tutorials/REEF3D_FNPF/9_1 Regular Wave Propagation` of the code repository. A 2nd-order Stokes wave with height $H = 0.04\,\text{m}$ and wave length $L = 4.0\,\text{m}$ is generated at the left end of a 40 m long tank with a water depth of $d = 2.0\,\text{m}$ ($kd \approx 3.1$, i.e. close to deep water) and absorbed at the right end. The simulated surface elevation is compared with the wave theory at three gauges.

Step 1: Create the simulation folder

Create an empty folder, e.g. `first_case`, open a terminal in it and create the two input files as described below (or copy them from the tutorial folder).

Step 2: Grid and domain – control.txt

```
C 11 6
C 12 3
C 13 3
C 14 7
C 15 21
C 16 3

B 1 0.1
B 2 400 1 10
B 10 0.0 40.0 0.0 0.1 0.0 2.0

B 103 5
B 113 2.5
B 116 2.0

M 10 6
M 20 2
```

C 11 – C 16 Boundary conditions of the six sides of the domain: 1 = x_{min} (left), 2 and 3 = the two sides in y-direction, 4 = x_{max} (right), 5 = bottom, 6 = top. Here: wave generation (6) on the left, numerical beach (7) on the right, a wall (21) at the bottom and symmetry planes (3) elsewhere. For a 2D case, the y-sides are symmetry planes.

B 1, B 2 Cell size (0.1 m) and number of cells in x, y and z ($400 \times 1 \times 10$). If B 2 is given, it determines the grid. One cell in y-direction makes the case 2D.

B 10 Rectangular domain: x from 0 to 40 m, y from 0 to 0.1 m, z from 0 to 2 m.

B 103, B 113, B 116 Vertical grid stretching (point focus with sinh, stretching factor 2.5, focus at $z = 2.0\,\text{m}$), so that the ten vertical cells are concentrated near the free surface. For the sigma-grid models FNPF and NHFLOW, the vertical grid is stretched between the bed and the free surface; the water depth itself comes from F 60 in `ctrl.txt`.

M 10, M 20 Decompose the domain into 6 subdomains with decomposition method 2.

Run **DiveMESH** in the folder. It creates `DIVEMesh_Grid` with the files `grid-000001.dat` to `grid-000006.dat`. The folder `DIVEMesh_Paraview` can be opened in ParaView to check the grid and the decomposition.

Step 3: Physics, numerics and output – ctrl.txt

```
A 10 3
A 310 3
A 311 4
A 343 0

B 90 1
B 92 4
B 91 0.04 4.0
B 96 4.0 8.0
B 98 2
B 99 1

F 60 2.0

N 41 40.0
N 47 1.0

M 10 6

P 10 1
P 30 0.5
P 50 15.0 0.005
P 50 20.0 0.005
P 50 25.0 0.005
P 51 15.0 0.005
P 51 20.0 0.005
P 51 25.0 0.005
P 52 0.005
P 53 1
P 54 100

W 22 -9.81
```

A 10 3 Selects REEF3D::FNPF. A 310 and A 311 select 3rd-order Runge-Kutta time stepping and 5th-order WENO for the free surface boundary conditions; A 343 0 switches wetting and drying off (there is no coastline in this case).

B 90 – B 99 Wave generation: B 90 1 activates wave generation, B 92 4 selects 2nd-order Stokes waves, B 91 gives wave height and wave length in m (B 93 would give height and period instead). B 96 sets the lengths of the generation zone (4 m, one wave length) and of the numerical beach (8 m, two wave lengths). B 98 2 and B 99 1 select the relaxation methods for generation and absorption.

F 60 Still water level $z = 2.0$ m, i.e. the water depth.

N 41, N 47 Simulate 40 s with a CFL safety factor of 1.0 (the default is 0.3; FNPF is typically run with larger values than CFD).

M 10 Number of processes, the same as in `control.txt`.

P 10, P 30 Write 3D results in VTU format every 0.5 s. Without an output interval (P 20, P 30 or similar) no field output is written.

P 50, P 51 Wave gauges at $x = 15, 20, 25$ m: P 50 records the wave theory, P 51 the simulated surface elevation.

P 52 – P 54 Write the free surface along the line $y = 0.005$ m every 100 iterations, including the theoretical surface (P 53 1).

W 22 Gravitational acceleration in z-direction.

Step 4: Run REEF3D

```
mpirun -n 6 REEF3D
```

REEF3D first prints `read ctrl`, then the iteration number, the simulated time and the time step. The simulation ends after 40 s of simulated time. If your computer has fewer than six cores, set M 10 to a smaller number in both input files and run DIVEMesh again.

Step 5: Check the results

After the run, the simulation folder contains, among others:

REEF3D_FNPF_VTU the 3D fields. Open the `.pvtu` files in ParaView (they combine the per-process `.vtu` files) and colour by `eta` or the velocity.

REEF3D_FNPF_WSF the gauge time series: `REEF3D-FNPF-WSF-HG.dat` (simulated, P 51) and `REEF3D-FNPF-WSF-HG-THEORY.dat` (theory, P 50).

REEF3D_FNPF_WSFLINE the free surface along the tank at the P 54 interval.

Plot the simulated and the theoretical elevation at the three gauges (see Section 9.5 for a Python example). After the waves have reached the gauges, both curves should agree closely. This comparison of P 50 and P 51 is a quick check that wave generation, grid resolution and absorption are adequate, and it is worth repeating for every new wave case.

3.3 Structure of ctrl.txt

The commands in `ctrl.txt` are grouped by their letter. Table 3.1 shows where to look for which setting. Commands that are not given keep their default value, so a `ctrl.txt` file only needs to contain the settings that differ from the defaults. In the reference chapter (Chapter 6), four markers next to each command show for which module it is valid: S (SFLOW), F (FNPF), N (NHFLOW) and C (CFD).

Group	Content	Typical first commands
A	module selection and module-specific numerics (Section 6.1)	A 10 (module), wave breaking, wetting and drying
B	boundary conditions, inflow and outflow, wave generation and absorption (Section 6.2)	B 90 – B 99 (waves), B 60 (flow)
C	concentration / pollutant transport (Section 6.3)	C 10
D	discretization of the CFD equations (Section 6.4)	D 10, D 20, D 30
E	discrete element method (DEM) for rigid particles, input file `dem.txt` (Section 6.5)	E 10
F	free surface and initial water level (Section 6.6)	F 60 (still water level), F 30 (level set)
G	grid: mesh refinement G 1 – G 47 (SFLOW, FNPF, NHFLOW, CFD, SEASTATE), curvilinear and multi-block grids G 50 – G 54 (Section 6.7.2), geometry (level sets of the topography and solids) G 505 – G 541 (Section 6.7)	G 1, G 10 (refinement), G 50, G 540, G 541
H	heat transfer (Section 6.8)	H 10
I	initialization and restart (Section 6.9)	I 10, I 40, I 45 (restart from checkpoints)
L	Lagrangian particle tracking in NHFLOW and CFD, e.g. marine plastics (Section 6.10)	L 10, L 21 – L 23
M	parallel computing (Section 6.11)	M 10 (number of processes), M 40, M 41 (decomposition by REEF3D)
N	solvers, time stepping and run length (Section 6.12)	N 41 (simulated time), N 47 (CFL factor)
P	output: 3D fields, gauges, probes, forces, checkpoints (Section 6.13)	P 10, P 30, P 51, P 48 (checkpoints)
Q	particle model for sediment, CPM (Section 6.14)	Q 10
S	sediment transport (Section 6.15)	S 10
T	turbulence modelling in CFD (Section 6.16)	T 10
W	fluid and material properties, gravity (Section 6.17)	W 1, W 22
X	floating bodies (6DOF) and moorings (Section 6.18)	X 10
Z	fluid-structure interaction, structural solver REEF3D::CORAL (Section 6.19)	Z 10, Z 40 (CORAL)

Table 3.1: Command groups in `ctrl.txt`.

3.4 Setting Up Your Own Case

The easiest way to set up a new case is to start from the tutorial that is closest to your problem (Chapters 10 to 13) and to change it step by step. Go through the following checklist:

1. **Module**: A 10 (Section 2.1).
2. **Domain and grid** (`control.txt`): B 10 for the domain, B 1 or B 2 for the resolution, stretching with B 101 – B 126 where needed, solids from primitives (S commands of DIVEMesh) or STL files, bathymetry from data points. Check the grid in `DIVEMesh_Paraview` before running REEF3D.
3. **Boundaries** (`control.txt`): C 11 – C 16 for the six sides (inflow, outflow, wave generation, numerical beach, wall, symmetry plane).
4. **Water level and flow**: F 60 for the still water level; for currents or open channel flow, the inflow and outflow settings in the B section (B 60 and following) and the outflow water level F 62 with its ramp F 64.
5. **Waves**: B 90 1, the wave theory B 92, wave height and length (B 91) or period (B 93), or a spectrum for irregular waves, the zone lengths B 96 and the methods B 98 / B 99. The tutorials use a generation zone of about one wave length and a numerical beach of about two wave lengths.
6. **Run length and time step**: N 41 (simulated time) or N 45 (maximum iterations), whichever is reached first ends the run; N 47 is the CFL safety factor. Reduce N 47 if a simulation becomes unstable.
7. **Output**: P 10 with an interval (P 20 iterations or P 30 seconds; SFLOW uses P 182) for field data, P 51 wave gauges, probes and force output. Output files can become large; choose the intervals accordingly.
8. **Parallelization**: M 10 in both files, at most the number of physical cores.

Grid resolution

The tutorials keep the grids small so that they run on a laptop; they are not grid-converged. Table 3.2 lists their horizontal resolution as a starting point. For every new case, check the sensitivity of the results to the grid, e.g. by comparing the P 51 gauges with the P 50 theory for a regular wave or by repeating the simulation with a finer grid.

3.5 Common Pitfalls and Troubleshooting

Tutorial	Module	Δx [m]	L/Δx	kd
8_1 Stokes wave propagation	SFLOW	0.02	200	0.79
9_1 Regular wave propagation	FNPF (10 cells vertically)	0.10	40	3.14
9_2 Submerged bar	FNPF (10 cells vertically)	0.04	120	0.52
2 Submerged bar	NHFLOW (3 cells vertically)	0.029	128	0.67
11_6 Rectangular wave tank	CFD	0.05	80	0.79
11_10 Submerged bar	CFD	0.01	373	0.67

Table 3.2: Horizontal resolution of selected wave tutorials: cell size Δx, cells per wave length $L/\Delta x$ and relative water depth kd at the wave generation.

Symptom	Cause and remedy
REEF3D stops with `Inconsistent M 10 parameter`	The number of MPI processes, M 10 in `ctrl.txt` and M 10 in `control.txt` differ. Make them equal and run DIVEMesh again after changing `control.txt`, or let REEF3D decompose the global grid file of DIVEMesh for the processes of the run (M 40, Section 8.3).
REEF3D stops with `Could not open DIVEMesh grid file`	DIVEMesh has not been run in this folder, or REEF3D is started from a different folder.
REEF3D stops after `read ctrl` with `missing input parameter in ctrl.txt`	A command has fewer values than required. Check the number of values against Chapter 6. If the incomplete command is on the last line of the file, REEF3D does not stop and the command keeps its default value.
A setting seems to have no effect	The command was not read. Unknown command numbers and lines starting with a lower-case letter (e.g. `b 120`) are ignored without a warning. Check spelling, letter case and the number of values. Also check whether the setting is changed automatically (Section 3.6) or whether the command is valid for the selected module (S F N C markers).
A comment is interpreted as a command	Comments must start with /. Free text without a leading / is read token by token; a capital letter in it can be taken as a command. Use // for all comments.
An integer command gets a decimal value	For `int` commands a value like `2.5` is cut to its integer part.
No result files are written	By default no periodic field output is written. Set P 10 together with an interval (P 20 or P 30; for SFLOW P 181 or P 182).
Old and new results are mixed	Result folders from a previous run are not deleted: files with the same name are overwritten, others remain. Delete or move old `REEF3D_*` folders before a new run.
The simulation becomes unstable or produces NaN	Reduce the CFL factor N 47, check the grid quality and resolution near the free surface and structures, check that the wave is resolved (Table 3.2) and that it is not too steep for the chosen wave theory, and check that the generation and absorption zones fit into the domain.
Waves are reflected at the end of the tank	The numerical beach is too short. Use a beach length (B 96) of at least about two wave lengths or active absorption (B 99 with the corresponding boundary condition in `control.txt`).
The wave height decreases along the tank	The grid is too coarse for the wave length, or the vertical resolution near the free surface is insufficient. Refine the grid (Table 3.2) and compare P 51 with the P 50 theory.

3.6 Inputs that are Changed Automatically

After reading `ctrl.txt`, REEF3D adjusts some settings to keep the input consistent (file `control_parse.cpp`). These changes are not reported in the terminal output. The changes from the row N 10 onwards are made when `ctrl.txt` is read (A 212, N 10) or at start-up by the modules themselves; except for A 212 and G 52, a message is printed. The affected entries in Chapter 6 mention them as well.

Condition	**Automatic change**
B 98 1	changed to B 98 2
NHFLOW (A 10 5) and B 98 3	changed to B 98 4 (Dirichlet generation with active absorption)
NHFLOW (A 10 5) and B 98 5	changed to B 98 3
X 10 $>$ 0 (6DOF)	D 22 is set to 2; N 40 3 and 23 are changed to 4 (only with G 1 0), N 40 13 to 44
N 40 14	changed to N 40 44
F 80 $>$ 0 (VOF) and F 35 $>$ 0	F 35 is set to 0
I 10 1	I 11 = 1, I 12 = 2, I 13 = 1
I 10 2	I 11 = 2, I 12 = 2, I 13 = 1
I 40 $>$ 0 (restart)	I 10, I 11, I 12 and I 13 are set to 0
T 10 0	I 13 is set to 0
S 10 $\geq$ 1 or topography in the grid file	P 27 is set to 1 (bed printed to the VTU files)
A 550 1 or A 560 $>$ 0	A 512 is set to 2 (implicit diffusion)
I 30 2 (masked full initialization) and A 10 $\neq$ 3	I 30 is set to 1
I 45 $>$ 1	restart from the checkpoint of that iteration, otherwise as I 45 1
N 10 $\geq$ 10 in a build without HYPRE (the default build, Section 7.1.4)	N 10 is set to 1 (REEFMG)
file `REEF3D_Checkpoint/rebalance` present (rebalance restart, G 46)	I 45, M 40 and M 41 are set to 1: restart from the latest checkpoint on a new decomposition with the weighted partition of the load map
NHFLOW (A 10 5), A 520 2 and X 330 $>$ 0 with a moving membrane	A 520 is set to 1 (full projection)
G 1 $>$ 0 for a case outside the scope of the mesh refinement (see G 1, e.g. NHFLOW with A 580 1, E 10 $>$ 0, L 10 $>$ 0, A 520 3, or A 550 1 with A 510 3; CFD outside the set-up listed in Section 6.7.1)	the mesh refinement is switched off; on a multi-block grid (G 50), a case outside the scope stops the run with a message; in SEASTATE (A 10 7), an invalid refinement setting (e.g. G 1 with A 750 or A 770 1, G 40 $\neq$ 0) stops the run with an input error instead
FNPF (A 10 3), A 328 1 and N 10 $\neq$ 1	the cross-derivative terms stay explicit, as for A 328 0
NHFLOW (A 10 5), A 529 1 with N 10 $\neq$ 1, G 50 $>$ 0, G 1 $>$ 0 or membranes (X 330)	the sigma cross terms of the Poisson equation stay explicit, as for A 529 0
NHFLOW (A 10 5), A 561 $\neq$ 0 or A 563 $>$ 0 without A 560 1, 2, 21 or 22	A 561 and A 563 are set to 0
multi-block grid (G 50 1 with blocks)	G 52 is set to 1
B 98 6 – 8 or B 99 6 – 8 in a module or with a wave type that does not support the option	changed to the fallback given under B 98 and B 99, e.g. B 99 6 to 2 in CFD, B 98 7 to 2 in FNPF with a wavemaker other than the piston
CFD (A 10 6), B 268 1 without the Runge-Kutta momentum solver	B 268 is set to 0 (explicit resistance)
A 212 not given	A 212 is set to 1 if A 260 $>$ 0, else to 0

Condition	Automatic change
B 89 1 with a wave type other than 5, 31, 41 or 51 (also of a B 500 source)	B 89 is set to 0
B 530 not given (-1)	set to 2 when a B 523 background carries a current, to 1 when it only sets a level, and to 0 otherwise (no background, no waves, B 89 1 or a wave type without support)
NHFLOW (A 10 5), B 99 3 or 4 with an outflow boundary at x_{max}	the side x_{max} becomes an absorbing Riemann edge with still water outside (B 520), unless a B 520 edge is given there
S 62 2 and A 10 $\neq$ 5	S 62 is set to 1
NHFLOW (A 10 5), X 15 8 (sharp hull) with X 10 $>$ 0 and A 520 2	A 520 is set to 1 (full projection)
NHFLOW (A 10 5), A 559 $\neq$ 0 without A 560 2 or 22	A 559 is set to 0 (EARSM off)
NHFLOW (A 10 5), A 559 1 and A 563 $>$ 0	A 563 is set to 0 (the EARSM replaces the QCR)
runtime FNPF coupling (B 92 62): source and receiver models with G 46 $>$ 0	G 46 is set to 0 (no rebalance stop in coupled runs)

Removed options The following options have been removed from the code. If they are given in `ctrl.txt`, the run stops with a message, which in most cases names a replacement: A 10 4 (REEF3D::PTF, use FNPF or CFD), F 80 1 and 3 (algebraic VOF, use F 80 4 or the level set F 30), F 85 51 – 53 (HRIC, modified HRIC and CICSAM), N 40 5 (use N 40 3 or 13), T 10 12 (EARSM, use T 10 2) and A 260 5 (second SFLOW k-ω implementation, use A 260 2). The SEASTATE wind file `seastate-wind.dat` has been retired: the wind field of A 730 2 is that of the meteorological forcing W 200 (Section 6.17.1), and A 730 2 with W 200 0 stops the run; if the old file is present, the message gives the conversion `python3 tools/met_forcing.py seastate seastate-wind.dat -o met-forcing.r3m` (then set W 200 2). Unknown values of T 10, T 12 and, for LES, T 21 also stop the run with a message. Some former options have been merged into others and are accepted without a message: N 40 14 (low-storage Runge-Kutta with its own time loop) is changed to N 40 44 (see the table above), and the uniform-grid WENO variants T 12 55 and H 15 9 now run the non-uniform grid schemes of T 12 5 and H 15 4. Commands that are no longer used at all (retired commands, e.g. F 84 or S 83) do not stop the run: their values are skipped and REEF3D prints `ctrl.txt:  X N is no longer used and is ignored`; they are listed under "How ctrl.txt is read" at the beginning of Chapter 6.

4. Theory and Numerical Methods

This chapter summarizes the governing equations and the main numerical methods of the four phase-resolving hydrodynamic modules and of the spectral wave model SEASTATE (Section 4.6). It is meant as a compact background for choosing and understanding the input commands, not as a complete derivation; the commands that select a method are given in parentheses, e.g. (A 311), and are described in Chapter 6. For details and validation, see the cited publications. Section 4.1 describes the methods that are shared by several modules (grids, WENO discretization, Runge-Kutta time stepping, linear solvers, mesh refinement, wave generation and absorption); the following sections describe each module in turn.

Throughout the chapter, $x_i = (x, y, z)$ are the Cartesian coordinates with z pointing upwards, $u_i = (u, v, w)$ the velocities, t the time, ρ the density, ν and ν_t the kinematic and eddy viscosity, p the pressure and g the gravitational acceleration. d is the still water depth, η the free surface elevation and $h = \eta + d$ the total water depth. In REEF3D::CFD, ϕ denotes the level set function; in REEF3D::FNPF, Φ is the velocity potential.

4.1 Common Numerical Methods

The four hydrodynamic modules share the grid, the parallelization, the high-order discretization schemes, the linear solvers and the wave generation library, which are summarized here.

4.1.1 Grid, data layout and parallelization

All modules use a rectilinear grid generated by DIVEMesh; SFLOW and NHFLOW can also run on a curvilinear or multi-block horizontal grid (G 50, see below). The cell sizes can vary in each direction (stretching with B 101–B 126 in the DIVEMesh `control.txt`), and all schemes use the local cell sizes. CFD uses a staggered grid: u, v and w are stored at the cell faces in their direction, while p, the level set function ϕ and the turbulence quantities are stored at the cell centres. NHFLOW and SFLOW are finite volume methods with cell-centred unknowns and fluxes at the faces. FNPF stores Φ at the horizontal cell centres and at the vertical grid nodes, so that the bed and the free surface are node levels. For the σ-grid models NHFLOW and FNPF, the vertical node coordinates from DIVEMesh are normalized to $\sigma \in [0, 1]$ from the bed to the free surface, keeping the vertical stretching (B 103, B 113, B 116).

All fields have three layers of ghost cells in each direction, which is enough for the WENO stencils. The ghost cells hold the boundary conditions at walls, inflow and outflow, and the values of the neighbouring subdomains. DIVEMesh splits the grid into M 10 subdomains, one per MPI process, and the ghost cells are updated by MPI exchange with the neighbours after each field update. M 10 must be identical in DIVEMesh, in `ctrl.txt` and in the MPI start command (Section 6.11).

4.1.2 Spatial discretization: WENO schemes

Convection terms and free surface gradients are discretized with the fifth-order weighted essentially non-oscillatory (WENO) scheme of Jiang and Shu [1996]. The scheme is fifth-order accurate on a smooth solution. Near steep gradients, such as the free surface, breaking fronts or bores, it falls back to the smoothest third-order sub-stencil and does not create spurious oscillations. For a positive face velocity $u_{i+1/2} > 0$, the face value of a variable φ is a weighted combination of three candidate stencils:

$$\begin{aligned}\varphi_{i+1/2} = {} & \omega_1\Big(\tfrac{1}{3}\varphi_{i-2} - \tfrac{7}{6}\varphi_{i-1} + \tfrac{11}{6}\varphi_i\Big) + \omega_2\Big(-\tfrac{1}{6}\varphi_{i-1} + \tfrac{5}{6}\varphi_i + \tfrac{1}{3}\varphi_{i+1}\Big) \\ & + \omega_3\Big(\tfrac{1}{3}\varphi_i + \tfrac{5}{6}\varphi_{i+1} - \tfrac{1}{6}\varphi_{i+2}\Big), \end{aligned} \tag{4.1}$$

$$\begin{aligned} IS_1 &= \tfrac{13}{12}\left(\varphi_{i-2} - 2\varphi_{i-1} + \varphi_i\right)^2 + \tfrac{1}{4}\left(\varphi_{i-2} - 4\varphi_{i-1} + 3\varphi_i\right)^2, \\ IS_2 &= \tfrac{13}{12}\left(\varphi_{i-1} - 2\varphi_i + \varphi_{i+1}\right)^2 + \tfrac{1}{4}\left(\varphi_{i-1} - \varphi_{i+1}\right)^2, \\ IS_3 &= \tfrac{13}{12}\left(\varphi_i - 2\varphi_{i+1} + \varphi_{i+2}\right)^2 + \tfrac{1}{4}\left(3\varphi_i - 4\varphi_{i+1} + \varphi_{i+2}\right)^2, \end{aligned} \tag{4.2}$$

$$\omega_k = \frac{\alpha_k}{\alpha_1 + \alpha_2 + \alpha_3}, \quad \alpha_k = \frac{C_k}{\left(\varepsilon + IS_k\right)^2}, \quad (C_1, C_2, C_3) = (0.1, 0.6, 0.3), \quad \varepsilon = 10^{-6}, \tag{4.3}$$

where IS_k are the smoothness indicators of the stencils. For $u_{i+1/2} < 0$ the stencil is mirrored about the face. Eqs. (4.1)–(4.3) are the uniform-grid form. All WENO5 options use a non-uniform grid version (the former uniform-grid implementations behind T 12 55 and H 15 9 have been removed; these options now select the non-uniform grid schemes of T 12 5 and H 15 4), in which the stencil coefficients, the C_k and the coefficients of IS_k are computed once from the local cell sizes [Shu, 1997]; on a uniform grid they reduce to the values above. Eq. (4.3) gives the classical WENO-JS weights of Jiang and Shu [1996]. Two alternative sets of nonlinear weights use the same stencils and smoothness indicators. The WENO-Z weights of Borges et al. (2008), $\alpha_k = C_k\Big(1 + \left(\tau_5/(IS_k + \varepsilon)\right)^2\Big)$ with $\tau_5 = |IS_1 - IS_3|$, are closer to the ideal weights in smooth regions and less dissipative at extrema. The TENO5 weights of Fu, Hu and Adams (2016) either keep a stencil with its ideal weight C_k or discard it, depending on a cutoff C_T. The weights are selected per module: A 213 (SFLOW, cutoff A 222) defaults to WENO-Z; A 527 (NHFLOW, cutoff A 528), A 318 (FNPF, cutoff A 319), D 12 (CFD momentum, cutoff D 13) and F 37 (CFD level set and reinitialization, cutoff F 38) default to WENO-JS.

There are two forms of the scheme. The conservative **WENO FLUX** form discretizes the convection term as $-(u_{i+1/2}\varphi_{i+1/2} - u_{i-1/2}\varphi_{i-1/2})/\Delta x_i$, with the face velocities interpolated according to D 11 (FOU or CDS2). The non-conservative **WENO HJ** (Hamilton–Jacobi) form [Jiang and Peng, 2000] discretizes $u\,\partial\varphi/\partial x$. It applies the same weights to the one-sided differences $(\varphi_{i-2} - \varphi_{i-3})/\Delta x, \ldots, (\varphi_{i+2} - \varphi_{i+1})/\Delta x$ in place of $\varphi_{i-2}, \ldots, \varphi_{i+2}$, which gives the upwind derivative $(\partial\varphi/\partial x)^-$ for $u > 0$ (mirrored for $u < 0$). In CFD, D 10 (momentum), F 35 (level set), C 15 (concentration) and H 15 (heat) select 4 WENO5 FLUX or 5 WENO5 HJ. The other options are the third-order variants WENO3 FLUX/HJ (7, 8) and lower-order schemes: first-order upwind FOU (1), central CDS2 (2), QUICK (3), CDS4 (6), TVD limiter schemes (10–16) and CDS6 (60). In NHFLOW (A 514 4, 5) and SFLOW (A 211 4), WENO reconstructs the left and right face states for the HLL-type fluxes. In FNPF, A 311 4/5 applies WENO5 to the horizontal gradients in the free surface boundary conditions.

4.1.3 Time integration and adaptive time stepping

The semi-discrete equations $\partial\varphi/\partial t = \mathcal{L}(\varphi)$ are advanced with the total variation diminishing (TVD) Runge–Kutta schemes of Shu and Osher [1988], in third order

$$\varphi^{(1)} = \varphi^n + \Delta t\,\mathcal{L}(\varphi^n), \qquad \varphi^{(2)} = \tfrac{3}{4}\varphi^n + \tfrac{1}{4}\varphi^{(1)} + \tfrac{1}{4}\Delta t\,\mathcal{L}(\varphi^{(1)}),$$
$$\varphi^{n+1} = \tfrac{1}{3}\varphi^n + \tfrac{2}{3}\varphi^{(2)} + \tfrac{2}{3}\Delta t\,\mathcal{L}(\varphi^{(2)}), \tag{4.4}$$

or in second order, $\varphi^{n+1} = \frac{1}{2}\varphi^n + \frac{1}{2}\varphi^{(1)} + \frac{1}{2}\Delta t\,\mathcal{L}(\varphi^{(1)})$ with the same first stage. The scheme is selected with N 40 (CFD), A 210 (SFLOW), A 310 (FNPF; option 4 is a classical fourth-order Runge–Kutta scheme) and A 510 (NHFLOW). In CFD and NHFLOW the pressure projection is carried out in every stage. The relaxation zones are applied after every stage.

With N 48 1 the time step follows from a stability criterion multiplied by $c =$ N 47. In CFD (N 50 1), convection, diffusion and source terms are included:

$$\Delta t = c \min_{\text{cells}} \frac{2}{\left(\dfrac{|\mathbf{u}|_{\max}}{\Delta x} + \dfrac{6\,\nu_{\text{eff}}}{\Delta x^2}\right) + \sqrt{\left(\dfrac{|\mathbf{u}|_{\max}}{\Delta x} + \nu_{\text{eff}}\right)^2 + \dfrac{4\,|S_{\max}|}{\Delta x}}}. \tag{4.5}$$

Here $|\mathbf{u}|_{\max} = \sqrt{u_{\max}^2 + v_{\max}^2 + w_{\max}^2}$ is formed from the global maxima of the velocity components, $\Delta x = \min(\Delta x_i, \Delta y_j, \Delta z_k)$ is the local cell size, $\nu_{\text{eff}} = \nu + \nu_t$, and $S_{\max}$ is the largest source term of the momentum equations, including gravity. N 50 2 evaluates Eq. (4.5) for each direction separately, with $u_{\max}$ and Δx_i, $v_{\max}$ and Δy_j, and $w_{\max}$ and Δz_k. The other modules use wave celerity based criteria:

- NHFLOW: $\Delta t = c \min\Big(\Delta x/(u_{\max} + \sqrt{g h_{\max}}),\ \Delta x/w_{\max}\Big)$, with the maximum total water depth $h_{\max}$. A 533 1 (and porous media, B 200) adds a vertical Courant number and a source term criterion.
- FNPF: $\Delta t = c \min\Big(\Delta x/\max(u_{\max}, \sqrt{g d_{\max}})\Big)$, where Δx is the horizontal cell size.
- SFLOW: $\Delta t = 2c \min\Big(\Delta x/(|U| + \sqrt{gh}),\ \Delta y/(|V| + \sqrt{gh})\Big)$, taken over the wet cells, so N 47 is half the Courant number.

In FNPF and NHFLOW the adaptive time step is limited by N 49. N 48 0 selects the fixed time step N 49.

4.1.4 Linear solvers

Each module solves an elliptic equation per time step or stage: the pressure Poisson equation of the projection method in CFD (D 30), the Poisson equation for the non-hydrostatic pressure in NHFLOW, the Laplace equation for Φ in FNPF and the equation for the non-hydrostatic (dynamic) pressure in SFLOW. N 10 selects the solver and N 11 the preconditioner, N 44 sets the tolerance and N 46 the maximum number of iterations. The default is the in-house solver REEFMG (N 10 1), which is available in every build. It is a BiCGStab iteration

preconditioned by a multigrid that coarsens only in x and y and solves each vertical grid line exactly (tridiagonal). It is therefore insensitive to strong vertical stretching. REEFMG is designed for the FNPF Laplace and NHFLOW pressure equations and also handles the CFD pressure and, in a 2D version, the SFLOW pressure (N 12–N 15). In builds with the optional HYPRE library [Falgout et al., 2006, Cen, 2015] (`make HYPRE=1`, Section 7.1.4), the following solvers are available as well; without HYPRE, N 10 $\geq$ 10 is replaced by N 10 1. On the structured interface (Struct, N 10 11–19; SFLOW uses its 2D version) there are Krylov solvers with the geometric multigrid preconditioners PFMG and SMG, e.g. BiCGStab with PFMG (N 10 14, N 11 11). The matrix interface (AIJ, N 10 21–25) provides BoomerAMG, and the semi-structured interface (SStruct, N 10 31–38) provides SysPFMG.

4.1.5 Mesh refinement

SFLOW, FNPF and NHFLOW can refine the horizontal grid locally with the block-structured mesh refinement REEFAMR (G 1 – G 47, Section 6.7.1). Each refined level halves the horizontal cell size of the level below (and, with G 6 1, the σ-layer thickness); the refined region is built from tiles (G 4) and divided into rectangular patches, on which the unchanged kernels of the module run. The cells around a patch are filled before every Runge–Kutta stage from neighbouring patches of the same level or by prolongation from the coarser level, the fine solution is restricted to the coarse cells below it, and in SFLOW and NHFLOW the fluxes are matched at the coarse-fine faces. With G 7 0 (default), all levels advance with one global time step, and the elliptic equations (SFLOW and NHFLOW pressure, FNPF Laplace equation) are solved on the composite grid. With G 7 1, every level takes two steps per step of the next coarser level (subcycling), with refluxing at the coarse-fine faces and, for the non-hydrostatic pressure, a solution per level followed by a synchronisation projection. The refined region is given by boxes (G 10), by flow criteria in SFLOW (G 20, G 22) and in FNPF and NHFLOW (G 20 – G 23) or by a zone that follows a moving body (G 12, G 13); it is not created in the G 11 boxes, in the relaxation zones (including those of B 520) and close to in- and outflow boundaries, and the refinement is switched off with a message for cases outside its scope (see G 1). In CFD, the patches are 3D boxes, and every refined level halves the cell size in every direction with more than one cell. The refined region is given by boxes (G 15) and by the interface, solid and vorticity criteria (G 24 – G 26); the pressure is solved as one composite projection over the leaf cells of all levels, and floating bodies are cast onto every grid (G 42). On several MPI ranks, the patches can be placed for the load of the ranks (G 40, G 41, G 44), the load can be reported and written as a load map for a weighted partition (G 43, G 45), and a run can stop and restart itself from a checkpoint on a new decomposition when the measured load is uneven (G 46, G 47; Section 8.4). In SEASTATE, the refinement is static: the patches are built once at the start from the boxes and the depth and coast criteria, and it is available in stand-alone runs only (Section 6.7.1).

4.1.6 Curvilinear and multi-block grids

SFLOW and NHFLOW can run on a curvilinear, boundary-fitted horizontal grid (G 50, Section 6.7.2), e.g. along a river corridor. The nodes come from DIVEMesh (G 50 1) or from the file `curvgrid.dat` (G 50 2), and the DIVEMesh grid is the logical box of the

curvilinear grid; the vertical grid is unchanged. The unknowns stay cell-centred with Cartesian velocity components, and the fluxes of every face are computed along its normal with the face lengths and cell areas of the quadrilateral cells, so that a lake at rest stays at rest on any grid. A multi-block grid consists of several curvilinear blocks with point-matching interfaces (DIVEMesh R 20 1). The blocks are run as level-0 grids of the mesh refinement, and the cells beyond their faces are filled from the neighbour blocks before every stage. In NHFLOW, the composite pressure equation is preconditioned with block-local multigrid V-cycles (G 51) and a coarse correction across the blocks (G 53); faces of the blocks can be opened as inflow or outflow with G 54, with a constant discharge or water level or with time series of them (types 3 and 4), and in NHFLOW the open boundary methods of B 520 can be placed on the faces of a block (B 522). With the semi-implicit free surface (A 534 1), the free surface solve over all blocks has a coarse space of its own (G 57). REEF3D places the blocks on the MPI ranks itself: with G 55 it cuts large blocks into slices (or, with G 58 1, into tiles) so that they spread over the ranks, and with G 56 the cut and the placement (largest load first, improved by moving and swapping blocks between the ranks) follow the load of the cells, in which dry land can count less; the interfaces of the pieces are rebuilt and checked, and G 54 and B 522 keep the block numbers of DIVEMesh. FNPF and CFD do not use the curvilinear grid and run on the Cartesian grid of DIVEMesh.

4.1.7 Wave generation and absorption

B 90 1 switches on the numerical wave tank. The wave theory is set with B 92, the wave parameters with B 91 or B 93, and the water depth for the theory with B 94. Waves are generated with relaxation zones or Dirichlet conditions (B 98) and absorbed with a numerical beach or active absorption (B 99), see Fig. 4.1.

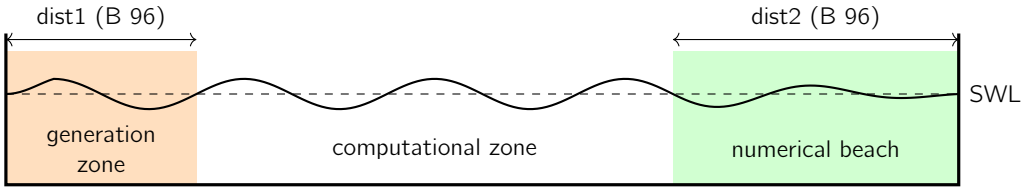

Figure 4.1: Numerical wave tank with relaxation zones for wave generation (B 98 2) and absorption (B 99 1, 2). B 108 and B 107 place the zones freely.

Relaxation method (B 98 2, B 99 1/2). Inside the zones, the solution is blended with a target solution after every Runge–Kutta stage [Jacobsen et al., 2012]:

$$\varphi = \Gamma(\tilde{x})\,\varphi_{\text{computed}} + (1-\Gamma(\tilde{x}))\,\varphi_{\text{target}}, \qquad \Gamma(\tilde{x}) = 1 - \frac{e^{\tilde{x}^{3.5}}-1}{e-1}, \qquad \tilde{x}\in[0,1], \tag{4.6}$$

so $\Gamma = 1$ keeps the computed solution and $\Gamma = 0$ imposes the target. In the generation zone, $\tilde{x} = 1 - x_g/L_g$, where x_g is the distance from the inlet (or from the B 108 line) and L_g is dist1 of B 96 (or d of B 108). The target is the wave theory multiplied by the ramp function B 101. The theory is thus fully imposed at the boundary and hands over smoothly to the computed solution. Waves reflected back towards the inlet are absorbed in the same zone. In the numerical beach, the target is still water: zero velocity, the still water level and hydrostatic pressure. Here $\tilde{x} = (L_b - x_b)/(a\,L_b)$, where x_b is the distance from the outlet (or from the B

107 line), L_b is dist2 of B 96 (or d of B 107), $a = 1$ for B 99 2 and $a = 2$ for the softer B 99 1. Additional wave trains with their own theory, direction, phase, origin and time window (B 500 – B 504) can be superposed on the B 92 wave in the target, and further generation or beach zones can be placed along any domain edge (B 520 – B 524, Section 6.2). Each module relaxes the quantities it advances in time: in CFD u_i, ϕ and p; in NHFLOW the water level, the velocities and p; in SFLOW η and the depth-averaged velocities; in FNPF η and $\tilde{\Phi}$. Typical zone lengths are one wavelength for generation and two wavelengths for the beach.

Dirichlet generation and active absorption (B 98 3/4, B 99 3–5). With Dirichlet generation, the wave theory sets η and the velocities in the inflow ghost cells, and no extra domain length is needed. With active absorption at the inlet (B 98 4), the inflow velocity is corrected with the difference between the target elevation η_T and the computed elevation η_M at the boundary, using shallow water theory: $U_c = (\eta_T - \eta_M)\sqrt{g/d}$. The active beach sets the outflow velocity to $U_c = \eta_M\sqrt{g/d}$ (B 99 3), or to $U_c = \eta_M\,\omega\cosh(k(d+z))/\sinh(kd)$ with the frequency and wave number of the input wave (B 99 4). B 99 5 distributes U_c over the depth like a flap hinged at z = B 123. The active beach also needs the matching boundary condition in DIVEMesh.

Combined and two-band absorption, SIRW (B 98, B 99 6–8). The combined methods (6) put a relaxation zone in front of an absorbing boundary: the zone relaxes towards the target plus the deviation from it low-passed over the time scale B 527 (generation) or towards the low-passed state over B 526 (beach), so that short waves are damped in the zone while long waves pass to the absorbing edge behind it (FNPF, NHFLOW; the beach also in SFLOW). The two-band active absorption (7, FNPF) splits the deviation of the surface elevation at the boundary with a first-order low pass into a long part, absorbed with the shallow water velocity, and a short part, absorbed with the linear velocity profile of the reference period B 528. The SIRW absorption (8, FNPF and NHFLOW) separates the wave travelling towards the boundary in real time from two gauges in front of it with FIR filters (Frigaard and Brorsen 1995; B 534, B 535) and absorbs it with its linear velocity profile; the remaining deviation is absorbed with the shallow water velocity. Where a module or wave type does not support an option, the run continues with a fallback and a message (Section 6.2).

Wave theories (B 92, full list and compatibility table in Section 6.2): shallow water, linear and deep water waves, Stokes waves of second and fifth order, cnoidal and solitary waves, piston and flap wavemakers, first- and second-order irregular and focused waves from a spectrum (B 85, e.g. Pierson–Moskowitz or JONSWAP), wave reconstruction, steady surface gravity waves (FNPF) and hydrodynamic coupling.

Hydrodynamic coupling (HDC) (B 92 61). Instead of a wave theory, the input comes from a previous large-scale simulation, typically FNPF (or SFLOW or NHFLOW) driving a local NHFLOW or CFD model. The large-scale run writes state files (P 40, area restricted with P 43), which DIVEMesh converts into the HDC input of the receiving model. There, η, u, v and w are interpolated linearly in time between the stored steps and linearly in space. The vertical positions are rebuilt from the stored σ levels, $z = \sigma(\eta + d - z_b) + z_b$ with the bed elevation z_b. The interpolated fields serve as the target of the relaxation zone (B 98 2) or of the Dirichlet inflow. I 231–I 233 shift the input, and B 125 and B 127 allow 2D input for a 3D domain.

Runtime coupling (B 92 62). The large-scale FNPF model and the receiving model run at the same time in one MPI job, each in its own folder with its own grid and output (Section 8.5);

no state files and no DIVEMesh step are needed. At the start, every rank of the receiver tells the FNPF ranks which FNPF columns cover its subdomain; after every FNPF time step, each FNPF rank sends η and u, v, w (and Φ for an FNPF receiver) on the σ nodes of these columns, labelled with the FNPF time. The receiver interpolates cubically (I 242 3) or linearly (I 242 1) in time between the levels, bilinearly in the horizontal and, in each FNPF column, linearly in its own σ coordinate, $z = \sigma(\eta + d - z_b) + z_b$. FNPF runs at most four levels ahead of the receiver. The receiver point (x, y) is mapped into the wave frame of B 105 (angle and origin) and shifted by I 231–I 233 into the FNPF frame, and the velocities are rotated back; a 2D FNPF model drives a 3D receiver with long-crested waves in the B 105 direction. The coupled fields are the target of the relaxation zones (B 98 2 and the edge zones B 520 on any side of the domain, so that waves scattered by a structure leave the domain while the incident waves pass), of the Dirichlet inflow and of the full initialization (I 30 1; with I 241 a hot start from a later FNPF time). An FNPF receiver (nesting) relaxes η and Φ_s towards the source, and an FNPF receiver can itself send on to a third model (I 246, e.g. regional FNPF $\rightarrow$ local FNPF $\rightarrow$ CFD).

Two feedback paths make the coupling two-way. With I 243 1 the 6DOF bodies of a CFD or NHFLOW receiver drive the FNPF bodies of the same number (X 10): the FNPF bodies follow the pose and velocity of the receiver, extrapolated from its last two samples to every FNPF stage time, so that the radiated waves in FNPF follow the motion computed in CFD. With I 244 a CFD or NHFLOW receiver samples its free surface (CFD: the zero level set and the velocity half a cell below it; NHFLOW: η and the velocity of the top σ cell) at the FNPF columns inside a box; FNPF relaxes η and Φ_s towards these values in every Runge-Kutta stage, with a weight rising from 0 at the edge of the box to 1 at the distance I 245. The target of Φ_s is integrated along the FNPF x lines from $\partial\Phi_s/\partial x = u_s + w_s\,\partial\eta/\partial x$, starting from the FNPF values at both ends of a line and blended linearly between them. Waves reflected or scattered by a structure in CFD thus enter FNPF and leave the CFD domain through its relaxation zones, which relax towards the FNPF field that now contains them. With feedback, FNPF waits at the start of each of its steps for a sample at least as recent as the level the receiver is waiting for; the feedback lags by one to two FNPF steps (I 247 0). With I 247 1 or 2, FNPF waits until the receiver has reached the FNPF time, which halves the lag; the receiver then waits for every FNPF step, which costs little when FNPF is much cheaper than the receiver. I 247 2 extrapolates the samples quadratically, with samples about as far apart as the extrapolation reaches, so that the noise of the samples is not amplified.

4.1.8 Key input commands

M 10	number of MPI processes/subdomains, as in DIVEMesh
D 10, F 35, C 15, H 15	CFD convection (4 WENO5 FLUX, 5 WENO5 HJ), D 11
A 211, A 311, A 514	WENO reconstruction/gradients in SFLOW, FNPF, NHFLOW
A 213, A 318, A 527	WENO weights (JS, Z, TENO5) in SFLOW, FNPF, NHFLOW
D 12, F 37	WENO weights of CFD momentum, level set
N 40, A 210, A 310, A 510	time schemes of CFD, SFLOW, FNPF, NHFLOW
N 47, N 48, N 49, N 50	CFL factor, adaptive on/off, fixed/maximum Δt, CFD criterion
N 10, N 11, N 44, N 46	solver, preconditioner, tolerance, max. iterations; N 12–15 REEFMG
B 90, B 92, B 91/93, B 94	wave tank, wave theory, wave parameters, depth for the theory
B 96, B 107, B 108	zone lengths, free placement of beach and generation zones
B 500–504, B 520–524	extra wave sources, edge relaxation zones
G 1–G 31	mesh refinement (all modules; CFD: G 15, G 24–26)
G 40–G 47	load balancing, load map, rebalancing; G 42 CFD bodies
G 50–G 54	curvilinear and multi-block grids (SFLOW, NHFLOW)
B 98, B 99, B 101/102	wave generation, numerical beach, ramp function
B 526–528, B 534–535	B 98, B 99 6–8: low pass, period, SIRW gauges
B 92 61, I 231–233, B 125	hydrodynamic coupling, input offsets, 2D slice
B 92 62, I 241–I 247	runtime coupling with FNPF: time offset, interpolation, two-way, chains, feedback lag

4.2 REEF3D::CFD

REEF3D::CFD (A 10 6) solves the incompressible Reynolds-averaged Navier-Stokes (RANS) equations for two immiscible phases, water and air, on a staggered rectilinear grid. The free surface is captured with the level set method, and solids and the bed enter through immersed boundary methods. The model and its validation as a numerical wave tank are described in Bihs et al. [2016a]. The shared building blocks (WENO, TVD Runge-Kutta, adaptive time stepping, linear solvers, wave generation and absorption) are described in Section 4.1.

4.2.1 Governing equations

Both phases are described by one set of equations with variable density ρ and kinematic viscosity ν. Continuity and momentum read

$$\frac{\partial u_i}{\partial x_i} = 0, \tag{4.7}$$

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[(\nu + \nu_t)\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right] + g_i + f_i, \tag{4.8}$$

with the velocities u_i, the pressure p, the eddy viscosity ν_t from the turbulence model, the gravity vector g_i = (W 20, W 21, W 22) and additional source terms f_i: a constant pressure gradient (W 29), the direct forcing of solids and floating bodies, porous media resistance, wall functions and, if W 5 > 0, surface tension with the continuum surface force approach [Brackbill et al., 1992]. Without a turbulence model (T 10 0), $\nu_t = 0$ and the equations are the Navier-Stokes equations.

The phases are distinguished by the level set function ϕ, which is the signed distance to the interface, positive in water (phase 1) and negative in air (phase 2). The material properties are blended across the interface with a smoothed Heaviside function:

$$\rho = \rho_1 H(\phi) + \rho_2 (1 - H(\phi)), \qquad \nu = \nu_1 H(\phi) + \nu_2 (1 - H(\phi)), \tag{4.9}$$

$$H(\phi) = \begin{cases} 0 & \phi < -\epsilon, \\ \frac{1}{2}\left(1 + \frac{\phi}{\epsilon} + \frac{1}{\pi}\sin\left(\frac{\pi\phi}{\epsilon}\right)\right) & |\phi| \le \epsilon, \\ 1 & \phi > \epsilon, \end{cases} \tag{4.10}$$

with ρ_1 = W 1, ν_1 = W 2, ρ_2 = W 3 and ν_2 = W 4. The half-thickness of the interface is $\epsilon = \mathrm{F}45 \cdot \overline{\Delta x}$ (default F 45 = 2.1), where $\overline{\Delta x}$ is the local cell size in the direction of the interface normal $\mathbf{n} = \nabla\phi/|\nabla\phi|$, $\overline{\Delta x} = \sqrt{n_x^2 \Delta x^2 + n_y^2 \Delta y^2 + n_z^2 \Delta z^2}$ (x and z only in 2D). On a uniform grid this is the cell size for every orientation of the interface; for a free surface on a grid stretched in the vertical direction it is the vertical cell size Δz, so that grid stretching towards the free surface is taken into account. Where the density is needed at a cell face (pressure gradient and pressure Poisson equation), the level set is first averaged to the face and H is evaluated there. This avoids averaging the density itself across the density jump [Bihs et al., 2016a].

4.2.2 Pressure-velocity coupling

The velocity components are stored at the cell faces, and the pressure, the level set and the turbulence quantities at the cell centres (staggered grid). Velocity and pressure are coupled with Chorin's projection method [Chorin, 1968] in incremental pressure-correction form (D 30 1). In every Runge-Kutta stage with coefficient α ($\alpha = 1, \frac{1}{4}, \frac{2}{3}$ for the three stages of the third-order scheme), an intermediate velocity u_i^* is computed from Eq. (4.8) with the pressure p^{old} of the previous stage. The pressure correction p' then follows from a Poisson equation with variable density:

$$\frac{\partial}{\partial x_i}\left(\frac{1}{\rho}\frac{\partial p'}{\partial x_i}\right) = \frac{1}{\alpha \Delta t}\frac{\partial u_i^*}{\partial x_i}. \tag{4.11}$$

Pressure and velocity are corrected as

$$p = p^{\mathrm{old}} + p', \qquad u_i = u_i^* - \frac{\alpha \Delta t}{\rho}\frac{\partial p'}{\partial x_i}, \tag{4.12}$$

which makes u_i divergence-free. Eq. (4.11) is solved with the solver selected in N 10 and N 11 (Section 4.1), and the pressure level is fixed with B 30 – B 33. With D 30 10, no Poisson equation is solved: the pressure is computed hydrostatically from the level set function, and only this pressure gradient enters the velocity update. With W 32 1 or 2, the air phase is treated as isentropically compressible, e.g. for trapped air pockets in wave impacts: the Poisson equation obtains the term $\frac{1}{\gamma\, p_{abs}}\frac{Dp}{Dt} = -\partial u_i/\partial x_i$ in the air, with the ratio of specific heats γ (W 33) and the absolute pressure p_{abs} = W 36 + p, using the local (W 32 1) or the material (W 32 2) time derivative. The water stays incompressible and the density of the air is not changed. In a band of air next to the interface (W 34), the compressibility is switched off, and this is compensated in the adjacent layer (W 35). W 32 needs the projection method and the level set method and cannot be combined with W 30.

4.2.3 Free surface: level set method

The free surface is the zero contour of the level set function ϕ [Osher and Sethian, 1988], which is advected with the flow:

$$\frac{\partial \phi}{\partial t} + u_j \frac{\partial \phi}{\partial x_j} = 0. \tag{4.13}$$

The convection term is discretized with the fifth-order WENO scheme in Hamilton-Jacobi form (F 35 5, default), with the nonlinear weights of F 37 (WENO-JS by default). In the coupled momentum time schemes (N 40 = 2, 3, 4, 22, 23, 24, 33), Eq. (4.13) is advanced inside every Runge-Kutta stage of the momentum equations, followed by an update of density and viscosity. In the separate schemes (N 40 = 12, 13, 44), the level set has its own time scheme F 30. In both cases F 30 > 0 activates the two-phase properties.

Since advection destroys the signed distance property on which Eq. (4.10) relies, ϕ is reinitialized with the PDE

$$\frac{\partial \phi}{\partial \tau} + S(\phi)\left(\left|\frac{\partial \phi}{\partial x_j}\right| - 1\right) = 0, \qquad S(\phi) = \frac{\phi}{\sqrt{\phi^2 + \left|\partial\phi/\partial x_j\right|^2 \overline{\Delta x}^2}}, \tag{4.14}$$

in pseudo time τ [Sussman et al., 1994]. The smoothed sign function $S(\phi)$ follows Peng et al. [1999]; here $\overline{\Delta x}$ is the local average cell size. The gradient is discretized with a Godunov-type upwind selection of fifth-order WENO derivatives, and Eq. (4.14) is integrated with the third-order TVD Runge-Kutta scheme (F 40 3). The pseudo time step is $\Delta\tau = \mathrm{F}43 \cdot \min(\Delta x, \Delta y, \Delta z)$ (default F 43 = 0.55). F 44 reinitialization steps are carried out per time step (default 3). In the coupled scheme N 40 3 the number is set internally to two, two and three steps in the three Runge-Kutta stages. At the start, ϕ is initialized from F 51 – F 72 and reinitialized with $2L/(\mathrm{F}43\,\overline{\Delta x})$ steps, where L is the maximum domain length or F 42. Optionally, the level set is shifted by a constant after advection and reinitialization so that the water volume is conserved (F 46, F 47).

As an alternative, the volume of fluid (VOF) method with the geometric PLIC reconstruction of the interface can be activated with F 80 4; the former algebraic compressive VOF schemes have been removed. The VOF implementation is in beta; the open channel flow and wave generation algorithms are designed for the level set method.

4.2.4 Turbulence modelling

The turbulence model is selected with T 10. The RANS model mainly used for wave and coastal applications is the k-ω model [Wilcox, 1994] (T 10 2):

$$\frac{\partial k}{\partial t} + u_j \frac{\partial k}{\partial x_j} = \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\nu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right] + P_k - \beta_k\, k\,\omega, \tag{4.15}$$

$$\frac{\partial \omega}{\partial t} + u_j \frac{\partial \omega}{\partial x_j} = \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\nu_t}{\sigma_\omega}\right)\frac{\partial \omega}{\partial x_j}\right] + \alpha\,\frac{\omega}{k}\,P_k - \beta\,\omega^2, \tag{4.16}$$

with the production $P_k = \nu_t\, 2S_{ij}S_{ij}$, where $S_{ij} = \frac{1}{2}\left(\partial u_i/\partial x_j + \partial u_j/\partial x_i\right)$. The closure coefficients in the code are $\alpha = 5/9$, $\beta = 3/40$, $\beta_k = c_\mu = 0.09$ and $\sigma_k = \sigma_\omega = 2$. Both equations are solved implicitly in time, and their convection terms are discretized with T 12.

The eddy viscosity is $\nu_t = k/\omega$. With T 34 1 (default), it is bounded by a strain-based limiter [Durbin, 2009]:

$$\nu_t = \max\!\left(\min\!\left(\frac{k}{\omega},\ \mathrm{T}31\,\frac{k}{|S|}\right),\ 10^{-4}\,\nu\right), \qquad |S| = \sqrt{2S_{ij}S_{ij}}, \tag{4.17}$$

with T 31 = $1/\sqrt{3}$ by default. This prevents the over-production of turbulence under propagating waves. T 41 activates the stabilization of Larsen and Fuhrman [2018] for nearly potential flow regions, $\nu_t \le (k/\omega)\, c_\mu \alpha\, (2\Omega_{ij}\Omega_{ij})/(\lambda_2 \beta\, 2S_{ij}S_{ij})$, where Ω_{ij} is the rotation tensor and λ_2 = T 42.

To avoid spurious turbulence production at the density jump of the free surface, T 36 > 0 overwrites ω in a band $|\phi| < \epsilon_t$ around the interface with a free surface value based on Naot and Rodi [1982]:

$$\omega = \delta(\phi)\,\frac{c_\mu^{-1/4}\sqrt{k}}{\kappa\, y'}, \qquad \delta(\phi) = \frac{1}{2\epsilon_t}\left(1 + \cos\!\left(\frac{\pi\phi}{\epsilon_t}\right)\right), \tag{4.18}$$

with $\kappa = 0.4$, $\epsilon_t = \mathrm{T}38 \cdot \overline{\Delta x}$ and the virtual origin y' = T 37. T 36 2 adds the inverse wall distance to $1/y'$, and with T 36 3, y' is T 37 times the local water depth. T 39 bounds ν_t near the interface from below by a Smagorinsky-type eddy viscosity. T 45 adds a buoyancy term to the k-equation, and T 33 adds the isotropic part of the Reynolds stress, $\frac{2}{3}\partial k/\partial x_i$, to the momentum equations.

Further options of T 10 are the k-ϵ model (1; $c_{1\epsilon} = 1.44$, $c_{2\epsilon} = 1.92$, $\sigma_k = 1.0$, $\sigma_\epsilon = 1.3$), an explicit algebraic Reynolds stress model based on k-ω (12) and URANS variants (21, 22). For T 10 22, ν_t is multiplied by $\min\!\left(1,\ \mathrm{T}23\, c_\mu\, \Delta\omega/\sqrt{k}\right)$ with $\Delta = (\Delta x \Delta y \Delta z)^{1/3}$. For large eddy simulation, T 10 31 selects the Smagorinsky model [Smagorinsky, 1963], $\nu_t = (C_s\Delta)^2|\tilde{S}|$ with $C_s = 0.2$ and the filtered velocity field (filter T 21), and T 10 33 the WALE model [Nicoud and Ducros, 1999] with $C_w = 0.6$.

4.2.5 Boundaries, solids and topography

Ghost cells. The outer boundaries of the domain and the cells that DIVEMesh excludes from the grid are treated with the ghost cell method. The ghost cell values are extrapolated along the grid line normal to the boundary, following the local directional ghost cell approach of Berthelsen and Faltinsen [2008]. The extrapolation uses Lagrange polynomials through the interior values and the boundary value at the wall location. B 20 selects slip or no-slip conditions for the velocities, B 23 extrapolation or reflection, and B 29 the image point distance for the Dirichlet extrapolation of B 20 3.

Direct forcing. Solids and the bed topography from DIVEMesh are described by their own signed distance functions, ϕ_s for solids and ϕ_b for the topography. These are reinitialized with the same PDE as the free surface level set (G 540, Section 6.7). The no-slip condition is imposed with a continuous direct forcing immersed boundary method [cf. Fadlun et al., 2000]. After the predictor step of every Runge-Kutta stage, the forcing

$$f_i = H_s\,\frac{u_{s,i} - u_i^*}{\alpha \Delta t} \tag{4.19}$$

is added to the intermediate velocity before the projection step. H_s is the smoothed Heaviside function of Eq. (4.10), evaluated for $-\min(\phi_s, \phi_b)$ (only ϕ_s or only ϕ_b if there is no topography forcing or no solid, respectively) at the velocity location with the half-width $\mathrm{X}\,41 \cdot \overline{\Delta x}$ (default X 41 = 0.6), and $u_{s,i}$ is the velocity of the solid (zero for fixed solids). With B 20 2 and B 21 1 (defaults), the forcing in the transition layer outside the body is weighted with the components of the surface normal, which preserves the tangential velocity. D 22 selects how the implicit diffusion treats neighbour cells inside direct-forcing regions, and X 48 extrapolates ϕ, k and ω into the solid.

Wall functions. With B 10 1, the wall shear stress from the rough-wall logarithmic law is applied as a momentum sink in the cells next to ghost cell walls and direct-forcing surfaces:

$$f_i = -\frac{|u_i|\,u_i}{(u^+)^2\,\Delta n}, \qquad u^+ = \frac{1}{\kappa}\ln\!\left(\frac{30\,y}{k_s}\right), \qquad y = \max\!\left(\frac{\Delta n}{2},\ \frac{k_s}{30}\right), \tag{4.20}$$

where u_i is a velocity component tangential to the wall, Δn the cell size normal to the wall, $\kappa = 0.4$ and k_s the equivalent sand-grain roughness (B 50 – B 56). For the k-ω model, the wall value $\omega = \sqrt{k}/(\kappa\, y\, c_\mu^{1/4})$ is used, and the production of k in the wall cells is computed from the log-law wall shear stress.

4.2.6 Discretization summary

Momentum convection uses the conservative fifth-order WENO scheme (D 10 4, default) of Jiang and Shu [1996] (WENO-JS weights; WENO-Z or TENO5 with D 12), with the convecting velocities interpolated to the faces according to D 11; lower-order, TVD and central schemes are alternatives. Diffusion is implicit by default (D 20 2), which removes the viscous time step restriction. Only the part $\partial/\partial x_j[(\nu+\nu_t)\partial u_i/\partial x_j]$ is implicit; the transposed part is explicit. D 20 1 is explicit and D 20 3 Crank-Nicolson (3D). The time scheme N 40 defaults to the third-order TVD Runge-Kutta scheme with the level set coupled into each stage (3). Alternatives are a separate level set step (13), low-storage variants (4, 44) and a conservative formulation with face densities (33). The time step follows the CFL criterion with N 48 1 and N 47 (N 50), see Section 4.1.

4.2.7 Multiphysics coupling

Floating bodies (X). With X 10 1, rigid bodies move in six degrees of freedom (X 11). Each body is represented by a level set function from an STL file or primitives. Force and moment are integrated from the pressure and viscous stresses over the triangulated body surface, and the Newton-Euler equations are advanced within the Runge-Kutta stages of the fluid solver. The body velocity is imposed with the direct forcing of Eq. (4.19) (X 41, X 14). With X 10 >0, N 40 3 is changed to 4 (not with the mesh refinement, G 1 >0), N 40 13 to 44, and D 22 is set to 2.

Sediment transport (S, Q). With S 10 1, the bed is the zero contour of the topography level set $\phi_b = z - z_b(x, y)$. It enters the flow through the direct forcing described above. The bed level z_b is updated with the Exner equation

$$(1-n)\frac{\partial z_b}{\partial t} + \frac{\partial q_{b,x}}{\partial x} + \frac{\partial q_{b,y}}{\partial y} = 0, \tag{4.21}$$

where n is the bed porosity (S 24) and q_b the bedload transport from the formula selected in S 11 with the bed shear stress of S 16. With suspended load (S 12), the erosion and deposition fluxes are added. S 10 2 treats the bed as a porous VRANS region, and Q 10 1 or 2 replaces the Exner model by the particle model of Section 6.14.

Porous media (B 270 – B 295). In porous zones (porosity n, grain size d_{50}), the volume-averaged (VRANS) form of the equations is solved. Its main terms are a linear and a nonlinear resistance force [see Jensen et al., 2014]:

$$f_i = -a\,u_i - b\,|u_i|\,u_i, \qquad a = \alpha_p \frac{(1-n)^2}{n^3}\frac{\nu}{d_{50}^2}, \qquad b = \beta_p\left(1 + \frac{7.5}{KC}\right)\frac{1-n}{n^3}\frac{1}{d_{50}}, \tag{4.22}$$

where the coefficients α_p and β_p are given with the porous objects, KC is set with B 264 (no correction by default), and the added mass coefficient B 260 scales the time derivative inside the porous media, as $C_m = c$ or, by default, as $C_m = c(1-n)/n$ (B 266). By default, the resistance is treated point-implicitly in the Runge-Kutta stages (B 268 1), which is unconditionally stable. VRANS is switched on automatically when porous objects are defined. B 295 controls the turbulence source terms in porous zones.

Heat and concentration (H, C). H 10 solves an advection-diffusion equation for the temperature, and the fluid properties then follow from H 3 (tabulated properties or the Boussinesq approximation). C 10 solves a transport equation for a concentration c. The concentration changes density and viscosity, $\rho = (\rho_1 + c\,\mathrm{C}1)H + (\rho_2 + c\,\mathrm{C}3)(1-H)$, and correspondingly for ν with C 2 and C 4.

4.2.8 Solution procedure

Figure 4.2 summarizes the solution procedure of REEF3D::CFD as it is executed by the code for the default time scheme N 40 3. After the control file and the grid have been read and all solver objects have been initialized, every time step first updates the inflow and wave generation, turbulence and sediment transport, and then advances the level set function and the velocities together in the three stages of the TVD Runge-Kutta scheme. In each stage, the pressure projection of Eqs. (4.11) and (4.12) follows the momentum predictor and the direct forcing of solid and floating bodies, before the fluid properties are updated with the new level set. The adaptive time step for the next step is computed at the end of each time step, after which the output is written.

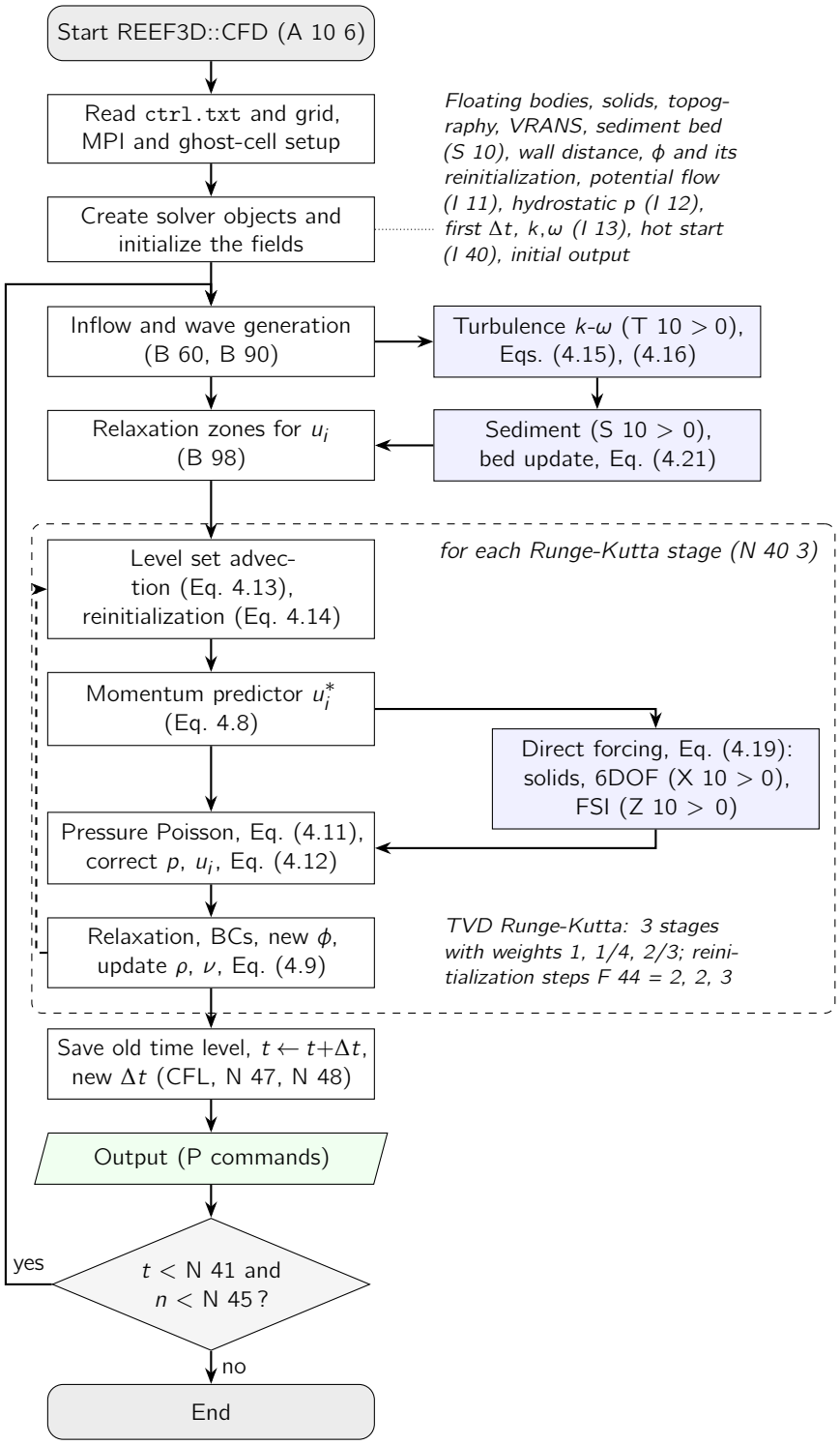

Figure 4.2: Solution procedure of REEF3D::CFD for the default time scheme N 40 3, in which the level set equation is advanced within each Runge-Kutta stage of the momentum equations (the dashed arrow marks the next stage), and optional models are drawn as shaded boxes. With X 10 > 0 the scheme N 40 3 is replaced by its low-storage variant N 40 4 with the same stage order; for the separate schemes N 40 12, 13 and 44 the level set is instead advanced with its own time scheme (F 30) and reinitialized before the turbulence step.

4.2.9 Key input commands

A 10 6	select REEF3D::CFD
W 1 – W 5, W 20 – W 22	fluid properties, surface tension, gravity
D 10, D 11, D 12, D 20	momentum convection, convecting velocities, WENO weights, diffusion
D 30, B 30 – B 33	pressure algorithm, pressure reference
N 10, N 11	Poisson solver and preconditioner
N 40, N 47, N 48, N 50	momentum time scheme, adaptive time step
F 30, F 35, F 37, F 45	level set time and convection scheme, WENO weights, interface half-thickness ϵ
F 40, F 42 – F 44	reinitialization
F 46, F 47	level set volume correction
F 80	VOF-PLIC (beta)
T 10, T 12	turbulence model, convection scheme of k and ω
T 31, T 34, T 41, T 42	eddy viscosity limiter, k-ω stabilization
T 36 – T 39	free surface turbulence damping
B 10, B 50 – B 56	wall functions, roughness
B 20 – B 29, X 41, G 540	ghost cells, direct forcing, solid and bed level sets
G 1, G 15, G 24 – G 26	mesh refinement (Section 6.7.1)
X 10	6DOF floating bodies
S 10, Q 10	sediment transport, particle model
L 10 – L 62	Lagrangian particle tracking (Section 6.10)
B 260 – B 295	VRANS porous media; B 266 added mass form, B 268 drag treatment
W 32 – W 36	isentropic air compressibility
H 10, C 10	heat transfer, concentration

4.3 REEF3D::NHFLOW

REEF3D::NHFLOW (A 10 5) solves the incompressible Navier-Stokes equations for a single-valued free surface on a three-dimensional σ-grid. The hydrostatic part of the pressure is treated together with the free surface in a shock-capturing Godunov-type finite volume scheme, while the non-hydrostatic (dynamic) pressure is obtained from a Poisson equation. The method is described in Bihs and Wang [2025] and Bihs et al. [2026]; this section summarises the formulation as it is implemented in the `nhflow_*` classes.

4.3.1 Governing equations

The flow is governed by continuity and momentum conservation,

$$\frac{\partial u_i}{\partial x_i} = 0, \tag{4.23}$$

$$\frac{\partial u_i}{\partial t} + \frac{\partial (u_i u_j)}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[(\nu + \nu_t)\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right] + g_i. \tag{4.24}$$

The free surface $z = \eta(x, y, t)$ is single-valued. The still water level is F 60, the bed lies at $z = -d(x, y)$ below it, and $h = \eta + d$ is the total water depth. The pressure is split into a hydrostatic and a non-hydrostatic part,

$$p = \rho g (\eta - z) + p_{nh}, \tag{4.25}$$

where $g = |\text{W } 22|$ and ρ is W 1. In the following, p stands for p_{nh} only. With the meteorological forcing (W 200, Section 5.11.1), the wind stress acts on the top σ-layer and the gradient of the air pressure in every layer; the Coriolis force (W 211) adds $f\,hv$ and $-f\,hu$ to the x- and y-momentum of every layer. Integration of (4.23) over the water column together with the kinematic conditions at the free surface and at the bed,

$$w\big|_{\eta} = \frac{\partial \eta}{\partial t} + u\frac{\partial \eta}{\partial x} + v\frac{\partial \eta}{\partial y}, \qquad w\big|_{-d} = -u\frac{\partial d}{\partial x} - v\frac{\partial d}{\partial y}, \tag{4.26}$$

gives the free surface equation

$$\frac{\partial \eta}{\partial t} + \frac{\partial}{\partial x}\int_{-d}^{\eta} u\,dz + \frac{\partial}{\partial y}\int_{-d}^{\eta} v\,dz = 0. \tag{4.27}$$

Since d does not change in time (except with sediment transport), the code advances h with (4.27) and sets $\eta = h - d$.

4.3.2 Sigma-coordinate transformation

The vertical coordinate is mapped onto $\sigma \in [0, 1]$ from the bed to the free surface,

$$\sigma = \frac{z + d(x, y)}{h(x, y, t)}, \tag{4.28}$$

so that every water column has the same number of cells, independent of the local depth (Figure 4.3). The vertical node distribution is taken from the DIVEMesh grid and normalised to $[0, 1]$, so vertical stretching towards the free surface is set in DIVEMesh (B 103, B 113, B 116); the number of vertical cells is the number of cells of the DIVEMesh grid in z-direction. Derivatives are transformed with the chain rule,

$$\frac{\partial}{\partial x} \to \frac{\partial}{\partial x} + \sigma_x\frac{\partial}{\partial \sigma}, \quad \frac{\partial}{\partial y} \to \frac{\partial}{\partial y} + \sigma_y\frac{\partial}{\partial \sigma}, \quad \frac{\partial}{\partial z} \to \sigma_z\frac{\partial}{\partial \sigma}, \quad \frac{\partial}{\partial t} \to \frac{\partial}{\partial t} + \sigma_t\frac{\partial}{\partial \sigma}, \tag{4.29}$$

with the metric coefficients

$$\sigma_x = \frac{1-\sigma}{h}\frac{\partial d}{\partial x} - \frac{\sigma}{h}\frac{\partial \eta}{\partial x}, \quad \sigma_y = \frac{1-\sigma}{h}\frac{\partial d}{\partial y} - \frac{\sigma}{h}\frac{\partial \eta}{\partial y}, \quad \sigma_z = \frac{1}{h}, \quad \sigma_t = -\frac{\sigma}{h}\frac{\partial \eta}{\partial t}. \tag{4.30}$$

The horizontal free surface gradients in (4.30) are discretised as selected with A 524. The metric terms are updated at every Runge-Kutta stage and are set to zero in dry columns.

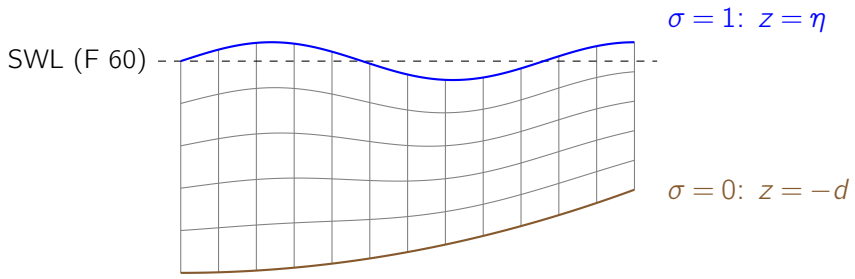

Figure 4.3: Sketch of the NHFLOW σ-grid: each water column is divided into the same number of layers between the bed and the free surface.

Applying the transformation to (4.23)–(4.24) and multiplying by h gives the conservative form solved by NHFLOW. Local continuity in the σ-layer reads

$$\frac{\partial h}{\partial t}+\frac{\partial (hu)}{\partial x}+\frac{\partial (hv)}{\partial y}+\frac{\partial \omega}{\partial \sigma}=0, \tag{4.31}$$

where ω is the vertical velocity relative to the moving σ-surfaces (scaled with h). It is not a prognostic variable: the code integrates (4.31) from the bed ($\omega=0$) upwards layer by layer, using the same numerical fluxes of hu and hv as the free surface equation; ω is also zero at the free surface. Summing (4.31) over all layers gives the discrete form of (4.27), $\partial h/\partial t=-\sum_k \Delta\sigma_k\,[\partial(hu)_k/\partial x+\partial(hv)_k/\partial y]$.

The momentum equations are written for the conserved variables $\mathbf{U}=(hu,hv,hw)^T$, located at the cell centres,

$$\frac{\partial \mathbf{U}}{\partial t}+\frac{\partial \mathbf{F}}{\partial x}+\frac{\partial \mathbf{G}}{\partial y}+\frac{\partial \mathbf{H}}{\partial \sigma}=\mathbf{S}, \tag{4.32}$$

$$\mathbf{F}=\begin{pmatrix} hu^2+\frac{1}{2}g\eta^2+g\eta d \\ huv \\ huw \end{pmatrix},\quad \mathbf{G}=\begin{pmatrix} hvu \\ hv^2+\frac{1}{2}g\eta^2+g\eta d \\ hvw \end{pmatrix},\quad \mathbf{H}=\begin{pmatrix} \omega u \\ \omega v \\ \omega w \end{pmatrix}, \tag{4.33}$$

$$\mathbf{S}=\begin{pmatrix} g\eta\dfrac{\partial d}{\partial x}-\dfrac{h}{\rho}\left(\dfrac{\partial p}{\partial x}+\sigma_x\dfrac{\partial p}{\partial \sigma}\right) \\ g\eta\dfrac{\partial d}{\partial y}-\dfrac{h}{\rho}\left(\dfrac{\partial p}{\partial y}+\sigma_y\dfrac{\partial p}{\partial \sigma}\right) \\ -\dfrac{1}{\rho}\dfrac{\partial p}{\partial \sigma} \end{pmatrix}+\mathbf{S}_D+\mathbf{S}_{ext}. \tag{4.34}$$

The hydrostatic pressure gradient $gh\partial\eta/\partial x$ is split into the flux part $\partial(\frac{1}{2}g\eta^2+g\eta d)/\partial x$ and the bed slope source $g\eta\,\partial d/\partial x$. In the code the bed slope term is evaluated with the same face depths $d_{i\pm1/2}$ (arithmetic mean of the neighbouring columns) that enter the flux, so that for a lake at rest (η = const, $u_i=0$) the flux difference and the source cancel exactly. $\mathbf{S}_D$ contains the viscous and turbulent diffusion of hu_i in σ-coordinates (below), and $\mathbf{S}_{ext}$ collects the optional sources: direct forcing of solids and floating bodies, VRANS porous media (B 200 ff.), wind forcing (A 570 ff.), bed and wall friction (A 519) and the relaxation zones of the wave generation.

4.3.3 Shock-capturing finite volume scheme

Reconstruction. The face values left and right of each cell face are reconstructed from the cell averages of η, u, v, w, hu, hv and hw. The free surface is reconstructed through η (not h); the face depth is then $D_{L,R}=\max(\eta_{L,R}+d_{i+1/2}$, A 544). The scheme is chosen with A 514: options 1–3 use a limited linear (MUSCL-type) reconstruction with the van Leer, Superbee or van Albada limiter; with A 514 4 (default) the horizontal reconstruction is the fifth-order WENO scheme in its formulation for non-uniform grids, by default with the WENO-JS weights [Jiang and Shu, 1996] (A 527 0; 1 WENO-Z, 2 TENO5 with the cutoff A 528; see Section 4.1), and the vertical one is linear with the Superbee limiter; A 514 5 applies WENO in all three directions. The limited slopes are set to zero next to dry cells.

Riemann solver. The horizontal numerical fluxes are computed with the HLL approximate Riemann solver (A 511 1) [Harten et al., 1983, Toro, 2009],

$$\mathbf{F}_{i+1/2} = \begin{cases} \mathbf{F}_L & \text{if } S_L \geq 0, \\ \dfrac{S_R \mathbf{F}_L - S_L \mathbf{F}_R + S_L S_R (\mathbf{U}_R - \mathbf{U}_L)}{S_R - S_L} & \text{if } S_L < 0 < S_R, \\ \mathbf{F}_R & \text{if } S_R \leq 0, \end{cases} \tag{4.35}$$

where $\mathbf{F}_{L,R}$ are the physical fluxes evaluated with the reconstructed states. The same formula is used for the fluxes of the free surface equation (flux hu, jump $D_R - D_L$). The wave speed estimates are those of the two-rarefaction approximation for shallow water [Toro, 2009], with $c_{L,R} = \sqrt{g D_{L,R}}$:

$$\begin{aligned} u^* &= \tfrac{1}{2}(u_L + u_R) + c_L - c_R, & c^* &= \tfrac{1}{2}(c_L + c_R) + \tfrac{1}{4}(u_L - u_R), \\ S_L &= \min(u_L - c_L,\, u^* - c^*), & S_R &= \max(u_R + c_R,\, u^* + c^*). \end{aligned} \tag{4.36}$$

At a wet-dry face the dry-bed estimates are used, e.g. $S_L = u_R - 2c_R$, $S_R = u_R + c_R$ for a dry left cell. With A 511 2 the HLLC solver is used instead, which restores the contact wave with the speed $S^* = u^*$ and the star states $D_K(S_K - u_K)/(S_K - S^*)$ times the transported velocity ($K = L, R$); next to dry cells it falls back to HLL. The vertical fluxes $\mathbf{H}$ are upwinded with the sign of ω at the σ-face, using the vertically reconstructed velocities. Because the method is conservative and shock-capturing, bores and broken waves are represented as moving discontinuities without additional treatment.

Diffusion. Viscous and turbulent diffusion (A 512) is applied to hu_i with the Laplacian transformed to σ-coordinates, including the metric terms $(\sigma_x^2 + \sigma_y^2 + \sigma_z^2)\partial^2/\partial\sigma^2$ and the mixed derivatives. The implicit option (A 512 2) solves the resulting system for each velocity component with BiCGStab; A 513 sets the treatment next to solid walls. A 512 is set to 2 automatically when a turbulence model is used.

4.3.4 Non-hydrostatic pressure

After the momentum predictor, the non-hydrostatic pressure enforces the continuity equation (4.23). The pressure is stored at the vertical cell faces (the σ-nodes), the velocities at the cell centres. A 520 selects the scheme: 0 is hydrostatic (no pressure solution), 1 is a projection method (PJM) which solves for the full p each Runge-Kutta stage from a predictor without non-hydrostatic pressure gradient, and 2 (default) is an incremental pressure-correction method [Zijlema et al., 2005]: the predictor includes the gradient of the old pressure p^n, a correction p' is solved, and $p^{n+1} = p^n + p'$. For option 2, the Poisson equation in σ-coordinates reads

$$\begin{aligned} &\frac{\partial^2 p'}{\partial x^2} + \frac{\partial^2 p'}{\partial y^2} + (\sigma_x^2 + \sigma_y^2 + \sigma_z^2)\frac{\partial^2 p'}{\partial \sigma^2} + 2\sigma_x \frac{\partial^2 p'}{\partial x\, \partial \sigma} + 2\sigma_y \frac{\partial^2 p'}{\partial y\, \partial \sigma} + (\sigma_{xx} + \sigma_{yy})\frac{\partial p'}{\partial \sigma} \\ &\qquad = \frac{\rho}{\alpha \Delta t}\left(\frac{\partial u^*}{\partial x} + \sigma_x \frac{\partial u^*}{\partial \sigma} + \frac{\partial v^*}{\partial y} + \sigma_y \frac{\partial v^*}{\partial \sigma} + \sigma_z \frac{\partial w^*}{\partial \sigma} \right), \end{aligned} \tag{4.37}$$

where u_i^* is the predicted velocity and α the weight of the current Runge-Kutta stage. The mixed derivatives are evaluated explicitly on the right-hand side (deferred correction), so that the matrix has a seven-point stencil. This lagged treatment can diverge where the free surface is very steep, e.g. at a run-up front. With A 529 1, the mixed derivatives are assembled into the matrix instead (15-point operator, 9-point in 2D), as A 328 1 in FNPF; this requires REEFMG (N 10 1) on a Cartesian grid without mesh refinement. Next to porous cells, the cross terms are left out unless A 530 1 is set. In the bottom layer, the vertical interpolation of the horizontal velocities in the divergence is reduced over steep slopes with A 522. The velocities are then corrected with

$$(hu)^{n+1} = (hu)^* - \alpha\Delta t\,\frac{h}{\rho}\left(\frac{\partial p'}{\partial x} + \sigma_x\frac{\partial p'}{\partial \sigma}\right), \quad (hv)^{n+1} = (hv)^* - \alpha\Delta t\,\frac{h}{\rho}\left(\frac{\partial p'}{\partial y} + \sigma_y\frac{\partial p'}{\partial \sigma}\right),$$
$$(hw)^{n+1} = (hw)^* - \frac{\alpha\Delta t}{\rho}\frac{\partial p'}{\partial \sigma}. \tag{4.38}$$

At the free surface the Dirichlet condition $p = 0$ is imposed. At the bed, at closed side walls and next to solid or dry cells the coupling to the neighbouring ghost node is removed from the matrix; with A 520 1 the ghost value is replaced by the pressure of the current cell (zero gradient). At the outflow boundary $p = 0$ is prescribed, at the inflow either $p = 0$ (wave generation) or a zero gradient (B 76). The pressure is only solved in "deep" cells: wet columns that are deeper than A 545·A 544 and not directly next to dry cells. In the shallow swash zone the model therefore reduces to the hydrostatic, shock-capturing formulation. The linear system is solved with the solver selected by N 10, by default REEF3D's semicoarsening multigrid REEFMG (N 10 1) or, in builds with HYPRE, a HYPRE solver (N 10 11–19 Struct, 21–25 AIJ, 31–38 SStruct, preconditioner N 11), see Section 4.1. Option A 520 3 is an alternative non-hydrostatic scheme with its own flux formulation.

4.3.5 Wetting and drying, breaking waves and turbulence

Wetting and drying. A column is dry when $h \leq$ A 544 (default 0.001 m); its depth is then kept at A 544 and its velocities are set to zero. With A 540 1 (default) a dry cell only becomes wet when a wet neighbour has a higher free surface, following the approach of Roeber et al. [2010]; A 540 2 decides from the depth threshold alone. Isolated wet cells are dried. With A 540 1, the free surface and depth of the wet cell are used on both sides of a wet-dry face and the face velocities are set to zero; in the Riemann solver the dry-bed wave speeds are used. The Froude number limiter $|hu_i| \leq$ A 531·$h\sqrt{gh}$ (area A 532) keeps the velocities bounded in very shallow water.

Wave breaking. Breaking bores are captured by the Godunov-type scheme itself, which dissipates energy at the wave front without a breaking criterion. In addition, A 550 1 activates a breaking-wave viscosity. Breaking is detected (A 551) either in shallow water from the rise rate of the free surface, $\partial\eta/\partial t >$ A 554·$\sqrt{gh}$, or in deep water from the free-surface slope $|\partial\eta/\partial x|$ or $|\partial\eta/\partial y| >$ A 555. In the flagged columns and their direct neighbours the artificial viscosity A 557 is added to the diffusion coefficient (half the value in the second ring of neighbours); A 552 additionally filters the water depth in breaking columns. The breaking viscosity acts through the implicit diffusion, so A 550 1 sets A 512 to 2.

Turbulence. A 560 selects the turbulence closure: the k-ϵ (1) or k-ω (2) model, their URANS-filtered variants (21, 22, filter coefficient A 568), or the Smagorinsky LES model (31). The transport equations of the two-equation models are solved implicitly on the σ-grid, with the same closure coefficients as in CFD (Section 4.2); options A 564–A 567 control the eddy viscosity limiter, the stabilisation in nearly potential flow regions, buoyancy and the free surface condition for ϵ/ω. The eddy viscosity ν_t enters the implicit diffusion of (4.32). For shallow flows with large horizontal eddies, A 561 splits the eddy viscosity into a vertical part, the ν_t of the two-equation model, and a horizontal part that contains the Smagorinsky viscosity of the horizontal strain (A 562), so that the large horizontal eddies are resolved on the grid (horizontal large eddy simulation, HLES). A 563 extends the Reynolds stresses with the quadratic constitutive relation (QCR) of Spalart (2000), whose anisotropic normal stresses drive secondary currents of the second kind, e.g. in corners and narrow channels. Both need the k-ϵ or k-ω model (A 560 1, 2, 21, 22) and are switched off with a message otherwise. The k-ω model (A 560 2, 22) can be extended by the explicit algebraic Reynolds stress model (EARSM) of Wallin and Johansson [2000] (A 559 1): the Reynolds stress anisotropy follows from the strain and rotation tensors, its linear part gives the eddy viscosity of the implicit diffusion, and the remaining anisotropy is added as an explicit momentum source that drives secondary currents like the QCR, which it replaces (A 563 is then switched off). With other turbulence models, A 559 is switched off with a message.

4.3.6 Time stepping

The momentum equations and the free surface equation are integrated together with the second- (A 510 2, default) or third-order (A 510 3) TVD Runge-Kutta scheme (Section 4.1). Each stage consists of: update of the σ-metrics, reconstruction and wave speeds, free surface update with (4.27) and wetting-drying, computation of ω, momentum predictor with the HLL fluxes and sources, implicit diffusion, direct forcing, pressure solution and velocity correction, and finally the relaxation zones of the wave generation and absorption (B 98, B 99; Section 4.1). With adaptive time stepping (N 48 1) the time step is

$$\Delta t = \text{N } 47 \cdot \min\left(\frac{\Delta x}{|u|_{max} + \sqrt{g\, h_{max}}},\ \frac{\Delta x}{|w|_{max}} \right), \tag{4.39}$$

with Δx the smallest horizontal cell size, limited from above by N 49. A 533 1 adds the vertical Courant number $h\Delta\sigma/|w|_{max}$ and a source-term criterion.

Semi-implicit free surface. For long hydrostatic simulations such as tides, rivers and coastal flows, A 534 1 replaces the explicit free surface by a semi-implicit θ method (Casulli and Cattani 1994; θ = A 535). The explicit terms are integrated with the SSP Runge-Kutta predictor of A 510, and the free surface elevation follows from one 2D linear system per time step; the vertical diffusion and advection, the bed friction and the pressure gradient are implicit in each water column. With A 537 1, the system is solved with the nonlinear water depth $\max(\eta+d, 0)$ and iterated on the wet and dry pattern (Casulli 2009), which keeps the volume exact at the wet-dry front. A 538 adds an implicit mass flux dissipation for transcritical flow, and A 539 the non-orthogonal part of the free surface gradient on curvilinear grids. In thin water at wet-dry fronts, the velocities of the predictor are desingularised, $u = \sqrt{2}\, h\,(hu)/\sqrt{h^4 + h_v^4}$

with h_v = A 546 (Kurganov and Petrova 2007), and A 548 can take the still water depth of a face from the higher bed of its two cells (Stelling and Duinmeijer 2003); without a turbulence model, A 547 gives a parabolic algebraic eddy viscosity from the bed (and wind) friction velocity. The time step is then limited by the advective Courant number and the explicit horizontal diffusion, not by the gravity wave speed, and capped by A 536. The scheme requires the hydrostatic pressure (A 520 0); it is not available with mesh refinement.

Open boundaries. In addition to the wave generation and absorption zones, NHFLOW can impose background currents and tides (B 510 – B 515) and open boundary conditions along whole domain sides: Riemann, Flather, clamped level and clamped discharge edges (B 520 methods 3–6, discharge B 525). With the active beach B 99 3 or 4, the outflow side x_{max} becomes an absorbing Riemann edge. See Section 6.2.

4.3.7 Floating bodies and sediment transport

Fixed structures (A 581–A 590) and floating bodies (X group) are represented by a direct forcing immersed boundary method: a smoothed Heaviside function H of the body's signed distance, with an interface width of A 526 times the horizontal cell size, gives the forcing term $f_i = H(u_{i,body} - u_i)/(\alpha \Delta t)$, which is added to hu_i before and after the pressure solution. With A 517 1 only the normal velocity is enforced in the interface band, so that the tangential velocity is preserved. For floating bodies (X 10 1), the pressure and viscous forces are integrated over the triangulated body surface and the rigid body motion is integrated with the scheme of A 510; A 516 sets the treatment of the free surface inside the body. Alternatively, A 599 1 uses a distributed Lagrange multiplier method. Sediment transport (S 10 1, or S 10 12 for the fully coupled RK2 version) uses the bed shear stress from the NHFLOW velocity field (wall law B 22 with S 16 1) in the Exner-based morphology model of REEF3D; the updated bed changes d, and thereby the σ-grid, at every time step.

4.3.8 Solution procedure

The solution procedure of REEF3D::NHFLOW, as implemented in the NHFLOW driver and the Runge-Kutta momentum classes, is shown in Fig. 4.4. The free surface and the momentum equations are advanced together in each Runge-Kutta stage: the free surface is updated first with the HLL fluxes of the reconstructed state, then the momentum predictor is computed on the new water depth and projected with the non-hydrostatic pressure. Turbulence and sediment transport are solved once per time step before the Runge-Kutta stages, and the new time step size is computed at the end of the step.

4.3.9 Key input commands

Command	Meaning

A 10 5	select REEF3D::NHFLOW
A 510	time scheme: 2nd- or 3rd-order TVD Runge-Kutta
A 511	Riemann solver: 1 HLL, 2 HLLC
A 512, A 513	diffusion: off / explicit / implicit; wall treatment
A 514	reconstruction: 1–3 TVD (van Leer, Superbee, van Albada), 4 WENO horizontal, 5 WENO 3D
A 527, A 528	WENO weights: 0 WENO-JS (default), 1 WENO-Z, 2 TENO5 with cutoff A 528
A 518, A 519	bed boundary condition (slip/no-slip), bed and wall friction
A 520	pressure: 0 hydrostatic, 1 projection, 2 pressure correction, 3 alternative scheme
A 529, A 530	implicit σ cross terms of the Poisson equation; cross terms in porous cells
A 534–A 539	hydrostatic semi-implicit free surface: on/off, θ, maximum Δt, wetting-drying iteration, mass flux dissipation, non-orthogonal correction
A 546–A 548	semi-implicit: thin-water velocities, algebraic eddy viscosity, still water depth of the faces
A 522, A 524	bed slope factor in the Poisson right-hand side; discretisation of σ-metrics
A 531, A 532	Froude number limiter and its area
A 540, A 544, A 545	wetting-drying scheme, dry depth threshold, shallow (hydrostatic) cell criterion
A 550–A 557	breaking detection and breaking viscosity
A 559–A 568	turbulence model and its options; A 559 EARSM, A 561, A 562 HLES, A 563 QCR
A 570–A 578	wind forcing
W 200–W 227, W 211	meteorological forcing (wind, air pressure), Coriolis force (Section 5.11.1)
A 581–A 599	direct forcing solids, STL geometry, DLM
B 98, B 99	wave generation and absorption (Section 4.1)
B 510–B 515, B 520–B 525	background currents and tides, open boundary edges (Section 6.2)
B 103, B 113, B 116	vertical grid stretching (DIVEMesh)
F 60	still water level
N 10, N 11	linear solver and preconditioner for the pressure
N 47, N 48, N 49	CFL number, adaptive time stepping, maximum/fixed time step
X 10	floating bodies (6DOF)
S 10	sediment transport
L 10 – L 62	Lagrangian particle tracking (Section 6.10)

4.4 REEF3D::FNPF

REEF3D::FNPF (A 10 3) solves the fully nonlinear potential flow equations on a σ-coordinate grid that follows the bed and the free surface [Bihs et al., 2020]. The breaking wave detection

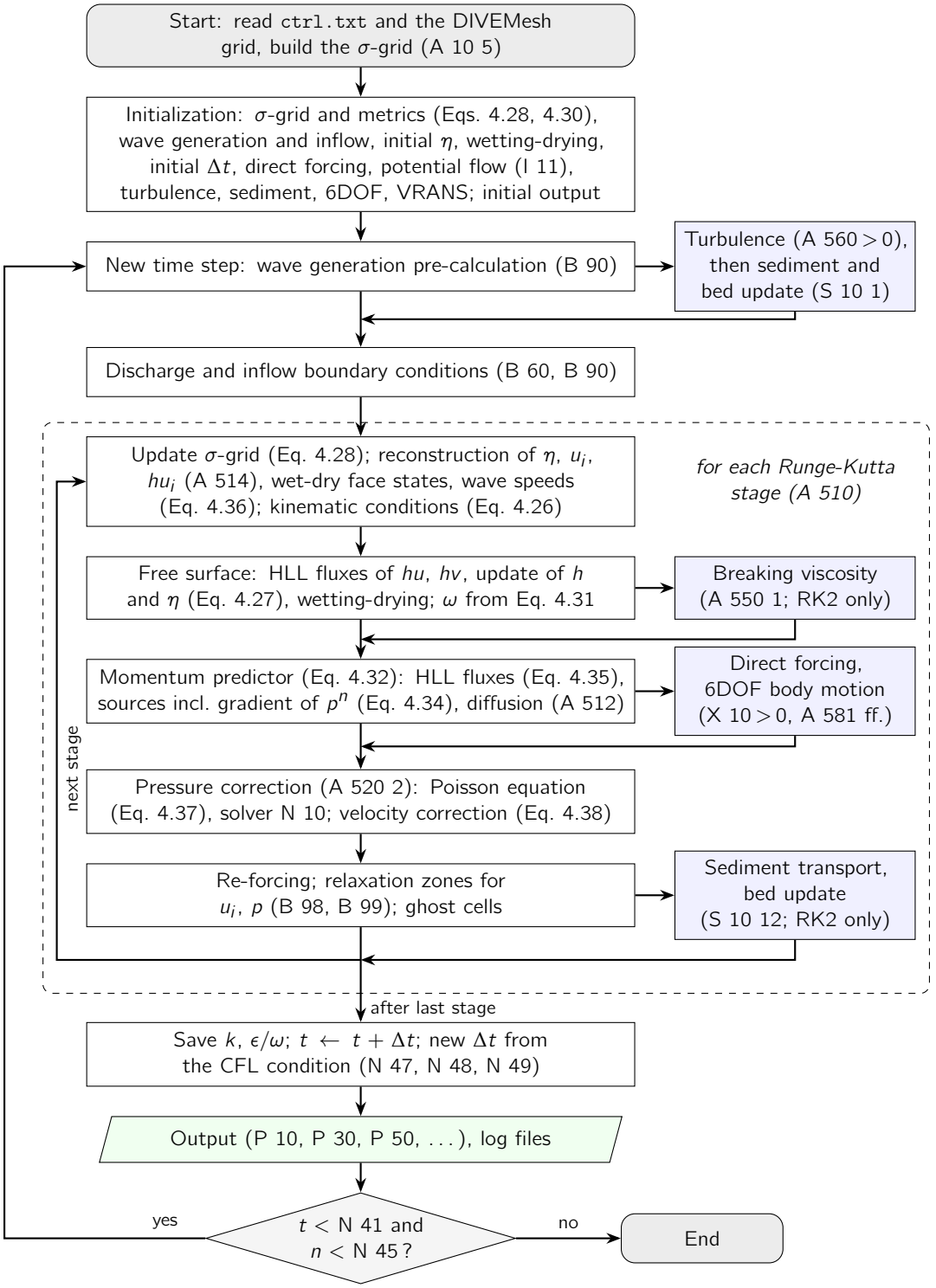

Figure 4.4: Solution procedure of REEF3D::NHFLOW for the default second-order Runge-Kutta scheme (A 510 2) and pressure-correction method (A 520 2); with A 510 3 there are three stages and neither the breaking viscosity nor the stage-wise sediment update (S 10 12) is called, and with A 520 0 or 1 the pressure step is skipped or replaced by the projection method. The loop also ends when the sediment time exceeds S 19.

and the coastline algorithm for complex topographies are described in Wang et al. [2022]. The flow is assumed inviscid and irrotational, and the free surface single-valued. The unknowns are the free surface elevation $\eta(x,y,t)$ and the velocity potential at the free surface $\tilde{\Phi}(x,y,t)$, both stored on the horizontal (2D) grid, and the velocity potential $\Phi(x,y,\sigma,t)$ in the water column, which is obtained from a Laplace equation in every Runge-Kutta stage.

4.4.1 Governing equations

With z measured from the still water level F 60 and the still water depth $d(x,y)$ (bed at $z=-d$), the velocity potential satisfies the Laplace equation in the fluid domain $-d \le z \le \eta$:

$$\frac{\partial^2\Phi}{\partial x^2}+\frac{\partial^2\Phi}{\partial y^2}+\frac{\partial^2\Phi}{\partial z^2}=0. \tag{4.40}$$

At the bed, the normal velocity vanishes (Neumann condition including the bed slope):

$$\frac{\partial\Phi}{\partial z}+\frac{\partial d}{\partial x}\frac{\partial\Phi}{\partial x}+\frac{\partial d}{\partial y}\frac{\partial\Phi}{\partial y}=0 \qquad \text{at } z=-d. \tag{4.41}$$

At the free surface, the kinematic and dynamic boundary conditions are written in terms of η, the surface potential $\tilde{\Phi}=\Phi(x,y,\eta,t)$ and the vertical velocity at the surface $\tilde{w}=\partial\Phi/\partial z|_{z=\eta}$ (Zakharov form):

$$\frac{\partial\eta}{\partial t}=-\frac{\partial\eta}{\partial x}\frac{\partial\tilde{\Phi}}{\partial x}-\frac{\partial\eta}{\partial y}\frac{\partial\tilde{\Phi}}{\partial y}+\tilde{w}\left(1+\left(\frac{\partial\eta}{\partial x}\right)^2+\left(\frac{\partial\eta}{\partial y}\right)^2\right), \tag{4.42}$$

$$\frac{\partial\tilde{\Phi}}{\partial t}=-\frac{1}{2}\left(\left(\frac{\partial\tilde{\Phi}}{\partial x}\right)^2+\left(\frac{\partial\tilde{\Phi}}{\partial y}\right)^2\right)+\frac{1}{2}\tilde{w}^2\left(1+\left(\frac{\partial\eta}{\partial x}\right)^2+\left(\frac{\partial\eta}{\partial y}\right)^2\right)-g\eta. \tag{4.43}$$

The gravitational acceleration g is taken from W 22. With A 314 1 the linearised conditions $\partial\eta/\partial t=\tilde{w}$ and $\partial\tilde{\Phi}/\partial t=-g\eta$ are used instead (default A 314 2: fully nonlinear). Optional wind forcing (A 370–A 373) adds a surface pressure or stress term to Eq. (4.43). The velocities follow from the potential, $u_i=\partial\Phi/\partial x_i$; they are needed for output, wave kinematics and hydrodynamic coupling, but not for the time integration itself.

4.4.2 σ-coordinate transformation and Laplace solver

The vertical coordinate is mapped onto $\sigma\in[0,1]$, with $\sigma=0$ at the bed and $\sigma=1$ at the free surface:

$$\sigma=\frac{z+d(x,y)}{\eta(x,y,t)+d(x,y)}=\frac{z+d}{h}, \qquad h=\eta+d. \tag{4.44}$$

The σ-levels are the normalised vertical grid of DIVEMesh, so the number of vertical cells and the vertical stretching are controlled by B 2 and B 103, B 113, B 116 (typically a refinement towards the free surface); the vertical extent given in B 10 is irrelevant. The grid moves with η and is updated after every stage. Applying the chain rule to Eq. (4.40) gives, with derivatives now taken at constant σ,

$$\frac{\partial^2\Phi}{\partial x^2}+\frac{\partial^2\Phi}{\partial y^2}+\left(\sigma_x^2+\sigma_y^2+\sigma_z^2\right)\frac{\partial^2\Phi}{\partial\sigma^2}+2\sigma_x\frac{\partial^2\Phi}{\partial x\,\partial\sigma}+2\sigma_y\frac{\partial^2\Phi}{\partial y\,\partial\sigma}+\left(\sigma_{xx}+\sigma_{yy}\right)\frac{\partial\Phi}{\partial\sigma}=0, \tag{4.45}$$

with the metric terms

$$\sigma_x = \frac{(1-\sigma)\, d_x - \sigma\, \eta_x}{h}, \qquad \sigma_z = \frac{1}{h}, \qquad \sigma_{xx} = \frac{(1-\sigma)\, d_{xx} - \sigma\, \eta_{xx} - 2\sigma_x\,(d_x + \eta_x)}{h}, \tag{4.46}$$

and the corresponding expressions in y ($d_x = \partial d/\partial x$, $\eta_x = \partial \eta/\partial x$ etc.). The first derivatives η_x, d_x are computed with the scheme of A 311 (Section 4.4.3), the second derivatives η_{xx}, d_{xx} with second-order (A 312 2) or fourth-order (A 312 3) central differences. The bed metrics can be smoothed with A 309 passes of a 1-2-1 filter applied to the bed elevation, and A 317 1 (default) uses a symmetric bed gradient including the bed curvature terms d_{xx}, d_{yy}.

The boundary conditions become $\Phi = \tilde{\Phi}$ at $\sigma = 1$ (Dirichlet) and, from Eq. (4.41),

$$\frac{\partial \Phi}{\partial \sigma} = -\frac{d_x\, \partial\Phi/\partial x + d_y\, \partial\Phi/\partial y}{\sigma_z + d_x \sigma_x + d_y \sigma_y} \qquad \text{at } \sigma = 0, \tag{4.47}$$

which is imposed through ghost points below the bed, with second-order central differences for $\partial\Phi/\partial x$ and $\partial\Phi/\partial y$.

Eq. (4.45) is discretised with second-order central differences on the non-uniform σ grid (7-point stencil, fnpf_laplace_cds2). The mixed derivatives $\partial^2\Phi/\partial x\partial\sigma$ and $\partial^2\Phi/\partial y\partial\sigma$ are evaluated explicitly from the current values of Φ and moved to the right-hand side; the $\sigma_{xx}\,\partial\Phi/\partial\sigma$ term and the bed condition enter the matrix, which is therefore not symmetric. At closed lateral boundaries and next to dry cells, $\partial\Phi/\partial n = 0$ is used; at Dirichlet wave generation or active absorption boundaries (B 98 3, 4, B 99 3, 4) the prescribed horizontal velocity enters as a Neumann condition, extrapolated to first or second order (A 329). Dry columns are removed from the system (identity rows). The linear system is solved with the solver chosen in N 10 and N 11 (Section 4.1): by default REEF3D's own semicoarsening multigrid-preconditioned BiCGStab REEFMG (N 10 1, controlled by N 12–N 15), or, in builds with HYPRE, e.g. HYPRE's Struct BiCGStab with the PFMG geometric multigrid preconditioner (N 10 14, N 11 11). The solution of the previous stage is the initial guess.

After the last stage, the velocities in the water column are computed from Φ: $u = \partial\Phi/\partial x + \sigma_x\,\partial\Phi/\partial\sigma$, $v = \partial\Phi/\partial y + \sigma_y\,\partial\Phi/\partial\sigma$, $w = \sigma_z\,\partial\Phi/\partial\sigma$, with fourth-order central differences in the interior (second order in σ for the metric terms and at the free surface, where $w = \tilde{w}$). Velocities in columns next to a dry cell are set to zero.

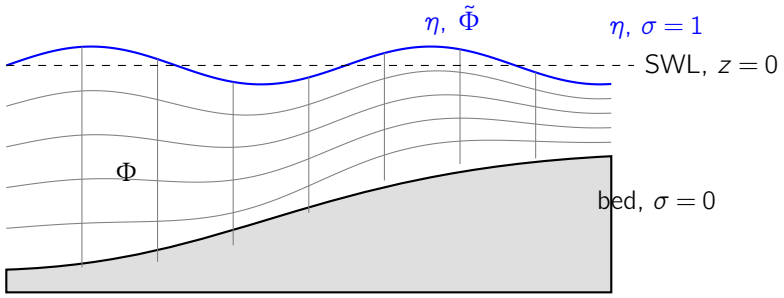

Figure 4.5: Sketch of the σ grid of REEF3D::FNPF: the vertical grid lines connect bed and free surface; η and $\tilde{\Phi}$ live on the horizontal grid at $\sigma = 1$, Φ in the water column.

4.4.3 Time integration of the free surface boundary conditions

Eqs. (4.42) and (4.43) are integrated with the third-order TVD Runge-Kutta scheme (A 310 3, default, Section 4.1); a classical fourth-order Runge-Kutta scheme is available with

A 310 4. In every stage, the right-hand sides of Eqs. (4.42) and (4.43) are evaluated, the breaking and coastline damping (Section 4.4.4) is applied and η, $\tilde{\Phi}$ are updated; then the wave generation and beach relaxation, the wetting-drying and the coastline relaxation are applied, breaking is detected, the horizontal gradients and the σ grid are updated, and finally the Laplace equation is solved with $\Phi = \tilde{\Phi}$ at $\sigma = 1$ to obtain $\tilde{w}$ for the next stage. The horizontal gradients $\partial\eta/\partial x_i$ and $\partial\tilde{\Phi}/\partial x_i$ are discretised with the scheme selected in A 311: second-, fourth- or sixth-order central differences (1, 2, 6), third-order WENO (3) or the default fifth-order WENO (4, 5), written for non-uniform grids, with the nonlinear weights of A 318 (WENO-JS by default). The WENO stencil for $\tilde{\Phi}_x$ is upwinded with the sign of a central estimate of $\tilde{\Phi}_x$. In the kinematic condition, η_x is upwinded with the characteristic speed $\tilde{\Phi}_x - 2\tilde{w}\,\eta_x$ of Eq. (4.42) (A 315 1, default); A 315 2 additionally uses symmetric η gradients for the geometric terms (metrics, breaking, wind, dynamic condition), and A 315 0 is the legacy treatment. The wetting-drying versions of the schemes are selected automatically with A 343 ≥ 1 (A 316 controls the WENO5 fallback at the shoreline).

The vertical velocity at the free surface is obtained from the Laplace solution as $\tilde{w} = \sigma_z\,\partial\Phi/\partial\sigma|_{\sigma=1}$, using a one-sided backward difference in σ whose order follows A 311: 3 points (second order) for CDS2, 5 points (fourth order) for CDS4 and WENO5 (with a reduction of the stencil where not enough wet points are available), 7 points for CDS6 and 5-point finite difference weights on the actual grid for WENO3. The one-sided stencil is applied to Φ and to the σ coordinates of the grid points, so it remains consistent on stretched grids.

The time step is adaptive (N 47, N 48, Section 4.1), based on the maximum of the surface velocity and the shallow water celerity:

$$\Delta t = \text{N 47} \cdot \min\left(\frac{\Delta x}{\max\left(|u|_{max},\ \sqrt{g\,d_{max}}\right)}\right), \tag{4.48}$$

limited by N 49. Since the phase speed of the waves determines the time step, FNPF is typically run with larger CFL numbers than CFD (e.g. N 47 1.0).

4.4.4 Wave breaking

Potential flow cannot represent the overturning of a breaker, so breaking is modelled by detecting breaking cells and dissipating energy there (A 350). The breaking detection is controlled by A 351:

- shallow water (depth-induced) breaking, A 351 1 or 3: the surface rises faster than a fraction of the shallow water celerity, $\partial\eta/\partial t > \alpha\sqrt{gh}$ with α = A 354 (default 0.6), where $\partial\eta/\partial t$ is evaluated from the change of η over the current stage;
- deep water (steepness-induced) breaking, A 351 2 or 3: the free surface slope exceeds a threshold, $|\partial\eta/\partial x| >$ A 355 or $|\partial\eta/\partial y| >$ A 355 (default 1.25).

With the base algorithm (A 350 1), the flagged region is widened by up to two cells, the flags are collected over the stages of one time step, and an artificial viscosity $\nu_b =$ A 365

(default 1.86 m^2/s) is assigned to the breaking cells and their direct neighbours, with $0.5\,\nu_b$ in the next ring of cells. The dissipation is applied as an implicit horizontal diffusion step to $\tilde{\Phi}$ and/or η (A 357) before the Runge-Kutta update of each stage:

$$\frac{f^* - f}{\alpha_{RK}\,\Delta t} = \nu_f \left(\frac{\partial^2 f^*}{\partial x^2} + \frac{\partial^2 f^*}{\partial y^2} \right), \qquad f \in \{\eta, \tilde{\Phi}\}, \quad \nu_\eta = \text{A 366} \cdot \nu_b, \quad \nu_{\tilde{\Phi}} = \nu_b, \tag{4.49}$$

where α_{RK} is the weight of the stage (1, 1/4, 2/3 for RK3). The 2D system is solved with a BiCGStab solver and skipped when $\nu_b = 0$ everywhere. Optionally (A 352), η and $\tilde{\Phi}$ are additionally smoothed in the breaking cells with a local 1-2-1 filter with predictor-corrector iterations (A 361 outer, A 362 inner iterations).

With A 350 2, an eddy viscosity in the spirit of Kennedy et al. [2000] is used: depth-induced breaking starts when $\partial\eta/\partial t > 0.65\sqrt{gh}$ and stops when $\partial\eta/\partial t < 0.15\sqrt{gh}$ (fixed values in the code, A 354 is not used), and the viscosity is $\nu_b = \delta_b^2\, h\, |\partial\eta/\partial t|$ with $\delta_b =$ A 365; for steepness-induced breaking (onset at A 355, cessation below A 356·A 355) $\nu_b = \delta_b^2 h$. The viscosity is limited to 10 m^2/s and applied with Eq. (4.49).

A 350 3 implements an energy-dissipation-based model following Romero et al. [2012]: from a local wavenumber k (estimated from the curvature of η), a local steepness $S = kA$ and the linear phase speed c, breaking occurs for $S > S_0$ and ceases for $S < 0.5 S_0$; the viscosity is obtained by matching the dissipation to $b(S)\, c^5/(g L_b)$ with the breaking strength $b = a(S - S_0)^{5/2}$ and the breaking length $L_b = \alpha_L 2\pi/k$, ramped up over $3\sqrt{h/g}$. The coefficients are fixed in the code ($a = 0.5$, $S_0 = 0.08$, $\alpha_L = 0.35$, $\nu_b \le 10\,m^2/s$); A 351–A 356 are not used by this option.

A 350 4 tracks the individual wave crests along the x-direction (in 3D along every grid row) and estimates the crest speed C from the crest positions over time. A crest starts breaking when the ratio of the horizontal surface velocity u_s at the crest to the crest speed, $B = u_s/C$, exceeds the threshold A 359 (default 0.85), following the universal onset criterion of Barthelemy et al. (2018), and stops breaking when the largest slope of its front face falls below the angle A 360. The dissipation rate per unit crest length of a breaking crest is the larger of the bore dissipation of a moving hydraulic jump, A 364·$g\,C\,H^3/(4h_c)$ with the height H from the crest to the trough ahead and the depth h_c at the crest, and, with A 367 1, the deep-water dissipation $b\,C^5/g$ with the breaking strength $b = 0.034\,(\Gamma - 0.30)^{2.5}$ of Derakhti et al. (2018), where Γ is the growth rate of B at the onset scaled with the crest period. The viscosity of Eq. (4.49) is set on the front face, from the crest to the trough ahead, such that the damping step dissipates this rate. No breaking is detected in the relaxation zones. With P 59 1, the tracked crests are written to a log file.

4.4.5 Coastline and wetting-drying

FNPF does not resolve the swash zone like a shallow water model. Instead, a cell is dry when its water depth is below A 344 (default 0.001 m), and the waves are damped before they reach the coastline, so that complex coastlines, islands and skerries can be included directly from the topography without special meshing [Wang et al., 2022]. A 343 selects the treatment:

- A 343 1 (default), static coastline: the wet and dry cells are determined once from the still water depth. In dry cells $\tilde{\Phi} = 0$ and the Laplace equation is not solved. Wet

cells whose depth drops below A 344 during a drawdown are clamped to a thin film ($h = 1.1 \cdot$A 344) with $\tilde{\Phi}$ averaged from the wet neighbours; with A 337 1 (default) the clamp volume is redistributed (A 334) and one-sided gradients are used at the coast (A 336).

- A 343 2 and 3, dynamic wetting-drying (runup and rundown, or rundown only): the wet/dry flags are updated once per time step. A cell dries when $h \leq$ A 344 and it has been wet for at least A 330 steps; it rewets when a wet neighbour is higher by at least A 331·A 344, taking $\tilde{\Phi}$ from its wet neighbours. A 335 limits the runup elevation, and a viscosity band of A 332 cells with $\nu \geq$ A 346 is applied along the moving front. Option 2 uses no coastline relaxation.

For the static coastline (and A 343 3), the waves are absorbed in a relaxation zone along the coastline. At the start of the simulation, a signed distance function d_c to the coastline is computed from the wet/dry mask by fast sweeping (A 339 1, default) or by PDE reinitialisation (A 339 0); isolated wet areas not connected to the main water body can be dried out first (A 338). Within the zone width L_c = A 341·Δx or L_c = A 342 (absolute width, has priority), the free surface quantities are relaxed after every stage,

$$f \leftarrow r(\hat{x})\, f + (1 - r(\hat{x}))\, f_{ref}, \qquad r(\hat{x}) = 1 - \frac{e^{\hat{x}^{3.5}} - 1}{e - 1}, \qquad \hat{x} = \max\left(0,\ 1 - \frac{d_c}{L_c}\right), \quad (4.50)$$

so that $r = 0$ at the coastline and $r = 1$ at the edge of the zone. This is applied to η and/or $\tilde{\Phi}$ (A 347). The target f_{ref} is zero by default; with A 326 > 0 it is a running mean of f over the time scale A 326, which lets wave set-up and the slow drift of $\tilde{\Phi}$ pass through the zone. The velocities and $\tilde{w}$ are damped in the same way over a zone of width $3L_c$ with exponent 1.5. With A 337 1, the coastline viscosity A 346 is added to ν_b inside the zone, rising to A 346 towards the coast, and in thin-film cells; it is applied through Eq. (4.49) and therefore requires A 350 ≥ 1. For a full-domain initialisation (I 30 1) the zone is widened by the factor A 349 in the first time step.

4.4.6 Wave generation, absorption and coupling

Waves are generated and absorbed with the methods of Section 4.1. With the relaxation method (B 98 2), η and $\tilde{\Phi}$ are relaxed towards the wave theory (B 92) in the generation zone, and in the numerical beach (B 99 1, 2) towards zero (A 348 selects η and/or $\tilde{\Phi}$); only wave theories that provide the potential are suitable (marked in the B 92 table). With Dirichlet generation (B 98 3, 4) and active absorption (B 99 3, 4) the horizontal velocity of the wave theory is imposed as a Neumann condition in the Laplace equation.

Because FNPF is cheap compared with CFD, a typical workflow is to propagate the waves over a large domain in FNPF and to write the free surface, the surface potential and the velocities to state files (P 40, P 44 1). After preprocessing with DIVEMesh (H options), these drive a smaller REEF3D::NHFLOW or REEF3D::CFD domain around a structure through the hydrodynamic coupling (HDC, B 92 61), with Dirichlet or relaxation wave generation (B 98) at the coupling boundary. Alternatively, the FNPF model runs in the same MPI job as the NHFLOW or CFD model and passes the waves at runtime (B 92 62, Section 4.1.7), also in two-way coupling of the floating bodies and the free surface.

With I 30 1, the whole domain is initialized with the wave theory at $t = 0$. With I 30 2, every wave component is first multiplied by an exposure mask for its propagation direction, so that areas in the lee of islands and headlands start with reduced or no waves. The masks are computed on a coarse grid (I 33) for I 32 direction bins by sweeping along the propagation direction from the upwave boundary over the wet cells, with a lateral spreading by diffraction of variance $\kappa L_p s$ (κ = I 34, peak wave length L_p, distance s along the ray); they are mixed across neighbouring directions (I 35) and smoothed over the wet cells (I 31).

4.4.7 Solution procedure

The solution procedure of REEF3D::FNPF is shown in Fig. 4.6. After the σ grid, the wet/dry mask with the coastline distance and the wave generation have been initialised, every time step advances η and $\tilde{\Phi}$ with the Runge-Kutta scheme of A 310. In each stage the free surface boundary conditions are integrated explicitly, relaxation, wetting-drying and breaking are applied to the new surface values, and the σ grid is updated before the Laplace equation is solved for Φ, which provides $\tilde{w}$ for the next stage.

4.4.8 Key input commands

Command	Meaning
A 10 3	select REEF3D::FNPF
A 309	number of smoothing passes for the bed
A 310	time scheme: 3 TVD-RK3 (default), 4 RK4
A 311	horizontal gradients in the FSBCs: CDS2/4/6, WENO3, WENO5 (default 5)
A 312	second derivatives of η, d in the σ metrics: CDS2 (default), CDS4
A 314	linear (1) or fully nonlinear (2, default) free surface boundary conditions
A 315–A 317	gradient treatment of η, WENO5 at the shoreline, bed metrics
A 329	extrapolation order for Dirichlet wave generation and active absorption
A 326, A 341, A 342, A 347, A 349	coastline relaxation: reference time scale, zone width (relative, absolute), variables, initialisation factor
A 330–A 339, A 343, A 344	wetting-drying: method (A 343), dry threshold (A 344), dynamic front parameters, coastline distance
A 346	coastline damping viscosity
A 348	variables relaxed in the numerical beach
A 350, A 351	breaking model (0 off, 1 base, 2 Kennedy, 3 Romero, 4 crest tracking) and detection (shallow/deep)
A 359, A 360, A 364, A 367	crest tracking (A 350 4): onset $B = u_s/C$, termination angle, bore dissipation, deep-water breaking strength
A 352, A 361, A 362	additional filtering in breaking cells
A 354, A 355, A 356	breaking thresholds: $\partial\eta/\partial t$ factor, slope onset, slope cessation
A 357, A 365, A 366	breaking viscosity: applied to $\eta/\tilde{\Phi}$, value, factor for η

Command	Meaning
A 370–A 373	wind forcing
B 2, B 103, B 113, B 116	number of vertical cells and vertical stretching of the σ grid (DIVEMesh)
B 92, B 98, B 99	wave theory, generation and absorption method
I 30, I 31–I 35	full initialization, exposure masks of I 30 2
F 60	still water level
N 10, N 11	solver and preconditioner for the Laplace equation (REEFMG N 10 1, default; HYPRE)
N 47, N 48, N 49	CFL number, adaptive time stepping, maximum time step
P 40, P 44	state files for hydrodynamic coupling and nesting

4.5 REEF3D::SFLOW

REEF3D::SFLOW (A 10 2) solves the depth-averaged shallow water equations (SWE) on a two-dimensional horizontal grid. A depth-averaged non-hydrostatic pressure can be added to represent frequency dispersion, so that waves from shallow to intermediate water depth can be computed [Stelling and Zijlema, 2003, Jeschke et al., 2017, Wang et al., 2020]. Bed friction, depth-averaged turbulence models, wetting and drying, wave breaking and bedload sediment transport are included.

Note. The numerical scheme of SFLOW is under active development. This section describes the current source code: a cell-centred finite volume scheme with HLL fluxes. It differs from the staggered finite difference discretization described by Wang et al. [2020]. The Boussinesq option A 220 4 is new. Details may change in future versions.

4.5.1 Governing equations

The bed is at $z = -d(x, y)$ below the still water level, η is the free surface elevation and $h = \eta + d$ the total water depth. u, v are the depth-averaged horizontal velocities and $\bar{w}$ is the depth-averaged vertical velocity. Continuity and horizontal momentum read

$$\frac{\partial h}{\partial t} + \frac{\partial (hu)}{\partial x} + \frac{\partial (hv)}{\partial y} = 0, \tag{4.51}$$

$$\frac{\partial (hu)}{\partial t} + \frac{\partial}{\partial x}\left(hu^2 + \tfrac{1}{2} g\eta^2 + g\eta d\right) + \frac{\partial (huv)}{\partial y} = g\eta \frac{\partial d}{\partial x} - \frac{1}{\rho}\left(\frac{\partial (h\bar{p})}{\partial x} - p_b \frac{\partial d}{\partial x}\right) - \frac{\tau_{bx}}{\rho} + D_x, \tag{4.52}$$

and correspondingly for hv. With the meteorological forcing (W 200) and the Coriolis force (W 211), the right-hand sides also contain the wind stress, the air pressure gradient and the Coriolis terms $f\,hv$ and $-f\,hu$ of Eq. (5.68) (Section 5.11.1). The hydrostatic pressure is written in the pre-balanced form $\frac{1}{2} g\eta^2 + g\eta d$ with the bed slope source $g\eta\,\partial d/\partial x$. Together they equal $gh\,\partial\eta/\partial x$, and still water over an uneven bed stays exactly at rest in the discrete equations (well-balanced scheme). $\bar{p}$ is the depth-averaged non-hydrostatic pressure and p_b its value at the bed (Section 4.5.2). With A 220 0 both are zero and the equations are the nonlinear SWE.

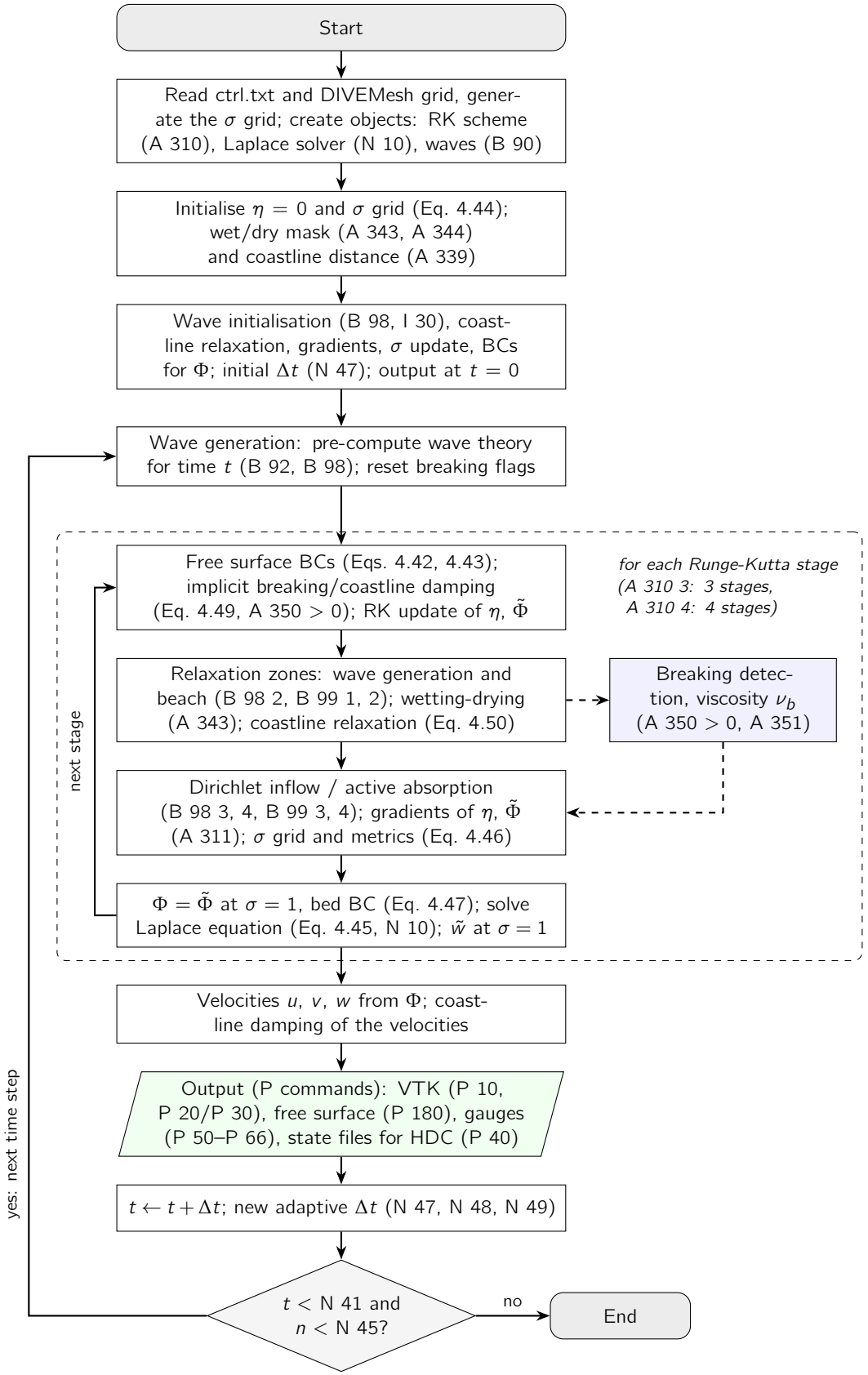

Figure 4.6: Solution procedure of REEF3D::FNPF. Shown is the default RK3 path (A 310 3); with A 310 4 the same steps are carried out in four stages, but the wetting-drying is applied after the breaking detection.

Friction, diffusion and turbulence. With A 218 1, the Manning bed shear stress

$$\frac{\tau_{b,i}}{\rho} = \frac{g n^2}{h^{1/3}} u_i \sqrt{u^2+v^2}, \qquad n = \frac{k_s^{1/6}}{20}, \tag{4.53}$$

is applied, with the roughness k_s from B 50. W 90 1 adds a Bingham-type bed stress for non-Newtonian flows. Horizontal diffusion is switched on with A 212 (1 explicit, 2 implicit):

$$D_x = h\nu_e\left(2\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 v}{\partial x \partial y}\right), \qquad \nu_e = \nu + \nu_t \text{ (+A 250 in breaking cells with A 246 2).} \tag{4.54}$$

The eddy viscosity ν_t comes from the depth-averaged turbulence model A 260: k-ϵ (1), k-ω (2), Prandtl mixing length (3) with $l = 0.4 \cdot$A 261$\cdot h$, or a parabolic profile (4) $\nu_t =$ A 262$\cdot u_* h$. The bed shear stress of the k-ϵ, k-ω and parabolic models follows Eq. (4.53) with $g = |$W 22$|$. The eddy viscosity acts on the velocities only when A 212 is on; when A 212 is not given, it is set to 1 with a turbulence model (A 260 > 0) and to 0 otherwise.

4.5.2 Non-hydrostatic extension

The non-hydrostatic pressure vanishes at the free surface, and the vertical velocity varies linearly over the depth, $\bar{w} = (w_s + w_b)/2$, with the surface value w_s and the kinematic bed value $w_b = -u\,\partial d/\partial x - v\,\partial d/\partial y$. The depth-integrated continuity equation then gives the constraint

$$h\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) + 2\left(\bar{w} + u\frac{\partial d}{\partial x} + v\frac{\partial d}{\partial y}\right) = 0, \tag{4.55}$$

and the depth-averaged vertical momentum equation is

$$\frac{\partial (h\bar{w})}{\partial t} + \frac{\partial (hu\bar{w})}{\partial x} + \frac{\partial (hv\bar{w})}{\partial y} = \frac{p_b}{\rho} + D_w. \tag{4.56}$$

The convective terms are included with A 214 1 and omitted with A 214 0. The bed pressure depends on the vertical pressure profile (A 220):

- **Linear (A 220 1)**, as in the one-layer Keller-box approach of Stelling and Zijlema [2003]: $p_b = 2\bar{p}$.
- **Quadratic (A 220 2)** following Jeschke et al. [2017]:

  $$p_b = \tfrac{3}{2}\bar{p} + \frac{\rho h}{4}\varphi, \qquad \varphi = \frac{Dw_b}{Dt} = -\frac{Du}{Dt}\frac{\partial d}{\partial x} - \frac{Dv}{Dt}\frac{\partial d}{\partial y} - u^2\frac{\partial^2 d}{\partial x^2} - v^2\frac{\partial^2 d}{\partial y^2} - 2uv\frac{\partial^2 d}{\partial x \partial y}. \tag{4.57}$$

  The bed acceleration φ is evaluated explicitly in each Runge-Kutta stage. On a flat bed, $\varphi = 0$.
- **Quadratic with improved dispersion (A 220 3)**, with the parameter $\alpha \geq 1$ (A 224, default 1.159). The pressure source in Eq. (4.56) becomes $\frac{3}{2\alpha}\bar{p}/\rho + h\varphi/4$. In the horizontal momentum, φ is multiplied by α and the explicit term $Bg\nabla(h^3\nabla^2\eta)$, $B = (\alpha - 1)/3$, is added. $\alpha = 1$ recovers option 2.

Pressure projection. In every Runge-Kutta stage, provisional values $(hu)^*$, $(hv)^*$, $(h\bar{w})^*$ are computed without $\bar{p}$ and then corrected:

$$(hu)^{s+1} = (hu)^* - \frac{\Delta t_s}{\rho}\left(\frac{\partial(h\bar{p})}{\partial x} - p_b\frac{\partial d}{\partial x}\right), \qquad (h\bar{w})^{s+1} = (h\bar{w})^* + \Delta t_s\left(\frac{c_w\bar{p}}{\rho} + \delta\,\frac{h\varphi}{4}\right), \tag{4.58}$$

with $c_w = 2$, $\delta = 0$ (linear) or $c_w = 3/(2\alpha)$, $\delta = 1$ (quadratic). Δt_s is the effective time step of the stage. Inserting Eq. (4.58) into the constraint (4.55) gives the Poisson-type equation

$$-\frac{h}{\rho}\left(\frac{\partial^2\bar{p}}{\partial x^2} + \frac{\partial^2\bar{p}}{\partial y^2}\right) + \frac{2c_w}{\rho h}\bar{p} = -\frac{1}{\Delta t_s}\left[h\left(\frac{\partial u^*}{\partial x} + \frac{\partial v^*}{\partial y}\right) + 2\left(\bar{w}^* + u^*\frac{\partial d}{\partial x} + v^*\frac{\partial d}{\partial y}\right)\right] - \delta\,\frac{\varphi}{2}. \tag{4.59}$$

The left-hand side is discretized as a five-point stencil with the cell depth h, the gradients with central differences. The equation is solved with the 2D version of REEFMG (N 10 1, default) or, in builds with HYPRE, with HYPRE Struct (N 10 11–19, preconditioner N 11), with the stopping criterion N 44, with $\bar{p} = 0$ at in- and outflow boundaries and $\partial\bar{p}/\partial n = 0$ at walls. $\bar{p}$ is only solved in wet cells with wet neighbours that are not breaking and not excluded by A 221 0 (cells close to the shoreline or with $h \le 10\cdot$A 244). Elsewhere $\bar{p} = 0$ and the flow is hydrostatic.

Boussinesq equations (A 220 4, new). Instead of a pressure equation, this option solves the fully nonlinear Boussinesq equations of FUNWAVE-TVD in the conservative form of Shi et al. [2012]:

$$\frac{\partial\eta}{\partial t} + \frac{\partial M_j}{\partial x_j} = 0, \qquad \frac{\partial V_i}{\partial t} + \frac{\partial}{\partial x_j}\left(\frac{M_i M_j}{h}\right) + \frac{\partial}{\partial x_i}\left(\tfrac{1}{2}g\eta^2 + g\eta d\right) = g\eta\frac{\partial d}{\partial x_i} + \psi_i - \frac{\tau_{b,i}}{\rho} + D_i, \tag{4.60}$$

with $M_i = h(u_{\alpha,i} + u_{2,i})$ and $V_i = h(u_{\alpha,i} + V'_{1,i})$. $u_{\alpha,i}$ is the velocity at the reference level $z_\alpha = \zeta d + (1+\zeta)\eta$, $\zeta = -0.53$. The dispersive velocities $u_{2,i}$ and $V'_{1,i}$ depend on $\nabla(\nabla\cdot\mathbf{u}_\alpha)$ and $\nabla(\nabla\cdot(d\,\mathbf{u}_\alpha))$, and ψ_i collects the remaining dispersive terms, all as given by Shi et al. [2012]. ψ_i is explicit. After each stage, $u_{\alpha,i}$ is recovered from V_i by a line-implicit solve per component, with lagged cross derivatives. M_i enters the HLL fluxes, while the dissipative part of the HLL flux uses V_i, as in FUNWAVE-TVD. The dispersive terms are switched off smoothly in breaking cells (A 246), where $|\eta|/d \ge 0.8$, next to the shoreline and next to in- and outflow boundaries. N 47 is limited to 0.25.

4.5.3 Discretization

Finite volume scheme. SFLOW uses a cell-centred finite volume method on the horizontal, possibly non-uniform grid. The prognostic variables h, hu, hv, $h\bar{w}$ (or V_i for A 220 4) and the pressure $\bar{p}$ are stored at the cell centres. Face velocities are only computed as diagnostic quantities for the sediment, turbulence and output modules. At each face, left and right states of η, u, v and the conserved variables are reconstructed as selected with A 211: first order (0), TVD with the van Leer (1), Superbee (2) or van Albada (3) limiter, or fifth-order WENO (4, default; Section 4.1), by default with the WENO-Z weights (A 213 1; 0 WENO-JS, 2 TENO5 with the cutoff A 222). The limiters fall back to first order next to dry cells. The

face depth is $D = \max(\eta_{face} + d_{face}, \text{A 244})$, with d_{face} the mean of the two adjacent cells. The HLL flux [Harten et al., 1983] is

$$F = \begin{cases} F_L, & S_L \geq 0, \\ \Big(S_R F_L - S_L F_R + S_L S_R (q_R - q_L)\Big)/(S_R - S_L), & S_L < 0 < S_R, \\ F_R, & S_R \leq 0, \end{cases} \tag{4.61}$$

with the conserved variable q and the physical flux F. Between wet cells, the signal speeds are $S_L = \min(u_L - \sqrt{gD_L}, u_* - c_*)$ and $S_R = \max(u_R + \sqrt{gD_R}, u_* + c_*)$, with $u_* = \frac{1}{2}(u_L + u_R) + \sqrt{gD_L} - \sqrt{gD_R}$ and $c_* = \frac{1}{2}(\sqrt{gD_L} + \sqrt{gD_R}) + \frac{1}{4}(u_L - u_R)$. Next to a dry cell, dry-bed speeds are used, e.g. $S_L = u_R - 2\sqrt{gD_R}$ for a dry left side. The bed slope source uses the same d_{face} as the flux, which keeps the scheme well balanced. At walls, the mass flux is zero and the momentum flux is $\frac{1}{2}g\eta^2 + g\eta d$ of the wall cell. At in- and outflow boundaries, the flux of the ghost cell state is used. The discharge is limited to $|hu| \leq$ A 531$\cdot h\sqrt{gh}$. With A 225 2 (HLLC), the fluxes of the tangential momentum components (v on the x-faces, u on the y-faces, and $h\bar{w}$ on both) are the HLL mass flux of the face times the tangential velocity upwind of it, which resolves shear layers and contact waves with less numerical diffusion; the mass and normal momentum fluxes stay HLL. On curvilinear grids, HLL is used.

Time stepping. The SSP Runge-Kutta schemes of Section 4.1 are used, third order (A 210 3, default) or second order (A 210 2). Each stage computes $q^{(s+1)} = a\,q^n + (1-a)\Big(q^{(s)} + \Delta t\, L(q^{(s)})\Big)$, with $a = 0, \frac{3}{4}, \frac{1}{3}$ (RK3) or $a = 0, \frac{1}{2}$ (RK2), and includes the pressure projection with $\Delta t_s = (1-a)\Delta t$. The adaptive time step (N 48 1) is

$$\Delta t = 2 \cdot \text{N 47} \cdot \min\left(\frac{\Delta x}{|u| + \sqrt{gh}}, \frac{\Delta y}{|v| + \sqrt{gh}}\right) \tag{4.62}$$

over all wet cells. A 219 2 adds the limit $\Delta x/|u|$, and for A 220 3 the explicit dispersion term requires $\Delta t \leq 1.8/\Big[(\Delta x^{-2} + \Delta y^{-2})\sqrt{gBh^3}\Big]$. N 48 0 uses the fixed time step N 49.

Semi-implicit free surface. For long hydrostatic simulations such as rivers and tides, A 226 1 replaces the explicit HLL scheme by a semi-implicit θ method in the manner of Casulli and Cattani (1994), with θ = A 227. The advection and diffusion terms are integrated explicitly with the SSP Runge-Kutta predictor of A 210, the bed friction implicitly, and the free surface elevation follows from a linear system with the face discharges (solver N 10). With A 229 1 (default), the system is solved with the nonlinear wet volume $\max(\eta + d, 0)$ of each cell and iterated on the wet and dry pattern (Casulli 2009), so that the volume stays exact at the wet-dry front. A 231 adds an implicit mass flux dissipation for hydraulic jumps and transcritical flow. The water depth of a face is its still water depth plus η upwind of the face discharge (Stelling and Duinmeijer 2003), with the still water depth of the higher bed next to a dry cell, and in thin water the velocities of the predictor are desingularised, $u = \sqrt{2}\,h(hu)/\sqrt{h^4 + h_v^4}$ with h_v = A 232 (Kurganov and Petrova 2007). The gravity wave speed then does not limit the time step, which follows from the Courant number N 47 of the flow velocity and is capped with A 228. A 226 1 requires the hydrostatic pressure (A 220 0) and is not available with mesh refinement, curvilinear grids or floating bodies.

Wetting and drying. A cell is dry when h falls below A 244. Its depth is then set to A 244 and its momentum to zero. With A 243 2 (default), a dry cell becomes wet again only when a wet neighbour has a higher free surface. At wet-dry faces, the face states are taken from the wet cell with zero velocity, and no mass flux enters a dry cell.

Wave breaking. With A 246 1, a cell breaks when $\partial\eta/\partial t >$ A 247$\cdot\sqrt{gh}$. A 246 2 also flags free surface slopes $|\partial\eta/\partial x| >$ A 355 and adds the breaking viscosity A 250. With A 248 1, breaking continues while $\partial\eta/\partial t >$ A 249$\cdot\sqrt{gh}$. A 242 flags further shallow cells near the shoreline. In breaking cells, the non-hydrostatic pressure (or the Boussinesq dispersion) is switched off. The breaker then travels as a bore of the SWE and loses energy through the shock-capturing HLL flux.

Waves are generated and absorbed with relaxation zones or Dirichlet boundaries (B 98, B 99; Section 4.1), with the input velocities averaged over B 160 + 1 points over the depth.

4.5.4 Sediment transport

With S 10 1, the bed elevation z_b follows the Exner equation for bedload transport,

$$\frac{\partial z_b}{\partial t} = -\frac{\text{S 35}}{1-n}\left(\frac{\partial q_{b,x}}{\partial x} + \frac{\partial q_{b,y}}{\partial y}\right), \tag{4.63}$$

with the porosity n (S 24) and the morphological factor S 35. The sediment module is shared with NHFLOW and CFD (Section 6.15): bedload formula S 11, slope reduction of the critical shear stress S 80, Exner formulation S 31, non-equilibrium transport S 33 and sandslide S 90. In SFLOW, the bed shear stress is computed from the depth-averaged velocity: S 16 1 uses the Manning form $\tau_b = \rho g n^2 |\mathbf{u}|^2 / h^{1/3}$, S 16 2 and 3 the depth-averaged log-law $\tau_b = \rho g |\mathbf{u}|^2 / C^2$ with $C = 2.5 \ln(12h/k_s)$, and S 16 4 $\tau_b = 0.3\rho k$. After each bed update, d is recomputed while h is kept, so that η follows the bed. With S 64, a part of the transport capacity is carried as suspended load, which relaxes along the flow direction over the adaptation length of the suspension (Galappatti and Vreugdenhil 1985; S 65); the Exner fluxes carry the sum of bed load and suspended load. The transverse bed slope effect (S 86 – S 89; Talmon, van Mierlo and Struiksma 1995) turns the bed load down the transverse slope, and the spiral flow correction (S 95 – S 97) turns it towards the centre of curvature of the streamlines in bends, optionally with the adaptation of the spiral flow intensity (Kalkwijk and Booij 1986).

4.5.5 Solution procedure

The solution procedure of REEF3D::SFLOW is shown in Fig. 4.7. After the initialization, each time step starts with the wave generation and the in- and outflow boundary values. The Runge-Kutta stages then compute the HLL fluxes, update the free surface and the momentum and solve the non-hydrostatic pressure. Turbulence and sediment transport follow once per time step, before the new time step size is computed and the results are written.

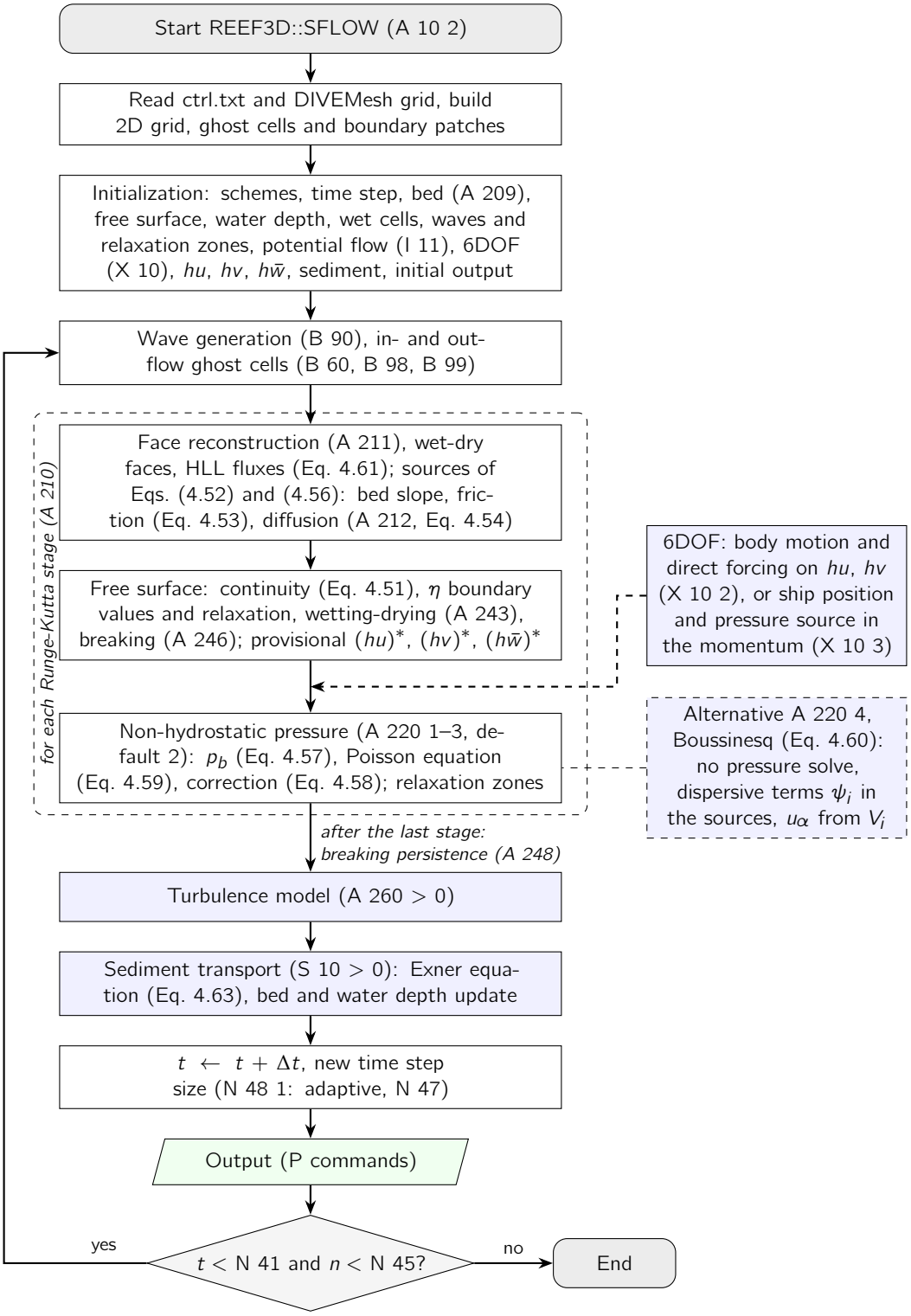

Figure 4.7: Solution procedure of REEF3D::SFLOW with the default quadratic non-hydrostatic pressure (A 220 2); with A 220 0 the pressure step is skipped, and with the Boussinesq option A 220 4 (dashed box) the dispersion enters through the momentum sources and the recovery of u_α instead of a pressure equation. The time step size is computed at the end of each step (initially during the initialization), and with sediment transport the loop also stops at the sediment time S 19.

4.5.6 Key input commands

Command	Meaning
A 10 2	select SFLOW
A 210	Runge-Kutta scheme: 2 (RK2) or 3 (RK3, default)
A 211	face reconstruction: 0 first order, 1–3 TVD limiters, 4 WENO
A 213, A 222	WENO weights: 0 WENO-JS, 1 WENO-Z (default), 2 TENO5 with cutoff A 222
A 225	Riemann solver: 1 HLL (default), 2 HLLC
A 226–A 232	semi-implicit free surface: on/off, θ, maximum Δt, wetting-drying iteration, mass flux dissipation, thin-water velocities
A 212	horizontal diffusion: 0 off, 1 explicit, 2 implicit (default 1 with A 260 $>$ 0, else 0)
A 214	convection of $\bar{w}$ in the non-hydrostatic schemes
A 218, B 50	Manning bed friction and roughness k_s
A 220	0 hydrostatic, 1 linear, 2 quadratic, 3 quadratic with A 224, 4 Boussinesq (new)
A 221, A 242	hydrostatic treatment of cells near the shoreline
A 243, A 244	wetting and drying algorithm and threshold
A 246–A 250	breaking criterion, persistence and breaking viscosity
A 260–A 264	turbulence model (1 k-ϵ, 2 k-ω, 3 mixing length, 4 parabolic) and coefficients
A 531	discharge (Froude number) limiter
B 98, B 99, B 160	wave generation, absorption, depth averaging of the input wave
N 10, N 11, N 44	solver, preconditioner and stopping criterion for $\bar{p}$
N 47, N 48, N 49	time step factor, adaptive or fixed time step
S 10, S 11, S 16	sediment transport, bedload formula, bed shear stress
S 64, S 65; S 86–S 89; S 95–S 97	suspended part of the transport; transverse bed slope effect; spiral flow
W 200–W 227, W 211	meteorological forcing (wind, air pressure) and Coriolis force for storm surge (Section 5.11.1)
P 112–P 115	maxima of the run (water level, depth, speed, flooding)

4.6 REEF3D::SEASTATE

REEF3D::SEASTATE (A 10 7) is a phase-averaged spectral wave model for wave fields at the scale of coasts, harbours and shelf seas. It does not resolve the free surface. It computes the evolution of the wave action density spectrum on the two-dimensional horizontal grid with the physics of third-generation models such as SWAN [Booij et al., 1999, Holthuijsen, 2007]. SEASTATE runs stand-alone or inside SFLOW and NHFLOW, where it supplies the wave-induced forces and currents. It also provides 2D spectra for the phase-resolving wave generation of FNPF and NHFLOW. The commands are described in Section 6.1.4.

Note. SEASTATE is a new module and is under active development. This section describes the current source code. Details may change in future versions.

4.6.1 Governing equations

The sea state is described by the wave action density $N(x,y,\sigma,\theta,t) = E(\sigma,\theta)/\sigma$. E is the variance density of the surface elevation ($\mathrm{m}^2\,\mathrm{s/rad}^2$), σ the relative (intrinsic) radian frequency and θ the direction of propagation, counter-clockwise from the $+x$ axis. In the presence of a depth-averaged current $\mathbf{U} = (U,V)$, the action is conserved, and N follows the wave action balance

$$\frac{\partial N}{\partial t} + \frac{\partial (c_x N)}{\partial x} + \frac{\partial (c_y N)}{\partial y} + \frac{\partial (c_\sigma N)}{\partial \sigma} + \frac{\partial (c_\theta N)}{\partial \theta} = \frac{S_{tot}}{\sigma}. \tag{4.64}$$

S_{tot} is the sum of the source terms in energy units (Section 4.6.4). The wave number k follows from the linear dispersion relation $\sigma^2 = gk\tanh(kd)$ with the water depth d. It is computed from the explicit approximation of Guo (2002), $k = k_0\big(1-\exp(-x^{2.4908})\big)^{-1/2.4908}$ with $k_0 = \sigma^2/g$ and $x = \sigma\sqrt{d/g}$, followed by Newton iterations to machine precision ($k = k_0$ for $k_0 d > 30$). The propagation velocities in geographic and spectral space are

$$c_x = c_g\cos\theta + U, \qquad c_y = c_g\sin\theta + V, \qquad c_g = \frac{1}{2}\Big(1 + \frac{2kd}{\sinh 2kd}\Big)\frac{\sigma}{k}, \tag{4.65}$$

$$\begin{aligned} c_\theta = {} & \frac{\sigma}{\sinh 2kd}\Big(\sin\theta\frac{\partial d}{\partial x} - \cos\theta\frac{\partial d}{\partial y}\Big) \\ & + \sin\theta\cos\theta\Big(\frac{\partial U}{\partial x} - \frac{\partial V}{\partial y}\Big) + \sin^2\theta\frac{\partial V}{\partial x} - \cos^2\theta\frac{\partial U}{\partial y}, \end{aligned} \tag{4.66}$$

$$\begin{aligned} c_\sigma = {} & \frac{k\sigma}{\sinh 2kd}\Big(\frac{\partial d}{\partial t} + U\frac{\partial d}{\partial x} + V\frac{\partial d}{\partial y}\Big) \\ & - c_g k\Big(\cos^2\theta\frac{\partial U}{\partial x} + \sin\theta\cos\theta\Big(\frac{\partial U}{\partial y} + \frac{\partial V}{\partial x}\Big) + \sin^2\theta\frac{\partial V}{\partial y}\Big). \end{aligned} \tag{4.67}$$

c_θ describes depth- and current-induced refraction (A 713), c_σ the frequency shift by unsteady depth and current gradients (A 714). The gradients are central differences, one-sided at the edge of the domain. Next to land or dry cells, c_θ and c_σ are set to zero, as in SWAN. The current is prescribed (A 720, A 721) or comes from the host model (A 753 2).

The integral parameters are computed from the moments $m_n = \iint \sigma^n E\, d\sigma\, d\theta$, with a σ^{-4} tail above the highest frequency: $H_s = 4\sqrt{m_0}$, the mean frequencies $\sigma_{01} = m_1/m_0$ and $\tilde{\sigma} = m_0/m_{-1}$, and the mean wave number $\tilde{k} = \big(m_0^{-1}\iint k^{-1/2} E\, d\sigma\, d\theta\big)^{-2}$.

4.6.2 Spectral and geographic grid

The A 701 frequencies between f_{min} and f_{max} (A 702) are distributed logarithmically, $\sigma_l = 2\pi f_{min} r^l$ with $r = (f_{max}/f_{min})^{1/(n_\sigma - 1)}$. The bin faces lie at the geometric midpoints, so that $\Delta\sigma_l = \sigma_l(\sqrt{r} - 1/\sqrt{r})$ is proportional to σ_l. The A 703 directions cover the full circle with $\Delta\theta = 2\pi/n_\theta$. With A 715, the directions in a given sector are subdivided for a finer directional resolution, and the directional fluxes and integrals are weighted with the individual bin widths; A 716 sets this sector automatically around the directions of the parametric boundary spectrum. Energy that leaves the frequency range is lost (outflow only at both ends), and the directional space is periodic. With A 717, only the directions of a sector are computed (as SWAN CGRID SECTOR); the directional space is then not periodic, and energy refracted beyond the ends of the sector leaves the spectral grid.

The geographic grid is the 2D grid of DIVEMesh (as for SFLOW). Cells with a water depth of at least A 705 are active. Land and dry cells absorb all incoming wave energy. The spectra are stored in single precision in square tiles of A 704 × A 704 cells. Memory is allocated only for tiles that contain active cells, so that land areas need no spectral memory. All computations use double precision.

4.6.3 Numerical method

Implicit finite volumes. Eq. (4.64) is integrated over the cells of the four-dimensional space (x, y, σ, θ) with the implicit Euler (BDF1) scheme in time and first-order upwind fluxes in all four dimensions. The face velocities are the means of the two adjacent cells, e.g. $c_{x,i+1/2} = \frac{1}{2}(c_{g,i} + c_{g,i+1})\cos\theta + \frac{1}{2}(U_i + U_{i+1})$. The discrete equation of a bin of cell c reads

$$\left(\frac{1}{\Delta t} + D_c + \sum_{f=1}^{8} \frac{c_f^{out}}{\Delta_f}\right) N_c - \sum_{f=1}^{8} \frac{c_f^{in}}{\Delta_f} N_f = \frac{N_c^n}{\Delta t} + P_c. \tag{4.68}$$

The sums run over the eight faces of the four-dimensional cell, two in each of x, y, σ and θ. $c_f^{out} = \max(0, c_f)$ and $c_f^{in} = \max(0, -c_f)$ are the parts of the outward face velocity c_f that leave and enter the cell, N_f is the action density of the neighbour bin across face f, and Δ_f is the cell width Δx, Δy, $\Delta\sigma_l$ or $\Delta\theta_m$. $P \geq 0$ and $D \geq 0$ are the production and damping parts of the source terms (Eq. 4.69). The diagonal is positive and dominates the off-diagonal entries, which are all non-positive. The system matrix is therefore an M-matrix, and the solution stays non-negative, $N \geq 0$, for any time step. The scheme has no stability limit. Nonstationary runs (A 700 1) use the time step A 706. In stationary runs (A 700 2), $1/\Delta t = 0$. With the deep-water physics (A 732 1 or 2), the source terms of the wind sea are lagged by one iteration, and A 706 is used as a pseudo time step relative to the previous iterate to damp the iterations.

Gauss-Seidel sweeps. The system is solved with Gauss-Seidel iterations in geographic space, as in SWAN. One iteration consists of four sweeps over the grid, one for each directional quadrant: the sweep for $0° \leq \theta < 90°$ runs in increasing i and j, the one for $90° \leq \theta < 180°$ in decreasing i and increasing j, and so on. In each cell, only the directions of the current quadrant are solved. Their upwind geographic neighbours have already been updated in the same sweep, so that without currents the waves cross the whole domain in one iteration. In each cell and for each frequency, the directions of the quadrant form a tridiagonal system in θ, which is solved directly (Thomas algorithm). The directions of the other quadrants enter with their latest values. Without frequency shift, the frequencies are independent. With frequency shift, they are solved from low to high: the inflow from $l-1$ takes the value just solved and the inflow from $l+1$ the latest value. On several MPI ranks, the sweeps of all ranks form one sweep by default (A 781 1): the cells are solved in tiles, and a tile at a rank boundary waits for the latest values of the rank upstream, so that the result is that of one rank. With A 781 0, and where the pipeline is not available (e.g. the level-by-level sweeps of the mesh refinement), each rank sweeps from the halo values of the last exchange, which lag by one sweep. Nonstationary runs make A 707 iterations per time step (by default 1; 2 with the FAS correction or the surfbeat model, and on several ranks without the pipeline). Stationary runs iterate until the relative change of H_s (A 799 0), or the estimated distance

$|\Delta H_s|/(1-\rho)$ to the converged solution with the contraction ρ of the last two iterations (A 799 1), is below A 708 in all cells or in the percentage A 709 of them. In stationary runs, each cell can be solved several times per sweep with updated source terms (A 738, tolerance A 739). With A 758, a coarse-grid correction (full approximation scheme, FAS) is made every n iterations: the residuals of the fine cells are restricted to a coarse grid, which is iterated, and the coarse change is passed back to the fine cells. This carries the information across the domain faster and reduces the number of iterations. The coarse grid is level 0 with mesh refinement, otherwise a V-cycle over up to six coarse grids of 2×2 cells each.

Options. A 796 2 replaces the first-order geographic fluxes by the second-order upwind scheme of SWAN (SORDUP), $F_{i+1/2} = c_{x,i+1/2}\big(\frac{3}{2}N_i - \frac{1}{2}N_{i-1}\big)$ for $c_{x,i+1/2} > 0$. The upwind cells are treated with their latest values. This scheme is not monotone, and N is clipped at zero. With the spectral sparsity A 795 (stationary runs), bins with less than this fraction of the cell energy are set to zero, and only the window of directions that can receive energy is solved. A 798 distributes the sweeps over several threads in tiles of up to 16×16 cells: a tile waits for its upwind tiles, which gives the same result as one thread. With mesh refinement, the patches are threaded as well.

4.6.4 Source terms

Every source term in action units, $S = S_{tot}/\sigma$, is split into a production and a damping part (Patankar):

$$S(N) = P - D\,N, \qquad P \geq 0, \quad D \geq 0. \tag{4.69}$$

Positive contributions go to P. Negative contributions $S < 0$ give $D = -S/N$. For the quadruplets and the Battjes-Janssen breaking, the term is Newton-linearised around the latest spectrum: with $\partial S/\partial N < 0$, P is increased by $-(\partial S/\partial N)N$ and D by $-\partial S/\partial N$, as in SWAN. P goes to the right-hand side and D to the diagonal of Eq. (4.68).

Wind input (A 730, A 731, A 734). The friction velocity is $u_* = \sqrt{C_D}\,U_{10}$ with the drag coefficient of Wu [1982], $C_D = 1.2875 \cdot 10^{-3}$ for $U_{10} < 7.5\,\mathrm{m/s}$ and $(0.8 + 0.065\,U_{10}) \cdot 10^{-3}$ above. The linear growth after Cavaleri and Rizzoli [1981], with the filter of Tolman (1992), starts the growth from calm conditions:

$$S_{in,lin} = \frac{\alpha_{lin}}{2\pi g^2 \sigma}\Big(u_* \max(0, \cos(\theta - \theta_w))\Big)^4 \exp\Big(-(\sigma^*_{PM}/\sigma)^4\Big), \qquad \sigma^*_{PM} = \frac{g}{28 u_*}, \tag{4.70}$$

with α_{lin} = A 734. With A 732 1, the exponential growth of Komen et al. (1984) is added (with A 732 2, that of Yan (1987), as SWAN GEN3 WESTH): $S_{in,exp} = \max\Big(0,\, 0.25\,\frac{\rho_a}{\rho_w}\,(28\,\frac{u_*}{c}\cos(\theta - \theta_w) - 1)\Big)\sigma N$ with $\rho_a/\rho_w = 1.28/1025$ and $c = \sigma/k$.

Whitecapping (A 732 1, 2). With A 732 1, the dissipation of Komen et al. (1984), with the SWAN coefficients $C_{ds} = 2.36 \cdot 10^{-5}$ and $\tilde{s}^2_{PM} = 3.02 \cdot 10^{-3}$, is

$$S_{wc} = -C_{ds}\Big(\frac{\tilde{s}}{\tilde{s}_{PM}}\Big)^4 \Big(\frac{k}{\tilde{k}}\Big)^2 \tilde{\sigma}\,N, \qquad \tilde{s} = \tilde{k}\sqrt{m_0}. \tag{4.71}$$

With A 732 2, the saturation-based whitecapping of van der Westhuysen et al. (2007) is used: where the saturation $B_k = c_g k^3 E(\sigma)$ exceeds the threshold B_r, the dissipation is proportional to $C_{ds2}(B_k/B_r)^{p/2}$ (C_{ds2} and B_r from A 757), blended with a non-breaking dissipation of Komen type below the threshold. With A 732 1 or 2, the change of N per iteration is limited to $\gamma\alpha_{PM}/(2\sigma k^3 c_g)$, $\alpha_{PM} = 0.0081$, γ = A 735 (Ris 1997).

Quadruplets (A 733 1). The four-wave interactions are computed with the Discrete Interaction Approximation (DIA) of Hasselmann et al. (1985), with $\lambda = 0.25$, $C = 3\cdot 10^7$ and the shallow water scaling $R = 1 + \frac{5.5}{x}\left(1 - \frac{5x}{6}\right)\exp(-\frac{5x}{4})$, $x = \max(0.75\,\tilde{k}d, 0.5)$, as SWAN. The DIA needs uniform directions (no A 715 or A 716).

Triads (A 744 1). The Lumped Triad Approximation of Eldeberky (1996) transfers energy from $f/2$ to f. In terms of $E(f,\theta)$,

$$S^+(f) = \max\Big[0,\, \alpha_{EB}\, c\, c_g\, J^2\, |\sin\beta|\Big(E^2(f/2) - 2E(f/2)E(f)\Big)\Big],$$
$$S_{nl3}(f) = S^+(f) - 2S^+(2f), \tag{4.72}$$

with α_{EB} = A 745 and the interaction coefficient J of Madsen and Sørensen (1993). The biphase is $\beta = -\frac{\pi}{2} + \frac{\pi}{2}\tanh(U_{r,crit}/U_r)$ with the Ursell number $U_r = gH_s/(2\sqrt{2}\sigma_{01}^2 d^2)$. Triads act only for $U_r \geq$ urslim and for sum frequencies $\sigma <$ cutfr σ_{01} (A 736).

Depth-induced breaking (A 740, A 741). With A 740 1, the model of Battjes and Janssen (1978) gives the fraction of breaking waves Q_b from $(1-Q_b)/\ln Q_b = -(H_{rms}/H_m)^2$, with $H_{rms} = \sqrt{8m_0}$ and $H_m = \gamma d$. The dissipation is distributed in proportion to the spectrum:

$$S_{br} = -\frac{\alpha}{\pi}\, Q_b\, \sigma_{01}\, \frac{H_m^2}{H_{rms}^2}\, N. \tag{4.73}$$

With A 737 1, the total energy of a cell is limited to $m_0 \leq (\gamma d)^2/4$ before the source terms are evaluated, as in SWAN. The surfbeat model uses Roelvink (1993) with A 740 2: $Q_b = 1 - \exp\Big(-(H/(\gamma d))^n\Big)$, n = A 746, and the dissipation rate $D = 2\alpha f_{rep} Q_b H/d$ with $H = \sqrt{8E}$ of the wave group.

Bottom friction (A 742, A 743). The JONSWAP formulation of SWAN has the damping rate $D = C_b(\sigma/\sinh kd)^2/g^2$, with the coefficient C_b from A 743.

Vegetation (A 754–A 756). With A 754 1, the dissipation of Dalrymple et al. (1984) in the form of Suzuki et al. (2011) (SWAN VEGETATION) acts on all bins with the rate

$$D_v = \sqrt{\frac{2}{\pi}}\, g^2 C_D b_v N_v \left(\frac{\tilde{k}}{\sigma_{01}}\right)^3 \frac{\sinh^3(\tilde{k}h_v) + 3\sinh(\tilde{k}h_v)}{3\tilde{k}\cosh^3(\tilde{k}d)} \sqrt{m_0}, \tag{4.74}$$

with the vegetation height $h_v \leq d$, stem diameter b_v, stem density N_v and drag coefficient C_D (A 755, or N_v per cell from A 756). With A 754 2, the dissipation of Jacobsen et al. (2019) is computed per frequency from the orbital velocities $\Omega_l(z) = \sigma_l \cosh(k_l z)/\sinh(k_l d)$ (z above the bed): $D_{v,l} = \sqrt{2/\pi}\, g^{-1} C_D b_v N_v \int_0^{h_v} \Omega_l^2 \Big(\sum_{l'} \Omega_{l'}^2 E_{l'}\Big)^{1/2} dz$, integrated with Simpson's rule.

Diffraction (A 718, A 719). The phase-decoupled approximation of Holthuijsen et al. (2003) uses the diffraction parameter

$$\delta = \frac{\nabla \cdot \left(c_g / k \, \nabla \sqrt{E} \right)}{k \, c_g \sqrt{E}}, \qquad C_a = \sqrt{1+\delta} \quad (C_a = 1 \text{ for } \delta \le -1). \tag{4.75}$$

The geographic group velocity is scaled with C_a, the depth refraction term in Eq. (4.66) as well, and c_θ gains $c_g(-\sin\theta\,\partial C_a/\partial x + \cos\theta\,\partial C_a/\partial y)$, which turns the waves into the shadow zones. The energy is smoothed before the derivatives are taken (A 719). C_a is computed from the total energy of the cell (A 718 1, as SWAN) or per frequency (A 718 2).

Obstacles and reflection (A 722–A 726). A line obstacle blocks the cell faces it crosses. The inflow through a blocked face is multiplied with K_t^2, and K_r^2 of the outflow is reflected into the direction $2\alpha - \theta$, specular at the obstacle line with orientation α or spread with $\cos^p$ around it (A 726). The rest is dissipated. K_t is constant, follows the formula of Goda for dams, $K_t = \frac{1}{2}\left(1 - \sin\left(\frac{\pi}{2\alpha_G}(R_c/H_s + \beta_G)\right)\right)$ with $\alpha_G = 2.6$, $\beta_G = 0.15$ and the freeboard R_c, or comes from A 725: d'Angremond et al. (1996) for rubble-mound breakwaters, a table per frequency, or a porous structure after Madsen (1974) with the resistance of van Gent [1995]. Faces between active and land cells reflect K_r^2 of the outflow at the local coastline direction with A 723 and A 724.

4.6.5 Boundary conditions

On the four domain sides (A 712), the inflow spectrum is a boundary spectrum (option 1), the spectrum of the boundary cell itself (zero gradient, option 2), or zero (open, option 0). Outgoing waves always leave the domain. The boundary spectrum (A 711) is a parametric spectrum, $E(\sigma,\theta) = S(\sigma)D(\theta)$, built from B 85 (Pierson-Moskowitz, JONSWAP or Torsethaugen), B 93, B 88 and the spreading function B 130 ($\cos^s$ or Mitsuyasu) around B 131, a SWAN spectral file, or a time series of SWAN spectra at several locations, interpolated in space and time. The directional spreading is normalised on the discrete directions, so that the energy is kept. With a computational sector (A 717), it is normalised over the full circle, and the part of the spectrum outside the sector is not imposed. Wind fields (A 730 2) and the boundary series are interpolated linearly in time from the date A 780 on. A 710 sets the initial spectrum.

4.6.6 Coupling with SFLOW and NHFLOW

With A 750 1, SEASTATE runs inside SFLOW (Section 4.5) or NHFLOW (Section 4.3). The host passes the total water depth, the free surface and the depth-averaged current (A 753). A wave step is computed every A 706 seconds of host time, and cells of the initial wave domain dry and re-wet with the water depth of the host. The wave force is multiplied with the ramp $\min(1, t/T)$, T = A 752, and added to the momentum equations of the host. In NHFLOW, it is applied uniformly over all σ-layers. The coupling is not available with mesh refinement, on curvilinear and multi-block grids (G 50), with the semi-implicit free surface (SFLOW A 226 1, NHFLOW A 534 1) and with the checkpoints (P 48, I 45). Two formulations are available (A 751):

Radiation stress (A 751 1). Following Longuet-Higgins and Stewart (1964), the force is $F_i = -\partial S_{ij}/\partial x_j$ with

$$S_{ij} = g \iint E\left[n\frac{k_i k_j}{k^2} + \left(n - \tfrac{1}{2}\right)\delta_{ij}\right] d\sigma\, d\theta + g \int R\,\frac{k_i k_j}{k^2}\, d\theta, \qquad n = c_g k/\sigma, \tag{4.76}$$

including the roller energy R of the surfbeat model. The host velocity is then the mass transport velocity, and the Eulerian current $\mathbf{U} - \mathbf{M}/h$ is passed back to SEASTATE, with the Stokes transport

$$\mathbf{M} = g \iint E\,\frac{\mathbf{k}}{\sigma}\, d\sigma\, d\theta + 2g \int \frac{R}{c}\,\frac{\mathbf{k}}{k}\, d\theta. \tag{4.77}$$

Vortex force (A 751 2). The depth-averaged vortex force formulation of McWilliams et al. (2004) and Smith (2006) is used with an Eulerian host velocity:

$$\mathbf{F} = \mathbf{F}_d + B\,\nabla d - \nabla Q + \chi(M_y, -M_x) - \mathbf{U}\,\nabla\cdot\mathbf{M}, \qquad \frac{\partial \eta}{\partial t} + \nabla\cdot(h\mathbf{U}) = -\nabla\cdot\mathbf{M}, \tag{4.78}$$

with $Q = g\iint E(n - \frac{1}{2})\, d\sigma\, d\theta$, $B = g\iint E\, k/\sinh(2kd)\, d\sigma\, d\theta$ and the vorticity $\chi = \partial V/\partial x - \partial U/\partial y$. The dissipation force $\mathbf{F}_d = -g\iint S_{net}\,\mathbf{k}\, d\sigma\, d\theta$ uses the net source terms S_{net} without the wind input, in action units. With the roller, the roller dissipation replaces the wave breaking in $\mathbf{F}_d$. $\nabla\cdot\mathbf{M}$ is computed in flux form with zero transport through land and domain edges, so that the mass source integrates to zero. The term $-\mathbf{U}\,\nabla\cdot\mathbf{M}$ balances the momentum of the mass source in the conservative momentum equation of the host.

4.6.7 Surfbeat

The surfbeat model (A 770 1) follows XBeach (Roelvink 1993, Reniers et al. 2004). It solves Eq. (4.64) for the wave-group energy on one representative frequency $f_{rep} = 1/T_{rep}$ (A 771, default $T_{m-1,0}$ of the boundary spectrum) and the A 703 directions, without frequency shift and with the Roelvink breaking (A 740 2). The geographic advection is first-order implicit or Crank-Nicolson with a van Leer limited second-order correction (A 775). After each step, $\sqrt{8E}$ is limited to A 747 times the water depth. At the x_- side, a random-phase wave train with the components $a_m = \sqrt{2S(f_m)\Delta f}$, $\Delta f = 1/T_{rec}$ (A 772, seed A 773), is built from the boundary spectrum. Its envelope $Z = \sum_m a_m \exp\!\big(i(2\pi f_m t - k_m y \sin\theta_m + \varphi_m)\big)$ gives the incoming group energy $E(t, y) = |Z|^2/2$, distributed over the directions with the mean directional distribution of the spectrum.

Roller (A 748, A 749). The roller energy $R(\theta)$ is transported with the phase speed plus the current:

$$\frac{\partial R}{\partial t} + \frac{\partial\big((c\cos\theta + U)R\big)}{\partial x} + \frac{\partial\big((c\sin\theta + V)R\big)}{\partial y} = D_w(\theta) - \frac{2g\beta R}{c}. \tag{4.79}$$

The source D_w is the breaking dissipation of the wave groups, and the roller dissipation $2g\beta R/c$ follows Reniers et al. (2004) with the front slope β = A 749. The equation is solved with the same implicit upwind sweeps as the wave action.

Long waves (A 774). The bound long wave of the wave groups, $\zeta_b = \sum_{m<n} D_{mn} a_m a_n \cos\Delta\psi_{mn}$, follows from the second-order difference interactions of Herbers et al. (1994), with the phase difference $\Delta\psi_{mn}$ of the components and the coefficients D_{mn} converted to surface elevation as in XBeach. It enters the host model through a weakly reflective boundary after Van Dongeren and Svendsen (1997),

$$u_b = \frac{q_b}{h} - \sqrt{\frac{g}{h}}\,(\eta - \zeta_b), \tag{4.80}$$

with the flux q_b of the bound wave, so that outgoing long waves are absorbed.

4.6.8 Handover and mesh refinement

Handover (A 760, A 761). At given points, the 2D spectrum is written in the format of the 2D spectrum input of the wave generation (B 85 11), as $S(\omega,\theta) = E(\sigma,\theta)\Delta\sigma/\Delta\omega$ on the direction sector A 761 around the mean direction. Every wave component of the phase-resolving model (FNPF or NHFLOW) then carries the energy of its SEASTATE bin.

Mesh refinement (G 1). Stand-alone runs can use static REEFAMR patches (Section 6.7.1) with the refinement ratio 2. They are defined with refinement boxes (G 10, G 11), the depth, coastline and bathymetry-gradient criteria A 791–A 793 and the water width criterion A 762, and no patches are placed within A 794 cells of the forced sides. Each patch has its own spectra and solver. The bed of the patches comes from the bathymetry raster (A 790 1) or is interpolated from level 0. The levels are swept per directional quadrant from coarse to fine and back, with constant prolongation into the ring cells of the patches and restriction of the fine cells, or (A 797 1) in one composite sweep, in which every covered cell sweeps its finer cells when the sweep reaches it. The composite sweep needs the iterations of a uniform grid and, on several MPI ranks, is pipelined across the ranks (A 781). With FAS (A 758), level 0 is the coarse grid of the patches of all levels, and A 798 also shares the patches out to the threads.

4.6.9 Key input commands

Command	Meaning
A 10 7	select SEASTATE (stand-alone)
A 700	nonstationary (1) or stationary (2)
A 701–A 703, A 715	frequencies, frequency range, directions, fine direction sector
A 716, A 717	automatic fine direction sector, computational direction sector
A 704, A 705	tile size of the spectral storage, minimum water depth
A 706, A 707	time step (coupling interval), iterations per time step
A 708, A 709, A 799	stationary convergence criterion, percentage of cells, convergence test
A 710–A 712	initial spectrum, boundary spectrum, boundary type of the sides

Command	Meaning
A 713, A 714	refraction, frequency shift
A 718, A 719	diffraction and smoothing
A 720, A 721	prescribed current
A 722–A 726	obstacles, reflecting coasts, structures, diffuse reflection
A 730–A 735	wind, deep-water physics (Komen, van der Westhuysen), quadruplets, linear growth, limiter
A 757	van der Westhuysen whitecapping coefficients
A 736, A 744, A 745	triads (LTA)
A 737, A 740, A 741, A 746	depth-induced breaking (Battjes-Janssen, Roelvink)
A 738, A 739	source iterations (stationary)
A 742, A 743	bottom friction (JONSWAP)
A 754–A 756	vegetation
A 750–A 753	coupling with SFLOW or NHFLOW, wave force, ramp, feedback
A 760, A 761	handover spectra for the wave generation (B 85 11)
A 747–A 749, A 770–A 775	surfbeat, roller, long waves
A 780	start date of the forcing files
A 790–A 794, A 797, A 762, G 1	bathymetry raster and mesh refinement, water width criterion
A 758	FAS coarse-grid correction
A 781	sweeps pipelined across the MPI ranks
A 795, A 796, A 798	spectral sparsity, second-order fluxes, threads
B 85, B 88, B 93, B 130, B 131	parametric spectrum and directional spreading

5. Multiphysics

Besides the flow of water and air, REEF3D contains a number of models for the physical processes that interact with the flow: rigid floating bodies with mooring lines and aquaculture nets, flexible structures, the structural solver REEF3D::CORAL, sediment transport and scour, particle-laden flows, porous media and vegetation, heat and concentration transport, non-Newtonian rheology, Lagrangian particle tracking, sea ice and wind forcing. These multiphysics models are not separate solvers. They are coupled to the hydrodynamic modules of Chapter 4, receive the velocities, the pressure and the free surface from them and act back on the flow through source terms, boundary conditions or a change of the geometry. This chapter summarizes the governing equations and the numerical methods of these models in the same compact form as Chapter 4; the shared numerics (grid, WENO schemes, time stepping, linear solvers) are described there and are not repeated. The commands that activate a model are given in parentheses and are described in Chapter 6.

5.1 Overview

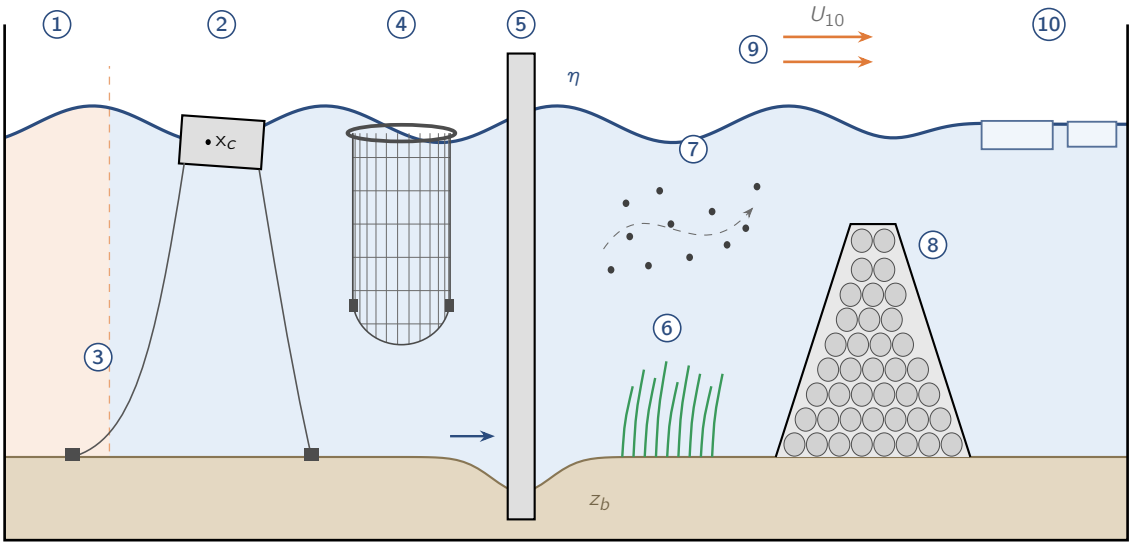

Figure 5.1: Multiphysics models of REEF3D in a schematic coastal wave tank: (1) wave generation zone (Section 4.1.7), (2) floating body with the centre of gravity $\mathbf{x}_c$ (Section 5.2), (3) mooring lines and (4) aquaculture net cage (Section 5.3), (5) pile with a scour hole in the sand bed z_b (Section 5.5; particle-based sediment: Section 5.6), (6) vegetation patch (Section 5.7; flexible vegetation as rod trees: Section 5.4), (7) drifting Lagrangian particles (Section 5.9), (8) porous rubble-mound breakwater (Section 5.7), (9) wind forcing (Section 5.11) and (10) ice floes (Section 5.10).

Figure 5.1 shows the multiphysics models in a schematic coastal numerical wave tank. The models are coupled to the hydrodynamics in different ways:

- In NHFLOW and CFD, rigid bodies, nets, flexible strips, rod trees and FEM solids are represented on the fixed grid and exchange forces with the flow through forcing terms in the momentum equations; in FNPF, a floating body enters through the boundary condition on its wetted surface. The body or structure is then moved with its own equation of motion (Sections 5.2 to 5.4).

- The structural solver REEF3D::CORAL is a multibody solver of its own (Z 40, `coral.txt`); it drives the 6DOF bodies of the fluid solver or receives their loads, and it can sample the fluid velocity and free surface for its own members and mooring lines (Section 5.12).
- Sediment transport changes the bed level z_b and thus the geometry of the flow domain (Section 5.5); the CPM (MP-PIC) model instead represents the sediment as computational parcels (Section 5.6), and the DEM resolves individual rigid particles (Section 6.5).
- Porous media and vegetation modify the governing equations through the porosity and additional resistance terms (VRANS, Section 5.7).
- Heat and concentration are transported scalars that change the density and viscosity of the fluid (Section 5.8).
- Lagrangian particles are passive: they are moved by the flow but do not act back on it (Section 5.9).
- Sea ice floes and wind act on the free surface through an additional surface pressure or stress (Sections 5.10 and 5.11).
- The spectral wave model REEF3D::SEASTATE can run inside SFLOW or NHFLOW (A 750) and passes the wave forcing, as radiation stress or vortex force, to the flow (Section 4.6).

Table 5.1 lists in which hydrodynamic modules each model is available and which command activates it. Not every model is implemented in every module: the σ-grid and depth-averaged modules do not resolve the air phase, and some models have so far only been implemented in one module. The validity of each command is also given in the reference entries of Chapter 6.

The main restrictions follow from the governing equations of the modules. FNPF solves the Laplace equation for the potential Φ and therefore has no viscous stresses, turbulence or transport equations; its multiphysics models (floating bodies, moorings, ice floes and wind) act through the boundary conditions on the body surface and on the free surface. SFLOW is depth-averaged, so that bedload transport, the rheology of the depth-averaged flow, one-way body motion and the meteorological forcing with the Coriolis force are available, but not the three-dimensional models. NHFLOW and CFD resolve the vertical flow structure and provide most of the models; the two-phase CFD module is required for heat and concentration transport, flexible strips, vegetation and the particle-based sediment model, while the membranes are only implemented in NHFLOW.

5.2 Floating Bodies (6DOF)

Floating rigid bodies are handled by one kernel shared by all modules (body state, surface triangulation, moorings, nets, time integration); only the body representation in the fluid and the loads are module specific. The module is switched on with X 10. **CFD** supports two-way coupling by direct forcing (X 10 1), with several bodies (X 20). **NHFLOW** supports two-way (1) and one-way coupling (2), a moving pressure field for ship waves (3) and prescribed

Table 5.1: Availability of the multiphysics models in the hydrodynamic modules (●available, ○not available) and the command that activates the model. Letters refer to the notes below the table.

Model	SFLOW	FNPF	NHFLOW	CFD	Commands	Sec.
Floating bodies (6DOF)	●[a]	●[b]	●	●	X 10	5.2
Ship module (hull loads, propeller, rudder, MMG)	○	●	●	●	X 350 (`ship.dat`)	5.2
Mooring lines	○	●	●	●	X 310	5.3
Nets	○	○[c]	●	●	X 320	5.3
FSI: flexible strips	○	○	○	●	Z 10	5.4
FSI: rod trees	○	○	●	●	Z 20 (`rodtree.dat`)	5.4
Membranes (closed flexible net cages)	○	○	●	○	X 330 (`membrane.dat`)	5.4
FSI: FEM solids	○	○	●	●	Z 30 (`fem.dat`)	5.4
Structural solver CORAL	○	●[d]	●[d]	●[d]	Z 40 (`coral.txt`)	5.12
Sediment: bedload	●	○	●	●	S 10, S 11	5.5
Sediment: suspended load	○	○	●	●	S 12	5.5
CPM (MP-PIC) sediment particles	○	○	○	●	Q 10 1/2, Q 11	5.6
DEM particles (rigid, shaped)	○	○	●	●	E 10 (`dem.txt`)	6.5
Porous media (VRANS)	○	○	●	●	B 201, B 202, B 210–B 230 (NHFLOW); B 270–B 291 (CFD); B 260 (both)	5.7
Vegetation	○	○	○	●	B 308, B 310–B 322	5.7
Heat, concentration, stratification	○	○	○	●	H 10, C 10	5.8
Non-Newtonian rheology	●	○	○	●	W 90, W 110	5.8
Lagrangian particles	○	○	●	○	L 10	5.9
Sea ice floes	○	●	○	○	A 380 (`ice_floes.dat`)	5.10
Wind forcing	●[e]	●	●	○	A 370 (FNPF), A 570 (NHFLOW), W 200 (SFLOW, NHFLOW)	5.11
Meteorological forcing (wind, air pressure), Coriolis force	●	○	●	○	W 200 – W 227, W 211	5.11.1

[a] SFLOW: one-way coupling only, the body moves with the prescribed or initial velocities without hydrodynamic feedback (X 10 2), or acts as a moving pressure field on the free surface for ship waves (X 10 3).
[b] FNPF: several bodies (X 20), coupled through their added mass; all geometries (X 110 – X 180) are triangulated; moorings and a power take-off with controllers (X 500 – X 508) are supported.
[c] FNPF: nets (X 320) are ignored with a warning.
[d] CORAL: two-way coupling through the 6DOF bodies (X 10 1) and one-way body loads in all three modules; the sampled fluid kinematics for the CORAL members and mooring lines in FNPF and NHFLOW only.
[e] SFLOW: wind only through the meteorological forcing W 200.

wavemakers (4). **FNPF** (X 10 > 0) resolves the body in the Laplace problem with implicit added mass, with several bodies (X 20) coupled through their added mass, and supports a power take-off (X 500), and **SFLOW** supports one-way coupling (2) and the moving pressure field (3). NHFLOW and SFLOW use one body. With REEF3D::CORAL (Z 40, Section 5.12), the 6DOF bodies of FNPF, NHFLOW and CFD can instead follow the dynamics of a CORAL model. Contacts between the bodies and DEM particles (Section 6.5) are not implemented. The body is a closed surface triangulation from primitives or an STL file (Section 6.18); mass, centre of gravity and inertia follow from it and the density X 21, unless given with X 22–X 24 (Fig. 5.2).

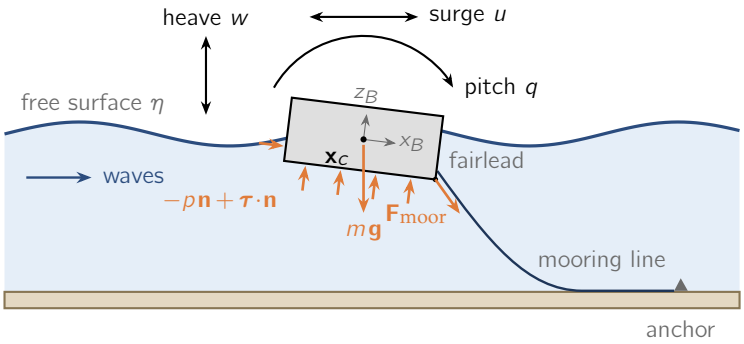

Figure 5.2: Floating body in waves: centre of gravity $\mathbf{x}_C$ with body axes, surface loads, gravity and a mooring line at the fairlead. Surge, heave and pitch are the 2D degrees of freedom; 3D adds sway v, roll p and yaw r (X 11).

5.2.1 Rigid-body equations

The orientation is represented by the unit quaternion $\mathbf{e} = (e_0, e_1, e_2, e_3)$, which avoids the singularities of Euler angles (these are only used for the initial rotation X 101 and the output). With

$$\mathbf{E} = \begin{pmatrix} -e_1 & e_0 & -e_3 & e_2 \\ -e_2 & e_3 & e_0 & -e_1 \\ -e_3 & -e_2 & e_1 & e_0 \end{pmatrix}, \qquad \mathbf{G} = \begin{pmatrix} -e_1 & e_0 & e_3 & -e_2 \\ -e_2 & -e_3 & e_0 & e_1 \\ -e_3 & e_2 & -e_1 & e_0 \end{pmatrix}, \qquad \mathbf{R} = \mathbf{E}\mathbf{G}^T, \tag{5.1}$$

$\mathbf{R}$ rotates body-frame vectors into the inertial frame. The linear momentum $\mathbf{P} = m\mathbf{u}_c$ and the position $\mathbf{x}_c$ (inertial frame) and the angular momentum $\mathbf{h}$ and the quaternion $\mathbf{e}$ (body frame) are integrated in the Hamiltonian form

$$\frac{d\mathbf{P}}{dt} = \mathbf{F}, \qquad \frac{d\mathbf{x}_c}{dt} = \frac{\mathbf{P}}{m}, \qquad \frac{d\mathbf{e}}{dt} = \frac{1}{2}\mathbf{G}^T\mathbf{I}_B^{-1}\mathbf{h}, \qquad \frac{d\mathbf{h}}{dt} = 2\dot{\mathbf{G}}\mathbf{G}^T\mathbf{h} + \mathbf{R}^T\mathbf{M}, \tag{5.2}$$

with the body-frame inertia tensor $\mathbf{I}_B$ about $\mathbf{x}_c$. With $\boldsymbol{\omega} = \mathbf{R}\mathbf{I}_B^{-1}\mathbf{h}$, a body point moves with $\mathbf{u}_b = \mathbf{u}_c + \boldsymbol{\omega} \times (\mathbf{x} - \mathbf{x}_c)$, and the surface points are moved rigidly, $\mathbf{x} = \mathbf{x}_c + \mathbf{R}\mathbf{x}_0$. Force and moment (about $\mathbf{x}_c$) are collected in the inertial frame:

$$\begin{aligned} \mathbf{F} &= \mathbf{F}_{\text{hyd}} + m\mathbf{g} + \sum \mathbf{F}_{\text{moor}} + \sum \mathbf{F}_{\text{net}} - \mathbf{C}_t\mathbf{u}_c, \\ \mathbf{M} &= \mathbf{M}_{\text{hyd}} + \sum (\mathbf{x}_f - \mathbf{x}_c) \times \mathbf{F}_{\text{moor}} + \sum \mathbf{M}_{\text{net}} - \mathbf{C}_r\boldsymbol{\omega}, \end{aligned} \tag{5.3}$$

with $\mathbf{g}$ = (W 20, W 21, W 22), the linear damping coefficients $\mathbf{C}_t$ (X 26) and $\mathbf{C}_r$ (X 25), the fairlead positions $\mathbf{x}_f$ and the mooring and net loads of Section 5.3. X 11 selects for each degree of freedom whether it is solved (1), fixed (0) or prescribed (2); in 2D, sway, roll and yaw are switched off.

Prescribed motion. Prescribed degrees of freedom follow constant velocities (X 210, X 211) or a motion file (X 240, X 241) with a ramp (X 205, X 206). In NHFLOW, X 10 2 imposes the motion on the flow without feedback, and X 10 4 moves a piston (X 170) or flap (X 172) wavemaker through the same direct forcing.

Time integration. Eq. (5.2) is advanced stage by stage with the Runge–Kutta scheme of the fluid solver (CFD: X 10 >0 changes N 40 3 to N 40 4; NHFLOW: A 510, Eq. 4.4; FNPF: A 310; SFLOW: A 210), normalizing the quaternion after every stage. The coupling is explicit in CFD and NHFLOW; the direct forcing uses the stage weight α_s.

5.2.2 Hydrodynamic loads

With X 60 1, force and moment are integrated over the surface triangles,

$$\mathbf{F}_{\mathrm{hyd}} = \sum_{\triangle} (-p\mathbf{n} + \boldsymbol{\tau}\cdot\mathbf{n})\, A_{\triangle}, \qquad \mathbf{M}_{\mathrm{hyd}} = \sum_{\triangle} (\mathbf{x}_{\triangle} - \mathbf{x}_c) \times (-p\mathbf{n} + \boldsymbol{\tau}\cdot\mathbf{n})\, A_{\triangle}, \tag{5.4}$$

with the normal $\mathbf{n}$ into the fluid and the area $A_{\triangle}$ (X 60 2: triangulation of the body level set, CFD and NHFLOW). In **CFD**, p is sampled X 42 cell sizes off the wall, and the viscous stress (X 38, Section 6.18) is computed from the velocity X 43 cell sizes off the wall. In **NHFLOW** and **FNPF**, each triangle is clipped at the local free surface $z = d + \eta(x, y)$, and the wetted part is integrated with the edge midpoints, which is exact for the hydrostatic pressure on a planar facet. NHFLOW uses $p = p_{nh} + \rho g \max(0,\, d + \eta - z)$ (plus a skin friction with X 38 1), FNPF the Bernoulli pressure of Eq. (5.7).

5.2.3 Coupling with the hydrodynamic modules

CFD: level set of the body and direct forcing. After every position update, the signed distance ϕ_b of the body (negative inside) is rebuilt from the triangulation by ray casting (X 188). The body velocity is imposed on the predicted velocity $\mathbf{u}^*$ of each stage by the continuous direct forcing

$$f_i = H(\phi_b)\,\frac{u_{b,i} - u_i^*}{\alpha_s \Delta t}, \qquad H = \begin{cases} 1 & -\phi_b > \epsilon \\ 0 & -\phi_b < -\epsilon \\ \frac{1}{2}\left(1 - \dfrac{\phi_b}{\epsilon} - \dfrac{1}{\pi}\sin\dfrac{\pi\phi_b}{\epsilon}\right) & \text{else} \end{cases}, \tag{5.5}$$

at the staggered velocity positions, with ϵ = X 41 times the mean cell size. With X 14 1, the forcing outside the body is weighted with $|n_i|$ of the level set normal.

NHFLOW: ray casting on the σ-grid. The level set of the body is built on the moving σ-grid (Fig. 5.3): vertical rays through the triangles flag the cell centres of each column as inside or outside, and the signed distance is computed in a band around the surface. Eq. (5.5) is applied to the cell-centred velocities with ϵ = A 526 cell sizes, as no-slip (X 15 0), $|n_i|$-weighted (1) or free-slip forcing (2); X 15 3 forces only the cells inside the body and takes the momentum fluxes across the body surface from the fluid side (for hulls moving at speed), X 15 4 is the free-slip forcing with the smooth transition of option 1, and X 15 5 option 1 with a sharp switch at the body surface. With X 15 6, the faces between body and fluid cells also act as reflective walls moving with the body in the HLL fluxes, and X 15 7 combines these walls with the forcing of option 3 for bodies at speed in waves. X 15 8 uses no direct forcing at all: the hull is a sharp internal boundary of the σ-grid whose faces are blocked in the fluxes and in the pressure equation, with the enclosed water moving with the hull (the pressure scheme A 520 2 is then changed to 1). With X 13 1, the water level in the columns pierced by the body is the harmonic extension of the surrounding free surface, as in FNPF; with the prototype X 13 2, the water columns under the hull end at the hull bottom instead. With X 16, the body is a moving porous medium with a Darcy–Forchheimer resistance on the velocity relative to $\mathbf{u}_b$.

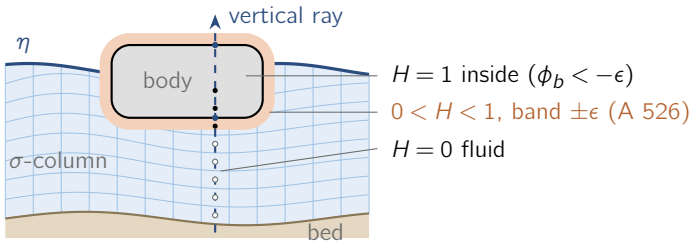

Figure 5.3: NHFLOW body on the σ-grid: vertical rays flag the cell centres inside the body (filled); the signed distance in a band defines the Heaviside function H of the forcing.

FNPF: resolved body with implicit added mass. The body is a boundary of the Laplace problem for Φ (Section 4.4). A vertical ray per σ-column flags the σ-nodes inside the body; these are decoupled from the Laplace matrix, and every fluid/body face carries the staircase Neumann condition

$$\left.\frac{\partial \Phi}{\partial x_i}\right|_{\text{face}} = \mathbf{u}_b \cdot \mathbf{e}_i, \qquad \mathbf{u}_b = \mathbf{u}_c + \boldsymbol{\omega} \times \mathbf{r}, \quad \mathbf{r} = \mathbf{x} - \mathbf{x}_c, \tag{5.6}$$

with the face direction $\mathbf{e}_i$. In the body footprint, η and $\tilde{\Phi}$ are harmonically extended from the surrounding free surface. The pressure follows from the Bernoulli equation,

$$p = -\rho\left(\psi + \tfrac{1}{2}|\nabla\Phi|^2\right) + \rho g\,(d - z), \qquad \psi = \frac{\partial \Phi}{\partial t}, \tag{5.7}$$

sampled half a mean cell off the hull. ψ is solved with the same Laplace operator; its hull condition contains the unknown body acceleration $\mathbf{a} = (\dot{\mathbf{u}}_c, \dot{\boldsymbol{\omega}})$, so ψ is split into

$$\begin{aligned}
&\psi = \psi_0 + \sum_{j=1}^{6} a_j \psi_j, \qquad \mathbf{N} = (\mathbf{n}, \mathbf{r} \times \mathbf{n}), \\
&\psi_0 = \frac{\partial \tilde{\Phi}}{\partial t} - \frac{\partial \Phi}{\partial z}\frac{\partial \eta}{\partial t} \text{ (free surface)}, \quad \frac{\partial \psi_0}{\partial n} = \Big(\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})\Big) \cdot \mathbf{n} \text{ (hull)}, \\
&\psi_j = 0 \text{ (free surface)}, \quad \frac{\partial \psi_j}{\partial n} = N_j \text{ (hull)},
\end{aligned} \tag{5.8}$$

where the m-terms of the hull condition are neglected. ψ_0 gives the load $\mathbf{F}_0$, and the unit modes give the instantaneous added-mass matrix $A_{ij} = -\rho \int_S \psi_j N_i \, dS$ (symmetrized) over the wetted hull. The body is advanced with

$$\begin{pmatrix} m\mathbf{I} + \mathbf{A}_{tt} & \mathbf{A}_{tr} \\ \mathbf{A}_{rt} & \mathbf{I}_I + \mathbf{A}_{rr} \end{pmatrix} \begin{pmatrix} \dot{\mathbf{u}}_c \\ \dot{\boldsymbol{\omega}} \end{pmatrix} = \begin{pmatrix} \mathbf{F} \\ \mathbf{M} - \boldsymbol{\omega} \times (\mathbf{I}_I \boldsymbol{\omega}) \end{pmatrix}, \tag{5.9}$$

with $\mathbf{I}_I = \mathbf{R}\mathbf{I}_B\mathbf{R}^T$ and $\mathbf{F}$, $\mathbf{M}$ from Eq. (5.3) with $\mathbf{F}_{\mathrm{hyd}} = \mathbf{F}_0$; the implicit added mass keeps the coupling stable for light bodies. All body geometries are triangulated (primitives X 110 – X 164 including the cylinders X 131 – X 133 and the wedge X 153, and STL files X 180), and nets are ignored. With several bodies (X 20), Eq. (5.9) is solved for all bodies together, including the added-mass cross terms between them. X 27 extends the footprint of a surface-piercing body to the columns in which the free surface lies only a few σ-layers above the hull. For surface-piercing hulls moving at speed, X 17 relaxes η in the columns next to the body footprint, and X 18 removes the running mean of the time derivative of the potential following the body points near the waterline, which would otherwise give a spurious mean load (e.g. a yaw moment in a steady turn). X 47 filters η and $\tilde{\Phi}$ in a few columns around the footprint with the 5-point filter of Longuet-Higgins and Cokelet (1976) against the two-cell mode fed by the waterline. To save Laplace solves, the added mass can be updated only every X 28 time steps or after a motion beyond the thresholds X 29, the body load solves can use a relaxed tolerance (X 30) and an extrapolated initial guess (X 35), and with the mesh refinement zones around the bodies the unit modes can be solved on the patches of the body only (X 36, calibrated every X 37 updates).

FNPF and NHFLOW: power take-off. With X 500, a power take-off acts between the body and the ground, e.g. for a heaving point absorber. It is a prismatic joint with the axis and attachment point X 501 (default: heave at the centre of gravity), and its coordinate q and velocity $\dot{q}$ drive a linear damper (X 502), a spring-damper (X 503) and end-stops (X 504). The generator force can be limited in force and power (X 505) and controlled: tuned passive damping from the added mass and radiation damping of the table `pto_hydro.dat` (X 506), latching (X 507) and declutching (X 508). The resulting force acts along the joint axis and adds a force and a moment to Eq. (5.3). With X 500 1 it is added explicitly like the mooring forces; with X 500 2 the Jacobians of the stiff end-stops and of the latch are added to the added-mass system of Eq. (5.9) (linearly implicit), while dampers and springs stay explicit. The PTO coordinate, force, absorbed power and energy are written to `REEF3D_FNPF_6DOF` (NHFLOW: `REEF3D_NHFLOW_6DOF`). When the body follows the dynamics of CORAL (Z 40), X 500 is not used; the PTO is then part of the CORAL model.

SFLOW: one-way forcing and moving pressure fields. With X 10 2, the depth-averaged velocities under the body footprint are forced to the body velocity as in Eq. (5.5), with a 2D footprint level set. With X 10 3 (SFLOW and NHFLOW, ship waves), the body is a moving surface pressure $p_s(x, y, t)$ (X 400–X 410) with the momentum source $-(h/\rho)\nabla p_s$, moving with the prescribed velocities.

Ship module. With X 350 1, semi-empirical hull loads, propulsion and steering are added to the two-way coupled body (X 10 1) in FNPF, NHFLOW and CFD, with the parameters from the file `ship.dat` (Section 6.18). The hull loads are the ITTC-1957 frictional resistance with a form factor, roll damping and a cross-flow drag. The propeller is a body-force actuator disk with the Hough-Ordway radial distribution of thrust and torque from the open-water curves $K_T(J)$ and $K_Q(J)$; in NHFLOW and CFD, thrust and torque act on the fluid as a momentum source, otherwise the thrust acts on the hull only. The rudder follows the MMG standard method with a fixed, zig-zag or autopilot rudder angle, and the MMG manoeuvring model (Yasukawa and Yoshimura 2015) can replace or complement the surge, sway and yaw loads of the flow solver. The ship loads are written to `REEF3D_*_6DOF`.

5.2.4 Key input commands

X 10, X 11, X 20	6DOF mode, degrees of freedom, number of bodies (CFD, FNPF)
X 21–X 24	density, mass, centre of gravity, inertia
X 25, X 26	rotational and translational damping
X 110–X 164, X 180–X 190	primitives, STL geometry, refinement/remeshing
X 500–X 508	power take-off, limits, controllers (FNPF, NHFLOW)
X 17, X 18	FNPF moving hulls: relaxation next to the footprint, running mean of χ
X 27	FNPF footprint of the body in the σ-layers
X 28–X 30, X 35–X 37	FNPF added mass update, tolerance and initial guess of the body load solves
X 47	FNPF filter at the body footprint
X 13	NHFLOW water level in the columns pierced by the body
Z 40	bodies driven by CORAL (Section 5.12)
X 350	ship module (hull loads, propeller, rudder, MMG), input `ship.dat`
X 101–X 103	initial rotation and velocities
X 38, X 42, X 43, X 60	viscous force model, sampling distances, STL/LSM integration
X 14, X 41, X 188	CFD forcing: tangential velocity, layer width, ray casting
X 15, X 16, A 526	NHFLOW forcing type, porous body, layer width
X 205–X 241	ramps, prescribed velocities, motion files
X 170, X 172	NHFLOW wavemakers (X 10 4)
X 400–X 410	moving pressure field (X 10 3)
N 40, A 510, A 310, A 210	time schemes of CFD, NHFLOW, FNPF, SFLOW
X 19, X 50	output interval, VTP/STL output

5.3 Moorings and Nets

Mooring lines and nets add their loads to Eq. (5.3), evaluated in every Runge–Kutta stage. Moorings (X 310) are available in CFD and NHFLOW (X 10 1) and in FNPF, nets (X 320) in CFD and NHFLOW, where they are fixed structures with X 10 0. SFLOW supports neither.

5.3.1 Mooring models

A line (X 311) is defined by the anchor $\mathbf{x}_a$, the fairlead $\mathbf{x}_f$, the mass per length in air w_a and density ρ_c (submerged weight $w = w_a\, g\,(\rho_c - \rho)/\rho_c$), the axial stiffness EA, the diameter d_c, the length L and N_e elements. It breaks at the tension X 314 or the time X 315.

Quasi-static catenary (X 310 1). The elastic catenary with a part resting on the flat seabed (no friction) gives the fairlead forces H and V (Fig. 5.4). With the span $l_h = |\mathbf{x}_f - \mathbf{x}_a|_{xy}$ and the height $\Delta z = z_f - z_a$,

$$l_h = L - \frac{V}{w} + \frac{H}{w}\operatorname{asinh}\!\left(\frac{V}{H}\right) + \frac{HV}{EAw}, \tag{5.10}$$

$$\Delta z = \frac{H}{w}\left(\sqrt{1+\left(\frac{V}{H}\right)^2} - 1\right) + \frac{V^2}{2EAw}. \tag{5.11}$$

solved with Newton's method; $\mathbf{F}_{\text{moor}} = -(H\,\mathbf{e}_h + V\,\mathbf{e}_z)$ with the horizontal unit vector $\mathbf{e}_h$ from anchor to fairlead.

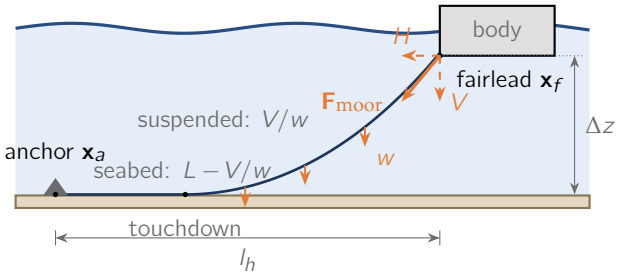

Figure 5.4: Quasi-static catenary (X 310 1) with seabed part, submerged weight w and fairlead forces H, V.

Spring (X 310 4). A spring (X 312) carries $T = \max(T_0 + k\,(|\mathbf{x}_f - \mathbf{x}_a| - L_0), 0)$ (stiffness k, pre-tension T_0, initial length L_0).

Quasi-static bar model (X 310 2). N_e bars of the unstretched length $l_0 = L/N_e$ with the directions $\mathbf{f}_j$ and tensions T_j satisfy knot equilibrium and geometric closure,

$$T_{j+1}\mathbf{f}_{j+1} - T_j\mathbf{f}_j = w\,\bar{l}_j\,\mathbf{e}_z - \mathbf{R}_j, \quad \sum_j l_j\,\mathbf{f}_j = \mathbf{x}_f - \mathbf{x}_a, \quad |\mathbf{f}_j| = 1, \quad l_j = l_0\left(1 + \frac{T_j}{EA}\right), \tag{5.12}$$

with the mean bar length $\bar{l}_j$ at knot j and a Morison-type drag $\mathbf{R}_j$, solved iteratively; the body force is $-T\mathbf{f}$ of the last bar.

Dynamic model (X 310 3). The line is a geometrically exact rod ($A = w_a/\rho_c$, no bending stiffness) solved with the rod solver of Section 5.4 and RADAU5 [Hairer and Wanner, 1996] once per time step. The loads per unit length are

$$\mathbf{f} = \tfrac{1}{2}\rho d_c\,(C_{dt}|u_t|u_t\,\mathbf{t} + C_{dn}|\mathbf{u}_n|\mathbf{u}_n) + \rho\frac{\pi d_c^2}{4}\,(\dot{\mathbf{u}}_f + C_{mt}a_t\,\mathbf{t} + C_{mn}\mathbf{a}_n) - w\,\mathbf{e}_z + \mathbf{f}_{\text{bed}}, \tag{5.13}$$

with the tangent $\mathbf{t}$, the relative velocity $\mathbf{u}_f - \dot{\mathbf{c}}$ and acceleration, $C_{dt} = 0.5$, $C_{dn} = 2.5$, $C_{mt} = 0$, $C_{mn} = 3.8$ [Morison et al., 1950], and a spring–damper seabed.

5.3.2 Nets

Nets (X 320–X 322: solidity S_n, twine diameter d_t and length l_t, knot diameter d_k) are meshes of knots and bars (twines). The load is computed for screens spanned by neighbouring bars (Fig. 5.5), with area A_s, normal $\mathbf{n}_s$ and inflow $\mathbf{u}_{\mathrm{rel}}$ at the angle θ to $\mathbf{n}_s$ [Kristiansen and Faltinsen, 2012]:

$$\mathbf{F}_s = \tfrac{1}{2}\rho A_s |\mathbf{u}_\infty|^2 (C_D\,\mathbf{n}_d + C_L\,\mathbf{n}_l), \qquad \mathbf{n}_d = \frac{\mathbf{u}_{\mathrm{rel}}}{|\mathbf{u}_{\mathrm{rel}}|}, \quad \mathbf{n}_l = \frac{(\mathbf{n}_d \times \mathbf{n}_s)\times \mathbf{n}_d}{|(\mathbf{n}_d \times \mathbf{n}_s)\times \mathbf{n}_d|}, \tag{5.14}$$

$$C_D = C_{D0}\left[(1 - a_3 - a_5)\cos\theta + a_3 \cos 3\theta + a_5 \cos 5\theta\right],$$
$$C_L = C_{L0}\left[b_2 \sin 2\theta + b_4 \sin 4\theta + b_6 \sin 6\theta\right], \tag{5.15}$$

with $a_3 = 0.0129$, $a_5 = 0.0151$, $b_2 = 1.2044$, $b_4 = 0.1466$, $b_6 = 0.135$,

$$C_{D0} = C_{\mathrm{cyl}} \frac{S_n(2 - S_n)}{2(1 - S_n)^2}, \quad C_{L0} = \frac{C_{D0}/2 - C_{T,\pi/4}}{\sqrt{2}}, \quad C_{T,\pi/4} = \frac{\pi C_{D0}\cos^2(\pi/4)}{8 + C_{D0}\cos^2(\pi/4)}. \tag{5.16}$$

where $C_{\mathrm{cyl}} = 1.1 + 4\,Re^{-1/2}$, $Re = d_t|\mathbf{u}|/(\nu(1 - S_n))$, and $|\mathbf{u}_\infty|$ follows from momentum theory. Y 1 selects the model of Martin et al. [2022].

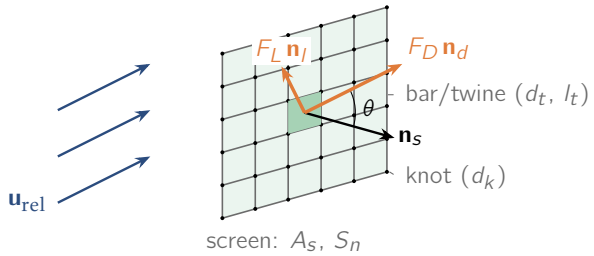

Figure 5.5: Net panel of knots and bars; screen (shaded) load from θ and Eq. (5.15).

Net structural models.

- **Quasi-static bar net** (X 320 2, 3): static knot equilibrium under weight, screen and twine loads, with the bar tensions as unknowns [Martin et al., 2020]; no load is returned to the body.
- **Dynamic bar net** (X 320 12–14): the knots carry the lumped twine mass plus added mass and move under

$$(m_k + m_{a,k})\,\ddot{\mathbf{x}}_k = \textstyle\sum_b T_b\,\mathbf{e}_b + \mathbf{F}_k,$$
$$\mathbf{F}_k = -g\,(m_k - \rho V_k)\,\mathbf{e}_z + (1 + C_a)\rho V_k \dot{\mathbf{u}}_f + \textstyle\sum_s \mathbf{F}_s + \mathbf{F}_{d,k} - \mathbf{r}\dot{\mathbf{x}}_k, \tag{5.17}$$

  with the bar directions $\mathbf{e}_b$, $C_a = 1$, knot and sinker drag $\mathbf{F}_{d,k}$ (X 323) and relaxation $\mathbf{r}$ (X 325). The tensions T_b of the inextensible bars follow from an accelerated Newton method [Martin and Bihs, 2021a]; the top-bar tensions give $\mathbf{F}_{\mathrm{net}}$, $\mathbf{M}_{\mathrm{net}}$.
- **Rigid sheets** (X 320 4): moving with the body, loads added to the body.

Coupling to the fluid. The screen load $\mathbf{F}_k$ at the centroid $\mathbf{X}_k$ of each net triangle [Martin et al., 2020] is spread onto the grid as a volume force with the four-point cosine kernel [Peskin, 2002]:

$$\mathbf{f}(\mathbf{x}) = -\sum_k \frac{\mathbf{F}_k}{\rho}\,\frac{\delta_h(\mathbf{x}-\mathbf{X}_k)}{\Delta x\,\Delta y\,\Delta z}, \qquad \varphi(r) = \begin{cases} \frac{1}{4}\left(1+\cos\frac{\pi r}{2}\right) & |r| \le 2 \\ 0 & \text{else} \end{cases},$$
$$\delta_h = \varphi\!\left(\frac{x-X_k}{\Delta x}\right)\varphi\!\left(\frac{y-Y_k}{\Delta y}\right)\varphi\!\left(\frac{z-Z_k}{\Delta z}\right). \tag{5.18}$$

It enters the momentum equations in the next predictor step: the net acts as a porous, momentum-absorbing surface not resolved by the grid.

5.3.3 Key input commands

X 310, X 311, X 312	mooring model, line definition, spring definition
X 313	rotate the fairleads with the initial body rotation
X 314, X 315	breaking tension and breaking time of the lines
X 320, X 321, X 322	net type, net material, net dimensions and position
X 323, X 324, X 325	sinkers, knot probes, net time step and relaxation (dynamic nets)
Y 1	simulation-based screen force model

5.4 Flexible Structures (FSI)

REEF3D has two models for flexible, slender structures, both treated as rods and coupled to the fluid through Lagrangian points and a regularized delta function: **flexible strips** (Z 10 1, Z 11, Z 12), thin blades resolved by the grid, modelled as a geometrically exact Cosserat rod with direct-forcing immersed boundary coupling (REEF3D::CFD only); and **rod trees** (Z 20 1, Z 21, input file `rodtree.dat`), branched soft corals, gorgonians or flexible vegetation thinner than the cells, with Morison-type loads spread as a body force (unresolved, actuator-line type coupling; REEF3D::CFD and REEF3D::NHFLOW). Neither is available in REEF3D::FNPF or REEF3D::SFLOW. Impermeable membranes (REEF3D::NHFLOW) and deformable FEM solids (REEF3D::CFD and REEF3D::NHFLOW) are summarized at the end of this section.

5.4.1 Flexible strips: geometrically exact rod

Each strip (Z 11) is a rod of length L, width W and thickness T, clamped at its bottom point and initially pointing in $+z$, with the width in y. It is described by the centre line $\mathbf{c}(s,t)$, $s \in [0, L]$, and a unit quaternion $\mathbf{q}(s,t)$ for the orientation of the cross-section (rotation matrix $\mathbf{R}(\mathbf{q})$), which avoids the singularities of Euler angles at large rotations [Martin and Bihs, 2021b]. The translational balance is

$$\rho_s A\,\frac{\partial^2 \mathbf{c}}{\partial t^2} = \frac{\partial \mathbf{n}}{\partial s} + \rho_s A\,\mathbf{f}_{\text{ext}}, \qquad \mathbf{n} = \mathbf{R}(\mathbf{q})\,\mathbf{n}^0, \tag{5.19}$$

with the material density ρ_s, $A = WT$, the internal force $\mathbf{n}$ and the external load per unit mass $\mathbf{f}_{\text{ext}}$. Body-frame force and moment follow from a linear visco-elastic law,

$$\mathbf{n}^0 = \mathbf{C}_\varepsilon\left(\mathbf{R}^T\frac{\partial \mathbf{c}}{\partial s} - \mathbf{R}_0^T\frac{\partial \mathbf{c}_0}{\partial s}\right) + \mathbf{D}_\varepsilon\dot{\boldsymbol{\varepsilon}}, \qquad \mathbf{C}_\varepsilon = \operatorname{diag}(EA,\ GA,\ GA), \tag{5.20}$$

$$\mathbf{m}^0 = \mathbf{C}_\kappa(\boldsymbol{\kappa} - \boldsymbol{\kappa}_0) + \mathbf{D}_\kappa\dot{\boldsymbol{\kappa}}, \qquad \mathbf{C}_\kappa = \operatorname{diag}(GI_x,\ EI_y,\ EI_z), \tag{5.21}$$

with E, $G = E/(2(1+\nu_s))$, I_x, I_y, I_z from Z 11, the damping matrices $\mathbf{D}_\varepsilon$, $\mathbf{D}_\kappa$ from Z 12, the reference configuration (index 0) and the curvature $\boldsymbol{\kappa}$ from the quaternion product of neighbouring cross-sections. The rotational balance is a second-order equation for $\mathbf{q}$ whose term $-|\dot{\mathbf{q}}|^2\mathbf{q}$ keeps $|\mathbf{q}| = 1$. The rod is discretized with finite differences on N elements (positions at the nodes, quaternions at the element centres), clamped at the bottom and free at the top, and integrated with the implicit Runge-Kutta method RADAU5 with step size control [Hairer and Wanner, 1996].

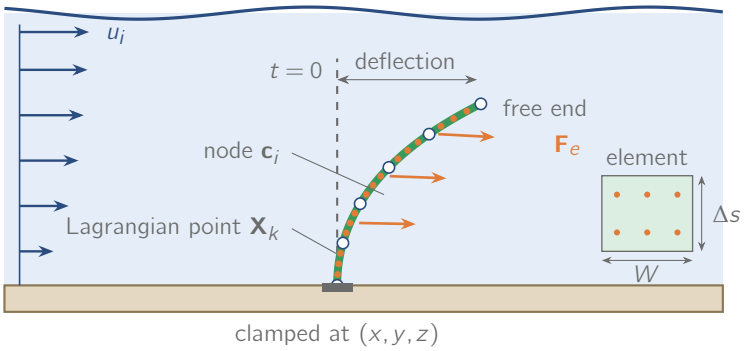

Figure 5.6: Flexible strip (Z 11) in a current: rod nodes $\mathbf{c}_i$ (white), Lagrangian points (orange) over the length and width of each element (inset), and the fluid load $\mathbf{F}_e$ bending the strip from its initial position (dashed).

5.4.2 Flexible strips: immersed boundary coupling

Each element carries $n_l \times n_w$ Lagrangian points $\mathbf{X}_k$ over its length and width, spaced close to the local cell size Δx_b, with the area ΔA_k and the fluid volume $\Delta V_k = \Delta A_k\,\Delta x_b$ (Fig. 5.6); they move rigidly with the element. Interpolation and spreading use the three-point kernel of Roma et al. [1999] at the staggered velocity positions,

$$\varphi(r) = \begin{cases} \frac{1}{3}\left(1 + \sqrt{1-3r^2}\right), & |r| \le \frac{1}{2}, \\ \frac{1}{6}\left(5 - 3|r| - \sqrt{1-3(1-|r|)^2}\right), & \frac{1}{2} < |r| \le \frac{3}{2}, \\ 0, & \text{otherwise}, \end{cases} \qquad \delta_h(\mathbf{x}-\mathbf{X}) = \frac{\varphi\left(\frac{x-X}{\Delta x}\right)\varphi\left(\frac{y-Y}{\Delta y}\right)\varphi\left(\frac{z-Z}{\Delta z}\right)}{\Delta x\,\Delta y\,\Delta z}. \tag{5.22}$$

In every Runge-Kutta stage (coefficient α) of the CFD momentum equations, the fluid velocity $\mathbf{U}_k$ is interpolated to the points and the rod is advanced over $\alpha\Delta t$. Its external load contains the buoyancy-reduced gravity $(1-\rho_f/\rho_s)\,\mathbf{g}$ and the fluid load from the momentum balance of the fluid carried by the points of element e: with $m_e = \sum_k \rho_f \Delta V_k$, the momentum $\mathbf{P}_e = m_e\bar{\mathbf{u}}_e + \boldsymbol{\omega}_e \times \mathbf{R}\mathbf{s}_0$ in the structural motion ($\mathbf{s}_0$ static moment of the points) and $\mathbf{P}_e^* = \sum_k \rho_f \Delta V_k \mathbf{U}_k$,

$$\mathbf{F}_e = -\frac{\mathbf{P}_e - \mathbf{P}_e^n}{t - t^n} - \frac{\mathbf{P}_e^n - \mathbf{P}_e^*}{\alpha\Delta t}, \tag{5.23}$$

with t the time within the RADAU5 integration; the moment follows analogously (switched off with Y 2 1). The structural point velocity $\mathbf{U}_k^s$ then gives the direct-forcing term $\mathbf{f}_k = (\mathbf{U}_k^s - \mathbf{U}_k)/(\alpha\Delta t)$, which is spread to the grid, $f_i(\mathbf{x}) = \sum_k f_{k,i}\,\delta_h(\mathbf{x}-\mathbf{X}_k)\,\Delta V_k$, and added as source term in Eq. (4.8). With the k-ω model (T 10 2), ω is set in the kernel footprint to represent the wall friction of the strip. STL output follows P 20/P 30, log files X 19.

5.4.3 Rod trees for soft corals

With Z 20 1, colonies are read from `rodtree.dat`: blocks `colony ... end` with `node` (id x y z r), `edge` (parent, child), `clamp`, `material` (E ν_s ρ_s), `hydro` (C_{dn} C_{dt} C_a [C_m], default $C_m = 1 + C_a$), `damping` (β), `refine` and optionally `instance`, `polyps`, `polyp_response`; global keywords `integrator`, `substeps`, `reaction`.

The tree is a discrete Cosserat rod of rigid, slightly conical cylinder elements (length l, mean radius r, mass m, centre $\mathbf{c}_e$, orientation $\mathbf{q}_e$), connected by elastic 6-DOF joints at the shared nodes; branch points and the root (joint to the fixed bed) are ordinary joints (Fig. 5.7). With the gap $\mathbf{g}$ between the attachment points and the rotation vector $\boldsymbol{\theta}$ of the relative rotation with respect to the rest state,

$$\mathbf{F}_j = -k\,(\mathbf{g} + \beta\dot{\mathbf{g}})\,, \qquad \mathbf{T}_j = -\mathbf{K}\,(\boldsymbol{\theta} + \beta\,\boldsymbol{\omega}_{\mathrm{rel}})\,, \qquad \mathbf{K} = \frac{EI}{l_j}\left(\mathbf{I} - \mathbf{t}\mathbf{t}^T\right) + \frac{G\,2I}{l_j}\,\mathbf{t}\mathbf{t}^T, \tag{5.24}$$

with $k = EA/l_j$, $A = \pi r_j^2$, $I = \pi r_j^4/4$ at the joint radius r_j, the rest axis $\mathbf{t}$ and $l_j = (l_a + l_b)/2$ ($l/2$ at a root). Each element (volume V, $D = 2r$) is loaded with the relative velocity $\mathbf{u}_r = \mathbf{u}_f - \dot{\mathbf{c}}_e$, split into the normal part $\mathbf{u}_n = (\mathbf{I} - \mathbf{t}\mathbf{t}^T)\mathbf{u}_r$ and the tangential part $\mathbf{u}_t$ [Morison et al., 1950]:

$$\mathbf{F}_e = (m - \chi\rho_f V)\,\mathbf{g} + c_n|\mathbf{u}_n|\mathbf{u}_n + c_t|\mathbf{u}_t|\mathbf{u}_t + \chi\rho_f V\,(\mathbf{I} - \mathbf{t}\mathbf{t}^T)\,(C_m\mathbf{a}_f - C_a\ddot{\mathbf{c}}_e)\,, \tag{5.25}$$

$$c_n = \tfrac{1}{2}\rho_f\,l\,\chi\left(C_{dn}D + C_{dp}b_p\right), \qquad c_t = \tfrac{1}{2}\rho_f\,l\,\chi\left(C_{dt}\pi D + k_t C_{dp} b_p\right), \tag{5.26}$$

with the submerged fraction χ. The optional polyp layer (height h_p, frontal solidity ϕ_p) widens the element by $b_p = 2\,e_p h_p \phi_p$; with `polyp_response` U_r ΔU_r τ_p its extension e_p relaxes with time scale τ_p to a target that decreases linearly from 1 at $|\mathbf{u}_n| = U_r$ to 0 at $U_r + \Delta U_r$. By default the tree is advanced with a linearly implicit Euler step (added mass in the mass matrix, drag and gyroscopic terms linearized) with adaptive sub-steps; `integrator explicit` selects symplectic Euler with sub-cycling.

5.4.4 Rod trees: coupling to CFD and NHFLOW

Each element carries $n_q = \lceil l/h \rceil$ Lagrangian points along its axis (h smallest horizontal cell size, $1 \le n_q \le 64$); the structure is replicated on all MPI ranks. In every Runge-Kutta stage, the fluid velocity is sampled at the points (CFD: trilinear, points with $\phi \ge 0$; NHFLOW: bilinear horizontally and linear in the σ column, wet points only); $\mathbf{u}_f$ is the mean over the submerged points and χ their fraction. The reaction $-$(drag + inertia) (`reaction full`, default), $-$drag or zero (one-way coupling) is distributed over the submerged points and spread onto the grid as an acceleration with the cosine kernel [Peskin, 2002]

$$\varphi(r) = \tfrac{1}{4}\left(1 + \cos\tfrac{\pi r}{2}\right) \quad \text{for } |r| < 2, \qquad \varphi(r) = 0 \quad \text{otherwise.} \tag{5.27}$$

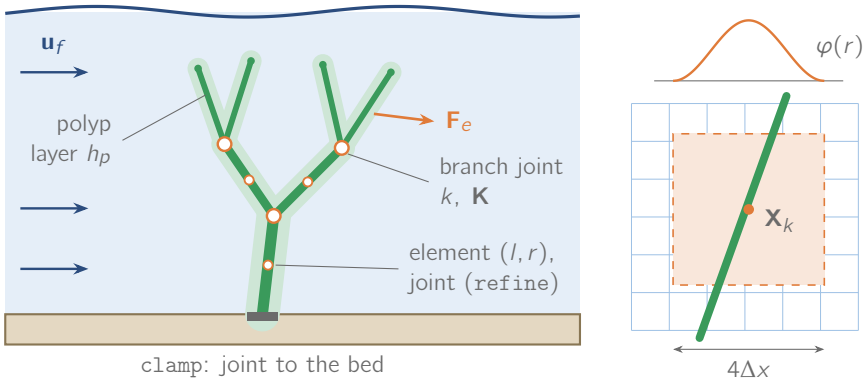

Figure 5.7: Rod tree (Z 20 1). Left: chains of rigid cylinder elements (green) connected by elastic joints (orange), root clamped to the bed, optional polyp layer (light green). Right: the reaction at a Lagrangian point $\mathbf{X}_k$ is spread over the 4×4 cell support of the cosine kernel $\varphi(r)$.

In REEF3D::CFD it enters the forcing fields f_i; in REEF3D::NHFLOW it is added to U,V,W and UH,VH,WH, with the kernel renormalized over the wet cells of the σ column so that no force is lost at the bed or the free surface. The fluid-side drag is treated point-implicitly, which keeps dense canopies stable. The structure is advanced once per time step, in the final stage, with the fluid acceleration $\mathbf{a}_f$ from a backward difference (staggered explicit coupling). Tip displacement, forces, submerged elements and polyp extension per colony are written to `REEF3D_RodTree/`, VTP files every Z 21 seconds.

5.4.5 Impermeable membranes (NHFLOW)

In REEF3D::NHFLOW, closed flexible net cages and other impermeable membranes are modelled with X 330 1 and the input file `membrane.dat` (box or cylinder membranes and their parameters, Section 6.18). The triangulated membrane acts on the fluid as a thin anisotropic porous layer with a hydraulic resistance (porous jump). The resistance is integrated implicitly and also enters the pressure Poisson equation, so that the projection does not let flow pass through the membrane. A membrane is fixed, moves rigidly with the floating body (X 10), or is a flexible mass-spring membrane whose top edge follows the floating body; its loads act on the body as external forces. For moving membranes, the pressure scheme A 520 2 is changed to 1. Water levels inside and outside, leakage, forces and tension are written to `REEF3D_NHFLOW_Membrane`.

5.4.6 FEM solids (CFD, NHFLOW)

In REEF3D::CFD and REEF3D::NHFLOW, deformable, failing and collapsing solid structures and rigid debris are computed with an explicit finite element module that is two-way coupled with the flow (Z 30 1, input file `fem.dat`, Section 6.19). The solids are meshed with 8-node hexahedra on a voxel lattice from boxes and STL surfaces, with elastic, elasto-plastic, concrete damage (optionally reinforced) or rigid material models, which can be given by presets for common concretes, steels, aluminium, timber, rocks, masonry and rubber, and with element erosion, debris and contact with the ground and between fragments. In every Runge-Kutta

stage, the surface velocity of the solid is imposed on the fluid by direct forcing; in the final stage, the fluid loads are transferred to the solid, which is advanced over the fluid time step with subcycling (staggered coupling). In NHFLOW, the cells inside the structure are solid over the whole water column, so a structure that stands out of the water cannot be overtopped. VTU output is written every Z 31 seconds to `REEF3D_FEM`.

5.4.7 Key input commands

Z 10 1	flexible strips, resolved immersed boundary coupling (CFD)
Z 11	strip definition: bottom point, L, W, T, ρ_s, E, I_x, I_y, I_z, ν_s, N
Z 12	structural damping of the strips, translational and rotational
Y 2	1: coupling moments on the strips off
Z 20 1	rod trees (soft corals, vegetation), unresolved coupling, input `rodtree.dat` (CFD, NHFLOW)
Z 21	rod-tree VTP print interval [s]
X 330	impermeable membranes, input `membrane.dat` (NHFLOW)
Z 30, Z 31	FEM solids, input `fem.dat`, VTU print interval [s] (CFD, NHFLOW)
T 10 2	k-ω model; ω forcing at the strips
X 19; P 20, P 30	strip log file interval; STL output interval

5.5 Sediment Transport

The morphodynamic model computes the bed level change due to bedload and suspended load. It is shared by REEF3D::SFLOW, REEF3D::NHFLOW and REEF3D::CFD: τ_b comes from the respective flow solution, while the transport formulas, the Exner equation, the sandslide and the time stepping are common code. REEF3D::FNPF has no sediment transport. The model is based on Bihs [2011] and Bihs and Olsen [2011]; scour applications are given e.g. by Afzal et al. [2015]. The sediment is characterized by d_{50} (S 20), ρ_s (S 22) and the bed porosity n (S 24), with $s = \rho_s/\rho$ (Fig. 5.8). All options are listed in Section 6.15. In REEF3D::CFD, the sediment can alternatively be represented by computational parcels (CPM, Section 5.6); individual shaped rigid particles such as armour stones are resolved with the DEM (Section 6.5) in CFD and NHFLOW.

5.5.1 Bed representation and module support

The bed level $z_b(x,y)$ is a two-dimensional field on the horizontal grid. In **CFD** the bed is the zero contour of the level set $\phi_b = z - z_b(x,y)$, which enters the flow through the direct forcing of Section 4.2 and is recomputed and reinitialized after each bed update; with S 10 2 the sediment is additionally a porous VRANS region (S 24, S 26). In **NHFLOW** z_b becomes the bed of the σ-grid, and in **SFLOW** d is recomputed while h is kept, so that η follows the bed. Table 5.3 lists the options per module.

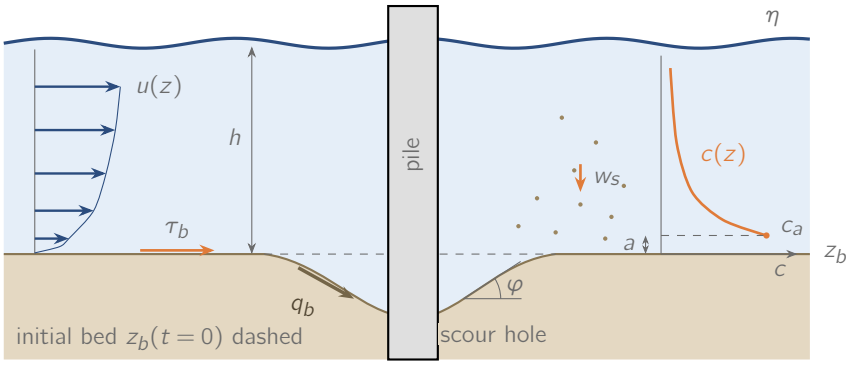

Figure 5.8: Sediment transport around a pile. The bed shear stress τ_b of the near-bed flow drives the bedload flux q_b along the bed, and the scour hole slopes are limited by the angle of repose φ. The suspended concentration $c(z)$ decreases from the reference concentration c_a at the reference level a upwards; the grains settle with w_s.

Table 5.3: Sediment transport options per module.

	SFLOW	NHFLOW	CFD
Exner model with bedload (S 10 1)	●	●	●
bed update within the RK2 stages (S 10 12, with A 510 2)	○	●	○
porous VRANS sediment bed (S 10 2)	○	○	●
bed shear stress options S 16	1–4	1–4, 6, 9–11	1–11
inertia and shallow water blending of τ_b (S 37, S 38)	○	●	○
suspended load (S 12)	○	●	●
suspended part of the transport (S 64)	●	○	○
transverse bed slope effect (S 86)	●	●	●
spiral flow correction (S 95)	●	○	○
Exner sub-iterations (S 27)	○	○	●
graded sediment (S 51)	●	●	●
particle model CPM (Q 10 1, 2, Section 5.6)	○	○	●

5.5.2 Bed shear stress

The default (S 16 1) in CFD and NHFLOW is the rough-wall logarithmic law

$$\tau_b = \rho\left(\frac{\kappa\,|\mathbf{u}_p|}{\ln(30\,\Delta z_p/k_s)}\right)^2, \qquad \kappa = 0.4, \qquad k_s = \text{S } 21 \cdot d_{50}, \tag{5.28}$$

with the near-bed velocity $\mathbf{u}_p$ at the height Δz_p above the bed (CFD: interpolated at $z_b{+}1.6\,\Delta z$, $\Delta z_p = \Delta z$; NHFLOW: lowest σ-cell, Δz_p half its height, smooth-wall law with B 22 1). In SFLOW, S 16 1 is the Manning form $\tau_b = \rho g\, n_M^2 |\mathbf{u}|^2/h^{1/3}$, $n_M = k_s^{1/6}/20$. The other S 16 options (k-based, viscous, depth-averaged Chezy/Manning, transitional roughness) are listed in Section 6.15. S 36 1 adds the bedform roughness of van Rijn [1984b] to k_s, giving $k_{s,eff}$; in NHFLOW, S 37 adds an inertia contribution and S 38 blends with the Manning form in very shallow water. In CFD and NHFLOW, S 16 9 takes the larger of the log-law stress and the wave friction stress $\tau_w = \frac{1}{2}\rho f_w |\mathbf{u}_w|^2$ with $f_w = \min(1.39(A/z_0)^{-0.52}, 0.3)$ of Soulsby (1997), from the oscillatory part $\mathbf{u}_w$ of the near-bed velocity and its orbital amplitude A for the wave period S 58. S 16 10 and 11 combine the log law, with the velocity at the reference height S 63, with the turbulent kinetic energy, $\max(\tau_{log}, 0.3\rho k)$ and $\sqrt{\tau_{log} \cdot 0.3\rho k}$.

5.5.3 Critical shear stress on sloping beds

The Shields parameter and the critical shear stress are

$$\theta = \frac{\tau_b}{(\rho_s - \rho)\, g\, d_{50}}, \qquad \tau_{cr} = r\, \theta_{cr}\, (\rho_s - \rho)\, g\, d_{50}, \tag{5.29}$$

with θ_{cr} of a flat bed (S 30, default 0.047) [Shields, 1936]. The factor r (S 80) accounts for gravity on a sloping bed [Bihs and Olsen, 2011]; with the longitudinal slope angle θ_l (positive for a bed descending in the flow direction), the transverse slope angle α and the angle of repose φ, S 80 4 reads [Fredsøe and Deigaard, 1992]

$$r = \cos\theta_l \Big(1 - \frac{\tan\theta_l}{\tan\varphi}\Big) \cos\alpha \sqrt{1 - \frac{\tan^2\alpha}{\tan^2\varphi}}. \tag{5.30}$$

S 80 1 follows Kovacs and Parker [1994], S 80 2 and 3 Dey [2001, 2003]. r is limited to $0.01 \le r \le 1.5$ (S 84 beyond φ); a descending bed gives $r < 1$, an adverse slope $r > 1$. φ varies around S 81 by up to S 82 with the slope direction.

5.5.4 Bedload transport

The equilibrium bedload capacity $q_{b,e}$ (S 11) of van Rijn [1984a] (S 11 1) and Meyer-Peter and Müller [1948] (S 11 2) reads

$$q_{b,e} = 0.053 \sqrt{(s-1)g}\; d_{50}^{1.5}\, \frac{T^{2.1}}{D_*^{0.3}}, \qquad T = \frac{\theta - \theta_{cr} r}{\theta_{cr} r}, \qquad D_* = d_{50} \left(\frac{(s-1)g}{\nu^2}\right)^{1/3}, \tag{5.31}$$

$$q_{b,e} = 8\, (\theta - r\theta_{cr})^{1.5} \sqrt{(s-1) g\, d_{50}^3}, \tag{5.32}$$

and $q_{b,e} = 0$ for $\theta < r\theta_{cr}$. S 11 3 is Engelund and Fredsøe [1976] in the form of Fredsøe and Deigaard [1992], S 11 4 Engelund-Hansen and S 11 5 Einstein-Brown. The bedload moves in the near-bed flow direction $\hat{\mathbf{s}} = \mathbf{u}_p / |\mathbf{u}_p|$; S 85 1 scales its magnitude with $1 - \min(1,\, 1.3\, \hat{\mathbf{s}} \cdot \nabla z_b)$. With S 86 1, gravity turns the bedload down the transverse bed slope, $\mathbf{q}_b = q_b \big(\hat{\mathbf{s}} - \nabla z_b / f(\theta)\big)$ with $f(\theta) = A_{sh}\, \theta^{B_{sh}} (d_{50}/h)^{C_{sh}}$ (S 87 – S 89; Talmon, van Mierlo and Struiksma 1995). In SFLOW, S 95 turns the transport direction towards the centre of curvature of the streamlines by the angle δ of the spiral flow in bends, $\tan\delta = \alpha_l h / R_s$ (Engelund 1974; S 96), with S 95 2 with the adaptation of the spiral flow intensity along the flow (Kalkwijk and Booij 1986; S 97). With S 33 > 0 (non-equilibrium), q_b adapts to the capacity over the length L_s,

$$q_b + L_s\, \hat{\mathbf{s}} \cdot \nabla q_b = q_{b,e}, \tag{5.33}$$

with L_s from S 40 or the saltation length of van Rijn [1984a]; S 33 0 gives $q_b = q_{b,e}$.

5.5.5 Suspended load

With S 12 1 (CFD and NHFLOW), the volumetric concentration c follows

$$\frac{\partial c}{\partial t} + \frac{\partial}{\partial x_j}\Big(\big(u_j - w_{s,eff}\, \delta_{j3}\big)\, c\Big) = \frac{\partial}{\partial x_j}\left((\nu + \nu_t) \frac{\partial c}{\partial x_j}\right), \qquad w_{s,eff} = w_s \Big(1 - \frac{c}{0.635}\Big)^{m_s}, \tag{5.34}$$

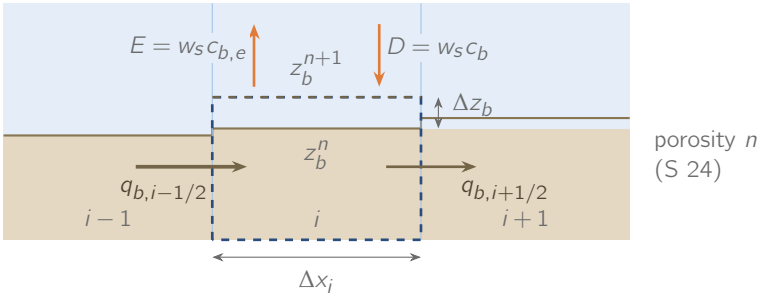

Figure 5.9: Exner balance of the bed column i: the difference of the bedload fluxes through the faces and the net erosion $E - D$ change the sediment volume $(1-n)\,\Delta z_b\,\Delta x_i$ of the column.

with hindered settling (m_s depends on $Re_p = w_s d_{50}/\nu$) and w_s from S 23 or computed (S 25 1, including van Rijn [1984b]). It is solved implicitly with first-order upwind convection; in the cell above the bed the exchange flux $w_s(c_{b,e} - c)$ is an implicit source. The equilibrium concentration $c_{b,e}$ follows from the reference concentration of van Rijn [1984b] at the level a,

$$c_a = \min\left(0.015\,\frac{d_{50}}{a}\,\frac{T^{1.5}}{D_*^{0.3}},\ 0.05\right), \tag{5.35}$$

with $a = \max(k_s, 0.01h, 2d_{50})$, at most $h/2$, transferred with the Rouse profile to the first cell centre. The near-bed concentration c_b is the concentration of that cell, so both concentrations of the exchange flux refer to the same height. S 62 selects the coupling to the bed (Exner source $E - D$, or in NHFLOW the depth-integrated suspended flux added to q_b). In the depth-averaged SFLOW, S 64 carries a part f_s of the transport capacity q_e as suspended load, which relaxes along the flow over the adaptation length of the suspension (Galappatti and Vreugdenhil 1985), $q_s + L_s\,\hat{\mathbf{s}}\cdot\nabla q_s = f_s q_e$ with $L_s = h|\mathbf{u}|/(\text{S 65}\ w_s)$; the bed load is $(1-f_s)\,q_e$, and the Exner fluxes carry the sum of both.

5.5.6 Exner equation

The bed level follows from the sediment mass balance of a bed column (Fig. 5.9),

$$(1-n)\,\frac{\partial z_b}{\partial t} = -f_m\left[\frac{\partial(q_b s_x)}{\partial x} + \frac{\partial(q_b s_y)}{\partial y} + w_s\,(c_{b,e} - c_b)\right], \tag{5.36}$$

with the morphological factor f_m (S 35) and $(s_x, s_y) = \hat{\mathbf{s}}$. The right-hand side is zero outside S 71 $\le x \le$ S 72 and in non-erodible zones (S 74), and it is multiplied by the relaxation function (S 73, S 75). S 31 and S 32 select the divergence form and its reconstruction (Section 6.15). The bed is updated explicitly, $z_b^{n+1} = z_b^n + \Delta t_{sed}\,(\partial z_b/\partial t)^n$, or with S 10 12 (NHFLOW) by a two-stage Runge-Kutta scheme within the flow stages.

Sandslide. Where the slope between neighbouring columns exceeds the angle of repose, $\Delta z_b > \tan\varphi\,\Delta x$, the sandslide S 90 moves sediment downslope conserving its volume, repeated up to S 91 times until no slope exceeds φ.

5.5.7 Morphological time stepping

The sediment module is decoupled in time from the flow: it starts after S 41 (S 43, S 45, S 47) and is called at the interval S 42 (S 44, S 46, S 48). Each call advances the bed by

$$\Delta t_{sed} = \text{S 17} \cdot \min\left(\text{S 13},\ \frac{\text{S 14}\,\Delta x}{\max|\partial z_b/\partial t|}\right) \qquad \text{(S 15 0)}, \tag{5.37}$$

and the run stops when the morphological time $t_{sed} = \sum \Delta t_{sed}$ reaches S 19. **Morphological acceleration** is obtained with f_m (S 35) in Eq. (5.36), or by advancing the bed by S 27·Δt_{sed} per call (CFD) while the flow advances only by the call interval. In both cases the flow must stay close to equilibrium with the bed; S 104 limits the sediment time step so that the bed forms move relative to the flow at most at S 104 times the near-bed flow speed.

5.5.8 Graded sediment

With S 51, the bed has up to 20 size fractions d_k in an active layer over a substrate. The capacity of fraction k is $q_{b,e,k} = F_k\, q_{b,e}(d_k)$ with the fraction F_k in the active layer, $\theta_k = \theta\, d_{50}/d_k$ and $\theta_{cr,k} = \xi_k\, r\, \theta_{cr}$, with the hiding factor ξ_k of Wu et al. [2000] (S 54 1, S 55). Eq. (5.36) is solved per fraction and updates the active layer composition. Not available with S 10 12.

5.5.9 Key input commands

S 10	on (1), porous VRANS bed (2, CFD), RK2 coupled (12, NHFLOW)
S 20, S 22, S 24	grain size d_{50}, sediment density, bed porosity
S 11, S 18	bedload formula, dynamic friction coefficient for S 11 3
S 16, S 21, S 36	bed shear stress, roughness factor, bedform roughness
S 58, S 63	wave period of S 16 9, reference height of S 16 10, 11
S 30, S 80–S 84	critical Shields parameter, slope reduction, angle of repose
S 33, S 40, S 85	non-equilibrium bedload, adaptation length, slope correction
S 86–S 89, S 95–S 97	transverse bed slope, spiral flow (SFLOW)
S 64, S 65	suspended part of the transport (SFLOW)
S 12, S 23, S 25, S 62	suspended load, settling velocity, coupling to the bed
S 31, S 32, S 100–S 102	Exner formulation and discretization, bed filter
S 13–S 15, S 17, S 19, S 29	sediment time step, factor, end time, ramp
S 35, S 27, S 103, S 104	morph. factor, Exner sub-iterations, celerity and acceleration limits
S 41–S 48	start and interval of the sediment calls
S 90–S 94	sandslide
S 50, S 71–S 78	bed boundary conditions, erodible regions, relaxation zones
S 51–S 56	graded sediment

5.6 Particle Model for Sediment (CPM, MP-PIC)

REEF3D::CFD contains a particle-based alternative to the Exner model, the continuum particle method (CPM), activated with S 10 1 (or S 10 2) together with Q 10 1 or 2; it is not available

in SFLOW, NHFLOW and FNPF. The sediment below the initial bed ($\phi_b < 0$) is represented by Q 24 parcels per cell (Fig. 5.10), each representing $N_p = \theta_0 V_{bed}/(N_{cells}\,\mathrm{Q}\,24\,\pi d_{50}^3/6)$ grains of size d_{50} and density ρ_s, where V_{bed} and N_{cells} are the volume and number of the seeded cells and θ_0 is the solid fraction of the seeded bed (Q 45, default $1-n$). The particle volume fraction θ_p follows from distributing the parcel volumes to the cell centres with trilinear weights (smoothed with Q 27). The bed level is the iso-surface $\theta_p = \mathrm{Q}\,26\,(1-n)$; it is updated from the parcels and is the topography level set ϕ_b seen by the flow solver. The parcels are computational parcels of identical grains; shaped rigid particles such as armour stones, which are resolved individually with their contacts, are modelled with the DEM (Section 6.5). All options are listed in Section 6.14.

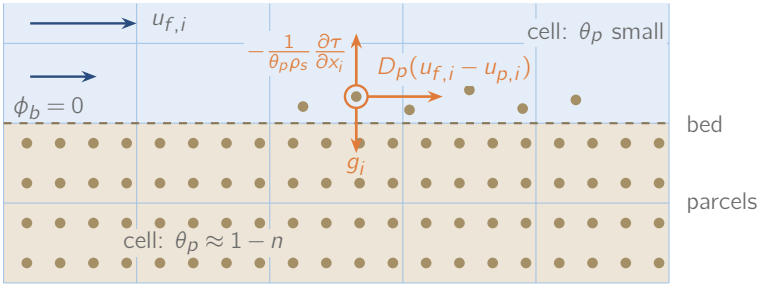

Figure 5.10: Continuum particle method: parcels seeded below the bed $\phi_b = 0$. A parcel is accelerated by the drag of the fluid velocity interpolated to its position, by gravity, by the pressure gradients and by the gradient of the inter-particle stress $\tau(\theta_p)$, which keeps the packing below the close-packing limit.

With Q 11 2, the parcels follow the multiphase particle-in-cell (MP-PIC) formulation of Snider [2001]:

$$\frac{du_{p,i}}{dt} = D_p\big(u_{f,i}-u_{p,i}\big) - \frac{1}{\rho_s}\frac{\partial p}{\partial x_i} + g_i - \frac{1}{\theta_p\rho_s}\frac{\partial \tau}{\partial x_i}, \qquad \frac{dx_{p,i}}{dt} = u_{p,i}, \tag{5.38}$$

with $u_{f,i}$ and p interpolated to the parcel. With S 10 1, the bed is a solid boundary for the fluid, and the fluid velocity at the parcels near the bed surface is taken from the logarithmic law of the wall with $k_s = \mathrm{S}\,21 \cdot d_{50}$. The drag coefficient D_p per unit parcel mass follows from the drag law Q 51 (Andrews and O'Rourke, or Gidaspow) with the relative velocity $|u_f - u_p|$ and the fluid fraction $\theta_f = \max(1-\theta_p, 1-\theta_{max})$ (θ_{max} from Q 32); the drag is treated point-implicitly. The inter-particle stress τ (Q 12) is either the model of Snider [2001],

$$\tau = \frac{P_s\,\theta_p^{\beta}}{\max(\theta_{cs}-\theta_p,\ \epsilon\,(1-\theta_p))}, \tag{5.39}$$

with P_s, β, ϵ and θ_{cs} from Q 14 – Q 17 (Q 12 1), which rises steeply towards close packing θ_{cs} and pushes the parcels apart, or, by default, the packed-bed model (Q 12 2): the effective stress of the contact network from the submerged weight of the bed, the contact pressure of Johnson and Jackson (Q 32 – Q 35) and Coulomb friction with the $\mu(I)$ rheology (Q 13, Q 36 – Q 38). The parcels are advanced in every flow time step with adaptive sub-steps, limited by the particle Courant number S 14 and the wave speed of the stress (Q 18) and to at most Q 28 sub-steps, with explicit Euler (Q 10 1) or two-stage Runge-Kutta (Q 10 2) sub-steps. With Q 19 1, a parcel enters a cell only if that cell has free volume. The plain model Q 11 1 uses a two-stage Runge-Kutta scheme with the sediment time step of S 13 – S 15 and without τ: a parcel moves where the drag exceeds a resistance force scaled by Q 30.

By default (Q 50 0), the coupling is one-way in momentum: no drag reaction is added to the fluid, and the fluid sees the sediment through ϕ_b (S 10 1) or as a VRANS porous region with

the porosity $1-\theta_p$ (S 10 2). These cells get no Darcy–Forchheimer resistance, since the grain drag is that of the particle coupling; porous structures (B 270 – B 291) are kept and take precedence. With Q 50 1 (two-way coupling, Q 11 2), the reaction of the parcel drag is added to the fluid momentum equations as a point-implicit source term, the drag uses the interstitial fluid velocity $u_{f,i}/(1-\theta_p)$, and the mixture continuity $\partial u_i/\partial x_i = -\partial(\theta_p u_{p,i})/\partial x_i$ enters the pressure Poisson equation. With the hydrostatic pressure scheme D 30 10, the fluid density is blended to ρ_s below the bed. τ_b (S 16) and τ_{cr} (S 80) are computed for output only, and the sandslide (S 90) is not applied.

Graded sediment and near-bed processes. With a graded bed (S 51), all parcels keep the same solid volume, but each parcel carries the diameter of one fraction, drawn with the volume fractions of the active layer. Further cells, also above the bed, can be seeded with the boxes Q 110. For MP-PIC (Q 11 2) and an impermeable bed (S 10 1), several near-bed processes can be added. The sub-grid bedload layer (Q 58) replaces the resolved near-bed drag: the parcels resting on the bed surface are picked up per bed column by a bedload closure (grain velocity, critical Shields number, hop length and further parameters, Q 59 – Q 68), hop along the bed and are deposited again; with Q 58 2, parcels are also released into the suspension with the van Rijn reference concentration. With Q 58 3 or 4 (hybrid), the suspended load is the concentration of the Eulerian suspended sediment solver instead: per bed column, the first fluid cell exchanges $w_s(c_{be}-c_1)$ with the bed, and the accumulated net exchange moves whole parcels between the bedload layer and the concentration, so that the sediment volume is conserved (Q 58 4 without the onset criterion of the suspension). The bed level seen by the fluid is smoothed over the columns (Q 72); with Q 74 1, the bed grains see no resolved flow at all, Q 75 adds a sand slide of the bed surface, and Q 76 and Q 77 fix the parcel bed upstream and downstream of given positions. Turbulent dispersion (Q 52) moves the parcels by a random walk with the eddy diffusivity ν_t/Sc_t (Q 53), so that the ensemble of suspended parcels gives the Rouse profile. Q 54 adds a lift force on the exposed grains, and Q 55 a dilatancy of the sheared granular layer, which loosens the packing of the contact network with the inertial number. With Q 69, the seepage flow in the bed is computed from Darcy's law, with an optional excess pore pressure at the bottom (Q 70); the seepage force acts on the parcels and loads or unloads the contact network, so that an upward gradient can fluidise the bed.

5.6.1 Key input commands

S 10, Q 10	sediment transport on, particle model and sub-step scheme (CFD only)
Q 11	plain particle transport (1) or MP-PIC (2)
Q 12, Q 13	inter-particle stress model, friction of the packed bed
Q 14–Q 17	Snider stress parameters P_s, β, ϵ, θ_{cs} (Q 12 1)
Q 32–Q 38	packing limit, contact pressure and $\mu(I)$ friction (Q 12 2)
Q 18, Q 28, S 14	sub-step limits: stress wave speed, maximum number, Courant number
Q 19	grid-limited step (packing limit on the grid)
Q 24, Q 29, Q 45	parcels per cell (must be set), seeding type, seeded solid fraction
Q 26	solid fraction threshold of the bed interface
Q 50, Q 51	one- or two-way fluid coupling, drag law
S 20, S 22, S 24	grain size, sediment density, porosity (parcel size, packing limit)
Q 30	friction factor of the resistance force (Q 11 1)
S 13–S 15, S 41–S 47	sediment time step (Q 11 1), start of the particle transport
Q 73	relaxation zones for the particle forces
S 51, Q 110	graded sediment (parcel diameters), seeding boxes
Q 52, Q 53	turbulent dispersion of the parcels, turbulent Schmidt number
Q 54, Q 55	lift on exposed grains, dilatancy
Q 58–Q 68	sub-grid bedload layer and its closure; Q 58 3, 4 hybrid Eulerian suspension
Q 72, Q 74, Q 75	smoothing of the bed level, bed grains without resolved flow, sand slide
Q 76, Q 77	fixed parcel bed upstream and downstream
Q 69, Q 70	seepage flow in the bed, excess pore pressure at the bottom
Q 180–Q 182	particle output

5.7 Porous Media and Vegetation

Breakwaters, rock filters, porous beds and vegetation contain pores and stems far smaller than a grid cell. REEF3D therefore solves the volume-averaged RANS (VRANS) equations, in which the solid skeleton is described by the porosity n (pore volume per total volume) and its effect on the flow by resistance and added mass terms. VRANS porous media are available in REEF3D::CFD (B 270 – B 291, porous sediment bed S 10 2) and REEF3D::NHFLOW (B 201 – B 230, porous floating bodies X 16, and the volume of unresolved DEM particles with E 28, Section 6.5), the vegetation model in REEF3D::CFD only (B 307 – B 322); REEF3D::FNPF and REEF3D::SFLOW have neither. Porous regions are defined with input commands, not in the DIVEMesh grid. Flexible vegetation is treated with the rod models of Section 5.4.

5.7.1 VRANS equations in REEF3D::CFD

In the porous zones, u_i is the filtered (Darcy) velocity, so that continuity keeps its form $\partial u_i/\partial x_i = 0$. In cells with $n < 1$, the momentum equations are

$$(1+c_A)\frac{\partial u_i}{\partial t} + \frac{u_j}{n}\frac{\partial u_i}{\partial x_j} = n\left(-\frac{1}{\rho}\frac{\partial p}{\partial x_i} + g_i + f_i - a\,u_i - b\,|u_i|\,u_i\right) + \frac{\partial}{\partial x_j}\left[(\nu+\nu_t)\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right], \tag{5.40}$$

which reduces to Eq. (4.8) for $n = 1$. The pore velocity u_j/n convects the momentum (the convecting face velocities of the flux schemes are divided by n). Pressure gradient, gravity, external forces f_i and resistance are multiplied by n, which also enters the pressure Poisson equation and the velocity correction:

$$\frac{\partial}{\partial x_i}\left(\frac{n}{(1+c_A)\,\rho}\frac{\partial p'}{\partial x_i}\right) = \frac{1}{\alpha\Delta t}\frac{\partial u_i^*}{\partial x_i}, \qquad u_i = u_i^* - \frac{n\alpha\Delta t}{(1+c_A)\,\rho}\frac{\partial p'}{\partial x_i}. \tag{5.41}$$

The whole momentum increment of a Runge-Kutta stage, including the implicit diffusion, is divided by $1 + c_A$, with $c_A = c\,(1-n)/n$ (B 266 1, default; Darcy-velocity form of Jensen et al. [2014], continuous at the surface of the porous medium) or the constant $c_A = c$ (B 266 0), and c = B 260 (default 0.34). The resistance coefficients follow van Gent [1995] as used by Jensen et al. [2014]:

$$a = \alpha\,\frac{(1-n)^2}{n^3}\,\frac{\nu}{d_{50}^2}, \qquad b = \beta\left(1 + \frac{7.5}{KC}\right)\frac{1-n}{n^3}\,\frac{1}{d_{50}}, \tag{5.42}$$

with the dimensionless coefficients α, β, the grain size d_{50}, the local ν and the constant KC = B 264 (default 10^{20}, i.e. no correction). $b|u_i|u_i$ is evaluated component-wise; n, d_{50}, α, β are cell-centred and averaged arithmetically to the faces. With B 268 1 (default), the resistance is treated point-implicitly in the Runge-Kutta stages: it is unconditionally stable, and the steady state is the Darcy-Forchheimer balance independent of Δt; B 268 0 is the explicit source, stable only for $\Delta t < 2(1 + c_A)/(n(a + 2b|u|))$. Without the Runge-Kutta momentum solver, the explicit source is used.

Porous geometries. Each object has its own n, d_{50}, α, β: boxes (B 270), vertical cylinders (B 274), wedges (B 281, B 282) and inclined plates (B 291). A cell belongs to an object if its centre lies inside, and later objects overwrite earlier ones, e.g. a core wedge inside a larger armour-layer wedge (Fig. 5.11). Outside, $n = 1$ and $\alpha = \beta = 0$. With S 10 2, the cells below the bed become porous (n = S 24, d_{50} = S 20, α,β = S 26) and follow the bed evolution (Section 5.5). B 240 – B 243 give a simple Darcy-Forchheimer box, $f_i = -D\,\nu\,u_i - \frac{1}{2}C\,|u_i|u_i$, without porosity scaling.

Turbulence. With B 295 1 (default), source terms keep the turbulence in the porous zone at the level of the pore flow, with $|u|^2 = u_i u_i$:

$$k_\infty = 3.7\,(1-n)\,n^{3/2}\,|u|^2, \qquad \epsilon_\infty = 39\,(1-n)^{5/2}\,n^2\,\frac{|u|^3}{d_{50}}. \tag{5.43}$$

$n\epsilon_\infty$ is added to the k-equation and $n\,c_{2\epsilon}\epsilon_\infty^2/k_\infty$ ($c_{2\epsilon} = 1.92$) to the ϵ-equation, with corresponding terms for k-ω. B 295 2 sets $\nu_t = 10^{-3}\nu$ in the porous cells.

5.7.2 VRANS in REEF3D::NHFLOW

All NHFLOW porous objects share the core values n_p, d_{50}, α_p, β_p of B 201: boxes, cylinders, spheres and wedges (B 210 – B 219) and an STL geometry (B 230). As for solid structures,

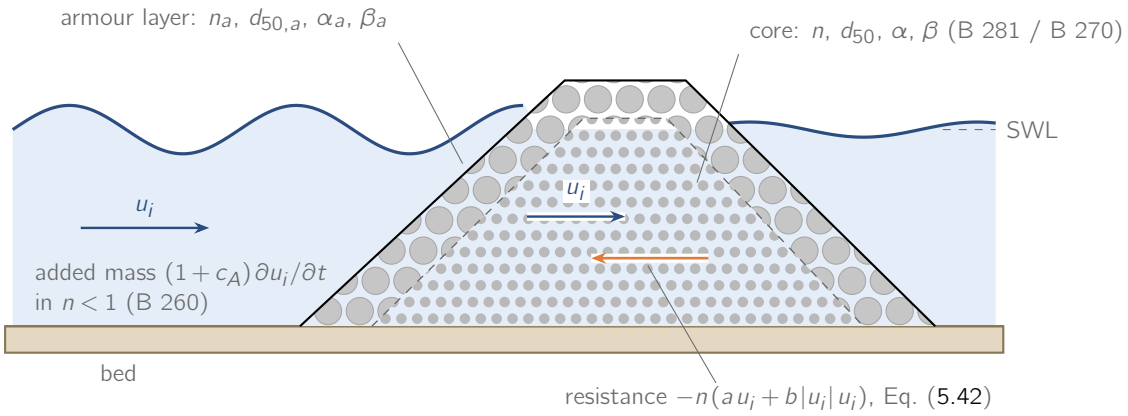

Figure 5.11: Rubble-mound breakwater with armour layer and core as porous VRANS zones in a wave tank. Each zone has its own porosity n, grain size d_{50} and resistance coefficients α, β; inside, the filtered velocity u_i is retarded by the linear and nonlinear resistance and the added mass term.

a signed distance function ϕ_p is computed, and with the smoothed Heaviside function $H_p = H(-\phi_p)$ of Eq. (4.10) (half-width B 209 times the horizontal cell size)

$$n = H_p\, n_p + (1 - H_p), \qquad \alpha = H_p\, \alpha_p, \qquad \beta = H_p \beta_p. \tag{5.44}$$

In the equations of Section 4.3, the fluxes of the free surface equation (4.27) are divided by n, the convective fluxes $hu_i u_j$ by n^2, and the momentum increment of each stage, including the non-hydrostatic pressure correction, is multiplied by the inertia factor selected with B 265: $1/(1 + c_A(1-n)/n^2)$ after van Gent [1995] (B 265 0, default), or $n/(1+\tilde{c}_A)$ with $\tilde{c}_A = c_A(1-n)/n$ (B 265 1, VARANS for the superficial velocity, Liu et al. 1999), with c_A = B 260. The free surface equation uses the porosity of the surface cell for the storage term; with B 265 1, long waves in a porous layer without resistance travel at $\sqrt{gh/(1+\tilde{c}_A)}$, while with B 265 0 they are faster than in open water. B 265 1 is recommended with the DEM particle porosity E 28, which is multiplied with the porosity of the VRANS structures. The resistance acts on the momentum hu_i,

$$f_i = -H_p\, h\,(a\,u_i + b_{nh}|u_i|\,u_i)\,, \qquad b_{nh} = \beta\left(1 + \frac{7.5}{KC}\right)\frac{1-n}{n^4}\,\frac{1}{d_{50}}, \tag{5.45}$$

with a from Eq. (5.42). By default (B 204 1), it is treated point-implicitly together with the stage update, which keeps a fine-grained core stable; B 204 0 is explicit. Layers of different material, e.g. the armour and filter layers of a breakwater, are given with B 202 (thickness, n, d_{50}, α, β), counted from the exposed surface of the structures inwards, with the core values of B 201 below the last layer; the properties are blended at the layer boundaries over B 209 cell sizes, and with B 203 1 (default) the faces of a structure resting on the bed do not count as exposed surface. B 208 1 outputs the total force on the structure (resistance, added mass and the part $(1-n)$ of the pressure gradient). There are no turbulence source terms.

Porous floating bodies. With X 16 (n_{fb}, d_{50}, α, β), the porosity $n = 1 - H_{fb}(1 - n_{fb})$ follows the moving NHFLOW body, and the resistance of Eq. (5.42) acts on the velocity relative to the body. It is integrated implicitly over the substep, so that large coefficients remain stable, and its reaction is added to the body forces (Section 5.2).

5.7.3 Vegetation

The vegetation model (CFD, activated with B 307 or B 308) represents plants as rigid vertical cylinders with N stems per unit area, diameter D and drag coefficient C_d in boxes (B 310) and wedges (B 321, B 322). The momentum equations receive the Morison-type force per unit mass

$$f_i = -\frac{1}{2} C_d \, N \, D \, |u_i| \, u_i \; - \; C_M \, \frac{\pi D^2}{4} \, N \, \frac{\partial u_i}{\partial t}, \tag{5.46}$$

with C_M = B 309 (default 2.0) and the time derivative from the stored velocities. The drag follows the local velocity and vanishes above the canopy (Fig. 5.12). With B 308 1 (default), the stems block the volume through the porosity $n = 1 - N\pi D^2/4$, which enters Eqs. (5.40) and (5.41) as for porous media. B 295 1 adds turbulence sources proportional to $C_d D N$, B 295 2 sets $\nu_t = 10^{-3}\nu$ in the canopy.

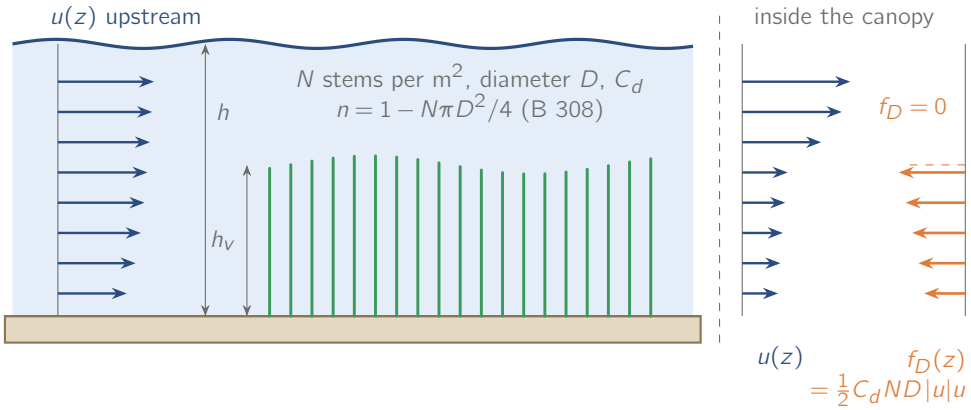

Figure 5.12: Vegetation field of rigid stems (B 310) in a current. Left: approach flow and canopy; right: the velocity inside the canopy is reduced, and the drag force per unit volume, Eq. (5.46), follows the local velocity over the canopy height h_v and vanishes above it.

5.7.4 Key input commands

B 260	added mass coefficient c_A (CFD, NHFLOW)
B 266, B 268	added mass form, point-implicit resistance (CFD)
B 264	constant KC number of the nonlinear resistance (CFD, NHFLOW)
B 270, B 274, B 281, B 282, B 291	porous box, vertical cylinder, wedges, plate with n, d_{50}, α, β (CFD)
B 295	turbulence treatment in porous media and vegetation (CFD)
B 240 – B 243	simple Darcy-Forchheimer box (CFD)
S 10 2, S 20, S 24, S 26	porous sediment bed (CFD)
B 201	n, d_{50}, α, β for all NHFLOW porous objects (core)
B 202 – B 204	NHFLOW porous layers, faces on the bed, point-implicit resistance
B 210 – B 219, B 230	NHFLOW porous primitives and porous STL geometry
B 208, B 209	NHFLOW porous force output, Heaviside half-width
X 16	porous floating body (NHFLOW)
B 307, B 308, B 309	vegetation model on, stem porosity, inertia coefficient C_M (CFD)
B 310, B 321, B 322	vegetation box and wedges with N, D, C_d (CFD)

5.8 Heat, Concentration, Stratification and Rheology

REEF3D::CFD can transport a temperature T (H group) or a concentration c (C group) that change the density and viscosity of the fluid, for stratified flows, plumes and density currents, and phase 1 can be a non-Newtonian material (W 90) such as mud or debris. These models require the level set free surface (F 30 > 0); REEF3D::SFLOW has only a Bingham-type bed friction, and REEF3D::NHFLOW and REEF3D::FNPF have no heat or concentration transport.

5.8.1 Temperature and concentration

With H 10 > 0, the temperature T in °C is advected and diffused,

$$\frac{\partial T}{\partial t} + u_j \frac{\partial T}{\partial x_j} = \frac{\partial}{\partial x_j}\left[\left(\alpha_T + \frac{\nu_t}{\sigma_T}\right)\frac{\partial T}{\partial x_j}\right], \qquad \alpha_T = \alpha_1 H(\phi) + \alpha_2 (1 - H(\phi)), \tag{5.47}$$

with the diffusivities of water and air (H 1, H 2, phase assignment H 9) and $\sigma_T = 0.9$. H 10 selects the time scheme, H 15 the convection scheme; diffusion is implicit. H 3 1 (default) interpolates $\rho(T)$ and $\nu(T)$ of water and air from tables and blends them with Eq. (4.9). With H 3 2, the viscosities stay W 2 and W 4 and the densities follow

$$\rho_k = \rho_{0,k}\,(1 - \beta_k\,(T - T_{0,k}))\,, \qquad k = 1, 2, \tag{5.48}$$

with $\rho_{0,k}$ = W 1, W 3, $T_{0,k}$ from H 50 and β_k = H 4 (default $1/T_{0,k}$ in K).

With C 10 > 0, a concentration c is transported with the diffusivity $\nu + \nu_t$ (Schmidt number 1),

$$\frac{\partial c}{\partial t} + u_j \frac{\partial c}{\partial x_j} = \frac{\partial}{\partial x_j}\left[(\nu + \nu_t)\,\frac{\partial c}{\partial x_j}\right], \tag{5.49}$$

with the convection scheme C 15; C 75 defines inclined slabs of c_1 with an initial velocity, e.g. for a turbidity current on a slope. Density and viscosity change linearly in both phases:

$$\begin{aligned} \rho &= (\rho_1 + c\,\mathrm{C}1)\,H(\phi) + (\rho_2 + c\,\mathrm{C}3)\,(1 - H(\phi))\,, \\ \nu &= (\nu_1 + c\,\mathrm{C}2)\,H(\phi) + (\nu_2 + c\,\mathrm{C}4)\,(1 - H(\phi))\,. \end{aligned} \tag{5.50}$$

With C 1 = 10 kg/m^3 (default), $c = 1$ gives a salt-water-like excess density of 1 %.

Stratified flow. T and c act on the flow only through the variable density in Eqs. (4.8), (4.11) and (4.12); no separate Boussinesq term is added, and horizontal density gradients drive the motion (Fig. 5.13). The face densities follow from the face-averaged level set and T or c. T 45 adds buoyancy to the k-equation of the k-ω model. A high-order convection scheme (H 15 or C 15 = 4 or 5) keeps the density interface sharp.

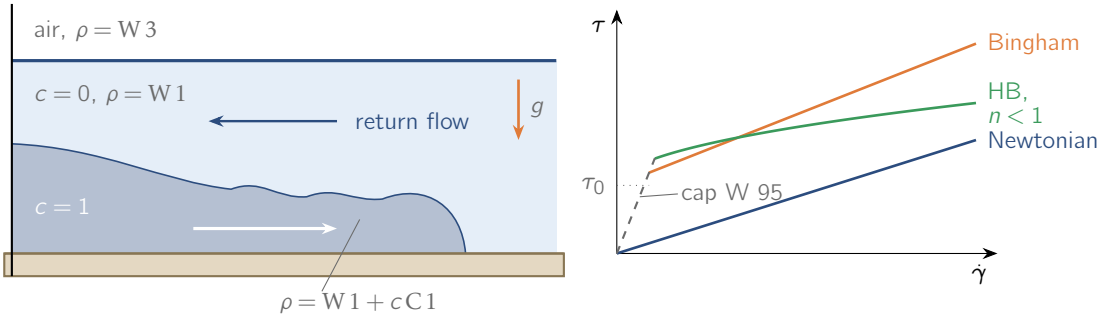

Figure 5.13: Left: a density current in REEF3D::CFD. A dense fluid ($c = 1$, density from Eq. 5.50) spreads along the bed under the lighter water, driven only by the variable density in the momentum and pressure equations; the displaced water returns above it. Right: shear stress over shear rate for a Newtonian fluid, a Bingham fluid (W 98 = 1) and a shear-thinning Herschel-Bulkley fluid with yield stress τ_0; below the grey dashed line, the viscosity is capped at ν_0 (W 95).

5.8.2 Non-Newtonian rheology

With W 90 1, the viscosity of phase 1 follows the Herschel-Bulkley model [Herschel and Bulkley, 1926], $\tau = \tau_0 + K\dot{\gamma}^m$, with the shear rate $\dot{\gamma} = \sqrt{2 S_{ij} S_{ij}}$, the yield stress τ_0, K = W 97 and m = W 98, regularized by the cap ν_0 = W 95; phase 2 keeps ν_2 = W 4:

$$\nu_1 = \min\left(\frac{1}{\rho}\left(\frac{\tau_0}{\dot{\gamma}} + K\dot{\gamma}^{m-1}\right),\ \nu_0\right). \tag{5.51}$$

$m = 1$ gives a Bingham fluid, $\tau_0 = 0$ a power-law fluid. In unyielded regions the viscosity reaches ν_0, and the material moves as a very viscous fluid (Fig. 5.13, right). The yield stress is W 96 (W 101 0) or follows a pressure-dependent Mohr-Coulomb law (W 111) with the angle of repose φ and the cohesion c_0 (W 102), for W 101 1 and 2

$$\tau_0 = \max(0,\ p\tan\varphi + c_0), \qquad \tau_0 = (p\tan\varphi + c_0)\left(1 - e^{-m_\tau \dot{\gamma}}\right), \tag{5.52}$$

where m_τ = W 103 gives a Papanastasiou regularization [Papanastasiou, 1987]; W 101 3 – 5 use the buoyant pressure and an excess pore pressure (W 104). W 110 1 (default) puts yield stress and viscous part into the viscosity of Eq. (5.51); W 110 2 – 7 move them into explicit momentum source terms acting in phase 1 only.

For debris flows (W 90 2), $\nu_1 = (a_2\mu_2 + (1 - a_2)\mu_3)/\rho_1$ mixes a Herschel-Bulkley slurry, $\mu_2 = k|\dot{\gamma}|^{m-1} + \tau_y/|\dot{\gamma}|$ with $\tau_y = \tau_0 C^2 e^{22CP_1}$ (W 105) and $k = b\tau_y$ (W 106), capped by W 95, and a gravel phase $\mu_3 = \min\left(\mu_{min} + p\sin\delta\,(1 - e^{-m_y\dot{\gamma}})/\dot{\gamma},\ \mu_{max}\right)$ (W 106, W 107), with a_2 = W 108.

SFLOW. In REEF3D::SFLOW, W 90 1 subtracts $\tau_{b,i}/\rho$ with the Bingham-type bed shear stress

$$\tau_{b,i} = \left(1.5\,\tau_0 + 3\,K\,\frac{|\mathbf{U}|}{h}\right)\frac{U_i}{|\mathbf{U}|}, \tag{5.53}$$

from the depth-averaged momentum, with τ_0 from W 96 or W 101 1 with the bed pressure.

5.8.3 Key input commands

H 10, H 15	heat transfer on and time scheme, convection scheme (CFD)
H 1, H 2, H 9	thermal diffusivities of water and air, phase assignment
H 3, H 4	property model (tables or Boussinesq), expansion coefficients
H 50 – H 58, H 61 – H 66	initial temperature field, fixed boundary temperatures
C 10, C 15	concentration on and time scheme, convection scheme (CFD)
C 1 – C 4	density and viscosity change per unit concentration
C 50 – C 58, C 75	initial concentration field, inclined slabs with initial velocity
T 45	buoyancy term in the k-equation
W 90	non-Newtonian rheology of phase 1 (CFD; SFLOW bed friction)
W 95 – W 98	viscosity cap ν_0, yield stress τ_0, consistency K, exponent m
W 101 – W 104, W 111, W 112	pressure-dependent yield stress models
W 105 – W 108	three-phase debris mixture (W 90 2)
W 110	rheology as viscosity or as momentum source term

5.9 Lagrangian Particle Tracking

Passive Lagrangian particles, e.g. floating or buoyant (macro)plastics, drifters or larvae, can be tracked in the flow field of REEF3D::NHFLOW (L 10 1). They are moved by the resolved flow but do not act back on it. Each release (point L 21, line L 22, box L 23) carries its own rise or settling velocity, windage coefficient and particle mode, so that several particle classes can be simulated together. The options, the seeding and the output are described in Section 6.10.

The particle velocity is the sum of the fluid velocity, the rise velocity w_s ($w_s > 0$ buoyant, $w_s < 0$ settling) and, at the free surface, the windage:

$$\frac{d\mathbf{x}_p}{dt} = \mathbf{v}_p = \mathbf{u}(\mathbf{x}_p, t) + w_s\,\mathbf{e}_z + \delta_s\, c_w U_{10}\,(\cos\alpha_w,\ \sin\alpha_w,\ 0)^T, \tag{5.54}$$

with $\delta_s = 1$ at the free surface and 0 otherwise, the windage coefficient c_w and the wind U_{10}, α_w of L 41 (default: A 571, Section 5.11). $\mathbf{u}$ is interpolated from the cell centres (bilinear in the horizontal, linear along the σ-levels and extrapolated to η), and Eq. (5.54) is integrated with Heun's method with the time step of the flow solver:

$$\mathbf{x}^* = \mathbf{x}_p^n + \Delta t\,\mathbf{v}_p(\mathbf{x}_p^n, t^n), \qquad \mathbf{x}_p^{n+1} = \mathbf{x}_p^n + \frac{\Delta t}{2}\Big(\mathbf{v}_p(\mathbf{x}_p^n, t^n) + \mathbf{v}_p(\mathbf{x}^*, t^{n+1})\Big). \tag{5.55}$$

Since NHFLOW resolves the individual waves, the Stokes drift follows from the orbital motion itself (Fig. 5.14); no Stokes drift parameterisation is added.

Unresolved dispersion is modelled by a random walk added after the Heun step, with the horizontal diffusivity K_h (L 31) and standard normal random numbers R (seed L 11):

$$\Delta x = \sqrt{2K_h\Delta t}\,R_x, \qquad \Delta y = \sqrt{2K_h\Delta t}\,R_y. \tag{5.56}$$

For three-dimensional particles (mode 0) a vertical random walk (L 32) uses either a constant $K_{v,0}$ (L 33) or $K_v(z) = K_{v,0} + \nu_t/Sc_t$ (Sc_t: L 34) with the scheme of Visser [1997], which avoids the artificial accumulation of particles in regions of low diffusivity:

$$\Delta z = K_v'(z)\,\Delta t + \sqrt{2\,K_v\Big(z + \tfrac{1}{2}K_v'(z)\Delta t\Big)\Delta t}\,R_z. \tag{5.57}$$

A particle is in the water column (state 0), at the free surface (1, subject to windage), stranded in a dry cell (2) or deposited on the bed (3) (Fig. 5.14). Surface-trapped particles (mode 1) stay at $z = \eta$. At the bed particles are reflected or, with L 52 1 and $w_s < 0$, deposited (no resuspension); stranded particles refloat when the cell becomes wet again (L 51 0) or stay beached (L 51 1); at the lateral boundaries they are reflected or removed (L 53).

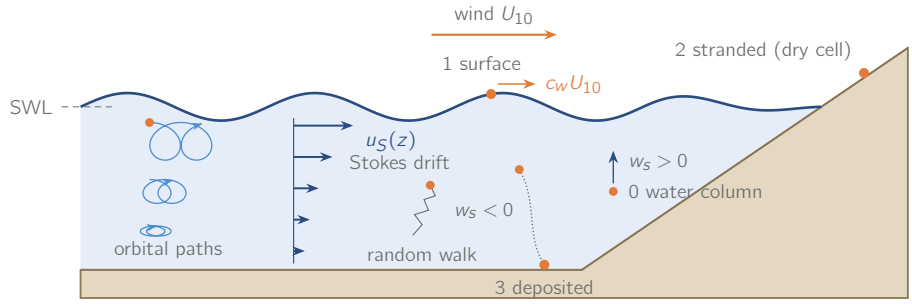

Figure 5.14: Lagrangian particles in NHFLOW: orbital paths with the resulting Stokes drift, windage of floating particles, rise and settling velocity, random walk, and the particle states 0–3.

5.9.1 Key input commands

Command	Meaning
L 10	particle tracking on (NHFLOW, A 10 5)
L 11	random seed of the random walk
L 21, L 22, L 23	point, line and box releases: position, number, release window, w_s, c_w, mode
L 31	horizontal diffusivity K_h
L 32, L 33, L 34	vertical random walk: off / constant $K_{v,0}$ / $K_{v,0} + \nu_t/Sc_t$; $K_{v,0}$; Sc_t
L 41	U_{10} and direction for the windage (default: A 571)
L 51, L 52, L 53	beaching, bed contact, lateral domain boundaries
L 61, L 62	print interval; output format (VTP, CSV track file)

5.10 Sea Ice (FNPF)

REEF3D::FNPF (only) can simulate rigid ice floes that interact with the waves, with each other and with fixed obstacles (A 380 1). The floes are coupled to the potential flow through a compliant lid, contacts are resolved with a non-smooth contact dynamics method, and floes can break by flexural failure or by splitting in contact (Fig. 5.15). A floe is a prism of thickness h_i and density ρ_i (default 917 kg/m^3) over a convex planform; single floes, random floe fields and fixed obstacles are defined in the floe input file.

5.10.1 Compliant lid

Under the footprint of a floe the free surface is loaded with the pressure

$$p_{ice} = \varphi \max\left(0,\ \rho_i g h_i + k\,(\eta - z_b) + c\left(\frac{\partial \eta}{\partial t} - \frac{\partial z_b}{\partial t}\right)\right), \tag{5.58}$$

with the lid stiffness $k = \alpha \rho_w g$ (α: A 381, default 20), the damping $c = 2\zeta\sqrt{k \rho_i h_i}$ (ζ: A 382) and the height $z_b(x, y, t)$ of the (tilted) floe bottom. It enters Eq. (4.43) as $-p_{ice}/\rho_w$, like the wind pressure (Section 5.11). The weight term gives the exact static draft; the unilateral spring-dashpot enforces the rigid-body kinematics up to $O(1/\alpha)$ and lets the floe lift off. The cell coverage φ is tapered across each edge over the half width δ (A 386, A 389) with an area-preserving sine ramp, since the single-valued σ-grid surface does not tolerate a step of the size of the draft. The lid stiffens the surface ($g_{eff} = g(1+\alpha)$), so the adaptive time step (N 48) is reduced by $1/\sqrt{1+\alpha}$.

5.10.2 Floe dynamics

The lid pressure integrated over the footprint plus the ice-water skin drag (C_d: A 383) give the force and moment on the floe:

$$\mathbf{F} = \sum_{\text{footprint}} \left[p_{ice} A \Big(-\frac{\partial \eta}{\partial x},\ -\frac{\partial \eta}{\partial y},\ 1 \Big)^T + \rho_w C_d \varphi A |\Delta \mathbf{u}| \, \Delta \mathbf{u} \right], \qquad \mathbf{M} = \sum_{\text{footprint}} \mathbf{r} \times d\mathbf{F}, \tag{5.59}$$

with the cell area A, the lever arm $\mathbf{r}$ from the centre of gravity and the horizontal velocity $\Delta \mathbf{u}$ of the surface flow relative to the floe. The slope term is the reaction of the pressure on the fluid, so horizontal momentum is conserved; radiation damping and added mass are provided by the potential flow. The rigid-body equations (orientation as unit quaternion $\mathbf{q}$, $\mathbf{I} = \mathbf{R}\mathbf{I}_b\mathbf{R}^T$) are

$$\frac{d\mathbf{x}_c}{dt} = \mathbf{u}_c, \qquad \frac{d\mathbf{q}}{dt} = \frac{1}{2}(0, \boldsymbol{\omega}) \otimes \mathbf{q}, \qquad m\frac{d\mathbf{u}_c}{dt} = \mathbf{F} - mg\,\mathbf{e}_z, \qquad \mathbf{I}\frac{d\boldsymbol{\omega}}{dt} = \mathbf{M} - \boldsymbol{\omega} \times \mathbf{I}\boldsymbol{\omega}. \tag{5.60}$$

They are advanced within the same Runge-Kutta stages as the free surface (A 310), which avoids the added-mass instability of a staggered coupling. In a 2D flume only surge, heave and pitch are active.

5.10.3 Non-smooth contact dynamics

After each time step the contacts between floes, fixed obstacles and, with A 385 1, the domain walls are resolved in the horizontal plane (surge, sway, yaw) with a non-smooth contact dynamics method of Moreau-Jean type [Jean, 1999]. Each contact point has the normal $\mathbf{n}$, the tangent $\mathbf{t}$ and the gap g_c; the relative normal velocity v_n and the normal impulse P_n satisfy the Signorini condition with Coulomb friction (μ, e: A 384),

$$P_n \geq 0, \quad v_n^+ - b \geq 0, \quad P_n (v_n^+ - b) = 0, \qquad |P_t| \leq \mu P_n, \tag{5.61}$$

with the target velocity

$$b = \begin{cases} -g_c/\Delta t & g_c > 0 \text{ (speculative contact)} \\ \max\Big(-e\, v_n^-,\ \beta \max(0, -g_c - g_s)/\Delta t \Big) & g_c \leq 0 \end{cases} \tag{5.62}$$

Speculative contacts may close the gap within the step but not more, so that floes do not tunnel through each other; for closed contacts e is the restitution coefficient and the Baumgarte term ($\beta = 0.2$) removes penetrations beyond the slop g_s. The impulses follow from projected Gauss-Seidel iterations (A 388), $P_n \leftarrow \max(0,\ P_n + m_n(b - v_n))$ and P_t clipped to $|P_t| \leq \mu P_n$, with the effective masses m_n, m_t of the pair (infinite for obstacles and walls); for obstacles the contact force is logged as ice load.

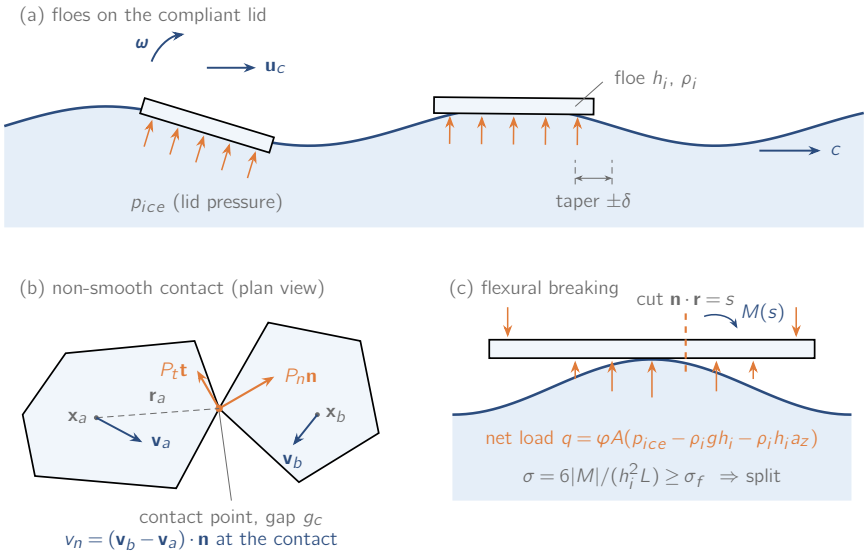

Figure 5.15: Sea ice in REEF3D::FNPF: (a) rigid floes on the compliant lid, loaded by the lid pressure p_{ice} with the tapered footprint edge; (b) planar non-smooth contact between two convex floes with normal and friction impulses; (c) flexural breaking: bending moment of the net load q about a candidate cut line.

5.10.4 Breaking

With A 390 a floe can split along a straight cut into two pieces (not narrower than D_{min}, A 393) that keep the rigid-body velocity field of the parent, which conserves mass, momentum and angular momentum. *Flexural failure*: for candidate cut lines $\mathbf{n}\cdot\mathbf{r} = s$ (A 395) the bending moment of the rigid floe follows from the net load on one side of the cut, including the inertia load with the local vertical acceleration a_z,

$$q = \varphi A(p_{ice} - \rho_i g h_i - \rho_i h_i a_z), \qquad M(s) = \sum_{\mathrm{n}\cdot\mathrm{r}>s} q(\mathbf{n}\cdot\mathbf{r} - s), \qquad \sigma = \frac{6|M|}{h_i^2 L}, \tag{5.63}$$

with the chord L of the cut. The floe breaks along the cut with the largest $\sigma \geq \sigma_f$ (A 391); for floes much longer than the flexural length this is an upper bound of the stress. *Contact splitting* (3D only): a floe splits along the contact normal when the pair normal force, averaged over the time t_c (A 396), exceeds the splitting load of linear elastic fracture mechanics [Bhat, 1988, Lu et al., 2015],

$$F_s = C\,K_{IC}\,h_i\sqrt{D}, \qquad D = 2\sqrt{A_f/\pi}, \tag{5.64}$$

with the fracture toughness K_{IC}, a calibration coefficient C (A 392) and the floe area A_f.

5.10.5 Key input commands

Command	Meaning
A 380	ice floes on, read `ice_floes.dat` (FNPF, A 10 3)
A 381, A 382	lid stiffness $\alpha = k/(\rho_w g)$ (default 20); lid damping ratio ζ (default 0.2)
A 383	ice-water drag coefficient C_d (default 0.0055)

A 384	contact friction μ and restitution e (default 0.3, 0.0)
A 385	domain boundaries act as walls for the floes (default 1)
A 386, A 389	footprint sub-samples per cell direction (4); edge taper half width in cells (2.0)
A 387	floe print interval (s, < 0: as the free surface output)
A 388	contact solver iterations (30)
A 390	breaking, bit flags: 1 flexural, 2 contact splitting, 4 spalling (7: all)
A 391, A 392	flexural strength σ_f (0.5 MPa); K_{IC} and C for splitting
A 393, A 394	minimum piece size D_{min} (< 0: 4 mean cells); check interval in steps
A 395, A 396	cut directions and cut positions per direction (12, 64); force averaging time t_c (0.05 s)
A 397 – A 402	Young's modulus and Poisson ratio for the flexural stress, crushing strength, Weibull scatter of the strength, cracks per check and per splitting event, spalling (Section 6.1)
A 403	lid time step factor
A 310, N 48	FNPF time scheme (floes use the same RK stages); adaptive time step

5.11 Wind Forcing

Wind forcing is available in REEF3D::FNPF (A 370 – A 378) and REEF3D::NHFLOW (A 570 – A 578). For storm surge, SFLOW and NHFLOW are driven by the wind and air pressure of a meteorological field with the Coriolis force (W 200 – W 227, Section 5.11.1). The first command selects the drag coefficient $C_d(U_{10})$ for the wind speed U_{10} at 10 m height and direction α_w (A 371, A 571), which gives the surface stress $\tau = \rho_a C_d U_{10}^2$ and the friction velocity $u_*^2 = C_d U_{10}^2$ (ρ_a: W 3):

$$10^3 C_d = \begin{cases} 0.75 + 0.067\,U_{10} \quad (U_{10} < 41 \text{ m/s}), & 3.5 & \text{Garratt [Garratt, 1977]} \\ 1.2875 \quad (U_{10} < 7.5 \text{ m/s}), & 0.8 + 0.065\,U_{10} & \text{Wu [Wu, 1982]} \\ 0.63 + 0.066\,U_{10} & & \text{Smith and Banke [Smith and Banke, 1975]} \\ 0.55 + 2.97\,\tilde{U} - 1.49\,\tilde{U}^2, & \tilde{U} = U_{10}/31.5 \text{ m/s} & \text{Zijlema [Zijlema et al., 2012]} \end{cases} \tag{5.65}$$

The type of forcing is selected with A 373 / A 573 (option 2 is deprecated and treated as 1).

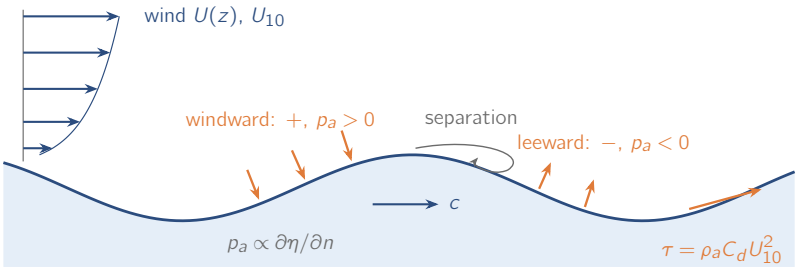

Figure 5.16: Wind forcing: logarithmic wind profile over the waves, surface stress τ, and the air pressure in phase with the slope, positive on the windward and negative on the leeward face (Jeffreys sheltering with flow separation behind steep crests).

Surface stress / wind setup (type 1). In NHFLOW the stress acts on the top σ-layer, so that the depth-integrated momentum receives τ/ρ_w. The potential flow cannot carry a shear stress; in FNPF the stress is spread over the reference depth h_{ref} (A 377) and applied as an

along-wind potential, which gives the setup $g\,\partial\eta/\partial s = \tau/(\rho_w h_{ref})$ in the along-wind direction s.

Wave growth by surface pressure (types 3 and 4). A zero-mean air pressure in phase with the along-wind slope $\partial\eta/\partial n$ transfers energy to the waves:

$$p_a = \beta\,\rho_a u_*^2 \frac{\partial\eta}{\partial n} \quad \text{(type 3)}, \qquad p_a = s_J\,\rho_a \max(U_{10}-c,0)^2\; w\,\frac{\partial\eta}{\partial n} \quad \text{(type 4)}. \tag{5.66}$$

Type 3 follows the shear-flow instability mechanism of Miles [1957] with the energy growth rate $\gamma/\omega = \beta(\rho_a/\rho_w)(u_*/c)^2$; $\beta \approx 32$ (A 375) reproduces the growth rate of Plant [1982]. Type 4 is the sheltering mechanism of Jeffreys [1925] in the modified form of Touboul et al. [2006] and Kharif et al. [2008]: air flow separation on the leeward side of steep crests (Fig. 5.16), so the pressure acts only on faces steeper than the critical slope S_c; the switch w ramps linearly from 0 at $0.8S_c$ to 1 at S_c (A 376: s_J, S_c, c; $c \le 0$: celerity of the generated wave). Because the input grows like $(u_*/c)^2$, the slope is low-pass filtered (A 378 passes of a 1-2-1 filter) so that grid-scale waves are not forced most strongly. In FNPF p_a enters the dynamic free surface boundary condition as $-p_a/\rho_w$; in NHFLOW it acts in every layer through its horizontal gradient. The forcing can be restricted to a rectangular area (A 372, A 572) with an optional $\cos^2$ downwind decay (A 374, A 574).

5.11.1 Meteorological forcing and storm surge (SFLOW, NHFLOW)

For storm surge, wind-driven circulation and tides with Coriolis force, REEF3D::SFLOW and REEF3D::NHFLOW can be driven by the 10 m wind $\vec{U}_{10}$ and the air pressure p_a at mean sea level of a meteorological field (W 200 – W 227, Section 6.17.1). In NHFLOW this replaces the wind forcing A 570 above; the same wind field is the wind input of SEASTATE with A 730 2 (stand-alone or coupled, A 750). Once per time step, every cell takes $\vec{U}_{10}$ and p_a from the source of W 200: uniform values (1, W 201, W 202), the gridded file `met-forcing.r3m` (2, bilinear in space and linear in time), a parametric tropical cyclone vortex from `met-vortex.trk` (3) or the blend of a background file, a moving nest `met-nest.r3m` and the vortex (4). Model time 0 is the date-time W 203 or the first record of the input.

Surface stress and pressure gradient. The surface stress

$$\vec{\tau}_s = r(t)\,\rho_a\,C_d(|\vec{U}_{10}|)\,|\vec{U}_{10}|\,\vec{U}_{10}, \qquad r(t) = \tfrac{1}{2}\left(1-\cos\frac{\pi t}{T}\right)\ (t<T),\ 1 \text{ otherwise}, \tag{5.67}$$

with ρ_a = W 3 and the ramp time T = W 204, uses one of the drag laws of W 207: the laws 1 – 4 of Eq. (5.65), the Garratt law capped at W 213 (5, default $2.5\cdot10^{-3}$, reached at 26.1 m/s: the saturation of the drag in hurricane winds) or the Zijlema law that beyond its peak at 31.4 m/s does not fall below W 214 (6). By default (W 207 0), the Garratt law is used for uniform and gridded fields and the capped Garratt law for the vortex and the blend. The depth-integrated momentum equations of the wet cells receive

$$\frac{\partial(hu)}{\partial t} = \ldots + \frac{\tau_{s,x}}{\rho_w} - \frac{h}{\rho_w}\,r(t)\frac{\partial p_a}{\partial x} + f\,hv, \qquad \frac{\partial(hv)}{\partial t} = \ldots + \frac{\tau_{s,y}}{\rho_w} - \frac{h}{\rho_w}\,r(t)\frac{\partial p_a}{\partial y} - f\,hu, \tag{5.68}$$

with ρ_w = W 1. The pressure gradient (W 208) is taken with central differences. In NHFLOW the surface stress acts in the top σ-layer, the pressure gradient and the Coriolis force in every

layer. At the open boundaries (B 520 methods 3 – 5) and in the relaxation and beach zones, the target water level is raised by the inverse barometer $\eta_{ib} = -r(t)\,(p_a - p_{ref})/(\rho_w g)$ with p_{ref} = W 209 (W 210), so that a low pressure system enters the domain with its static response.

Coriolis force. W 211 adds the Coriolis terms of Eq. (5.68) with $f = 2\Omega\sin\varphi$, $\Omega = 7.2921159 \cdot 10^{-5}$ 1/s, independent of W 200: on an f-plane with f = W 212 or from the latitude φ_0 of W 205 (W 211 1), or with the latitude of every cell (W 211 2). The terms are explicit in the Runge-Kutta stages.

Geographic frame. W 205 and W 206 place the grid on the earth with a spherical transverse Mercator projection (R = 6371 km) about the longitude and latitude (λ_0, φ_0), whose model coordinates are W 206, with the model x-axis rotated counter-clockwise from east. The frame is needed for the vortex, which is given in longitude and latitude, and for the latitude of the cells. Gridded files are written in model coordinates; their frame is stored in the file and compared with W 205, W 206.

Parametric vortex. With the distance r from the storm centre, the central and outer pressure p_c, p_n, $\Delta p = p_n - p_c$ and the radius of maximum wind R_m of the track record, the gradient wind V_g and the pressure follow from the model of W 215:

$$\text{Holland (1980):}\quad V_g = \sqrt{\frac{B\,\Delta p}{\rho_a}\left(\frac{R_m}{r}\right)^B e^{-(R_m/r)^B} + \left(\frac{rf}{2}\right)^2} - \frac{rf}{2}, \qquad p_a = p_c + \Delta p\, e^{-(R_m/r)^B},$$

$$\text{GAHM (Gao 2018):}\quad V_g = \sqrt{V_{max}^2\left(1 + \frac{1}{Ro}\right)e^{\phi\left(1-(R_m/r)^{B_g}\right)}\left(\frac{R_m}{r}\right)^{B_g} + \left(\frac{rf}{2}\right)^2} - \frac{rf}{2}, \qquad p_a = p_c + \Delta p\, e^{-\phi(R_m/r}$$

with $Ro = V_{max}/(|f|R_m)$ and R_m, B_g, ϕ fitted per quadrant and isotach, interpolated in r between the isotachs and in azimuth between the quadrants (the default, W 215 2). The model of Holland, Belanger and Fritz (2010) (W 215 3) gives the surface wind directly, $V = V_{ms}\left[(R_m/r)^{b_s} e^{1-(R_m/r)^{b_s}}\right]^x$ with $x = 0.5$ inside R_m, rising linearly to x_n at r_n outside. The surface wind is V_g reduced to 10 m (W 216), turned inwards by the inflow angle (W 218), plus the storm translation scaled with V_g/V_{max} (W 219), times the 1-min to 10-min factor W 217; in the southern hemisphere the vortex turns clockwise. In the blend (W 200 4), every layer replaces the one below with a weight that tapers in space and time (nest: W 221; vortex: the radii W 222 and W 224; shape W 225); the vortex pressure can be rescaled to merge into the background on the outer ring (W 223), the vortex can carry the large-scale flow of the background (W 226), and the nest records can be interpolated in time relative to the storm centre (W 227).

Input preparation. The repository tools convert the meteorological data: `tools/met_forcing.py` writes `met-forcing.r3m` from ERA5, NORA3 or BARRA-C2 NetCDF files (`grid`), from a time series table (`uniform`) or from an old SEASTATE wind file (`seastate`), and `met-nest.r3m` from a moving nest such as HAFS (`nest`); `tools/met_track.py fit` fits the vortex parameters of all three models from an ATCF best track or IBTrACS and writes `met-vortex.trk`. Both take the frame of W 205, W 206 as `-origin LON0 LAT0 -rotation ROT`. The free surface output of SFLOW and NHFLOW contains the wind, the air pressure and the surface stress, and the SFLOW maxima (P 112) record the highest water level, the maximum wind and the minimum air pressure of the run.

5.11.2 Key input commands

Command	Meaning
A 370 / A 570	wind forcing on, C_d: 1 Garratt, 2 Wu, 3 Smith and Banke, 4 Zijlema (FNPF / NHFLOW)
A 371 / A 571	U_{10} and direction
A 372 / A 572	forcing area
A 373 / A 573	type: 1 surface stress (NHFLOW) or wind setup (FNPF), 3 Miles/Plant, 4 modified Jeffreys
A 374 / A 574	$\cos^2$ downwind decay over the forcing area
A 375 / A 575	Miles/Plant coefficient β (default 32)
A 376 / A 576	Jeffreys: sheltering coefficient s_J, critical slope S_c, phase speed c (0.5, 0.3, 0.0)
A 377	reference depth h_{ref} of the FNPF wind setup
A 378 / A 578	1-2-1 filter passes on the slope (default 4)
L 41	wind for the windage of Lagrangian particles (Section 5.9)
W 200	meteorological forcing (SFLOW, NHFLOW; SEASTATE A 730 2): 1 uniform, 2 `met-forcing.r3m`, 3 vortex, 4 blend
W 201, W 202	uniform wind and air pressure
W 203, W 204	date-time of model time 0, ramp time
W 205, W 206	geographic frame (transverse Mercator)
W 207, W 213, W 214	drag law, cap and floor of the drag coefficient
W 208 – W 210	air pressure gradient, inverse barometer at the open boundaries
W 211, W 212	Coriolis force: f-plane or per cell, f
W 215 – W 219	vortex model and surface wind
W 220 – W 227	blend of background, nest and vortex

5.12 Structural Solver REEF3D::CORAL

REEF3D::CORAL (**C**oupled **O**ffshore **R**otor **A**nalysis and **L**oads) is the structural solver of REEF3D for fixed and floating offshore structures, wind turbines and wave energy converters. It holds everything that moves, deforms or carries a load: rigid bodies connected by joints, flexible beam structures and Craig-Bampton superelements, springs and foundation matrices, pile foundations in layered soil, mooring lines, power take-offs (PTO), wind turbine rotors with their drivetrain and controller, and the engineering hydrodynamic and aerodynamic loads on them (Morison loads, strip theory, the Cummins equation with a hydrodynamic database, quadratic drag, blade element momentum). Unlike the models of the previous sections, CORAL is a complete multibody solver of its own with its own input file `coral.txt` in the simulation folder. It runs in two ways:

- **coupled with REEF3D** (Z 40 1): the CORAL model is advanced in step with the fluid time of REEF3D::FNPF, REEF3D::NHFLOW or REEF3D::CFD. The 6DOF bodies of the fluid solver can be driven by CORAL (two-way), their hull loads can act on CORAL bodies and flexible members (one-way), and the Morison members and dynamic mooring lines of CORAL can take the velocity and free surface of the resolved fluid (one-way). REEF3D::SFLOW is not coupled to CORAL;

- **standalone** (executable `CORAL`, no fluid grid): with CORAL's own linear waves and engineering hydrodynamics, or replaying the fluid input recorded in a coupled run.

The source code is organized in the folders of `src/coral/` (core, structure, hydro, mooring, pto, soil, wind, aero, control, io, linear, reef3d, app). Ice loads, a panel solver, load case batches and post-processing (rainflow, damage-equivalent loads) are planned; their folders exist but contain no code yet, and `coral.txt` has no keywords for them. Without Z 40, REEF3D runs as before.

5.12.1 Multibody model and equations of motion

The model consists of rigid bodies with six degrees of freedom each, with position $\mathbf{x}$ of the centre of gravity and rotation matrix $\mathbf{R}$ (body to global), the velocity $\mathbf{v}$ in the global frame and the angular velocity $\boldsymbol{\omega}$ in the body frame, followed by scalar generalized coordinates (e.g. the modal amplitudes of a superelement). The nodes of beam structures are small rigid bodies as well. The bodies are connected to each other or to the `ground` by holonomic constraints $\mathbf{g}(\mathbf{q}) = 0$ with the Jacobian $\mathbf{G}$, $\dot{\mathbf{g}} = \mathbf{G}\boldsymbol{\nu}$: fixed (6 equations), spherical (3), revolute (5, rotation about the axis $\mathbf{e}$), prismatic (5, translation along $\mathbf{e}$), cylindrical (4) and planar (3). The equations of motion are the index-3 differential-algebraic system

$$\mathbf{M}(\mathbf{q})\,\dot{\boldsymbol{\nu}} + \mathbf{h}(\mathbf{q},\boldsymbol{\nu}) = \mathbf{f}(\mathbf{q},\boldsymbol{\nu},\dot{\boldsymbol{\nu}},t) - \mathbf{G}^T(\mathbf{q})\,\boldsymbol{\lambda}, \qquad \mathbf{g}(\mathbf{q}) = 0, \tag{5.69}$$

with the Lagrange multipliers $\boldsymbol{\lambda}$ (the constraint forces are $-\mathbf{G}^T\boldsymbol{\lambda}$). Per body, $\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{h}$ is $m\dot{\mathbf{v}}$ and $\mathbf{J}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega}$ (inertia tensor $\mathbf{J}$ about the centre of gravity, body frame), including point masses attached at offsets. The generalized forces $\mathbf{f}$ are the sum of all loads of the model (gravity, structure, hydrodynamics, mooring, PTO, soil, fluid); a load may depend on positions, velocities and accelerations (added mass).

Time integration. Eq. (5.69) is integrated with the Lie group generalized-α method (Chung and Hulbert 1993; Arnold and Brüls 2007; Brüls, Cardona and Arnold 2012) on $\mathbb{R}^3 \times SO(3)$ for every body and on $\mathbb{R}^n$ for the generalized coordinates:

$$\begin{aligned} \mathbf{q}_{n+1} &= \mathbf{q}_n \circ \exp\Big(h\boldsymbol{\nu}_n + h^2\big(\tfrac{1}{2} - \beta\big)\mathbf{a}_n + h^2\beta\,\mathbf{a}_{n+1}\Big), \\ \boldsymbol{\nu}_{n+1} &= \boldsymbol{\nu}_n + h(1-\gamma)\,\mathbf{a}_n + h\gamma\,\mathbf{a}_{n+1}, \\ (1-\alpha_m)\,\mathbf{a}_{n+1} + \alpha_m\,\mathbf{a}_n &= (1-\alpha_f)\,\dot{\boldsymbol{\nu}}_{n+1} + \alpha_f\,\dot{\boldsymbol{\nu}}_n, \end{aligned} \tag{5.70}$$

with $\alpha_m = (2\rho_\infty - 1)/(\rho_\infty + 1)$, $\alpha_f = \rho_\infty/(\rho_\infty + 1)$, $\gamma = \frac{1}{2} + \alpha_f - \alpha_m$ and $\beta = \frac{1}{4}(\gamma + \frac{1}{2})^2$ from the spectral radius at infinite frequency ρ_∞ (`time rho_inf`, default 0.8; $\rho_\infty = 1$ has no numerical damping, but leaves the multipliers undamped). The unknowns $\dot{\boldsymbol{\nu}}_{n+1}$ and $\boldsymbol{\lambda}_{n+1}$ are solved by Newton's method (tolerance `time tol`, default 10^{-12}, at most 30 iterations). The Jacobian combines the analytic tangents of the loads that have one (beams, superelements, soil springs) with a finite-difference Jacobian of all other loads, whose columns are grouped by a distance-2 colouring of the body coupling graph, so that a frame of n nodes needs a few dozen residual evaluations instead of $6n$. The Jacobian is kept over iterations and steps while Newton converges fast (modified Newton; `full_newton` rebuilds it in every iteration). Up to 400 unknowns the linear systems are solved by dense LU, above by sparse LU (COLAMD ordering). A step that fails is repeated with $2, 4, \ldots, 64$ sub-steps; if these fail too, the run stops.

Static equilibrium and linear analysis. With `static`, the equilibrium $\mathbf{f}(\mathbf{q},0,0,t)-\mathbf{G}^T\boldsymbol{\lambda}=0$, $\mathbf{g}=0$ is computed before the time integration by Newton's method, falling back to Newton damped by the natural monotonicity test (Deuflhard) from the start configuration. With `modes` n, the model is linearized about the current configuration, $\mathbf{M}\,\delta\ddot{\mathbf{q}}+\mathbf{C}\,\delta\dot{\mathbf{q}}+\mathbf{K}\,\delta\mathbf{q}=0$ (including the geometric stiffness of the constraint forces and $\mathbf{A}_\infty$ of a hydrodynamic database, without the radiation memory), reduced to the null space of $\mathbf{G}$, and the first n undamped modes ($\mathbf{K}\boldsymbol{\phi}=\omega^2\mathbf{M}\boldsymbol{\phi}$) or, with `damped`, the damped modes with their damping ratios are written to `modes.dat` (frequency, period, damping and the split of the modal kinetic energy over the six global directions) and `mode_shapes.dat`.

5.12.2 Structural elements and foundations

Springs, foundation matrices and point loads. A spring-damper between two points (`spring`, e.g. a tether) has the tension $T=T_0+k\,(L-L_0)+c\,\dot{L}$, optionally tension only. A linear 6×6 element (`linear6`) acts between a body and the ground about a reference point, $\mathbf{F}=-\mathbf{K}\boldsymbol{\xi}-\mathbf{C}\dot{\boldsymbol{\xi}}$ with $\boldsymbol{\xi}=[\mathbf{p}-\mathbf{p}_0;\ \log(\mathbf{R}\mathbf{R}_0^T)]$, for linearized moorings or foundation matrices (diagonal or full). `force` applies a constant force and moment with an optional ramp, optionally released at a given time (`until`, e.g. a free decay from the static equilibrium under the load).

Beam frames. A `member` between two points is divided into n co-rotational 3D beam elements (Crisfield; Battini) whose nodes are named `<member>:0` … `<member>:`n; members meeting at a point share the node, and a member end can sit on a rigid body or on the ground. Large displacements and rotations with small strains are described in the element frame that follows the chord and the mean of the end triads; there the linear Timoshenko stiffness of the section gives

$$N=\frac{EA\,u}{L_0},\qquad T=\frac{GJ(\theta_{b,x}-\theta_{a,x})}{L_0},$$
$$M_a=\frac{EI\,((4+\Phi)\,\theta_a+(2-\Phi)\,\theta_b)}{L_0\,(1+\Phi)},\qquad \Phi=\frac{12\,EI}{GAL_0^2},\tag{5.71}$$

about both local bending axes (M_b symmetric; $GA=0$: Euler-Bernoulli), with stiffness-proportional damping $\beta\mathbf{K}$. The forces on the nodes are $-\mathbf{B}^T\mathbf{f}$ with the analytic strain matrix $\mathbf{B}$, and the tangent is analytic. The mass is lumped on the nodes. Sections are thin-walled tubes from D, t, E, G, ρ (`section tube`) or given by their properties, and members can be tapered between two sections.

Craig-Bampton superelements. Members listed in a `superelement` are assembled from linear Timoshenko elements with consistent mass instead. Ends on the ground are clamped, ends on rigid bodies form the interface (rigid transition piece). The interior motion is reduced to the static (Guyan) shapes of the interface motion $\mathbf{u}_R$ and n fixed-interface modes $\mathbf{q}$,

$$\mathbf{u}_I=\boldsymbol{\Phi}\,\mathbf{u}_R+\boldsymbol{\Psi}\,\mathbf{q},\qquad \boldsymbol{\Phi}=-\mathbf{K}_{II}^{-1}\mathbf{K}_{IR},\qquad \mathbf{K}_{II}\boldsymbol{\psi}=\omega^2\mathbf{M}_{II}\boldsymbol{\psi},\tag{5.72}$$

with modal damping ratio ζ on the modes and stiffness-proportional damping on the Guyan part. The modal amplitudes are generalized coordinates of the model. Superelements use a fixed reference frame (small interface rotations) and are meant for fixed-bottom substructures (jackets, tripods, monopiles).

Pile foundations. A `soil` profile below a mudline consists of sand, clay or linear Winkler layers (`soil_layer`). `pile` puts nonlinear elastic springs on the nodes of a frame member below the mudline, times their tributary length: lateral p-y springs (API sand after Reese, Cox and Koop; soft clay after Matlock, static or cyclic), axial t-z springs, a tip Q-z spring (compression only) and the torsion from the shaft friction, following API RP 2GEO / RP 2A, or only their initial slopes (`linear`). The springs have no hysteresis.

5.12.3 Hydrodynamic loads, moorings and power take-off

Waves. Without a fluid grid, the incident waves are a sum of linear components (`waves`): regular waves, single components, or JONSWAP and Pierson-Moskowitz spectra with seeded random phases (default 200 components between $0.5\omega_p$ and $3\omega_p$), on finite or infinite depth, with an optional start ramp. The still water level is at $z = 0$ of CORAL's frame; above it the kinematics of $z = 0$ are used.

Morison loads. Slender cylinders on rigid bodies (`morison ... body`) and Morison loads distributed along frame and superelement members (`morison ... member`) are integrated over strips with the load per unit length normal to the axis

$$\mathbf{f} = \rho A C_m \mathbf{a}_w - \rho A C_a \mathbf{a}_s + \tfrac{1}{2}\rho C_d D\,|\mathbf{u}_r|\mathbf{u}_r, \qquad A = \frac{\pi D^2}{4}, \qquad \mathbf{u}_r = \mathbf{u}_w + \mathbf{U}_c - \mathbf{u}_s, \quad (5.73)$$

[Morison et al., 1950] with the wave velocity and acceleration $\mathbf{u}_w$, $\mathbf{a}_w$, a uniform current $\mathbf{U}_c$ and the structure velocity and acceleration $\mathbf{u}_s$, $\mathbf{a}_s$ (defaults $C_d = 1$, $C_a = 1$, $C_m = 1 + C_a$). Only the part below the instantaneous surface carries load; on members, a strip cut by the surface counts with its wet part, the buoyancy of the submerged strips is added, and a flooded member adds its inner water as node mass.

Hydrodynamic database (Cummins equation). For bodies with a linear potential-flow database (`hydro`, CORAL text format or WAMIT output `.1`, `.3`, `.hst`), the load at the reference point of each database body is

$$\mathbf{F} = \rho g V \mathbf{e}_z - \mathbf{C}\boldsymbol{\xi} + \mathbf{F}_{exc}(t) - \mathbf{A}_\infty \ddot{\boldsymbol{\xi}} - \int_0^t \mathbf{K}(t-\tau)\,\dot{\boldsymbol{\xi}}(\tau)\,d\tau - \mathbf{B}_{lin}\dot{\boldsymbol{\xi}} - \mathbf{Q}\,|\dot{\boldsymbol{\xi}}|\,\dot{\boldsymbol{\xi}},$$
$$\mathbf{K}(t) = \frac{2}{\pi}\int_0^\infty \mathbf{B}(\omega)\cos(\omega t)\,d\omega, \qquad \boldsymbol{\xi} = [\mathbf{p} - \mathbf{p}_0;\ \log(\mathbf{R}\mathbf{R}_0^T)], \quad (5.74)$$

with buoyancy, hydrostatic stiffness $\mathbf{C}$, the excitation from the incident waves and $\mathbf{X}(\omega,\beta)$, the infinite-frequency added mass (from the database or Ogilvie's relation), the radiation memory by convolution or by a state-space realization (`state_space`), and additional linear and quadratic damping. A separate quadratic drag (`drag`) $F_i = -Q_i|u_i|u_i$, or $Q = \rho C_d A/2$ along an axis, adds e.g. the viscous damping that a potential-flow solver such as FNPF leaves out.

Strip theory for floaters. `strip_member` describes the columns and pontoons of a rigid floater by strips with nonlinear hydrostatics: the buoyancy of the wet part of each strip below the instantaneous wave elevation, the Morison load of Eq. (5.73) normal to the member, and at closed ends below the surface (`end1`, `end2`) the dynamic pressure of the waves, the added mass $C_{a,end}\,\rho D^3/6$ and a drag along the axis (e.g. heave plates). The wave kinematics are taken at the undisplaced strips once per time, or at the displaced ones with `exact_kinematics`.

Mooring lines. A quasi-static line (`mooring ... quasi_static`) is an elastic catenary (Jonkman 2007), evaluated for the current fairlead position, with a seabed portion and seabed friction `CB` at a ground end on the seabed; otherwise it hangs free between its ends. A dynamic line (`mooring ... dynamic`) is a lumped-mass line with N segments of the MoorDyn type: axial stiffness EA or a nonlinear stress-strain table (`EA_curve`) and axial damping `BA` (< 0: the damping ratio of the axial mode of a segment), carried only while a segment is taut (`BA_slack`: the damping in every segment, as in MoorDyn); an optional bending stiffness EI for cables and risers, with the ends pinned unless clamped to their body (`clamp_a`, `clamp_b`); normal and tangential drag and added mass, the Froude-Krylov inertia of the water, waves and a uniform current or a current profile over the height (`current_profile`), with the submerged fraction of each node at the free surface; and seabed contact with a penalty spring `kb` and damping `cb`. The seabed friction (default none) has the kinetic coefficients `mu` across and `mu_axial` along the line (elliptic limit): each node in contact sticks to an anchor through a shear spring and damper (`ks`, `cs`) up to `mu_static` times the kinetic limit and then slides, so that a line under a load below the static limit does not creep; with `v_eps` > 0, the regularized Coulomb law $\mu F_n \tanh(v/v_\varepsilon)$ without static friction is used instead. The seabed is flat at the height of the ground end of the line (`seabed` z or `no_seabed` to change it; a line between two bodies has none unless given) or follows the `bathymetry`: a plane, a grid of heights, or the bed of the resolved fluid (FNPF, NHFLOW with `fluid_kinematics`). Line properties can be named (`line_type`), and a line can consist of several sections in series (`section`, e.g. chain-wire-chain), joined at junction points without mass. Dynamic lines can also end on free points (`point`: bridle nodes, clump weights, buoys with mass, volume, drag, added mass and seabed contact); the lines connected through points form a mooring system that is advanced together. A dynamic line breaks at a given time or when the tension of an end segment reaches a limit (`break_time`, `break_tension`, `break_end`): the end lets go of its body or point and moves free. The inner nodes are advanced with an explicit fourth-order Runge-Kutta scheme in sub-steps inside the multibody step; the line starts in the static equilibrium of its segments (`no_static`: from the catenary) and can be settled in the water at rest before the run (`settle`). The dynamic lines are validated against MoorDyn v2.

Power take-off and prescribed motion. A PTO (`pto`) acts in a joint as an axial force or, with `rotational`, as a torque, with the joint coordinate q:

$$F = F_0 - k\,(q - q_0) - c\,\dot{q} - F_c \tanh(\dot{q}/v_\varepsilon), \qquad |F| \le F_{max}, \qquad |F\,\dot{q}| \le P_{max}, \tag{5.75}$$

plus end stops (push-only penalty spring-dampers beyond q_{min}, q_{max}). Drive-train losses, a damper `c_loss` and a Coulomb friction `coulomb_loss`, act in every state, are not limited, and their power is dissipated, not absorbed. Latching holds the joint with a stiff latch element for t_{hold} after each zero crossing of $\dot{q}$, declutching switches the generator off for t_{hold}.

The absorbed power is $P = -F\dot{q}$. The same PTO library drives the FNPF and NHFLOW 6DOF power take-off X 500 – X 510 (Section 5.2). `drive` prescribes a joint coordinate $q_p(t) = q_0 + a\,r(t)\sin(2\pi t/T + \varphi)$ with a start ramp $r(t)$ through a stiff spring-damper, e.g. for forced oscillations of a body in the resolved fluid. `jforce` prescribes a force along a joint axis, or a torque about it, from a table t, F (scaled, with a start ramp, zero outside the table unless `periodic`), e.g. a multisine force for the system identification of a wave energy converter.

5.12.4 Wind, rotors and controllers

Wind. The wind inflow (`wind`) is a steady wind (`wind steady`: speed U at the height z_{ref}, direction, power-law or logarithmic shear, veer, upflow, start ramp) or a uniform wind in time from an InflowWind uniform wind file (`wind file`). With an IEC 61400-1 turbine class (`wind iec`), the deterministic events (extreme operating gust, direction change, coherent gust, vertical and horizontal shear; `wind event`) and IEC Kaimal turbulence with the exponential coherence (`wind turbulence kaimal`, normal, extreme or extreme-wind turbulence model, generated on a grid around the hub by Veers' method, frozen and convected with the hub speed, optionally written as a TurbSim file) can be added. Alternatively, a TurbSim full field is read (`wind turbsim`). `wind_probe` writes the inflow at a point fixed in space or on a body.

Rotors. A `rotor` is a hub turning in a revolute joint (*name*`_shaft`) on the nacelle body (or the ground), with the blade stations of an AeroDyn 15 blade file and the airfoil polars of AeroDyn airfoil files. The aerodynamic loads are computed by quasi-steady blade element momentum theory at each station from its actual motion and orientation (Ning's single-equation method, Prandtl tip and hub loss, Buhl's high-induction correction, tangential induction, Pitt-Peters skewed wake), with the tower potential flow; the air density is `rho_air`. The blades are rigid on the hub with their distributed mass from an ElastoDyn blade file, or sit on pitch bearings driven by second-order actuators (`pitch_joints`), or are flexible co-rotational beams from the structural table (`blade_elements`). The drivetrain has a gearbox ratio, a generator inertia, an optional torsion spring and damper of the low-speed shaft, an efficiency and a generator torque law (none, constant, induction-generator slip, $k\omega^2$).

Controller. A `controller` of the NREL 5 MW baseline type acts on a rotor once per step: variable-speed generator torque on the filtered generator speed (regions 1, 1.5, 2 with $k\omega^2$, 2.5 and 3 with constant power, torque and rate limits) and gain-scheduled PI collective pitch with pitch and rate limits. The defaults are those of the NREL 5 MW turbine; with `tune`, the gains are computed for the rotor from CORAL's steady blade element momentum at the rated power and rotor speed.

5.12.5 Coupling with the hydrodynamic modules

With Z 40 1, rank 0 reads `coral.txt` and broadcasts it; every MPI rank holds the same model and advances it with the same input from the fluid, broadcast from rank 0, so that the

ranks stay bit for bit identical (a difference is reported as a warning). Each fluid time step from t^n to $t^n + \Delta t$ is covered by CORAL steps of at most the `time dt` of `coral.txt`; the `end` time is not used. Positions exchanged with the fluid are in the frame of the fluid grid, which is CORAL's frame shifted by `fluid_coupling origin`. The 6DOF bodies of the fluid solver (Section 5.2) are numbered from 0 in the order of their definition; the fluid solver keeps their geometry, the body boundary condition and the integration of the hull loads (Fig. 5.17).

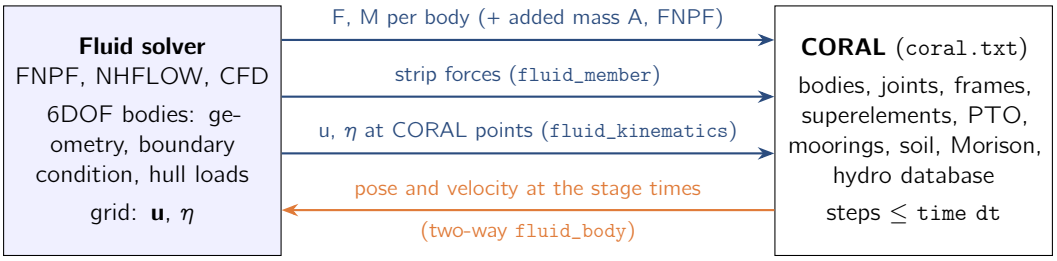

Figure 5.17: Data exchanged between the fluid solver and REEF3D::CORAL per fluid time step (blue: fluid to CORAL, one-way; orange: back to the fluid, two-way bodies only).

Two-way: fluid bodies driven by CORAL. With `fluid_body` (without `one_way`), CORAL replaces the rigid-body equations of a REEF3D 6DOF body (X 10 1): the fluid solver hands over the force and the moment about the centre of gravity without the weight, CORAL steps the whole model (joints, PTO, moorings, structure) and returns the pose and velocity of the body at the Runge-Kutta stage times of the fluid, interpolated from the recorded motion of the step (cubic Hermite in position and rotation). Mass and inertia are taken from the REEF3D body unless `own_mass`; the PTO X 500 and the mooring X 310 of the REEF3D body are not used, they belong to the CORAL model. Between the fluid samples, the load is a Lagrange polynomial in time of order `fluid_coupling order` (default 2) through the last samples. With the predictor (default), the load of the next fluid stage, computed at a pose CORAL predicts, enters the CORAL step as a provisional sample, so that the step interpolates instead of extrapolating. **FNPF** delivers the instantaneous added mass $\mathbf{A}$ of its unit-mode solves (Section 5.2) and the load $\mathbf{F}$ without the acceleration part, so that

$$\mathbf{F}_{total}(t) = \mathbf{F}(t) - \mathbf{A}\,\mathbf{a} \tag{5.76}$$

is implicit in CORAL's Newton iterations and the coupling is stable for light bodies. With mesh refinement (G 1), the loads and the added mass of all grids are used; with subcycling (G 7 1), every step of the finest level is a CORAL step. **NHFLOW** (one body) and **CFD** (several bodies, X 20) have no instantaneous added mass: their loads contain the reaction to the body acceleration one stage late, which makes the staggered coupling of light bodies unstable. The artificial added mass $\mathbf{A}_s$ (`fluid_body ... added_mass`, diagonal, global axes about the centre of gravity) makes it implicit with compensation,

$$\mathbf{F}_{total}(t) = \mathbf{P}(t) - \mathbf{A}_s\,\mathbf{a}, \tag{5.77}$$

where $\mathbf{P}$ is the polynomial through the samples $\mathbf{F}_k + \mathbf{A}_s\mathbf{a}_k$ with the accelerations $\mathbf{a}_k$ the fluid saw, so that the extra term vanishes where CORAL's motion is the fluid's. In NHFLOW, about half to once the added mass of the body's motion stabilized a body nine times lighter than its added mass; in CFD, about once the added mass was needed. For bodies not much lighter than their added mass, $\mathbf{A}_s$ should be left out. NHFLOW integrates its loads at the

end of each stage (also with X 13 and with G 7 1), CFD and FNPF at the start; in FNPF all or none of the bodies must be fluid bodies.

One-way: fluid loads on the structure. With `fluid_body ... one_way`, REEF3D keeps moving its 6DOF body itself (fixed with X 11 0, or prescribed) and only its loads act on a CORAL body, at the REEF3D centre of gravity; `excitation` subtracts the load of the start (still water), e.g. when a hydrodynamic database supplies the hydrostatics. With `fluid_member`, the hull loads of such a body are binned in strips along a frame or superelement member (default two per element) and applied along it, so that e.g. a resolved monopile in FNPF, NHFLOW or CFD loads a flexible CORAL tower. The pressure on each triangle of the load integration is integrated exactly over its parts in each strip (CFD and NHFLOW need the STL load integration X 60 1). FNPF and CFD deliver the loads of the start of the fluid step, so CORAL runs one fluid step behind; NHFLOW delivers those of its end.

One-way: resolved kinematics. With `fluid_kinematics` (FNPF and NHFLOW), the Morison members and the dynamic mooring lines take the velocity, the acceleration and the free surface of the resolved fluid instead of CORAL's waves. The fluid is interpolated at their points (strip centres, line nodes) once per fluid time level (FNPF: σ-grid velocities, the top cell sampled just below the surface; NHFLOW: cell centres; above the surface the surface velocity, below the bed zero), reduced over the ranks, and interpolated in time by a Lagrange polynomial of the given order (default 2); the acceleration is its time derivative. Points outside the fluid domain and times before the first sample use the `waves` of `coral.txt`, or still water. These loads do not act back on the fluid. If REEF3D's still water level F 60 is not at CORAL's $z = 0$ while dynamic lines or Morison members do not sample the fluid, a warning asks for `fluid_coupling origin 0 0` F 60 or `fluid_kinematics`; a gravity different from W 20 – W 22 is reported as well.

Checkpoints and replay. In the checkpoints of REEF3D (P 48, restart I 45 1, rebalancing G 46), the complete CORAL state (bodies, joint coordinates, the integrator with its Jacobian, every load with its history, the sampled kinematics and the tracks of the fluid bodies) is stored; a restart skips the static equilibrium, the modes and the first output and continues the time series and the journal. With `fluid_coupling record` *file*, every input from the fluid is written to the binary journal `REEF3D_CORAL/`*file*, which the standalone executable replays without the fluid, e.g. for a changed structure, PTO or controller study on the same one-way fluid loads (for two-way bodies the replay only repeats the recorded run).

5.12.6 Standalone executable

The standalone executable `CORAL` is built with `make coral` (`bin/CORAL`) or the CMake target `CORAL`; it is not part of the REEF3D executable and needs no MPI. `CORAL` [*input file*] (default `coral.txt`) reads the model, computes the static equilibrium and the modes if asked, integrates from `start` to `end` with steps `dt`, and writes the time series and, with `output vtp`, the structure for ParaView to `coral_vtp/structure_*.vtp` (bodies as vertices, beam and superelement elements and mooring lines as lines). With `end` not after `start`, only the analyses run. `CORAL` *input file* `replay` *journal* runs the case on a journal recorded by a coupled run. In the standalone run, fluid bodies receive no fluid loads.

5.12.7 Input file `coral.txt`

`coral.txt` contains one item per line; # starts a comment and a line ending in \ continues on the next line. Each item starts with a keyword, mostly followed by a name and key-value pairs in any order; bodies, joints, members and loads are referred to by name (`ground` is the fixed frame, nodes of frame members are `<member>:<i>`). Units are SI, angles in degrees, z points up, and the still water level is at $z = 0$. Unknown keywords, missing values and unknown names stop the run with the line number. Table 5.9 lists the keywords; optional parts are in brackets.

Table 5.9: Keywords of `coral.txt`.

Keyword	Parameters and meaning
Run settings	
`gravity`	g_x g_y g_z (default 0 0 −9.81), or `off`
`rho`	water density for Morison loads, mooring lines and drag (1025)
`rho_air`	air density for the aerodynamic loads (1.225)
`time`	`dt` h `end` t [`start` t_0] [`rho_inf` ρ_∞ (0.8)] [`tol` (10^{-12})] [`full_newton`]; coupled: h is the maximum CORAL step, `end` is not used; without `time`: $h = 0.01$
`output`	`dt` h (0: every step) [`file` *name*] (default `coral_results.dat`); `output vtp` Δt: VTP files (standalone)
`static`	static equilibrium before the time integration
`modes`	n [`damped`]: linear analysis, `modes.dat`, `mode_shapes.dat`
Bodies and joints	
`body`	*name* `mass` m [`inertia` I_{xx} I_{yy} I_{zz} [I_{xy} I_{xz} I_{yz}]] [`cog` x y z] [`rpy` roll pitch yaw] [`velocity` ...] [`omega` ...] [`offset` dx dy dz]; inertia about the CoG in body axes; `offset` displaces the body after the set-up (decay tests)
`joint`	*name* `fixed`|`spherical`|`revolute`|`prismatic`|`cylindrical`|`planar` *body a body b* `point` x y z [`axis` e_x e_y e_z (default z)]
`point_mass`	*body* `mass` m [`inertia` I_{xx} I_{yy} I_{zz}] [`at` x y z]
Structure and soil	
`section`	*name* `tube D` D `t` t `E` E `G` G `rho` ρ, or *name* `EA`, `EIy`, `EIz`, `GJ`, `m` [`GAy`, `GAz`, `rIx`, `rIy`, `rIz`, `D`]
`member`	*name* `p1` x y z `p2` x y z [`n` elements (10)] `section` s [`section2` s_2 (tapered)] [`orient` x y z] [`beta` β] [`end1`, `end2` *body*|`ground`]
`superelement`	*name* `members` m_1 ... [`modes` n (10)] [`zeta` ζ] [`beta` β]
`spring`	*name* `a` *body* x y z `b` *body* x y z `k` k [`c` c] [`L0` L_0] [`T0` T_0] [`tension_only`]
`linear6`	*name* `body` *b* `point` x y z [`k` $k_1..k_6$] [`c` $c_1..c_6$] [`K` 36 values] [`C` 36 values]
`force`	*name* `body` *b* [`point` x y z] [`F` f_x f_y f_z] [`M` m_x m_y m_z] [`ramp` t] [`until` t (released then)]
`soil`	*name* `mudline` z
`soil_layer`	*soil* `sand top` d `bottom` d `phi` [`k`, `gamma`, `delta`, `K`, `Nq`, `cyclic`]; *soil* `clay top bottom su` [`su_bottom`, `gamma`, `eps50`, `J`, `residual`, `cyclic`]; *soil* `linear top bottom kh` [`kv`, `kq`] (depths below the mudline)
`pile`	*frame member* `soil` *soil* `D` D [`py`] [`tz`] [`qz`] [`linear`] (all three kinds unless some are listed)
Waves and hydrodynamics	
`waves`	`regular H` H `T` T, `component a` a `w` ω [`direction`] [`phase`]; `jonswap`|`pm Hs Tp` [`gamma`, `wmin`, `wmax`, `n` (200), `seed`, `direction`]; `waves depth` d (≤ 0: infinite); `waves ramp` t

Keyword	Parameters and meaning
`morison`	*name* `body` *b* `p1 p2 D` [`Cd` (1), `Ca` (1), `Cm` ($1 + C_a$), `n` strips (20), `current`]; *name* `member` *m* [`D`, `Cd`, `Ca`, `Cm`, `n` strips per element (2), `current`, `no_buoyancy`, `flooded` D_{in}]
`hydro`	*name* `file` *db* | `wamit` *root* `ulen` *L*, `bodies` b_1 ... [`state_space`] [`memory` *T* Δt] [`no_radiation`, `no_excitation`, `no_hydrostatics`, `no_buoyancy`] [`damping`|`drag` *mode value*] [`volume` *i V*] [`ref` *i x y z*]
`strip_member`	*name* `body` *b* `p1 p2 D` [`Cd` (1), `Ca` (1), `Cm` ($1+C_a$), `n` strips (20), `end1`, `end2`, `Ca_end` (1), `Cd_end` (1), `no_buoyancy`, `current`, `exact_kinematics`]: strip theory with nonlinear hydrostatics
`drag`	*name* `body` *b* [`point`] `Q` $Q_1..Q_6$ | `Cd` C_d `area` *A* [`axis` (default *z*)] [`rho`]
Mooring	
`mooring`	*name* `quasi_static`|`dynamic` `a` *body x y z* | `a` *point*, `b` likewise; `L` `d` `mass` (per length) `EA` | `type` *line type* `L` | `section` *line type L* n_{seg} [`section` ...] (several: dynamic lines); [`EA_curve` *file*, `BA`, `BA_slack`, `CB`, `Cdn` (1.2), `Cdt` (0.05), `Can` (1), `Cat` (0), `kb` ($3\cdot10^6$), `cb` ($3\cdot10^5$), `mu` (0), `mu_axial` (`mu`), `mu_static` (1), `ks` (`kb`), `cs` (`cb`), `v_eps` (0: stick-slip), `EI` (0), `clamp_a` [d_x d_y d_z], `clamp_b` [...], `nseg` (20), `seabed` *z* | `no_seabed`, `dt_line`, `settle` *T*, `no_static`, `current`, `current_profile` *file*, `exact_kinematics`, `outputs` `p`|`t`|`pt`, `break_time` *t*, `break_tension` *T*, `break_end` `a`|`b`]
`line_type`	*name* `d` `mass` `EA` | `EA_curve` *file* [line properties as for `mooring`]: properties for `type` and `section`
`point`	*name x y z* [`mass`, `volume`, `CdA`, `Ca`, `radius`, `kb`, `cb`, `mu`, `mu_static`, `ks`, `cs`, `v_eps`, `seabed` *z* | `no_seabed`]: free point of a mooring system (bridle node, clump weight, buoy)
`bathymetry`	`plane` z_0 [s_x s_y] | `grid` *file* | `fluid`: seabed of the dynamic lines and points (not of lines with `seabed` *z* and of quasi-static lines)
Power take-off	
`pto`	*name* `joint` *j* [`rotational`] [`k`, `c`, `q0`, `F0`, `coulomb` F_c, `v_eps` (10^{-3}), `F_max`, `P_max`, `c_loss`, `coulomb_loss`] [`stops` q_{min} q_{max} k_{stop} [c_{stop}]] [`latching` t_{hold} c_{latch} [k_{latch}]] [`declutching` t_{hold}]
`drive`	*name* `joint` *j* `sine` *a T* [`phase`, `ramp`, `q0`, `rotational`, `k` (10^7), `c` (10^4)]
`jforce`	*name* `joint` *j* `file` *table* (*t F*) [`scale` (1), `ramp`, `periodic`, `rotational`]
Wind, rotors and controllers	
`wind`	`steady` `U` *U* | `file` *file* [`z_ref`, `L_ref`, `direction`, `shear` α | `log` z_0, `veer`, `upflow`, `origin` *x y*, `ramp`]; `iec` `I`|`II`|`III` `A+`|`A`|`B`|`C` `z_hub` *z* `D` *D*; `event` `eog`|`edc±`|`ecd±`|`ews_v±`|`ews_h±` `t0` *t*; `turbulence` `kaimal` [`ntm`|`etm`|`ewm`] [`ny`, `nz` (15), `width`, `height` (1.1*D*), `T`, `dt` (0.05), `seed`, `coherence_vw`, `scale_hub`, `write` *file*]; `turbsim` *file* [`direction`, `origin`, `x0`]
`wind_probe`	*name x y z* [`body` *b*]: inflow channels at the point
`rotor`	*name* `hub` *x y z* `axis` a_x a_y a_z `R_hub` *r* `blade` *file* `airfoils` *n* $file_1$ ... [`blades` (3), `structure` *file*, `hub_mass`, `hub_inertia`, `precone`, `azimuth`, `ccw`, `nacelle` *body*, `rpm`, `pitch` | `pitch_file`, `gearbox`, `gen_inertia`, `shaft` *k c*, `efficiency`, `generator` `none`|`constant` *T*|`slip` *slope rpm*|`k_omega2` *K* [T_{max}], `no_skew`, `no_tip_loss`, `no_hub_loss`, `no_tangential`, `no_induction`, `no_drag_induction`, `tower` *base top* D_{base} D_{top} [`tower_body`], `pitch_joints` [`pitch_freq` (2), `pitch_zeta` (0.7)], `blade_elements` *n* [`blade_beta`, `ea_per_mass`, `gj_factor`]]; needs `wind`
`controller`	*name* `rotor` *r* [`tune` `P_rated` `rpm_rated` [`omega_n` (0.6), `zeta` (0.7), `write` *file*]] [`K`, `rpm_cutin`, `rpm_rgn2`, `rpm_rated_torque`, `P_rated`, `slip`, `T_max`, `T_rate`, `pitch_rgn3`, `rpm_ref`, `KP`, `KI`, `theta_K`, `theta_min`, `theta_max`, `theta_rate`, `corner`, `dt`, `no_pitch`, `no_torque`]

Keyword	Parameters and meaning
Coupling with REEF3D (Z 40 1)	
`fluid_body`	*body* [`n` *k*] [`own_mass`] [`added_mass` $a_1..a_6$] [`one_way` [`excitation`]]; *k*: REEF3D 6DOF body (from 0, default: order of the `fluid_body`/`fluid_member` lines)
`fluid_member`	*name* `member` *m* [`n` *k*] [`strips` n_s] [`order` 0|1|2 (2)] [`excitation`]
`fluid_kinematics`	[`order` 0|1|2 (2)]: Morison members and dynamic lines sample the fluid (FNPF, NHFLOW)
`fluid_coupling`	[`order` 0|1|2 (2)] [`no_predictor`] [`origin` *x y z* (0 0 0)] [`record` *file*]

A minimal coupled input for a heaving float driven by FNPF with a linear PTO in a vertical prismatic joint reads

```
time dt 0.005 end 30
output dt 0.05
body float mass 500 inertia 50 50 50 cog 0 0 -0.1
joint heave prismatic ground float point 0 0 -0.1 axis 0 0 1
pto gen joint heave c 2000
fluid_body float
```

where mass and inertia of `float` are replaced by those of the REEF3D body (X 21 – X 24) at its first step because `own_mass` is not given. The pose of a fluid body is CORAL's (`cog`, `rpy`), and a warning is printed when it differs from the REEF3D body.

5.12.8 Output

Coupled with REEF3D, rank 0 writes to the folder `REEF3D_CORAL/`: the time series (`output file`, default `coral_results.dat`), `modes.dat` and `mode_shapes.dat` if asked, and the journal of `fluid_coupling record`. The time series has a header line naming the columns and one line per output time: t, for each body the position of the centre of gravity, roll, pitch and yaw in degrees, the velocity and the angular velocity (global), followed by the channels of all loads, e.g. *pto*_`q`, `_qd`, `_F`, `_P`, `_E` of a PTO (with losses also `_Ploss`, `_Eloss`), the end forces, the maximum and end tensions of mooring lines (`_broken` with a failure, the node positions and segment tensions with `outputs`), the positions of the mooring points, the loads of strip members, the rotor channels (azimuth, speed, thrust, torque, power, C_p, C_t, tip speed ratio, pitch, blade root moments, drivetrain) and those of the controller, and the fluid loads `fluid_`*body*`_Fx` ... `_Mz` of the fluid bodies. The standalone executable writes the same files to the current folder and the VTP files to `coral_vtp/`; the coupled run writes no VTP files of the CORAL structure; with the LAGOON output of the 6DOF bodies (P 18), the mooring lines of CORAL are stored as polylines with their tension.

5.12.9 Key input commands

Z 40 1	REEF3D::CORAL structural solver, input `coral.txt` (FNPF, NHFLOW, CFD)
`time`, `output`	maximum CORAL step, generalized-α settings; time series
`body`, `joint`, `point_mass`	rigid bodies and joints
`section`, `member`, `superelement`	beam frames, Craig-Bampton superelements
`spring`, `linear6`, `force`	springs and tethers, 6×6 matrices, constant loads
`soil`, `soil_layer`, `pile`	pile foundations (p-y, t-z, Q-z)
`waves`, `morison`, `hydro`, `drag`	waves, Morison loads, hydrodynamic database, quadratic drag
`mooring`, `line_type`, `point`, `bathymetry`	quasi-static and dynamic mooring lines, line properties, free points, seabed
`strip_member`	strip theory with nonlinear hydrostatics for floaters
`pto`, `drive`, `jforce`	power take-off, prescribed joint motion and force
`wind`, `wind_probe`, `rho_air`	wind inflow
`rotor`, `controller`	wind turbine rotor with drivetrain, turbine controller
`static`, `modes`	static equilibrium, linear analysis
`fluid_body`, `fluid_member`	two-way and one-way coupling with the 6DOF bodies
`fluid_kinematics`	Morison members and dynamic lines sample the resolved fluid (FNPF, NHFLOW)
`fluid_coupling`	order of the fluid loads in time, predictor, origin of CORAL in the fluid grid, journal
X 10, X 11, X 20, X 60	6DOF bodies of the fluid solver: two-way needs X 10 1; X 11 0 for fixed one-way bodies; CFD bodies; X 60 1 for `fluid_member`
F 60, W 20 – W 22	still water level and gravity of REEF3D (checked against `coral.txt`)
P 48, I 45, G 46	checkpoints, restart and rebalancing include the CORAL state

6. Functions

REEF3D reads its input from the file `ctrl.txt`, which has to be located in the folder from which REEF3D is started (the simulation folder, see Chapter 8). Each command consists of a capital letter, a number and the corresponding values, separated by blanks, e.g.

```
A 10 6          // CFD module
N 41 30.0       // maximum simulated time
P 51 5.0 0.5    // wave gauge at x = 5.0 m, y = 0.5 m
```

In the command descriptions, the type of each value is given in front of it: for int an integer is expected, for double a floating-point number. The commands can be given in any order. If a command that allows only one instance is given more than once, the last occurrence is used.

How ctrl.txt is read

The file is read token by token, and the following rules follow from the parser:

- **Comments.** A token starting with / makes the parser skip the rest of the line. Comments are therefore written as // and can stand on their own line or after the values of a command. Text that is not preceded by / is not a comment: lower-case letters and digits are skipped, but a capital letter can be interpreted as the start of a command, which usually stops REEF3D (see below). Always start comments with //; then capital letters in comments are no problem.
- **Unknown commands are silently ignored.** A letter or number that does not correspond to an existing command (for example a typo such as `P 03 0.1` instead of `P 30 0.1`) is skipped without a warning, together with its values, and the default value of the intended command is used. Check the command numbers carefully, and check the terminal output and the log files if the simulation does not behave as expected.
- **Missing values stop REEF3D.** If a command is given with fewer values than required, the parser cannot continue and REEF3D stops with the message `!!! missing input parameter in ctrl.txt !!!`. The same happens when a capital letter appears in text that is not preceded by /. Exception: if the value is missing in the very last line of the file, the end of the file is reached and the command silently keeps its default value. A few commands (B 91, B 93, B 96) additionally stop with `wrong input error` if too many values are given.
- **Types.** Give integer values for int commands; a decimal number is not rejected, only its integer part is used.
- **Retired commands.** The commands listed below are no longer used. They are still accepted so that older `ctrl.txt` files can be run unchanged: the rest of the line is skipped, whatever values it contains, and REEF3D prints `ctrl.txt: X N is no longer used and is ignored` (for example `ctrl.txt: F 84 is no longer used and is ignored`). These commands have no entries in this chapter; remove them from `ctrl.txt` to silence the message.

A: 215, 216, 230, 241, 321, 322, 323, 340, 345, 353, 358, 363, 368, 410, 440, 523, 525, 541, 542, 543, 553, 556, 569, 594

B: 122, 234, 267

C: 5, 9

D: 31, 37

F: 31, 32, 33, 34, 36, 39, 84, 89, 350

G: 510, 511, 512, 520, 521, 522, 530

I: 240

N: 24, 60

P: 11, 39, 99, 120, 132, 312

Q: 20, 22, 23, 43, 101, 102, 120, 121, 122, 183, 200, 201, 202

S: 60, 66, 79, 83

T: 43

Y: 5, 50

Some input values are changed automatically after reading, either because an option requires another setting or because two options exclude each other. For example, the level set convection scheme F 35 is set to 0 when VOF is selected with F 80 $>$ 0, and the time scheme N 40 = 3 is changed to 4 when floating bodies are activated with X 10 $>$ 0 (without mesh refinement, G 1 = 0). Such automatic changes are noted in the descriptions of the affected commands.

Notation

The validity of each command for the four hydrodynamic modules is marked by the circles in front of the command, in the columns S (SFLOW), F (FNPF), N (NHFLOW) and C (CFD): a filled circle ● means that the command is used by that module, an empty circle ○ that it has no effect. For example, ● ○ ○ ● denotes a command that is valid for SFLOW and CFD only. The module itself is selected with A 10. The circles do not include REEF3D::SEASTATE (A 10 7): its commands A 700 – A 799 (Section 6.1.4) carry no validity circles, and the other commands that SEASTATE uses are listed there and mentioned in their descriptions.

Commands marked with an asterisk ($*$), e.g. **P 51** $*$, can be issued multiple times; each line creates one additional instance (for example one wave gauge per line).

The default value given for each command is used when the command is not specified in `ctrl.txt`.

6.1 A :: Hydrodynamic models

The commands in this section are used to select the hydrodynamic model that is to be activated for the simulation. This will determine the governing equations that will be used for the simulations. The different models available in REEF3D are:

- **S** : SFLOW: A depth-averaged shallow water equations model, that solves the non-hydrostatic pressure.
- **F** : FNPF: This model solves the Laplace equation.
- **N** : NHFLOW: This model solves the Navier-Stokes in three dimensions on a sigma grid, and is suitable for large scale wave modeling with a single-value free surface.
- **C** : CFD : This solves the 3D RANS equations with two-phase flow.

In addition, REEF3D::SEASTATE (A 10 7) is a phase-averaged spectral wave model that solves the wave action balance on the 2D horizontal grid. It runs stand-alone or coupled with SFLOW or NHFLOW. Its commands (A 700–A 799) are described in Section 6.1.4; the validity circles below do not include it.

Valid for:
S F N C

6.1.1 SFLOW

SFLOW can be coupled with the spectral wave model REEF3D::SEASTATE (A 750 1), which provides the wave forcing and receives the SFLOW water level and currents, see Section 6.1.4. The coupling is not available on curvilinear grids (G 50). With the semi-implicit free surface (A 226 1), the surfbeat mode (A 770 1) is not available.

The atmospheric forcing (wind stress and air pressure gradient, W 200) and the Coriolis force (W 211) act as sources in the SFLOW momentum equations, with the explicit and with the semi-implicit free surface (A 226 1), see Section 6.17. On curvilinear grids (G 50), they are evaluated at the physical cell centres.

SFLOW also runs on curvilinear and multi-block horizontal grids (G 50). Options that are not available there are noted in the entries below; the run then stops with a message.

● ● ● ●

A 10 int Hydrodynamic model

Selects the governing equations and the numerical grid used for the whole simulation. SFLOW and SEASTATE run on a 2D horizontal grid, FNPF and NHFLOW on a sigma grid, and CFD on the fixed Cartesian grid. The former potential flow solver PTF (A 10 4) has been removed: with A 10 4, the run stops with a message.

2 SFLOW

3 FNPF

5 NHFLOW

6 CFD

7 SEASTATE: phase-averaged spectral wave model (Section 6.1.4)

default: 6

● ○ ○ ○

A 209 int Number of smoothing sweeps for the bed

Number of passes of a 1-2-1 type filter (0.5 centre, 0.125 per neighbour) applied to the bed elevation at initialisation. Use 0 to keep the bed exactly as read.

default: 1

● ○ ○ ○

A 210 int Time scheme for SFLOW momentum equations

Any value other than 2 selects the 3rd-order scheme. The same choice is used for the 6DOF time integration in SFLOW. With the semi-implicit free surface (A 226 1), it selects the SSP Runge-Kutta predictor of the advection and diffusion terms. On multi-block grids (G 50), only option 3 is available: the run stops with a message. The mesh refinement (G 1) also needs option 3: with option 2, the refinement is switched off with a message.

2 2nd-order Runge-Kutta

3 3rd-order Runge-Kutta

default: 3

● ○ ○ ○

A 211 int Reconstruction scheme for SFLOW momentum equations

SFLOW uses HLL or HLLC fluxes (A 225). This command selects how the face values for the fluxes are reconstructed. The limiters fall back to first order next to dry cells. Options 0–3 also use first-order upwind convection for the turbulence quantities, option 4 WENO. With the semi-implicit free surface (A 226 1), it reconstructs the velocities upwind of the face discharges in the advection terms.

0 first-order

1 van Leer TVD

2 Superbee TVD

3 van Albada TVD

4 WENO reconstruction

default: 4

● ○ ○ ○

A 213 int Nonlinear weights of the WENO reconstruction

Only used with the WENO reconstruction (A 211 4) of the face values for the HLL fluxes. Option 1 uses the WENO-Z weights with $p = 2$ (Borges et al. 2008). Option 2 uses the TENO5 weights (Fu, Hu and Adams 2016): each of the three stencils is either kept with its ideal weight or discarded, depending on the cutoff A 222.

0 WENO-JS

1 WENO-Z

2 TENO5

default: 1

● ○ ○ ○

A 212 int Turn on diffusion for SFLOW velocities

Needed when a turbulence model (A 260) or the breaking viscosity (A 246 2) should act on the velocities. When A 212 is not given, it is set to 1 if a turbulence model is selected (A 260 > 0) and to 0 otherwise. With A 212 0 and a turbulence model, a warning is printed. With option 1 and the adaptive time step (N 48 1), the time step is also limited by the explicit diffusion criterion $\Delta t \leq 0.5/\left(\nu_{eff}\,(1/\Delta x^2 + 1/\Delta y^2)\right)$, with ν_{eff} including the eddy viscosity and the breaking viscosity A 250. Dry neighbour cells are treated with a zero normal gradient (free slip) in the diffusion terms. Option 2 is not available on curvilinear grids (G 50): the run stops with a message.

0 OFF

1 explicit

2 implicit

default: 1 with A 260 > 0, else 0

● ○ ○ ○

A 214 int Turn on convection for vertical SFLOW velocity w_s

Only used with the non-hydrostatic pressure schemes (A 220 1–3).

0 OFF

1 ON

default: 1

● ○ ○ ○

A 217 int Boundary conditions at walls

Ghost-cell treatment of the depth-averaged velocities at solid walls.

1 slip

2 no-slip

default: 2

● ○ ○ ○

A 218 int Turn on bed roughness

Adds Manning bed friction $\tau_b/\rho = g n^2 |u| u / h^{1/3}$ with $n = k_s^{1/6}/20$. The roughness k_s (m) is taken from B 50. When B 50 is given with A 218 0, a warning is printed. With the semi-implicit free surface (A 226 1), the bed friction is treated implicitly.

0 OFF

1 ON

default: 0

● ○ ○ ○

A 219 int Additional time step constraint

With option 2, the time step is also limited by the advective Courant number $\Delta x/|u|$, in addition to the default $\Delta x/(|u| + \sqrt{gh})$ criterion. Not used with the semi-implicit free surface (A 226 1), see A 228.

1 OFF

2 ON

default: 1

● ○ ○ ○

A 220 int Pressure scheme for SFLOW

Option 3 uses the dispersion parameter A 224 and adds an extra time step limit. Option 4 solves the Boussinesq equations in the FUNWAVE-TVD formulation with hydrostatic pressure. Its dispersive terms are explicit, so N 47 is reduced to 0.25 if set higher. Values of 5 and above fall back to option 1; on curvilinear grids (G 50), they stop the run. On multi-block grids (G 50) and with the semi-implicit free surface (A 226 1), only option 0 is available: the run stops with a message. Non-hydrostatic pressure is only solved in wet, deep (A 221) and non-breaking cells (A 242, A 246).

0 hydrostatic

1 linear non-hydrostatic pressure

2 quadratic non-hydrostatic pressure

3 quadratic non-hydrostatic pressure with improved dispersion (A 224)

4 Boussinesq equations (FUNWAVE-TVD formulation)

default: 2

● ○ ○ ○

A 221 int Non-hydrostatic pressure in very shallow regions

With option 0, cells within two or three cells of a dry cell, and cells with water depth ≤ 10·A 244, are excluded from the non-hydrostatic pressure solution. They are treated as hydrostatic. With option 1, all wet cells are included.

0 exclude near-dry cells

1 all wet cells

default: 1

● ○ ○ ○

A 222 double TENO5 cutoff C_T

Used with A 213 2. A stencil is discarded when its normalised smoothness measure falls below C_T. Smaller values discard fewer stencils, i.e. the scheme stays closer to the linear upwind-biased scheme. Dimensionless.

default: $1.0 \cdot 10^{-5}$

● ○ ○ ○

A 223 double Blending of current and previous free surface for the hydrostatic pressure gradient at patch boundaries
Weight θ of the current free surface: $\theta\eta^{n+1} + (1-\theta)\eta^n$. Range 0–1. Currently not used: the hydrostatic pressure gradient at the SFLOW patches it was meant for was never active and has been removed with the patchBC revision (October 2026).

default: 0.5

● ○ ○ ○

A 224 double Dispersion parameter α for A 220 3
Values below 1.0 are set to 1.0. A value of 1.0 gives the standard quadratic pressure. The default 1.159 is the optimised value. For $\alpha > 1$, the time step is also limited by the explicit dispersion correction.

default: 1.159

● ○ ○ ○

A 225 int Riemann solver for SFLOW
With option 2 (HLLC), the fluxes of the tangential momentum components (v on the x-faces, u on the y-faces, and the vertical velocity of the non-hydrostatic schemes on both) are the HLL mass flux of the face times the reconstructed tangential velocity upwind of it. This resolves shear layers and contact waves with less numerical diffusion. The mass and normal momentum fluxes stay HLL. On curvilinear grids (G 50), option 2 is ignored and HLL is used. Not used with the semi-implicit free surface (A 226 1).

1 HLL

2 HLLC

default: 1

● ○ ○ ○

A 226 int Free surface time integration for SFLOW
Option 0 is the explicit HLL scheme with the Runge-Kutta scheme A 210. Option 1 is a semi-implicit θ method (A 227): the water level follows from a linear system with the face discharges (solver N 10), and the advection and diffusion terms are integrated explicitly with an SSP Runge-Kutta predictor (A 210). The gravity wave speed then does not limit the time step (A 228), which suits long simulations such as rivers and tides. The bed friction (A 218) is treated implicitly. Option 1 requires the hydrostatic pressure (A 220 0; note that the default is A 220 2). It is not available with the mesh refinement (G 1), on curvilinear and multi-block grids (G 50), with

patch boundary conditions (B 411 – B 422), with 6DOF and direct forcing (X 10 > 0) or with the SEASTATE surfbeat mode (A 750 1 with A 770 1): the run stops with a message. With the SEASTATE coupling (A 750 1), the wave force acts in the predictor, and the Stokes mass source of the vortex force formulation (A 751 2) is added to the water level system. The atmospheric forcing (W 200) and the Coriolis force (W 211) act in the predictor. In thin water, the velocities of the predictor are desingularised (A 232). The volume and the open boundary discharges are written to `REEF3D_SFLOW_Log/REEF3D_SFLOW_SI.dat`.

0 explicit

1 semi-implicit θ method

default: 0

● ○ ○ ○

A 227 double Implicitness θ of the semi-implicit free surface

Used with A 226 1. Weight of the new time level in the free surface gradient of the momentum equations and in the face discharges of the continuity equation, e.g. $\theta\, q^{n+1} + (1-\theta)\, q^{n}$. Range 0.5–1; values closer to 1 are more dissipative. Values below 0.5 are unstable for large time steps; a warning is printed.

default: 0.55

● ○ ○ ○

A 228 double Maximum time step for the semi-implicit free surface

Used with A 226 1 and the adaptive time step (N 48 1). The time step follows from the Courant number N 47 of the flow velocity alone (without the gravity wave speed) and, with A 212 1, the explicit diffusion limit, and is capped at A 228 (s). The first time step is A 228. With 0, the cap is the explicit gravity wave limit, so the semi-implicit scheme runs with about the explicit time step; a message is printed.

default: 0.0 (cap at the explicit gravity wave limit)

● ○ ○ ○

A 229 int Wetting-drying iteration for the semi-implicit free surface

Used with A 226 1. With option 1, the water level system is solved with the nonlinear wet volume $\max(\eta + d, 0)$ of each cell (Casulli 2009): starting from all cells wet, the wet pattern is updated and the system solved again, at most 20 times, until the pattern no longer changes. This keeps the volume exact at the wet-dry front. With option 0, the linear system is solved once with all cells wet.

0 OFF

1 ON

default: 1

● ○ ○ ○

A 231 double Mass flux dissipation for transcritical flow

Used with A 226 1. Adds the implicit dissipation $K =$ A 231$\cdot(|u| + \sqrt{gh})\,\Delta x/2$ to the face discharges in the water level system, i.e. the Rusanov dissipation of the continuity equation. It is ramped up from zero at Froude number 0.5 to full strength at Froude number 1 and is meant for hydraulic jumps and transcritical flow. Dimensionless factor; use 0 to turn it off.

default: 0.0 (OFF)

● ○ ○ ○

A 232 double Depth for the desingularised velocities of the semi-implicit free surface

Used with A 226 1. In wet cells with water depth h below A 232 (m), the velocities of the predictor stages are computed from the discharge uh as $u = \sqrt{2}\,h\,(uh)/\sqrt{h^4 + h_v^4}$ with $h_v =$ A 232 (Kurganov and Petrova 2007), instead of $(uh)/h$. This is continuous at h_v and lets u go to zero with h, so that cells that are just wetting do not receive unphysically large velocities. Use 0 to turn it off.

default: 0.05

● ○ ○ ○

A 240 int Free surface scheme for SFLOW

Only option 1 is implemented. The free surface solver is created only for values ≥ 1, so do not set this to 0.

1 ON

default: 1

● ○ ○ ○

A 242 int Hydrostatic pressure for shallow areas

Marks shallow cells as "breaking", which switches off the non-hydrostatic pressure there. Option 1 marks cells within two cells of a dry cell and cells with water depth below 0.01 m. Option 2 marks cells within two cells of a dry cell and cells at steep bed slopes ($|\partial d/\partial x| > 2$, including three neighbours on each side). Option 3 is option 2 plus cells with water depth below 0.02 m.

0 OFF

1 near dry cells and very shallow water

2 near dry cells and steep bed slopes

3 as 2, plus very shallow water

default: 0

● ○ ○ ○

A 243 int Wetting and drying algorithm

Option 1 decides wet or dry from the water depth threshold A 244 alone. Option 2 also requires a dry cell to have a higher, wet neighbour before it can become wet again. Option 2 helps to maintain mass conservation, e.g. in open channel flow. With option 0, the water depth is clipped at A 244 but all cells stay wet.

0 OFF

1 depth criterion

2 depth criterion and wetting from higher wet neighbours

default: 2

● ○ ○ ○

A 244 double Absolute wetting-drying threshold

Water depth (m) below which a cell is dry. It is also the minimum depth used in divisions throughout SFLOW.

default: 0.001

● ○ ○ ○

A 246 int Breaking wave algorithm

Breaking cells are treated hydrostatically. A cell breaks when $\partial\eta/\partial t >$ A 247 $\cdot\sqrt{gh}$. Option 2 also flags cells with free-surface slope $|\partial\eta/\partial x| >$ A 355, and adds the breaking viscosity A 250 in the diffusion terms, which requires A 212 1 or 2.

0 OFF

1 ON

2 ON, with slope criterion and breaking viscosity

default: 1

● ○ ○ ○

A 247 double Breaking wave threshold parameter

Factor α in the onset criterion $\partial\eta/\partial t > \alpha\sqrt{gh}$ (A 246).

default: 0.6

● ○ ○ ○

A 248 int Breaking wave persistence algorithm

With option 0, the breaking flag is reset every time step. With option 1, a breaking cell keeps breaking while $\partial\eta/\partial t >$ A 249 $\cdot\sqrt{gh}$.

0 OFF

1 ON

default: 0

● ○ ○ ○

A 249 double Breaking wave persistence parameter

Factor β in the cessation criterion $\partial\eta/\partial t > \beta\sqrt{gh}$ used with A 248 1.

default: 0.3

● ○ ○ ○

A 250 double Breaking wave viscosity

Additional viscosity (m^2/s) added in breaking cells. Only used with A 246 2 and diffusion turned on (A 212 1 or 2).

default: 1.86

● ○ ○ ○

A 251 double Initial free surface slope in x-direction

Initialises the free surface as $\eta = -s(x - x_{min})$ with slope s (m/m). Use it, e.g., to start an open channel flow close to equilibrium. Inactive when not given.

default: not set (flat free surface)

● ○ ○ ○

A 260 int Turbulence model

The eddy viscosity acts on the velocities only when diffusion is turned on (A 212), which is the default when a turbulence model is selected. The bed shear stress in the k-ϵ, k-ω and parabolic models uses the same Manning friction as A 218, with the roughness k_s from B 50. The former alternative k-ω implementation (A 260 5) has been removed: with A 260 5, the run stops with a message. On curvilinear grids (G 50), only options 0, 3 and 4 are available: options 1 and 2 stop the run with a message. On these grids and with the mesh refinement (G 1), the eddy viscosity of options 3 and 4 is computed by each grid within the explicit diffusion (A 212 1).

0 OFF

1 k-ϵ model

2 k-ω model

3 Prandtl length scale model

4 Parabolic eddy-viscosity model

default: 0

● ○ ○ ○

A 261 double Length scale factor for the Prandtl length scale model
The mixing length is $l = 0.4 \cdot$A 261$\cdot h$. Used with A 260 3.

default: 0.267

● ○ ○ ○

A 262 double Factor for parabolic eddy-viscosity model
Coefficient α_t of the parabolic eddy-viscosity model (A 260 4).

default: 0.0667

● ○ ○ ○

A 263 double Factor for eddy-viscosity limiter
In the k-ϵ model (A 260 1), the eddy viscosity is limited to $\nu_t \leq$ A 263$\cdot k/S$, where S is the strain rate.

default: 10.0

● ○ ○ ○

A 264 double $c_{\epsilon\Gamma}$ coefficient
Coefficient of the bottom-shear production term in the ϵ-equation of the depth-averaged k-ϵ model (A 260 1). The bed source of the ω-equation of the k-ω model (A 260 2) is derived from the same equilibrium and also uses A 264. The default 3.6 follows Rastogi and Rodi (1978).

default: 3.6

6.1.2 FNPF

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A 309 int Number of smoothing sweeps for the bed
Number of passes of a 1-2-1 type filter applied to the bed elevation when the sigma grid is set up. Use 0 to keep the bed unfiltered.

default: 1

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A 310 int Time scheme for FNPF algorithm

Floating bodies in FNPF and the ice floes (A 380) are advanced within the same Runge-Kutta stages as the free surface.

3 3rd-order TVD Runge-Kutta

4 4th-order Runge-Kutta

default: 3

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A 311 int Spatial discretization for KFSFBC and DFSFBC

Discretization of the horizontal gradients in the free-surface boundary conditions. Options 4 and 5 are the same scheme. The wetting-and-drying version is selected by A 343.

0 OFF

1 CDS2

2 CDS4

3 WENO3

4 WENO5

5 WENO5

6 CDS6

default: 5

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A 312 int Spatial discretization for second-order derivatives in the sigma coordinate system

2 CDS2

3 CDS4

default: 2

○ ● ○ ○

A 313 int Spatial discretization for the bed boundary condition

Discretization of the bed gradients in the Neumann condition for the velocity potential at the bed.

2 CDS2

3 CDS4

default: 3

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A 314 int Linear or fully nonlinear free-surface boundary conditions

1 linear

2 fully nonlinear

default: 2

○ ● ○ ○

A 315 int Free-surface gradient treatment

Option 0 is the legacy scheme: $\partial\eta/\partial x$ is upwinded by the sign of the potential gradient and used everywhere. Option 1 upwinds the gradient in the kinematic boundary condition by $\partial H/\partial\eta_x$, using a symmetric stencil at zero speed. Option 2 is option 1 plus symmetric gradients for the geometry terms (sigma metrics, breaking, wind, dynamic boundary condition). Option 3 is option 1 with a Godunov gradient in the kinematic boundary condition: from the left- and right-biased gradients, the one that minimises or maximises the Hamiltonian is chosen, including the sonic point. This avoids the steepening of grid-scale dips and spikes of the free surface, e.g. next to moored bodies. Away from sonic points and for small slopes, it is identical to option 1.

0 legacy

1 upwinded kinematic FSBC

2 as 1, plus symmetric geometry gradients

3 as 1, with the Godunov gradient in the kinematic FSBC

default: 1

○ ● ○ ○

A 316 int WENO5 near the shoreline

Only used with wetting-and-drying (A 343 ≥ 1) and WENO5 (A 311 4, 5). Where the stencil is not fully wet, option 1 falls back to a first-order gradient from wet neighbours. Option 0 is the legacy zero gradient.

0 legacy (zero gradient)

1 first-order fallback from wet neighbours

default: 1

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A 317 int Bed metrics

Option 1 computes a symmetric bed gradient and includes the bed curvature terms. Option 0 is the legacy treatment: a left-biased bed gradient without wetting-drying, and zero bed curvature.

0 legacy

1 symmetric bed gradient and bed curvature

default: 1

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A 318 int Nonlinear weights of the WENO5 free-surface gradients

Only used with WENO5 (A 311 4, 5), with and without wetting-drying. Option 1 uses the WENO-Z weights with $p = 2$ (Borges et al. 2008). Option 2 uses the TENO5 weights (Fu, Hu and Adams 2016): each of the three stencils is either kept with its ideal weight or discarded, depending on the cutoff A 319.

0 WENO-JS

1 WENO-Z

2 TENO5

default: 0

○ ● ○ ○

A 319 double TENO5 cutoff C_T

Used with A 318 2. A stencil is discarded when its normalised smoothness measure falls below C_T. Smaller values discard fewer stencils, i.e. the scheme stays closer to the linear upwind-biased scheme. Dimensionless.

default: $1.0 \cdot 10^{-5}$

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A 324 int Picard iterations for the Laplace cross-derivative terms

The mixed derivatives $2\sigma_x \partial^2\Phi/\partial x\partial\sigma$ (and y) of the sigma-transformed Laplace equation are not in the 7-point matrix, but are evaluated on the right-hand side with the current potential. With 0, they are lagged by one Runge-Kutta stage; this error grows with the wave slope. With A 324 > 0, the assembly and solve are repeated with the new potential at most A 324 times, until the change is below A 325 times the change of the first solve. Each iteration costs one extra Laplace solve.

0 OFF (lagged by one stage)

> 0 maximum number of Picard iterations

default: 0

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A 325 double Picard convergence criterion for the Laplace equation

Used with A 324 > 0. The iterations stop when the maximum change of the potential is below A 325 times the maximum change of the first solve (relative, dimensionless).

default: $1.0 \cdot 10^{-3}$

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A 326 double Coastline relaxation reference time scale

Only used with the static coastline (A 343 1). If > 0, η and ϕ in the coastline relaxation zone are relaxed towards their running mean over this time scale (s) instead of towards zero.

default: 0.0 (relax to zero)

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A 327 int Vertical derivative of the potential at the free surface

Discretization of the one-sided derivative $\partial\Phi/\partial\sigma$ at the free-surface node, used for the vertical velocity in the free-surface boundary conditions. Option 1 computes the stencil weights on the actual positions of the sigma nodes (Fornberg 1988), which is exact for polynomials in σ for any vertical grid stretching. Option 0 is the legacy stencil in grid-index space, which loses accuracy on strongly stretched vertical grids (e.g. few layers in deep water with B 103 5). The stencil has 3 nodes for A 311 1, 5 for A 311 2, 4, 5 and 7 for A 311 6, and is reduced next to inactive nodes. Not used with A 311 3 (WENO3), which has its own stencil.

0 legacy

1 Fornberg weights on the sigma nodes

default: 1

○ ● ○ ○

A 328 int Implicit cross-derivative terms of the Laplace equation

With option 1, the mixed derivatives of the sigma-transformed Laplace equation (see A 324) are assembled into the matrix, giving a 15-point operator (9-point in 2D) instead of the 7-point matrix with the cross terms on the right-hand side. The system is solved with the BiCGStab of REEFMG, preconditioned by a V-cycle on the 7-point part. Only couplings to nodes that are not unknowns (free surface, walls, dry cells) stay on the right-hand side. Requires REEFMG (N 10 1); with other solvers the cross terms stay explicit. With the mesh refinement (G 1), the cross terms are also implicit on the refined patches.

0 explicit (right-hand side)

1 implicit

default: 0

○ ● ○ ○

A 329 int Order of velocity potential extrapolation for Dirichlet wave generation and active wave absorption

1 1st-order

2 2nd-order

default: 1

○ ● ○ ○

A 330 int Dynamic wetting-drying: minimum wet time
Minimum number of time steps a rewetted cell stays wet. Used with A 343 2, 3.

default: 10

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A 331 double Dynamic wetting-drying: rewetting hysteresis
Rewetting margin as a factor of A 344. Used with A 343 2, 3.

default: 1.0

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A 332 int Dynamic wetting-drying: width of the front viscosity band
Cells within this many cells of a dry cell get at least the damping viscosity A 346. Requires A 350 1. Use 0 to turn it off. Used with A 343 2, 3.

default: 2

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A 334 int Wetting-drying: redistribute clamp volume
Redistributes the water volume removed by the wet-dry clamp to wet neighbours. Used with A 343 2, 3, and with A 343 1 if A 337 1.

0 OFF

1 ON

default: 1

○ ● ○ ○

A 335 double Dynamic wetting-drying: maximum runup elevation
Maximum bed elevation above the still water level (m) that can become wet. Used with A 343 2. The default sentinel means no limit.

default: $1.0 \cdot 10^{20}$ (no limit)

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A 336 int Wetting-drying: one-sided free surface gradients at the front

Computes the free surface gradients at the wet-dry front from wet cells only. Used with A 343 2, 3, and with A 343 1 if A 337 1.

0 OFF

1 ON

default: 1

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A 337 int Static coastline: improved treatment

For A 343 1, option 1 averages the ϕ clamp over wet cells only and redistributes the clamp volume (A 334). It also uses wet-only free surface gradients with $\eta_{xx} = 0$ at the coast (A 336). The damping viscosity A 346 is ramped up towards the coastline inside the coastline relaxation zone (A 341, A 342) and applied in full in thin-film cells (water depth $<$ 10·A 344), where a drawdown has exposed the bed. Option 2 is the same, but applies A 346 only in the thin-film cells and leaves the coastline zone undamped (thin coastline). The viscosity requires A 350 1. Option 0 is the legacy treatment.

0 legacy

1 ON

2 ON, damping viscosity only in thin-film cells

default: 1

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A 338 int Remove isolated wet areas

At coastline initialisation, dries out wet areas that are not connected to the main water body.

0 OFF

1 ON

default: 1

○ ● ○ ○

A 339 int Coastline distance initialisation method

0 PDE reinitialisation (legacy)

1 fast sweeping

default: 1

○ ● ○ ○

A 341 double Coastline relaxation zone width as factor of the cell size

The width of the relaxation zone along the coastline is A 341 times the horizontal cell size. The value 0.0 means no coastline relaxation zone. A 342 takes priority when set.

default: 0.0

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A 342 double Coastline relaxation zone absolute width

Width of the coastline relaxation zone (m). When > 0, it overrides A 341.

default: 0.0

○ ● ○ ○

A 343 int Wetting-and-drying

Option 1 keeps the initial coastline fixed and damps η and ϕ in the coastline relaxation zone (A 341, A 342, A 347). Options 2 and 3 move the wet-dry front dynamically (A 330–A 336). Option 2 uses no coastline damping.

0 OFF

1 static coastline

2 dynamic runup and rundown

3 dynamic rundown only

default: 1

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A 344 double Wetting-drying threshold value

Water depth (m) below which a cell is dry.

default: 0.001

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A 346 double Damping viscosity at the coastline

Viscosity (m^2/s) applied within the coastline relaxation zone, modulated by the relaxation function. With dynamic wetting-drying, it is applied in the front band (A 332). It is applied through the breaking viscosity solve, so it requires A 350 1 or 4.

default: 1.86

○ ● ○ ○

A 347 int Coastline relaxation is used for

1 η and ϕ
2 η
3 ϕ
default: 1

○ ● ○ ○

A 348 int Numerical beach relaxation is used for

1 η and ϕ
2 η
3 ϕ
default: 1

○ ● ○ ○

A 349 double Coastline relaxation zone factor for full initialisation
In the first time step of a full-domain wave initialisation (I 30 1 or 2), the coastline relaxation zone width (A 341, A 342) is multiplied by this factor.

default: 10.0

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A 350 int Breaking wave algorithm
Breaking is modelled with an artificial viscosity applied to η and ϕ (A 357) in an implicit damping step. With options 1 and 2, the breaking cells are detected with A 351 and the viscosity is set with A 365. Options 3 and 4 match the dissipation of the damping step to an estimated breaking dissipation rate.

Option 4 tracks the wave crests along the x-direction (in 3D along every grid row) and estimates the crest speed C from their positions over time. A crest starts breaking when $B = u_s/C$ exceeds A 359, where u_s is the horizontal surface velocity at the crest (Barthelemy et al. 2018), and stops when the largest slope of its front face falls below A 360. The dissipation rate of a breaking crest is the larger of the bore dissipation (A 364) and, with A 367 1, the deep-water dissipation of Derakhti et al. (2018). The viscosity on the front face, from the crest to the trough ahead, is set such that the damping step dissipates this rate; it is limited to $|C|$ times the length of the front face. No breaking is detected inside the wave generation and numerical beach relaxation zones (B 96). The coastline viscosity A 346 and the filter A 352 are used as with option 1. With P 59 1, the tracked crests are written to `REEF3D_FNPF_Breaking_Log/REEF3D-FNPF-Breaking-Crests.dat`.

The mesh refinement (G 1) is only available with options 0–3: with option 4, the refinement is switched off with a message.

0 OFF

1 base version (Baquet)

2 Kennedy

3 Romero (energy dissipation based)

4 crest tracking with the onset criterion of Barthelemy et al. (2018)

default: 0

○ ● ○ ○

A 351 int Breaking wave detection

Shallow water: $\partial\eta/\partial t >$ A 354$\cdot\sqrt{gh}$. Deep water: free-surface slope $>$ A 355. Used by A 350 1 and 2 (A 350 2 uses fixed thresholds in shallow water, see A 354). Not used by A 350 3. With A 350 4, the onset follows from A 359; options 1 and 3 there only add the cells of the shallow-water criterion to the filter A 352, shoreward of the deepest breaking onset so far.

0 OFF

1 shallow water

2 deep water

3 shallow and deep water

default: 3

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A 352 int Additional filtering for viscosity-based breaking

Filters η and ϕ in breaking cells for additional robustness (iterations: A 361, A 362), for breaking in the regions below. Only used with A 350 1 and 4. With A 350 4, any value > 0 filters the front faces of the breaking crests (± 2 cells) and, with A 351 1 or 3, the cells of the shallow-water criterion (A 354) shoreward of the deepest breaking onset so far.

0 OFF

1 shallow water

2 deep water

3 shallow and deep water

default: 0

○ ● ○ ○

A 354 double Breaking wave threshold parameter shallow water

Factor α in $\partial\eta/\partial t > \alpha\sqrt{gh}$. Used by A 350 1 and, for the filter zone (A 352), by A 350 4. The Kennedy algorithm (A 350 2) uses the fixed onset factor 0.65 and stops breaking when $\partial\eta/\partial t < 0.15\sqrt{gh}$.

default: 0.6

● ● ○ ○

A 355 double Breaking wave slope threshold deep water α

Breaking starts where the free-surface slope $|\partial\eta/\partial x|$ exceeds this value (A 350 1, 2). SFLOW also uses it with A 246 2.

default: 1.25

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A 356 double Breaking wave slope cessation threshold β

Breaking stops where the slope drops below β·A 355. Only used by the Kennedy algorithm (A 350 2); the base version (A 350 1) has no cessation criterion, the breaking flags are reset every time step.

default: 0.1

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A 357 int Breaking wave viscosity is applied to

With A 350 3 and 4, the choice also enters the matching of the viscosity to the breaking dissipation rate.

1 η and ϕ

2 ϕ

3 η

default: 1

○ ● ○ ○

A 359 double Breaking onset threshold of $B = u_s/C$

Used with A 350 4. A tracked crest starts breaking when the ratio B of the horizontal surface velocity u_s at the crest to the crest speed C exceeds this value (Barthelemy et al. 2018). C is estimated from at least four samples of the crest position, so a crest travels about one cell before it can break. Dimensionless.

default: 0.85

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A 360 double Breaking termination angle

Used with A 350 4. A breaking crest stops breaking when the largest slope of its front face, between the crest and the trough ahead, falls below tan(A 360). It is not checked in the time step of the onset. Angle in degrees.

default: 8.0

○ ● ○ ○

A 361 int Breaking wave filter algorithm: outer iterations
Used with A 352 (A 350 1, 4).

default: 5

○ ● ○ ○

A 362 int Breaking wave filter algorithm: inner iterations
Used with A 352 (A 350 1, 4).

default: 2

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A 364 double Coefficient of the bore dissipation
Used with A 350 4. The dissipation rate per unit crest length and density of a breaking crest is at least A 364·$g C H^3/(4h_c)$, the energy loss of a moving hydraulic jump, where C is the crest speed, H the height from the crest to the trough ahead and h_c the water depth at the crest.

default: 1.0

○ ● ○ ○

A 365 double Breaking wave viscosity
With A 350 1, the viscosity (m^2/s) in the breaking cells and their direct neighbours; the next ring of cells receives half of it. With A 350 2, the mixing length coefficient δ_b of the eddy viscosity $\nu = \delta_b^2 h|\partial\eta/\partial t|$ (shallow-water breaking) or $\nu = \delta_b^2 h$ (deep-water breaking), limited to 10 m^2/s and spread over ±2 cells. Not used by A 350 3 and 4, which compute the viscosity from the breaking dissipation rate.

default: 1.86

○ ● ○ ○

A 366 double Breaking wave viscosity factor for η
Factor applied to the breaking viscosity when damping η (A 357 1, 3). With A 350 3 and 4, it also enters the matching of the viscosity to the breaking dissipation rate.

default: 1.0

○ ● ○ ○

A 367 int Deep-water breaking strength

Used with A 350 4. With option 1, the dissipation rate of a breaking crest is at least bC^5/g with the breaking strength $b = 0.034\,(\Gamma - 0.30)^{2.5}$ for $\Gamma > 0.30$ (Derakhti, Banner and Kirby 2018). $\Gamma = T_b\,\mathrm{d}B/\mathrm{d}t$ is taken at the onset (A 359), with $T_b = L/C$ and L the distance between the troughs on both sides of the crest. With option 0, only the bore dissipation (A 364) is used.

0 OFF

1 ON

default: 1

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A 370 int Wind forcing: drag coefficient formulation

Turns on wind forcing in the dynamic free-surface boundary condition and selects the drag coefficient $C_d(U_{10})$. Wind speed and direction are set with A 371, the type of forcing with A 373.

0 OFF

1 Garratt

2 Wu

3 Smith and Banke

4 Zijlema

default: 0

○ ● ○ ○

A 371 double U_{10}, double direction

Wind speed at 10 m height (m/s) and the direction it blows towards (deg), measured counter-clockwise from the positive x-axis.

default: 0.0 ; 0.0

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A 372 double x_s, double x_e, double y_s, double y_e Wind forcing area

Rectangular area (m) where the wind forcing is applied. When not given, the whole domain is forced.

default: not set (whole domain)

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A 373 int Type of wind forcing

Option 1 gives the wind setup from the surface stress $\tau = \rho_a C_d U_{10}^2$, spread over the reference depth A 377. Option 2 is deprecated and treated as 1. Option 3 applies a surface pressure in phase with the slope, $p_a = \beta \rho_a u_*^2 \partial\eta/\partial n$ with β = A 375. Option 4 applies $p_a = s\rho_a (U_{10} - c)^2 \partial\eta/\partial n$ only on faces steeper than the critical slope (A 376).

1 wind setup (surface stress)

2 deprecated, same as 1

3 Miles/Plant wave growth

4 modified Jeffreys sheltering

default: 1

○ ● ○ ○

A 374 int Wind decay

Applies a $\cos^2$ downwind decay of the pressure forcing over $[x_s, x_e]$. Requires A 372.

0 OFF

1 ON

default: 0

○ ● ○ ○

A 375 double Miles/Plant wind input coefficient β

Used with A 373 3. $\beta \approx 32$ reproduces the growth rate of Plant (1982).

default: 32.0

○ ● ○ ○

A 376 double s, double slope$_c$, double c Jeffreys parameters

Used with A 373 4: sheltering coefficient s, critical slope, and wave phase speed c (m/s). For $c \leq 0$, the celerity of the generated wave is used.

default: 0.5 ; 0.3 ; 0.0

○ ● ○ ○

A 377 double Reference depth for wind setup

Depth h_{ref} (m) over which the surface stress is spread for A 373 1. For values ≤ 0, the mean still water depth of the forcing area is used.

default: 0.0

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A 378 int Low-pass filter passes on the wind forcing slope

Number of 1-2-1 filter passes on the free-surface slope used for A 373 3, 4.

default: 4

FNPF sea ice

The commands A 380 – A 403 control the rigid ice floes of REEF3D::FNPF: the compliant-lid coupling to the free surface, the floe contacts and the floe breaking. See Section 5.10 for the model and the symbols used below.

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A 380 int Ice floes

Turns on the ice floes and reads them from the file `ice_floes.dat`. The file has one entry per line; # starts a comment, lengths are in m and angles in deg:

`density` ρ_i ice density (kg/m^3) for the entries that follow (default 917)

`rect` x_c y_c h L_x L_y **yaw** rectangular floe with thickness h

`poly` x_c y_c h D n **yaw** regular n-gon with circumscribed diameter D

`convex` h n x_1 y_1 `...` x_n y_n convex floe from world vertices (the convex hull is taken)

`field` x_s x_e y_s y_e h **conc** D_{min} D_{max} n_{min} n_{max} **seed** random field of convex floes with n_{min} to n_{max} vertices and diameters uniform in $[D_{min}, D_{max}]$, placed without overlap until the area concentration conc is reached

`obstacle` n x_1 y_1 `...` x_n y_n fixed convex polygon, contact only; the ice load on it is logged

`disp` dz **roll pitch** initial offset of the previous floe from its equilibrium

`vel` u v initial drift velocity (m/s) of the previous floe

In a 2D flume, only `rect` is meaningful: y_c and yaw are ignored, and L_y is the floe width the pressure acts on (use $L_y = 1$ for results per unit width).

0 OFF

1 ON

default: 0

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A 381 double Lid stiffness α

Stiffness of the compliant lid, $k = \alpha \rho_w g$. The floe follows the rigid-body kinematics up to $O(1/\alpha)$. Larger values stiffen the free surface under the floes and reduce the time step (A 403).

default: 20.0

○ ● ○ ○

A 382 double Lid damping ratio ζ

Damping of the lid as a fraction of the critical damping of each floe on the lid spring, $c = 2\zeta\sqrt{k\rho_i h}$.

default: 0.2

○ ● ○ ○

A 383 double Ice-water drag coefficient C_d

Skin drag between the floe bottom and the surface flow relative to the floe. Dimensionless. Use 0 to turn it off.

default: 0.0055

○ ● ○ ○

A 384 double μ, double e Contact friction and restitution

Coulomb friction coefficient μ and restitution coefficient e of the contacts between floes, obstacles and walls.

default: 0.3 ; 0.0

○ ● ○ ○

A 385 int Domain boundaries act as walls for the floes

With option 1, the floes are kept inside the domain by contacts with the domain boundaries.

0 OFF

1 ON

default: 1

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A 386 int Footprint supersampling

Number of sub-samples per cell direction used to compute the floe coverage φ of each cell.

default: 4

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A 387 double Floe print interval

Time interval (s) for the floe output. For values < 0, the floes are printed with the free-surface print settings (P 180 – P 182).

default: -1.0 (as the free-surface output)

○ ● ○ ○

A 388 int Contact solver iterations

Number of projected Gauss-Seidel iterations of the non-smooth contact solver.

default: 30

○ ● ○ ○

A 389 double Footprint edge taper half width

Half width δ of the smooth, area-preserving taper of the floe coverage across the floe edge, in multiples of the mean horizontal cell size. A sharp edge (0) gives a step of the size of the draft in the free surface, which the sigma grid does not tolerate once the floe moves.

default: 2.0

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A 390 int Ice breaking

Bit flags for the breaking mechanisms; add the values to combine them (e.g. 7 for all). Contact splitting is only active in 3D. Spalling requires crushing (A 398 > 0). The check interval is set with A 394, the minimum piece size with A 393.

0 OFF

1 flexural failure (A 391, A 395, A 397, A 399, A 400)

2 contact splitting (A 392, A 396, A 401)

4 spalling at crushing contacts (A 398, A 402)

default: 0

○ ● ○ ○

A 391 double Flexural strength σ_f

Flexural strength (Pa) for flexural failure (A 390 1). With A 399, it is the mean of the Weibull distribution for a floe of the reference area.

default: $0.5 \cdot 10^6$

○ ● ○ ○

A 392 double K_{IC}, double C Contact splitting parameters

Fracture toughness K_{IC} (Pa m$^{0.5}$) and calibration coefficient C of the splitting load $F_s = C\,K_{IC}\,h\sqrt{D}$ (A 390 2).

default: $1.0 \cdot 10^5$; 1.0

○ ● ○ ○

A 393 double Minimum floe size after breaking D_{min}

Pieces narrower than D_{min} (m) are not created. For values ≤ 0, four mean horizontal cell sizes are used.

default: -1.0 (four mean cell sizes)

○ ● ○ ○

A 394 int Breaking check interval

The breaking criteria are checked every A 394 time steps.

default: 1

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A 395 int n_{dir}, int n_{bin} Flexural cut lines

Number of cut directions and number of candidate cut positions (beam bins) per direction for flexural failure. In 2D, one direction is used.

default: 12 ; 64

○ ● ○ ○

A 396 double Contact force averaging time t_c

Averaging time (s) of the pair contact force that is compared with the splitting load (A 390 2).

default: 0.05

○ ● ○ ○

A 397 double E, double ν Elastic ice properties

Young's modulus E (Pa) and Poisson ratio ν for the flexural stress. For $E > 0$, the bending moment follows from a free-free beam on an elastic foundation with the plate stiffness $Eh^3/(12(1-\nu^2))$. For $E \leq 0$, rigid-floe statics are used.

default: $5.0 \cdot 10^9$; 0.3

○ ● ○ ○

A 398 double Ice crushing strength σ_c

Caps the contact force at $\sigma_c hw$ (Pa, with the contact width w). The crushing contacts are logged. Required for spalling (A 390 4). Use 0 to turn crushing off.

default: 0.0 (OFF)

○ ● ○ ○

A 399 double m, double A_{ref}, int seed Weibull flexural strength

Gives each floe a random flexural strength $\sigma_f (A_{ref}/A_f)^{1/m} (-\ln u)^{1/m} / \Gamma(1+1/m)$, with the Weibull modulus m, the reference area A_{ref} (m^2), the floe area A_f, u uniform in $(0,1]$ and σ_f from A 391. For $m \leq 0$, all floes have the strength σ_f, without scatter and size effect.

default: 0.0 ; 1.0 ; 1

○ ● ○ ○

A 400 int Maximum flexural cracks per floe and check

With values > 1, several local stress maxima above the strength can break a floe in one check.

default: 1

○ ● ○ ○

A 401 int Radial cracks per contact splitting event

Number of cracks, fanned around the contact normal, when a floe splits in contact (A 390 2).

default: 1

○ ● ○ ○

A 402 double L_{sp}, double C_{loc}, double T Spalling parameters

Used with A 390 4 and A 398 > 0. When the crushed depth reaches the spall length L_{sp} (m, ≤ 0: floe thickness), the tip of the floe breaks off. Only local chips are created: the chord of the cut must not exceed C_{loc} times the chip depth. Chips narrower than D_{min} (A 393) are removed as rubble, whose lid pressure fades out over the time T (s).

default: -1.0 ; 3.0 ; 1.0

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A 403 double Lid time step factor

The lid stiffens the free surface under the floes. The time step is limited to $\Delta t \leq$ A 403/ω_{lid}, with the frequency ω_{lid} of the fastest lid-stiffened surface mode on the grid (including the lid damping). A 403 1 is about 60 % of the RK3 stability limit. For values ≤ 0, the previous limit is used: the FNPF time step times $1/\sqrt{1+\alpha}$ (A 381). In addition, the floes do not move more than half a cell per time step.

default: 1.0

6.1.3 NHFLOW

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A 509 int Number of smoothing sweeps for the bed

Number of passes of a 1-2-1 type filter applied to the bed elevation at initialisation. Use 0 to keep the bed unfiltered.

default: 1

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A 510 int Time scheme for NHFLOW momentum equations

The same choice is used for the 6DOF time integration in NHFLOW. With the semi-implicit free surface (A 534 1), it selects the SSP-RK2 or SSP-RK3 predictor of the explicit terms.

2 2nd-order Runge-Kutta

3 3rd-order Runge-Kutta

default: 2

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A 511 int Flux scheme for NHFLOW momentum equations

Ignored with A 520 3, which uses its own flux scheme. With membranes (X 330) in a current (B 60 > 0), HLLC is recommended: HLL smears the shear at the membrane layer and overestimates the drag. On a curvilinear grid (G 50 > 0), only HLL is available; option 2 stops the run with a message.

1 HLL

2 HLLC

default: 1

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A 512 int Diffusion treatment

If used, implicit is recommended. If a turbulence model (A 560 > 0) or wave breaking (A 550 1) is used, the code sets A 512 2 automatically. The breaking wave viscosity (A 550 1, A 557) only acts with implicit diffusion. With explicit diffusion, the time step is additionally limited by the diffusion stability criterion, independent of N 47.

0 OFF

1 explicit

2 implicit

default: 0

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A 513 int Wall treatment for diffusion

Boundary values used for the explicit and implicit diffusion (A 512 1, 2) next to solid walls and direct-forcing bodies. Option 1 uses the cell value (zero gradient). Option 2 uses the ghost-cell values. With implicit diffusion, option 2 uses the mirrored cell value at direct-forcing bodies (tangential components symmetric, normal component antisymmetric), so that they act as slip walls like the domain walls.

1 zero gradient

2 ghost-cell values

default: 1

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A 514 int Reconstruction scheme for NHFLOW momentum equations

With option 4, WENO is used in the horizontal and a limited linear reconstruction in the vertical. Option 5 uses WENO in all three directions.

1 van Leer TVD

2 Superbee TVD

3 van Albada TVD

4 WENO reconstruction (horizontal)

5 WENO reconstruction (horizontal and vertical)

default: 4

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A 515 int Free surface treatment at the Dirichlet wave generation boundary

Sets the free surface in the inflow ghost cells and the water depth used for the prescribed discharges UH, VH, WH. Options 1, 2 and 4 copy the interior free surface into the ghost cells. Option 1 computes the discharges with the still water depth, option 2 with the actual water depth. Option 3 sets the theoretical wave elevation and water depth in the ghost cells and uses them for the discharges. Option 4 is mass-conservative: the wave kinematics are sampled at the still-water sigma levels and the wave discharge is taken over the still water depth, so that no net second-order (Stokes) mass flux enters the domain. A current is taken over the actual depth, and the ghost-cell velocity is $U = UH/D$ with the actual depth D. For plain Dirichlet generation without absorption (B 98 5 in the input, see B 98), option 4 also imposes the prescribed discharge exactly as the mass flux across the inflow face.

1 zero gradient, discharge from the still water depth

2 zero gradient, discharge from the actual water depth

3 prescribed wave elevation

4 mass-conservative

default: 1

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A 516 int Free surface treatment at forcing object

The free surface is also kept fixed inside the body, as with options 2 and 4, when X 10 4 or X 240 21 is used.

1 free moving

2 fixed inside

3 enforcing zero gradient, free inside

4 enforcing zero gradient, fixed inside

default: 1

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A 517 int Preserve tangential velocities for forcing

0 OFF

1 ON

default: 1

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A 518 int Bed boundary condition

1 slip

2 no-slip

default: 2

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A 519 int Bed roughness for velocities

Option 1 applies log-law bed friction in the bottom layer, with roughness k_s (m) from B 50, or from the sediment settings if sediment transport is active. Option 2 applies log-law friction at every wall face of a cell, on the velocity components tangential to that face: at the bed with the bed roughness, at solid side walls, solids above the cell and forcing bodies with k_s from B 57. The domain ends in x-direction are open boundaries and never walls; the lateral domain sides are walls. With a turbulence model (A 560), the side and top walls receive turbulence wall functions only with option 2; otherwise they are slip walls. Option 2 is not available with mesh refinement (G 1, the refinement is switched off), on a curvilinear or multi-block grid (G 50 $>$ 0) or with the semi-implicit free surface (A 534 1); the last two stop the run with a message. The bed friction of option 1 is point-implicit in time, so that it brings the velocity at most to rest in thin layers at wet-dry fronts. With A 534 1, the bed friction of option 1 is implicit in the column step.

0 OFF

1 bed friction

2 bed and wall friction

default: 0

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A 520 int Pressure scheme for NHFLOW

Option 3 uses its own flux scheme and ignores A 511. Membranes (X 330) require option 1 or 2. With a moving membrane (structure rigid or flexible), option 2 is changed to option 1 automatically, as well as with the sharp hull (X 15 8) of a floating body (X 10 $>$ 0). Sharp membranes (X 330, mobility sharp) require option 1. Mesh refinement (G 1) supports options 0–2. Option 3 is not available on a curvilinear grid (G 50 $>$ 0). The semi-implicit free surface (A 534 1) requires option 0.

0 hydrostatic

1 PJM for non-hydrostatic pressure

2 Pressure correction for non-hydrostatic pressure

3 alternative non-hydrostatic scheme (YL)

default: 2

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A 521 int Fluxes at direct-forcing interfaces

With option 0, velocities and fluxes are set to zero on cell faces between fluid cells and direct-forcing solids or floating bodies. At fixed direct-forcing solids (A 580 1, A 581–A 590), the continuity fluxes of the free surface equation through the faces of the solid cells are also set to zero, so that no water passes into or through the solid. Any other value turns this off.

0 ON

1 OFF

default: 0

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A 522 double Bed slope factor in the non-hydrostatic pressure

In the bottom layer, the vertical interpolation weight of the horizontal velocities in the pressure solution (A 520 1, 2) is reduced by $\max(1-\text{A } 522|\partial d/\partial x|, 0)$. This improves robustness over steep bed slopes.

default: 5.0

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A 524 int Horizontal gradients for the sigma transformation

Discretization of the free surface and bed gradients in the sigma coordinate metrics.

1 central differences

2 van Leer limited

3 van Leer limited, from the filtered free surface

default: 2

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A 526 double Direct forcing interface width factor

The smoothed direct forcing interface has a width of A 526 times the horizontal cell size. It applies to fixed solids (A 580 1, A 581–A 590) and floating bodies, including porous floating bodies (X 16).

default: 1.1

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A 527 int Nonlinear weights of the WENO reconstruction

Only used with the WENO reconstruction (A 514 4, 5), for all reconstructed quantities. The corresponding switches are A 213 for SFLOW and D 12 for CFD. Option 1 uses the WENO-Z weights with $p = 2$ (Borges et al. 2008). Option 2 uses the TENO5 weights (Fu, Hu and Adams 2016): each of the three stencils is either kept with its ideal weight or discarded, depending on the cutoff A 528.

0 WENO-JS

1 WENO-Z

2 TENO5

default: 0

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A 528 double TENO5 cutoff C_T

Used with A 527 2. A stencil is discarded when its normalised smoothness measure falls below C_T. Smaller values discard fewer stencils, i.e. the scheme stays closer to the linear upwind-biased scheme. Dimensionless.

default: $1.0 \cdot 10^{-5}$

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A 529 int Implicit sigma cross terms of the pressure Poisson equation

Treatment of the mixed derivatives $2\sigma_x \partial^2 P/\partial x \partial\sigma + 2\sigma_y \partial^2 P/\partial y \partial\sigma$ of the sigma-transformed Poisson equation for the non-hydrostatic pressure (A 520 1–3). With option 0, they are added to the right-hand side with the pressure of the last solve; this lagged treatment can diverge where the free surface is very steep, e.g. at a run-up front on a slope. With option 1, they are assembled into the matrix, giving a 15-point operator (9-point in 2D) instead of the 7-point one, as A 328 1 in FNPF. The system is solved with the BiCGStab of REEFMG, preconditioned by a V-cycle on the 7-point part. Only couplings to pressure nodes that are not unknowns (free surface, walls, dry or shallow columns, below the bed) stay on the right-hand side. Option 1 requires REEFMG (N 10 1) on a Cartesian grid (G 50 0) without mesh refinement (G 1) and membranes (X 330); otherwise a message is printed and the cross terms stay explicit.

0 explicit (right-hand side)

1 implicit

default: 0

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A 530 int Sigma cross terms of the pressure Poisson equation in porous cells

With option 0, the sigma cross terms (see A 529) are left out at pressure nodes next to porous cells (porosity < 1: VRANS structures, porous bodies, DEM). With option 1, they are also included there.

0 OFF

1 ON

default: 0

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A 531 double Froude number limiter

Limits the discharge in each wet cell to $|uh| \leq$ A 531$\cdot h\sqrt{gh}$. Next to dry cells, a factor 0.1 is applied when B 60 is used (NHFLOW: B 60 1, also in the shallow cells of A 545). In NHFLOW, the area is set by A 532. Also used by SFLOW.

default: 3.0

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A 532 int Froude number limiter area

Option 2 applies the limiter only in shallow cells near dry cells (see A 545).

0 OFF

1 all wet cells

2 shallow cells only

default: 1

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A 533 int Vertical velocity time step control

Adds the vertical Courant number $\Delta z/|w|$ and a source-term criterion to the time step control.

0 OFF

1 ON

default: 0

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A 534 int Hydrostatic semi-implicit free surface

Solves the hydrostatic NHFLOW equations with a semi-implicit free surface (θ method, Casulli and Cattani 1994), intended for long simulations such as tides, rivers and coastal flows. In each time step, the explicit terms are integrated with the SSP-RK2 or SSP-RK3 predictor of A 510, and the free surface elevation follows from one 2D linear system. The vertical diffusion, the vertical advection, the bed friction (A 519 1) and the pressure gradient are implicit in each water column. The time step is then limited only by the advective Courant number (N 47) and the explicit horizontal diffusion, not by the gravity wave speed; it is capped by A 536. The volume and the open boundary discharges are written to `REEF3D_NHFLOW_Log/REEF3D_NHFLOW_SI.dat`. Requires the hydrostatic pressure (A 520 0). Works on Cartesian, curvilinear and multi-block grids (G 50); on multi-block grids, G 57 adds a coarse space across the blocks to the preconditioner of the free surface solve. The run stops with a message for mesh refinement (G 1), A 519 2, VRANS (B 200), 6DOF bodies (X 10 $>$ 0, X 13 $>$ 0), membranes (X 330), FEM coupling (Z 30), the SEASTATE surfbeat model (A 750 1 with A 770 1) and patch boundary conditions (B 411–B 422). The SEASTATE coupling (A 750 1) is available: the wave force acts in the predictor and, with A 751 2, the Stokes mass source enters the free surface system. The meteorological forcing and the Coriolis force (W 200, W 211, see Section 6.17) are available. Wave breaking (A 550) is not: a message is printed and the run continues without breaking. Explicit diffusion (A 512 1) limits the time step; use A 512 2. See also A 546–A 548.

0 OFF (explicit)

1 ON

default: 0

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A 535 double Implicitness θ of the semi-implicit free surface

Used with A 534 1. The free surface gradient and the divergence of the discharges are weighted with θ at the new and $1-\theta$ at the old time level. $\theta = 1$ is fully implicit. Values below 0.5 are unstable for large time steps; a message is printed.

default: 0.55

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A 536 double Maximum time step of the semi-implicit free surface

Used with A 534 1. Upper limit (s) of the time step, which otherwise follows from the advective Courant number (N 47) and the explicit horizontal diffusion. The time step grows by at most 20 % per step. With 0.0, the time step stays at the explicit gravity wave limit $N47 \cdot \Delta x/\sqrt{g h_{max}}$ and a message is printed.

default: 0.0

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A 537 int Wetting and drying iteration of the semi-implicit free surface

Used with A 534 1. With option 1, the free surface system is solved with the nonlinear water depth $\max(\eta + d, 0)$ by a Newton-type iteration (Casulli 2009): the system is solved again with the new wet and dry pattern until it no longer changes, at most 20 times. With option 0, the system is solved once per time step.

0 OFF

1 ON

default: 1

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A 538 double Mass flux dissipation of the semi-implicit free surface

Used with A 534 1, for hydraulic jumps and transcritical flow. Adds the dissipative mass flux $-K\,\partial\eta/\partial n$ with $K =$ A 538$\cdot(|u| + \sqrt{gh})\,\Delta x/2$ to the discharges across the faces, implicit in the free surface system. It is switched on gradually between the Froude numbers 0.5 and 1. Use 0.0 to turn it off.

default: 0.0

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A 539 int Non-orthogonal correction of the free surface gradient on curvilinear grids

Used with A 534 1 on a curvilinear grid (G 50 $>$ 0). The orthogonal part of the free surface gradient across a face is implicit in the free surface system. With option 1, the non-orthogonal part is added explicitly (deferred correction) from the cell gradients of the free surface.

0 OFF

1 ON

default: 0

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A 540 int Wetting-drying scheme

Option 1 lets a dry cell become wet only from a higher, wet neighbour, and uses the wet-dry face flux treatment. Option 2 decides wet or dry from the depth threshold A 544 alone.

1 depth criterion and wetting from higher wet neighbours

2 depth criterion

default: 1

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A 544 double Absolute wetting-drying threshold

Water depth (m) below which a cell is dry. It is also the minimum depth used in divisions throughout NHFLOW.

default: 0.001

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A 545 double Shallow water criterion factor

Cells with water depth $\leq$ A 545·A 544, or within two (x-direction) or three (y-direction) cells of a dry cell, are "not deep". There, the non-hydrostatic pressure is not solved, and A 532 2 applies the Froude number limiter.

default: 10.0

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A 546 double Desingularisation depth of the semi-implicit free surface

Used with A 534 1. In wet cells with a water depth h below this value (m), the velocities of the momentum predictor are computed from the discharges as $u = \sqrt{2}\,h(uh)/\sqrt{h^4 + h_v^4}$ with h_v = A 546 (Kurganov and Petrova 2007) instead of $u = (uh)/h$, so that thin films at wet-dry fronts do not get large velocities. Use 0.0 to turn it off.

default: 0.05

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A 547 double c, double ν_0 Algebraic eddy viscosity of the semi-implicit free surface

Used with A 534 1 without a turbulence model (A 560 0). The eddy viscosity of every layer follows the parabolic profile $\nu_t = c\,\kappa\,(u_{*b} + u_{*s})\,z'(1 - z'/h) + \nu_0$, with the height z' of the layer centre above the bed, the water depth h and $\kappa = 0.4$. The bed friction velocity u_{*b} comes from the log law of the bottom layer with the bed roughness k_s, as the bed friction of A 519 1. The surface friction velocity u_{*s} comes from the wind stress of the meteorological forcing (W 200, see Section 6.17) and is zero without it. Dry columns get ν_0 (m^2/s). The vertical part is implicit in the column step, the horizontal part explicit in the predictor, where it limits the time step. A value $c > 0$ or $\nu_0 > 0$ switches the diffusion on, also with A 512 0. Use 0.0 for both to turn it off; with A 560 > 0, the turbulence model is used instead.

default: 0.0 ; 0.0

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A 548 int Still water depth at faces of the semi-implicit free surface

Used with A 534 1. Still water depth of a face between two wet cells. Option 0 uses the mean of the two cells; over a bed step, the face then has the depth of the deep side, and a thin film on the high side can drain and flood by turns. Option 1 uses the smaller depth of the two cells, i.e. the higher bed (Stelling and Duinmeijer 2003). Option 2 uses the smaller depth only where one of the two cells holds less than 10·A 544 of water (shoreline films), and otherwise the mean, so that the conveyance of channels is kept.

0 mean of the two cells

1 smaller depth (higher bed)

2 smaller depth at shoreline films only

default: 0

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A 550 int Breaking wave algorithm

Breaking cells are detected with A 551 and receive the viscosity A 557. The viscosity is added in the implicit diffusion, so A 550 1 sets A 512 2 automatically. Breaking is computed with the second- and third-order Runge-Kutta schemes (A 510 2, 3); with mesh refinement (G 1) only with A 510 2. It is not available with the semi-implicit free surface (A 534 1): a message is printed and the run continues without breaking.

0 OFF

1 base version (Baquet)

default: 0

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A 551 int Breaking wave detection

Shallow water: $\partial\eta/\partial t >$ A 554·$\sqrt{gh}$. Deep water: free-surface slope $>$ A 555.

0 OFF

1 shallow water

2 deep water

3 shallow and deep water

default: 3

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A 552 int Additional filtering for viscosity-based breaking

Filters the water depth in breaking cells for additional robustness, for breaking in:

0 OFF

1 shallow water

2 deep water

3 shallow and deep water

default: 0

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A 554 double Breaking wave threshold parameter shallow water
Factor α in $\partial\eta/\partial t > \alpha\sqrt{gh}$.

default: 0.6

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A 555 double Breaking wave slope threshold deep water
Breaking starts where the free-surface slope exceeds this value.

default: 1.25

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A 557 double Breaking wave viscosity
Artificial viscosity (m^2/s) in breaking cells and their direct neighbours (A 550 1). The next ring of cells receives half of this value.

default: 1.86

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A 559 int Explicit algebraic Reynolds stress model (EARSM)
Extends the k-ω model (A 560 2, 22) by the explicit algebraic Reynolds stress model of Wallin and Johansson (2000). The Reynolds stress anisotropy follows from the strain and rotation tensors, normalized with the turbulent time scale $\tau = \max(1/(C_\mu\omega), 6\sqrt{\nu/(C_\mu k\omega)})$. Its linear part gives the eddy viscosity $\nu_t = -\beta_1 k\tau/2$ of the implicit diffusion; the extra anisotropy is added as an explicit momentum source, lagged by one time step, and drives secondary currents like the QCR (A 563). The eddy viscosity limiters (A 564) and the stabilization (A 565) are not used; with A 560 22, the URANS filter (A 568) acts on ν_t and the extra stresses. With other turbulence models, A 559 is switched off with a message. A 559 1 switches the QCR (A 563) off.

0 OFF

1 ON

default: 0

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A 560 int Turbulence model

Any turbulence model switches on implicit diffusion (A 512 2). The URANS variants scale the eddy viscosity with a filter based on the local cell size and coefficient A 568. Other values stop the program with an error message. Mesh refinement (G 1) supports A 560 0, 1, 2, 21 and 22; with a turbulence model, not with subcycling (G 7 1), A 561 or A 563. The horizontal/vertical eddy viscosity split (A 561) and the quadratic constitutive relation (A 563) are available for the k-ϵ and k-ω models. The k-ω model can be extended by the EARSM (A 559).

0 OFF

1 k-ϵ model

2 k-ω model

21 k-ϵ model, URANS filtered

22 k-ω model, URANS filtered

31 LES, Smagorinsky

default: 0

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A 561 int Horizontal and vertical eddy viscosity (HLES)

Splits the eddy viscosity of the momentum equations into a vertical and a horizontal part, for a horizontal large eddy simulation of shallow flows (as Hinterberger, Fröhlich and Rodi 2007). The stresses with a vertical component (xz, yz, zz and the whole w equation) use the eddy viscosity ν_t of the k-ϵ or k-ω model. The horizontal stresses (xx, yy, xy) use ν_h, which includes the Smagorinsky viscosity $\nu_{sgs} = (C_s\Delta_h)^2|S_h|$ of the horizontal strain, with $\Delta_h = \sqrt{\Delta x \Delta y}$ and C_s, ν_b from A 562. With option 1, $\nu_h = \nu_t + \nu_{sgs} + \nu_b$, and the turbulence production of the RANS model only comes from the vertical shear. With option 2, $\nu_h = \nu_{sgs} + \nu_b$ and the RANS production is unchanged. With a ν_h well below the RANS value, the large horizontal eddies of the shear layers are resolved on the horizontal grid. ν_h is written to the VTU files as `eddyv_h`. Requires A 560 1, 2, 21 or 22; with other turbulence models, A 561 and A 563 are switched off with a message. Other values stop the run. Not available with mesh refinement (G 1).

0 OFF

1 $\nu_h = \nu_t + \nu_{sgs} + \nu_b$, RANS production from the vertical shear

2 $\nu_h = \nu_{sgs} + \nu_b$

default: 0

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A 562 double C_s, double ν_b Horizontal Smagorinsky constant and background viscosity for A 561

Smagorinsky constant C_s of the horizontal subgrid viscosity and background horizontal eddy viscosity ν_b (m^2/s), used with A 561 1, 2.

default: 0.15 ; 0.0

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A 563 double Coefficient of the quadratic constitutive relation (QCR)

With a value $C_{cr1} > 0$, the Reynolds stresses of the k-ϵ and k-ω models are extended by the quadratic constitutive relation of Spalart (2000), $\tau_{ij} \to \tau_{ij} - C_{cr1}(O_{ik}\tau_{jk} + O_{jk}\tau_{ik})$ with the normalized rotation tensor O_{ik}. The resulting anisotropy of the normal stresses drives secondary currents of the second kind, e.g. in corners and narrow channels. The extra stresses are explicit and lagged by one time step. Spalart (2000) uses $C_{cr1} = 0.3$. Use 0.0 to turn it off. Requires A 560 1, 2, 21 or 22; not available with mesh refinement (G 1). Switched off with a message when the EARSM is used (A 559 1).

default: 0.0

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A 564 int Eddy viscosity limiter

Option 1 is the realizability limiter $\nu_t \leq$ T 31·k/S. Option 2 is the SST/Bradshaw limiter $\nu_t \leq a_1 k/(F_2 S)$. With both options, the turbulence length scale is also limited to κh, with the local water depth h, by a lower bound on ϵ or ω.

0 OFF

1 strain-based (realizability)

2 SST/Bradshaw

default: 1

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A 565 int Turbulence model stabilization

Stabilization of the eddy viscosity in nearly potential flow regions, following Larsen and Fuhrman (2018). Uses T 42. Applies to the k-ω and k-ϵ models.

0 OFF

1 ON

default: 0

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A 566 int Add buoyancy term to k-equation

Buoyancy production from the air-water density difference (W 1, W 3) in the top cell layer. A sink is treated implicitly and k is kept non-negative.

0 OFF

1 ON

default: 0

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A 567 int Free surface boundary condition for turbulent dissipation

Sets ϵ or ω in the top layer from a mixing length based on T 37, for the k-ϵ and the k-ω model. Option 1 keeps the former model-specific length: T 37·h for k-ϵ and T 37 (m) for k-ω, with the local water depth h. Option 2 uses the length T 37 (m) and option 3 the length T 37·h for both models.

0 OFF

1 ON, model-specific length

2 ON, length T 37

3 ON, length T 37·h

default: 0

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A 568 double URANS filter coefficient

Coefficient of the URANS eddy viscosity filter for A 560 21 and 22.

default: 1.0

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A 570 int Wind forcing: drag coefficient formulation

Turns on wind forcing and selects the drag coefficient $C_d(U_{10})$. Wind speed and direction are set with A 571, the type of forcing with A 573. Not available on a curvilinear grid (G 50 $>$ 0). Not together with the meteorological forcing (W 200, see Section 6.17); the run stops with a message.

0 OFF

1 Garratt

2 Wu

3 Smith and Banke

4 Zijlema

default: 0

○ ○ ● ○

A 571 double U_{10}, double direction

Wind speed at 10 m height (m/s) and the direction it blows towards (deg), measured counter-clockwise from the positive x-axis. Also used for the windage of Lagrangian particles if L 41 is not given and A 570 $>$ 0.

default: 0.0 ; 0.0

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A 572 double x_s, double x_e, double y_s, double y_e Wind forcing area
Rectangular area (m) where the wind forcing is applied. When not given, the whole domain is forced.

default: not set (whole domain)

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A 573 int Type of wind forcing
Option 1 applies the surface shear stress $\tau = \rho_a C_d U_{10}^2$ in the top sigma layer. Option 2 is deprecated and treated as 1. Options 3 and 4 apply a surface pressure that acts in all layers: Miles/Plant (A 575) or modified Jeffreys (A 576). The slope used for options 3 and 4 is filtered with A 578 passes.

1 surface shear stress

2 deprecated, same as 1

3 Miles/Plant wave growth

4 modified Jeffreys sheltering

default: 1

○ ○ ● ○

A 574 int Wind decay
Applies a $\cos^2$ downwind decay of the wind forcing over $[x_s, x_e]$ for all forcing types. Requires A 572.

0 OFF

1 ON

default: 0

○ ○ ● ○

A 575 double Miles/Plant wind input coefficient β
Used with A 573 3: $p_a = \beta \rho_a u_*^2 \partial\eta/\partial n$.

default: 32.0

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A 576 double s, double slope$_c$, double c Jeffreys parameters
Used with A 573 4: sheltering coefficient s, critical slope, and wave phase speed c (m/s). For $c \leq 0$, the celerity of the generated wave is used.

default: 0.5 ; 0.3 ; 0.0

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A 578 int Low-pass filter passes on the wind forcing slope
Number of 1-2-1 filter passes on the free-surface slope used for A 573 3, 4.

default: 4

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A 580 int Solids of the grid file
Treatment of the solid entities (S) of the DIVEMesh grid file (`DIVEMesh_Grid/grid-geometry.dat`) in NHFLOW. With option 0, the solids raise the bed level. With option 1, they are represented as immersed solids by direct forcing, in the same way as A 581–A 590, and the bed level only follows the topography (T). Option 1 cannot be combined with mesh refinement (G 1). Immersed solids (A 580 1, A 581–A 590, A 599 1) are not available on a curvilinear grid (G 50 $>$ 0); the run stops with a message.

0 solids are part of the bed

1 immersed solids (direct forcing)

default: 0

○ ○ ● ○

A 581 * double x_s, double x_e, double y_s, double y_e, double z_s, double z_e Direct forcing rectangular box
Fixed solid box (m) represented by direct forcing. It can be given several times. The fixed solids A 581–A 590 can be combined with mesh refinement (G 1), where every grid level forces them itself (refinement around them with G 12), but not with subcycling (G 7 1) or A 516 2, 4; then the refinement is switched off with a message.

default: not set (no object)

○ ○ ● ○

A 583 * double x_c, double z_c, double y_s, double y_e, double r Direct forcing cylinder in y-direction
Horizontal cylinder along y with centre (x_c, z_c), extent y_s to y_e and radius r (m).

default: not set (no object)

○ ○ ● ○

A 584 * double x_c, double y_c, double z_s, double z_e, double r Direct forcing cylinder in z-direction
Vertical cylinder with centre (x_c, y_c), extent z_s to z_e and radius r (m). These cylinders are also used by the DLM method (A 599).

default: not set (no object)

○ ○ ● ○

A 585 * double x_s, double y_s, double z_s, double r_s, double x_e, double y_e, double z_e, double r_e Direct forcing cylinder with flexible orientation
Cylinder or cone between the start and end points (m), with end faces orthogonal to the axis and radii r_s and r_e (m) at the two ends. Suitable, e.g., for jacket members.

default: not set (no object)

○ ○ ● ○

A 586 * double x_c, double y_c, double z_c, double r Direct forcing sphere
Sphere with centre (x_c, y_c, z_c) and radius r (m).

default: not set (no object)

○ ○ ● ○

A 587 * double x_s, double x_e, double y_s, double y_e, double z_s, double z_e Direct forcing wedge object in x-direction
Wedge within the given bounding box (m), oriented in x-direction.

default: not set (no object)

○ ○ ● ○

A 588 * double x_s, double x_e, double y_s, double y_e, double z_s, double z_e Direct forcing wedge object in y-direction
Wedge within the given bounding box (m), oriented in y-direction.

default: not set (no object)

○ ○ ● ○

A 589 * double x_s, double x_e, double y_s, double y_e, double z_s, double z_e Direct forcing wedge object in z-direction
Wedge within the given bounding box (m), oriented in z-direction.

default: not set (no object)

○ ○ ● ○

A 590 int Direct forcing solid from STL file

Reads the file 'solid.stl' and represents the geometry by direct forcing. It can be scaled, translated and rotated with A 591–A 593.

IMPORTANT: the STL file needs to be in ASCII format! Most CAD programs export to a binary STL file. It is possible to use e.g. Paraview for STL conversion from binary to ASCII.

0 OFF

1 ON

default: 0

○ ○ ● ○

A 591 double s_x, double s_y, double s_z Scale STL geometry

Scaling factors in x-, y- and z-direction for the A 590 geometry. They are applied before translation and rotation.

default: 1.0 ; 1.0 ; 1.0

○ ○ ● ○

A 592 double dx, double dy, double dz Translate STL geometry

Shift (m) of the A 590 geometry, applied after scaling.

default: 0.0 ; 0.0 ; 0.0

○ ○ ● ○

A 593 double x_o, double y_o, double z_o, double ϕ, double θ, double ψ Rotate STL geometry

Rotates the A 590 geometry about the origin (x_o, y_o, z_o) (m) by the angles ϕ, θ, ψ (deg) about the x-, y- and z-axes. Applied after scaling and translation.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ● ○

A 599 int Use the distributed Lagrange multiplier method instead of direct forcing

With option 1, the direct forcing for solids and floating bodies is turned off and the DLM method is used for the A 584 cylinders. The DLM method only supports the A 584 cylinders; other solid objects (A 580, A 581–A 590) and floating bodies are not represented.

0 OFF (direct forcing)

1 ON (DLM)

default: 0

6.1.4 SEASTATE

REEF3D::SEASTATE (A 10 7) is a phase-averaged spectral wave model. It does not resolve the free surface. Instead, it computes the wave action density $N(\sigma,\theta) = E(\sigma,\theta)/\sigma$ in every cell of the 2D horizontal grid from the wave action balance equation

$$\frac{\partial N}{\partial t} + \frac{\partial (c_x N)}{\partial x} + \frac{\partial (c_y N)}{\partial y} + \frac{\partial (c_\sigma N)}{\partial \sigma} + \frac{\partial (c_\theta N)}{\partial \theta} = \frac{S}{\sigma},$$

with $c_x = c_g \cos\theta + U$ and $c_y = c_g \sin\theta + V$ from the linear dispersion relation, depth and current refraction c_θ (A 713) and the frequency shift c_σ (A 714). The physics follows SWAN: wind input, whitecapping (Komen or van der Westhuysen) and quadruplets (A 729–A 735, A 757; the wind field can come from the meteorological forcing W 200, see Section 6.17), depth-induced breaking, bottom friction, triads and vegetation (A 736, A 737, A 740–A 756), phase-decoupled diffraction (A 718, A 719), line obstacles, structures and reflecting coasts (A 722–A 726). The equation is solved with implicit finite volumes, first-order upwind in all four dimensions (second order in space with A 796 2), by Gauss-Seidel iterations with one sweep per directional quadrant. A coarse-grid correction (FAS, A 758) can speed up the convergence, and on several MPI ranks the sweeps are pipelined across the ranks (A 781), so that the result is that of one rank. The source terms are split into a production and an implicit damping part, so the scheme is stable and keeps $N \geq 0$ for any time step.

Spectral grid: the A 701 frequencies between A 702 f_{min} and f_{max} are spaced logarithmically, the A 703 directions cover the full circle (optionally refined in a sector with A 715, or automatically around the parametric boundary spectrum with A 716). With A 717, only the directions of a sector are computed. All directions of SEASTATE are Cartesian directions of propagation in degrees, counter-clockwise from the $+x$ axis: B 131, A 715, A 717, A 731 and the direction in the output.

Geographic grid: SEASTATE runs on the 2D grid of DIVEMesh, like SFLOW. The bed comes from the 2D grid (or from `seastate-bathy.dat` with A 790 1), and the still water level is F 60, which has to be given in stand-alone runs. Cells with a water depth of at least A 705 are active, land and dry cells absorb the waves. The domain sides take a boundary spectrum, zero gradient or nothing (A 711, A 712). Static mesh refinement is available in stand-alone runs with G 1 (see Section 6.7.1), using G 3, G 4 (even, at least 4), G 10, G 11, G 40 0 and the criteria A 762 and A 791–A 794. The patches are written to `REEF3D_SEASTATE_AMR`.

Time stepping: A 700 selects the mode. Nonstationary runs (A 700 1) advance the spectra with the time step A 706. Stationary runs (A 700 2) iterate a stationary solution in each time step of length A 706 until it has converged (A 707–A 709, A 738, A 739, A 799). Time-dependent forcing (A 711 3, A 730 2) then gives a sequence of stationary solutions, one per A 706 interval, as in the quasi-stationary mode of SWAN. A run ends at N 41 (s) or after N 45 time steps. For a single stationary solution set N 45 1. P 12 sets the terminal output interval.

Checkpoints and rebalance: stand-alone runs write checkpoints (P 48) with the spectra of all grids and restart from the latest one (I 45 1), also on another number of processes and another decomposition (Section 8.4). Stationary runs write them every P 48 iterations within a time step, and the restart continues the step at the iteration of the checkpoint. By default,

only the latest checkpoint of a run is kept (P 49). The measured work of the cells gives the load map (G 45) and, on several MPI ranks, the rebalance on a new decomposition (G 46, G 47). Not with the coupling (A 750 1) or the surfbeat model (A 770 1).

Modes of use:

- stand-alone (A 10 7);
- coupled with SFLOW (A 10 2) or NHFLOW (A 10 5) through A 750 1. The host model owns the grid, the time step and the time loop. It passes the water depth, the water level and the current to SEASTATE (A 753) and receives the wave force as radiation stress or vortex force (A 751). A wave step is computed every A 706 seconds of the host time;
- surfbeat: a wave-group action balance on one representative frequency (A 770–A 775);
- handover: 2D spectra at given points (A 760, A 761) for the irregular wave generation of FNPF or NHFLOW (B 85 11).

The commands of this subsection are used only by SEASTATE, in stand-alone runs and when SEASTATE runs inside SFLOW or NHFLOW (A 750 1). They therefore have no validity circles.

Other commands used by SEASTATE: B 85 (1 PM, 2 JONSWAP, 3 Torsethaugen), B 93 (H_s, T_p), B 88 (γ), B 130 (spreading: 0 none, 1 $\cos^s$, 2 Mitsuyasu), B 131 (main direction, deg), B 134 and B 135 for the parametric spectra of A 710 1, A 711 1 and the surfbeat boundary; F 60; G 1, G 3, G 4, G 10, G 11, G 40; G 45 – G 47 (load map and rebalance); I 45, P 48 (checkpoints); N 41, N 45; P 10 (VTP output for P 10 $>$ 0), P 12, P 15, P 181, P 182; M 10, M 40, M 41; W 200 – W 227, the meteorological forcing of SFLOW and NHFLOW (Section 6.17), which gives the wind field of A 730 2; W 207, W 213, W 214 (drag law of A 729 1).

Input files (in the simulation folder):

- `seastate-boundary.spc`: SWAN 2D spectral file. A 711 2 uses the first location and the first time, A 711 3 uses a time series of spectra at many locations.
- the files of the meteorological forcing (W 200, Section 6.17), e.g. `met-forcing.r3m`: wind field in space and time (A 730 2). The former wind file `seastate-wind.dat` is retired (see A 730).
- `seastate-bathy.dat`: bathymetry raster (A 790 1). Line 1 is a title, line 2 holds `nx ny`, the number of raster nodes, and line 3 holds `x0 y0 dx dy`, the first node and the spacing (m). Then follow `ny` lines of `nx` values: the bed level z (m, positive up, same vertical datum as F 60), row j at $y_0 + j\,dy$ from south to north. A $ starts a comment. Each cell takes the mean of the raster nodes inside it. If wet and dry nodes are mixed, the cell takes the majority group and its mean bed.
- `seastate-vegetation.dat`: number of vegetation stems per m^2 (A 756 1), in the same raster format as the bathymetry file.

- `seastate-obstacle-`*n*`.dat`: transmission and reflection per frequency for obstacle *n* (A 725 *n* 2).

Output:

- `REEF3D_SEASTATE_VTP`: 2D VTP files `REEF3D-SEASTATE-<n>-<rank>.vtp` with a `.pvtp` index. They hold the point data Hs, Tm01 ($2\pi m_0/m_1$), Tm-10 ($2\pi m_{-1}/m_0$), Tp (discrete peak period), dir (mean direction, deg), spread (directional spread after Kuik, deg), depth and wet. Additional fields: U10x and U10y with a wind field (A 730 2); roller, Dr and Dw with the surfbeat roller; and Fx, Fy (wave force) and MStx, MSty (Stokes transport) in coupled runs. Nodal values average the active neighbour cells only. Stand-alone runs print with P 181 (iterations) or P 182 (s). With the defaults of P 181 and P 182, only the initial state is written. Coupled runs print after every wave step (surfbeat: P 181 or P 182) with their own file numbering.
- `REEF3D_SEASTATE_Log`: `REEF3D_SEASTATE_integral.dat` with one line per time step: step, time, total energy, maximum and mean H_s, minimum action density and solver iterations. Stationary runs also write the convergence history `REEF3D_SEASTATE_convergence.dat`, the surfbeat model writes its boundary files, and G 45 or G 46 the load map `REEF3D_SEASTATE_loadmap.dat`.
- `REEF3D_SEASTATE_AMR`: VTR files of the refined patches with a `.vtm` index per print, the patch log and the convergence history (G 1).
- `REEF3D_SEASTATE_Spectra`: handover spectra `spectrum-file-2d_P<n>.dat` and the matching wave generation settings `seastate-wavegen_P<n>.txt` (A 760).

The run stops with an input error message when a SEASTATE command has an invalid value or invalid combination, and with an emergency stop when H_s becomes NaN.

A 700 int Mode

In stationary mode, the time derivative is dropped and every time step (A 706) is iterated to a stationary solution. In stationary runs with the deep-water physics (A 732 1 or 2), A 706 is used as a pseudo time step that damps the iterations. The surfbeat model (A 770 1) is nonstationary.

1 nonstationary

2 stationary

default: 1

A 701 int Number of frequencies

Number of frequency bins, spaced logarithmically between A 702 f_{min} and f_{max}: $f_l = f_{min} r^l$ with $r = (f_{max}/f_{min})^{1/(n-1)}$. At least 2. With the surfbeat model (A 770 1), this grid is only used for the boundary spectrum.

default: 32

A 702 double f_{min}, double f_{max} Frequency range

Lowest and highest frequency of the spectral grid (Hz), $0 < f_{min} < f_{max}$. Energy outside this range leaves the grid (outflow only at both ends). When the iterations of a stationary run with the deep-water physics end without convergence while cells have their peak within a factor 1.5 of f_{max}, a note is printed: the young wind sea at upwind shores may not settle, and a higher f_{max} (e.g. 2 Hz) lets it converge.

default: 0.04 ; 1.0

A 703 int Number of directions

Number of direction bins over the full circle, $\Delta\theta = 360^\circ/n$, with the first direction at 0° ($+x$). At least 4. The surfbeat model (A 770 1) uses the same directions on its single representative frequency.

default: 36

A 704 int Tile size of the spectral storage

The action density is stored block-sparse in square tiles of $n \times n$ cells. Memory is only allocated for tiles that contain at least one active cell, so land and dry areas need no spectral memory. This affects memory and performance, not the results. At least 1.

default: 16

A 705 double Minimum water depth

Minimum water depth (m) of an active (wet) cell. Shallower cells are treated as land and absorb the waves. In coupled runs (A 750 1), the cells of the initial active area dry and re-wet with the water depth of the host model, and a cell that dries loses its spectrum. Also the dry threshold of the bathymetry raster (A 790 1): a raster node is dry above F 60 − A 705.

default: 0.05

A 706 double Time step

Time step (s) of stand-alone runs. The implicit scheme has no stability limit. In stationary runs (A 700 2), this is the interval between the stationary solutions and, with the deep-water physics (A 732 1 or 2), the pseudo time step of the iterations. In coupled runs (A 750 1), it is the coupling interval: a wave step is computed every A 706 seconds of host time and the wave force is held constant in between. With the surfbeat model, a wave step is computed every host time step.

default: 60.0

A 707 int Iterations per time step

Maximum number of iterations (four quadrant sweeps each) per time step. With 0, the code chooses: 50 for stationary runs; for nonstationary runs 1, or 2 when running on several MPI ranks (M 10 $>$ 1) or with the surfbeat model, so that the halo values lagged after the first sweep are corrected. When the sweeps are pipelined across the ranks (A 781 1), nonstationary runs without surfbeat and without FAS (A 758) make 1 iteration again, as on one rank. With the FAS correction (A 758), nonstationary runs need at least 2 iterations. Stationary runs stop earlier when the convergence criterion (A 708, A 709) is met.

default: 0

A 708 double Stationary convergence criterion

Stationary iterations stop when, in every active cell, the change of H_s in the last iteration is below this fraction of H_s (relative to at least 1% of the largest H_s of the domain). With A 799 1, the test uses the estimated distance of H_s to the solution instead of the change. With mesh refinement, the leaf cells of all levels are tested.

default: 0.001

A 709 double Percentage of converged cells

Percentage of the active cells that must meet A 708 (SWAN `npnts`, e.g. 99.5) for a stationary solution to count as converged. With 100, all cells must meet it. With mesh refinement, the cells are weighted with their area. Range $(0, 100]$.

default: 100.0

A 710 int Initial spectrum

With option 1, every active cell starts with the same parametric spectrum $E(\sigma,\theta) = S(\sigma)D(\sigma,\theta)$, built from B 85 (1 PM, 2 JONSWAP, 3 Torsethaugen), B 93 (H_s, T_p), B 88 (JONSWAP γ), B 130 (0 unidirectional, 1 $\cos^s$, 2 Mitsuyasu), B 131 (main direction, deg) and B 134, B 135. The spreading is normalised on the discrete directions, so the energy of S is kept. A spreading narrower than one bin puts all energy into the bin of the main direction. The surfbeat model starts from rest (A 710 0).

0 zero (calm)

1 parametric spectrum

default: 0

A 711 int Boundary spectrum

Spectrum that enters through the sides with A 712 1.

0 none

1 parametric spectrum, as for A 710 1 (B 85, B 93, B 88, B 130, B 131, B 134, B 135)

2 SWAN 2D spectral file `seastate-boundary.spc`, first location and first time, the same on all sides. Frequencies AFREQ or RFREQ (Hz), directions CDIR (Cartesian) or NDIR (nautical, converted with $\theta = 270^\circ - \theta_{naut}$), QUANT VaDens or EnDens (divided by $\rho g = 1025 \cdot 9.81$), FACTOR, ZERO and NODATA blocks. The spectrum is interpolated onto the spectral grid (linear in f, zero outside the frequency range of the file, periodic in θ).

3 time series of spectra at many locations from `seastate-boundary.spc`. The format is the same as for option 2, with LOCATIONS in model coordinates (m; LONLAT is not accepted) and a TIME block (`YYYYMMDD.HHMMSS`) per record; a file without TIME is one record at time 0. Every boundary cell of the sides with A 712 1 takes the inverse-distance weighted spectrum of the two nearest locations, linear in time and constant beyond the first and last record. Model time 0 is A 780, or the first time in the file.

default: 0

A 712 int x_-, int x_+, int y_-, int y_+ Boundary conditions of the domain sides

Type of each of the four domain sides. Waves leave through every side. The surfbeat model needs $x_- = 1$. With mesh refinement, no patches are placed within A 794 level-0 cells of the sides with option 1 or 2.

0 open, no incoming waves

1 boundary spectrum (A 711)

2 zero gradient: the incoming spectrum is the spectrum of the boundary cell itself

default: 1 ; 0 ; 0 ; 0

A 713 int Refraction

Turning of the wave directions (c_θ) caused by depth gradients and current shear. The gradients are central differences and one-sided at the domain edge. Next to land or dry cells, refraction and the frequency shift are switched off, as in SWAN.

0 OFF

1 ON

default: 1

A 714 int Frequency shift

Shift of the relative frequency (c_σ) caused by a water depth changing in time (coupled runs) or along a current, and by current gradients. Not used by the surfbeat model, which has a single frequency.

0 OFF

1 ON

default: 1

A 715 double θ_1, double θ_2, int k Fine direction sector

Each of the A 703 directions whose centre lies in the sector from θ_1 counter-clockwise to θ_2 (deg, Cartesian) is divided into k bins. Use this for a finer directional resolution around the main wave direction, e.g. for long swell. $k = 1$ switches it off. Not available with the surfbeat model or with the DIA quadruplets (A 732 1 or 2 with A 733 1), and not together with A 716, which sets the sector itself. With the spectral sparsity (A 795), at most 250 directions per quadrant are allowed.

default: 0.0 ; 0.0 ; 1

A 716 double w, double margin Automatic fine direction sector

Sets the fine direction sector of A 715 from the parametric boundary spectrum (A 711 1, with B 85, B 93, B 130, B 131, B 134). The sector covers the directions around B 131 in which the directional distribution, weighted with the energy of the frequencies, is at least 10^{-4} of its maximum (B 130 0: B 131 only), widened by the margin (deg) on both sides for refraction and diffraction. Each of the A 703 directions in the sector is divided into $k = \lceil \Delta\theta / w \rceil$ bins, so that the bins there are at most w (deg) wide. $w = 0$ switches it off. Not together with A 715, and not with the surfbeat model or the DIA quadruplets (A 732 1 or 2 with A 733 1).

default: 0.0 ; 15.0

A 717 double θ_1, double θ_2 Computational direction sector

Only the directions whose centres lie in the sector from θ_1 counter-clockwise to θ_2 (deg, Cartesian) are computed, as SWAN CGRID SECTOR. The other directions hold no energy, the part of the boundary and initial spectra outside the sector is not imposed, and energy refracted beyond the ends of the sector leaves the spectral grid. With uniform directions (no A 715 or A 716), only the directions of the sector are stored, which saves memory and time. The command switches the sector on. The gap outside the sector must not lie inside one 90° quadrant (possible only for sectors wider than 270°). Not available with the surfbeat model.

default: not set (full circle)

A 718 int Diffraction

Phase-decoupled diffraction after Holthuijsen et al. (2003), as SWAN DIFFRACTION. The diffraction parameter is $\delta = \nabla \cdot (c_g / k \nabla \sqrt{E}) / (k c_g \sqrt{E})$ and $C_a = \sqrt{1 + \delta}$ (1 where $\delta \leq -1$). C_a scales the geographic propagation velocities and turns the wave directions towards the shadow zones behind obstacles and islands. The energy is smoothed before the derivative is taken (A 719). The parameter is updated from the latest spectra before every iteration. With mesh refinement, it is computed on every level. Not available with the surfbeat model.

0 OFF

1 from the total energy of the cell with its mean k and c_g, one C_a for all frequencies (as SWAN)

2 per frequency

default: 0

A 719 int Smoothing steps for diffraction

Number of smoothing steps applied to the energy before the diffraction parameter is computed (A 718). Each step is $E - 0.2\sum(E - E_{nb})$ over the active neighbours (SWAN `smpar` 0.2, `smnum`). With 0, the number is $0.4(L/\Delta x)^2$ (between 1 and 400), from the mean wavelength L of the domain and the smallest cell size Δx, so that the smoothing covers a fixed fraction of the wavelength. With mesh refinement, a given value n is used as $n \cdot 4^{level}$ on the patches.

default: 0

A 720 int Prescribed current

Current for stand-alone runs. Option 1 sets a current U in x-direction that varies linearly in x (A 721) and $V = 0$. It acts through the propagation velocities, the refraction (A 713) and the frequency shift (A 714). In coupled runs, it is replaced by the current of the host model with A 753 2.

0 none

1 U linear in x (A 721)

default: 0

A 721 double U_s, double U_e, double x_s, double x_e Linear current

Current of A 720 1: $U = U_s$ (m/s) for $x \le x_s$, $U = U_e$ for $x \ge x_e$, and linear in between (x in m). With $x_e \le x_s$, U jumps from U_s to U_e at x_s.

default: 0.0 ; 0.0 ; 0.0 ; 0.0

A 722 * double x_s, double y_s, double x_e, double y_e, double K_t, double K_r, double z_c Line obstacle

Line obstacle from (x_s, y_s) to (x_e, y_e) (m), e.g. a breakwater, as SWAN OBSTACLE. A cell face is blocked when the obstacle crosses the line between the two cell centres. K_t and K_r are wave-height coefficients. Through a blocked face, K_t^2 of the energy flux is transmitted. K_r^2 of the flux leaving the cell is reflected into the direction mirrored at the obstacle line (specular, or diffuse with A 726). The rest, $1 - K_t^2 - K_r^2$, is dissipated. With $K_t < 0$, the transmission over a dam or low-crested breakwater follows Goda (SWAN DAM GODA, α 2.6, β 0.15). It uses the freeboard $R_c = z_c -$ water level, where z_c is the crest level (m, same vertical datum as F 60 and the bed), and the larger H_s of the two cells. z_c is only used for Goda and for A 725 type 1. Conditions: $K_t \le 1$, $0 \le K_r \le 1$, and $K_t^2 + K_r^2 \le 1$. Obstacles are numbered from 1 in the order of input (A 725, A 726). Obstacles also bound the smoothing and the derivatives of the diffraction. Not available with the surfbeat model. It can be given several times.

default: not set (no obstacle)

A 723 double K_r, double p Reflecting coasts

Every face between an active cell and a land or dry cell inside the domain reflects K_r^2 of the energy flux leaving the active cell (K_r is a wave-height coefficient in $[0,1]$, 0 switches it off). With $p = 0$, the reflection is specular at the local coastline direction (A 724). With $p > 0$, it is diffuse, spread with $\cos^p$ around the specular direction (as SWAN RDIFF). Faces with an obstacle (A 722) keep the obstacle. The coast faces are rebuilt when cells dry or re-wet. Not available with the surfbeat model.

default: 0.0 ; 0.0

A 724 int Cells for the coastline direction

Reflecting coasts (A 723) are oriented along the coastline. Its direction is the principal direction of the midpoints of the land faces within this many cells, averaged over the coastal cells within the same distance. This way a straight coast on the staircase of cells reflects at its own direction. With 0, each face reflects at its own orientation (staircase). Range 0 to 3.

default: 3

A 725 * double n, double type, double a, double b, double c Structure

Replaces the constant transmission of the A 722 obstacle number n (counted from 1) with a structure formula. It can be given several times.

1 rubble-mound breakwater after d'Angremond et al. (1996), as SWAN DAM DANGREMOND: a is the seaward slope (deg, between 0 and 90) and b the crest width (m). The crest level is z_c and the reflection K_r of A 722. K_t follows from the freeboard, H_s and the peak period of the cell with the larger H_s.

2 K_t and K_r per frequency from the file `seastate-obstacle-`n`.dat`, with lines f (Hz), K_t, K_r (wave-height coefficients; $ starts a comment). Linear in f, constant beyond the ends (as SWAN TRANS1D, with a reflection per frequency).

3 porous breakwater through the water column after Madsen (1974), with the resistance of van Gent (1995): a is the width (m), b the porosity (between 0 and 1) and c the stone diameter D_{50} (m). K_t and K_r are computed per frequency from the incident spectrum, for normal incidence.

default: not set

A 726 * double n, double p Diffuse reflection at an obstacle

The reflection at the A 722 obstacle number n (counted from 1) is spread with $\cos^p$ around the specular direction (as SWAN RDIFF). With $p = 0$, it stays specular. It can be given several times.

default: not set (specular)

A 729 int Drag of the wind input

Drag coefficient $C_d(U_{10})$ of the friction velocity $u_* = \sqrt{C_d}\,U_{10}$ of the wind input (A 730, A 732, A 734). Option 0 uses the drag of Wu (1982), as SWAN. Option 1 uses the drag law W 207 of the meteorological forcing with its cap W 213 and floor W 214 (Section 6.17), e.g. a drag that saturates in hurricane winds, as for the storm surge of a coupled run. With W 207 0, the law follows W 200 (Garratt for W 200 0–2, Garratt capped at W 213 for the vortex and the blend W 200 3, 4). Other values stop the run with a message.

0 Wu (1982)

1 drag law of the meteorological forcing (W 207)

default: 0

A 730 int Wind

Wind input. Needs the deep-water physics (A 732 1 or 2), which also sets the exponential wind growth. Not available with the surfbeat model. With option 2, the wind field is that of the meteorological forcing W 200 (uniform W 201, met file `met-forcing.r3m`, vortex or blend; Section 6.17), the same as for the storm surge of a coupled SFLOW or NHFLOW run. It is evaluated at the cell centres and is not ramped (W 204 ramps only the surface stress of the hydrodynamics). Model time 0 is then that of the met forcing (see A 780). Option 2 with W 200 0 stops the run with a message. The former wind file `seastate-wind.dat` is retired; if it is present, the message says to convert it with `python3 tools/met_forcing.py seastate seastate-wind.dat -o met-forcing.r3m` and to set W 200 2. The wind components are added to the VTP output as U10x and U10y.

0 none

1 uniform and constant (A 731)

2 field in space and time of the meteorological forcing (W 200)

default: 0

A 731 double U_{10}, double direction Uniform wind

Wind speed at 10 m height (m/s) and the direction the wind blows to (deg, counter-clockwise from $+x$) for A 730 1.

default: 0.0 ; 0.0

A 732 int Deep-water physics

Option 1 switches on the third-generation deep-water physics of Komen et al. (1984), as SWAN GEN3 KOMEN. This includes the whitecapping (SWAN coefficients $C_{ds} = 2.36 \cdot 10^{-5}$, $\tilde{s}^2_{PM} = 3.02 \cdot 10^{-3}$), the exponential wind input (with A 730, drag of Wu 1982 or of A 729), the quadruplets (A 733) and the action density limiter (A 735). Option 2 follows van der Westhuysen et al. (2007), as SWAN GEN3 WESTH: the exponential wind input of Yan (1987) and the saturation-based whitecapping. Where the saturation $B_k = c_g k^3 E(\sigma)$ exceeds the threshold B_r, the dissipation is proportional

to $C_{ds2}(B_k/B_r)^{p/2}$; the two parts are blended with $f_{br} = \frac{1}{2}(1+\tanh(10(\sqrt{B_k/B_r}-1)))$, and below the threshold a non-breaking dissipation of Komen type ($C_{ds} = 3\cdot 10^{-5}$) acts. C_{ds2} and B_r are set with A 757. The quadruplets, the limiter and the linear growth are the same as with option 1. In stationary runs with option 1 or 2, A 706 is the pseudo time step of the iterations. Not available with the surfbeat model.

0 OFF

1 Komen

2 van der Westhuysen

default: 0

A 733 int Quadruplets

Nonlinear four-wave interactions with A 732 1 or 2. They use the Discrete Interaction Approximation (Hasselmann et al. 1985, SWAN coefficients $\lambda = 0.25$, $C = 3\cdot 10^7$, with shallow-water scaling). The DIA needs uniform directions (no A 715 or A 716).

0 OFF

1 DIA

default: 1

A 734 double Linear wind growth

Coefficient of the linear wind growth after Cavaleri and Malanotte-Rizzoli (1981) with the filter of Tolman (1992), as SWAN. It starts the wave growth from calm conditions. Used with wind (A 730). 0 switches it off.

default: 1.5e-3

A 735 double Action density limiter

With A 732 1 or 2, the change of N per iteration is limited to $\gamma\alpha_{PM}/(2\sigma k^3 c_g)$ with $\alpha_{PM} = 0.0081$ and γ given here (SWAN LIMITER, Ris 1997). This is only active while the wind sea is far from its balance (large time steps, early stationary iterations). Breaking, friction and triads are not limited. With source iterations (A 738), the change from the spectrum of the cell before its first solve is limited. 0 switches it off.

default: 0.1

A 736 double cutfr, double urcrit, double urslim Triad parameters

Parameters of the LTA triads (A 744 1), named as in SWAN. cutfr is the highest sum frequency relative to σ_{01} (> 1). urcrit is the critical Ursell number of the biphase (> 0). urslim is the lowest Ursell number at which triads act (≥ 0). SWAN versions 40.x–41.31 used urcrit 0.2 and urslim 0.01.

default: 2.5 ; 0.63 ; 0.1

A 737 int Maximum energy with Battjes-Janssen breaking

With Battjes-Janssen breaking (A 740 1), the total energy (variance m_0) of a cell is limited to $(\gamma d)^2/4$, i.e. $H_{m0} = 4\sqrt{m_0} \leq 2\gamma d$ with γ of A 741. The limit is applied before the source terms are evaluated, as in SWAN.

0 OFF

1 ON

default: 1

A 738 int Source iterations per cell

Stationary runs (A 700 2) only. Maximum number of times that each cell is solved per sweep, with the source terms evaluated again from the spectrum just solved. The source iterations stop when the estimated distance of the cell energy to its balance is below A 739. If the iterations do not contract, the cell keeps the spectrum of its first solve. This can reduce the number of iterations with strong source terms. At least 1.

default: 1

A 739 double Tolerance of the source iterations

Relative tolerance of the source iterations of A 738: the estimated distance of the energy of the cell to its balance, as a fraction of the cell energy.

default: 1.0e-4

A 740 int Depth-induced breaking

Battjes and Janssen (1978) as in SWAN: the fraction of breaking waves Q_b follows from $H_{rms} = \sqrt{8E_{tot}}$ and $H_m = \gamma d$, the dissipation rate is $\alpha/\pi\, Q_b \sigma_{01}\, H_m^2/H_{rms}^2$ (Newton-linearised as in SWAN). Roelvink (1993) is the breaking model of the surfbeat mode (A 770 1, the only breaking allowed there): $Q_b = 1 - \exp(-(H/(\gamma h))^n)$ and $D/E = 2\alpha f_{rep} Q_b H/h$ with $H = \sqrt{8E}$ of the wave group. α and γ are set with A 741 for both models, n with A 746. With option 1, A 737 also limits the total energy of a cell.

0 off

1 Battjes-Janssen

2 Roelvink (surfbeat, A 770 1 only)

default: 0

A 741 double α, double γ Breaking coefficients

Proportionality coefficient α of the dissipation and breaker index γ (maximum wave height to water depth ratio) for A 740 1 and A 740 2. Both must be positive.

default: 1.0 0.73

A 742 int Bottom friction

JONSWAP bottom friction as in SWAN: dissipation rate $C_b/g^2\,(\sigma/\sinh(kd))^2$ with the coefficient C_b from A 743.

0 off

1 JONSWAP

default: 0

A 743 double JONSWAP bottom friction coefficient

Coefficient C_b in m^2/s^3 for A 742 1.

default: 0.038

A 744 int Triad wave-wave interactions

Lumped Triad Approximation (LTA) of Eldeberky (1996) as the original SWAN LTA, with the interaction coefficient of Madsen and Sørensen (1993). The cut-off frequency and the Ursell number limits are set with A 736, the proportionality coefficient with A 745. Not available with the surfbeat mode (A 770 1).

0 off

1 LTA

default: 0

A 745 double Triad proportionality coefficient

Proportionality coefficient α_{EB} of the LTA triads (A 744 1).

default: 0.05

A 746 double Roelvink breaking exponent

Exponent n in $Q_b = 1 - \exp(-(H/(\gamma h))^n)$ of the Roelvink breaking of the surfbeat mode (A 740 2, A 770 1).

default: 10.0

A 747 double Surfbeat: maximum wave height to water depth ratio

In the surfbeat mode (A 770 1), the energy of the wave groups is limited after every step so that $H = \sqrt{8E}$ does not exceed A 747 times the water depth. Use 0 to switch the limit off.

default: 2.0

A 748 int Surfbeat: roller model

Roller energy balance of the surfbeat mode (A 770 1), per direction, transported with the phase speed plus the current. Its source is the Roelvink breaking dissipation (A 740 2, the roller is off without it), its dissipation is $2g\beta R/c$ (Reniers et al. 2004) with β from A 749. With the coupling (A 750 1), the roller adds to the radiation stress and to the wave mass flux, and with the vortex force (A 751 2) the roller dissipation replaces the breaking dissipation in the wave force.

0 off

1 on

default: 1

A 749 double Surfbeat: roller front slope

Slope β of the roller front in the roller dissipation (A 748 1). Must be positive.

default: 0.1

A 750 int Coupling with SFLOW or NHFLOW

Runs SEASTATE inside an SFLOW (A 10 2) or NHFLOW (A 10 5) simulation on the same 2D grid. The host model owns the time loop: a wave step is made every A 706 seconds (the coupling interval; the wave forcing is held constant in between), with the surfbeat mode (A 770 1) every host time step. SEASTATE returns the wave force (A 751, ramped up over A 752) to the momentum equations of the host (in NHFLOW uniform over the depth) and the host returns water level and currents (A 753). Not with mesh refinement (G 1) or the bathymetry raster (A 790 1). With the semi-implicit free surface (SFLOW A 226 1, NHFLOW A 534 1), the coupling is available without the surfbeat model (A 770 1). The coupling is not available yet on curvilinear and multi-block grids (G 50) and in the checkpoints (P 48 writes none, I 45 1 stops). The SEASTATE VTP files then also contain the wave force (`Fx`, `Fy`) and the wave mass flux (`MStx`, `MSty`).

0 off

1 coupling with the host model

default: 0

A 751 int Wave forcing of the host model

Formulation of the wave force acting on SFLOW or NHFLOW with A 750 1. Option 1: the radiation stress gradient $F_i = -\partial S_{ij}/\partial x_j$ (Longuet-Higgins and Stewart 1964) including the roller; the host velocity is the mass transport velocity, and $U - M/h$ is passed back to SEASTATE as the current. Option 2: the depth-averaged vortex force (McWilliams et al. 2004, Smith 2006), consisting of the dissipation force (breaking, friction and nonlinear transfers; the wind input only when the host model gets the full wind stress, W 200 or NHFLOW A 570, so that the momentum taken up by the waves is subtracted from the mean flow), a bottom-slope term, the wave pressure gradient and the vortex force $J = \chi(M_y, -M_x)$; the host velocity is Eulerian and $-\nabla \cdot M$ of the Stokes transport M is added to the continuity equation of the host.

1 radiation stress

2 vortex force

default: 1

A 752 double Ramp-up time of the wave forcing

The wave force (and the Stokes mass source with A 751 2) applied to the host model is multiplied by $\min(1, t/T)$ with T = A 752 in s. In the surfbeat mode the same ramp is applied to the wave groups and the long waves entering at the x- side. Use 0 for no ramp.

default: 0.0

A 753 int Feedback from the host model to SEASTATE

Host fields passed to SEASTATE before each wave step with A 750 1. With the water level, the cells of the initial wave domain dry and re-wet with the water depth of the host (threshold A 705). The current is the depth-averaged velocity of SFLOW or NHFLOW (Eulerian, see A 751).

0 none

1 water level

2 water level and currents

default: 2

A 754 int Vegetation

Wave damping by vegetation as in SWAN, one vegetation layer with the properties of A 755 (or the stem density of A 756). Option 1: Dalrymple et al. (1984) and Mendez and Losada (2004) in the form of Suzuki et al. (2011) (SWAN VEGETATION 1), from the total energy, mean wave number and frequency. Option 2: per frequency, Jacobsen et al. (2019) (SWAN option 2). Not available with the surfbeat mode (A 770 1).

0 off

1 Dalrymple / Suzuki et al. (2011)

2 per frequency, Jacobsen et al. (2019)

default: 0

A 755 double h_v, double b_v, double N_v, double C_D Vegetation properties

Vegetation height h_v in m (limited to the water depth), stem diameter b_v in m, number of stems per m^2 N_v and drag coefficient C_D for A 754. Height, diameter and number of stems must be positive when A 754 is used (N_v is ignored with A 756 1).

default: 0.0 0.0 0.0 1.0

A 756 int Vegetation density field

With option 1, the number of stems per m^2 is read per cell from the raster file `seastate-vegetation.dat` (same format as `seastate-bathy.dat`, see A 790; the cell value is the mean of the raster nodes inside the cell, also on refined patches) instead of the uniform value of A 755.

0 uniform density from A 755

1 density field from `seastate-vegetation.dat`

default: 0

A 757 double C_{ds2}, double B_r Westhuysen whitecapping coefficients

Coefficient C_{ds2} (SWAN cds2) and saturation threshold B_r (SWAN br) of the saturation-based whitecapping of van der Westhuysen et al. (2007) (A 732 2). Both must be positive.

default: 5.0e-5 ; 1.75e-3

A 758 int n, int m FAS coarse-grid correction

Full approximation scheme (FAS): every n iterations (nonstationary: iterations of the time step), the residuals of the fine cells are restricted to a coarse grid, which is iterated m times, and the coarse change is passed back to the fine cells (multiplicative, between 1/4 and 4). This carries the information across the domain faster and reduces the number of iterations. With mesh refinement (G 1), level 0 is the coarse grid of the patches of all levels. Without mesh refinement, a V-cycle runs over up to six coarse grids of 2×2 cells each: $m/2$ iterations before and $m/2$ after the next coarser grid, $4m$ on the coarsest. Nonstationary runs correct within the step but not after its last iteration, so they need at least 2 iterations per step (A 707). Without mesh refinement, not available with diffraction (A 718), vegetation (A 754) or the coupling (A 750 1). Not available with the surfbeat model. $n = 0$ switches it off.

default: 0 ; 4

A 760 * double x, double y Handover point for the phase-resolved wave generation

At the point (x, y) in m, the 2D spectrum of the cell containing the point (on the finest grid with mesh refinement) is written after every step to `REEF3D_SEASTATE_Spectra/spectrum-file-2d_` in the format of the 2D spectrum input of the wave generation (B 85 11, `spectrum-file-2d.dat`): directions in rad, counter-clockwise from $+x$, frequencies ω in rad/s and $S(\omega, \theta)$ scaled so that every wave component carries the energy of its SEASTATE bin. Only the direction sector A 761 around the mean direction is written. A companion file `seastate-wavegen_P<n>.txt` lists the location, water depth, H_s of the full spectrum and of the sector, T_p, the mean direction and the ctrl.txt commands for the wave generation. Can be given several times. Not with the surfbeat mode (A 770 1).

default: not set (no handover)

A 761 double Half-width of the handover direction sector

Half-width in degrees of the direction sector written at the handover points (A 760), around the grid direction closest to the mean wave direction. Must be positive.

default: 90.0

A 762 int Mesh refinement: water width criterion

With mesh refinement (G 1), cells are refined where the water is narrower than this many cells of their level, e.g. in channels, fjords and harbour entrances. The water width of a level-0 cell is the shortest of the wet runs through it along x, y and the two diagonals; runs that reach the edge of the domain do not count. Use 0 to switch the criterion off.

default: 0

A 770 int Surfbeat mode

Wave-group action balance of XBeach type (Roelvink 1993, Reniers et al. 2004) on one representative frequency (A 771) and the A 703 directions, nonstationary (A 700 1) with the time step A 706 or, with the coupling (A 750 1), the time step of the host model. At the x- side (which must have A 712 1), a random-phase wave train is built from the boundary spectrum (A 711 1 or 2, on the spectral grid A 701 – A 702) and gives the time series of the wave-group energy (A 772, A 773) and of the incoming bound long wave (A 774), which is imposed in the host model with an absorbing-generating boundary. Breaking with A 740 2, roller A 748. Requires A 710 0, no wind or deep-water physics (A 730 0, A 732 0), no triads (A 744 0), no handover (A 760), no mesh refinement, fine direction sector, diffraction, obstacles or vegetation. Not covered by the checkpoints (P 48 writes none, I 45 1 stops).

0 off

1 on

default: 0

A 771 double Surfbeat: representative period

Representative period T_{rep} in s of the surfbeat mode (A 770 1). With 0, the mean period $T_{m-1,0}$ of the boundary spectrum is used.

default: 0.0

A 772 double Surfbeat: record length of the boundary time series

Length T_{rec} in s of the wave train generated at the x- side (A 770 1); the frequency spacing of its components is $1/T_{rec}$ and the time series repeat after T_{rec}.

default: 3600.0

A 773 int Surfbeat: random seed

Seed of the random phases and directions of the wave components at the x- side (A 770 1). The same seed gives the same boundary series on all ranks and in repeated runs.

default: 1

A 774 int Surfbeat: long waves at the offshore boundary

Long-wave boundary condition of the host model (SFLOW or NHFLOW, A 750 1) at the x- side in the surfbeat mode. With option 1, the bound long wave of the wave groups (second-order difference interactions, Herbers et al. 1994) enters and outgoing long waves are absorbed (Van Dongeren and Svendsen 1997); with option 0, the boundary only absorbs.

0 absorbing only

1 incoming bound long wave, absorbing outgoing waves

default: 1

A 775 int Surfbeat: geographic advection of the wave groups

Option 1 is first-order upwind, implicit in time. Option 2 is Crank-Nicolson in time with a second-order (van Leer limited) correction of the geographic fluxes, which keeps the wave groups from smearing out over long distances.

1 first-order upwind

2 second order (van Leer)

default: 2

A 780 double Start date and time of the forcing files

Date and time YYYYMMDD.HHMMSS of the model time 0 in the time-dependent forcing files (boundary spectra A 711 3). The forcing is interpolated linearly in time and held constant beyond the ends of the files. With 0, the model starts at the first time in the files. With the wind of the meteorological forcing (A 730 2 with a met file, vortex or blend, W 200 2–4, or with W 203 given), model time 0 is that of the met forcing (W 203); A 780 must then be 0 or the same date and time, otherwise the run stops with a message.

default: 0.0

A 781 int Sweeps pipelined across the MPI ranks

With option 1 and several MPI ranks (M 10 $>$ 1), the Gauss-Seidel sweeps of all ranks form one sweep: the cells are solved in tiles, and a tile at a rank boundary waits for the latest values of the rank upstream, so that every cell sees its neighbours as on one rank and the result is that of one rank. With A 707 0, nonstationary runs then make 1 iteration per step as on one rank (2 with FAS, A 758, or the surfbeat model). This is

used on grids without mesh refinement and with the composite sweep of the refinement (A 797 1); the level-by-level sweeps (A 797 0) and the second-order surfbeat advection (A 775 2) are not pipelined. It needs aligned ranks (the same rows and columns on both sides of a rank boundary); otherwise, and with option 0, each rank sweeps from the halo values of the last exchange, which lag by one sweep. A message on the terminal tells which is used.

0 each rank from the halo of the last exchange

1 pipelined across the ranks

default: 1

A 790 int Bathymetry raster

With option 1, the bed level of all SEASTATE grids (level 0 and the refined patches, G 1) is taken from the raster file `seastate-bathy.dat` instead of the bed of the 2D grid, so that refined patches see the bathymetry and coastline at their own resolution. The file contains a title line, the number of nodes n_x n_y, the first node and spacing x_0 y_0 dx dy in m, then n_y lines of n_x bed levels z in m (positive up, same datum as F 60), from south to north; $ starts a comment. A cell takes the mean bed of the majority (wet or dry) of the raster nodes inside it. Stand-alone runs (A 10 7) only. With option 0, the patches interpolate the bed from level 0.

0 bed from the 2D grid

1 bed from `seastate-bathy.dat`

default: 0

A 791 double Mesh refinement: depth criterion

With mesh refinement (G 1), cells with a water depth below this value in m are refined, evaluated on each level for the next finer one. Use 0 to switch the criterion off.

default: 0.0

A 792 int Mesh refinement: coastline criterion

With mesh refinement (G 1), cells within this many cells (1 to 3) of a dry or land cell are refined. Use 0 to switch the criterion off.

default: 0

A 793 double Mesh refinement: bathymetry gradient criterion

With mesh refinement (G 1), a cell is refined where the relative depth change to a neighbour cell, $|d_{nb} - d| / \max(d, d_{nb})$, exceeds this value. Use 0 to switch the criterion off.

default: 0.0

A 794 int Mesh refinement: unrefined band at the boundaries

With mesh refinement (G 1), no refinement within this many level-0 cells of the sides with a boundary spectrum or zero gradient (A 712 1 or 2), so that the boundary forcing acts on level 0.

default: 4

A 795 double Spectral sparsity threshold

Stationary runs (A 700 2) only: spectral bins holding less than this fraction of the energy of the cell are set to zero, and each sweep solves per frequency only the window of directions that can receive energy, which speeds up spectra concentrated in a narrow sector. At most 250 directions per quadrant. Must be in $[0, 1)$; use 0 to switch it off.

default: 0.0

A 796 int Geographic advection scheme

Scheme of the geographic fluxes of the action balance. Option 2 is the second-order upwind scheme of SWAN (SORDUP), $F = c\,(3/2\,N_{up} - 1/2\,N_{upup})$, first order where a face does not have two active upwind cells; it is not monotone and the solution is clipped at zero. Not with the surfbeat mode (A 770 1, see A 775).

1 first-order upwind

2 second-order upwind

default: 1

A 797 int Mesh refinement: sweep across the levels

Solution order of the refined grids (G 1). Option 0 sweeps the grids level by level per directional quadrant (a V-cycle of sweeps, the patches take the outflow of each other). Option 1 makes one composite sweep of level 0 in which every covered cell sweeps its finer cells where the sweep reaches it, so that the waves cross the whole hierarchy in one sweep and the refined grids need the iterations of a uniform grid. On several MPI ranks, only the composite sweep is pipelined across the ranks (A 781 1) and gives the result of one rank.

0 level by level

1 composite sweep

default: 0

A 798 int Threads per MPI rank

Number of threads per MPI rank. The sweeps of the implicit solver are shared out in tiles of up to 16×16 cells: a tile waits for its upwind tiles, so the result is that of one thread. With mesh refinement (G 1), level 0 and the patches of at least 64 cells are threaded, and on a level with at least 2 A 798 patches the patches themselves are shared out to the threads where their dependencies allow it. The kinematics, the wave parameters and the initial spectra are threaded as well. Not used for the second-order surfbeat advection (A 770 1 with A 775 2).

default: 1

A 799 int Stationary convergence test

Convergence measure per cell for the stationary iterations (A 700 2), compared with A 708 (and A 709). Option 0 uses the relative change of H_s in the last iteration. Option 1 uses the estimated distance of H_s to the converged solution, $|\Delta H_s|/(1-\rho)$ with the contraction ρ of the last two iterations of the cell (at most 0.99), relative to H_s.

0 change of H_s per iteration

1 estimated distance to the solution

default: 0

6.2 B :: Boundary Conditions

Valid for:

S F N C

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B 10 int Wall functions for the velocities

Applies a logarithmic wall law to the tangential velocities at walls and bottom, using the roughness from B 50–B 56.

0 OFF

1 ON

default: 0

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B 11 int Wall functions for the turbulence model

Wall-law boundary conditions for the turbulence quantities (k, ϵ/ω, eddy viscosity) at walls and bottom in NHFLOW. When OFF, the turbulence quantities are set to zero in the wall ghost cells.

0 OFF

1 ON

default: 1

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B 20 int Slip or no-slip boundary conditions for velocities
Ghost-cell treatment of the velocities at walls and at the bottom topography.

1 slip (zero gradient)

2 no-slip

3 no-slip via Dirichlet image-point extrapolation (see B 29)

4 no-slip at walls, slip at the bottom topography

default: 2

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B 21 int Preserve tangential velocities for forcing
Used by the direct forcing immersed boundary method, independent of the wall ghost-cell condition B 20. When ON, the tangential velocities at the solid surface are preserved in the forcing; when OFF, all velocity components inside the solid are simply forced to zero.

0 OFF

1 ON

default: 1

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B 22 int Smooth or rough wall law for the bed shear stress
Wall law used to compute the bed shear stress for sediment transport in NHFLOW when S 16 1 is selected. The roughness is the effective bed roughness of the sediment model. The current part of S 16 9 always uses the rough wall law.

1 smooth wall law (iterative)

2 rough wall law

default: 2

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B 23 int Extrapolation or reflection for ghostcell updates

1 extrapolation

2 reflection

default: 1

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B 29 double Image point distance for ghost-cell Dirichlet extrapolation
Distance of the image point from the wall as a fraction γ of the cell size, used by the Dirichlet ghost-cell extrapolation (e.g. B 20 3). Range $0 < \gamma \leq 1$.

default: 0.5

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B 30 int Type of pressure reference point
Defines how the pressure level is fixed. The reference value is given in B 31; how it is applied is set by B 33.

0 OFF

1 fixed location (see B 32)

2 at the free surface above the inlet: the gage x- and y-location is set automatically to the inlet cell with the smallest x- and y-coordinates (outlet with the largest x-coordinate when B 99 $\geq$ 3), the z-location follows the free surface

3 atmospheric pressure B 31 imposed at the top boundary (lid)

4 normalize the pressure so that its mean value at the free surface (zero level set contour) is zero

default: 0

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B 31 double Pressure reference value
Pressure in Pa at the reference point (B 30 1, 2) or at the lid (B 30 3).

default: 0.0

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B 32 double *x*, double *y*, double *z* Location of the fixed pressure reference point
Coordinates in m, used with B 30 1. If B 32 is not given, the location is set automatically to the inlet cell with the smallest x- and y-coordinates and the highest z-coordinate (outlet with the largest x-coordinate when B 99 $\geq$ 3). For B 30 2 and 3 the location is always set automatically and B 32 is overwritten.

default: 0.0 ; 0.0 ; 0.0 (determined automatically, see above)

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B 33 int Type of pressure reference gage normalization

Virtual normalization only stores the gage offset and subtracts it in the output, inline normalization subtracts it from the pressure field in every time step. For certain configurations, inline pressure normalization causes instabilities.

1 virtual

2 inline

default: 1

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B 50 double Global wall roughness

Equivalent sand-grain roughness k_s in m, used for the bed and all walls unless overwritten by B 51–B 56 (CFD) or B 57 (NHFLOW side walls). In SFLOW it is the bed roughness used for the bottom friction (A 218 1). The bed friction on the momentum needs A 218 1 in SFLOW and A 519 1 or 2 in NHFLOW; if B 50 is set without it, a warning is printed and B 50 acts only in the wall functions of the turbulence model (and the sediment model).

default: 0.0001

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B 51 double Wall 1 roughness

Roughness k_s in m for the boundary side facing the negative x-direction (x_{min}). A value ≤ 0 means not set, B 50 is used.

default: -1.0 (not set, B 50 is used)

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B 52 double Wall 2 roughness

Roughness k_s in m for the boundary side facing the positive y-direction (y_{max}).

default: -1.0 (not set, B 50 is used)

○ ○ ○ ●

B 53 double Wall 3 roughness

Roughness k_s in m for the boundary side facing the negative y-direction (y_{min}).

default: -1.0 (not set, B 50 is used)

○ ○ ○ ●

B 54 double Wall 4 roughness
Roughness k_s in m for the boundary side facing the positive x-direction (x_{max}).

default: -1.0 (not set, B 50 is used)

○ ○ ○ ●

B 55 double Wall 5 roughness
Roughness k_s in m for the boundary side facing the negative z-direction (bottom).

default: -1.0 (not set, B 50 is used)

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B 56 double Wall 6 roughness
Roughness k_s in m for the boundary side facing the positive z-direction (top).

default: -1.0 (not set, B 50 is used)

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B 57 double Side wall roughness for NHFLOW
Roughness k_s in m for the side walls, the top and solid bodies in the NHFLOW wall law: the momentum wall friction with A 519 2 and the turbulence wall functions of the k-ϵ and k-ω models at these walls (also only with A 519 2), as well as the faces of sharp thin bodies (X 330). The bottom uses B 50.

default: 0.0001

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B 60 int Enable ioFlow for open channel flow
This parameter is important for open channel flow calculations. The ioFlow module makes sure that the inflow discharge as given in W 10 is constant throughout the computation even though the inflow water level may change. This is the counterpart to B 90 for open channel flow. When selecting the hydrograph inflow option, make sure that a file with the name "hydrograph.dat" is present in the simulation folder. When selecting the hydrograph outflow option, make sure that a file with the name "hydrograph_out.dat" is present in the simulation folder. The data format of the hydrograph file consists of two columns, the first one gives the time in s and the second column the discharge in m^3/s.

0 OFF

1 constant inflow

2 hydrograph inflow

3 hydrograph outflow

4 hydrograph inflow and outflow

default: 0

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B 61 int Inflow profile for ioFlow

Velocity profile at the inflow boundary for B 60. The logarithmic profiles use the roughness B 50. "Phase 1" refers to the water phase.

1 constant velocity

2 logarithmic profile bed

3 constant velocity only for phase 1

4 logarithmic profile only for phase 1, bed

5 logarithmic profile only for phase 1, sidewalls and bed

default: 2

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B 62 int Distribution of the inflow discharge across the inflow face

For the discharge inflow of SFLOW and NHFLOW (B 60). With option 1 the depth-averaged inflow velocity of each wet column follows the conveyance of a Manning strip, $u = U_k h^{2/3}$ with the local water depth h, and U_k is scaled so that the total discharge is unchanged; shallow cells at the edges of the inflow then get smaller velocities. In NHFLOW with the logarithmic profiles (B 61 2, 4, 5), the profile of each column is scaled so that its depth average follows $h^{2/3}$.

0 as B 61 (uniform depth-averaged velocity across the face)

1 by conveyance, $u \sim h^{2/3}$

default: 0

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B 71 * double value, double distance, double line angle, double line x-origin, double line y-origin

Relaxation of the free surface towards a fixed water level for the initialization only (first time step), for open channel flow with B 60. value: water level z-coordinate in m; distance: length of the relaxation zone in m, measured from the line; line angle in deg; x- and y-origin of the line in m. Multiple zones are possible.

default: na

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B 75 int Outflow boundary condition
Boundary condition for the velocities at the outflow boundary.

1 zero gradient

2 convection condition

3 u-velocity scaled so that the outflow discharge matches the inflow discharge W 10 (scaling limited to 0.1–2.0), convection condition for v and w

default: 1

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B 76 int Pressure inflow boundary conditions

0 hydrostatic pressure

1 zero gradient pressure

default: 1

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B 77 int Pressure outflow boundary conditions
Option 2 controls the outflow water level with F 62. SFLOW only uses option 10. In CFD, the pressure in the outflow ghost cells is set in the same way for ioflow (B 90 0), iowave (B 90 1) and the pressure Poisson equation.

0 zero gradient pressure

1 controlled outflow with pressure condition: hydrostatic pressure from the outflow water level (F 60, F 62) for F 50 = 2 and 3; zero gradient for F 50 = 1 and 4 and with a relaxation beach (B 90 $>$ 0 with B 99 = 1, 2)

2 controlled outflow with free surface condition (see F 62), zero gradient pressure

10 free stream outflow (zero pressure in the water, zero gradient in the air)

default: 1

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B 81 double x_f, double y_f, double t_f Focus location and focus time for focused waves
x- and y-location in m and time in s at which the wave components of the focused wave theories (B 92 41–43) are in phase. For a 2D simulation set y_f to 0.0.

default: 0.0 ; 0.0 ; 0.0

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B 82 int Type of focusing amplitude calculation

Amplitudes A_i of the wave components of focused waves. For NewWave the amplitudes are scaled with the spectrum so that their sum equals half of the wave height given in B 91 or B 93. Options 11–14 are identical to 1–4 but shift all phases by π, so that a wave trough instead of a crest is focused. The phase shift is only applied for unidirectional focused waves (B 130 0).

1 NewWave

2 Spectrum, $A_i = \sqrt{2S(\omega_i)\Delta\omega_i}$

3 Constant wave steepness spectrum, see B 83

4 Constant wave amplitude, $A_i = H/2$

11–14 as 1–4, trough focused

default: 1

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B 83 double Gain factor per wave component for constant wave steepness spectrum focused waves

The amplitudes of the wave components are calculated using $A_i = \frac{B83}{k_i}$ (B 82 3). The frequency distribution is calculated as defined in B 84. For the Gaussian wavepackets (B 85 6) this value is used as the width of the Gaussian distribution in rad/s.

default: 0.0025

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B 84 int Frequency spectrum discretisation method

1 Peak enhance method

2 Equal energy method

3 Uniform distribution

default: 1

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B 85 int Wave spectrum for irregular waves

For the spectrum file, provide a "spectrum-file.dat" with the two columns "ω" and "S". For the 2D spectrum file, provide a "spectrum-file-2d.dat": the header line starts with a label followed by the directions in rad, each following row gives the angular frequency in rad/s and the spectral values for each direction; the number of frequencies and directions then replaces B 86 and B 133. The file directions are used directly as the directions of propagation of the wave components (counter-clockwise, in the wave generation frame of B 105), and B 130 must be > 0. Each (frequency, direction) pair

gives one component with $A_i = \sqrt{2S\Delta\omega\Delta\beta}$. The JONSWAP-type spectra use γ from B 88.

REEF3D::SEASTATE (Section 6.1.4) writes the spectra at its handover points (A 760) in this format (folder REEF3D_SEASTATE_Spectra, file spectrum-file-2d_Pn.dat, to be renamed to "spectrum-file-2d.dat") for the irregular wave generation with B 85 11. SEASTATE also uses B 85 1–3 with B 88, B 93 and B 130–B 135 for its parametric spectrum.

1 Pierson-Moskowitz

2 JONSWAP

3 Torsethaugen

4 Wavepackets

5 Wavepackets for steep waves

6 Gaussian Wavepackets

10 spectrum file

11 2D frequency-direction spectrum file

21 Goda JONSWAP

22 TMA

default: 1

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B 86 int Number of regular waves for irregular wave generation
Number of frequency components used to discretise the spectrum. With directional spreading (B 130) the total number of components is B 86 × B 133.

default: 10

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B 87 double ω_s, double ω_e Frequency range for irregular wave generation
Start and end angular frequency in rad/s of the discretised spectrum. When not given, the model will calculate the values automatically.

default: not set (calculated automatically)

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B 88 double Peak enhancement factor γ
Used by the JONSWAP, Goda JONSWAP and TMA spectra (B 85 2, 21, 22).

default: 3.3

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B 89 int Wave generation optimization through space-time decomposition

Splits the input wave into precomputed spatial and temporal parts, which makes wave generation with many components considerably cheaper. Supported for the wave theories B 92 5, 31, 41 and 51; with any other wave type, also of an additional wave source (B 500), a message is printed and the program runs with B 89 0. Additional wave sources of these types are decomposed together with the B 92 wave. In CFD and FNPF it applies to relaxation generation (B 98 2), in NHFLOW to relaxation and Dirichlet generation; FNPF Dirichlet and active generation (B 98 3, 4) run with B 89 0 and a message is printed. Not available in SFLOW. With zone sources (B 524) the sources of the zone are selected per cell. Cannot be combined with waves on the background (B 530 1 or 2); the program stops with an input error, and the automatic B 530 stays off.

0 OFF

1 ON

default: 0

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B 90 int Enable ioWave for numerical wave tank

This parameter turns on the numerical wave tank. It is the counterpart to B 60 for wave simulations. A wave tank setup needs at least: the wave theory (B 92), the wave parameters (B 91 or B 93), the wave generation method (B 98), the numerical beach (B 99) and, for relaxation zones, their lengths (B 96).

0 OFF

1 ON

default: 0

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B 91 double wave height, double wave length

Wave input with the wave length known: H in m and L in m. The wave period is calculated from the dispersion relation of the selected wave theory. For irregular, focused and reconstructed waves (B 92 31–53) the first value is the significant wave height H_s and the second the peak wave length, from which the peak period is calculated with linear dispersion. Use either B 91 or B 93; if both are given, B 93 is used. Exactly two values must be given, otherwise the program stops with an input error.

default: 0.0 ; 0.0

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B 92 int Wave type

Different wave theories can be used as input for the numerical wave tank. It is important to check beforehand whether the selected theory fits the given wave conditions consisting of wave height, wave length/wave period (B 91 or B 93) and still water depth (B 94). The table below lists which generation methods (B 98) are available for each theory and model.

For the piston and flap wavemaker theories with η timeseries input (20, 24), provide a "wavemaker_eta.dat" input file with two columns "t" and "η".

For the piston and flap wavemaker theories (21, 22), provide a "wavemaker.dat" input file with two columns "t" and "X" or "t" and "β", depending on input from B 116. The vertical extent of the paddle is set with B 110 (piston), B 111 (flap) or B 112 (double-hinged flap). A second-order correction of the paddle signal can be added with B 113 and B 114, and in FNPF and NHFLOW the moving-paddle boundary terms with B 119.

For the double-hinged flap wavemaker, two "X" or two "β" columns are expected. For giving two "X" columns, this will be the total excitation at the end of each flap. For giving two "β" columns, the second angle is the additional excitation of the second flap.

For the hydrodynamic coupling HDC (61), the input data is prepared with DIVEMesh (H 10) from the results of the source simulation; NHFLOW and CFD read it from the folder REEF3D_CFD_HDC_Input, FNPF from REEF3D_FNPF_HDC_Input. The program stops with a message if the files are missing.

For the runtime coupling (62), the FNPF source runs in the same MPI job (Section 8.5); the coupling is set with B 105, I 231–I 233, I 241–I 247.

For the wave reconstruction, provide a "waverecon.dat" input file with three columns giving "A" and "ω" and phase "ϕ" for each of the harmonic wave components. An optional fourth column gives the direction of each component in deg from the x-axis of the wave generation frame (B 105; the first line decides whether the column is read), and the components are evaluated as a directional sea. Without it all components travel in the B 105 direction.

Wave theories which prescribe the potential directly are marked with a $*$ and are suitable for relaxation zone wave generation in FNPF. Those wave theories without $*$ require B 98 3 or B 98 4 for FNPF.

Further wave trains can be superposed on the B 92 wave with B 500–B 505 (additional wave sources), and relaxation zones and open boundary conditions along the domain edges with their own selection of wave sources can be defined with B 520–B 525. In NHFLOW, a tidal or current background can be added with B 510–B 515, and the waves can be adapted to it with B 530.

1 Shallow Water Waves

2 Linear Waves $*$

3 Deep Water Waves $*$

4 2*nd*-order Stokes Waves $*$

5 5*th*-order Stokes Waves (Fenton) $*$

6 Shallow Water Cnoidal Waves

7 1st-order Cnoidal Waves

8 5th-order Cnoidal Waves

9 1st-order Solitary Wave

10 3rd-order Solitary Wave

11 5th-order Stokes Waves (Skjelbreia)

20 Piston Wavemaker based on η timeseries

21 Piston Wavemaker

22 Flap Wavemaker

23 Double-hinged Flap Wavemaker

24 Flap Wavemaker based on η timeseries

31 1st-order Irregular Waves $*$

32 2nd-order Irregular Waves (Longuet-Higgins)

33 2nd-order Irregular Waves (Schäffer) $*$

41 1st-order Focused Waves $*$

42 2nd-order Focused Waves (Longuet-Higgins)

43 2nd-order Focused Waves (Schäffer) $*$

51 Wave Reconstruction with 1st-order Irregular Waves $*$

52 Wave Reconstruction with 2nd-order Irregular Waves (Longuet-Higgins)

53 Wave Reconstruction with 2nd-order Irregular Waves (Schäffer) $*$

61 Hydrodynamic Coupling HDC (into FNPF, NHFLOW or CFD)

62 Runtime Coupling with an FNPF model in the same MPI job (into FNPF, NHFLOW or CFD)

70 Steady Surface Gravity Wave SSGW $*$ (number of Fourier modes B 170)

default: 0

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B 93 double wave height, double wave period

Wave input with the wave period known: H in m and T in s. The wave length is calculated from the dispersion relation of the selected wave theory. For irregular, focused and reconstructed waves (B 92 31–53) the first value is the significant wave height H_s and the second the peak period T_p. REEF3D::SEASTATE uses B 93 as H_s and T_p of its parametric spectrum (Section 6.1.4). Use either B 91 or B 93; if both are given, B 93 is used. Exactly two values must be given, otherwise the program stops with an input error.

default: 0.0 ; 0.0

Wave Generation Compatibility					
B 92	Wave Type	SFLOW	FNPF	NHFLOW	CFD
1	Shallow Water Waves	D + R	D	D + R	D + R
2	Linear Waves	D + R	D + R	D + R	D + R
3	Deep Water Waves	D + R	D + R	D + R	D + R
4	2*nd*-order Stokes Waves	D + R	D + R	D + R	D + R
5	5*th*-order Stokes Waves (Fenton)	D + R	D + R	D + R	D + R
6	Shallow Water Cnoidal Waves	D + R	D	D + R	D + R
7	1*st*-order Cnoidal Waves	D + R	D	D + R	D + R
8	5*th*-order Cnoidal Waves	D + R	D	D + R	D + R
9	1*st*-order Solitary Wave	D + R	D	D + R	D + R
10	3*rd*-order Solitary Wave	D + R	D	D + R	D + R
11	5*th*-order Stokes Waves (Skjelbreia)	D + R	D	D + R	D + R
20	Piston Wavemaker, η timeseries	D	D	D	D
21	Piston Wavemaker	D	D	D	D
22	Flap Wavemaker	D	D	D	D
23	Double-hinged Flap Wavemaker	D	D	D	D
24	Flap Wavemaker, η timeseries	D	D	D	D
31	1*st*-order Irregular Waves	D + R	D + R	D + R	D + R
32	2*nd*-order Irregular Waves (Longuet-Higgins)	D + R	D + R	D + R	D + R
33	2*nd*-order Irregular Waves (Schäffer)	D + R	D + R	D + R	D + R
41	1*st*-order Focused Waves	D + R	D + R	D + R	D + R
42	2*nd*-order Focused Waves (Longuet-Higgins)	D + R	D + R	D + R	D + R
43	2*nd*-order Focused Waves (Schäffer)	D + R	D + R	D + R	D + R
51	Wave Reconstruction with 1*st*-order Irregular Waves	D + R	D + R	D + R	D + R
52	Wave Reconstruction with 2*nd*-order Irregular Waves (Longuet-Higgins)	D + R	D + R	D + R	D + R
53	Wave Reconstruction with 2*nd*-order Irregular Waves (Schäffer)	D + R	D + R	D + R	D + R
61	Hydrodynamic Coupling HDC		D + R	D + R	D + R
62	Runtime Coupling (FNPF source)		D + R	D + R	D + R
70	Steady Surface Gravity Wave SSGW		R	R	

Table 6.1: Wave generation compatibility for B 92, D = Dirichlet/Neumann generation (B 98 3, 4, 5); R = Relaxation generation (B 98 2).

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B 94 double Water depth d for wave theory

Water depth in m used to evaluate the wave theory. If not given, d is set to the still water level F 60, which requires the bed at the wave generation to be located at $z = 0.0$ m.

default: not set (still water level F 60)

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B 95 double Step-size consistent relaxation zones

For relaxation wave generation (B 98 2) in SFLOW and NHFLOW. The relaxation zones act in every Runge-Kutta stage. With B 95 0, the wave targets are evaluated once per time step at its start and the relaxation factor r of a zone does not depend on the time step, so that the generated wave depends on the number of time steps per wave period. With B 95 $\neq$ 0, the targets (and the ramp B 101) are evaluated at the time of each stage output (RK3: $t+\Delta t$, $t+\Delta t/2$, $t+\Delta t$; RK2: $t+\Delta t$, $t+\Delta t$), and the relaxation factors of the generation and beach zones are replaced by $r^{\Delta t/\Delta t_{ref}}$. A positive value is the reference time step Δt_{ref} in s; with a negative value (B 95 -1), Δt_{ref} is the time step of the finest level ($\Delta t/2^{G1}$ with mesh refinement subcycling G 7 1, otherwise Δt). Over a fixed time, the zones then relax by the same amount whatever the time step.

0 OFF, targets once per time step (as before)

>**0** reference time step Δt_{ref} in s

<**0** (e.g. -1) reference time step of the finest refinement level

default: 0.0

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B 96 double dist1, double dist2 Lengths of the relaxation zones

Length in m of the wave generation relaxation zone (dist1), measured from the start of the domain (x_{min}), and length in m of the numerical beach relaxation zone (dist2), measured from the end of the domain (x_{max}). The relaxation zones are only active for relaxation generation (B 98 2) and relaxation beaches (B 99 1 or 2). If B 108 or B 107 are given, they replace the default zones and the respective value of B 96 is not used. A typical choice, used in the tutorials, is a generation zone of about one wavelength and a numerical beach of about two wavelengths, e.g. B 96 4.0 8.0 for L = 4.0 m. Exactly two values must be given, otherwise the program stops with an input error. The combined generation and beach (B 98 6, B 99 6) also use these lengths. On a curvilinear grid (G 50, NHFLOW) the B 96 zones (an x-range of the box) are not available; give the zones with B 107 and B 108 in physical coordinates.

default: 0.0 ; 0.0

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B 97 int Waves on a current for NHFLOW

With B 97 1, the NHFLOW numerical beach relaxation zones (B 99 1 or 2, B 107, B 520 method 2) relax the horizontal velocity towards the inflow current velocity U_i = W 10 / inflow cross-section area (ramped with B 101) instead of towards zero; the water level is still relaxed to the still water level. Intended for waves on a current with relaxation wave generation (B 98 2) together with B 60 1 and the discharge W 10: the generation zone then relaxes towards the wave velocity plus U_i, and the inflow ghost cells take over the velocity of the relaxation zone (zero gradient). Not available on a curvilinear grid (G 50).

0 OFF, beach relaxes the velocity to zero

1 ON, beach relaxes the velocity to the inflow current

default: 0

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B 98 int Wave Generation Method

Relaxation generation ramps the solution towards the wave theory inside the generation zone (B 96, B 108) following the relaxation function by Jacobsen et al. [2012]; the zone also absorbs waves reflected back towards the inlet. For the Dirichlet type, waves are generated by assigning the values for the free surface and the velocity at the inflow boundary only, which needs no extra domain length. Active absorption additionally corrects the inflow velocity with the difference between the target and the computed free surface elevation using shallow water theory, $U_c = (\eta_T - \eta_M)\sqrt{g/d}$, so that reflected waves leave the domain through the inlet.

Overrides (control_parse.cpp): B 98 1 (the former two-zone relaxation method by Afshar [2010]) is silently changed to B 98 2. For NHFLOW (A 10 5), B 98 3 is changed to B 98 4, i.e. Dirichlet generation in NHFLOW always includes active absorption, and B 98 5 is changed to B 98 3 to obtain plain Dirichlet generation without absorption. In SFLOW, options 3 and 4 are identical (Dirichlet). In FNPF, option 4 adds active absorption: for the wave theories, the deviation of the computed from the target surface elevation leaves the domain with the shallow-water velocity $-(\eta_M - \eta_T)\sqrt{g/h}$ on top of the target velocity (zero gradient water level at the inlet); for the piston wavemaker (B 92 21), the target is the linear surface elevation of the paddle signal at the first cell (progressive and evanescent modes); for the other wavemakers option 4 is identical to option 3. On a curvilinear grid (G 50), wave generation is not available in SFLOW, and NHFLOW Dirichlet or active generation (B 98 3, 4) cannot be combined with a current (W 10). In CFD with B 98 2 and W 10 > 0, the discharge of W 10 is imposed at the inlet in addition to the waves. In NHFLOW, the mass-conservative Dirichlet BC A 515 4 imposes the prescribed mass flux exactly at the inflow face only for plain Dirichlet generation, i.e. with B 98 5 in the input (changed to 3 internally); with B 98 3 or 4 (active absorption) this flux correction is not applied.

Options 6–8 are converted at start-up (iowave); where a module or wave type does not support them, a message is printed and the program runs with the fallback named for each option.

Option 6, combined generation, for FNPF and NHFLOW with a wave theory (not the wavemakers B 92 20–29): a relaxation zone of length dist1 (B 96) relaxes towards the target plus the deviation from it, low-passed over the time scale B 527, so that free long waves pass the zone, and an absorbing generating edge at x_{min} lies behind it (FNPF: active generation as B 98 4; NHFLOW: a Riemann edge carrying the waves of all sources, see B 520 method 3, unless a B 520 edge is given at x_{min}). Otherwise B 98 2.

Option 7, two-band active absorption, for FNPF with a wave theory, or with the piston wavemaker (B 92 21) when the reference period B 528 is given (else B 98 4): the deviation from the target is split by a first-order low pass with the time scale B 527; the long part is absorbed with the shallow-water velocity, the short part with the linear velocity profile of the reference period (B 528). Otherwise B 98 4 (NHFLOW, CFD) or B 98 2 (SFLOW, FNPF with the other wavemakers).

Option 8, SIRW active absorption, for FNPF and NHFLOW: the wave reflected towards the wavemaker is separated in real time from two gauges in front of the boundary with FIR filters (Frigaard and Brorsen 1995; band and gauge positions B 534, reference period B 528) and absorbed with its linear velocity profile; the remaining deviation from the target is absorbed with the shallow-water velocity (in FNPF after a first-order low pass). FNPF runs active generation (B 98 4); NHFLOW with a wave theory generates through a Riemann edge at x_{min} whose outside state contains the predicted reflected wave (internally B 98 2, a relaxation zone only with B 96 dist1 > 0), with a wavemaker active generation (B 98 4) corrected by the velocity and the non-hydrostatic pressure of the predicted reflected wave. If neither the band nor a period is available, or the gauges lie outside the domain, SIRW is switched off with a message. Otherwise B 98 4 (CFD) or B 98 2 (SFLOW).

0 OFF

1 changed to 2

2 Relaxation Method

3 Dirichlet (NHFLOW: with active absorption, see above)

4 Dirichlet with Active Absorption, based on shallow water theory (CFD, NHFLOW, FNPF)

5 Dirichlet without absorption, NHFLOW only

6 combined generation: relaxation zone with low-passed deviation and absorbing edge (FNPF, NHFLOW)

7 two-band active absorption (FNPF)

8 SIRW active absorption with two gauges (FNPF, NHFLOW)

default: 0

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B 99 int Numerical Beach Method

Both relaxation methods dissipate the wave energy by ramping the free surface down to the still water level, the velocity to zero and the pressure to its hydrostatic distribution for still water conditions inside the beach zone (B 96, B 107). Typically the length of the

dissipating relaxation zone should be two wavelengths. Option 1 uses a softer relaxation function than option 2 (the relaxation coordinate is scaled by 1/2). The Active Wave Absorption avoids reflections by generating a wave opposite of the reflected wave at the outflow boundary, thus canceling it out. The method is more efficient in computational terms, as it does not require additional space. Option 4 uses the wave frequency and wave number of the input wave (B 91 or B 93). SFLOW supports options 1–4 and 6. In NHFLOW and SFLOW, options 3 and 4 turn the outflow boundary at x_{max} into an absorbing Riemann edge with still water outside (see B 520 method 3); option 4 is treated as option 3 there. An edge zone given with B 520 (methods 3–6) on the edge x_{max} takes precedence, and B 99 is then ignored at x_{max}.

Options 6–8 are converted at start-up (iowave); where a module does not support them, a message is printed and the program runs with the fallback named for each option.

Option 6, combined beach, for NHFLOW, SFLOW and FNPF: a relaxation beach of length dist2 (B 96) relaxes towards the state low-passed over the time scale B 526, so that short waves are damped in the zone and long waves pass to an absorbing edge at x_{max} behind it (as option 3). Otherwise (CFD) B 99 2.

Option 7, two-band active beach, for FNPF: the deviation of the surface elevation at the outflow is split by a first-order low pass with the time scale B 526; the long part is absorbed with the shallow-water velocity, the short part with the linear velocity profile of the reference period B 528. Otherwise B 99 3.

Option 8, SIRW active beach, for FNPF and NHFLOW: the wave travelling towards the boundary is separated in real time from two gauges in front of it with FIR filters (Frigaard and Brorsen 1995; band and gauge positions B 535, reference period B 528). FNPF absorbs it with its linear velocity profile and the remaining deviation with the shallow-water velocity; in NHFLOW the predicted outgoing wave is the outside state of the absorbing Riemann edge at x_{max}. If neither the band nor a period is available, or the gauges lie outside the domain, SIRW is switched off with a message. Otherwise B 99 3.
In order to use the Active Wave Absorption beach, keep in mind to use the active beach boundary condition in the C data set in DIVEMesh.

0 OFF

1 Relaxation Method 1

2 Relaxation Method 2

3 Active Wave Absorption, based on shallow water theory

4 Active Wave Absorption, based on intermediate water theory

5 Active Wave Absorption, using a flap wavemaker (see B 123)

6 combined beach: relaxation to the low-passed state and absorbing edge (NHFLOW, SFLOW, FNPF)

7 two-band active beach (FNPF)

8 SIRW active beach with two gauges (FNPF, NHFLOW)

default: 0

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B 101 int Ramp function for wave generation
Ramps the generated wave up smoothly from zero, $f = t/t_r - \frac{1}{\pi}\sin(\pi t/t_r)$ for $t < t_r$, with the ramp duration t_r from B 102.

0 OFF

1 ON, t_r = B 102 × wave period

2 ON, t_r = B 102 in s of simulation time

default: 0

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B 102 double Ramp function duration
Duration of the ramp in wave periods (B 101 1) or in s (B 101 2).

default: 1.0

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B 105 double angle β, double x_0, double y_0 Wave direction and origin for the generated wave
Propagation direction of the generated wave in deg (measured from the x-axis) and the x- and y-coordinates in m of the origin from which the wave phase distance is measured. If not given, the origin is the minimum x- and y-coordinate of the domain (0.0 ; 0.0 for HDC and the runtime FNPF coupling, B 92 61 and 62).

default: 0.0 ; domain x_{min} ; domain y_{min}

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B 106 * double angle β, double x, double y Origin of the wave generation relaxation zone
Angle in deg and x-, y-coordinates in m. In the current code the input has no effect on the relaxation zones; use B 108 to place wave generation zones.

default: B 105 angle ; domain x_{min} ; domain y_{min}

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B 107 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double d Numerical beach relaxation zones
The line from (x_{start}, y_{start}) to (x_{end}, y_{end}) in m defines where the waves are fully damped; the relaxation zone extends the acting distance d in m to both sides of this line, and the damping decreases with the distance from the line. As default, without any user input for B 107, a relaxation zone for the numerical beach is created at the end of the domain, i.e. at the maximum x-coordinate of the domain, with an orientation perpendicular to the x-axis over the full domain width, using the distance dist2 from B

96. If user input is provided for B 107, the default relaxation zone will not be created and the input from B 96 will not be used. The B 107 zones are active even without B 99 1 or 2. Multiple numerical beach zones are possible. Beach zones along a domain edge can also be given with B 520 and B 521; they are used in addition to the B 96 or B 107 zones.

default: not set (zone from B 96, see above)

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B 108 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double d Wave generation relaxation zones

The line from (x_{start}, y_{start}) to (x_{end}, y_{end}) in m defines where the wave theory is imposed fully; the relaxation zone extends the acting distance d in m to both sides of this line, and the influence of the wave theory decreases with the distance from the line. As default, without any user input for B 108, a relaxation zone for the wave generation is created at the beginning of the domain, i.e. at the minimum x-coordinate of the domain, with an orientation perpendicular to the x-axis over the full domain width, using the distance dist1 from B 96. If user input is provided for B 108, the default relaxation zone will not be created and the input from B 96 will not be used. Requires relaxation generation (B 98 2, or B 98 6). Multiple wave generation zones are possible. Generation zones along a domain edge can also be given with B 520 and B 521; they are used in addition to the B 96 or B 108 zones.

default: not set (zone from B 96, see above)

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B 110 double z_{start}, double z_{end} Vertical extent of the piston wavemaker

z-coordinates in m between which the piston velocity is applied (B 92 20, 21). Outside this range the velocity is zero.

default: not set (whole water column)

● ● ● ●

B 111 double z_{start}, double z_{end} Vertical extent of the flap wavemaker

z-coordinate of the hinge (z_{start}) and of the top of the flap (z_{end}) in m for B 92 22 and 24. Must be set when these theories are used.

default: 0.0 ; 0.0

● ● ● ●

B 112 double z_{start}, double z_{hinge_2}, double z_{end} Vertical extent of the double-hinged flap wavemaker

z-coordinates in m of the lower hinge, the second hinge and the top of the upper flap for B 92 23.

default: 0.0 ; 0.0 ; 1.0

● ● ● ●

B 113 int mode, int J Second-order wavemaker correction

Second-order wavemaker theory of Schäffer [1996] for the piston, flap and double-hinged flap wavemakers driven by "wavemaker.dat" (B 92 21, 22, 23). At start-up the first-order paddle signal is resampled, decomposed by FFT (at most 500 components, frequency band B 114), and the paddle displacement $X^{(2)}(t)$ that cancels the spurious free second-order waves is added to the input signal. For the double-hinged flap the correction is split over the two flaps in the proportion of the first-order motion. J is the number of evanescent modes per frequency (0: progressive modes only). The water depth is that of the wave theory (B 94); in NHFLOW it is the still water depth at the inflow boundary (e.g. a wavemaker pit), unless B 94 is given. The correction assumes that the paddle reaches above the still water level; a warning is written if B 111 or B 112 z_{end} lies below it. The applied maximum correction is written to the screen. In FNPF and NHFLOW, use together with B 119 1 to include the second-order paddle boundary terms consistently.

0 OFF

1 sub- and superharmonic correction

2 subharmonic correction only

3 superharmonic correction only

default: 0 ; 10

● ● ● ●

B 114 double f_{min}, double f_{max}, double $f_{2,min}$ Frequency band for the second-order wavemaker correction

Frequencies in Hz for B 113: first-order components between f_{min} and f_{max} are used ($f_{max} \leq 0$: chosen automatically; the same band is used for the paddle terms of B 119), and second-order components below $f_{2,min}$ are not corrected ($f_{2,min} \leq 0$: 0.1 times the peak frequency). The paddle terms of B 119 leave out difference frequencies below the same $f_{2,min}$ (0: 0.1 times the peak frequency, < 0: no cutoff).

default: 0.0 ; 0.0 ; 0.0

○ ○ ● ●

B 115 int Vertical velocity component for flap wavemaker kinematics

Includes the vertical velocity of the rotating flap (B 92 22, 23) in the generated velocities.

0 OFF

1 ON

default: 0

● ● ● ●

B 116 int Flap wavemaker kinematics input as paddle motion or angle

Meaning of the second (and third) column in "wavemaker.dat" for the flap wavemakers (B 92 22, 23).

1 paddle displacement $X(t)$ in m

2 paddle angle $\beta(t)$ in rad

default: 1

● ● ● ●

B 117 double Starting time shift for wavemaker kinematics input

Time in s: entries of "wavemaker.dat" or "wavemaker_eta.dat" before this time are skipped, and the remaining timeseries is shifted to start at $t = 0$.

default: 0.0

● ● ● ●

B 118 double Correction factor for the wavemaker velocities

Scaling factor applied to the paddle velocities of the wavemaker theories (B 92 21–24) and to the moving wavemaker objects in the 6DOF module. It was introduced to correct wavemaker generation in NHFLOW.

default: 1.0

○ ● ● ○

B 119 int Moving-paddle boundary condition for FNPF and NHFLOW wavemakers

For the piston and flap wavemakers (B 92 21, 22, 23) with Dirichlet generation in FNPF and NHFLOW, the inflow velocity $\phi_x = X_t$ at $x = 0$ is extended by the second-order Taylor terms of the moving paddle, $X_z\phi_z - X\phi_{xx}$. These terms are evaluated from the linear paddle solution of the first-order signal (progressive and B 113 J evanescent modes, band B 114), tabulated at start-up, and not from the computed flow. With B 119 1 the numerical paddle behaves like a physical moving paddle, and the B 113 correction then also cancels these terms.

0 OFF

1 ON

default: 0

● ● ● ●

B 120 double Phase angle shift for periodic regular waves

Phase shift in deg added to the phase of the regular wave theories.

default: -90.0°

○ ● ● ●

B 123 double z-coordinate of the hinge for flap-type active wave absorption

z-coordinate in m of the hinge for B 99 5. Below the hinge the absorbing velocity is zero, above it increases linearly towards the free surface.

default: 0.0

○ ● ● ●

B 125 double y-coordinate for the hydrodynamic coupling slice

Uses the 2D slice of the HDC input data at this y-coordinate in m for the whole coupling boundary (B 92 61), e.g. to drive a 3D domain with results from a 2D simulation.

default: not set (full 3D input)

○ ● ● ●

B 127 int Remove y-component from hydrodynamic coupling

Sets the y-velocity of the HDC input (B 92 61) to zero.

0 OFF

1 ON

default: 0

● ● ● ●

B 130 int Enable directional spreading for irregular waves

Generates short-crested waves by distributing each frequency component over B 133 directions between B 132 around the main direction B 131. The cosine-power function is $D = \frac{2}{\pi}\cos^s(\beta - \beta_0)$ with s = B 134; use B 134 2 for the cosine-squared function (the factor $\frac{2}{\pi}$ normalizes $s = 2$). The Mitsuyasu function uses s = B 134, or a frequency-dependent s from B 135 if B 134 ≤ 0. B 130, B 131, B 134 and B 135 also define the directional spreading of the parametric spectrum of REEF3D::SEASTATE (Section 6.1.4).

0 OFF

1 Cosine power (cosine squared with B 134 2)

2 Mitsuyasu

default: 0

● ● ● ●

B 131 double Main direction for multidirectional irregular waves
Main wave direction β_0 in deg.

default: 0.0°

● ● ● ●

B 132 double β_s, double β_e Start and end angle for directional spreading
Angles in deg.

default: -90.0° ; 90.0°

● ● ● ●

B 133 int Number of angle intervals for directional spreading
If directional spreading is active (B 130 $>$ 0) and B 133 $\leq$ 1, the number is set to 3.

default: 1

● ● ● ●

B 134 double Spreading parameter s
Exponent s of the directional spreading function (B 130). Must be set to a positive value for B 130 1. For B 130 2, a positive value gives a constant s (recommended, $s \leq 85$); a value ≤ 0 switches to the frequency-dependent s from B 135.

default: -1.0 (not set)

● ● ● ●

B 135 double Peak spreading parameter s_{max} for Mitsuyasu spreading
Used for B 130 2 when B 134 ≤ 0: $s = s_{max}(\omega/\omega_p)^5$ for $\omega \leq \omega_p$ and $s = s_{max}(\omega/\omega_p)^{-2.5}$ above. A code comment marks this option as not correct; use B 134 instead.

default: 10.0

● ● ● ●

B 136 int Type of multi-directional irregular wave calculation

1 DSM - double summation method
2 PSSM - Pascal's Single Summation Method
3 YSSM - Yu's Single Summation Method
4 EEM - equal energy method, to be used together with B 84 2

default: 1

● ● ● ●

B 138 int seed 1, int seed 2 Seed numbers for the random phase generation of multi-directional irregular waves
If not given, the random numbers are based on the computer clock.

default: not set (clock based)

● ● ● ●

B 139 int Seed number for the random phase generation of irregular waves
IMPORTANT: seed numbers are not machine transferable. In order to reliably repeat a given wave spectrum, the wave component file (see REEF3D_Log-Wave folder) needs to be re-used and renamed to "waverecon.dat" (B 92 51–53).

0 time based seed number

>**0** input based seed number

default: 0

● ○ ○ ○

B 160 int Number of vertical layers for depth-averaged velocities in the wave generation for REEF3D::SFLOW
The wave theory velocity is evaluated at B 160 + 1 points over the water depth and averaged.

default: 5

○ ● ● ●

B 170 int Number of Fourier modes for calculating the steady surface gravity wave (B 92 70)

default: 1024

○ ○ ○ ●

B 180 int Moving tank through modified gravity
Represents a translating tank (e.g. sloshing) by adding the tank acceleration from B 181–B 183 to the gravity vector W 20–W 22. Required for B 181–B 183; the rotations B 191 and B 192 switch this mode on by themselves. Cannot be combined with B 60 or B 90. The LAGOON store (P 18) is not written with B 180, B 191 or B 192.

0 OFF

1 ON

default: 0

○ ○ ○ ●

B 181 double amplitude, double frequency, double phase Harmonic tank motion in x-direction
Displacement amplitude a in m, frequency f in Hz and phase φ in deg of a harmonic motion; the corresponding acceleration $a(2\pi f)^2 \sin(2\pi f t + \varphi)$ is added to the gravity in x-direction. Requires B 180 1.

default: 0.0 ; 0.0 ; 0.0

○ ○ ○ ●

B 182 double amplitude, double frequency, double phase Harmonic tank motion in y-direction
As B 181, for the y-direction. Requires B 180 1.

default: 0.0 ; 0.0 ; 0.0

○ ○ ○ ●

B 183 double amplitude, double frequency, double phase Harmonic tank motion in z-direction
As B 181, for the z-direction. Requires B 180 1.

default: 0.0 ; 0.0 ; 0.0

○ ○ ● ●

B 191 double angle, double frequency, double y, double z Harmonic tank rotation around the x-axis
Rotation amplitude θ in deg and frequency f in Hz of the roll motion $\theta(t) = \theta \sin(2\pi f t)$, and the y- and z-coordinates in m of the rotation axis. Active between the times given in B 194.

default: 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ● ●

B 192 double angle, double frequency, double x, double z Harmonic tank rotation around the y-axis
Rotation amplitude in deg and frequency in Hz of the pitch motion, and the x- and z-coordinates in m of the rotation axis. Active between the times given in B 194.

default: 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ● ●

B 194 double t_s, double t_e Start and end time of the rotation
Time window in s in which the rotations B 191 and B 192 are applied.

default: $-1.0 \cdot 10^9$; $1.0 \cdot 10^9$ (always active)

○ ○ ● ○

B 201 double n, double d_{50}, double α, double β VRANS porous media parameters

Porosity n, mean grain diameter d_{50} in m and the linear and nonlinear resistance coefficients α and β for all NHFLOW VRANS objects (B 210–B 230). With porous layers (B 202), these are the values of the core below the layers.

default: 1.0 ; 0.01 ; 0.0 ; 0.0

○ ○ ● ○

B 202 * double t, double n, double d_{50}, double α, double β VRANS porous layers

Layers of different porous material in the NHFLOW VRANS objects (B 210–B 230), e.g. the armour and filter layers of a breakwater: thickness t in m, porosity n, mean grain diameter d_{50} in m and the resistance coefficients α and β. The layers are counted from the exposed surface of the porous structures inwards, in the order given; the first layer covers the depth 0–t_1 below the surface, the second t_1–$t_1 + t_2$, and so on; below the last layer the core values of B 201 apply. The properties are blended at each layer boundary with the smoothed Heaviside function of half width B 209 times the cell size. Faces of the structures on the domain boundary, internal faces between overlapping objects and, with B 203 1, faces resting on the bed do not count as exposed surface. A layer with $t \leq 0$ is ignored with a message. The layered porosity is written to the NHFLOW output ("porosity").

default: na

○ ○ ● ○

B 203 int VRANS surface faces resting on the bed

Only the exposed surface faces of the NHFLOW VRANS objects define the distance to the porous interface, i.e. the smoothed transition of the porosity (B 209), and the depth of the porous layers (B 202). With option 1, downward-facing faces (outward normal more than 45 deg below the horizontal) with a vertex or the centroid at or below the local bed level plus B 209 times the cell size are not exposed, so that the bottom of a structure placed on the bed does not create a porous interface; the underside of an elevated porous body stays exposed. The number of exposed faces is written to the screen at start-up.

0 all faces

1 faces resting on the bed are skipped

default: 1

○ ○ ● ○

B 204 int NHFLOW VRANS drag treatment

Time integration of the Darcy-Forchheimer resistance $-(A + B|u|)u$ of the NHFLOW VRANS objects. The explicit source is unstable when Δt times the resistance rate

becomes large, e.g. in a fine-grained core (B 201) below a coarse, permeable armour layer (B 202). The point-implicit treatment linearises the drag with the velocity of the stage and solves it together with the stage update; it is unconditionally stable and keeps the Darcy-Forchheimer balance in steady flow.

0 explicit

1 point-implicit with the stage update

default: 1

○ ○ ● ○

B 208 int Print VRANS porous media forces

Calculates the total force of the flow on the NHFLOW VRANS porous media and writes it to the folder REEF3D_NHFLOW_VRANS_Force.

0 OFF

1 ON

default: 0

○ ○ ● ○

B 209 double Thickness of porous Heaviside layer

Half thickness of the smoothed transition at the surface of the NHFLOW VRANS objects, as a factor times the horizontal cell size. The same width blends the properties at the boundaries of the porous layers (B 202) and is the tolerance of the bed test of B 203.

default: 1.1

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B 210 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} VRANS rectangular box

Coordinates in m. Porous parameters from B 201.

default: na

○ ○ ● ○

B 212 * double x_{center}, double z_{center}, double y_{start}, double y_{end}, double $radius$ VRANS cylinder in y-direction

Coordinates and radius in m. Porous parameters from B 201.

default: na

○ ○ ● ○

B 213 * double x_{center}, double y_{center}, double z_{start}, double z_{end}, double $radius$ VRANS cylinder in z-direction
Coordinates and radius in m. Porous parameters from B 201.

default: na

○ ○ ● ○

B 214 * VRANS cylinder with flexible orientation and front face orthogonal to the cylinder axis

double x_{start}, double y_{start}, double z_{start}, double $radius_{start}$, double x_{end}, double y_{end}, double z_{end}, double $radius_{end}$
Coordinates and radii in m. Porous parameters from B 201.

default: na

○ ○ ● ○

B 215 * double x_{center}, double y_{center}, double z_{center}, double $radius$ VRANS sphere
Coordinates and radius in m. Porous parameters from B 201.

default: na

○ ○ ● ○

B 217 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} VRANS wedge object in x-direction
Coordinates in m. Porous parameters from B 201.

default: na

○ ○ ● ○

B 218 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} VRANS wedge object in y-direction
Coordinates in m. Porous parameters from B 201.

default: na

○ ○ ● ○

B 219 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} VRANS wedge object in z-direction
Coordinates in m. Porous parameters from B 201.

default: na

○ ○ ● ○

B 230 int VRANS, read STL file and generate porous solid

Reads the geometry from a file named "solid.stl". The geometry is scaled, translated and rotated with A 591–A 593. Porous parameters from B 201.

IMPORTANT: the STL file needs to be in ASCII format! Most CAD programs export to a binary STL file. It is possible to use e.g. Paraview for STL conversion from binary to ASCII.

0 OFF

1 ON

default: 0

○ ○ ○ ●

B 240 * double C, double D, double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} Darcy porous media as rectangular box

Resistance coefficients C and D of the Darcy-Forchheimer source term and box coordinates in m. The active directions are set with B 241–B 243.

default: na

○ ○ ○ ●

B 241 int Porous media in x-direction for B 240

0 OFF

1 ON

default: 1

○ ○ ○ ●

B 242 int Porous media in y-direction for B 240

0 OFF

1 ON

default: 1

○ ○ ○ ●

B 243 int Porous media in z-direction for B 240

0 OFF

1 ON

default: 1

○ ○ ● ●

B 260 double Added mass coefficient c for VRANS porous media

Coefficient of the added mass (inertia) term in the VRANS equations. In NHFLOW all momentum tendencies inside the porous media are scaled with an inertia factor that depends on c and the porosity n (see B 265); in CFD the factor is $1/(1+C_m)$ inside the porous media, with $C_m = c(1-n)/n$ (B 266 1, default) or $C_m = c$ independent of the porosity (B 266 0).

default: 0.34

○ ○ ● ●

B 264 double KC number for the VRANS porous media resistance

Keulegan-Carpenter number used in the nonlinear resistance term, which is multiplied by $(1+7.5/KC)$. The default sentinel $1.0 \cdot 10^{20}$ means not set, i.e. no KC correction.

default: $1.0 \cdot 10^{20}$ (not set, no KC correction)

○ ○ ● ○

B 265 int Inertia factor for NHFLOW VRANS porous media

Selects the inertia (added mass) factor with which all momentum tendencies, including the non-hydrostatic pressure correction, are scaled in NHFLOW where the porosity $n < 1$, i.e. in the VRANS porous media (B 201–B 230) and in the porosity of DEM particles (E 28, see Section 6.5). c is the added mass coefficient B 260. The NHFLOW free surface equation uses the porosity of the surface cell for the storage term. With option 0, long waves in a porous layer travel faster than in open water; option 1 is consistent with this storage term, long waves then travel at $\sqrt{gh/(1+c_A)}$. Option 1 is recommended together with the DEM particle porosity (E 28).

0 $1/(1+c(1-n)/n^2)$ (van Gent 1995)

1 $n/(1+c_A)$ with $c_A = c(1-n)/n$ (VARANS for the superficial velocity, Liu et al. 1999)

default: 0

○ ○ ○ ●

B 266 int Added mass form for CFD VRANS porous media

Added mass coefficient C_m of the CFD VRANS porous media (B 270–B 291, porous sediment bed S 10 2), which scales the momentum tendencies with $1/(1+C_m)$ where the porosity $n < 1$; c is B 260. Option 1 (Jensen et al. 2014) is continuous at the surface of the porous media ($C_m \to 0$ for $n \to 1$).

0 $C_m = c$ wherever $n < 1$

1 $C_m = c(1-n)/n$

default: 1

○ ○ ○ ●

B 268 int CFD VRANS drag treatment

Time integration of the Darcy-Forchheimer resistance $(A+B|U|)u$ of the CFD VRANS porous media (B 270–B 291, porous sediment bed S 10 2), with the coefficients averaged from the cells to the velocity faces and $|U|$ the magnitude of the velocity vector. The explicit source is stable only for $\Delta t < 2/(n(A+2B|U|)/(1+C_m))$; the point-implicit treatment in the Runge-Kutta stages is unconditionally stable, and the steady state is the Darcy-Forchheimer balance independent of Δt. Option 1 needs the Runge-Kutta momentum solver (N 40 2, 12, 22, 44, and N 40 3, 4, 13, 23, 24, 33 without F 80 4); otherwise the explicit source is used and a message is printed.

0 explicit source

1 point-implicit in the Runge-Kutta stages

default: 1

○ ○ ○ ●

B 270 * VRANS porous media as rectangular box

double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double n, double d_{50}, double α, double β
Coordinates in m, porosity n, mean grain diameter d_{50} in m, resistance coefficients α and β.

default: na

○ ○ ○ ●

B 274 * VRANS porous media as vertical cylinder

double x_{center}, double y_{center}, double z_{start}, double z_{end}, double $radius$, double n, double d_{50}, double α, double β
Coordinates and radius in m, porous parameters as in B 270.

default: na

○ ○ ○ ●

B 281 * VRANS porous media as wedge in x-direction

double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double n, double d_{50}, double α, double β
Coordinates in m, porous parameters as in B 270.

default: na

○ ○ ○ ●

B 282 * VRANS porous media as wedge in y-direction

double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double n, double d_{50}, double α, double β
Coordinates in m, porous parameters as in B 270.

default: na

○ ○ ○ ●

B 291 * VRANS porous media as plate in x-direction

double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double $d_{thickness}$, double n, double d_{50}, double α, double β
Coordinates and plate thickness in m, porous parameters as in B 270.

default: na

○ ○ ● ●

B 295 int VRANS source treatment for turbulence model
With option 1, the CFD vegetation zones (B 310–B 322) also add the turbulence production of the vegetation drag to k, ϵ and ω (Lopez and Garcia 2001).

0 OFF

1 VRANS source terms

2 turn off turbulence inside porous media

default: 1

○ ○ ○ ●

B 307 int Number of smoothing iterations for the vegetation parameters
Smooths the vegetation density N, diameter D, drag coefficient C_d and the porosity of the vegetation zones (B 310–B 322) B 307 times after their initialization. Giving B 307 in the control file also switches on the vegetation model, like B 308.

default: 0

○ ○ ○ ●

B 308 int Consider vegetation porosity effects on fluid acceleration
When ON, the porosity inside the vegetation zones is set to $1 - N\pi D^2/4$. Giving B 308 in the control file also switches on the vegetation model, which is required for B 310–B 322.

0 OFF

1 ON

default: 1

○ ○ ○ ●

B 309 double Inertia coefficient C_M for vegetation

default: 2.0

○ ○ ○ ●

B 310 * VRANS vegetation as rectangular box

double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double N, double D, double C_d
Coordinates in m, N the number of cylinders per unit area in $1/m^2$, D the diameter of the cylinders in m and C_d the drag coefficient. Requires B 308 to activate the vegetation model.

default: na

○ ○ ○ ●

B 321 * VRANS vegetation as wedge in x-direction

double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double N, double D, double C_d
Parameters as in B 310. Requires B 308.

default: na

○ ○ ○ ●

B 322 * VRANS vegetation as wedge in y-direction

double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double N, double D, double C_d
Parameters as in B 310. Requires B 308.

default: na

● ○ ○ ●

B 411 * int ID, double Q patchBC discharge

Discharge in m^3/s for the patch with this ID (see B 440–B 442). Makes the patch an inlet. The velocity normal to the face is $U_n = Q/A$ with the wetted area A of the patch (CFD: from the level set, SFLOW: from the water depth); it is imposed in the wetted part of the patch, the gas phase is closed there. Q is positive in the positive coordinate direction of the face normal, i.e. inflow at the faces 1, 3, 5 and outflow at the faces 2, 4, 6. The flow direction is normal to the face unless set with B 416 or B 417. Not together with B 412, B 414, B 415, B 418 or B 421. The prescribed and the measured discharge, velocity, wetted area and water level of each patch are written to the screen output.

default: na

○ ○ ○ ●

B 412 * int ID, double p patchBC pressure

Pressure in Pa for the patch with this ID. Makes the patch an outlet with zero gradient velocity. With a free surface (level set or VOF), p is the pressure at the free surface and the hydrostatic pressure is added in the ghost cells: $p + \rho g\,(h - z)$, with the water level h from B 413/B 422 or, without these, from the local free surface (level set). Without a free surface, p is the pressure at the patch. The pressure next to the patch is relaxed to this value by the pressure correction. Not together with an inlet condition (B 411, B 414, B 415, B 421). Note: before the October 2026 revision the value was imposed as a constant over the depth and had no effect on the flow; for a water level controlled outlet use B 413.

default: na

● ○ ○ ●

B 413 * int ID, double h patchBC waterlevel

Water level (free surface z-coordinate) in m for the patch with this ID. Sets the level set (CFD) or the free surface and water depth (SFLOW) in the ghost cells of the patch. Without an inlet condition the patch is an outlet with the hydrostatic pressure below h (see B 412). Together with an inlet condition (B 411, B 414, B 415, B 421) it fixes the water level at the inflow. Not together with B 422.

default: na

● ○ ○ ●

B 414 * int ID, double U patchBC velocity perpendicular to face

Velocity in m/s normal to the patch with this ID, positive in the positive coordinate direction of the face normal. Makes the patch an inlet. The flow direction is normal to the face unless set with B 416 or B 417.

default: na

● ○ ○ ●

B 415 * int ID, double U, double V, double W patchBC velocity components

Velocity components in m/s for the patch with this ID, imposed as given in the ghost cells (SFLOW: U, V). Makes the patch an inlet. Not together with B 411, B 414, B 416, B 417 or B 421.

default: na

● ○ ○ ●

B 416 * int ID, double α patchBC horizontal inflow angle

Horizontal inflow angle in deg for the patch with this ID, measured counter-clockwise (seen from above) from the face normal: 0 deg is perpendicular to the face. The velocity normal to the face from B 411, B 414 or B 421 is kept, the tangential velocity is $U_n \tan\alpha$. For the side faces 1–4, $|\alpha| < 84$ deg. Not together with B 417. Note: before the October 2026 revision the angle was measured from the face (90 deg perpendicular).

default: na

● ○ ○ ●

B 417 * int ID, double N_x, double N_y, double N_z patchBC flow direction

Flow direction vector (any length) for the patch with this ID. The velocity normal to the face from B 411, B 414 or B 421 is kept and the velocity is aligned with the vector: $\mathbf{u} = U_n \mathbf{N}/(\mathbf{N}\cdot\mathbf{n})$, with the face normal $\mathbf{n}$. The vector must not be (nearly) parallel to the face. Not together with B 416.

default: na

● ○ ○ ●

B 418 * int ID, int flag patchBC free stream outflow

Free stream outflow for the patch with this ID: the patch is an outlet as a patch without flow values (zero gradient velocity and water level, hydrostatic pressure from the local free surface). Not together with an inlet condition. The condition is activated for the ID whenever the command is given.

0 OFF

1 ON

default: 0

● ○ ○ ●

B 421 * int *ID*, int flag patchBC discharge hydrograph

Requires a file named e.g. "hydrograph_Q_1.dat", where the number is the patchBC ID, with the columns time in s and discharge in m^3/s. The hydrograph is activated for the ID whenever the command is given. Makes the patch an inlet as B 411, the values are interpolated linearly in time and kept constant before the first and after the last time. Not together with B 411.

0 OFF

1 ON

default: 0

● ○ ○ ●

B 422 * int *ID*, int flag patchBC waterlevel hydrograph

Requires a file named e.g. "hydrograph_FSF_1.dat", where the number is the patchBC ID, with the columns time in s and water level in m. The hydrograph is activated for the ID whenever the command is given. As B 413 with the water level interpolated linearly in time. Not together with B 413.

0 OFF

1 ON

default: 0

● ○ ○ ●

B 440 * int *ID*, int *face*, double x_{start}, double x_{end}, double y_{start}, double y_{end} patchBC as line

The patch boundary condition turns wall faces (code 21 in DIVEMesh, C 11–C 16) on the selected side into inlets or outlets. Patches are addressed with the ID; all geometries B 440–B 442 with the same ID form one patch. Each patch is one of two kinds: an inlet with the velocity prescribed (B 411, B 414, B 415, B 421) and zero gradient pressure, or an outlet with the pressure prescribed (B 412, B 413, B 418, B 422) and zero gradient velocity. A patch without flow values is a free outlet with the hydrostatic pressure of the local free surface. A water level (B 413, B 422) can be given for both kinds. Combinations that prescribe velocity and pressure, or the same quantity twice, stop the program with an error message. The face selects the side of the boundary: 1 = x_{min}, 2 = y_{max}, 3 = y_{min}, 4 = x_{max}, 5 = bottom, 6 = top. A wall face belongs to the patch when its face centre lies within x_{start}–x_{end} and y_{start}–y_{end} (all z); for a patch on face 1 at $x = 0$ choose e.g. $x_{start} = -0.1$, $x_{end} = 0.1$. Coordinates in m. A face belongs to one patch only, the first geometry in the order B 440, B 441, B 442 that contains it. The patches and their number of faces are written to the screen output at start-up; a patch without faces gives a warning. The patch BC is available for SFLOW (A 10 2, faces 1–4) and CFD (A 10 6); not on a curvilinear grid (G 50) in SFLOW.

default: na

● ○ ○ ●

B 441 * int ID, int $face$, double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} patchBC as rectangular box

As B 440, with the face centre within the box given in m. In SFLOW the z-range is not used.

default: na

● ○ ○ ●

B 442 * int ID, int $face$, double x_M, double y_M, double z_M, double $radius$ patchBC as circle

As B 440, with the patch as a circle with center and radius in m: the face plane lies within one cell of the center and the face centre within the radius. In SFLOW the z-coordinate is not used.

default: na

● ● ● ●

B 500 * int ID, int $type$, double H, double T Additional wave source

Defines a wave train that is superposed (summed) on the B 92 wave in the wave generation and relaxation zones. The B 92 wave is source 1, additional sources need an $ID \geq 2$. $type$ is the wave theory as in B 92, H the wave height in m and T the wave period in s (for irregular and focused waves H_s and T_p). The spectrum shape, directional spreading and the water depth are taken from the input of the B 92 wave (B 84–B 88, B 94, B 130–B 136). Wavemaker theories (B 92 20–24), HDC and the runtime FNPF coupling (61, 62) and type 0 are not allowed, and at most one nonlinear wave theory (other than B 92 1, 2, 3, 31, 41, 51) may be used among all sources, including the B 92 wave. With B 89 1, the additional sources are decomposed together with the B 92 wave (types 5, 31, 41, 51; otherwise the program runs with B 89 0). Direction, phase and time window are set with B 501, the origin with B 502, the frame of both with B 505 and the seed of irregular sources with B 504. The equivalent input of all sources is written to the screen at start-up.

default: na

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B 501 * int ID, double β, double φ, double t_s, double t_e, double t_r Direction, phase and time window of an additional wave source

For the B 500 source with this ID: propagation direction β in deg relative to the direction of the B 92 wave (B 105; measured from the x-axis with B 505 1), phase shift φ in deg (applied as a time shift of $\varphi/360 \cdot T$), start and end time t_s and t_e in s of the time window in which the source is active, and the duration t_r in s of a cosine ramp at the start and, if t_e is given, at the end of the window ($t_r = 0$: no ramp). t_e must be larger than t_s.

default: 0.0 ; 0.0 ; 0.0 ; $1.0 \cdot 10^{20}$; 0.0

● ● ● ●

B 502 * int *ID*, double x_0, double y_0 Origin of an additional wave source

Origin in m of the B 500 source with this ID, given in the wave generation frame of the B 92 wave, i.e. relative to the B 105 origin and along and across the B 105 direction, or in domain coordinates with B 505 1.

default: 0.0 ; 0.0

● ● ● ●

B 504 * int *ID*, int *seed* Seed number of an additional irregular wave source

Seed for the random phases of the irregular B 500 source with this ID (replaces B 139 and B 138 for this source). Without B 504 the source uses the seeds of the B 92 wave.

default: 0 (seeds of the B 92 wave)

● ● ● ●

B 505 * int *ID*, int *frame* Frame of an additional wave source

Frame in which the direction (B 501) and the origin (B 502) of the B 500 source with this ID are given. With the global frame, the direction is measured from the x-axis and the origin is given in domain coordinates; they are converted into the generation frame of the B 92 wave at start-up.

0 generation frame of the B 92 wave (B 105)

1 global frame

default: 0

● ○ ● ○

B 510 * int *ID*, int *mode*, double *dir*, double t_r Tidal or current background

Defines a background state with this ID (≥ 1) for the boundary zones of NHFLOW and SFLOW; it is assigned to a zone with B 523. *dir* is the direction of propagation of the tide in deg (from the x-axis), t_r the duration in s of a cosine spin-up ramp $r(t)$ applied to the whole background. The background water level is $\eta_b = r(t)\eta_0 + \eta_t$ and the depth averaged current $U_b = r(t)U + \eta_t\sqrt{g/h}\cos(dir)$, $V_b = r(t)V + \eta_t\sqrt{g/h}\sin(dir)$, with the constant offsets η_0, U, V of B 514, the tide η_t and the local still water depth h. With mode 1, $\eta_t = r(t)\sum a\cos(2\pi t/T - \varphi - ks)$ with the constituents of B 511. With mode 2, η_t is interpolated linearly in time from the file "background-*ID*.dat" with the lines "t η" or "t η U V" (time in s, at least two lines, increasing times, comments after #); with four columns, the file gives the current directly. Mode 3 is a constant background from B 514 only. With mode 4, $\eta_t = r(t)\eta_f(t,x,y)$ varies in space as well: the file "background-*ID*.dat" starts with a line "points N", followed by N lines "x y"

(model coordinates in m, an ordered polyline, e.g. along the open edges), and then the lines "$t\ \eta_1\ \ldots\ \eta_N$" or "$t\ \eta_1\ U_1\ V_1\ \ldots\ \eta_N\ U_N\ V_N$" (at least two lines, increasing times, comments after #); a position is projected onto the nearest segment of the polyline and the values are interpolated linearly between its two points and linearly in time, and held beyond the ends. With U and V, the file gives the current directly, as in mode 2. Mode 4 cannot be combined with B 515. Without B 515 the background is uniform in space ($s = 0$). The background settings are written to the screen at start-up.

1 harmonic tide (B 511)

2 time series file

3 constant (B 514)

4 time series at points (file)

default: na

● ○ ● ○

B 511 * int ID, double a, double T, double φ Harmonic constituent of a background
Amplitude a in m, period T in s (> 0) and phase φ in deg of one tidal constituent $a\cos(2\pi t/T - \varphi)$ of the B 510 background with this ID, which must have mode 1. Repeat the command for several constituents.

default: na

○ ○ ● ○

B 513 * int ID, int $profile$, double par Vertical profile of the background current
Vertical profile of the current of the B 510 background with this ID, scaled so that its depth average equals the background current. It is used at the inflow of the edges, in the relaxation and beach zones and for waves on the background (B 530). Log-law: $\ln(1 + z/z_0)$ with the roughness length $z_0 = par$ in m (> 0), z the height above the bed. Power law: $(z/h)^{1/n}$ with $n = par$ ($n = 7$ if $par \leq 0$).

0 uniform

1 log-law

2 power law

default: 0 (uniform)

● ○ ● ○

B 514 * int ID, double η_0, double U, double V Constant level and current of a background
Constant water level offset η_0 in m and constant current components U and V in m/s of the B 510 background with this ID. They are added to the tide of modes 1, 2 and 4 and are the only contribution for mode 3.

default: 0.0 ; 0.0 ; 0.0

● ○ ● ○

B 515 * int ID, double x_0, double y_0, double h_{ref} Progressive background

The tide of the B 510 background with this ID (mode 1 or 2; modes 3 and 4 stop the program with a message) travels as a long wave in the direction dir of B 510, starting from (x_0, y_0) in m, with the celerity $c = \sqrt{g h_{ref}}$. $s = (x - x_0)\cos(dir) + (y - y_0)\sin(dir)$ is the distance along dir; a harmonic constituent gets the wave number $k = 2\pi/(Tc)$, a time series is evaluated at $t - s/c$. h_{ref} in m; $h_{ref} \leq 0$ uses the still water level F 60.

default: not set (uniform background)

● ● ● ●

B 520 * int ID, int $method$, int $priority$ Boundary zone along a domain edge

Defines a boundary zone with this ID; its geometry is given with B 521 (required), its wave sources with B 524 and its background with B 523. Relaxation and beach zones (methods 1 and 2) are used in addition to the zones of B 96, B 107 and B 108. A wave generation zone requires relaxation generation (B 98 2, or B 98 6), otherwise the program stops with an input error. A beach zone is active even without B 99 1 or 2 and uses the softer relaxation function of B 99 1 when B 99 1 is selected; in NHFLOW it relaxes the water level as well as the velocities. Where generation zones overlap, the zone with the highest $priority$ selects the wave sources of the target; the zones of B 96 and B 108 have priority 0, and for equal priority they come first, followed by the B 520 zones in the order given. The mesh refinement (G 1) of SFLOW, FNPF and NHFLOW does not refine inside B 520 relaxation and beach zones.

Methods 3–6 are open boundary conditions for NHFLOW and SFLOW only, which set the velocities and the water level in the ghost cells of the whole domain side given by B 521 (one such zone per side). The side must be an inflow or outflow boundary in DIVEMesh; a side in y-direction (only in 3D) must be an outflow boundary. They need a background (B 523), except the clamped discharge edge with B 525. The Riemann edge combines the incoming characteristic of the background (plus the waves of its B 524 sources, ramped with B 101) with the outgoing characteristic of the interior, so that outgoing waves leave the domain (SFLOW: the waves averaged over B 160 + 1 levels); in addition, in NHFLOW at x_{min} the non-hydrostatic pressure of the incoming waves is imposed. The Flather edge keeps the interior water depth and sets the inward normal velocity to $u_b - \sqrt{g/h}(\eta - \eta_b)$, with the normal background velocity u_b and the interior water level η. The clamped level edge imposes the background water level with the normal velocity of the interior; the clamped discharge edge imposes the normal velocity of the background current, or the discharge of B 525, with the water depth of the interior. At inflow, the tangential velocity is that of the background (and waves), at outflow that of the interior. Flather and clamped edges carry the background only, waves enter through a Riemann edge or a relaxation zone. On a multi-block grid (G 50 1), edges on a face of a block 1, 2, ... are placed with B 522.

With the atmospheric forcing of SFLOW and NHFLOW (W 200 > 0) and its inverse barometer switched on (W 210 1), the water level targeted by the Riemann, Flather and clamped level edges (methods 3–5) and by the relaxation and beach zones (also those of B 96, B 107 and B 108) is raised by the inverse barometer $\eta_{ib} = -r(p_a - p_{ref})/(\rho_w g)$

of the local air pressure p_a (reference pressure p_{ref} = W 209, ramp r of W 204, water density W 1), so that a low pressure enters the domain with its static response.

1 wave generation (relaxation towards the wave theory)

2 numerical beach

3 Riemann edge (NHFLOW, SFLOW)

4 Flather edge (NHFLOW, SFLOW)

5 clamped level edge (NHFLOW, SFLOW)

6 clamped discharge edge (NHFLOW, SFLOW)

default: na

● ● ● ●

B 521 * int ID, int $edge$, double s_0, double s_1, double d Geometry of a B 520 zone

The zone with this ID lies along the domain edge $edge$ and reaches the distance d in m (positive) from the edge into the domain; as for B 107 and B 108, the relaxation weight depends on the distance from the edge. s_0 and s_1 in m give the extent along the edge, measured from the edge's start (y_{min} for edges 1, 2; x_{min} for edges 3, 4); with $s_1 \leq s_0$ the zone covers the whole edge. For the edge methods 3–6 of B 520 only $edge$ is used: they act on the whole domain side (with B 522, on the whole face of the block).

1 x_{min}

2 x_{max}

3 y_{min}

4 y_{max}

default: na

○ ○ ● ○

B 522 * int ID, int b Edge zone on a face of a block

On a multi-block grid (G 50 1), places the B 520 zone with this ID on a face of block b (1, 2, ..., the block numbers of DIVEMesh; the zones of block 0, the domain of the model, need no B 522). Only the edge methods 3–6 of B 520 are available, on the $-i$ face (B 521 edge 1) or the $+i$ face (edge 2) of the block; the zone carries its background (B 523) or the discharge of B 525 and no wave sources (B 524). The face must have no interface and must not be opened with G 54. With G 55, a block with such a zone is cut along i only, so that its $-i$ and $+i$ faces stay whole. NHFLOW only; a zone without B 520, on block 0, with another method, edge or module, or on a face that is not available stops the run with a message.

default: na (zone on block 0)

● ○ ● ○

B 523 * int *ID*, int *background* Background of a B 520 zone

Assigns the B 510 background with the ID *background* to the B 520 zone with this ID. A relaxation zone then relaxes towards the background plus the ramped waves (B 101 acts on the waves only), a beach zone towards the background level and current instead of still water (with B 97 1 the beach velocity follows B 97), a Riemann or Flather edge uses the background as the outer state, a clamped level edge its level and a clamped discharge edge its normal current. NHFLOW and SFLOW only.

default: na (still water)

● ● ● ●

B 524 * int *ID*, int *source* Wave source of a B 520 zone

Restricts the target of the wave generation zone with this ID to the given wave source (1: the B 92 wave, ≥ 2: B 500 sources). Repeat the command to use several sources in one zone. Generation zones without B 524 use the sum of all sources; also with B 89 1. A Riemann edge (B 520 method 3, NHFLOW and SFLOW) carries the waves of its B 524 sources and no waves without B 524; beach zones and the other edges cannot have sources.

default: na (all sources)

● ○ ● ○

B 525 * int *ID*, double Q, double t_r Discharge of a clamped discharge edge

Discharge Q in m^3/s into the domain through the clamped discharge edge (B 520 method 6) with this ID, ramped up with a cosine ramp over t_r in s. The normal velocity at the edge is Q divided by the wet cross-section (water depth times cell width, summed along the edge); it replaces the current of a background.

default: na

● ● ● ○

B 526 double Low-pass time scale of the combined and two-band beach

Time scale τ in s of the first-order low pass of the combined beach (B 99 6: NHFLOW, SFLOW, FNPF) and of the two-band active beach (B 99 7: FNPF). Waves much longer than τ pass the relaxation zone of B 99 6 to the absorbing edge, shorter waves are damped in the zone; a much smaller τ slows the long waves in the zone and reflects them. A value ≤ 0 uses a quarter of the peak period of the B 92 wave (B 99 6) or of the reference period B 528 (B 99 7).

default: 0.0 (automatic, see above)

○ ● ● ○

B 527 double Low-pass time scale of the combined and two-band generation

Time scale τ in s of the first-order low pass of the deviation from the target in the combined generation (B 98 6: FNPF, NHFLOW) and in the two-band active absorption at the wavemaker (B 98 7: FNPF). A value ≤ 0 uses a quarter of the peak period of the B 92 wave (B 98 6) or of the reference period B 528 (B 98 7).

default: 0.0 (automatic, see above)

○ ● ● ○

B 528 double Reference period of the two-band and SIRW absorption

Period T_r in s for the two-band absorption (B 98 7, B 99 7: FNPF), whose short part is absorbed with the linear velocity profile of T_r, and for the automatic band and gauge positions of the SIRW absorption (B 98 8, B 99 8: FNPF, NHFLOW; see B 534). A value ≤ 0 uses the peak period of the B 92 wave. Required for B 98 7 with the piston wavemaker (B 92 21), which has no wave period.

default: 0.0 (peak period of the B 92 wave)

● ○ ● ○

B 529 int Edge mass balance log

Every B 529 time steps, a line with the time, the water volume V of the domain, dV/dt over the interval, the mean volume flux into the domain through each open edge (B 520 methods 3–6, from the continuity flux of the solver at the boundary face), their sum and the residual dV/dt minus the sum is written to REEF3D_Log/REEF3D-NHFLOW-iowave-mass-balance.dat (SFLOW: REEF3D_SFLOW_Log/REEF3D-SFLOW-iowave-mass-balance.dat). The residual contains the contribution of relaxation and beach zones and of the time discretisation. Requires a zone with a background or an edge (B 520–B 523).

0 OFF

>**0** output interval in time steps

default: 0

● ○ ● ○

B 530 int *mode*, int *N* Waves on the background

Adapts the generated waves to the background of their zones (B 523). Every N time steps, the background level and current are averaged over the generation zone columns (B 98 2) and Riemann edges that carry the source, and for each wave component the wave number k is recomputed on the effective depth $h_{eff} = h + \eta_b$, with h the depth of the wave theory (B 94). With mode 2, k follows from the Doppler-shifted dispersion relation $\omega = \sigma + kU_n$, with the absolute frequency ω of the input wave and the background current U_n in the direction of the component; with a vertical current

profile (B 513, NHFLOW), U_n is weighted with the wave kinematics (Stewart and Joy 1974). Components blocked by an opposing current are faded out, and a message is printed. The change of k is blended in over the N steps. Available in NHFLOW and SFLOW with a background, for linear waves (B 92 2), 5th-order Stokes waves (B 92 5; k is then refined with their 5th-order dispersion relation including the current) and spectral irregular waves (B 92 31 with B 85 other than 4–6), and not with B 89 1. If B 530 is not given, mode 2 is selected when a zone's background carries a current (a tide or time series, or U, $V \neq 0$), mode 1 when it only sets a level, and B 530 is off otherwise (also without waves, for other wave types and with B 89 1).

0 OFF

1 waves on h_{eff}

2 waves on h_{eff} with Doppler shift

default: automatic (see above) ; 10

○ ● ● ○

B 534 double d, double Δx, double f_{low}, double f_{high} SIRW absorption at the wavemaker

Gauges and frequency band of the SIRW absorption at x_{min} (B 98 8): distance d in m of the first gauge from the boundary, spacing Δx in m of the two gauges, and the band f_{low}–f_{high} in Hz in which the reflected wave is separated by the FIR filters. The surface elevation at the gauges is averaged over y. Values ≤ 0 are set automatically from the reference period T_r (B 528, else the peak period of the B 92 wave): $f_{low} = 0.4/T_r$, $f_{high} = 3/T_r$, $\Delta x = 0.4\,L(f_{high})$ and $d = 0.7\,L(f_{low})$ (Frigaard and Brorsen 1995), with the linear wave length L at the depth of the wave theory. If a gauge lies outside the domain, SIRW is switched off with a message.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 (automatic)

○ ● ● ○

B 535 double d, double Δx, double f_{low}, double f_{high} SIRW absorption at the beach

As B 534, for the SIRW active beach at x_{max} (B 99 8), with the gauges in front of the outflow boundary.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 (automatic)

○ ○ ● ○

B 536 int Target of the NHFLOW wavemaker absorption

Target surface elevation η_T of the active absorption at the NHFLOW piston wavemaker (B 92 21 with B 98 3 or 4). The velocity correction is $(\eta_T - \eta_M)\sqrt{g/d}$ (see B 98).

0 shallow-water estimate from the paddle velocity, $\eta_T = U\sqrt{h/g}$ (as before)

1 linear surface elevation of the paddle signal at the first cell (progressive and evanescent modes)

default: 0

6.3 C :: Concentration

Transport of a passive or density-active scalar concentration in the CFD solver. The concentration c modifies the fluid properties as $\rho = \rho_{1,2} + c \cdot C1/C3$ and $\nu = \nu_{1,2} + c \cdot C2/C4$ in phase 1 and phase 2, respectively. The concentration module is activated with C 10.
Valid for:

S	F	N	C
○	○	○	●

C 1 double Additional density from concentration in phase 1
Density increase per unit concentration in phase 1 (water), in kg/m^3: $\rho = W1 + c \cdot C1$.

default: 10.0 kg/m^3

S	F	N	C
○	○	○	●

C 2 double Additional viscosity from concentration in phase 1
Kinematic viscosity increase per unit concentration in phase 1, in m^2/s: $\nu = W2 + c \cdot C2$.

default: 0.0 m^2/s

S	F	N	C
○	○	○	●

C 3 double Additional density from concentration in phase 2
Density increase per unit concentration in phase 2 (air), in kg/m^3: $\rho = W3 + c \cdot C3$.

default: 10.0 kg/m^3

S	F	N	C
○	○	○	●

C 4 double Additional viscosity from concentration in phase 2
Kinematic viscosity increase per unit concentration in phase 2, in m^2/s: $\nu = W4 + c \cdot C4$.

default: 0.0 m^2/s

S	F	N	C
○	○	○	●

C 10 int Time scheme concentration function
Activates the concentration transport and selects its time integration. Concentration diffusion is controlled by D 20. The CFD mesh refinement (G 1, Section 6.7.1) supports only C 10 3 with C 15 0, 1 or 4 and D 20 0 – 2, without heat transfer (H 10), suspended sediment (S 12) and porous structures (B 200); otherwise the refinement is switched off with a message.

0 OFF

1 Adams-Bashforth

2 2nd-order TVD Runge-Kutta

3 3rd-order TVD Runge-Kutta

default: 0

○ ○ ○ ●

C 15 int Convection discretization for concentration function
Use together with C 10. When suspended sediment transport is active (S 12 $\geq$ 1), the concentration convection is replaced by an implicit WENO HJ scheme. With the CFD mesh refinement (G 1), only the options 0, 1 and 4 are supported (see C 10).

0 OFF

1 FOU

2 CDS2

3 QUICK

4 WENO5 FLUX

5 WENO5 HJ

6 CDS4

7 WENO3 FLUX

8 WENO3 HJ

10 High Resolution TVD scheme with Minmod limiter

11 High Resolution TVD scheme with van Leer limiter

12 High Resolution TVD scheme with Umist limiter

13 High Resolution TVD scheme with van Albada limiter

14 High Resolution TVD scheme with Superbee limiter

15 High Resolution scheme with Smart limiter

16 High Resolution TVD scheme with Limo3 limiter

42 High Resolution TVD scheme with Weller limiter

default: 0

○ ○ ○ ●

C 50 double c_1, double c_2 Initial concentration inside and outside area 1
The concentration is initialized to c_2 everywhere and set to c_1 inside the area defined by C 51 – C 58 and C 75.

default: 1.0 ; 0.0

○ ○ ○ ●

C 51 double Area 1 start, x-direction
Area 1 is the box C 51 $\leq x <$ C 54, C 52 $\leq y <$ C 55, C 53 $\leq z <$ C 56. The large default values mean that the box is not limited in this direction.

default: $-1.0 \cdot 10^7$ *m*

○ ○ ○ ●

C 52 double Area 1 start, y-direction

default: $-1.0 \cdot 10^7$ *m*

○ ○ ○ ●

C 53 double Area 1 start, z-direction

default: $-1.0 \cdot 10^7$ *m*

○ ○ ○ ●

C 54 double Area 1 end, x-direction

default: $1.0 \cdot 10^7$ *m*

○ ○ ○ ●

C 55 double Area 1 end, y-direction

default: $1.0 \cdot 10^7$ *m*

○ ○ ○ ●

C 56 double Area 1 end, z-direction

default: $1.0 \cdot 10^7$ *m*

○ ○ ○ ●

C 57 double *a*, double *b*, double *c*, double *d* Area 1 as half-space below a plane
Cells with $ax + by + cz < d$ are set to the concentration c_1 of C 50. Only active if at least one of the four values is positive.

default: 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ○ ●

C 58 double x_0, double y_0, double z_0, double r Area 1 as sphere

Center coordinates and radius of a sphere in m in which the concentration is set to c_1 of C 50. Only active for $r > 0$.

default: 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ○ ●

C 75 * double x_0, double z_0, double α, double s, double l, double v Area 1 as tilted box

Tilted rectangle in the x-z plane (extended over the whole y-direction) starting at the corner (x_0, z_0) in m, inclined by the angle α in deg, with length s in m along the inclination and thickness l in m normal to it. Inside, the concentration is set to c_1 of C 50 and the initial velocity is set to $u = v\cos\alpha$, $w = -v\sin\alpha$ with v in m/s. Can be used multiple times.

default: not set

6.4 D :: Discretization

Here the discretization of the convection and diffusion terms in the momentum equations of the CFD solver and the pressure algorithm are chosen. All convection methods are implemented in terms of conservative finite differences. The exception are the WENO HJ schemes, which are the non-conservative form of the WENO scheme.
Valid for:

S F N C

○ ○ ○ ●

D 10 int Convection discretization

Convection scheme for the momentum equations. WENO5 FLUX is the recommended default. The nonlinear weights of the WENO5 schemes (options 4 and 5) are selected with D 12.

0 OFF

1 FOU

2 CDS2

3 QUICK

4 WENO5 FLUX

5 WENO5 HJ

6 CDS4

7 WENO3 FLUX

8 WENO3 HJ

10 High Resolution TVD scheme with Minmod limiter

11 High Resolution TVD scheme with van Leer limiter

12 High Resolution TVD scheme with Umist limiter

13 High Resolution TVD scheme with van Albada limiter

14 High Resolution TVD scheme with Superbee limiter

15 High Resolution scheme with Smart limiter

16 High Resolution TVD scheme with Limo3 limiter

60 CDS6, sixth-order central scheme

default: 4

○ ○ ○ ●

D 11 int Convection velocities for momentum equations

Interpolation of the convecting velocities to the cell faces. Applies to the flux-based convection schemes (FOU, CDS2, QUICK, WENO FLUX, CDS4, CDS6, High Resolution) of D 10, F 35, F 85 and C 15.

1 FOU

2 CDS2

default: 2

○ ○ ○ ●

D 12 int Nonlinear weights of the WENO5 momentum convection

Only used with the WENO5 schemes for the momentum equations (D 10 = 4, 5). The corresponding switches are A 213 for SFLOW, A 318 for FNPF, A 527 for NHFLOW and F 37 for the level set. Option 1 uses the WENO-Z weights with $p = 2$ (Borges et al. 2008). Option 2 uses the TENO5 weights (Fu, Hu and Adams 2016): each of the three stencils is either kept with its ideal weight or discarded, depending on the cutoff D 13.

0 WENO-JS

1 WENO-Z

2 TENO5

default: 0

○ ○ ○ ●

D 13 double TENO5 cutoff C_T

Used with D 12 2. A stencil is discarded when its normalised smoothness measure falls below C_T. Smaller values discard fewer stencils, i.e. the scheme stays closer to the linear upwind-biased scheme. Dimensionless.

default: $1.0 \cdot 10^{-5}$

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D 20 int Diffusion discretization

Selecting the implicit treatment of the diffusion term in the momentum equations for explicit time stepping has the advantage of removing diffusion from the CFL criterion, which determines the time step size. As the diffusion term enters the CFL criterion in the order of $1/dx^2$, effectively larger time steps for D 20 = 2 will be used on especially finer grids. When RANS turbulence models are used, the effective diffusion can be several magnitudes higher than the molecular diffusion. Then implicit diffusion treatment will increase the time step significantly. D 20 also sets the diffusion treatment of the concentration (C 10). The turbulence quantities are diffused implicitly for any D 20 $>$ 0 and not at all for D 20 = 0. With D 20 = 0 and a turbulence model (T 10 $>$ 0), a warning is printed, since then the eddy viscosity does not enter the momentum equations.

0 OFF

1 explicit

2 implicit

3 implicit Crank-Nicolson (3D only)

default: 2

● ● ● ●

D 21 int Print out implicit diffusion solver time and iterations

Prints the solver iterations and time of the implicit diffusion step to the screen every P 12 iterations. In FNPF, it prints the iterations and time of the implicit breaking wave damping (A 350).

0 OFF

1 ON

default: 0

○ ○ ○ ●

D 22 int Wall boundary condition for implicit diffusion

Treatment of solid neighbour cells in the implicit momentum diffusion (D 20 = 2, not used with VOF-PLIC F 80 = 4). If X 10 $>$ 0 (6DOF), D 22 is set to 2 automatically. With D 22 = 1 and S 10 = 1, the direct-forcing flags are also updated after each bed change.

1 Coefficients towards solid cells and direct-forcing solid regions are replaced by the value of the cell itself (zero normal gradient); velocities inside direct-forcing solid regions are set to zero.

2 The wall ghost cells are coupled implicitly to the new velocities: the ghost cell value of the wall boundary condition (B 20, B 23, B 29), written as a linear combination of the two adjacent fluid cells, enters the matrix coefficients. Where this is not possible (e.g. prescribed inflow values), the current ghost cell value is used on the right-hand side.

default: 1

○ ○ ○ ●

D 23 int Time accuracy of the implicit momentum diffusion

Time integration of the implicit diffusion (D 20 = 2) within the Runge-Kutta stages of the momentum schemes N 40 = 2, 3, 4, 12, 13, 22, 23, 24, 33, 44 (not with VOF-PLIC, F 80 = 4).

1 first order: the diffusion is treated fully implicitly in every stage

2 second order: implicit-explicit (IMEX) combination, in which the implicit diffusion of each stage is complemented by the explicit diffusion of the earlier stage velocities

default: 2

○ ○ ○ ●

D 30 int Pressure algorithm

The projection method (PJM) solves a Poisson equation for the pressure correction every time step. Option 10 computes a hydrostatic pressure $p = \phi \rho |g|$ from the level set function without solving a Poisson equation. The isentropic air compressibility (W 32) requires D 30 1; otherwise the run stops. The CFD mesh refinement (G 1, Section 6.7.1) also requires D 30 1; otherwise the refinement is switched off with a message.

0 OFF

1 PJM CORR, projection method with pressure correction

10 hydrostatic pressure

default: 1

6.5 E :: DEM

REEF3D::DEM is a standalone discrete element module for rigid particles of arbitrary shape (spheres, boxes, cylinders, ellipsoids and closed STL surfaces), coupled to REEF3D::CFD and REEF3D::NHFLOW. It is independent of the 6DOF module (X 10) and of the sediment particle models (Q, S); contacts between DEM particles and 6DOF bodies are not implemented. The DEM is switched on with E 10 1, and the shapes, materials, walls and particles are read from the file `dem.txt` (Section 6.5.1).

Each shape has a signed distance function in its body frame (analytic for spheres, boxes and cylinders, a grid SDF for ellipsoids and STL shapes), surface nodes for the contact detection and a volume quadrature for the fluid loads. Contacts with other particles, with the planar walls of `dem.txt` and, optionally, with the REEF3D level sets (topography and solids in CFD, the bed and solids in NHFLOW) are hard contacts. They are solved as velocity impulses with Newton restitution and Coulomb friction (non-smooth contact dynamics, projected

Gauss-Seidel, E 13, E 14), and penetration is removed with split impulses (E 19). Friction and restitution of a contact are the means of the two materials. The DEM step is limited by the particle travel (E 20), not by a contact stiffness, so it normally runs with the fluid time step or a few substeps (E 16). The fluid coupling is selected with E 11: unresolved particles (smaller than the grid) get drag (Haider-Levenspiel with the Di Felice voidage correction, implicit in the DEM step), buoyancy, added mass and optionally the fluid acceleration force, and their reaction is spread to the fluid with a smoothing kernel; resolved particles (larger than the grid) are represented with direct forcing inside the smoothed particle indicator, and their hydrodynamic force follows from the forcing integral. In NHFLOW, resolved coupling is only used while a particle is fully submerged; a surface-piercing particle switches to unresolved loads with one-way coupling (fluid to particle). Particles are distributed over the MPI ranks; large particles are replicated on all ranks (E 24).

Gravity is taken from W 20 – W 22, so W 22 $= -9.81$ has to be set. For CFD cases in still water, the pressure should be initialized (I 10 1), otherwise the start-up flow disturbs the particles. In CFD, the reaction of the particles on the fluid is applied in all momentum schemes of N 40. With E 10 1, the NHFLOW and CFD mesh refinement (G 1, Section 6.7.1) is switched off. In 2D cases (one cell in y), the particles move in the x-z plane and keep their 3D mass; resolved particles should then be extruded shapes whose y extent equals the cell width. A particle that leaves the domain is deactivated. The particle volume can enter the NHFLOW VRANS terms as porosity (E 28); use it together with B 265 1 (Section 6.2).

Output: every E 15 seconds, the particle surfaces with velocity, id and coupling mode are written to `REEF3D_DEM_VTP/REEF3D-DEM-NNNNNN.vtp` (ParaView), and one line per particle is appended to `REEF3D_DEM/REEF3D-DEM-state.dat` with the columns time, id, x, y, z, quaternion q_w, q_x, q_y, q_z, u, v, w, ω_x, ω_y, ω_z, active flag, coupling mode (1 resolved), hydrodynamic force F_x, F_y, F_z and fluid velocity at the particle. With P 18 = 1 or 2, the particle surfaces are also written to the LAGOON store (P 18 = 1: instead of the VTP files); the state file is always written. Every P 12 iterations, the number of particles, substeps, contacts, solver iterations, residual, maximum penetration, kinetic energy and DEM time are printed to the terminal.
Valid for:

S	F	N	C
○	○	●	●

E 10 int DEM

Turns on the discrete element module. The particles are read from the file `dem.txt` in the case directory; the program stops if the file is missing or refers to an undefined shape or material. The particles are part of the checkpoints (P 48) of CFD and NHFLOW: a restart (I 45) continues with the particles of the checkpoint, also on another decomposition, and the DEM output files continue.

0 OFF

1 ON

default: 0

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E 11 int DEM fluid coupling

Coupling between the particles and the flow. The equivalent diameter d_{eq} is that of the sphere with the particle volume; the grid size Δx is the mean cell size (in NHFLOW the mean horizontal cell size).

0 dry: gravity and contacts only

1 unresolved: particles smaller than the grid

2 resolved: particles larger than the grid

3 hybrid: per particle, resolved if $d_{eq}/\Delta x \geq$ E 12, otherwise unresolved

default: 1

○ ○ ● ●

E 12 double Hybrid coupling threshold

With E 11 3, particles with $d_{eq}/\Delta x \geq$ E 12 are resolved. In NHFLOW, fixed particles larger than E 12 cells give no drag (E 29) and no porosity (E 28 2), only a contact surface.

default: 5.0

○ ○ ● ●

E 13 int Maximum number of contact solver iterations

Maximum number of symmetric projected Gauss-Seidel sweeps per DEM step.

default: 1000

○ ○ ● ●

E 14 double Contact solver tolerance

The iterations stop when the largest velocity change of a sweep, relative to max(v_{max}, 1 mm/s), is smaller than E 14.

default: $1.0 \cdot 10^{-5}$

○ ○ ● ●

E 15 double DEM print interval

Time interval in s for the VTP files and the state file. With 0, no particle output is written.

default: 0.1 s

○ ○ ● ●

E 16 int Minimum number of DEM substeps
Minimum number of DEM substeps per fluid time step. More substeps are taken automatically when required by E 20 (at most 1000).

default: 1

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E 17 double Added mass coefficient
Added mass coefficient C_a of unresolved particles.

default: 0.5

○ ○ ● ●

E 18 double Resolved coupling: interface half width
Half width of the smoothed Heaviside function of the particle indicator for resolved particles, in cells (at least 0.1).

default: 1.5

○ ○ ● ●

E 19 double Penetration correction factor
Fraction of the contact penetration removed per DEM step with split impulses, which move the particles without adding kinetic energy. The correction velocity is limited to 0.5 m/s.

default: 0.2

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E 20 double Maximum particle travel per DEM step
Maximum travel of a particle per DEM substep as a fraction of the smallest bounding radius of the moving particles (at least 0.01). The number of substeps per fluid time step is the larger of E 16 and $v_{max}\Delta t/(\mathrm{E}\,20 \cdot r_{min})$.

default: 0.25

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E 21 int Contact points per particle pair
Maximum number of contact points kept per particle pair and normal direction cluster (the deepest point and points spread over the contact patch). With 0, all contact points are kept.

default: 6

○ ○ ● ●

E 22 int Unresolved coupling: fluid acceleration force
Includes the force of the local fluid acceleration (pressure gradient and added mass of the fluid acceleration) on unresolved particles. Floating particles in waves need it to follow the orbital motion.

0 OFF

1 ON

default: 1

○ ○ ● ●

E 23 double Unresolved coupling: kernel radius
Radius of the kernel that spreads the reaction force of unresolved particles to the fluid, in cells. The radius is max(d_{eq}, E 23 Δx).

default: 2.0

○ ○ ● ●

E 24 double Distributed particles: maximum size
With several MPI ranks, particles with a bounding radius up to E 24 times the smallest subdomain extent are distributed: each is owned by the rank that holds its centroid and has copies on the neighbouring ranks. Larger particles are replicated on all ranks; keep their number small. With 0, all particles are replicated. Because contact forces in piles are statically indeterminate, the results depend slightly on the decomposition.

default: 0.25

○ ○ ● ●

E 25 double Unresolved coupling: solid volume fraction kernel
Radius of the kernel for the solid volume fraction of unresolved particles, in equivalent diameters. The radius is max(E 25 d_{eq}, E 23 Δx). The voidage around a particle (without its own share) enters its drag law.

default: 2.0

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E 26 int NHFLOW unresolved coupling: exchange points
With 1, the fluid velocity is averaged over the submerged volume quadrature points of each particle (the non-uniform part gives a drag torque), and the reaction is shared by these points and spread from each with a 4-point Peskin kernel. Only with E 11 1 or 3.

0 exchange at the centroid

1 exchange at the volume quadrature points

default: 0

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E 27 int NHFLOW unresolved coupling: point-implicit drag
The reaction of the particle drag on the fluid is linearized in the local fluid velocity and applied point-implicitly in every Runge-Kutta stage. This keeps dense beds stable at large time steps. Only with E 11 1 or 3.

0 OFF

1 ON

default: 0

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E 28 int NHFLOW: particle volume as porosity
The submerged volume of the unresolved particles is spread with the momentum kernel and gives the porosity $n = 1 - \alpha$ ($n \geq 0.3$), which enters the NHFLOW VRANS terms. With VRANS porous structures (B 200), the two porosities are multiplied. The porosity follows the particles with the relaxation time E 30. Only with E 11 1 or 3. Use it with B 265 1 (Section 6.2); with B 265 0 the VRANS inertia factor makes long waves in the particle layer too fast, and a note is printed.

0 OFF

1 moving particles

2 moving and fixed particles (fixed particles up to E 12 cells)

default: 0

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E 29 int NHFLOW unresolved coupling: drag of fixed particles
Fixed particles up to E 12 cells exert drag on the fluid, e.g. filter layers or fixed toes. Otherwise fixed particles only give a contact surface. Only with E 11 1 or 3.

0 OFF

1 ON

default: 0

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E 30 double NHFLOW: relaxation time of the particle porosity
Relaxation time in s with which the porosity of E 28 follows the particles. NHFLOW has no term for the time derivative of the porosity, and an instantaneous porosity (E 30 0) produces spurious flow around moving particles.

default: 1.0 s

6.5.1 DEM input file `dem.txt`

The file `dem.txt` has one keyword per line; # starts a comment, lengths are in m, angles in deg (rotation about x, then y, then z) and velocities in m/s. Particle positions are the centroids. Unknown keywords are reported and ignored.

`material` **id** ρ μ e material with density (kg/m^3), friction coefficient and restitution coefficient. Without any `material` line, material 0 with $\rho = 2650$, $\mu = 0.5$, $e = 0.3$ is used.

`wall_material` μ e friction and restitution of the walls and of the REEF3D level sets (default 0.5, 0.3)

`shape` **id** `sphere` r sphere with radius r

`shape` **id** `box` l_x l_y l_z box with edge lengths l_x, l_y, l_z

`shape` **id** `cylinder` r l cylinder with radius r and length l along the body z-axis

`shape` **id** `ellipsoid` a b c ellipsoid with semi-axes a, b, c

`shape` **id** `stl` **file [scale]** closed ASCII or binary STL surface, scaled with scale (default 1). It is moved to its centroid and rotated into its principal axes.

`node_spacing` f spacing of the surface contact nodes as a fraction of d_{eq} (default 0.15)

`sdf_resolution` n cells of the grid SDF (ellipsoid, STL) along the longest axis (default 32)

`quadrature` n volume quadrature points along the longest axis (default 6)

`grid_walls` **0|1** contacts with the REEF3D level sets: topography and solids (CFD), bed and solids (NHFLOW) (default 1)

`wall` n_x n_y n_z p_x p_y p_z plane wall through the point **p** with the normal **n** pointing into the domain

`particle` **shape mat** x y z **[**r_x r_y r_z **[**u v w**]]** moving particle with optional rotation and initial velocity

`fixed` **shape mat** x y z **[**r_x r_y r_z**]** immovable particle; with resolved coupling it acts as an obstacle

`block` **shape mat** n_x n_y n_z x_0 y_0 z_0 Δx Δy Δz **[rand [seed]]** regular array of $n_x \times n_y \times n_z$ moving particles starting at $\mathbf{x}_0$ with the spacings Δx, Δy, Δz; with rand 1 each particle gets a random orientation (random number seed: seed)

6.6 F :: Free Surface

Free surface capturing (level set and VOF) and the initial free surface of all models. The level set function ϕ is positive in phase 1 (water) and negative in phase 2 (air). The initial free surface commands (F 51 – F 72, F 112 – F 115) are applied in the order in which they are listed here; later commands overwrite earlier ones.
Valid for:

S	F	N	C
○	○	○	●

F 30 int Time scheme level set
Activates the level set method for two-phase flow. The level set time scheme is used for the segregated momentum schemes N 40 = 12, 13, 44. For the fully coupled schemes (N 40 = 2, 3, 4, 22, 23, 24, 33) the level set is advanced inside the momentum time scheme, but F 30 > 0 is still required for the two-phase fluid properties.

0 OFF

2 2nd-order TVD Runge-Kutta

3 3rd-order TVD Runge-Kutta

default: 3

○	○	○	●

F 35 int Convection discretization for level set
If F 80 > 0 (VOF), F 35 is set to 0 automatically. If F 85 > 0, the scheme of F 85 replaces the scheme selected here. The nonlinear weights of the WENO5 schemes (options 4 and 5) are selected with F 37. The CFD mesh refinement (G 1, Section 6.7.1) supports the options 1 – 8 and 10 – 16; otherwise the refinement is switched off with a message.

0 OFF

1 FOU

2 CDS2

3 QUICK

4 WENO5 FLUX

5 WENO5 HJ

6 CDS4

7 WENO3 FLUX

8 WENO3 HJ

10 High Resolution TVD scheme with Minmod limiter

11 High Resolution TVD scheme with van Leer limiter

12 High Resolution TVD scheme with Umist limiter

13 High Resolution TVD scheme with van Albada limiter

14 High Resolution TVD scheme with Superbee limiter

15 High Resolution scheme with Smart limiter

16 High Resolution TVD scheme with Limo3 limiter

42 High Resolution TVD scheme with Weller limiter

default: 5

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F 37 int Nonlinear weights of the WENO5 level set discretization

Used for the WENO5 level set convection (F 35 = 4, 5), the WENO5 convection of the three-phase level sets (F 305 = 4, 5) and the WENO HJ discretization of the level set reinitialization (F 40, F 310). It is not used for the schemes of F 85. The corresponding switch for the momentum equations is D 12. Option 1 uses the WENO-Z weights with $p = 2$ (Borges et al. 2008). Option 2 uses the TENO5 weights (Fu, Hu and Adams 2016): each of the three stencils is either kept with its ideal weight or discarded, depending on the cutoff F 38.

0 WENO-JS

1 WENO-Z

2 TENO5

default: 0

○ ○ ○ ●

F 38 double TENO5 cutoff C_T

Used with F 37 2. A stencil is discarded when its normalised smoothness measure falls below C_T. Smaller values discard fewer stencils, i.e. the scheme stays closer to the linear upwind-biased scheme. Dimensionless.

default: $1.0 \cdot 10^{-5}$

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F 40 int Reinitialization time scheme

PDE-based reinitialization of the level set function to a signed distance function. F 40 > 0 is also required for the level set initialization with F 51 – F 72 and F 112 – F 115. The CFD mesh refinement (G 1) requires F 40 3 with the level set; otherwise the refinement is switched off with a message.

0 OFF

3 3rd-order TVD Runge-Kutta

default: 3

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F 42 double Length for initial level set reinitialization

Length in m over which the level set is reinitialized at the start of the simulation; the number of initial reinitialization steps is $2\,L/(F\,43 \cdot dx)$. If F 42 ≥ 0 it overrides the default length L, which is the maximum domain length. Use it to reduce the initialization time in long domains. In FNPF, F 42 likewise replaces the domain length for the initial coastline distance (A 339).

default: -1.0 (use maximum domain length)

● ● ○ ●

F 43 double Factor for reinitialization time step size

Pseudo time step of the reinitialization as a fraction of the smallest local cell size: $\Delta\tau = F\,43 \cdot dx$. Also used for the PDE-based coastline reinitialization in FNPF (A 339 = 0), for the topography and solid reinitialization (G 540), for the level set of floating bodies in CFD (X 10) and for the level set of the 2D floating bodies in SFLOW (X 10).

default: 0.55

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F 44 int Number of reinitialization time steps

Number of reinitialization steps per time step (after the initial reinitialization). For the fully coupled momentum schemes (N 40 = 2, 3, 4, 22, 23, 24, 33), the reinitialization is done in every Runge-Kutta stage, with F 44 steps in the final stage and F 44 $-$ 1 steps (at least 1) in the other stages. For VOF-PLIC (F 80 = 4) the number of steps is set internally for each Runge-Kutta stage.

default: 3

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F 45 double Factor for the calculation of the interface thickness ϵ

Half-thickness of the smoothed interface as a multiple of the cell size, $\epsilon = F\,45 \cdot dx$, over which density and viscosity change between the phases. dx is the local cell size in the direction of the interface normal $\mathbf{n} = \nabla\phi/|\nabla\phi|$, $dx = \sqrt{n_x^2\Delta x^2 + n_y^2\Delta y^2 + n_z^2\Delta z^2}$: on a uniform grid this is the cell size for every orientation, for a free surface on a grid stretched in z it is the vertical cell size. At cell faces the mean of the two neighbouring cells is used. The same width is used for the heat, concentration and rheology fluid properties, the volume correction (F 46) and the active wave generation and absorption.

default: 2.1

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F 46 int Type of Picard iterations for volume correction
Mass conservation correction: the level set is shifted by a constant so that the volume of phase 1 matches the reference volume. The correction is applied once per time step after the level set convection and reinitialization (for the fully coupled momentum schemes after the final Runge-Kutta stage; for VOF-PLIC with F 92 = 1 after each redistancing) and once after the initial reinitialization. The number of iterations is given by F 47. The shift w is computed with Newton iterations, $w = (V_{ref} - V)/(dV/dw)$, where dV/dw is the volume integral of the smoothed delta function (the interface area), and the volume is integrated with the actual cell volumes.

0 OFF

2 Volume correction using the volume at the beginning of the time step

3 Volume correction using the total volume balance: the initial volume plus the accumulated inflow minus outflow discharge

default: 0

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F 47 int Number of Picard iterations for volume correction
Use together with F 46.

default: 10

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F 50 int Fix free surface at inflow or outflow
Selects at which open boundaries the level set function (CFD) or free surface elevation (SFLOW, NHFLOW) is prescribed; at the other open boundaries it is extrapolated.

1 Inflow fixed

2 Outflow fixed

3 Fix both

4 Fix none

default: 2

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F 51 double Phase 1 start, x-direction
Phase 1 (water) is initialized in the box F 51 $\leq x <$ F 54, F 52 $\leq y <$ F 55, F 53 $\leq z <$ F 56. The box is only used if at least one of F 51 – F 56 is given; the large default values mean that the box is not limited in this direction. Setting only F 56 therefore defines a horizontal free surface at z = F 56.

default: $-1.0 \cdot 10^{20}$ m

○ ○ ○ ●

F 52 double Phase 1 start, y-direction

default: $-1.0 \cdot 10^{20}$ m

○ ○ ○ ●

F 53 double Phase 1 start, z-direction

default: $-1.0 \cdot 10^{20}$ m

○ ○ ○ ●

F 54 double Phase 1 end, x-direction

default: $1.0 \cdot 10^{20}$ m

○ ○ ○ ●

F 55 double Phase 1 end, y-direction

default: $1.0 \cdot 10^{20}$ m

○ ○ ○ ●

F 56 double Phase 1 end, z-direction

default: $1.0 \cdot 10^{20}$ m

○ ○ ○ ●

F 57 double a, double b, double c, double d Phase 1 as half-space below a plane
Cells with $ax + by + cz < d$ are set to phase 1. Only active if at least one of the four values is positive.

default: 0.0 ; 0.0 ; 0.0 ; 0.0

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F 58 double x_m, double y_m, double z_m, double r Phase 1 as sphere
Center coordinates and radius in m of a sphere of phase 1, e.g. a droplet. Only active for $r > 0$.

default: 0.0 ; 0.0 ; 0.0 ; 0.0

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F 59 double x_m, double y_m, double z_{start}, double z_{end}, double r Phase 1 as vertical cylinder
Center coordinates, vertical extent and radius in m of a vertical cylinder of phase 1, e.g. a water column. Only active for $r > 0$.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

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F 60 double Initial still water level for the whole domain
Still water level in m (z-coordinate). In CFD it initializes the level set as $\phi = F\,60 - z$ and overrides F 51 – F 59. In SFLOW, FNPF and NHFLOW it is the reference water level, and also in REEF3D::SEASTATE (A 10 7, Section 6.1.4) when it runs uncoupled. It is also the starting value of the outflow water level ramp (F 64).

default: $-1.0 \cdot 10^{20}$ m (not set)

● ○ ● ●

F 61 double Water level at the inflow boundary
Water level in m (z-coordinate) prescribed at the inflow boundary. In SFLOW it sets the water depth in the inflow ghost cells. In SFLOW and NHFLOW it is also the inflow level of the river initialisation F 65 1.

default: $-1.0 \cdot 10^{20}$ m (not set)

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F 62 double Water level at the outflow boundary
Water level in m (z-coordinate) prescribed at the outflow boundary (use F 50 = 2 or 3). The outflow condition is only applied for F 62 > 0. Use F 64 to ramp up from F 60 to F 62 and F 63 to initialize a sloping free surface.

default: $-1.0 \cdot 10^{20}$ m (not set)

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F 63 double x_{start} for linear free surface interpolation
From x = F 63 in m to the end of the domain, the initial free surface is interpolated linearly from the still water level (F 60 or F 56) to the outflow water level F 62. Requires F 62.

default: $-1.0 \cdot 10^{20}$ m (not set)

● ○ ● ●

F 64 int Number of iterations for outflow water level ramp up
The outflow water level is ramped from F 60 to F 62 over F 64 time steps with a cosine function. 0 applies F 62 directly.

default: 0

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F 65 int River free surface initialisation along the i-direction
Initial free surface of a river whose water level follows the slope of the bed. The cross-sections are the grid lines i = const: on a Cartesian grid the river runs along x, on a curvilinear grid (G 50) along i from the inflow. The level is uniform in a cross-section; cells above it, and side pools below it that are not connected to the lowest cell of the cross-section, start dry. If F 60 is not given, F 62 is the reference water level. REEF3D stops if F 62 is missing, or F 61 (option 1) or W 10 (option 2). Not available on multi-block grids: a message is printed and the initialisation is skipped.

0 OFF

1 water level linear in the distance along the river, from F 61 at the inflow to F 62 at the outflow

2 steady 1D flow of the discharge W 10: standard-step backwater computation upstream from the level F 62 (at least the critical level) at the outflow, with the bed roughness B 50 (SFLOW: Manning $n = k_s^{1/6}/20$; NHFLOW: log law at the centre of the bottom layer, as A 519 1). A cross-section without a subcritical solution gets the critical level. The wet cells get the strip velocity of their column along the i-direction, so that every cross-section carries W 10 (NHFLOW: the same velocity on all layers); these velocities replace those of the potential flow initialisation (I 11).

default: 0

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F 70 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Phase 1 box
Rectangular box in m that is set to phase 1 (water). Can be used multiple times.

default: not set

○ ○ ○ ●

F 71 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Phase 2 box
Rectangular box in m that is set to phase 2 (air). Can be used multiple times.

default: not set

● ○ ● ●

F 72 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double h Region with prescribed initial water level

In the horizontal region given in m, the initial free surface is set to the water level h in m (z-coordinate). On a curvilinear grid (G 50), the cells whose centre lies in the region are set. Can be used multiple times, e.g. for dam break cases.

default: not set

○ ○ ○ ●

F 80 int Time scheme VOF

Activates the volume of fluid (VOF) method instead of the level set method; F 35 is then set to 0. The VOF implementation is in beta: the algorithms for open channel flow and wave generation are designed for the level set method. VOF-PLIC (option 4) works with N 40 = 3, 13, 23, 33. The algebraic VOF schemes (former options 1 and 3, with the compression factor F 84) have been removed; REEF3D stops with a message if F 80 = 1 or 3 is given. VOF cannot be combined with the isentropic air compressibility W 32 (the run stops), and the CFD mesh refinement (G 1, Section 6.7.1) is switched off with a message.

0 OFF

4 VOF-PLIC, geometric reconstruction of the interface (under development)

default: 0

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F 85 int Convection discretization VOF

Convection scheme of the former algebraic VOF methods. VOF-PLIC (F 80 = 4) transports the volume fraction geometrically and does not use F 85. If F 85 $>$ 0, the selected scheme replaces the level set convection scheme F 35. The schemes HRIC, HRIC modified and CICSAM (former options 51, 52, 53) have been removed; REEF3D stops with a message if they are selected.

0 OFF

1 FOU

2 CDS2

3 QUICK

4 WENO5 FLUX

5 WENO5 HJ

6 CDS4

10 High Resolution TVD scheme with Minmod limiter

11 High Resolution TVD scheme with van Leer limiter

12 High Resolution TVD scheme with Umist limiter

13 High Resolution TVD scheme with van Albada limiter

14 High Resolution TVD scheme with Superbee limiter

15 High Resolution scheme with Smart limiter

16 High Resolution TVD scheme with Limo3 limiter

42 High Resolution TVD scheme with Weller limiter

default: 0

VOF-PLIC settings (F 88 – F 101). The following commands are only used with F 80 = 4 (VOF with piecewise linear interface construction). This method is under development; the defaults are recommended.

○ ○ ○ ●

F 88 int VOF-PLIC interface normal calculation
Method to compute the interface normal vector for the PLIC reconstruction.

2 Weymouth

3 Wang

4 first-order

5 least squares

6 WENO

7 from the level set function

8 mass centre

9 ELVIRA 2D

10 MYC 2D

11 MYC 2D, version 2

12 MYC 2D, version 3

13 MYC 3D

14 MYC 3D, version 2

default: 10

○ ○ ○ ●

F 90 int VOF-PLIC transport scheme
Geometric transport of the VOF field.

4 COSMIC 2D, advected distance $u\Delta t$

5 as 4, with a two-step velocity estimate for the advected distance

default: 4

○ ○ ○ ●

F 91 double Density threshold factor for N 40 = 33

For the momentum scheme N 40 = 33 (with and without VOF-PLIC), the velocity in cells with density below $\rho_{thres} = F91 \cdot W1 + W3$ is blended with the previous velocity to stabilize low-density cells near the interface.

default: 0.1

○ ○ ○ ●

F 92 int VOF-PLIC density method

Calculation of density and viscosity from the VOF field. Options 22 and 32 are identical to 2 and 3, except that in the implicit PLIC diffusion (F 80 4 with D 20 2) the eddy viscosity is weighted with the volume fraction, so that turbulent viscosity only acts in the water phase.

1 from a redistanced level set function (with reinitialization, F 40)

2 VOF-based, cell-weighted average at the cell faces

3 VOF-based with sub-cell fractions

22 as 2, eddy viscosity weighted with the volume fraction

32 as 3, eddy viscosity weighted with the volume fraction

default: 2

○ ○ ○ ●

F 93 double VOF-PLIC air phase threshold

Cells with a volume fraction below F 93 are treated as pure air.

default: 0.001

○ ○ ○ ●

F 94 double VOF-PLIC water phase threshold

Cells with a volume fraction above F 94 are treated as pure water.

default: 0.999

○ ○ ○ ●

F 95 double VOF-PLIC water threshold for the phase marker

Cells with a volume fraction above F 95 are marked as water in the phase marker functions used for the reconstruction and the spray deletion (F 98).

default: 0.999

○ ○ ○ ●

F 96 double x_{start}, double x_{end} VOF-PLIC density softening region
Between x_{start} and x_{end} in m, the density method F 92 = 3 is replaced by the method F 92 = 2.

default: not set

○ ○ ○ ●

F 98 int VOF-PLIC spray deletion
Removes small isolated fluid fractions (spray) after each time step based on the phase marker (F 93, F 95).

0 OFF

1 ON

default: 1

○ ○ ○ ●

F 100 int VOF-PLIC initial inclined plane free surface
Initializes the volume fraction with a plane interface defined by F 101 (e.g. for run-up on a sloping beach). Cells cut by the plane get their geometric volume fraction. The option selects the coordinate direction along which the plane position e of F 101 is given; the plane passes through the point at distance e on this coordinate axis. Applied after F 51 – F 58 and F 60 and overwrites them.

0 OFF

1 position e given on the x-axis

2 position e given on the y-axis

3 position e given on the z-axis (any other value > 0 acts the same)

default: 0

○ ○ ○ ●

F 101 double n_x, double n_y, double n_z, double s, double e VOF-PLIC inclined plane parameters
Normal vector (n_x, n_y, n_z) of the plane for F 100 (normalized internally) and position e in m on the coordinate axis selected by F 100. Phase 1 (water) lies on the side opposite to the normal vector, i.e. the normal points into the air. The value s is not used.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

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F 112 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Phase 1 as wedge in x-direction
Inside the horizontal region given in m, phase 1 fills the space from $\min(z_{start}, z_{end})$ up to the line that rises linearly from z_{start} at x_{start} to z_{end} at x_{end}. Can be used multiple times.

default: not set

○ ○ ○ ●

F 113 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Phase 1 as wedge in y-direction
As F 112, with the sloping line from z_{start} at y_{start} to z_{end} at y_{end}. Can be used multiple times.

default: not set

○ ○ ○ ●

F 114 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Phase 1 as inverse wedge in x-direction
Inside the horizontal region given in m, phase 1 fills the space above the line from z_{start} at x_{start} to z_{end} at x_{end}, up to $\max(z_{start}, z_{end})$. Can be used multiple times.

default: not set

○ ○ ○ ●

F 115 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Phase 1 as inverse wedge in y-direction
As F 114, with the sloping line from z_{start} at y_{start} to z_{end} at y_{end}. Can be used multiple times.

default: not set

○ ○ ○ ●

F 150 int Benchmarks
Initializes the level set function and prescribes the velocity field according to standard interface advection benchmark cases.

0 OFF

1 Vortex

2 Slotted disc

3 3D deformation

4 3D Taylor-Green vortex

11 1D convection of step, triangle and sine profiles

21 Slotted disc in the y-z plane

22 Slotted disc in the x-y plane

default: 0

○ ○ ○ ●

F 151 int Inverse sign of the benchmark level set

Inverts the sign of the initial level set function of the 3D deformation benchmark (F 150 = 3).

0 OFF

1 ON

default: 0

Three-phase flow (F 300 – F 395). Two level set functions ϕ_1 and ϕ_2 describe three fluids: phase 1 (water, W 1, W 2) where $\phi_1 > 0$; where $\phi_1 < 0$, phase 3 (e.g. oil, W 6, W 7) where $\phi_2 > 0$ and phase 2 (air, W 3, W 4) where $\phi_2 < 0$. Both functions are initialized to -1 and then set by the commands below; the interface thickness follows F 45.

○ ○ ○ ●

F 300 int Three-phase flow level set time scheme

With F 300 > 0, the CFD mesh refinement (G 1, Section 6.7.1) is switched off with a message.

0 OFF

3 3rd-order TVD Runge-Kutta

default: 0

○ ○ ○ ●

F 305 int Three-phase flow level set convection discretization

The nonlinear weights of the WENO5 schemes (options 4 and 5) are selected with F 37.

0 OFF

1 FOU

2 CDS2

3 QUICK

4 WENO5 FLUX

5 WENO5 HJ

6 CDS4

10 – 16, 42 High Resolution TVD schemes, limiters as for F 35

default: 5

○ ○ ○ ●

F 310 int Three-phase flow reinitialization

0 OFF

3 3rd-order TVD Runge-Kutta

default: 3

○ ○ ○ ●

F 360 double Initial interface of ϕ_1 in x-direction
$\phi_1 = F\,360 - x$, i.e. phase 1 for $x <$ F 360 in m.

default: $-1.0 \cdot 10^{20}$ m (not set)

○ ○ ○ ●

F 361 double Initial interface of ϕ_1 in y-direction
$\phi_1 = F\,361 - y$, i.e. phase 1 for $y <$ F 361 in m.

default: $-1.0 \cdot 10^{20}$ m (not set)

○ ○ ○ ●

F 362 double Initial interface of ϕ_1 in z-direction
$\phi_1 = F\,362 - z$, i.e. phase 1 below $z =$ F 362 in m.

default: $-1.0 \cdot 10^{20}$ m (not set)

○ ○ ○ ●

F 369 * double x_0, double z_0, double α, double s, double l, double v Tilted box of phase 1 for ϕ_1
Tilted rectangle in the x-z plane as for C 75: corner (x_0, z_0) in m, angle α in deg, length s and thickness l in m. Inside, $\phi_1 > 0$ and the initial velocity is $u = v\cos\alpha$, $w = -v\sin\alpha$ with v in m/s. Can be used multiple times.

default: not set

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F 370 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Positive box for ϕ_1
Rectangular box in m where $\phi_1 > 0$ (phase 1). Can be used multiple times.

default: not set

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F 371 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Negative box for ϕ_1
Rectangular box in m where $\phi_1 < 0$. Can be used multiple times.

default: not set

○ ○ ○ ●

F 374 * double x_c, double z_c, double r Positive cylinder in y-direction for ϕ_1
Horizontal cylinder along the y-axis with center (x_c, z_c) and radius r in m: $\phi_1 > 0$ inside and $\phi_1 < 0$ outside (the whole domain is overwritten). Can be used multiple times.

default: not set

○ ○ ○ ●

F 375 * double x_c, double z_c, double r Negative cylinder in y-direction for ϕ_1
As F 374 with inverted sign: $\phi_1 < 0$ inside and $\phi_1 > 0$ outside. Can be used multiple times.

default: not set

○ ○ ○ ●

F 380 double Initial interface of ϕ_2 in x-direction
$\phi_2 = F\,380 - x$ in m.

default: $-1.0 \cdot 10^{20}$ m (not set)

○ ○ ○ ●

F 381 double Initial interface of ϕ_2 in y-direction
$\phi_2 = F\,381 - y$ in m.

default: $-1.0 \cdot 10^{20}$ m (not set)

○ ○ ○ ●

F 382 double Initial interface of ϕ_2 in z-direction
$\phi_2 = F\,382 - z$ in m.

default: $-1.0 \cdot 10^{20}$ m (not set)

○ ○ ○ ●

F 390 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Positive box for ϕ_2
Rectangular box in m where $\phi_2 > 0$. Can be used multiple times.

default: not set

○ ○ ○ ●

F 391 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Negative box for ϕ_2
Rectangular box in m where $\phi_2 < 0$. Can be used multiple times.

default: not set

○ ○ ○ ●

F 394 * double x_c, double z_c, double r Positive cylinder in y-direction for ϕ_2
As F 374, for ϕ_2. Can be used multiple times.

default: not set

○ ○ ○ ●

F 395 * double x_c, double z_c, double r Negative cylinder in y-direction for ϕ_2
As F 375, for ϕ_2. Can be used multiple times.

default: not set

6.7 G :: Grid: mesh refinement and geometry

The G commands act on the numerical grid. G 1 – G 47 control the patch-based mesh refinement REEFAMR of SFLOW, FNPF, NHFLOW, CFD and SEASTATE, the placement of its patches on the MPI ranks and the restart on a new decomposition for the measured load (Section 6.7.1). G 50 – G 58 select curvilinear and multi-block horizontal grids for SFLOW and NHFLOW and the cutting and placement of the blocks (Section 6.7.2). G 505 – G 541 control the treatment of the geometry level set functions of the bed topography

and of solids read from DIVEMesh, and of floating bodies represented on the horizontal grid (Section 6.7.3).
The mesh refinement commands were formerly A 270 – A 281 and the geometry commands G 5, G 40 and G 41. The new numbers are: A 270 → G 1, A 271 → G 2, A 272 → G 3, A 275 → G 4, A 280 → G 5, A 281 → G 6, A 276 → G 10, A 277 → G 11, A 278 → G 12, A 279 → G 13, A 273 → G 20, A 274 → G 22, and G 5 → G 505, G 40 → G 540, G 41 → G 541. The numbers G 40 and G 41 are now used again by the mesh refinement (load balancing); an old control file with G 40 or G 41 for the reinitialization of the geometry must use G 540 and G 541 instead. The validity circles do not include SEASTATE (A 10 7); its use of the commands is given in the text.
Valid for:
S F N C

6.7.1 Mesh refinement (SFLOW, FNPF, NHFLOW, CFD, SEASTATE)

The commands G 1 – G 47 control the patch-based mesh refinement (REEFAMR) of SFLOW, FNPF, NHFLOW, CFD and SEASTATE. Level 0 is the normal grid of the model. In SFLOW, FNPF, NHFLOW and SEASTATE, every refined level halves the horizontal cell size of the level below it (and, in FNPF and NHFLOW with G 6 1, the sigma layer thickness). In CFD (A 10 6), the patches are 3D boxes and every refined level halves the cell size in every direction that has more than one cell. The refined region is built from tiles of G 4 cells and divided into rectangular patches. By default, the patches are solved together with level 0 using one global time step; with G 7 1, every level takes two steps per step of the next coarser level. The refined region can be given by boxes (SFLOW, FNPF, NHFLOW: G 10; CFD: G 15), by flow criteria (SFLOW: G 20, G 22; FNPF and NHFLOW: G 20 – G 23; CFD: G 24 – G 26) and by a zone around a moving body (G 12, G 13; in NHFLOW also around the fixed solids A 581 – A 590). In SFLOW, FNPF and NHFLOW, there is no refinement in the G 11 boxes, within four level-0 cells of in- and outflow boundaries and in the wave generation and numerical beach relaxation zones (FNPF and NHFLOW: the B 96 lengths with B 98 2 and B 99 1 or 2, and the zones of B 520; SFLOW: the zones of B 98 2 and B 99 1 or 2, also when given by B 107, B 108 or B 520). In SFLOW, FNPF and NHFLOW, all levels are written as a `.vtm` multiblock file at the print interval of the model, together with a log file and the wave gauges (P 51, evaluated on the finest grid), in the folder `REEF3D_<M>_AMR`; with P 18, the grids are also written to the LAGOON store (with P 18 1 instead of the `.vtr` and `.vtm` files). The patches are part of the checkpoints (P 48, I 45). On several MPI ranks, the patches can be placed for the load of the ranks (G 40, G 41, G 44); the load of the ranks can be reported (G 43) and written as a load map for a weighted partition (G 45), and a run can stop and restart itself on a new decomposition when the measured load is uneven (G 46, G 47).
In CFD (A 10 6), G 1 $>$ 0 refines the G 15 boxes to the finest level, the cells flagged by G 24 – G 26 and the zone around the floating bodies (G 12, G 13); the flags and the zone are evaluated at the start and every G 2 steps, with the buffer G 3, the tile size G 4 and the hysteresis G 5. All levels take the time step of the finest level (G 7 0) or are subcycled (G 7 1); the pressure is solved as one composite projection over the leaf cells of all levels. The refinement of CFD needs: N 40 2 or 3; the level set (F 30) or single phase flow, no VOF (F 80) and no multiphase flow (F 300); laminar flow or T 10 1, 2, 21 or 22; no compressibility or rheology (W 30, W 32, W 90); heat or concentration, not both, only with H 10 3 or C 10 3, the convection H 15 or C 15 0, 1 or 4, H 3 1 for heat, no S 12, without porous structures and with D 20 0 – 2; no sediment (S 10 0, Q 10 0); no DEM (E 10 0); floating bodies X

10 0 or 1 and no structures (Z 10, Z 20, Z 30); porous structures B 200 0 or 1; solids and topography with G 505 0; D 30 1 – 3; B 30 0 – 4; the convection D 10 0 – 8, 10 – 16 or 60 and, with F 30, F 35 1 – 8 or 10 – 16 and F 40 3; the diffusion D 20 0 – 2 (3 in 3D); no partition and no periodic boundary in z. Otherwise the message `CFD AMR (G 1 n): needs ...; running without mesh refinement` is printed and the run continues on level 0. The solids and the topography of the grid file are cast on every patch at its resolution. The tiles whose patch would reach an inflow or outflow boundary or a face with a patch boundary condition (B 440 – B 442) stay on level 0. The relaxation zones of the wave generation (B 98 2) and of the numerical beach (B 99 1 or 2) are refined and relaxed on the patches as on level 0 when the waves come from a wave theory with B 95 0; with B 95 $\neq$ 0, a wavemaker (B 92 20 – 29) or zone sources, backgrounds or edges (B 520 – B 524), the zones stay on level 0 with a message. Floating bodies (X 10 1) are cast onto every grid (G 42) and their loads are taken from the finest grid; the point probes (P 61), pressure probes (P 64) and wave gauges (P 51) also read the finest grid at their position, and the forces on the solids (P 81) are integrated on the finest grid. External body dynamics with REEF3D::CORAL (Z 40) stop the run together with G 7 1. The patches are written as `.vtr` files with a `.vtm` multiblock file at the output times of the CFD output (P 10) into `REEF3D_CFD_AMR`, and with P 18 also to the LAGOON store (with P 18 1 instead). CFD does not use G 6, G 10, G 11, G 20 – G 23, G 30, G 31 and G 44.

In SEASTATE (A 10 7), the refinement is static: the patches are built once at the start from the boxes (G 10) and the depth and coast criteria A 791 – A 793, with the buffer G 3 and the tile size G 4. There is no refinement in the G 11 boxes and within A 794 level-0 cells of the sides with boundary spectra or zero gradient. The levels are swept level by level or in one composite sweep (A 797), see Section 6.1.4. SEASTATE does not use G 2, G 5 – G 7, G 12, G 13, G 15, G 20 – G 31 and G 41 – G 44, and G 40 must be 0; the load map and the rebalance (G 45 – G 47) are used in stand-alone runs, also without refinement. The refinement of SEASTATE is for stand-alone runs only: with the coupling to SFLOW or NHFLOW (A 750) or with the surfbeat model (A 770 1), G 1 $>$ 0 stops the run with an input error. The patches are written as `.vtr` files with a `.vtm` multiblock index, together with a log file, into `REEF3D_SEASTATE_AMR`.

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G 1 int Mesh refinement: number of refined levels

Turns on the mesh refinement with the given number of levels above level 0 (also in SEASTATE). In SFLOW, the refinement is only created when at least one of G 10, G 12, G 20 or G 22 is given; otherwise a message is printed. When the case is outside the scope of the refinement, a message is printed and the refinement is switched off. SFLOW: A 210 3 (with A 210 2 the refinement is switched off), A 220 0–4, A 260 0, 3 or 4, S 10 0 or 1 (with S 51 0, S 31 1 or 2 and S 32 1 or 2), X 10 0–3, W 90 0. FNPF: A 310 3, A 311 other than 3, A 350 0–3, A 380 0, X 10 0 or 1. NHFLOW: A 510 2 or 3, A 511 1 or 2, A 512 0–2, A 520 0–2 (hydrostatic, PJM or pressure correction; not option 3), A 519 0 or 1, A 550 0 (or 1 with A 510 2), A 560 0, 1, 2, 21 or 22 (a turbulence model only without G 7 1, A 561 and A 563), A 599 0, no VRANS porous media (B 210 – B 291), S 10 0 or 1 (with S 12 0, S 51 0, S 31 1 or 2 and S 32 1 or 2), X 330 0, Z 20 0, Z 30 0, and X 10 0–2 (bodies with X 60 1, X 16 0, X 320 0 and A 516 0, 1 or 3). In NHFLOW, the fixed direct forcing solids (A 581 – A 590) are forced on every grid and refined around with G 12; the refinement is

switched off with a message for these solids together with G 7 1 or A 516 2 or 4, for grid solids as immersed solids (A 580 1), for the sharp hull (X 15 8) unless there is one body with the zone G 12, A 520 1, a Cartesian grid and, on several MPI ranks, G 40 1, for DEM (E 10 $>$ 0, Section 6.5) and for Lagrangian particles (L 10 $>$ 0). In SFLOW and NHFLOW, the grid must have more than one cell in y-direction; in FNPF, a grid with one cell in y-direction must be Cartesian (no G 50). The semi-implicit free surface (SFLOW A 226 1, NHFLOW A 534 1) stops the run with G 1 $>$ 0. CFD: see above; G 1 $>$ 0 is used for A 10 6 only. SEASTATE: see above.

0 OFF

>0 number of refined levels

default: 0

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G 2 int Mesh refinement: regrid interval

Number of time steps between rebuilding the patches from the current refinement flags. In SFLOW, regridding is used with the flow criteria (G 20, G 22) and the body zone (G 12); with boxes only, the patches are static. In FNPF and NHFLOW, it is used with the body zone (G 12) and the flow criteria (G 20 – G 23), and the patch layout is kept as long as it covers the flagged tiles with at most 50 % excess. In CFD, it is used with the flow criteria (G 24 – G 26) and the body zone (G 12) and counts level-0 steps; with the boxes G 15 only, the patches are static. With the body zone, the CFD patch layout is also kept as long as it covers the zone with at most 50 % excess. With subcycling (G 7 1) in SFLOW, FNPF and NHFLOW, G 2 counts time steps of the finest level, i.e. the patches are rebuilt every G 2/$2^{\mathrm{G\,1}}$ level-0 steps (at least every level-0 step). Use 0 for static patches; the flow criteria are then evaluated once, at the start. Not used in SEASTATE (static patches).

default: 4

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G 3 int Mesh refinement: buffer cells

Number of cells around a cell flagged by a flow criterion that are also refined: SFLOW G 20 and G 22, FNPF and NHFLOW G 20 – G 23, CFD G 24 – G 26, SEASTATE A 791 – A 793.

default: 2

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G 4 int Mesh refinement: tile size

Width of the refinement tiles in cells of the refined level. The refined region is made of whole tiles. In SFLOW, FNPF and NHFLOW, values below 4 are set to 4 and odd values are increased by one. In CFD, the tiles have this width in every refined direction; values below 2 are set to 2 and odd values are increased by one. In SEASTATE, the value must be even and at least 4, otherwise the run stops with an input error.

default: 8

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G 5 int Mesh refinement: hysteresis

Number of regrids a refined tile is kept after it was last flagged, so that a flag that drops below the threshold for a moment does not remove and rebuild the patch. Range 0–250 in SFLOW, FNPF and NHFLOW. Not used in SEASTATE.

default: 4

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G 6 int Mesh refinement: vertical refinement

With option 1, every refined level also doubles the number of sigma layers of the level below it. The coarse nodes are nodes of the fine grid. In CFD, the patches are always refined in the vertical direction as well.

0 OFF

1 ON

default: 0

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G 7 int Mesh refinement: subcycling in time

With option 0, all levels advance with one global time step. With option 1, level l takes 2^l steps per level-0 step: every level takes two steps of half the step of the next coarser level, with the cells around its patches filled from the coarser level linearly in time. The level-0 time step is limited by the cells of every level as if they were 2^l times larger. In SFLOW and NHFLOW, the fluxes at the coarse-fine faces are corrected after the fine steps (refluxing), so that mass and momentum are conserved across the interface; the non-hydrostatic pressure (SFLOW A 220 1–3, NHFLOW) is solved per level and followed by a synchronisation projection. In CFD, the pressure of every level is projected on that level with the pressure of the coarser level at the patch edges, followed by one composite synchronisation projection over all levels at the end of the level-0 step; floating bodies (X 10 1) are advanced by the finest level. In FNPF and NHFLOW, subcycling with a moving body requires the body zone (G 12): the finest level advances the body, and the zone margin is at least 12 level-0 cells; without G 12, a message is printed and option 0 is used. In SFLOW, the moving body (X 10 2 or 3) is taken linear in time within the level-0 step. In NHFLOW with option 1, every grid solves its own implicit diffusion (G 31 is not used), and a turbulence model (A 560 $>$ 0) or fixed direct forcing solids (A 581 – A 590) switch the refinement off; with the sharp hull (X 15 8), option 0 is used with a message. Option 1 is not available on multi-block grids (Section 6.7.2), and in CFD not with external body dynamics (Z 40). With B 95 -1, the relaxation zones use the time step of the finest level. Not used in SEASTATE.

0 one time step for all levels

1 subcycling

default: 0

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G 10 * double x_s, double x_e, double y_s, double y_e Mesh refinement box
Rectangular area (m) that is refined to all G 1 levels. The level-0 cells whose centres lie inside the box are refined. Also used in SEASTATE. In CFD, the refinement boxes are given by G 15.

default: na ; na ; na ; na

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G 11 * double x_s, double x_e, double y_s, double y_e Box without mesh refinement
Rectangular area (m) where no refinement is created, also when it is flagged by G 10, G 12 or a flow criterion. Also used in SEASTATE and on the blocks of a multi-block grid (by the cell centres).

default: na ; na ; na ; na

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G 12 double Mesh refinement around the moving body
Refines a margin of this width (m) around the wetted part of the body. Requires a moving body: SFLOW X 10 2 or 3, FNPF X 10 1, NHFLOW X 10 1 or 2, CFD X 10 1; otherwise it is ignored (except for the fixed solids of NHFLOW, see below). In SFLOW, the zone is aligned with the direction of motion and reaches ahead of the bow by the distance travelled until the next regrid (G 2). In FNPF and NHFLOW (without G 13), the zone is a rectangle aligned with x and y and grown on every side by that distance, suitable for moored and oscillating bodies. In NHFLOW, the hull triangles are refined for the finest level; with the sharp hull (X 15 8), the margin is at least 4 level-0 cells. With subcycling (G 7 1) in FNPF and NHFLOW, the margin is at least 12 level-0 cells. In NHFLOW with fixed direct forcing solids (A 581 – A 590), the horizontal extent of every solid grown by this margin is also refined to the finest level as a static box; without G 12, a message is printed and the solids are not refined around. In CFD, the zone of a floating body is also limited in the vertical: the extent of the body grown by the margin and by its vertical travel until the next regrid; the cells within G 25 cells of a body can also be refined with G 25.

default: not set (OFF)

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G 13 double L, double α Mesh refinement wake wedge

Used with G 12. Adds a wedge with half angle α (deg) that reaches back from the bow by at most L (m), and only as far as the bow has travelled. In NHFLOW and CFD, $L > 0$ also aligns the zone with the direction of motion, as in SFLOW. The default angle is the Kelvin wake angle. Use $L = 0$ to turn the wedge off.

default: 0.0 ; 19.47

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G 15 * double x_s, double x_e, double y_s, double y_e, double z_s, double z_e CFD mesh refinement box

Box (m) that is refined to the finest level G 1 in CFD: the cells of the finest level whose centres lie inside the box are refined, and every coarser level covers them with a band of three cells. In a direction with only one cell, the box covers the whole extent. The boxes are static; they can be combined with the flow criteria G 24 – G 26.

default: na ; na ; na ; na ; na ; na

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G 20 double Mesh refinement: free surface jump criterion

A wet cell is refined when the free surface elevation differs by more than this value (m) from that of a wet neighbour cell. In NHFLOW, only wet cells with a water depth above A 545·A 544 are compared, in FNPF only wet cells with a water depth of at least 10·A 344; as the value is in m, a finer level only flags steeper or shorter waves. Use 0 to turn the criterion off.

default: 0.0 (OFF)

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G 21 double Mesh refinement: free surface curvature criterion

A wet cell is refined when the second difference of the free surface elevation along x or y, $|\eta_{i+1} - 2\eta_i + \eta_{i-1}|$, exceeds this value (m). Only wet cells with a water depth above A 545·A 544 (NHFLOW) or of at least 10·A 344 (FNPF) are used. Use 0 to turn the criterion off.

default: 0.0 (OFF)

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G 22 int Mesh refinement: shoreline criterion

SFLOW: with a value > 0, the cells on both sides of a change between wet and dry are refined if water deeper than 3·A 244 lies within two cells; their buffer covers the distance the shoreline can travel until the next regrid (at least G 3). NHFLOW (only with G 30 1) and FNPF (only with wetting and drying A 343 $\geq$ 1 and G 30 1; otherwise it is ignored with a message): a fluid cell is refined when a cell of the other wet state lies within n = G 22 cells (at most 3), so that the patches follow run-up and run-down.

0 OFF

> 0 ON (FNPF and NHFLOW: number of cells n, 1–3)

default: 0

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G 23 int Mesh refinement: breaking criterion

With option 1, breaking cells are refined, so that the breaking zone stays on the finest grid as it moves. NHFLOW: with the breaking wave algorithm A 550 1, the cells with breaking viscosity. FNPF: with wave breaking A 350 1 – 3, the cells flagged as breaking on their grid (the breaking detection and damping run on every grid); without A 350 1 – 3, G 23 is ignored with a message.

0 OFF

1 ON

default: 0

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G 24 double CFD mesh refinement: interface criterion

A cell is refined when the absolute value of the level set function, $|\phi|$, is below this many interface thicknesses ϵ (F 45) of its level, so that the free surface stays on the finest grid. Use 0 to turn the criterion off.

default: 0.0 (OFF)

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G 25 int CFD mesh refinement: solid criterion

A cell is refined when it lies within this many cells (of its level) of a solid, the topography or a floating body (X 10 1), measured with their level set functions. Use 0 to turn the criterion off.

0 OFF

> 0 number of cells

default: 0

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G 26 double CFD mesh refinement: vorticity criterion

A cell is refined when the magnitude of the vorticity, $|\nabla \times \mathbf{u}|$ from central differences of the cell-centred velocities, exceeds this value (1/s). Use 0 to turn the criterion off.

default: 0.0 (OFF)

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G 30 int Mesh refinement: wetting and drying in the patches

NHFLOW: with option 0, the patches are fully wet: no patch is created within 4 level-0 cells of a dry or shallow cell (water depth at most A 545·A 544), checked at every regrid; a refinement box (G 10) over dry cells loses these tiles as well. With option 1, the patches may cover dry and shallow cells: the wetting and drying of NHFLOW runs on every grid, and the coupling between the levels interpolates from wet cells only. FNPF with wetting and drying (A 343 1 – 3): with option 0, no patch is created within 4 level-0 cells (8 with A 343 2 or 3) of a dry cell, of a thin-film cell (water depth below 10·A 344) or of the coastline damping band (3 times A 341·Δx or A 342), checked at every regrid; with option 1, the patches may cover the static (A 343 1) or moving (A 343 2, 3) coastline. Required for the shoreline criterion (G 22) of FNPF and NHFLOW.

0 patches only in wet cells

1 wetting and drying in the patches

default: 0

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G 31 int Mesh refinement: composite implicit diffusion

Used with implicit diffusion (A 512 2), which also carries the breaking viscosity of A 550 1, and with a turbulence model with A 512 0. With option 1, the diffusion is solved once over the leaf cells of all grids together (composite). With option 0, every grid solves its own diffusion. With subcycling (G 7 1), option 0 is always used; on multi-block grids, option 1.

0 every grid solves its own diffusion

1 composite solve over all grids

default: 1

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G 40 int Mesh refinement: load balancing of the patches

Only used on several MPI ranks. With option 0, the patches are cut at the partition edges and every piece is computed by the rank of the level-0 cells below it. With option 1, the patches are not cut at the partition edges but split according to the work and placed on the ranks for an even load, with the work of a patch cell given by G 41 (and

G 44); a new placement is only made when it lowers the predicted maximum load by more than 10 % (with G 41 0 in SFLOW, FNPF and NHFLOW also when the measured imbalance of the last window is above 10 % and the new placement is better). In CFD, a level-1 patch with the finer patches inside it is placed as one unit on any rank; the outputs and the loads of the bodies are taken by the rank of the finest grid at the point. In SFLOW, option 1 is not available with a moving body (X 10 2 or 3); a message is printed and option 0 is used. On an NHFLOW multi-block grid with bodies (X 10 1 or 2), option 1 stops the run. In SEASTATE, G 40 must be 0.

0 patches on the rank of the level-0 cells below

1 patches placed for the load of the ranks

default: 0

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G 41 double Mesh refinement: work of a patch cell

Work of a patch cell relative to a level-0 cell, used for the placement of the patches (G 40 1), the predicted load (G 43) and the load map (G 45, G 46). With G 41 0 on several ranks, the work is measured during the run: in SFLOW, FNPF and NHFLOW, a cost model is fitted to the busy times of the ranks (their level-0 cells that are not solid, their patch cells and the parent cells they serve to patches on other ranks) at every regrid, and a static refinement is placed once more at step 15; in CFD, the work of the ranks is measured from their timers at every regrid with a report of the measured load, and the weight of the load map is fitted to the busy times of the ranks. Not to be confused with the former G 41, now G 541.

0 measured in the run

> 0 work of a patch cell in level-0 cells

default: 1.0

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G 42 double CFD mesh refinement: recast of floating bodies

With floating bodies (X 10 1), the level set of the bodies on a grid is only computed again (ray cast and reinitialization) when a vertex of a body has moved by more than G 42 times the cell size of the grid since its last cast; the stages in between keep the level set, with the forcing velocity of the stage. With 0, the bodies are cast again at every change of their position.

default: 0.0

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G 43 int Mesh refinement: load statistics

Every G 43 steps, the load of the ranks over the steps since the last report is written: busy and MPI time per rank, patch and regrid time, the predicted and the measured

load (maximum over mean of the ranks) and the measured cost of a patch cell. The files are `REEF3D_<M>_AMR_load.dat` and `REEF3D_<M>_AMR_load_ranks.dat` in `REEF3D_<M>_AMR`, and a line is printed. The MPI time is measured by wrapping the MPI calls (not with a build with `REEF3D_NO_MPI_CLOCK`). Use 0 to turn the statistics off.

0 OFF

> 0 interval in time steps

default: 0

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G 44 double Mesh refinement: weight of the patch phase in the placement
Used with G 40 1. The cost of a placement is the maximum work of the ranks plus G 44 times their maximum patch work, since the ranks synchronise between the level-0 phase and the patch phase of a step and the patch work needs a balance of its own. With 0, only the total work of the ranks is balanced.

default: 0.0

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G 45 int Mesh refinement: load map
Every G 45 steps, the load map is written: the work per step of every level-0 column (CFD: of every level-0 cell), i.e. its level-0 cells that are not solid plus the work of a patch cell (G 41, or measured) times the patch cells over it, averaged over the run (CFD: over the window of G 47 steps). The file `REEF3D_<M>_AMR_loadmap.dat` in `REEF3D_<M>_AMR` is read by the weighted partition of DIVEMesh (M 20 5, M 35 1; copied as `loadmap.dat`) and of REEF3D (M 41 1). Requires G 1 > 0; not written on multi-block grids. In stand-alone SEASTATE runs (A 10 7, also without G 1), the load map is the measured work of the cells (the time of their solves) per iteration of every level-0 column in microseconds, the patch cells over it included, averaged over the run, in `REEF3D_SEASTATE_Log/REEF3D_SEASTATE_loadmap.dat`. In CFD with the particle sediment model CPM (Q 10 1 or 2) without refinement (G 1 0), the load map is a map of the cells, 1 per fluid cell plus the measured cost of the parcel updates in the cell over the run, in `REEF3D_CFD_CPM_Particle/REEF3D_CFD_CPM_loadmap.dat`. Use 0 to turn it off.

0 OFF

> 0 interval in time steps

default: 0

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G 46 double Rebalance threshold

On several MPI ranks, the load of the ranks (maximum over mean of the busy times, the MPI waits taken out; in SEASTATE the measured time of the solves of the cells of the rank) is checked every G 47 steps. When it exceeds G 46, the run writes the load map of the mesh refinement (G 45; CFD with CPM: the map of the cells and parcels of G 45; without refinement otherwise, the partition uses the active cells only), a checkpoint (as P 48) and the marker file `REEF3D_Checkpoint/rebalance`, and stops with exit code 3. The next start of the same case (`tools/reef3d_rebalance.sh` repeats the command) sees the marker and restarts from the checkpoint (I 45 1) on a new decomposition made by REEF3D (M 40 1) with the weighted partition of the load map (M 41 1); this needs `DIVEMesh_Grid/grid-global.dat` (DIVEMesh M 50 1 or 2). There is no rebalance when the weighted partition equals the present one, in coupled runs (B 92 62, where G 46 is switched off with a message), when the run ends within the next G 47 steps, or when the case is not covered by the checkpoints (a message is printed). If the first check after a rebalance restart has not lowered the excess load (maximum over mean minus 1) by at least 10 % from the load that stopped the run, there is no further rebalance in this run. A failed restart removes the marker file. Stand-alone SEASTATE runs (A 10 7) rebalance as well: they write the load map (G 45) of the last check interval (G 47 steps or iterations) and stop only if the weighted partition lowers the excess load maximum over mean minus 1 by at least 10 % by this map and if the time saved over the rest of the run (stationary: the iterations estimated from the convergence) exceeds the cost of the restart (the set-up of the run and writing and reading the checkpoint); in stationary runs (A 700 2), the load is checked every G 47 iterations within a time step, but not within the last G 47 iterations of the step (A 707). Use 0 to turn it off.

0 OFF

> 0 threshold of the load maximum over mean

default: 0.0

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G 47 int Rebalance: check interval

Number of time steps between the checks of the load for G 46; the first check is made at step G 47. In CFD, it is also the averaging window of the load map (G 45, G 46). In stationary SEASTATE runs (A 700 2), the number of iterations within a time step between the checks.

default: 100

6.7.2 Curvilinear and multi-block grids

SFLOW and NHFLOW can run on a curvilinear (boundary-fitted) horizontal grid, for example along a river corridor (G 50 1 or 2). The DIVEMesh grid is then the logical box of the

curvilinear grid: its numbers of cells in x and y must equal the numbers of cells n_i and n_j of the curvilinear grid along i and j, otherwise the run stops; the vertical grid (NHFLOW: sigma layers) is unchanged. The state stays cell-centred with Cartesian velocity components; the fluxes of every face are computed along its normal, and the face lengths and cell areas come from the quadrilateral cells of the nodes, so that a lake at rest stays at rest on any grid. The bed level of a cell is the mean of the bed levels of its four nodes. The boundaries of the logical box keep their roles: the faces at $i = 0$ and $i = n_i$ are the former x-boundaries (in- and outflow), those at $j = 0$ and $j = n_j$ the former y-boundaries; at walls, the velocity normal to the face is mirrored. Inflow and outflow act along the face normals. Folded cells (area $\leq$ 0) stop the run; the grid quality (areas, non-orthogonality, size ratio of neighbour cells) is printed. Initial water level boxes (F 72) select the cells by their centroids.

The file `curvgrid.dat` (G 50 2), in the case folder, holds the numbers of cells `ni nj` and then, for every node, its coordinates and bed level `x y zb`, with $i = 0 \ldots$ ni running fastest and $j = 0 \ldots$ nj outermost, i.e. $(n_i + 1)(n_j + 1)$ nodes. With G 50 1, the nodes are taken from the CURV section of `DIVEMesh_Grid/grid-geometry.dat` (the river corridor grid of DIVEMesh, R 1 1).

A multi-block grid is a curvilinear grid of several blocks with point-matching interfaces, made by DIVEMesh (R 20 1) and given by the BLKS section of `grid-geometry.dat` together with G 50 1. Block 0 is the grid of the run (the CURV section) with a ring of cells around its faces with an interface (DIVEMesh R 21); the ring must hold at least the cells the solver reaches across a face, otherwise the run stops with a message. The blocks 1, 2, ... (with G 55, the pieces they are cut into) are run as level-0 grids of the mesh refinement (Section 6.7.1), whole on one rank each, placed for the load of the ranks: the largest block first onto the rank with the least load (its part of block 0 and the blocks it holds), the load counted in cells or, with G 56, weighted by the bed level of the cells; the cells beyond their faces are filled from the neighbour blocks before every stage. G 1 > 0 refines all blocks, with no refinement around singular vertices (three or five blocks around a node) and next to the open boundaries; the degenerate ring cells at singular vertices become solid. The faces of the blocks 1, 2, ... without an interface are walls unless they are opened with G 54. With a multi-block grid, G 52 is set to 1. Multi-block grids are not available with subcycling (G 7 1), river initialisation (F 65) and the load map (G 45).

Restrictions (the run stops with a message): SFLOW on a curvilinear grid: one cell in y-direction, A 220 $\geq$ 5, A 212 2, A 260 1 or 2, wave generation and absorption (B 90, B 98), patch boundary conditions (B 440 – B 442), SEASTATE coupling (A 750), I 11 1 and the semi-implicit free surface (A 226 1). SFLOW on a multi-block grid in addition: only A 210 3, A 220 0, A 260 0, 3 or 4, S 10 0 or 1 (S 51 0, S 31 1 or 2, S 32 1 or 2), X 10 0, 2 or 3 and W 90 0. NHFLOW on a curvilinear grid: one cell in y-direction, A 520 3, A 511 2, A 519 2, the LAGOON output of the mesh refinement or of the blocks (P 18), 6DOF bodies with X 60 2, X 38 or X 13, nets (X 320), membranes (X 330), FEM structures (Z 30), immersed solids (A 580 – A 599), wind forcing (A 570), SEASTATE coupling (A 750), relaxation zones given by B 96 (use B 107 and B 108 in physical coordinates), B 97 1, and Dirichlet or active wave generation (B 98 3, 4) on a current (W 10). NHFLOW on a multi-block grid in addition: the scope of the NHFLOW mesh refinement (G 1), breaking (A 550), the ship pressure (X 10 3) and bodies (X 10 1 or 2) with G 40 1. Sediment transport on a curvilinear grid: S 31 3, S 32 3 – 5 and S 10 2 stop the run.

G 50 int Curvilinear horizontal grid

Selects the nodes of the horizontal grid. With option 1 and a BLKS section in `grid-geometry.dat`, the grid is a multi-block grid. Without the CURV section (option 1) or with a missing or incomplete `curvgrid.dat` (option 2), the run stops with a message.

0 Cartesian grid of DIVEMesh

1 nodes of the CURV section of `DIVEMesh_Grid/grid-geometry.dat` (DIVEMesh R 1 1; multi-block with R 20 1)

2 nodes of `curvgrid.dat`

default: 0

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G 51 int Multi-block grid: passes of the block V-cycles

NHFLOW on a multi-block grid: number of passes of the block-local multigrid V-cycles in the preconditioner of the composite pressure solve; between the passes, the corrections of the neighbour blocks are filled in and the V-cycles run on the residual. Values below 1 are set to 1.

default: 2

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G 52 int Curvilinear grid: walls in both index directions

NHFLOW on a curvilinear grid. With option 1, walls at the *i*-faces act as the walls at the *j*-faces: slip walls with the normal velocity mirrored and a homogeneous Neumann condition for the pressure correction, so that a wall acts the same in any index direction (not at the in- and outflow and the wave generation). With option 0, the *i*-faces keep the walls of the Cartesian x-boundaries. Option 1 also makes the smoothing of the bed (A 509) independent of the index directions. On multi-block grids, G 52 is set to 1.

0 Cartesian x-walls

1 *i*-walls as *j*-walls

default: 0

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G 53 int Multi-block grid: coarse space across the blocks

NHFLOW on a multi-block grid: the preconditioner of the composite solves adds a coarse correction across the blocks, built from aggregates of G 53 × G 53 columns of every level-0 grid. If the coarse matrix is singular, it is switched off for that solve with a message. Use 0 to turn the coarse space off.

0 OFF

> 0 aggregate width in cells

default: 4

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G 54 * int b, int t, double v Multi-block grid: open boundary of a block

Opens a face of block b (1, 2, ...; block 0 has the boundaries of the model) of a multi-block grid. Type t = 1: inflow with the discharge v (m^3/s) through the $-i$ face of the block; t = 2: outflow with the water level v (m) at the $+i$ face; t = 3: inflow through the $-i$ face with the discharge time series of the file `discharge-<v>.dat`; t = 4: outflow at the $+i$ face with the water level time series of the file `waterlevel-<v>.dat`. For the types 3 and 4, v is the number of the file (a whole number ≥ 0); the file in the case folder holds lines `t value` with increasing times (# starts a comment), the value is interpolated linearly in time and held constant before the first and after the last time. With F 64, the water level of the outflow is ramped from F 60 over the first F 64 time steps. One line per face; the face must not have an interface and must not carry an edge zone of B 522. A block number outside the grid, another type, a face given twice, a face with an interface, a face cut into pieces (G 55) or, for the types 3 and 4, a missing time series file, a file without a line `t value` or with times that do not increase stop the run with a message. The block number is that of DIVEMesh, also when G 55 cuts the blocks. There is no refinement next to an open face.

1 discharge v in through the $-i$ face

2 water level v at the $+i$ face

3 discharge time series `discharge-<v>.dat` in through the $-i$ face

4 water level time series `waterlevel-<v>.dat` at the $+i$ face

default: na ; na ; na

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G 55 int Multi-block grid: cut the blocks into pieces

With $m > 0$, the blocks 1, 2, ... of a multi-block grid (G 50 1, DIVEMesh R 20 1) are cut into pieces at the start, so that they spread over the MPI ranks: every block is cut into pieces of at most (cells of the blocks 1, 2, ...) / (m times the number of ranks) cells (with G 56: of about that part of the load of the blocks), at least 16 cells deep. A block is cut into slices along i, or along j when it has more cells in j and no open face (G 54, B 522); with G 58 1, a block without open face is cut into tiles in both directions. A block with an open face is cut along i only, so that its $-i$ and $+i$ faces stay whole. Every piece is a block of its own; the interfaces between the pieces and to the neighbour blocks are rebuilt and checked, and the run stops with a message if they do not map back. The cuts are printed; if no block is larger than the limit, the blocks stay whole. G 54 and B 522 keep the block numbers of DIVEMesh. As the cut depends on the number of ranks, a restart from a checkpoint (I 45) needs the same G 55 and the same number of ranks.

0 the blocks of DIVEMesh as they are

$m > 0$ pieces of at most 1/(m·ranks) of the cells of the blocks 1, 2, ...

default: 0

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G 56 double *a*, double *dz* Multi-block grid: placement and cut of the blocks by their load
On a multi-block grid, the cells count with their load when the blocks are placed on the ranks and cut (G 55): a cell whose bed level (the mean of its four nodes) lies more than *dz* (m) above the still water level has the load *a*, every other cell 1. The still water level is F 60, or F 62 with F 65; without one of them, or with $a = 1$, every cell counts 1. With G 55, a block is then cut where its slices reach equal parts of its load. With G 56 on several ranks, the placement of the blocks (largest load first onto the rank with the least load) is improved by moving and swapping blocks between the rank with the largest load and the others, and the load of the ranks is printed.

default: not set (OFF; $a = 1.0$, $dz = 0.0$)

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G 57 int Multi-block grid: coarse space of the semi-implicit free surface
NHFLOW with the semi-implicit free surface (A 534 1) on a multi-block grid: the preconditioner of the free surface solve over all grids adds a coarse correction across the grids, built from aggregates of G 57 × G 57 columns of every grid; the V-cycle of every grid then works on the remaining residual. If the coarse matrix is singular, it is switched off for that solve with a message. Use 0 to turn the coarse space off.

0 OFF

> 0 aggregate width in cells

default: 0

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G 58 int Multi-block grid: cut the blocks into tiles
Used with G 55. With option 1, the blocks without open face (G 54, B 522) are cut into tiles in both index directions, about square in cells, instead of slices along one direction. Blocks with an open face are always cut along *i*.

0 slices along one direction

1 tiles in both directions

default: 0

6.7.3 Geometry

Treatment of the geometry level set functions of the bed topography and of solids read from DIVEMesh, and of floating bodies represented on the horizontal grid. These signed distance functions are kept as distance functions with the same PDE-based reinitialization as the free surface level set (F 43 sets its pseudo time step).

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G 505 int Solid cells from topography and solids

With G 505 = 1, the cell flags are recomputed from the topography and solid level sets whenever the direct forcing is updated (start, bed changes, moving bodies): cells that lie deeper inside the topography or a solid than the direct forcing half-width X 41·Δx are flagged as solid and taken out of the computation. With G 505 = 0, the cell flags are not changed and the topography and solids are represented by the direct forcing only. For S 10 = 2 with Q 11 = 2 and Q 50 = 1, the topography is taken out of the flags and the direct forcing at the start: cells that are solid only because of the topography become fluid cells, so that the bed does not block the flow. The CFD mesh refinement (G 1) needs G 505 = 0 with solids or topography.

0 OFF

1 ON

default: 0

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G 540 int Reinitialization time scheme for topography and solids

PDE-based reinitialization of the topography level set (used with sediment transport S 10 > 0 or a topography or solid read from DIVEMesh) and of the solid level set (solids read from DIVEMesh). At the start, the level sets are reinitialized over $2L/(F43 \cdot dx)$ steps with the length L of F 42; afterwards G 541 steps are used per call, e.g. after each bed change.

0 OFF

3 3rd-order TVD Runge-Kutta (any value > 0)

default: 3

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G 541 int Number of reinitialization steps for topography, solids and 2D floating bodies

Number of reinitialization steps per call after the initial reinitialization. Used for the topography and solid level sets in CFD (G 540) and for the horizontal level set of floating bodies (pressure field, ship waves) in SFLOW and NHFLOW. In FNPF, the coastline is only reinitialized at the start, so G 541 has no effect there. (Formerly G 41.)

default: 2

6.8 H :: Heat Transfer

Valid for:
S F N C

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H 1 double Thermal diffusivity of water

Thermal diffusivity in m^2/s of the water phase. Which level-set phase is water is set with H 9.

default: $1.4 \cdot 10^{-7}\ m^2/s$

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H 2 double Thermal diffusivity of air

Thermal diffusivity in m^2/s of the air phase. Which level-set phase is air is set with H 9.

default: $2.216 \cdot 10^{-5}\ m^2/s$

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H 3 int Fluid property calculation

How density and viscosity depend on the temperature when heat transfer is active (H 10 > 0, with a level-set free surface, F 30 > 0). With option 2 the density varies linearly with the temperature, $\rho = \rho_0 (1 - \beta(T - T_0))$, where ρ_0 is the reference density (W 1 or W 3), T_0 the reference temperature from H 50 and β the expansion coefficient from H 4; without H 4, $\beta = 1/T_0$ with T_0 in K. Viscosity stays constant (W 2, W 4).

1 density and viscosity interpolated from tabulated properties of water and air

2 Boussinesq approximation (density only)

default: 1

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H 4 double β_1, double β_2 Thermal expansion coefficients

Thermal expansion coefficients in $1/K$ of water (β_1) and air (β_2) for the Boussinesq approximation (H 3 = 2). Setting H 4 activates the use of these values; if H 4 is not given, $\beta = 1/T_0$ is used.

default: not set ($\beta = 1/T_0$)

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H 9 int Assignment of water and air to the two phases

Selects which level-set phase (phase 1: $\phi > 0$, phase 2: $\phi < 0$) is water and which is air. This determines which of H 1/H 2, H 4 and H 50 apply to which phase.

1 phase 1: water, phase 2: air

2 phase 1: air, phase 2: water

default: 1

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H 10 int Time scheme for heat transfer

Turns on the solution of the temperature transport equation and selects its time scheme. Requires a level-set free surface (F 30 > 0); the fluid properties then follow H 3. The CFD mesh refinement (G 1) supports heat transfer only with H 10 = 3, H 15 = 0, 1 or 4 and H 3 = 1, without concentration transport (C 10) and porous structures (B 200) and with D 20 = 0–2; otherwise the refinement is switched off with a message.

0 OFF

1 2nd-order Adams-Bashforth

2 2nd-order TVD Runge-Kutta

3 3rd-order TVD Runge-Kutta

4 3rd-order Runge-Kutta with Crank-Nicolson diffusion

default: 0

○ ○ ○ ●

H 15 int Convection discretization for heat transfer

Spatial discretization of the convection term of the temperature equation (H 10 > 0).

0 OFF

1 FOU

2 CDS2

3 QUICK

4 WENO5 FLUX

5 WENO5 HJ

6 CDS4

7 WENO3 FLUX

8 WENO3 HJ

9 same as 4 (the former uniform-grid WENO5 FLUX has been removed)

10 High Resolution TVD scheme with Minmod limiter

11 High Resolution TVD scheme with van Leer limiter

12 High Resolution TVD scheme with Umist limiter

13 High Resolution TVD scheme with van Albada limiter

14 High Resolution TVD scheme with Superbee limiter

15 High Resolution scheme with Smart limiter

16 High Resolution TVD scheme with Limo3 limiter

60 CDS6

default: 5

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H 50 double T_1, double T_2 Temperature 1 and temperature 2
Initial temperatures in $°C$. The whole domain is first set to T_2; cells inside the region defined by H 51–H 56, H 57 or H 58 are set to T_1. Since the default box of H 51–H 56 covers the whole domain, the domain starts at T_1 unless the box is narrowed. With H 3 = 2 the two values are also the reference temperatures T_0 of water (T_1) and air (T_2) for H 9 = 1, swapped for H 9 = 2.

default: 20.0 $°C$; 20.0 $°C$

○ ○ ○ ●

H 51 double Temperature 1 region start, x-direction
Lower x-bound in m of the box initialized with T_1 (H 50). Use together with H 52–H 56.

default: $-1.0 \cdot 10^7$ m (no bound)

○ ○ ○ ●

H 52 double Temperature 1 region start, y-direction
Lower y-bound in m of the box initialized with T_1 (H 50).

default: $-1.0 \cdot 10^7$ m (no bound)

○ ○ ○ ●

H 53 double Temperature 1 region start, z-direction
Lower z-bound in m of the box initialized with T_1 (H 50).

default: $-1.0 \cdot 10^7$ m (no bound)

○ ○ ○ ●

H 54 double Temperature 1 region end, x-direction
Upper x-bound in m of the box initialized with T_1 (H 50).

default: $1.0 \cdot 10^7$ m (no bound)

○ ○ ○ ●

H 55 double Temperature 1 region end, y-direction
Upper y-bound in m of the box initialized with T_1 (H 50).

default: $1.0 \cdot 10^7$ m (no bound)

○ ○ ○ ●

H 56 double Temperature 1 region end, z-direction
Upper z-bound in m of the box initialized with T_1 (H 50).

default: $1.0 \cdot 10^7$ m (no bound)

○ ○ ○ ●

H 57 double a, double b, double c, double d Temperature 1 region bounded by a plane
Cells with $ax + by + cz < d$ (x, y, z in m) are set to T_1 (H 50), in addition to the H 51–H 56 box. Active only if at least one of the four values is positive.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 (inactive)

○ ○ ○ ●

H 58 double x_0, double y_0, double z_0, double r Temperature 1 region as sphere
Cells inside the sphere with center (x_0, y_0, z_0) and radius r (all in m) are set to T_1 (H 50). Active for $r > 0$.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 (inactive)

○ ○ ○ ●

H 61 double Fixed temperature on surface 1
Sets a Dirichlet temperature in $°C$ on the domain boundary surface 1 (x_{min}). Surfaces are numbered 1: x_{min}, 2: y_{max}, 3: y_{min}, 4: x_{max}, 5: bottom (z_{min}), 6: top (z_{max}). Without the command the default boundary treatment for the temperature applies on this surface.

default: not set

○ ○ ○ ●

H 62 double Fixed temperature on surface 2
Dirichlet temperature in $°C$ on surface 2 (y_{max}), see H 61.

default: not set

○ ○ ○ ●

H 63 double Fixed temperature on surface 3
Dirichlet temperature in $°C$ on surface 3 (y_{min}), see H 61.

default: not set

○ ○ ○ ●

H 64 double Fixed temperature on surface 4
Dirichlet temperature in $°C$ on surface 4 (x_{max}), see H 61.

default: not set

○ ○ ○ ●

H 65 double Fixed temperature on surface 5
Dirichlet temperature in $°C$ on surface 5 (bottom, z_{min}), see H 61.

default: not set

○ ○ ○ ●

H 66 double Fixed temperature on surface 6
Dirichlet temperature in $°C$ on surface 6 (top, z_{max}), see H 61.

default: not set

6.9 I :: Initialization

Restart from checkpoints. SFLOW, FNPF, NHFLOW, CFD and SEASTATE (stand-alone, A 10 7) can write checkpoints (P 48) and restart from them (I 45), also on another number of processes and another domain decomposition. Every P 48 time steps, and with the optional second value of P 48 every Δt seconds of simulated time, a checkpoint is written into the folder `REEF3D_Checkpoint/<count>`, with the iteration number as eight-digit folder name (stationary SEASTATE runs, A 700 2: every P 48 iterations within a time step, into `REEF3D_Checkpoint/<count>_<iteration>` with the iteration of the step as six digits). SFLOW, FNPF, NHFLOW and CFD also write a checkpoint at the end of the run, unless the last step wrote one, so that a run can be continued with a later end time (N 41, N 45). A checkpoint holds `main.dat` (written by rank 0: the subdomains of the writing run, the times and counters of the run and of the outputs, the state of the 6DOF bodies and of REEF3D::CORAL), one file `piece-<rank>.dat` per process with its arrays including the ghost cells, copies of `ctrl.txt` and `control.txt`, and per process the table `outputs-<rank>.txt` with the sizes of its output files at the checkpoint. When all files are complete, the folder name is written into `REEF3D_Checkpoint/latest`, so a run that stops while writing leaves the previous checkpoint usable. The folders are listed in the order they were written in `REEF3D_Checkpoint/history`; only the newest P 49 checkpoints are kept (by default two, in SEASTATE one, since a spectrum per cell is large), the older ones are removed, the latest never. A run that receives the signal SIGTERM or SIGUSR1 (e.g. from a batch system before the end of the job) writes a checkpoint at the end of the current time step and stops; I 45 1 continues it. The checkpoint holds the complete state of the model: in CFD every field of the objects of the run (the restart must therefore use the same models and options), in

SFLOW, FNPF and NHFLOW the fields of the model and of its momentum, turbulence and pressure objects, in SEASTATE the spectra of the active cells of all grids by their global index, with the active ranges of the spectral sparsity (A 795) and, within a stationary time step, the last change of H_s of the convergence test (A 799 1), in all modules the patches of the mesh refinement and the blocks of a multi-block grid (G 1, G 50), the 6DOF bodies with their moorings, nets (X 320) and power take-off (X 500), REEF3D::CORAL (Z 40), sediment transport (S 10, including the parcels of the particle sediment model), DEM (E 10; NHFLOW, CFD), the NHFLOW long-run statistics (P 160) and the SFLOW maxima (P 112). With I 45 1 the run continues from the latest checkpoint, with I 45 <count> from the checkpoint of that iteration. The grid (number of cells) must be the same, the number of processes (M 10) and the decomposition (M 40, M 41) may differ; only a multi-block grid whose blocks are cut for the number of processes (G 55) needs the same G 55 and the same number of processes. On the same decomposition the restart continues bitwise; on another one the ghost cells are taken from the neighbouring subdomains and the internal state of the boundary conditions starts again. A stationary SEASTATE step continues at the iteration of the checkpoint. The output continues: numbered files continue their numbering, and the time series files (gauges, probes, logs, the output of REEF3D::CORAL) are continued from their size at the checkpoint, i.e. what the stopped run wrote after the checkpoint is cut off and a file that was only created after the checkpoint starts again. At the restart, the settings of `ctrl.txt` and `control.txt` that differ from the copies in the checkpoint are listed, with what is known of the change: P commands (output), M 10, N 41, N 45 and G 45 – G 47 are safe, N 47 – N 49 (time step) are used in practice, a change of A 10, X 10 or Z 40 is refused or does not fit, and other changes are not checked (the state comes from the checkpoint; a new setting may act from the restart on or be ignored). Not covered yet are SEASTATE coupled to a host model (A 750) and the SEASTATE surfbeat model (A 770 1), FSI and FEM structures (Z 10 in CFD, Z 30 in NHFLOW and CFD), 6DOF wavemaker objects (X 10 4), and in NHFLOW the Lagrangian particles (L 10) and membranes (X 330): P 48 then prints a message at the start and writes no checkpoint, and I 45 stops with a message. The checkpoints are also used by the automatic rebalance on the measured load (G 46, Section 6.7).

The older state files (P 40) are still written by all modules and are used for hydrodynamic coupling and nesting. The restart from a state file (I 40, I 41) is still available for CFD, but only on the same decomposition; for restarts the checkpoints are recommended.

Runtime coupling of FNPF with CFD, NHFLOW or FNPF (B 92 62). Besides the file-based hydrodynamic coupling (HDC, B 92 61: the FNPF run writes state files with P 40, DIVEMesh converts them into the input of the receiving run, which then reads them with the offsets I 231 – I 233 and I 241) and the CFD FlowFile inflow (P 230, I 230), an FNPF model and one or more receiving models can run side by side in one MPI job and exchange the waves during the run. Each model runs in its own folder with its own `ctrl.txt`, `control.txt` and DIVEMesh grid and writes its output there, e.g.

```
mpirun -np 4 -wdir fnpf REEF3D : -np 16 -wdir cfd REEF3D
```

The models of the job are numbered (app number) in the order of the command line, starting with 0. For launchers without `-wdir` or without MPI app numbers (e.g. `srun -multi-prog`), REEF3D takes the command line options `-dir <path>` (the folder of the model) and `-app`

<n> (its app number). The receiver, a CFD (A 10 6), NHFLOW (A 10 5) or FNPF model (A 10 3, nesting), sets wave generation (B 90) with B 92 62: its relaxation zones (B 98 2), Dirichlet inflow and full initialization (I 30 1) then use η and the velocities of the FNPF model (an FNPF receiver also the potential) instead of a wave theory. The FNPF source needs no coupling command: it is the model named with I 246 in the receiver, by default the lowest-numbered FNPF model that is no receiver itself. Several receivers can share one source, and an FNPF receiver can in turn be the source of another model (chain, e.g. regional FNPF $\rightarrow$ local FNPF $\rightarrow$ CFD with I 246 in the CFD model). The points of the receiver are mapped into the FNPF domain as for the HDC: with the angle and origin of B 105, then shifted by I 231 – I 233. The FNPF time is the wave time of the receiver plus I 241, and the FNPF data are interpolated in time between the FNPF time steps (I 242). A 2D FNPF model drives a 3D receiver with long-crested waves in the B 105 direction; a 3D FNPF model drives a 2D receiver with its line at y = I 232. After every time step, each FNPF process sends its part of the window that covers the subdomain of each receiver process; the FNPF model runs at most four time steps ahead of the receiver. If the FNPF model ends first, the receiver holds its last state; the FNPF model ends once all its receivers have ended. With a single model in the job, B 92 62 stops with a message. The rebalance stop (G 46) is switched off in coupled models. Two-way coupling (CFD and NHFLOW receivers): with I 243 the 6DOF bodies of the receiver move the FNPF bodies of the same number, so that the FNPF model radiates their waves, and with I 244 (CFD only) the FNPF free surface is relaxed towards the CFD free surface inside an overlap box. At the start of each of its time steps the receiver sends these data to the FNPF model, which waits for them and extrapolates the last two samples to its own stage times.

Valid for:

S	F	N	C
●	○	●	●

I 10 int Initialize everything

Shortcut that overrides I 11, I 12 and I 13: option 1 sets I 11 = 1, I 12 = 2, I 13 = 1; option 2 sets I 11 = 2, I 12 = 2, I 13 = 1. Option 1 also initializes the inflow velocities in the ioflow/iowave boundary routines at start-up. Option 2 (potential flow including the free surface) is available in CFD only. I 10 is reset to 0 when a restart is requested (I 40 $>$ 0), and I 13 is reset to 0 without turbulence model (T 10 = 0).

0 OFF

1 ON

2 ON, potential flow initialization including the free surface (CFD)

default: 0

●	○	●	●

I 11 int Initialize velocities with potential flow solver

Solves a potential flow problem once before the first time step and uses the resulting velocities as initial condition. The linear solver is the one selected with N 10; for NHFLOW use N 10 = 1 or 11–19. Set automatically by I 10; reset to 0 if I 40 $>$ 0. SFLOW on a curvilinear grid (G 50) stops with I 11 = 1.

0 OFF

1 ON

2 ON, potential flow in the water phase only, including the free surface (CFD)

default: 0

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I 12 int Initialize pressure

Initializes the pressure field hydrostatically, $p = \rho g h$, plus the reference pressure I 55. Skipped for options 1–3 if the numerical wave tank is fully initialized (I 30 = 1 together with B 90 $>$ 0). Set automatically to 2 by I 10; reset to 0 if I 40 $>$ 0.

0 OFF

1 hydrostatic, based on the vertical distance to the still water level

2 hydrostatic, based on the level-set function

3 hydrostatic, based on the vertical distance to the top of the domain (to the still water level with a free surface, F 30 $>$ 0)

default: 0

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I 13 int Initialize turbulence model

Calls the initialization of the selected turbulence model (T 10) before the first time step. Set automatically by I 10; reset to 0 if I 40 $>$ 0 or T 10 = 0.

0 OFF

1 ON

default: 0

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I 21 double Phase 2 velocities after potential flow initialization

Controls the velocities in the air phase (level set $\phi < 0$) after the potential flow initialization (I 11 $>$ 0).

0 keep the potential flow solution

1 set to zero

default: 0

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I 30 int Full initialization of the numerical wave tank

Initializes the free surface, the velocities (and in FNPF the velocity potential) in the whole domain with the wave theory selected in B 92 at $t = 0$, instead of only in the wave generation zone. Requires wave generation (B 90 > 0). Option 2 (FNPF only; the other modules use option 1) multiplies every wave component by an exposure mask for its propagation direction, so that sheltered areas, e.g. in the lee of islands and headlands, start with reduced or no waves instead of the undisturbed incident wave. For the masks, the wet and dry cells are gathered on a coarse grid (I 33); for each of I 32 direction bins a sweep along the propagation direction marks the wet cells that can be reached from the upwave boundary without crossing land, with a lateral spreading by diffraction (I 34). The masks are then mixed across neighbouring directions (I 35) and smoothed over the wet cells (I 31). Wave theories with a decomposition into components (B 92 5, 31, 41, 51) are masked per component, the others with one mask for the main direction (B 105). The energy-weighted mean mask is written by rank 0 into `REEF3D_FNPF_I30_mask.asc` (ESRI ASCII grid, -1 on land) in the simulation folder. The rest of the initialization is the same as for option 1.

0 OFF

1 ON

2 ON, with exposure masks (FNPF)

default: 0

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I 31 double Smoothing length of the exposure masks

Standard deviation in m of the smoothing of the exposure masks of I 30 2 over the wet cells; land does not pull the masks down along exposed coastlines. With a negative value the peak wave length is used, from the linear dispersion relation with the peak period (or period) of the wave and the water depth of the wave theory (B 94). 0 switches the smoothing off.

default: -1.0 (one peak wave length)

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I 32 int Number of direction bins of the exposure masks

Number of propagation directions, evenly spaced between the smallest and the largest direction of the wave components, for which the exposure masks of I 30 2 are computed. A component uses the masks of the two nearest directions, interpolated linearly. With a single wave direction one mask is used.

default: 32

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I 33 double Cell size of the exposure mask grid

Cell size in *m* of the coarse global grid on which the exposure masks of I 30 2 are computed. A coarse cell is wet if at least half of the FNPF cells in it are wet. With a value ≤ 0 the size is chosen automatically as the larger of twice the mean horizontal cell size and 1/1500 of the larger domain extent; it is never smaller than the mean horizontal cell size.

default: -1.0 (automatic)

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I 34 double Diffraction spreading coefficient of the exposure masks

Coefficient κ of the lateral spreading of the exposure masks of I 30 2 by diffraction: along the propagation direction the exposure spreads into the shadow with the variance $\sigma^2 = \kappa L_p s$, with the peak wave length L_p (see I 31) and the distance s along the ray, so that shadow edges widen like a Fresnel zone further downwave. The default $1/(2\pi)$ matches the diffraction at a knife edge. 0 switches the spreading off (geometric shadow).

default: 0.159155

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I 35 double Angular blur of the exposure masks

Standard deviation in *deg* of a Gaussian with which the masks of the direction bins of I 30 2 are mixed across neighbouring directions, so that the lee receives a broad mix of directions instead of a few beams. 0 switches the blur off.

default: 15.0 *deg*

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I 40 int Initialization from state file (restart)

Restarts a simulation from a CFD state file written with P 40 (one file per print-out, P 45 = 1), selected with I 41. The state files are written per MPI process, so the restart must use the same decomposition (M 10). Simulation time and print counters continue from the state file. After reading, the boundary conditions of pressure and velocities and (without 6DOF bodies) the time step are initialized again, and the first time step is reduced to one tenth. Any I 40 > 0 resets I 10–I 13 to 0. The state file contains only the main fields (see P 40). For a restart on any decomposition, with the complete state and for the other modules, use the checkpoints (P 48, I 45); with I 45 = 1 the checkpoint replaces the state read with I 40.

0 OFF

1 ON, from state file (requires I 41)

default: 0

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I 41 int State file number for restart

Number of the state file to restart from (I 40 = 1), i.e. the first number in the state file names `REEF3D-CFD-State-<number>-<rank>.r3d` in the folder `REEF3D_CFD_STATE`.

default: 0

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I 45 int Restart from a checkpoint

Restarts the simulation from a checkpoint written with P 48, on any number of processes and any decomposition of the same grid (see the introduction of this section). With 1 the checkpoint named in `REEF3D_Checkpoint/latest` is used (the file may also hold a bare iteration number, e.g. 1200), with a value > 1 the checkpoint `REEF3D_Checkpoint/<count>` of that iteration. The checkpoint of the restart becomes the newest of `REEF3D_Checkpoint/history`, so that later checkpoints of the earlier run are the first to be removed (P 49). Simulation time, iteration number, time step and the counters of the outputs continue from the checkpoint; the initial output is not written again, and the time series files are continued from their size at the checkpoint. The settings that differ from those of the checkpoint are listed. Also for stand-alone SEASTATE runs (A 10 7); a stationary step (A 700 2) continues at the iteration of the checkpoint. The run stops with a message if there is no complete checkpoint, if it was written on another grid or with other models (CFD), with another spectral grid (SEASTATE, A 701 – A 703), with other blocks of a multi-block grid (G 55 cuts the blocks for the number of processes: restart with the same G 55 and number of processes), or if the case contains a feature that the checkpoints do not cover. I 45 is set to 1 automatically for a rebalance restart (G 46), when the file `REEF3D_Checkpoint/rebalance` exists; a failed restart removes this file, so that `tools/reef3d_rebalance.sh` does not start the run again.

0 OFF

1 ON, from the latest checkpoint

$>$ **1** ON, from the checkpoint of this iteration

default: 0

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I 50 double Start time of the simulation

Initial value of the simulation time in *s*. With a restart (I 40 = 1 or I 45 = 1) the time from the state file or the checkpoint is used.

default: 0.0 *s*

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I 55 double Reference pressure
Constant pressure in Pa used as the initial pressure field and added to the hydrostatic pressure initialization (I 12).

default: 0.0 Pa

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I 56 int Zero pressure in the air phase
If ON, the hydrostatic pressure of the pressure initialization (I 12) is set to zero above the free surface (level set $\phi < 0$).

0 OFF

1 ON

default: 0

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I 58 double w_0, double r_2 Initial vertical velocity around the F 58 sphere
Sets the vertical velocity to w_0 in m/s inside the sphere defined by F 58 and decreases it linearly to zero between the F 58 radius and the radius r_2 in m (r_2 larger than the F 58 radius). Active for $r_2 > 0$, e.g. for droplet impact cases.

default: 0.0 ; 0.0 (inactive)

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I 230 int Read FlowFile for inflow boundary condition
Uses a FlowFile (velocities, pressure, level set over time) from a previous 2D CFD simulation, printed with P 230, as inflow boundary condition. The number is the ID of the FlowFile gage. When active, the iowave wave generation at the inflow (B 98) is bypassed, so B 98 should be set to 0. See also I 231.

default: 0 (OFF)

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I 231 double x-offset for hydrodynamic coupling input; vertical offset of FlowFile inflow
For hydrodynamic coupling (B 92 = 61) and runtime coupling (B 92 = 62), shift in m added to the x-coordinate when the input data is interpolated to the current grid (for 62 in the wave frame of B 105). For FlowFile inflow (I 230, CFD), vertical offset in m of the FlowFile profile, rounded to whole cells.

default: 0.0 m

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I 232 double y-offset for hydrodynamic coupling input

For hydrodynamic coupling (B 92 = 61) and runtime coupling (B 92 = 62), shift in *m* added to the y-coordinate when the input data is interpolated to the current grid (for 62 in the wave frame of B 105). For a 2D receiver and a 3D FNPF source, the y-coordinate of the FNPF line that drives the receiver.

default: 0.0 *m*

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I 233 double z-offset for hydrodynamic coupling input

For hydrodynamic coupling (B 92 = 61) and runtime coupling (B 92 = 62), shift in *m* added to the z-coordinate (relative to the still water depth) when 3D input data (from FNPF or NHFLOW) is interpolated to the current grid.

default: 0.0 *m*

● ● ● ●

I 241 double Time offset for hydrodynamic coupling input and FlowFile inflow

For hydrodynamic coupling (B 92 = 61) and runtime coupling (B 92 = 62), time in *s* added to the current wave time when the input data is read and interpolated in time, i.e. the coupling starts at time I 241 of the source simulation. With B 92 62 and I 30 1, the receiver starts from the FNPF state at time I 241 (hot start); the FNPF model runs alone up to this time. For FlowFile inflow (I 230, CFD), time in *s* added to the simulation time when the FlowFile is interpolated in time.

default: 0.0 *s*

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I 242 int Time interpolation of the runtime coupling

Interpolation in time between the levels received from the FNPF source (B 92 62): 1 linear, 3 cubic (Lagrange over four levels). Cubic needs one more level ahead and is the more accurate choice for one-way coupling; with feedback (I 243, I 244) linear interpolation gives a shorter lag.

default: 3

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I 243 int Bodies of the receiver drive the FNPF bodies (two-way)

With I 243 1, rank 0 of the CFD or NHFLOW receiver (B 92 62) sends the state of its 6DOF bodies (centre of gravity, velocity, angular velocity, rotation) at the start of each time step, transformed into the FNPF frame (B 105, I 231–I 233). The FNPF

bodies with the same numbers (X 10, X 20) follow them: their pose and velocity are extrapolated from the last two samples to every stage time of FNPF, and FNPF computes the radiated waves of this motion. The FNPF body must have the same geometry, and X 11 in FNPF must free the degrees of freedom in which the body moves. In a 2D receiver the body has to lie within the width of the domain.

default: 0 (OFF)

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I 244 double Overlap box of the two-way coupling

x_s, x_e, y_s, y_e in m (receiver coordinates) of a box inside a CFD or NHFLOW receiver (B 92 62) in which FNPF is relaxed towards the receiver solution: at the start of each step, the receiver samples η and the surface velocity at the FNPF columns inside the box (CFD: zero level set, velocity half a cell below the surface; NHFLOW: η, velocity of the top σ cell) and sends them to FNPF, which relaxes η and Φ_s there in every Runge-Kutta stage (Φ_s from the surface velocity along the FNPF x lines). Reflected and scattered waves of a structure in CFD thus enter the FNPF domain. The box must lie in the free part of the receiver domain, away from its relaxation zones that are driven by FNPF. For a 3D FNPF model and a 3D receiver, a box that extends beyond the side walls (y_s, y_e outside the domain) is ramped only in x. A 2D FNPF model and a 3D receiver: the receiver is averaged along the crest line of each FNPF column inside the box. The FNPF model needs B 90 1. Ramp width: I 245.

default: OFF

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I 245 double Ramp width of the overlap box

Distance in m from the edge of the I 244 box over which the relaxation weight rises from 0 to 1 (cosine ramp). 0: a quarter of the smaller side of the box.

default: 0.0 m

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I 246 int Source of the runtime coupling

App number (position in the mpirun command, from 0) of the FNPF model this receiver (B 92 62) takes its waves from. An FNPF receiver can be the source of another model, e.g. regional FNPF (app 0) → local FNPF (app 1, B 92 62) → CFD (app 2, B 92 62, I 246 1). -1: the lowest FNPF app that is no receiver.

default: -1

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I 247 int Feedback lag of the two-way coupling

Synchronisation of the FNPF source with the feedback of this receiver (I 243, I 244). 0: FNPF may run one level ahead of the last sample; the feedback is extrapolated linearly over up to two FNPF steps, and the receiver does not wait for FNPF. 1: FNPF steps only when the receiver has reached the FNPF time (or waits for a later level); the feedback is extrapolated over one FNPF step, and the receiver waits for each FNPF step. 2: as 1, with quadratic extrapolation through the last sample and the samples nearest to one and two extrapolation spans earlier. Options 1 and 2 pay off when FNPF is much cheaper than the receiver; when both cost about the same, keep 0. At the end of the run, both models report how long they waited for each other.

default: 0

6.10 L :: Lagrangian Particles (NHFLOW)

The L commands control the Lagrangian particle tracking of REEF3D::NHFLOW, which can be used for general particle tracing and for the transport of floating or buoyant marine (macro)plastics, in the spirit of OpenDrift. The particles are passive: they are moved by the flow but do not act back on it. The tracking is activated with L 10 1 and the particles are defined by one or more releases (L 21 – L 23). Each release carries its own rise or settling velocity, windage coefficient and particle mode, so that different particle classes can be simulated together. The particle tracking is not available in the other modules.

The deterministic particle velocity is the sum of the interpolated fluid velocity, the rise velocity w_s in the vertical and, for particles at the free surface, the windage $c_w U_{10}$ in the wind direction. It is integrated with Heun's method: the velocity at the old position and old time level is evaluated before the momentum step, and the corrector uses the new flow field after the momentum step, with the same time step as the flow solver. The fluid velocity is interpolated bilinearly in the horizontal over the surrounding cell-centre columns and linearly in the vertical within each column, using that column's σ-levels. Above the top cell centre the velocity is extrapolated linearly to the free surface, below the lowest cell centre it is held constant, and dry columns are excluded from the interpolation. Since NHFLOW resolves the individual waves, the Stokes drift follows from the tracking of the orbital velocities and no Stokes drift parameterization is added. The bed is the NHFLOW bed level (including the topography); particles above the free surface are moved onto it. For regular waves with $kh = 1.9$, the drift of surface particles agrees with linear Stokes theory within about 1%.

Turbulent dispersion is modelled as a random walk that is added after the Heun step: $\Delta x = \sqrt{2K_h \Delta t}\, R$ in each horizontal direction and, for a vertical diffusivity that varies over the depth, $\Delta z = K_v'(z)\Delta t + \sqrt{2K_v(z + K_v'\Delta t/2)\Delta t}\, R$ [Visser, 1997], where R is a standard normal random number. Particles that reach the free surface are placed on it (state 1) and are then subject to windage; surface-trapped particles (mode 1) always follow the free surface. At the bed, particles are reflected or deposited (L 52). Particles in dry cells are stranded (state 2) and either refloat when the cell becomes wet again or stay beached permanently (L

51). Particles that leave the computational domain through its lateral sides are removed or reflected (L 53).

In parallel runs, the particles are exchanged with the neighbouring processes after every time step. The release schedule, the particle ids and the positions of the box releases do not depend on the domain decomposition, so without random walk the results are identical for any number of processes. The random numbers are drawn per process (L 11), so trajectories with random walk depend on the decomposition, while their statistics do not.

Output is written into the folder `REEF3D_NHFLOW_Particles` with the interval L 61: `REEF3D-NHFLOW-Particles-<n>.vtp` for ParaView (point data: velocity, id, release, state, age; the simulation time is stored as TimeValue), `REEF3D-NHFLOW-Particles-Tracks.csv` with the columns t, id, release, x, y, z, u, v, w, state, and `REEF3D-NHFLOW-Particles-Log.dat` with the number of released particles, the number of particles in the domain, in the water column, at the surface, stranded and on the bed, and the cumulative number of particles that left the domain. Positions and velocities are written in world coordinates. The particle states are 0: water column, 1: free surface, 2: stranded in a dry cell, 3: deposited on the bed.

Example: floating macroplastics with 3% windage and a sinking fraction, released continuously over 600 s:

```
L 10 1
L 41 10.0 0.0
L 23 100.0 200.0 50.0 80.0 0.0 0.0 5000 0.0 600.0 0.0 0.03 1
L 23 100.0 200.0 50.0 80.0 -1.0 -1.0 2000 0.0 600.0 -0.01 0.0 0
L 31 0.05
L 61 5.0
L 62 3
```

Valid for:

S	F	N	C
○	○	●	○

L 10 int Lagrangian particle tracking

Activates the particle tracking for NHFLOW (A 10 5). The particles are defined with L 21 – L 23. The NHFLOW mesh refinement (G 1, Section 6.7.1) does not support particles and is switched off when L 10 1 is set. The particles are not part of the checkpoints (P 48, I 45): with L 10 1 no checkpoints are written.

0 OFF

1 ON

default: 0

○	○	●	○

L 11 int Random seed

Seed of the random numbers for the random walk (L 31, L 32). Each process uses the seed plus a process-dependent offset. Change the seed to obtain an independent realization of the random walk.

default: 1

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L 21 * double x, double y, double z, int n, double t_s, double t_e, double w_s, double c_w, int mode Point release
Releases n particles at the point (x, y, z) in m. The release times are distributed evenly over $[t_s, t_e)$ in s; with $t_e \leq t_s$ all particles are released at t_s. w_s is the rise velocity in m/s ($w_s > 0$: buoyant, $w_s < 0$: settling), c_w the windage coefficient as fraction of U_{10} (e.g. $0.01 - 0.05$ for floating plastics with emerged area). With mode 0 the particles move in three dimensions, with mode 1 they are trapped at the free surface and z is ignored. z is given in the absolute vertical coordinate of the grid (the same as for the probes, P 65); a particle released above the free surface is placed on it, one released below the bed is reflected into the water column (or deposited, L 52). x and y are given in world coordinates.

default: not set

○ ○ ● ○

L 22 * double x_s, double y_s, double z_s, double x_e, double y_e, double z_e, int n, double t_s, double t_e, double w_s, double c_w, int mode Line release
Releases n particles along the line from (x_s, y_s, z_s) to (x_e, y_e, z_e) in m, evenly spaced at the positions $s = (q + 0.5)/n$, $q = 0 \ldots n - 1$, along the line. For a release over a time window, the particles are released in this order, i.e. the line is seeded from start to end over the window. The remaining values are the same as for L 21.

default: not set

○ ○ ● ○

L 23 * double x_s, double x_e, double y_s, double y_e, double z_s, double z_e, int n, double t_s, double t_e, double w_s, double c_w, int mode Box release
Releases n particles at uniformly distributed random positions inside the box in m. The positions are reproducible and do not depend on L 11 or on the domain decomposition. The remaining values are the same as for L 21.

default: not set

○ ○ ● ○

L 31 double Horizontal diffusivity K_h
Constant horizontal diffusivity in m^2/s for the random walk. It represents the dispersion by processes that are not resolved by the grid. It does not act on stranded or deposited particles.

default: 0.0

○ ○ ● ○

L 32 int Vertical diffusion
Vertical random walk of the three-dimensional particles (mode 0). With option 2 the diffusivity is $K_v(z) = K_{v,0} + \nu_t / Sc_t$ with the background value L 33, the eddy viscosity ν_t of the NHFLOW turbulence model (A 560) and the turbulent Schmidt number L 34; the gradient of K_v is taken into account with the drift correction of Visser [1997].

0 OFF

1 constant diffusivity L 33

2 L 33 plus eddy viscosity / L 34

default: 0

○ ○ ● ○

L 33 double Vertical diffusivity $K_{v,0}$
Constant (L 32 1) or background (L 32 2) vertical diffusivity in m^2/s.

default: 0.0

○ ○ ● ○

L 34 double Turbulent Schmidt number Sc_t
Used with L 32 2.

default: 1.0

○ ○ ● ○

L 41 double U_{10}, double direction Wind for windage
Wind speed at 10 m height (m/s) and the direction it blows towards (deg), measured counter-clockwise from the positive x-axis, as for A 571. The windage $c_w U_{10}$ of each release acts on the particles at the free surface. If L 41 is not given and the NHFLOW wind forcing is active (A 570 $>$ 0), the values of A 571 are used. Note that the wind stress of A 570 – A 578 already drives a surface current in NHFLOW; c_w should therefore only represent the direct wind drag on the emerged part of the particles.

default: 0.0 ; 0.0 (or A 571)

○ ○ ● ○

L 51 int Beaching
Treatment of stranded particles, i.e. particles in dry cells (wetting and drying, A 540 – A 544). Stranded particles are placed on the bed and do not move.

0 stranded, refloat at the free surface when the cell becomes wet again

1 permanent beaching

default: 0

○ ○ ● ○

L 52 int Bed contact

Treatment of three-dimensional particles that reach the bed. Deposited particles stay on the bed; resuspension is not included.

0 reflection

1 deposition of settling particles ($w_s < 0$), reflection of all others

default: 1

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L 53 int Lateral domain boundaries

Treatment of particles that leave the horizontal extent of the computational domain. The boundary type from DIVEMesh is not taken into account. Removed particles are counted in the log file.

0 remove at all boundaries

1 reflect at all boundaries

default: 0

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L 61 double Print interval

Output interval of the particle files in s of simulated time. With a value ≤ 0 the particles are written every time step. The first output is written at the start of the simulation.

default: 1.0

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L 62 int Output format

The log file is always written. With P 18 the particles of options 1 and 3 are written into the LAGOON store `REEF3D_NHFLOW.lagoon`, instead of the VTP files (P 18 1) or in addition to them (P 18 2).

1 VTP files

2 CSV track file

3 VTP files and CSV track file

default: 1

6.11 M :: MPI

Valid for:

S	F	N	C
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M 10 int Number of processes for parallel computations

Must be set, also for serial runs (M 10 1). The value must equal the number of MPI processes REEF3D is started with (e.g. `mpirun -n 4` for M 10 4). If REEF3D reads the grid files of the processes written by DIVEMesh, it must also equal the M 10 used in DIVEMesh to decompose the grid; REEF3D compares all three at start-up and stops with the message "Inconsistent M 10 parameter" if they differ. This also applies to REEF3D::SEASTATE (A 10 7, Section 6.1.4). If REEF3D decomposes the global grid file itself (M 40), the M 10 of DIVEMesh is not used, and the number of processes can be changed without running DIVEMesh again.

default: 0 (not set)

● ● ● ●

M 40 int Domain decomposition by REEF3D

Selects where the grids of the processes come from. DIVEMesh writes either one grid file per process (`DIVEMesh_Grid/grid-<rank>.dat`) or, with DIVEMesh M 50 1 or 2, also the global grid file `DIVEMesh_Grid/grid-global.dat`. From the global grid file, rank 0 of REEF3D decomposes the grid for the M 10 processes of the run and sends every process its grid, the same grid DIVEMesh would write for M 10 processes. The decomposition method and its parameters are those of the DIVEMesh control file stored in the global grid file (DIVEMesh M 11–13, M 20, M 21, M 30–35, M 41, M 45); with M 41 1 the weighted partition with the load map is used instead. With DIVEMesh M 20 3 or 4, the numbers of subdomains of DIVEMesh M 30 must give M 10 processes, otherwise the run stops with a message. With option 0, the global grid file is only used if the grid file of the process is missing. Option 1 stops with a message if there is no global grid file. Set to 1 automatically for a rebalance restart (G 46).

0 grid files of the processes from DIVEMesh, otherwise the global grid file decomposed by REEF3D

1 always the global grid file, decomposed by REEF3D

default: 0

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M 41 int Weighted partition with the load map

When REEF3D decomposes the global grid file (M 40), option 1 uses the weighted partition of DIVEMesh (M 20 5) with the load map (M 35 1), whatever decomposition method the DIVEMesh control file sets. The subdomains stay a tensor product of cuts in x, y and z, placed so that the heaviest subdomain is as light as possible. The weight of a cell is taken from the load map, i.e. the measured work per time step of

the mesh refinement (G 45), read from `loadmap.dat` in the simulation folder or from `REEF3D_<M>_AMR/REEF3D_<M>_AMR_loadmap.dat` (SEASTATE: `REEF3D_SEASTATE_Log/REEF3D_SEASTATE_loadmap.dat`, the measured work of the cells per iteration); without a load map only the active cells count, and a message is printed. Set to 1 automatically for a rebalance restart (G 46).

0 decomposition of the DIVEMesh control file

1 weighted partition with the load map

default: 0

6.12 N :: Numerical Options

Valid for:

S	F	N	C
●	●	●	●

N 10 int Linear solver for the Poisson/Laplace equation

Selects the solver for the Laplace equation of the velocity potential (FNPF), the pressure equation (NHFLOW, CFD, non-hydrostatic SFLOW) and the potential flow initialization (I 11). The default is REEFMG (N 10 1), REEF3D's own multigrid solver, which is available in every build and in all four modules.

The HYPRE solvers (N 10 11–38) are only available if REEF3D is compiled with HYPRE support (`make HYPRE=1`); HYPRE is optional and not part of the default build. In a build without HYPRE, any N 10 $\geq$ 10 is replaced by N 10 1 at start-up and a message is printed, so input files written for the HYPRE solvers run with REEFMG. With HYPRE, use a matching preconditioner in N 11. The fastest HYPRE combination is the Struct BiCGStab solver with PFMG preconditioning (N 10 14, N 11 11); from the HYPRE AIJ solvers, PCG with BoomerAMG (N 10 21, N 11 21) is fastest. For more information on the HYPRE solvers and preconditioners, see the HYPRE user manual Cen [2015].

Module restrictions: SFLOW accepts 1 and 11–19 (HYPRE Struct 2D); without HYPRE it always uses REEFMG. The internal BiCGStab solver (3) with Jacobi preconditioning works for the CFD pressure only. HYPRE AIJ (21–25) and SStruct (31–38) work for FNPF, NHFLOW and CFD, but not for the NHFLOW potential flow initialization (I 11 1).

REEFMG (N 10 1) is a semicoarsening multigrid in x and y with exact vertical line relaxation, used as preconditioner for a BiCGStab iteration (N 11 is ignored). It solves the FNPF Laplace equation, the NHFLOW pressure, the CFD pressure (D 30 1–3), the potential flow initialization (I 11) and, in a 2D version, the SFLOW pressure and the free surface equation of the semi-implicit schemes of SFLOW (A 226) and NHFLOW (A 534); for these 2D equations, any N 10 $\neq$ 1 selects HYPRE Struct 2D in builds with HYPRE. It is controlled by N 12–N 15. The implicit cross-derivative terms of FNPF (A 328 1) and NHFLOW (A 529 1) are only available with REEFMG. REEFMG requires that the grid is not decomposed in the vertical direction, that all ranks at the same x (y) position have the same number of cells in x (y), and that there are no periodic

boundaries in the vertical direction; periodic boundaries in x and y are supported, except in the 2D version for SFLOW, which does not support periodic boundaries. Otherwise it stops with a message. For the potential flow initialization, REEFMG hands the solve to HYPRE in builds with HYPRE if its coarsest grid is too large to resolve the problem reliably (in SFLOW also if the solve diverges); without HYPRE it solves it with REEFMG anyway and prints a warning, and a diverged SFLOW solve is skipped (the initial field is kept).

1 REEFMG, BiCGStab with semicoarsening multigrid (SFLOW, FNPF, NHFLOW, CFD)

3 BiCGStab with Jacobi preconditioning (internal, CFD)

HYPRE solvers, only in builds with HYPRE (`make HYPRE=1`):

11 PCG (HYPRE Struct)

12 GMRES (HYPRE Struct)

13 LGMRES (HYPRE Struct)

14 BiCGStab (HYPRE Struct)

15 Hybrid-PCG (HYPRE Struct)

16 Hybrid-GMRES (HYPRE Struct)

17 Hybrid-BiCGStab (HYPRE Struct)

18 PFMG (HYPRE Struct, geometric multigrid)

19 SMG (HYPRE Struct, geometric multigrid)

21 PCG (HYPRE AIJ)

22 GMRES (HYPRE AIJ)

23 LGMRES (HYPRE AIJ)

24 BiCGStab (HYPRE AIJ)

25 AMG (HYPRE AIJ, BoomerAMG)

31 PCG (HYPRE SStruct)

32 GMRES (HYPRE SStruct)

33 LGMRES (HYPRE SStruct)

34 BiCGStab (HYPRE SStruct)

38 SysPFMG (HYPRE SStruct, geometric multigrid)

default: 1

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N 11 int Preconditioner for the Poisson/Laplace equation

Preconditioner for the HYPRE Krylov solvers selected in N 10, i.e. only used in builds with HYPRE (`make HYPRE=1`); use only options from the same solver family. The multigrid solvers used on their own (N 10 18, 19, 25, 38), REEFMG (N 10 1) and the internal BiCGStab (N 10 3) ignore N 11. With HYPRE Struct and PFMG (11), N 20 together with N 23, N 25 or N 26 can switch the preconditioner to SMG during the run.

0 OFF

10 Jacobi (HYPRE Struct, N 10 11–14)

11 PFMG (HYPRE Struct, N 10 11–17)

12 SMG (HYPRE Struct, N 10 11–17)

21 BoomerAMG (HYPRE AIJ, N 10 21–24)

31 SysPFMG (HYPRE SStruct, N 10 31–34)

32 Split with PFMG (HYPRE SStruct, N 10 31–34)

33 Split with SMG (HYPRE SStruct, N 10 31–34)

default: 11

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N 12 int REEFMG line smoother

Ordering of the vertical line relaxation in REEFMG (N 10 1).

0 one forward sweep before and one backward sweep after the coarse-grid correction

1 symmetric forward and backward sweeps

2 red-black line smoother (experimental, slower on CPUs)

default: 0

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N 13 int REEFMG maximum number of coarse levels

Limits the number of coarse levels (coarsening steps) of REEFMG (N 10 1). In each step both horizontal directions are halved while the mean cell aspect ratio $\Delta x/\Delta y$ stays between $1/\sqrt{2}$ and $\sqrt{2}$; otherwise only the finer direction is halved until the cells are nearly square again. A direction with fewer than 8 cells per process is still halved, down to one cell, as long as the other direction has at least 8. With 0 the grid is coarsened until no direction can be halved any more; in SFLOW the number of active levels is then also limited per solve by a screening length of the problem.

default: 0 (automatic)

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N 14 int REEFMG multigrid precision

Floating-point precision of the REEFMG multigrid hierarchy (N 10 1). With 32 the hierarchy is stored in single precision (about 25% less solver memory and a faster V-cycle); the BiCGStab iteration and the convergence test remain in double precision, so the converged result is unchanged. Any other value uses double precision. With mesh refinement (G 1, Section 6.7.1) N 14 also sets the precision of the multigrids of the composite-grid solves (SFLOW, FNPF, NHFLOW and CFD).

32 single precision

64 double precision

default: 32

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N 15 int REEFMG coarse-grid agglomeration

If ON, the coarsest REEFMG level (N 10 1) is gathered onto every rank and solved there, which replaces many halo exchanges by one all-gather per V-cycle and may pay off at high numbers of processes. It is refused if the gathered problem is too large, and it is not used with periodic boundaries.

0 OFF

1 ON

default: 0

○ ● ● ○

N 18 int Index ordering of the HYPRE Struct grid

Ordering of the grid axes passed to HYPRE Struct (N 10 11–19, builds with HYPRE only). Option 2 makes the vertical index the fastest-running one; it is implemented for the FNPF Laplace equation and the NHFLOW pressure only.

1 standard (x, y, z)

2 permuted (z, y, x)

default: 1

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N 20 int Iteration threshold for preconditioner switch

Only in builds with HYPRE: if a HYPRE Struct solve with PFMG preconditioning (N 11 11) needs more than N 20 iterations, the preconditioner is switched to SMG for the rest of the simulation. The switch is enabled per module with N 23 (FNPF), N 25 (NHFLOW) and N 26 (CFD).

default: 20

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N 23 int Preconditioner switch for FNPF
Enables the PFMG-to-SMG preconditioner switch (N 20) for FNPF.

0 OFF

1 ON

default: 0

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N 25 int Preconditioner switch for NHFLOW
Enables the PFMG-to-SMG preconditioner switch (N 20) for NHFLOW.

0 OFF

1 ON

default: 0

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N 26 int Preconditioner switch for CFD
Enables the PFMG-to-SMG preconditioner switch (N 20) for CFD.

0 OFF

1 ON

default: 1

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N 40 int Time scheme for the momentum equations
In the coupled schemes the free surface (level set or VOF) is advanced within the Runge-Kutta stages of the momentum equations; in the separate schemes the level set is advanced with its own time scheme (F 30 = 2 or 3). Options 22, 23 and 24 are identical to 2, 3 and 4. With PLIC (F 80 = 4), options 3, 23, 13 and 33 use their PLIC variants. Override: with 6DOF (X 10 > 0), N 40 = 3 or 23 is changed to 4 (not with mesh refinement, G 1 > 0) and N 40 = 13 to 44. The CFD mesh refinement (G 1) requires N 40 = 2 or 3 (otherwise it is switched off with a message), and the air compressibility W 32 stops with N 40 = 4, 24 or 44. The former option 5 (3rd-order Runge-Kutta with Crank-Nicolson diffusion) has been removed; selecting it stops the simulation with a message. The former option 14 (low-storage Runge-Kutta with its own time loop and momentum classes) has been removed as well; N 40 = 14 is changed to 44 automatically.

0 OFF

2 2nd-order TVD Runge-Kutta (momentum-free surface coupled)

3 3rd-order TVD Runge-Kutta (momentum-free surface coupled)

4 3rd-order low-storage Runge-Kutta (momentum-free surface coupled)

12 2nd-order TVD Runge-Kutta (separate free surface)

13 3rd-order TVD Runge-Kutta (separate free surface)

33 3rd-order TVD Runge-Kutta (momentum-free surface coupled), conservative formulation with face densities

44 3rd-order low-storage Runge-Kutta (separate free surface)

default: 3

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N 41 double Maximum simulated time

End time of the simulation in *s*. Of N 41 and N 45, whichever criterion is reached first ends the simulation. Also used by REEF3D::SEASTATE (A 10 7, Section 6.1.4), whose time step is set by its own commands instead of N 47 – N 49.

default: $1.0 \cdot 10^{19}$ *s* (no limit)

○ ○ ● ●

N 43 double Stopping criterion for the convection-diffusion solver

Convergence tolerance of the iterative solver for the implicit transport equations (implicit diffusion, turbulence, scalars), i.e. the internal BiCGStab solver and the HYPRE Struct/SStruct BiCGStab used for these equations.

default: $1.0 \cdot 10^{-5}$

● ● ● ●

N 44 double Stopping criterion for the Poisson/Laplace solver

Convergence tolerance of the solver selected in N 10 (pressure, Laplace and potential flow equations), of the SFLOW solvers and of the composite-grid solves with mesh refinement (G 1). The Laplace solves of the body loads in FNPF use X 30 $\times$ N 44.

default: $1.0 \cdot 10^{-8}$

● ● ● ●

N 45 int Maximum number of time steps

Maximum number of outer iterations (time steps). Of N 41 and N 45, whichever criterion is reached first ends the simulation. Also used by REEF3D::SEASTATE (Section 6.1.4).

default: $1.0 \cdot 10^{8}$ (effectively no limit)

● ● ● ●

N 46 int Maximum number of solver iterations

Maximum number of iterations of the linear solvers per solve. REEFMG and the pressure and diffusion solves of the NHFLOW mesh refinement (G 1) print a warning when the limit is reached. The SFLOW potential flow initialization and the CFD potential flow initialization in the water phase (I 11 2) use a fixed limit of 2500.

default: 250

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N 47 double Relaxation factor for time step size

Safety factor (Courant number) that multiplies the maximum stable time step from the CFL criterion for adaptive time stepping (N 48 1). In SFLOW with the Boussinesq option A 220 = 4 values above 0.25 are reduced to 0.25. In NHFLOW with explicit momentum diffusion (A 512 1) the diffusive limit of the time step is not reduced by N 47.

default: 0.3

● ● ● ●

N 48 int Adaptive time stepping

0 OFF, fixed time step N 49

1 ON, time step from the CFL criterion with N 47

default: 1

● ● ● ●

N 49 double Time step size for fixed time stepping

Time step in *s* for fixed time stepping (N 48 0). In FNPF and NHFLOW it is also the upper limit of the adaptive time step (not with the semi-implicit free surface of NHFLOW, A 534, which uses A 536 instead).

default: 1.0 *s*

○ ○ ○ ●

N 50 double Adaptive time stepping method

Formulation of the CFL criterion for adaptive time stepping in CFD (N 48 1).

1 standard, based on the velocity magnitude and the smallest cell size

2 component-wise, based on each velocity component and the cell size in that direction (less conservative)

default: 1

● ● ● ●

N 61 double Critical velocity for emergency stop

If the maximum velocity in any direction exceeds this value in m/s, the simulation stops and writes a final output file. In FNPF, velocities in the first five cells at x_{min} are additionally limited to 0.95 · N 61.

default: 500.0 m/s

6.13 P :: Printer

The P commands control the output of REEF3D. All output is written into folders inside the directory from which the simulation is started. The folder names start with `REEF3D_` followed by the name of the hydrodynamic model, which corresponds to the validity columns: SFLOW (S), FNPF (F), NHFLOW (N) and CFD (C), e.g. `REEF3D_CFD_VTU` or `REEF3D_FNPF_WSF`. In the descriptions below, `<M>` stands for this model name. Model-independent output goes to `REEF3D_Log` (log files, discharge gauges), `REEF3D_Log-Wave` (wave time series), `REEF3D_SOLID` and `REEF3D_FlowFile`. Every run also appends to the run log `REEF3D_Case/REEF3D_<M>_run.jsonl` (one JSON line at the start and end of the run, per output stream and per written output, which LAGOON reads, see P 18) and keeps a copy of the control file as `REEF3D_Case/ctrl-<run id>.txt`; the run id consists of the start time (UTC) and a random suffix.

Many outputs can be triggered either by the iteration count or by the simulation time. An iteration-based interval (e.g. P 20, P 41, P 54, P 181, P 191) writes a file every n^{th} time step. A time-based interval in seconds (e.g. P 30, P 42, P 55, P 182, P 192) writes a file at the first time step after each multiple of the interval, and once at the start of the simulation. The time-based setting takes precedence: the iteration-based output of the same kind is only active while the corresponding time-based value is negative. Negative defaults (e.g. -10 or -1.0) mean inactive. In addition, P 35, P 184, P 185, P 194 and P 195 define print-out windows with their own interval; they can be given several times and act on top of the regular intervals. P 34 and P 183 use the sediment (morphological) time instead of the hydrodynamic time. With the defaults, no periodic 3D field output is written: set P 20 or P 30 (and P 181 or P 182 for free-surface files and for SFLOW). Point gauges and probes are written every time step into one continuous file, line output (e.g. P 52, P 62, P 67, P 144) writes one new file per print-out.

Valid for:

S	F	N	C
●	●	●	●

P 10 int Format of the 3D field output (ParaView binary)

The 3D files are written into `REEF3D_<M>_VTU`, `REEF3D_<M>_VTS` or `REEF3D_<M>_VTR` for FNPF, NHFLOW and CFD. VTR is only available for CFD; in FNPF and NHFLOW option 3 switches the 3D output off. SFLOW writes 2D VTP files instead (`REEF3D_SFLOW_VTP_FSF` and `REEF3D_SFLOW_VTP_BED`, see P 181), which are written

for any P 10 > 0 (also the mesh refinement output of SFLOW). REEF3D::SEASTATE (A 10 7, Section 6.1.4) likewise writes its wave parameter maps as VTP files into `REEF3D_SEASTATE_VTP` for any P 10 > 0, with the interval of P 181 or P 182. With P 18 the VTU and VTS files can be written into the LAGOON store instead of or in addition to the files.

0 OFF

1 VTU (unstructured grid)

2 VTS (structured grid)

3 VTR (rectilinear grid, CFD only)

default: 1

● ● ● ●

P 12 int Terminal and log print frequency

Writes the iteration summary to the terminal and to the log files in `REEF3D_Log` (main, max, solver and volume log) every n^{th} iteration. Use larger values to reduce terminal output for long simulations. Also sets the interval of the terminal output of REEF3D::SEASTATE (Section 6.1.4).

default: 1

● ● ● ●

P 15 int Numbering of the output files

Applies to the 3D and VTP field files and to most other numbered output files, including the VTP files of REEF3D::SEASTATE (Section 6.1.4).

1 consecutive print-out counter

2 iteration number

default: 1

● ● ● ●

P 16 int Add time stamp to ParaView files

Writes the simulation time as `TimeValue` field data into the VTU/VTS/VTR and VTP files, so that ParaView shows the physical time instead of the file number, together with the run id of the run log (see the introduction of this section) as `RunId` field data. For the 3D files the sediment time is used when P 34 > 0.

0 OFF

1 ON

default: 1

● ● ● ●

P 18 int LAGOON store

Writes the field output into a LAGOON store, the data format of LAGOON, the post-processing and visualisation tool of REEF3D. The store is the folder `REEF3D_<M>.lagoon` in the simulation directory, a Zarr v3 store with one group per output stream and one block per MPI process. The arrays are stored as float32, compressed (byte shuffle and gzip) in shards of 16 outputs; each process writes its own block in a background thread while the solver goes on, and an output is committed once all processes have written it, so the store can be read during the run. A store holds one run: the store of an earlier run in the same folder is renamed to `REEF3D_<M>-<run id>.lagoon-earlier`. The outputs keep their print intervals. The store takes the 3D field files of P 10 = 1 or 2 (FNPF, NHFLOW and CFD; not VTR, and not with moving or tilted grids, B 180, B 191, B 192), the free surface VTP files (SFLOW, and P 180 for FNPF and NHFLOW), the bed VTP files (SFLOW, FNPF, NHFLOW, and the CFD topo files of P 190), the mesh refinement output (the `.vtr`/`.vtm` files of SFLOW, FNPF, NHFLOW and CFD with G 1 > 0, Section 6.7.1), the floating body VTP files (the mesh once, then the rigid-body motion per output) together with the mooring lines of REEF3D::CORAL (Z 40; polylines with their tension), the NHFLOW particles (L 62 = 1 or 3), the CFD CPM parcels (Q 10 $\geq$ 1; gathered on rank 0, set `cfd_particles` with the arrays of the VTP files: velocity, fluid velocity, radius, weight with Q 78, bed change, flag and parcel id), the DEM elements and the FNPF ice floes. On a curvilinear grid (G 50) the field and surface outputs carry the coordinates of every grid point; the mesh refinement output of SFLOW on a curvilinear grid is written as `.vts` files and not into the store, and NHFLOW on a curvilinear grid stops with P 18 > 0 together with mesh refinement or a multi-block grid. The CFD free surface files, the REEF3D::SEASTATE output, the state files, the checkpoints and all gauges and probes are written as before. If the store cannot be written, the files are written again from then on, also with option 1.

0 OFF

1 LAGOON store instead of the VTU/VTP files

2 LAGOON store as well as the VTU/VTP files

default: 0

○ ● ● ○

P 19 int Print interval for NHFLOW and FNPF floating body files

Selects which print-out interval is used for the floating body VTP/STL files in `REEF3D_<M>_6DOF_VTP` and `REEF3D_<M>_6DOF_STL`. In CFD the floating body files always follow the 3D interval (P 20, P 30, P 35), in SFLOW always the VTP interval (P 181, P 182, P 185).

1 3D (VTU) interval: P 20, P 30 or P 35

2 VTP interval: P 181, P 182 or P 185

default: 2

○ ● ● ●

P 20 int Print 3D results every n^{th} iteration

Writes the 3D field files (see P 10) into `REEF3D_<M>_VTU`. Only active if P 30 and P 34 are negative. Choose one of P 20, P 30, P 34 or P 35. For SFLOW use P 181. Also sets the interval of the floating body files in CFD (and NHFLOW and FNPF with P 19 = 1). With mesh refinement in FNPF and NHFLOW (G 1 > 0, Section 6.7.1), the free surface of level 0 and of the refined patches is written with the same interval (with P 180 = 1 with the interval of the free surface files instead, P 181 or P 182) as VTR files (VTS files on a curvilinear grid, G 50) with a `.vtm` multiblock index into `REEF3D_<M>_AMR`, in addition to the regular 3D files (into the LAGOON store with P 18). With the CFD mesh refinement, every refined patch is written with the same interval as a 3D VTR file `REEF3D_CFD_AMR/REEF3D-CFD-AMR-<n>-<patch>.vtr` (cell data: level set, pressure, velocity, level of the patch and a flag for the cells covered by a finer patch), together with the multiblock index `REEF3D-CFD-AMR-<n>.vtm`, to be opened next to the level-0 output (into the LAGOON store with P 18).

default: -10 (inactive)

○ ○ ○ ●

P 21 int Print time-averaged velocity, pressure and temperature

Adds `velocity_mean`, `pressure_mean` and, with heat transfer (H 10), `T_mean` to the CFD VTU files. Averaging starts at the time given with P 22.

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 22 double Start time for time averaging

Simulation time in s after which the averaging for P 21 starts. Set it after the initial transients have left the domain.

default: 0.0

● ● ● ●

P 23 int Print test array

Adds the internal `test` array to the 3D files and the VTP files. Used for debugging and development.

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 24 int Print density to the VTU file
Adds the density `rho` in kg/m^3. Ignored for multiphase flow (F 300 > 0), which writes its own density field.

0 OFF

1 ON

default: 0

○ ● ● ●

P 25 int Print solid to the VTU file
Adds the solid level set function `solid`. In NHFLOW, the `Heaviside` function of the solid is written as well.

0 OFF

1 ON

default: 0

○ ○ ● ●

P 26 int Print suspended sediment concentration to the VTU file
Adds `ST_conc`. Use together with sediment transport (S 10).

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 27 int Print topo to the VTU file
Adds the bed level set function `topo`. Set to 1 automatically when sediment transport is active (S 10 $\geq$ 1) or when a topography is read from the DIVEMesh grid file.

0 OFF

1 ON

default: 0

● ○ ● ●

P 28 int Print floating body level set function
Adds the floating body level set function (`floating` in the 3D files, `fb` in the SFLOW and NHFLOW free surface VTP files).

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 29 int Print wall distance to the VTU file
Adds `walldist`, the distance to the nearest wall in m.

0 OFF

1 ON

default: 0

○ ● ● ●

P 30 double Print 3D results every t seconds
Time-based alternative to P 20, writes into `REEF3D_<M>_VTU`. Only active if P 34 is negative. Choose one of P 20, P 30, P 34 or P 35. For SFLOW use P 182. Also used for the AMR output of FNPF, NHFLOW (without P 180) and CFD (see P 20).

default: -1.0 (inactive)

○ ○ ● ●

P 34 double Print 3D results every t seconds of sediment time
Uses the morphological (sediment) time instead of the hydrodynamic time. Only active if P 30 is negative. Choose one of P 20, P 30, P 34 or P 35.

default: -1.0 (inactive)

○ ● ● ●

P 35 * double t_{start}, double t_{end}, double Δt Print 3D results in a time window
Writes the 3D files every Δt seconds between t_{start} and t_{end} (all in s). Several windows can be defined; they act in addition to P 20 or P 30. In CFD the floating body files follow these windows as well.

default: na ; na ; na

● ● ● ●

P 40 int Print state files
Writes binary state files into `REEF3D_<M>_STATE` with the interval of P 41 or P 42. They are used for restart (see I 40 for full initialization from a state file), for hydrodynamic coupling (HDC, e.g. SFLOW or FNPF to CFD; see the H options in the DIVEMesh manual for building the HDC input from state files) and for grid nesting. The CFD

state file contains the velocities, pressure, level set function, k, eddy viscosity, ϵ or ω, topo, cbed and conc. With the particle sediment model (Q 10 $\geq$ 1), the particles are written into additional files `REEF3D-CFD-CPM-State-<n>-<rank>.r3d` in the same folder. Related commands: P 43 to P 47. For restarts, the checkpoints of P 48 are recommended: they hold the complete state of the run and can be read on any decomposition (see Section 6.9).

0 OFF

1 ON

default: 0

● ● ● ●

P 41 int Print state files every n^{th} iteration

Only used if P 42 is negative. With the default, a state file is written every iteration when P 40 = 1.

default: 1

● ● ● ●

P 42 double Print state files every t seconds

Time-based alternative to P 41. For hydrodynamic coupling, match the interval to the time step expected by the receiving simulation (see I 241).

default: -1.0 (inactive)

● ● ● ●

P 43 double x_{start}, double x_{end}, double y_{start}, double y_{end} Print area for state files

Restricts the state files of P 40 to the given horizontal area in m. Reduces the file size when only a part of the domain is needed, e.g. for HDC or nesting. Without P 43 the whole domain is written.

default: na ; na ; na ; na

○ ● ● ○

P 44 int Components of the FNPF and NHFLOW state files

For FNPF the options below apply. For NHFLOW, η and the velocities are always written; P 44 = 1 adds the pressure.

1 η, free surface potential, velocities (standard for HDC)

2 η, free surface potential, velocities, 3D potential

3 η, free surface potential (more efficient for FNPF nesting)

default: 1

● ● ● ●

P 45 int State file type

1 single file per print-out (numbered according to P 15)

2 continuous file (one file per processor)

default: 2

● ● ● ●

P 46 int i_{start}, int i_{end} Iteration window for state files
State files (P 40, P 41) are only written for iterations $i_{start} \leq n < i_{end}$.

default: na ; na (inactive)

● ● ● ●

P 47 double t_{start}, double t_{end} Time window for state files
Time-based state files (P 40, P 42) are only written for simulation times $t_{start} \leq t < t_{end}$ in s.

default: na ; na (inactive)

● ● ● ●

P 48 int n, double Δt Write a checkpoint every n^{th} iteration or every Δt seconds
Writes a checkpoint for the restart with I 45 at every iteration that is a multiple of n, into the folder `REEF3D_Checkpoint/<count>`, and updates `REEF3D_Checkpoint/latest` once it is complete (see Section 6.9 for the files and what they hold). The second value is optional: with $\Delta t > 0$ a checkpoint is also written at the first time step at or past each multiple of Δt seconds of simulated time; with P 48 0 Δt the checkpoints follow the time only. SFLOW, FNPF, NHFLOW and CFD also write a checkpoint at the end of the run, and a run that receives SIGTERM or SIGUSR1 writes one at the end of the current time step and stops. Each checkpoint holds the complete state; the newest P 49 checkpoints are kept, and the latest one is used for a restart (or the one selected with I 45). The time needed for writing is printed. Stand-alone SEASTATE runs (A 10 7) write a checkpoint every n time steps when nonstationary (A 700 1), and every n iterations within a time step when stationary (A 700 2; not after the last iteration of the step), then into `REEF3D_Checkpoint/<count>_<iteration>`. If the case contains a feature that the checkpoints do not cover (e.g. SEASTATE coupled to a host model, A 750, NHFLOW Lagrangian particles, L 10), a message is printed at the start and no checkpoint is written.

default: 0 ; 0.0 (OFF)

● ● ● ●

P 49 int Number of checkpoints kept

Only the newest P 49 checkpoints of P 48 are kept: after a checkpoint is complete, the older folders listed in `REEF3D_Checkpoint/history` (the checkpoints in the order they were written) are removed. The checkpoint named in `REEF3D_Checkpoint/latest` is never removed, and folders removed by hand are dropped from the list. With 0 all checkpoints are kept. Without P 49, two checkpoints are kept (so that one is left that an instability may not yet have reached), in stand-alone SEASTATE runs (A 10 7) one, since a spectrum per cell is large. After a restart from an earlier checkpoint (I 45), the later checkpoints of the earlier run are removed first.

0 all checkpoints are kept

default: 2 (SEASTATE: 1)

● ● ● ●

P 50 * double *x*, double *y* Wave gauge for the theoretical wave

Location in m. Records the free surface elevation of the wave theory selected for wave generation every time step into `REEF3D_<M>_WSF/REEF3D-<M>-WSF-HG-THEORY.dat`. Requires wave generation (B 90). Useful for comparing with P 51 at the same locations.

default: na ; na

● ● ● ●

P 51 * double *x*, double *y* Wave gauge

Location in m. Records the simulated free surface elevation every time step into `REEF3D_<M>_WSF/REEF3D-<M>-WSF-HG.dat`, one column per gauge. In CFD the surface is found from the level set function (from the VOF function for F 80 = 4). With mesh refinement (G 1 > 0, Section 6.7.1) the elevation is taken from the finest grid level holding each gauge (in CFD from the column of the finest patch at the free surface of level 0). In SFLOW, FNPF and NHFLOW these elevations are also written, together with the level that holds each gauge, into `REEF3D_<M>_AMR/REEF3D_<M>_AMR_gauges.dat`.

default: na ; na

● ● ● ●

P 52 * double *y* Free surface line in x-direction

y-location in m of a line along the x-direction. The free surface along the line is written into `REEF3D_<M>_WSFLINE`, one file per print-out with the interval of P 54 or P 55. In SFLOW, FNPF and NHFLOW the free surface is interpolated linearly in *y* between the two cell centres next to the line (the value of the cell holding the line next to a dry or solid cell).

default: na

● ● ● ●

P 53 int Add the theoretical wave to the free surface line files
Adds the free surface of the wave theory to the files of P 52 and P 56. Requires wave generation (B 90).

0 OFF

1 ON

default: 0

● ● ● ●

P 54 int Print line output every n^{th} iteration
Interval for the free surface lines (P 52, P 56), the CFD line probes (P 62), the SFLOW depth-averaged probes (P 63) and the NHFLOW profiles and line probes (P 67, P 144). Only used if P 55 is negative (P 63 always uses P 54).

default: 10

● ● ● ●

P 55 double Print line output every t seconds
Time-based alternative to P 54 for P 52, P 56, P 62, P 67 and P 144.

default: -1.0 (inactive)

● ● ● ●

P 56 * double x Free surface line in y-direction
x-location in m of a line along the y-direction. Written into `REEF3D_<M>_WSFLINE_Y` with the interval of P 54 or P 55. In SFLOW, FNPF and NHFLOW the free surface is interpolated linearly in x, as for P 52.

default: na

○ ● ○ ○

P 57 int Additional output for the FNPF wave gauges
Intended to add the rise velocity $\partial\eta/\partial t$ and the horizontal free surface velocity at the P 51 gauges, in `REEF3D_FNPF_WSF/REEF3D-FNPF-Rise-Velocity.dat` and `REEF3D_FNPF_WSF/REEF3D-FNPF-Horiontal-FSF-Velocity.dat`. Note: this output is currently not functional in the code. The value after P 57 is not read; the option is set by the number of P 57 lines in `ctrl.txt`, and only the file headers are written.

0 OFF

1 rise velocity

2 horizontal free surface velocity

3 both

default: 0

● ● ● ●

P 58 * double x, double y, double T Theoretical wave time series

Writes the free surface elevation of the wave theory at (x, y) in m from $t = 0$ to T in s, sampled every 0.1 s, into `REEF3D_Log-Wave/REEF3D-Wave-Timeseries-<n>.dat`. The series is computed once at the start of the simulation. Requires wave generation (B 90); not written for B 92 = 20, 21, 22, 23, 61 and 62. If no P 58 is given, one series at $(0, 0)$ with $T = 12800$ s is written.

default: 0.0 ; 0.0 ; 12800.0

○ ● ○ ○

P 59 int Print breaking wave log

Writes the time and the (x, y) location of every cell flagged as breaking, every time step, into `REEF3D_FNPF_Breaking_Log`. With the breaking model A 350 4, rank 0 also writes every time step the wave crests with a breaking indicator $B > 0.5$ or in the breaking state (time, crest position, elevation, crest speed C, particle speed u_s, B, state, wave height, local depth, dissipation parameters) into `REEF3D_FNPF_Breaking_Log/REEF3D-FNPF-Breaking-Crests.dat`.

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 61 * double x, double y, double z Point probe

Location in m. Records u, v, w, p, k, ϵ or ω and the eddy viscosity every time step into `REEF3D_CFD_ProbePoint/REEF3D-CFD-Probe-Point-<n>.dat`. With mesh refinement (G 1 > 0) the values are taken from the finest grid level holding the probe.

default: na ; na ; na

○ ○ ○ ●

P 62 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} Line probe

Start and end point of the line in m. Writes x, y, z, u, v, w, p, k, ω, eddy viscosity and level set function along the line into `REEF3D_CFD_ProbeLine`, one file per print-out with the interval of P 54 or P 55.

default: na ; na ; na ; na ; na ; na

● ○ ○ ○

P 63 * double x, double y Depth-averaged point probe
Location in m. Records the depth-averaged velocities U, V, W, the pressure, η and the bed level every P 54 iterations into `REEF3D_SFLOW_ProbePoint/REEF3D-SFLOW-Probe-Point-<n>.dat`. With mesh refinement (G 1 > 0) the values are taken from the finest grid level holding the probe.

default: na ; na

○ ○ ● ●

P 64 * double x, double y, double z Pressure probe
Location in m. Records the pressure every time step into `REEF3D_CFD_PressureProbe/REEF3D-CFD-Probe-Pressure-<n>.dat` (CFD) or `REEF3D_NHFLOW_ProbePoint/REEF3D-NHFLOW-Probe_Press-<n>.dat` (NHFLOW). In NHFLOW the recorded pressure is the sum of the non-hydrostatic pressure and the hydrostatic pressure $\rho g(d+\eta-z)$ at the probe, with the still water level d and the density W 1. With mesh refinement (G 1 > 0) the values are taken from the finest grid level holding the probe.

default: na ; na ; na

○ ● ● ●

P 65 * double x, double y, double z Velocity probe
Location in m. Records u, v, w (FNPF also the potential) every time step into `REEF3D_<M>_ProbePoint/REEF3D-<M>-Vel-Probe-<n>.dat`. With mesh refinement (G 1 > 0) in FNPF and NHFLOW the values are taken from the finest grid level holding the probe.

default: na ; na ; na

○ ● ● ●

P 66 * double x, double y, double z Velocity probe for the theoretical wave
Location in m. Records the velocities of the wave theory every time step into `REEF3D_<M>_ProbePoint/REEF3D-<M>-Vel-Probe-Theory-<n>.dat`. Requires wave generation (B 90).

default: na ; na ; na

○ ○ ● ○

P 67 * double x, double y Vertical velocity profile
Location in m. Writes the vertical profile of the horizontal velocity u over z into `REEF3D_NHFLOW_U-Profile/REEF3D-NHFLOW-U-Profile-<n>-<i>.dat` (with i the iteration number), one file per print-out with the interval of P 54 or P 55.

default: na ; na

○ ○ ○ ●

P 71 int Print viscosity to the VTU file
Adds the molecular `viscosity` field in m^2/s.

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 72 int Print VOF function to the VTU file
Adds the volume fraction as point data (`VOF`) and as cell data (`VOF_C`).

0 OFF

1 ON

default: 0

● ○ ○ ○

P 73 int Vertical coordinate of the SFLOW free surface VTP files
Selects the quantity used as the z-coordinate of the points in `REEF3D_SFLOW_VTP_FSF`. Options 1 and 2 show the water depths h_x and h_y at the cell faces used by the SFLOW fluxes, plotted on top of the bed. Any other value gives a flat surface at $z = 0$.

0 free surface elevation with wet-dry treatment: dry cells contribute an elevation below the bed to the interpolation, and points between four wet cells are kept at least A 244 above the bed

1 h_x + bed level

2 h_y + bed level

3 free surface elevation, interpolated without wet-dry treatment

default: 0

○ ○ ● ○

P 74 int Print sigma-coordinate vertical velocity to the VTU file
Adds `omega_sig`, the vertical velocity relative to the sigma levels in m/s.

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 75 int Print vorticity to the VTU file

0 OFF

1 ON

default: 0

● ○ ● ●

P 76 int Print sediment bedload: q_{be}, q_b
Written to the CFD VTU and topo VTP files (P 190) and to the bed VTP files of SFLOW and NHFLOW. Use together with sediment transport (S 10).

0 OFF

1 ON

default: 0

● ○ ● ●

P 77 int Print bed slope sediment parameters: $\alpha, \beta, \theta, \gamma, \phi$
Written to the same files as P 76.

0 OFF

1 ON

default: 0

● ○ ● ●

P 78 int Print sediment parameters: dh, bedchange, reduce, threshold, slideflag
Written to the same files as P 76.

0 OFF

1 ON

default: 0

● ○ ● ●

P 79 int Print bed shear stress
Writes the effective and the critical value of the selected quantity. Written to the same files as P 76.

0 OFF

1 bed shear stress

2 shear velocity

3 Shields parameter

default: 0

○ ● ● ○

P 80 int Print P 85 and P 88 results every n^{th} iteration

default: 1

○ ○ ● ●

P 81 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} Force calculation box

Box in m. Integrates pressure and viscous forces on the solid surfaces (solids and topography from DIVEMesh) inside the box and writes F_x, F_y, F_z in N every time step into `REEF3D_CFD_Force/REEF3D_CFD_Force-<n>.dat` or `REEF3D_NHFLOW_Force/REEF3D_NHFLOW_Force-<n>.dat`. The triangulated surface is written once at the start into `REEF3D_SOLID` (CFD) or `REEF3D_NHFLOW_SOLID`. With mesh refinement (G 1 > 0), the refined patches in the box triangulate the surface at their resolution, and every piece of the surface is integrated on the finest grid level at its position; the force file and the surface file are those of level 0. See also P 82, P 91 and P 92.

default: na ; na ; na ; na ; na ; na

○ ○ ● ●

P 82 int Add eddy viscosity to the viscous forces

Includes the turbulent eddy viscosity in the viscous part of the force calculation of P 81 and of the floating body forces (6DOF).

0 OFF

1 ON

default: 0

○ ● ● ○

P 85 * double x_c, double y_c, double r, double C_d, double C_m Morison force on a vertical cylinder

Centre and radius in m, drag and inertia coefficients. Computes the horizontal forces F_x, F_y in N from the Morison equation using the simulated kinematics (ALE approach) every P 80 iterations. Output: `REEF3D_FNPF_Force_ALE/REEF3D_FNPF_ALE_Force-<n>.dat` or `REEF3D_NHFLOW_Force_ALE/REEF3D_NHFLOW_ALE_Force-<n>.dat`.

default: na ; na ; na ; na ; na

○ ● ○ ○

P 88 * double x, double y Print wave kinematics
Location in m. Writes the time and, for each vertical level, z, u, $\partial u/\partial t$, v, $\partial v/\partial t$ (binary, double precision, after a one-line text header) every P 80 iterations into `REEF3D_FNPF_Kinematics/REEF3D_FNPF-Kinematics-<n>.dat`.

default: na ; na

○ ○ ● ●

P 91 double Pressure sampling distance for the force calculation
The pressure is interpolated at a distance of P 91 times the local cell size from the solid surface, along the surface normal. Used by P 81 and by the CFD floating body forces.

default: 0.25

○ ○ ○ ●

P 92 int Force from water or water and air combined
For P 81. With 0, only surface elements in water or close to the free surface (level set $> -1.6\Delta x$) contribute.

0 force only from water

1 force from water and air combined

default: 0

○ ○ ○ ●

P 101 double x_m, double y_m, double z_s, double z_e, double r_1, double r_2 Print sloshing force and moment
Activates the output of the pressure forces on the left and right tank walls and on the bottom and the moment, every time step, into `REEF3D_CFD_Force/REEF3D-CFD-Sloshing-Force.dat`. The moment is taken about the rotation point of B 192. The six values must be given but are currently not used in the calculation.

default: na ; na ; na ; na ; na ; na (inactive)

● ● ● ○

P 110 int Print significant wave height
Computes $H_s = 4\sqrt{\mathrm{var}(\eta)}$ in m at every surface point from the start time of P 111 and adds the field `Hs` to the VTU and free surface VTP files.

0 OFF

1 ON

default: 0

● ● ● ○

P 111 double Start time for the significant wave height calculation

Simulation time in s after which the statistics for P 110 are collected. Set it after the initial transients.

default: 0.0

● ○ ○ ○

P 112 int Print SFLOW maxima of the run

Records per cell, every time step from the start time P 113 on, the maxima needed for a storm surge or flood assessment: `max_eta` (highest water level while wet, in m relative to the still water level F 60, as the wave gauges P 51) and `time_max_eta` (its time in s), `max_depth` (largest water depth while wet, m), `max_speed` (largest depth-averaged speed while wet, m/s) and `time_max_speed`, `flood_duration` (time wet, s), `first_flood_time` (first time wet of a cell that was dry at P 113, s), `inundation` (0 never wet, 1 wet at P 113, 2 dry at P 113 and flooded later) and, with the meteorological forcing (W 200), `max_wind10` (largest 10 m wind, m/s) and `min_air_pressure` (lowest air pressure, Pa). A cell counts as wet if it is wet for the solver and its water depth exceeds P 114. Values that do not exist (e.g. `max_eta` of a cell never wet) are NaN. With mesh refinement (G 1 > 0) the maxima are those of level 0. The fields are written as cell data into `REEF3D_SFLOW_MAX/REEF3D-SFLOW-MAX.pvtp` with one piece `REEF3D-SFLOW-MAX-<rank>.vtp` per process (points at the bed), overwritten at every print-out of the free surface files (P 181, P 182), at an emergency stop and at the end of the run. The highest water level of the domain with its position and time is printed. The fields are part of the checkpoints (P 48), so a restart (I 45) continues them.

0 OFF

1 ON

default: 0

● ○ ○ ○

P 113 double Start time of the SFLOW maxima

Simulation time in s from which the maxima of P 112 and P 115 are recorded, e.g. after the spin-up. The wet state of the cells at this time sets their `inundation` class.

default: 0.0

● ○ ○ ○

P 114 double Flooding depth of the SFLOW maxima

Water depth in m above which a wet cell counts as wet for the maxima of P 112 and P 115, so that the thin water layer of dry cells (A 244) does not flood a cell.

default: 0.05

● ○ ○ ○

P 115 * double *x*, double *y* SFLOW maxima at points

Location in m, e.g. of a high-water mark. The maxima of P 112 of the cell holding each point are written into `REEF3D_SFLOW_MAX/REEF3D-SFLOW-MAX-points.dat`, one line per point with the number, *x*, *y*, the bed level and the fields of P 112 (`nan` for a point outside the domain, in a solid cell or without a value), with the same interval as P 112. P 115 records the maxima also without P 112, which then writes no VTP files.

default: na ; na

● ○ ● ●

P 121 * double *x*, double *y* Bed level gauge

Location in m. Records the bed level at every sediment time step into `REEF3D_<M>_Sediment/REEF3D-<M>-Sediment-Point.dat`. Use together with sediment transport (S 10).

default: na ; na

● ○ ● ●

P 122 int Maximum bed change gauge

Writes the maximum bed change into `REEF3D_<M>_Sediment/REEF3D-<M>-Sediment-Max.dat`.

0 OFF

1 ON

default: 0

● ○ ● ●

P 123 * double *y* Bed profile in x-direction

y-location in m of a line along the x-direction. The bed level along the line is written into `REEF3D_<M>_Sediment/Line`, one file per sediment time step.

default: na

● ○ ● ●

P 124 * double *x* Bed profile in y-direction

x-location in m of a line along the y-direction. Written into `REEF3D_<M>_Sediment/Line`, one file per sediment time step.

default: na

● ○ ● ●

P 125 * double *x*, double *y* Bed shear stress gauge

Location in m. Records the bed shear stress in N/m^2 into `REEF3D_<M>_Sediment/REEF3D-<M>-Sediment-Bedshear.dat`.

default: na ; na

● ○ ● ●

P 126 int Maximum bed shear stress print-out

Writes the maximum bed shear stress over time into `REEF3D_NHFLOW_Sediment/REEF3D-NHFLOW-Sediment-Bedshear-Max.dat` for NHFLOW and into `REEF3D_CFD_Sediment/REEF3D-CFD-Sediment-Bedshear-Max.dat` for the other models.

0 OFF

1 ON

default: 0

○ ○ ● ○

P 131 int Print maximum wet-dry state (inundation) into the free surface VTP file

Adds `wetdry_max`, which marks every cell that has been wet at least once. Requires P 180 = 1.

0 OFF

1 ON

default: 0

○ ○ ● ○

P 133 * double *y* Run-up gauge in x-direction

y-location in m of a cross section along the x-direction. Records the *x*-position and elevation of the wet-dry front (the wet cell furthest in positive x-direction) every time step into `REEF3D_NHFLOW_RUNUP/REEF3D-NHFLOW-runup-x.dat`.

default: na

○ ○ ● ○

P 134 * double *y* Maximum run-up gauge in x-direction

As P 133, but records the maximum run-up reached so far and the time it occurred into `REEF3D_NHFLOW_RUNUP/REEF3D-NHFLOW-runup-max-x.dat`. The file is overwritten every time step.

default: na

○ ● ○ ○

P 140 * double x_c, double y_c Run-up on a vertical cylinder

Centre of the cylinder in m, radius from P 141. Evaluates six empirical run-up formulas (R1 to R6, e.g. MacCamy and Fuchs, Hallermeier, Peng et al.) from the local wave kinematics every time step and writes them into `REEF3D_FNPF_Runup/REEF3D_Runup-<n>.dat`.

default: na ; na

○ ● ○ ○

P 141 double Cylinder radius for the run-up gauges

Radius in m, used for all P 140 gauges.

default: 0.0

○ ○ ● ○

P 144 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, int n Depth-averaged velocity line probe

Start and end point of the line in m and number of points along the line. Writes the depth-averaged velocities along the line into `REEF3D_NHFLOW_DEP-AVG-VEL-LINE`, one file per print-out with the interval of P 54 or P 55.

default: na ; na ; na ; na ; na

○ ○ ● ○

P 145 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, int n, double t_{start}, double t_{end} Depth- and time-averaged velocity line probe

As P 144, but the depth-averaged velocities are also averaged in time between t_{start} and t_{end} in s. One file is written at t_{end} into `REEF3D_NHFLOW_DEP_TIME_AVG_VEL_LINE`.

default: na ; na ; na ; na ; na ; na ; na

○ ○ ● ○

P 146 * double y, double t_{start}, double t_{end} Time-averaged free surface line in x-direction

y-location of the line in m and averaging window in s. One file is written at t_{end} into `REEF3D_NHFLOW_TIME_AVG_WSFLINE`.

default: na ; na ; na

○ ○ ● ○

P 147 * double x, double t_{start}, double t_{end} Time-averaged free surface line in y-direction
x-location of the line in m and averaging window in s. One file is written at t_{end} into `REEF3D_NHFLOW_TIME_AVG_WSFLINE_Y`.

default: na ; na ; na

○ ○ ● ○

P 148 * double x, double y, double t_{start}, double t_{end} Time-averaged horizontal velocity profile
Location in m and averaging window in s. Writes the time-averaged vertical profiles of u, v and of the horizontal velocity magnitude once at t_{end} into `REEF3D_NHFLOW_TIME_AVG_VEL_PROFILE`.

default: na ; na ; na ; na

○ ○ ○ ●

P 151 int Type of the DIVEMesh data array
When the DIVEMesh grid file contains an additional data array, it is read automatically and written to the CFD VTU files as `data`. P 151 selects how the values are interpreted.

1 scalar

2 signed distance function (converted from a surface elevation)

default: 1

○ ○ ○ ●

P 152 int Type of boundary condition for the DIVEMesh data array
Ghost cell treatment of the data array of P 151.

1 x-velocity

2 y-velocity

3 z-velocity

4 scalar with Neumann boundary conditions

default: 4

○ ○ ● ○

P 160 double t_0, double Δt_{out} Long-run statistics: harmonic analysis and time averages
From the simulation time t_0 in s on, every time step adds the water level η, the depth-averaged velocities u, v and the depth-integrated transports of every column to running sums, weighted with the time step. Every Δt_{out} s and at the end of the simulation (N 41 or N 45; with $\Delta t_{out} \leq 0$ only then), maps of the record $[t_0, t]$ are written

into REEF3D_NHFLOW_Tidal: REEF3D-NHFLOW-Tidal-<n>-<rank>-<grid>.vts per process and grid, the multiblock index REEF3D-NHFLOW-Tidal-<n>.vtm and REEF3D-NHFLOW-Tidal-<n>.txt with the record start and length and the resolved constituents (speed in deg/h, period in h). The harmonic analysis is a least-squares fit $x(t) = Z_0 + \sum_k a_k \cos\omega_k t + b_k \sin\omega_k t$ of η, u and v for the constituents M2, S2, K1, O1, M4, N2, K2, P1, Q1, MS4, MN4, M6 and 2N2; only the constituents that the record length resolves are used (P 161). For every resolved constituent C the maps contain C_eta_amp and C_eta_pha (amplitude in m and phase in deg of $\eta = A\cos(\omega t - g)$, relative to $t = 0$ of the run, without nodal corrections; with P 162 the amplitude $H = A/f$ and the Greenwich phase lag $G = g + V_0 + u$) and the tidal current ellipse C_major, C_minor (semi-axes in m/s, minor axis positive for anticlockwise rotation) and C_incl (inclination of the major axis in deg from the x-axis). The time averages are eta_mean, u_mean, v_mean (Eulerian residual current), qx_mean, qy_mean (residual transport), eta_max, eta_min (over the wet times), speed_max and wet_fraction (fraction of the time the column is wet), together with the bed level. Dry columns enter the fit with η at the bed level and zero velocity; use wet_fraction to judge intertidal areas. The maps cover level 0 and the blocks of a multi-block grid (G 50), not the refined patches of G 1. The record and its sums are part of the checkpoints (P 48), so a restart (I 45) continues the record on any decomposition; a checkpoint written without P 160 starts the record at the restart.

default: na ; na (OFF)

○ ○ ● ○

P 161 double Rayleigh criterion for the long-run statistics

A constituent of the harmonic analysis of P 160 is used if $f_k T \geq R$ and $|f_k - f_l| T \geq R$ for every constituent l before it in the list of P 160, with the frequencies f, the record length T and the criterion R = P 161. With the default, M2 and S2 are separated after about 14.8 days. Smaller values admit more constituents in short records, at the cost of a less reliable separation.

default: 1.0

○ ○ ● ○

P 162 int year, int month, int day, int hour, int minute Start date of the run for the harmonic analysis

Date and time (UTC) of $t = 0$ of the run. With P 162, the harmonic analysis of P 160 applies the nodal corrections and refers the phases to Greenwich: the amplitudes (of η and of the current ellipses) are divided by the nodal factor f at the middle of the record, and the phases become the Greenwich phase lags $G = g + V_0 + u$, with the astronomical argument V_0 at the start of the run and the nodal angle u at the middle of the record (following Foreman 1977 and t_tide/UTide; for 2N2 only the nodal correction of M2 is used). The values of f and $V_0 + u$ of the constituents are listed in the .txt files of P 160.

default: not set (phases relative to $t = 0$, no nodal corrections)

○ ○ ○ ●

P 166 int Print discharge of the P 167 cross sections to the terminal

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 167 * double x Discharge gauge

x-location in m of a cross section normal to the x-direction. The water discharge in m^3/s through the cross section is written every time step into `REEF3D_Log/REEF3D-discharge_x.dat`.

default: na

○ ○ ○ ●

P 168 * double x, double z_{start}, double z_{end} Discharge window gauge

As P 167, but only the part of the cross section between z_{start} and z_{end} in m is evaluated. Output: `REEF3D_Log/REEF3D-discharge_window_x.dat`.

default: na ; na ; na

○ ● ● ●

P 180 int Print free surface VTP files

Writes the free surface as 2D VTP files into `REEF3D_CFD_FSF`, `REEF3D_FNPF_VTP_FSF` or `REEF3D_NHFLOW_VTP_FSF`, with the interval of P 181, P 182, P 183, P 184 or P 185, and once at the end of the simulation. NHFLOW with sediment transport also writes the bed into `REEF3D_NHFLOW_VTP_BED`. With the meteorological forcing (W 200), the free surface files of NHFLOW and SFLOW also contain `wind10`, `air_pressure` and `wind_stress`. With mesh refinement in FNPF and NHFLOW (G 1 > 0), P 180 = 1 also sets the interval of the AMR free surface output (see P 20). SFLOW always writes its VTP files and does not need P 180. For the LAGOON store see P 18.

0 OFF

1 ON

default: 0

● ● ● ●

P 181 int Print VTP files every n^{th} iteration

Interval for the free surface files of P 180. For SFLOW this is the main output: the free surface and bed VTP files in `REEF3D_SFLOW_VTP_FSF` and `REEF3D_SFLOW_VTP_BED`, and with mesh refinement (G 1 > 0, Section 6.7.1) level 0 and the refined patches as VTR files with a `.vtm` multiblock index in `REEF3D_SFLOW_AMR`. REEF3D::SEASTATE (Section 6.1.4) writes its VTP files into `REEF3D_SEASTATE_VTP` with this interval when it runs stand-alone (A 10 7) and, coupled with SFLOW or NHFLOW (A 750 1), in surfbeat mode (A 770 1); otherwise the coupled SEASTATE writes them at every wave step. Also used for the floating body files in SFLOW, NHFLOW and FNPF (see P 19), for the AMR free surface output of FNPF and NHFLOW with P 180 = 1 (see P 20), for the SFLOW maxima (P 112, P 115) and, unless A 387 is set, for the FNPF ice floe files in `REEF3D_FNPF_ICE`. Only active if P 182 is negative (and P 183 for NHFLOW).

default: -10 (inactive)

● ● ● ●

P 182 double Print VTP files every t seconds

Time-based alternative to P 181, for the same files (free surface for CFD and FNPF, free surface and bed for SFLOW and NHFLOW, SFLOW AMR patches, floating bodies, FNPF ice floes and the REEF3D::SEASTATE VTP files, see P 181).

default: -1.0 (inactive)

○ ○ ● ○

P 183 double Print VTP files every t seconds of sediment time

As P 182, but based on the sediment (morphological) time. Only active if P 182 is negative.

default: -1.0 (inactive)

○ ● ● ●

P 184 * int i_{start}, int i_{end}, int Δi Print VTP files in an iteration window

Writes the free surface files of P 180 every Δi iterations between iteration i_{start} and i_{end}. Several windows can be defined; they act in addition to P 181 or P 182.

default: na ; na ; na

● ● ● ●

P 185 * double t_{start}, double t_{end}, double Δt Print VTP files in a time window

Writes the free surface files of P 180 every Δt seconds between t_{start} and t_{end} (in s). Several windows can be defined. In SFLOW only the floating body files use P 185.

default: na ; na ; na

○ ○ ● ○

P 186 * double t_{start}, double t_{end} Time-averaged free surface VTP file
Averages velocity, η and water level between t_{start} and t_{end} in s and writes one VTP file at t_{end} into `REEF3D_NHFLOW_VTP_TIME_AVG_FSF`. Several windows can be defined. Works independently of P 180.

default: na ; na

○ ○ ○ ●

P 190 int Print topo VTP files
Writes the bed surface as VTP files into `REEF3D_CFD_Topo`, with the interval of P 191, P 192, P 194 or P 195. Sediment fields are added with P 76 to P 79.

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 191 int Print topo VTP files every n^{th} iteration
Only active if P 192 is negative. Note: the code currently also requires P 182 to be negative.

default: -10 (inactive)

○ ○ ○ ●

P 192 double Print topo VTP files every t seconds
Time-based alternative to P 191. Avoid combining it with P 182, as both currently share one print timer.

default: -1.0 (inactive)

○ ○ ○ ●

P 194 * int i_{start}, int i_{end}, int Δi Print topo VTP files in an iteration window
Writes the topo files every Δi iterations between iteration i_{start} and i_{end}. Several windows can be defined.

default: na ; na ; na

○ ○ ○ ●

P 195 * double t_{start}, double t_{end}, double Δt Print topo VTP files in a time window
Writes the topo files every Δt seconds between t_{start} and t_{end} (in s). Several windows can be defined.

default: na ; na ; na

○ ○ ○ ●

P 230 * double x Print FlowFile
x-location in m of a vertical cross section in a 2D CFD simulation. The flow in the cross section is written every time step into `REEF3D_FlowFile` (one header file and one file per time step for each location). FlowFiles can be used as inflow boundary condition of another simulation, see I 230.

default: na

○ ● ○ ○

P 240 * double x Print PotentialFile
x-location in m for PotentialFile output from 2D FNPF simulations. Note: this output is currently not functional in the code.

default: na

○ ○ ○ ●

P 250 int Print CPM sediment variables to the VTU file
Adds `CPM_Ts` (solid volume fraction), `CPM_Ps` (particle normal stress), `CPM_Us`, `CPM_Ws` (particle velocity in x- and z-direction) and `CPM_Kt` (eddy diffusivity of the parcel dispersion, Q 52), and with the seepage flow (Q 69) `CPM_Pstar` (piezometric pressure $p^* = p - \rho_w \mathbf{g} \cdot \mathbf{x}$, the pore pressure in the bed). Only for the particle-based sediment model (S 10 ≥ 1 with Q 10 ≥ 1).

0 OFF

1 ON

default: 0

○ ● ○ ○

P 310 int Print breaking flag to the VTU file
Adds `breaking`, which marks the cells where wave breaking is detected.

0 OFF

1 ON

default: 0

○ ● ○ ○

P 311 int Print breaking viscosity to the VTU file
Adds `vb`, the artificial viscosity used by the breaking model, in m^2/s.

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 351 * double *x*, double *y* Wave gauge for multiphase level set 1
Location in m. For multiphase flow (F 300 > 0), records the interface of the first level set function every time step into `REEF3D_CFD_WSF/REEF3D-CFD-WSF-HG-1.dat`.

default: na ; na

○ ○ ○ ●

P 352 * double *x*, double *y* Wave gauge for multiphase level set 2
As P 351 for the second level set function. Output: `REEF3D_CFD_WSF/REEF3D-CFD-WSF-HG-2.dat`.

default: na ; na

6.14 Q :: Particles (CPM sediment)

The Q commands control the particle-based sediment transport model of REEF3D::CFD, the continuum particle method (CPM). It is activated with S 10 1 (impermeable bed) or S 10 2 (porous bed) together with Q 10 1 or 2; with Q 10 0 the Exner-based sediment model of section S is used. Instead of solving the Exner equation, the sediment below the initial bed is represented by computational parcels, which are seeded in all cells below the bed (Q 24, Q 29). Cells inside the seeding boxes Q 110 are seeded in the same way. Each parcel carries the grain size d_{50} (S 20) and density ρ_s (S 22) and represents $N_p = \theta_0 V_{bed}/(N_{cells}\, N_{ppc}\, \pi d_{50}^3/6)$ grains, where V_{bed} and N_{cells} are the volume and number of the seeded cells, N_{ppc} the number of parcels per cell (Q 24) and θ_0 the solid fraction of the seeded bed (Q 45, default $1-n$ with the porosity n of S 24). With a graded bed (S 51) all parcels keep the same solid volume, but each parcel gets the diameter of one fraction, drawn with the volume fractions $F_{a,k}$ of the active layer, so that the number of parcels of a fraction follows its volume fraction. The parcel volumes and velocities are distributed to the cell centres with trilinear weights, which gives the solid volume fraction θ (smoothed with Q 27) and the solid velocity on the grid. The bed level is the iso-surface $\theta = Q26\,(1-n)$ of the solid volume fraction; it is the topography level set seen by the flow solver and the bed level of the output (with the bedload layer Q 58 the flow solver sees a single valued bed level per column instead, see Q 63 and Q 72).

With Q 11 1 the parcels are moved by the drag of the fluid velocity against a resistance force (Q 30), with the sediment time step of S 13 – S 15. With Q 11 2 the parcels follow the multiphase particle-in-cell (MP-PIC) formulation

$$\frac{d\mathbf{u}_p}{dt} = D_p(\mathbf{u}_f - \mathbf{u}_p) - \frac{\nabla p}{\rho_s} + \mathbf{g} - \frac{\nabla \tau}{\theta \rho_s},$$

with the drag law of Q 51, the fluid pressure gradient (with S 10 1 the pore pressure inside the bed is hydrostatic, or follows the seepage flow of Q 69) and the inter-particle stress τ of Q 12. The drag is treated point-implicitly. The parcels are advanced in every flow time step with adaptive sub-steps (S 14, Q 18, Q 28), using explicit Euler (Q 10 1) or two-stage Runge-Kutta (Q 10 2) sub-steps, and optionally the grid-limited step of Q 19. With S 10 1 the bed is a solid boundary for the fluid and the fluid velocity at the parcels near the bed surface is taken from the logarithmic law of the wall with $k_s = S21 \cdot d_{50}$; deeper in the bed the pore water moves with the grains. With S 10 2 the fluid sees the bed as VRANS porous region with the porosity $1-\theta$ (without the resistance coefficients S 26), or, with the two-way coupling Q 50 1, through the drag of the parcels. Optional additions to the MP-PIC model are the turbulent dispersion of the parcels (Q 52, Q 53), lift on the exposed grains (Q 54), dilatancy of sheared layers (Q 55), ride-over of blocked grains (Q 56), Bagnold sheltering (Q 57) and, with S 10 1, the seepage flow in the bed (Q 69, Q 70).

With S 10 1 and Q 11 2 the near-bed transport can be modelled with the sub-grid bedload layer (Q 58). The bedload layer is a few grains thick, far below the grid scale. Its parcels are not moved by the resolved drag and the grid-limited step but by a closure per bed column: the bed shear stress follows from the log law above the bed (Q 66, Q 68), the Shields number includes the bed slope, grains are picked up from the exposed top layer with the rate $E = V_m^* u_b / L$, travel a hop of random length (mean Q 62 d) with the grain velocity u_b of Fernandez Luque and van Beek (Q 59, critical Shields number Q 60) and are deposited on the bed. The equilibrium moving volume $V_m^* = (\theta - \theta_c)\, d/\mu_s$ follows Bagnold. Below a mobile surface layer (Q 64) the bed of the parcels is jammed (Q 67, Q 69). With Q 58 2 parcels are also picked up into the suspension with the van Rijn (1984) reference concentration, and the resolved flow and the turbulent dispersion (Q 52) carry them until they settle back onto the bed. With Q 58 3 and 4 the suspended load is instead the Eulerian concentration of the suspended sediment solver, which exchanges whole parcels with the bed (hybrid model). The parcels still carry the sediment mass, so bed evolution, grading and slopes follow from the parcels.

Parcels are reflected at walls and removed when they leave the domain through an inflow or outflow boundary; the parcels in the first cell column at in- and outflow boundaries are fixed, as are all parcels upstream of x = Q 76 and downstream of x = Q 77. Periodic boundaries (DIVEMesh C 21, C 22) are supported: parcels that leave through a periodic side enter on the opposite side. The start of the particle transport follows S 41/S 43/S 45/S 47. The bed shear stress (S 16) and the slope reduction (S 80) are computed, but they do not enter the parcel motion; the sandslide (S 90) is not applied. The number of parcels, the sediment volume, the volume that left the domain, the moving volume, the volumetric transport rates per unit width and, with Q 58 3 and 4, the suspended sediment volume are written to `REEF3D_CFD_CPM_Particle/REEF3D-CFD-CPM-Log.dat`. The parcels are stored together with the CFD state files (P 40) in `REEF3D_CFD_STATE` and read for a hotstart (I 40); without a parcel state file they are seeded from the bed. They are also part of the checkpoints (P 48, I 45). Since the parcels are not spread evenly over the domain, the cost of their updates per cell can be

written as a load map (G 45, `REEF3D_CFD_CPM_Particle/REEF3D_CFD_CPM_loadmap.dat`) for the weighted partition of DIVEMesh or REEF3D (M 35 1, M 41 1), and the automatic rebalance (G 46) uses this map.

Valid for:

S	F	N	C
○	○	○	●

Q 10 int Sediment model type

Requires S 10 1 or 2. Options 1 and 2 select the particle model and the time scheme of the MP-PIC sub-steps (Q 11 2); the plain model Q 11 1 always uses a two-stage Runge-Kutta scheme. With the hydrostatic pressure scheme D 30 10 the fluid density is blended to the sediment density S 22 below the bed level.

0 Exner-based sediment model (section S)

1 particle model (CPM), explicit Euler sub-steps

2 particle model (CPM), two-stage Runge-Kutta (Heun) sub-steps

default: 0

○ ○ ○ ●

Q 11 int Particle transport model

1 plain particle transport: drag against the frictional resistance Q 30, two-stage Runge-Kutta time stepping with the sediment time step S 13 – S 15

2 MP-PIC: drag (Q 51), fluid pressure gradient, gravity and inter-particle stress (Q 12), adaptive sub-steps within the flow time step, time scheme Q 10

default: 1

○ ○ ○ ●

Q 12 int Inter-particle stress model

Normal stress τ of the particle phase for Q 11 2; it acts on the parcels with $-\nabla\tau/(\theta\rho_s)$.

0 OFF

1 Snider (2001): $\tau = P_s\theta^\beta/\max(\theta_{cs}-\theta,\ \epsilon(1-\theta))$ with the parameters Q 14 – Q 17

2 packed bed: effective stress of the contact network plus contact pressure (Q 32 – Q 35) and Coulomb friction (Q 13, Q 36 – Q 38). The submerged weight $(\rho_s-\rho_w)|g|\theta$ of the bed ($\theta \geq Q26\,(1-n)$) is integrated downwards from the top of the bed, which carries the grains of a bed at rest without compaction; with the seepage flow Q 69 the vertical gradient of the excess pore pressure is added to this load. The contact pressure $P_c = F_r(\theta-\theta_0)^{\eta_0}/(\theta_{max}-\theta)^{\eta_1}$ of Johnson and Jackson (1987) acts against compaction beyond $\theta_0 = 1-n$ (lowered in sheared layers with the dilatancy Q 55, with θ_{max} shifted by the same amount); it is continued linearly above $\theta_0 + 0.5(\theta_{max}-\theta_0)$.

default: 2

○ ○ ○ ●

Q 13 int Frictional stress for the packed bed model
Coulomb friction of each parcel on the bed below it (Q 12 2), treated implicitly. The normal load follows from the support by the contact network. The slip relative to the bed is stopped below the static friction coefficient μ_s (Q 36), otherwise it is reduced with $\mu(I) = \mu_s + (\mu_2 - \mu_s)/(I_0/I + 1)$ of the $\mu(I)$ rheology, with the inertial number $I = d_{50}\dot{\gamma}/\sqrt{P/\rho_s}$ (Q 37, Q 38). A slope fails at $\tan\beta = \mu_s$.

0 OFF

1 ON

default: 1

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Q 14 double Pressure constant P_s of the Snider stress model
In Pa, used with Q 12 1.

default: 10.0 Pa

○ ○ ○ ●

Q 15 double Exponent β of the Snider stress model
Used with Q 12 1.

default: 3.5

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Q 16 double Parameter ϵ of the Snider stress model
Small number that bounds the denominator of the stress near close packing. Used with Q 12 1.

default: $1.0 \cdot 10^{-6}$

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Q 17 double Critical solid volume fraction θ_{cs} of the Snider stress model
Close packing solid volume fraction at which the stress rises steeply. Used with Q 12 1.

default: 0.6

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Q 18 double Courant number for the particle stress wave speed

The MP-PIC sub-step (Q 11 2) is limited to $\Delta t_{sub} \leq Q18 \cdot h_{min}/c$, where $c = \sqrt{\max(d\tau/d\theta)/\rho_s}$ is the wave speed of the inter-particle stress and h_{min} the smallest cell size. In addition the particle Courant number S 14 limits the sub-step with the maximum parcel velocity.

default: 0.5

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Q 19 int Grid-limited step

Packing limit as kinematic constraint on the grid for Q 11 2: in each sub-step a parcel enters a new cell only if that cell has free volume, with the capacity $\max((\theta_0 + \theta_{max} - (1-n))V,\ \theta_0 V + V_{parcel})$ (Q 32), where $\theta_0 = 1 - n$ at rest and lower in sheared layers with the dilatancy Q 55. All moves are accepted first; a cell that ends up over capacity rejects its latest arrivals until it fits, which is repeated until no cell is over capacity. A parcel can thus enter a full cell while another one leaves it, so a full row of cells can shear. A rejected parcel stops at the cell face and its velocity into the full cell is removed (see also the ride-over Q 56). Cells inside solid bodies have no free volume.

0 OFF

1 ON

default: 1

○ ○ ○ ●

Q 24 int Number of particles per cell

Number of parcels seeded per cell below the initial bed. For the regular seeding types (Q 29 1 and 2) the parcels are placed on a regular sub-grid with $\mathbf{round}(Q24^{1/3})$ parcels per direction ($\mathbf{round}(Q24^{1/2})$ in 2D), e.g. Q 24 8 gives $2 \times 2 \times 2$ parcels; for Q 29 3 exactly Q 24 parcels are placed per cell. On non-uniform grids all parcels have the same volume: Q 24 holds for the smallest seeded cell, and larger cells get proportionally more parcels per direction (cell sizes should be multiples of the smallest one, otherwise a warning about the uneven packing is printed). If Q 24 is not set, one parcel per cell is used and a warning is printed.

default: 0 (one parcel per cell)

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Q 25 double Safety factor for the parcel allocation

Initial size of the parcel arrays relative to the number of seeded parcels. Values below 1.0 are set to 1.0.

default: 1.25

○ ○ ○ ●

Q 26 double Solid fraction threshold of the bed interface

The bed surface is the iso-surface $\theta = Q26\,(1-n)$ of the solid volume fraction, with the porosity n (S 24). With Q 12 2 the same threshold separates the bed, which carries the contact network stress and the friction, from suspended sediment.

default: 0.5

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Q 27 int Number of smoothing passes for the solid volume fraction

Passes of a conservative explicit filter (zero flux at walls) applied to the solid volume fraction θ after the parcel volumes are distributed to the grid.

default: 1

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Q 28 int Maximum number of particle sub-steps per flow time step

Used with Q 11 2. When the limit is reached, the last sub-step covers the remaining time of the flow time step and a warning is printed.

default: 100

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Q 29 int Particle seeding type

Parcels above the bed or inside solid bodies are removed after seeding.

1 regular spacing

2 regular spacing with random jitter of up to a quarter of the parcel spacing

3 fully random

default: 1

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Q 30 double Friction factor for the particle resistance force

Dimensionless friction coefficient of the resistance force $F_s = Q30\,(\rho_s - \rho_w)\,|g|\,\pi d_{50}^3/6$ of a grain. The driving drag force is reduced by F_s and set to zero where it is smaller than F_s. Used with Q 11 1.

default: 0.58

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Q 32 double Maximum solid volume fraction θ_{max}

Close packing limit. It is the singularity of the contact pressure (Q 12 2), the cell capacity of the grid-limited step (Q 19) and sets the minimum fluid fraction $1-\theta_{max}$ in the drag law, the two-way coupling and the VRANS porosity. Values ≤ 0 give $\theta_{max} = 1-n+0.035$ with the porosity n (S 24); $\theta_{max} > 1-n$ is required.

default: -1.0 $(1-n+0.035)$

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Q 33 double Contact pressure coefficient F_r

Coefficient in Pa of the contact pressure of Johnson and Jackson (1987) for Q 12 2; 0 switches the contact pressure off.

default: 0.05 Pa

○ ○ ○ ●

Q 34 double Contact pressure exponent η_0

Exponent of $(\theta-\theta_0)$ in the contact pressure for Q 12 2.

default: 3.0

○ ○ ○ ●

Q 35 double Contact pressure exponent η_1

Exponent of $(\theta_{max}-\theta)$ in the denominator of the contact pressure for Q 12 2.

default: 5.0

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Q 36 double Static friction coefficient μ_s

Friction coefficient of the packed bed (Q 12 2, Q 13 1); it sets the angle of repose $\arctan\mu_s$ and the threshold of motion of an exposed grain.

default: 0.63

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Q 37 double Dynamic friction coefficient μ_2

Limit of the friction coefficient $\mu(I)$ for large inertial numbers (Q 12 2, Q 13 1).

default: 1.13

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Q 38 double Inertial number scale I_0

Scale of the inertial number in the $\mu(I)$ friction law (Q 12 2, Q 13 1).

default: 0.6

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Q 44 int Fix all parcels

All parcels are held in place after seeding and the bed level is not updated. They still define the solid volume fraction, e.g. for the flow through a fixed porous bed with the two-way coupling Q 50 1.

0 OFF

1 ON

default: 0

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Q 45 double Solid volume fraction of the seeded bed

The number of grains per parcel is set such that the seeded cells have the solid volume fraction Q 45. Values ≤ 0 use $1-n$ with the porosity n (S 24).

default: -1.0 $(1-n)$

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Q 50 int Fluid coupling of the parcels

Two-way coupling requires Q 11 2 and starts when the parcels start to move (S 41). The reaction of the parcel drag is added to the fluid momentum equations as a point-implicit source term. With S 10 2 the drag uses the interstitial fluid velocity $\mathbf{u}/(1-\theta)$, and the mixture continuity $\nabla\cdot\mathbf{u} = -\nabla\cdot(\theta\mathbf{u}_p)$ enters the pressure Poisson equation; the bed is then neither a solid boundary nor a VRANS region for the fluid, the fluid flows through the parcels and feels their drag. With S 10 1 the bed stays a solid boundary with the wall function: only the grains above the bed and the exposed grains act on the fluid (the reaction of an exposed grain acts at the reference point 0.5 h above the bed), grains at rest belong to the bed roughness, and there is no mixture continuity source. Parcels hopping in the bedload layer (Q 58) do not act on the fluid.

0 one-way: the fluid sees the bed as solid boundary (S 10 1) or as VRANS porous region with the porosity $1-\theta$ (S 10 2)

1 two-way coupling

default: 0

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Q 51 int Drag law

Drag coefficient D_p per unit parcel mass, with the fluid volume fraction $\theta_f = \max(1-\theta, 1-\theta_{max})$.

1 Andrews and O'Rourke (1996): $D_p = 18\nu\rho_w/(\rho_s d_{50}^2)(\theta_f^{-2.65} + \frac{1}{6}Re_p^{2/3}\theta_f^{-1.78})$

2 Gidaspow (1994): Ergun for $\theta_f < 0.8$, Wen and Yu for $\theta_f \geq 0.8$

default: 1

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Q 52 int Turbulent dispersion of the parcels

Random displacement model for Q 11 2: in every sub-step the tentative parcel position is moved by $\nabla K\,\Delta t + \sqrt{2K\Delta t}\,\xi$ with a normally distributed random number ξ per direction, before the grid-limited step. The eddy diffusivity is $K = \nu_t/Sc_t$ (Q 53) of the fluid in the water; it is damped towards the bed and vanishes where the contact network carries the grains ($\theta \geq 0.5\,Q26\,(1-n)$), and with the bedload layer (Q 58) grains resting on the bed are not dispersed. With the settling velocity the ensemble of the parcels gives the Rouse profile of suspended sediment. The sub-step is limited to $\Delta t_{sub} \leq 0.125\,h_{min}^2/K_{max}$. The eddy viscosity ν_t comes from the turbulence model (T 10). The eddy diffusivity is written to the ParaView output with P 250 1 (CPM_Kt).

0 OFF

1 ON

default: 0

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Q 53 double Turbulent Schmidt number for the parcel dispersion

Schmidt number Sc_t in the eddy diffusivity $K = \nu_t/Sc_t$ of Q 52. Values ≤ 0 use 1.0.

default: 1.0

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Q 54 double Lift coefficient C_L for the exposed grains

Lift force on the exposed grains of the bed surface (S 10 1, Q 11 2) after Wiberg and Smith (1985), $F_L = 0.5\rho_w C_L A\,u_T^2$ with the velocity u_T at the top of the grain from the logarithmic law, acting along the bed normal. Not applied with the bedload layer Q 58. 0 switches the lift off.

default: 0.0

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Q 55 double Dilatancy coefficient

A sheared granular layer is looser than the bed at rest. With Q 55 > 0 the packing of the contact network follows the inertial number $I = d_{50}\dot{\gamma}/\sqrt{P/\rho_s}$ (Forterre and Pouliquen 2008), $\theta_0(I) = \max(Q26\,(1-n),\ (1-n) - Q55 \cdot I)$, with the shear rate $\dot{\gamma}$ of the solid velocity on the grid and the effective stress P of the contact network. The contact pressure (Q 12 2) and the capacity of the grid-limited step (Q 19) use $\theta_0(I)$; a bed at rest keeps $\theta_0 = 1 - n$. 0 switches the dilatancy off.

default: 0.0

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Q 56 double Ride-over factor

A grain that is stopped by the grid-limited step (Q 19 1) at a full cell while moving across the direction of gravity climbs over the grains ahead: its velocity against gravity is raised to at least Q 56 times the blocked velocity (Q 56 is the tangent of the pivot angle), and, with gravity in z-direction, it moves into the cell above its own cell if that cell has free volume. 0 switches the ride-over off.

default: 0.0

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Q 57 int Bagnold sheltering of the bed

For S 10 1 and Q 11 2 without the bedload layer (Q 58 0). Following Bagnold (1956) and Owen (1964) the moving grains carry part of the bed shear stress, $\tau_G = \sum \mu_s m g' f_m / A$ per bed column with the mobility $f_m = \min(1, |\mathbf{u}_p|/(0.1 u_*))$, averaged over 1 s and the neighbouring columns. An exposed grain at rest sees the near-bed velocity reduced with $\psi = \max(0, 1 - \tau_G/\tau_b)$, a moving grain the full velocity, so the entrainment stops when the moving grains carry the excess shear stress.

0 OFF

1 ON

default: 1

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Q 58 int Sub-grid bedload layer

Near-bed transport closure for S 10 1 and Q 11 2, see the introduction of this section. The parcels at rest on the bed surface see no resolved flow and no lift; they are picked up from the exposed top layer by the closure per bed column (Q 59, Q 60, Q 62, Q 65, Q 66, Q 68), hop along the bed with the grain velocity and are deposited at the end of the hop or when the column falls below the threshold of motion. The hops bypass the grid-limited step and are placed into the first cell with free volume. The bed below the mobile surface layer (Q 64) is jammed (Q 67). With Q 74 the bed grains see no

resolved flow at all, and Q 75 adds a sand slide of the bed surface. The fluid sees a single valued bed level per column from the parcels of the top cells, relaxed in time with Q 63 and smoothed over the columns with Q 72 (not with a domain decomposition in the vertical direction, then the iso-surface of the solid fraction is used). With the particle output (Q 180) the bed level and the bed shear stress of each column are written to `REEF3D_CFD_CPM_Particle/REEF3D-CFD-CPM-Bed-*.dat`.

0 OFF, resolved near-bed drag

1 bedload layer

2 bedload layer and pickup into the suspension: where $u_* > 4w_s/D_*$ ($D_* \leq 10$) or $0.4w_s$ ($D_* > 10$), parcels of the layer are released 0.5 h above the bed with the van Rijn (1984) reference concentration, $E_s = w_s c_a$; they are deposited when they settle back onto the bed

3 bedload layer and Eulerian suspended load (hybrid): the suspended load is the concentration of the implicit suspended sediment solver (convection with $w - w_s$, eddy diffusion, settling velocity w_s of Ferguson and Church). Per bed column the first fluid cell above the bed exchanges $w_s(c_{be} - c_1)$ with the bed, with the van Rijn (1984) reference concentration c_{be} (bed shear stress of the layer, onset of suspension as option 2, transferred to the cell centre with the Rouse profile) and the concentration c_1 of that cell, within S 71 – S 72 and only where the column has erodible parcels. The net exchange, multiplied by Q 65, is accumulated per column and moves whole parcels: a parcel of the layer leaves the bed into the concentration, or a new parcel is deposited at rest on the bed. The suspended transport starts with the parcel transport (S 41) and the sediment volume of parcels and concentration is conserved.

4 as 3, without the onset criterion of the suspension (pickup wherever the Shields number exceeds the critical one, as the Exner-based model)

default: 0

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Q 59 double Grain velocity coefficient of the bedload layer

Grain velocity $u_b = Q59\sqrt{Rgd}(\sqrt{\theta} - 0.7\sqrt{\theta_c})$ after Fernandez Luque and van Beek (1976), along the Shields vector, with $R = \rho_s/\rho_w - 1$. With the defaults the equilibrium transport rate follows Meyer-Peter and Müller within 8 % for $0.06 < \theta < 0.5$ (Wong and Parker 2006: 3.2). Used with Q 58.

default: 6.5

○ ○ ○ ●

Q 60 double Critical Shields number of the bedload layer

Critical Shields number θ_c of the grain size d_{50} for Q 58; a grain of diameter d uses $\theta_{c,d} = Q60\,(d/d_{50})^{-0.8}$ (hiding). It also sets the transport stage of the reference concentration for Q 58 2.

default: 0.047

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Q 62 double Mean hop length of the bedload layer

Mean hop length L in grain diameters for Q 58. The hop length of each picked-up parcel is drawn from an exponential distribution; it sets the pickup rate $E = V_m^* u_b / L$ and the adaptation length of the transport.

default: 50.0

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Q 63 double Relaxation time of the bed level seen by the fluid

Relaxation time in s of the bed level of the flow solver for Q 58. Single parcels moving in and out of the top cells make the bed level of a column jump; the relaxation keeps the fluid cells from switching on and off. Q 63 is a time of the bed (morphological time $Q65 \cdot t$): the relaxation rate per flow time step is $Q65\,\Delta t / Q63$.

default: 10.0 s

○ ○ ○ ●

Q 64 double Thickness of the mobile surface layer

Thickness ψ of the mobile surface layer of the bed in cells (vertical cell height) for Q 58. Below it the bed is jammed: the velocity of the parcels is set to zero, with a smooth transition between ψ and 2ψ below the bed surface. Slopes then fail by avalanches of the surface layer, held by the Coulomb friction (Q 13); deep failures need Q 67.

default: 0.6

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Q 65 double Morphological factor of the bedload layer

The pickup into the bedload layer, the hops of the layer and the exchange with the suspension (Q 58 $2-4$) are Q 65 times faster, so the bed evolves Q 65 times faster than the flow. The relaxation time Q 63 is given in this bed time; the sand slide Q 75 is instantaneous.

default: 1.0

○ ○ ○ ●

Q 66 double Reference height for the bed shear stress of the bedload layer

Height in cells above the bed where the velocity of the log law ($k_s = S21 \cdot d_{50}$) is taken for Q 58, interpolated between the cell centres and at most half way to a solid body or the free surface above. 0 uses the centre of the first fluid cell at least 0.5 h above the bed. A larger value helps where the first cell is under-resolved, e.g. in the gap below a pipeline. The same closure gives the Eulerian bed shear stress S 16 10 and 11 (reference height S 63).

default: 0.0

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Q 67 int Yield of the jammed bed

With Q 58 and S 10 1 the bed below the mobile surface layer (Q 64) is jammed only within the Mohr-Coulomb yield of its column: the lateral load F from the horizontal gradient of the effective stress, integrated from the top of the column, is compared with the frictional resistance $\mu_s P$ (Q 36). The jam is released smoothly between $F/(\mu_s P) = 0.95$ and 1.05, so slopes steeper than the angle of repose and free faces (e.g. the flank of a scour hole) fail as a whole and jam again below the angle of repose.

0 OFF

1 ON

default: 0

○ ○ ○ ●

Q 68 int Bed shear stress of the bedload layer

Used with Q 58.

0 log law

1 max(log law, $0.3\rho_w k$) with the turbulent kinetic energy of the first fluid cell (as S 16 2), for non-equilibrium flows such as wakes or gap jets

2 two velocity scales, $\sqrt{\tau_{log} \cdot 0.3\rho_w k}$: as the log law in equilibrium, zero where the mean velocity at the reference point vanishes (separation, reattachment), of the size of the k term in a recirculation; without a turbulence model the log law is used

default: 1

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Q 69 int Seepage flow in the bed

For S 10 1 and Q 11 2. The pore pressure in the bed follows Darcy's law in a rigid bed of uniform permeability: the piezometric pressure $p^* = p - \rho_w \mathbf{g} \cdot \mathbf{x}$ is harmonic in the bed, with p^* of the fluid at the bed surface, no flux through walls, solid bodies and the bottom (or the excess pressure Q 70 at the bottom). The Laplace equation is solved with SOR sweeps in every time step. The seepage force $-\nabla p^*$ acts on the parcels, the vertical gradient loads or unloads the contact network (Q 12 2), so that an upward gradient beyond the submerged weight fluidises the bed, and with Q 58 the bed is jammed only where the contact network stays loaded. With P 250 1 the field CPM_Pstar is added to the ParaView output.

0 OFF, hydrostatic pore pressure

1 ON

default: 0

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Q 70 double Excess pore pressure at the bottom

Excess piezometric pressure in Pa at the bottom of the domain for the seepage flow Q 69, relative to the mean p^* at the bed surface, e.g. for upflow tests. 0 gives a closed bottom.

default: 0.0 Pa

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Q 72 int Smoothing passes of the bed level seen by the fluid

Number of passes of a 1-2-1 filter over the neighbouring bed columns, applied to the relaxed bed level of the flow solver (Q 63) with the bedload layer Q 58. A fluid cell is bed or fluid by the sign of the topography level set at its centre, so a single column whose level crosses a cell centre by parcel noise blocks the whole cell and stalls the near-bed flow behind it. The filter removes this noise, not the bed forms: the level of a column is changed by at most half a cell height. At the domain sides (without a neighbouring subdomain and not periodic) the column is mirrored. 0 switches the filter off.

default: 2

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Q 73 * double value, double distance, double β, double x_0, double y_0 Relaxation zone for the particle forces along a line

The forces on the parcels are zero on the straight line through (x_0, y_0) in m and increase smoothly to their full value at the given distance in m from the line, so that the particles are held in place near e.g. the inflow boundary. The line angle β is given in degrees; $\beta = 0$ gives a line parallel to the y-axis. The first value is currently not used, but must be given. Several lines can be combined.

default: not set

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Q 74 int Bed grains without resolved flow

Used with the bedload layer Q 58. With Q 74 1 the parcels of the bed that were not released into the suspension see no resolved flow anywhere: no drag, no lift and no turbulent dispersion (Q 52); they are moved by the bedload layer, gravity and the packed bed only. With Q 74 0 only the grains resting within one cell above the bed level belong to the bed surface, and a bed grain higher above the bed level of the fluid (e.g. deposited on a crest) takes the full resolved drag.

0 OFF

1 ON

default: 0

○ ○ ○ ●

Q 75 double Sand slide slope of the bedload layer

Sand slide of the bed surface with the bedload layer Q 58 (S 10 1). Where the bed level of the layer drops from a column to its neighbour by more than Q 75 times their distance, the top grains of the higher column (at rest in the bed, not hopping) are moved onto the bed of the lower column until the slope is Q 75. Q 75 is the tangent of the angle of repose; the slope is taken from the bed level smoothed with two passes of a 1-2-1 filter over the columns of the subdomain. No grains slide into or out of solid bodies. The slide is instantaneous (also with Q 65), and the number of moved parcels is printed. 0 switches the sand slide off.

default: 0.0

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Q 76 double Fixed parcel bed upstream

All parcels with $x <$ Q 76 (in m) are fixed, as the parcels in the first cell column at the in- and outflow boundaries: the bed there is non-erodible, e.g. in a developing inflow.

default: $-1.0 \cdot 10^{20}$ m (not set)

○ ○ ○ ●

Q 77 double Fixed parcel bed downstream

All parcels with $x >$ Q 77 (in m) are fixed (non-erodible bed), see Q 76.

default: $1.0 \cdot 10^{20}$ m (not set)

○ ○ ○ ●

Q 78 int Parcel merge and split

Seeded parcels have the weight 1 (the parcel volume of Q 24); a parcel of weight w carries w times this volume. With Q 78 $N > 0$ each cell gets the target weight $w_c = 2^m$, the largest power of two with $w_c \le \theta_0 V/(V_1 N)$ (V the cell volume, V_1 the volume of a parcel of weight 1), so that a packed cell holds about N parcels. Once per step, grains at rest in the jammed bed (Q 58, S 10 1, not with Q 67 or Q 69) of equal weight, diameter and density below the target weight of their cell are merged in pairs at their mean position and velocity, and a parcel with $w > 1$ that moves (or is heavier than the target weight of its cell) is split into two halves until it is neither. The sediment volume and momentum are conserved. This reduces the number of parcels in large cells of a graded grid. The weight is added to the particle output (Q 180), and the numbers of merged and split parcels are printed. 0 switches it off.

default: 0

○ ○ ○ ●

Q 79 int Wall distance of the bed shear stress of the bedload layer

Reference for the distance of the log law in the bed shear stress of the bedload layer (Q 58, Q 66, Q 68). With option 1 the distance is measured from the face under the first fluid cell, the wall of the wall law of the fluid at the bed, and is at least half a cell height; the bed shear stress then does not change while the bed level moves within a cell. The Eulerian bed shear stress S 16 10 and 11 always measures the distance from the bed level set.

0 from the bed level set (the former behaviour)

1 from the face under the first fluid cell

default: 1

○ ○ ○ ●

Q 110 * double x_s, double x_e, double y_s, double y_e, double z_s, double z_e Seeding box for parcels

The cells whose centre lies inside the box (in m) are seeded with parcels in the same way as the cells below the initial bed (Q 24, Q 29, solid fraction Q 45), also above the bed, and these parcels are not removed after seeding. Cells inside solid bodies are not seeded. In 2D the y-limits are ignored. Several boxes can be defined.

default: not set

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Q 180 int Print particles

Writes the parcels as ParaView .vtp/.pvtp files to the folder REEF3D_CFD_CPM_Particle. The output interval is given with either Q 181 or Q 182. With the bedload layer (Q 58) the bed level and the bed shear stress of each bed column are written at the same times to `REEF3D-CFD-CPM-Bed-*.dat` in the same folder.

0 OFF

1 ON

default: 0

○ ○ ○ ●

Q 181 int Particle print interval in iterations

Prints the particles every Q 181 iterations. Only active if Q 181 > 0 and Q 182 < 0.

default: -10 (inactive)

○ ○ ○ ●

Q 182 double Particle print interval in time

Prints the particles every Q 182 seconds of simulation time. Only active if Q 182 > 0 and Q 181 < 0.

default: -1.0 *s* (inactive)

6.15 S :: Sediment

Valid for:

S	F	N	C
●	○	●	●

S 10 int Sediment transport module

Switches on morphodynamics. Any value > 0 also activates the topography handling (the internal flag G 501 is set to 1) and the printing of the bed (P 27 is set to 1). In REEF3D::CFD the particle-based sediment model is selected with Q 10 1 or 2 instead of the default Exner-based model (see section Q). On curvilinear grids (G 50) S 10 2, S 31 3 and S 32 3 – 5 stop the run. The mesh refinement and the multi-block grids of SFLOW and NHFLOW support S 10 1 with S 31 1 or 2, S 32 1 or 2 and a single fraction (no S 51; in NHFLOW also no suspended load S 12); otherwise the refinement is switched off with a message (a multi-block grid stops the run). The CFD mesh refinement is switched off with sediment transport. The sediment state (including the parcels of the particle model) is part of the checkpoints (P 48, I 45) of SFLOW, NHFLOW and CFD.

0 OFF

1 ON, impermeable sediment bed (SFLOW, NHFLOW, CFD)

2 ON, porous sediment bed with VRANS (CFD only, requires A 10 6; porosity S 24, VRANS coefficients S 26)

12 ON, fully coupled RK2 sediment time stepping (NHFLOW only): once the sediment transport has started (S 41), the bed is advanced in both Runge-Kutta stages of every flow time step with $\Delta t_{sed} = S17 \cdot \Delta t$; the interval criteria S 42 and the time step options S 13 – S 15 are not used

default: 0

● ○ ● ●

S 11 int Bedload transport formula

Equilibrium bedload transport capacity $q_{b,e}$ used in the Exner equation. All formulas use d_{50} (S 20), the sediment density (S 22) and the water density W 1; all except Einstein-Brown use the critical Shields parameter (S 30) as threshold.

0 OFF

1 van Rijn

2 Meyer-Peter Müller

3 Engelund and Fredsøe (dynamic friction coefficient S 18)

4 Engelund and Hansen (total load; friction coefficient from the depth-averaged velocity in SFLOW and from Engelund's resistance law $V/u_* = 6+2.5\ln(h/k_s)$ in NHFLOW and CFD)

5 Einstein-Brown, with Rubey's fall velocity factor F: $q_b = F\sqrt{Rgd_{50}^3}\,\Phi(\theta)$

default: 0

○ ○ ● ●

S 12 int Suspended sediment transport

Solves an implicit advection-diffusion equation for the suspended sediment concentration with the settling velocity from S 23/S 25. The reference concentration of van Rijn is evaluated at the reference height $a = \max(k_s, 0.01h, 2d_{50})$, at most $h/2$, and transferred with the Rouse profile to the centre of the first cell above the bed, where the bed exchanges sediment with the water column, $w_s(c_b - c_{b,e})$, with the concentration c_b of that cell. The exchange is only applied in water, on erodible cells and within S 71 – S 72. The coupling with the Exner equation is set with S 62. In NHFLOW the suspended sediment of columns that fall dry is deposited on the bed.

0 OFF

1 ON, van Rijn reference concentration

default: 0

● ○ ● ●

S 13 double Maximum sediment time step

Upper limit of the morphodynamic time step in s for S 15 0, and the fixed time step for S 15 2.

default: 10.0 s

● ○ ● ●

S 14 double Courant number for the sediment time step

The morphodynamic time step is decoupled from the hydrodynamic time step (see S 15). It is determined from a Courant criterion on the rate of bed elevation change, $\Delta t_{sed} = S14 \cdot \Delta x / \max|\partial z_b/\partial t|$. For the plain particle model (Q 11 1) the maximum parcel velocity is used instead of the bed change rate; for the MP-PIC particle model (Q 11 2) S 14 limits the parcel sub-steps within the flow time step, $\Delta t_{sub} \le S14 \cdot h_{min}/\max|\mathbf{u}_p|$ (see Q 18). Recommended range: 0.1 – 0.5.

default: 0.3

● ○ ● ●

S 15 int Sediment time step selection

All options are multiplied by the factor S 17 (Exner model only). The MP-PIC particle model (Q 11 2) always uses the flow time step with adaptive sub-steps; options $0-2$ apply to the plain particle model (Q 11 1). Not used with S 10 12, which advances the bed with S 17 times the flow time step.

0 adaptive: minimum of S 13 and the Courant criterion S 14

1 adaptive: minimum of the flow time step and the Courant criterion S 14

2 fixed time step S 13

3 flow solver time step (Exner model only)

4 flow time since the previous bed update, e.g. S 44 flow time steps (Exner model only); with S 17 it is the morphological factor on the flow time, independent of the interval S 42

default: 0

● ○ ● ●

S 16 int Bed shear stress formulation

The available options depend on the flow model; an option that is not available in the selected model gives zero bed shear stress. Log-law based options use the effective roughness $k_s = S21 \cdot d_{50}$ (see also S 36), $u/u_* = \ln(30z/k_s)/\kappa$ evaluated at the height z of the velocity sample and limited to $u/u_* \geq 1/\kappa$ in thin water. The water density W 1 is used.

1 wall function / log-law, velocity based (CFD: velocity at 1.6 Δz above the bed; NHFLOW: smooth or rough wall law selected with B 22; SFLOW: Manning formulation with $n = k_s^{1/6}/20$)

2 log-law, velocity based, limited from below by the turbulent kinetic energy (SFLOW: depth-averaged log-law, $U/u_* = 2.5\ln(12h/k_s)$)

3 viscosity based: $\tau = \rho(\nu + \nu_t)\, u/z$ with the velocity at the height z (CFD: $0.6\Delta z$, NHFLOW: half the first cell height) (SFLOW: depth-averaged log-law as option 2)

4 turbulent kinetic energy based: $\tau = 0.3\rho k$

5 geometric mean of options 7 and 4 (CFD only)

6 depth-averaged velocity, Chezy (CFD) or Manning (NHFLOW) formula (not SFLOW)

7 log-law with the grain roughness, horizontal velocity only (CFD only)

8 log-law with iterative roughness correction for transitional flow (CFD only)

9 wave friction (CFD and NHFLOW): $\max(\tau_{log}, \tau_w)$ with the log-law of option 1 (NHFLOW: rough wall) and the wave friction stress $\tau_w = 0.5\rho f_w |\mathbf{u}_w|^2$, $f_w = \min(1.39(A/z_0)^{-0.52}, 0.3)$, $z_0 = k_s/30$ (Soulsby 1997). $\mathbf{u}_w$ is the oscillatory part of the velocity sample (minus its running mean over one wave period), $A = \hat{U}T/(2\pi)$ the orbital amplitude of the sample and T the wave period S 58. Without a wave period it is option 1. Wave-current interaction after Soulsby (1997) and a phase lead of the wave stress are selected with S 59.

10 log-law with turbulent kinetic energy (CFD and NHFLOW): the closure of the bedload layer of the particle model (Q 66, Q 68 1), $\max(\tau_{log}, 0.3\rho k)$ with the horizontal velocity of the first cell centre at least half a cell above the bed (or at the reference height S 63) and k of the first fluid cell

11 as 10 with two velocity scales, $\sqrt{\tau_{log} \cdot 0.3\rho k}$ (as Q 68 2)

default: 1

● ○ ● ●

S 17 double Factor for the sediment time step
Dimensionless factor applied to the sediment time step selected with S 15 (for S 15 1, 3 and 4 it multiplies the flow time step). Values > 1 accelerate the morphodynamics relative to the hydrodynamics (morphological acceleration). Not used by the particle model and not applied when S 29 is used. With S 10 12 the sediment time step is S 17 times the flow time step.

default: 1.0

● ○ ● ●

S 18 double Dynamic friction coefficient μ_d for the Engelund and Fredsøe bedload formula
Only used with S 11 3.

default: 0.823

● ○ ● ●

S 19 double Maximum simulated sediment time
The simulation stops when the accumulated morphodynamic time reaches S 19 in s (in addition to the limits N 41 and N 45). The default is a large sentinel value, i.e. inactive.

default: $1.0 \cdot 10^{19}$ s

● ○ ● ●

S 20 double Sediment grain size d_{50}
Median grain diameter in m. It enters the transport formulas, the critical bed shear stress, the bed roughness (S 21) and, for the particle model, the parcel size.

default: 0.001 m

● ○ ● ●

S 21 double Roughness factor

The equivalent sand roughness of the sediment bed is $k_s = S21 \cdot d_{50}$. It is used in the bed shear stress calculation (S 16), with S 28 1 in the wall boundary condition of the flow over the bed and, for the MP-PIC particle model with S 10 1, in the logarithmic law for the fluid velocity at the parcels near the bed surface.

default: 3.0

● ○ ● ●

S 22 double Sediment density

default: 2650.0 kg/m^3

● ○ ● ●

S 23 double Sediment settling velocity

Settling velocity w_s in m/s used for suspended sediment transport. Only used if the automatic calculation is switched off (S 25 0).

default: 0.001 m/s

● ○ ● ●

S 24 double Porosity of the sediment bed

Volume fraction of pores n in the bed. It enters the Exner equation as $1/(1-n)$, the VRANS porosity for S 10 2, and the parcel size and packing limit of the particle model.

default: 0.5

● ○ ● ●

S 25 int Automatic settling velocity calculation

When ON, w_s is calculated from d_{50}, the sediment and water densities and the viscosity W 2: Stokes law for $d_{50} < 0.1\,mm$, the van Rijn formula for $0.1 \leq d_{50} < 1\,mm$ and $1.1\sqrt{g d_{50}(\rho_s/\rho_w - 1)}$ for $d_{50} \geq 1\,mm$.

0 OFF, use S 23

1 ON

default: 1

○ ○ ○ ●

S 26 double α, double β VRANS resistance coefficients for the porous sediment bed
Dimensionless linear (α) and non-linear (β) resistance coefficients of the VRANS porous media model inside the sediment bed. Only used with S 10 2 and A 10 6 by the Exner-based model; the porous bed of the particle model (Q 10 > 0) has no resistance coefficients.

default: 650.0 ; 2.2

○ ○ ○ ●

S 27 int Number of Exner sub-iterations per sediment step
Number of times the Exner equation (and the sandslide algorithm) is advanced with Δt_{sed} during one call of the sediment module, without updating the bed shear stress and transport rates in between.

default: 1

○ ○ ● ●

S 28 int Roughness of the sediment bed in the flow boundary condition
Selects the roughness used in the wall boundary condition of the hydrodynamics (velocities and turbulence) at the sediment bed when S 10 > 0.

0 $k_s = d_{50}$
1 $k_s = S21 \cdot d_{50}$
default: 1

● ○ ● ●

S 29 double t_s, double t_e, double Δt_s, double Δt_e Sediment time step ramp
Linear ramp of the maximum sediment time step from Δt_s to Δt_e (in s) between the sediment times t_s and t_e (in s). Before t_s the value Δt_s, after t_e the value Δt_e is used. The ramped value limits the Courant-based time step (S 14); S 15 and S 17 are then ignored. Exner model only.

default: not set (inactive)

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S 30 double Critical Shields parameter
θ_{cr} for incipient motion on a flat bed. The critical bed shear stress is $\tau_{cr} = r\,\theta_{cr}(\rho_s - \rho_w)g d_{50}$ with the slope reduction factor r (S 80).

default: 0.047

● ○ ● ●

S 31 int Type of Exner formulation

Determines how the transport direction enters the divergence of the bedload flux. Faces to cells without an erodible bed (structures, solids) and closed walls carry no bedload flux. The bend corrections S 86 and S 95 and the suspended part S 64 require option 1 or 2; option 3 stops the run with them and on curvilinear grids (G 50).

1 scalar q_b, upwinding with the flow direction at the cell faces (conservative)

2 vector flux $(q_b s_x, q_b s_y)$ with the cell-centred flow direction (conservative)

3 directional derivative of q_b along the cell-centred flow direction; not conservative where the transport direction turns, a warning is printed

default: 2

● ○ ● ●

S 32 int Exner equation discretization

Options 1, 2 and 4 are conservative; for 3 (with S 31 1) and 5 a warning is printed. On curvilinear grids (G 50) and with the mesh refinement or multi-block grids only options 1 and 2 are available.

1 FOU

2 CDS2

3 CDS2 HJ (advective form with S 31 1, not conservative)

4 WENO5 FLUX

5 WENO5 HJ (not conservative)

default: 1

● ○ ● ●

S 33 int Non-equilibrium bedload transport

With non-equilibrium transport the bedload flux q_b relaxes towards the transport capacity $q_{b,e}$ over the adaptation length L_s along the transport path, $q_b + L_s\,\hat{s}\cdot\nabla q_b = q_{b,e}$. The equation is solved with a Gauss-Seidel method in every sediment step (maximum number of sweeps S 49). For options 2 and 3, L_s is typically only a few grain diameters, so the result is practically identical to equilibrium transport on usual grids.

0 equilibrium transport, $q_b = q_{b,e}$

1 constant adaptation length L_s = S 40 in m

2 van Rijn (1984) saltation length $L_s = 3 d_{50} D_*^{0.6} T^{0.9}$

3 Phillips and Sutherland (1989) $L_s = 4000(\theta - \theta_{cr}) d_{50}$

4 depth scaled L_s = S 40 $\cdot h$

default: 0

● ○ ● ●

S 35 double Exner magnification factor
Dimensionless factor multiplying the bed change rate in the Exner equation (morphological acceleration). It is also included in the bed celerity limit S 103.

default: 1.0

● ○ ● ●

S 36 int Effective bed roughness calculation

0 grain roughness only, $k_s = S21 \cdot d_{50}$

1 grain roughness plus van Rijn's bedform roughness, based on the transport stage and the water depth

default: 0

○ ○ ● ○

S 37 int Inertia contribution to the bed shear stress
Adds a bed shear stress contribution from the horizontal flow acceleration, $\tau_{acc} = \rho_w c_m c_v c_n S21\, d_{50} |\partial \mathbf{u}/\partial t|$, to the effective bed shear stress (NHFLOW only).

0 OFF

1 ON

default: 0

○ ○ ● ○

S 38 int Blend bed shear stress with shallow water formulation
In very shallow water the bed shear stress is blended with a depth-averaged Manning-type formulation (NHFLOW only). The blending starts below the water depth $h = S39 \cdot A544$; below $0.5\, S39 \cdot A544$ only the shallow water formulation is used.

0 OFF

1 ON

default: 0

○ ○ ● ○

S 39 double Blending depth factor for S 38
Dimensionless factor on the wetting and drying threshold A 544 that sets the water depth where the shallow water blending of S 38 starts. With the defaults this is $100 \cdot 0.001\, m = 0.1\, m$.

default: 100.0

● ○ ● ●

S 40 double Adaptation length for non-equilibrium bedload transport
Adaptation length L_s in m for S 33 1, or dimensionless factor on the water depth for S 33 4.

default: 0.1

● ○ ● ●

S 41 int Type of sediment start criterion
Sediment transport starts once the criterion is reached; before that only the bed shear stress is calculated.

1 number of flow iterations (S 43)

2 flow simulation time (S 45)

3 flow simulation time divided by the wave period, t/T (S 47)

default: 1

● ○ ● ●

S 42 int Type of sediment interval criterion
Interval between two calls of the sediment module (Exner model only; the particle model and S 10 12 advance the bed in every flow time step).

1 number of flow iterations (S 44)

2 flow simulation time (S 46)

3 t/T (S 48)

default: 1

● ○ ● ●

S 43 int Number of flow iterations before sediment transport starts
Used with S 41 1. Independent of S 41, it is also the start of the erosion-deposition coupling of suspended sediments in the Exner equation (S 62 1).

default: 1000

● ○ ● ●

S 44 int Number of flow iterations between bed updates
Used with S 42 1.

default: 10

● ○ ● ●

S 45 double Flow simulation time before sediment transport starts
Used with S 41 2.

default: 1.0 *s*

● ○ ● ●

S 46 double Flow simulation time between bed updates
Used with S 42 2.

default: 1.0 *s*

● ○ ● ●

S 47 double t/T before sediment transport starts
Number of wave periods before sediment transport starts; used with S 41 3. Requires a wave generation with a defined wave period.

default: 1.0

● ○ ● ●

S 48 double t/T between bed updates
Number of wave periods between two bed updates; used with S 42 3.

default: 0.1

● ○ ● ●

S 49 int Maximum number of Gauss-Seidel sweeps for non-equilibrium bedload
Upper limit of sweeps per sediment step for S 33 > 0; the iteration stops earlier at convergence (relative tolerance 10^{-6}).

default: 200

● ○ ● ●

S 50 int Fixed bed level at inflow or outflow boundary
Boundary condition for the bed level at the inflow and outflow boundaries: a fixed bed does not change there, otherwise the bed level is extrapolated.

1 inflow fixed

2 outflow fixed

3 inflow and outflow fixed

4 none fixed

default: 4

● ○ ● ●

S 51 * double d_k, double $F_{a,k}$, double $F_{s,k}$ Sediment fraction for a graded bed
Each S 51 line defines one grain size class of a multi-fraction (graded) sediment bed: the diameter d_k in m, its initial volume fraction in the active layer $F_{a,k}$ and in the substrate $F_{s,k}$. Between 1 and 20 fractions are allowed; without S 51 the bed has a single fraction with d_{50} = S 20. The fractions of each layer are normalized to a sum of 1; if all $F_{s,k}$ are zero, the substrate gets the composition of the active layer. The bed consists of an active (mixing) layer of thickness S 52 on top of one well-mixed substrate layer of thickness S 53 (Hirano active-layer model). The bedload formula S 11 is evaluated once per fraction with d_k, i.e. with the Shields parameter $\theta_k = \tau_b/((\rho_s - \rho)g d_k)$ and the critical Shields parameter S 30 (including the slope reduction S 80) multiplied by the hiding/exposure factor (S 54), and the capacity is weighted with the local fraction in the active layer. The Exner equation, the sandslide and the bed relaxation zones act on each fraction, so that the volume of each fraction is conserved. Not available with the RK2 sediment solver S 10 12. With the particle model (Q 10 > 0) the fractions set the diameters of the parcels (see section Q). The fractions and the local d_{50}, d_m, d_{90} and substrate thickness are added to the sediment output (ST_Fk, ST_d50, ST_dm, ST_d90, ST_Hsub); the bed volume per fraction is logged in `REEF3D_Log/REEF3D_sediment_mixture.dat`.

default: not set (single fraction, S 20)

● ○ ● ●

S 52 double Active layer thickness
Thickness L_a in m of the active layer of the graded bed (S 51). With a value ≤ 0, $L_a = 2\,d_{90}$ of the initial active-layer composition.

default: 0.0 ($2\,d_{90}$)

● ○ ● ●

S 53 double Initial substrate layer thickness
Initial thickness in m of the substrate layer below the active layer of the graded bed (S 51). The substrate grows with deposition and shrinks with erosion.

default: 10.0

● ○ ● ●

S 54 int Hiding/exposure function
Correction of the critical Shields parameter of each fraction of the graded bed (S 51) for hiding of fine grains between coarse grains and exposure of coarse grains. With option 1, $\theta_{cr,k}$ is multiplied by $\xi_k = (p_{e,k}/p_{h,k})^{-m}$ with $p_{h,k} = \sum_j F_j d_j/(d_k + d_j)$, $p_{e,k} = 1 - p_{h,k}$ and the exponent m from S 55 [Wu et al., 2000].

0 OFF

1 Wu, Wang and Jia (2000)

default: 1

● ○ ● ●

S 55 double Hiding/exposure exponent m
Exponent of the hiding/exposure function S 54 1.

default: 0.6

● ○ ● ●

S 56 int Grain size for the bed roughness of a graded bed
Diameter d in the bed roughness k_s = S 21 $\cdot d$ when a graded bed (S 51) is used. Options 1–3 use the local grain size of the active layer, so that the roughness follows the bed sorting.

0 S 20

1 local d_{50}

2 local mean diameter d_m

3 local d_{90}

default: 0

○ ○ ○ ●

S 57 double Initial flat sediment bed level
Sets a horizontal sediment bed at the elevation z = S 57 in m over the whole domain, overriding the topography from the grid. The default sentinel value means not set.

default: $-1.0 \cdot 10^{20}$ m (not set)

○ ○ ● ●

S 58 double Wave period for the wave friction bed shear stress
Wave period T in s of the wave friction stress S 16 9 (and S 59). It sets the orbital amplitude $A = \hat{U}T/(2\pi)$ and the averaging time of the running mean that separates the current from the oscillatory velocity. Values ≤ 0 use the period of the wave generation.

default: 0.0 (period of the wave generation)

○ ○ ● ●

S 59 int mode, double φ Wave-current interaction of the wave friction bed shear stress
Combination of the log law and the wave friction stress for S 16 9. For modes 1 and 2 the current stress is the log law with the running mean of the velocity sample, $\tau_c = \rho(|\bar{\mathbf{u}}|/u^+)^2$, and the mean stress of Soulsby (1997, DATA13), $\tau_m = \tau_c[1 + 1.2(\tau_w/(\tau_c + \tau_w))^{3.2}]$ with the wave stress amplitude $\tau_w = 0.5\rho f_w \hat{U}^2$, acts in the direction of the current; the instantaneous wave stress vector $0.5\rho f_w |\mathbf{u}_w^*| \mathbf{u}_w^*$ is added to it. Without a wave period (S 58) the log law of the horizontal sample velocity is used. φ is the phase lead of the wave stress over the sample velocity in degrees, $\mathbf{u}_w^* = \mathbf{u}_w \cos\varphi + (d\mathbf{u}_w/dt) \sin\varphi/\omega$; it is used in all modes, 0 means no phase lead.

0 max(τ_{log}, τ_w) as described for S 16 9

1 max(τ_{log}, Soulsby DATA13): the log law where the current boundary layer reaches the sample, DATA13 under waves (recommended for wave-current interaction)

2 Soulsby DATA13

default: 0 ; 0.0

○ ○ ● ●

S 62 int Coupling of suspended sediment with the Exner equation

1 erosion-deposition term $w_s(c_b - c_{b,e})$ is added to the Exner equation (after S 43 iterations)

2 depth-integrated suspended load flux is added to the bedload flux (NHFLOW only, otherwise S 62 1 is used; with S 12 1 the suspended solver still exchanges sediment with the bed, so the suspended sediment is counted twice and a note is printed; S 62 1 is the conservative coupling)

default: 1

○ ○ ● ●

S 63 double Reference height for the bed shear stress S 16 10
Height in cells (of the first cell) above the bed where the velocity of the log law is taken for S 16 10 and 11, linear between the cell centres; in CFD at most half way to a solid body or the free surface above, in NHFLOW at most half the water depth. 0 uses the first cell centre (CFD: the first cell centre at least half a cell above the bed). A larger value helps where the first cell is under-resolved; it is the Eulerian counterpart of Q 66.

default: 0.0

● ○ ○ ○

S 64 double Suspended part of the transport formula
For SFLOW with a single fraction. The transport capacity q_e of S 11 is split into bed load $(1 - f_s)\, q_e$ (equilibrium or S 33) and suspended load with the part f_s = S 64,

which relaxes along the depth-averaged flow direction $\hat{s}$ over the adaptation length of the suspension (Galappatti and Vreugdenhil 1985), $q_s + L_s \hat{s} \cdot \nabla q_s = f_s q_e$ with $L_s = h|U|/(S65\, w_s)$ and the settling velocity w_s (S 23, S 25). The equation is solved with upwind Gauss-Seidel sweeps (at most S 49) in every sediment step. The Exner fluxes carry the sum of bed load and suspended load, so the bed volume stays conserved; S 86 and S 95 turn the total load. Requires A 10 2, $0 \leq$ S 64 ≤ 1, S 65 > 0, no S 51 and S 31 1 or 2, otherwise the run stops. 0 switches it off.

default: 0.0

● ○ ○ ○

S 65 double Concentration ratio for the adaptation length of S 64
Ratio of the near-bed to the depth-averaged concentration in the adaptation length $L_s = h|U|/(S65\, w_s)$ of the suspended load S 64; larger values give a shorter adaptation length. Must be > 0.

default: 1.0

● ○ ● ●

S 71 double Start of the erodible region in x-direction
The Exner equation is only solved for $S71 \leq x \leq S72$, and the slope reduction of the critical shear stress (S 80) is only applied there. The default sentinel means no limit. For smooth transitions S 73 or S 75 are preferred.

default: $-1.0 \cdot 10^{20}$ m

● ○ ● ●

S 72 double End of the erodible region in x-direction
See S 71.

default: $1.0 \cdot 10^{20}$ m

● ○ ● ●

S 73 * double z_b, double distance, double β, double x_0, double y_0 Relaxation zone for the sediment bed along a line
Inhibits sediment transport near a straight line through (x_0, y_0) in m and makes it smoothly transition to the active bed over the given distance in m. On the line the bed is held at the elevation z_b in m and the bed change, transport rates and bed shear stress are set to zero; they are blended with a smooth function to their computed values at the given distance. The line angle β is given in degrees; $\beta = 0$ gives a line parallel to the y-axis, i.e. the distance is measured in x-direction. Several lines can be combined.

default: not set

● ○ ● ●

S 74 * double x_s, double x_e, double y_s, double y_e Non-erodible rectangular zone
The bedload transport is set to zero inside the rectangle $x_s < x < x_e$, $y_s < y < y_e$ (in m). Several zones can be defined.

default: not set

● ○ ● ●

S 75 * double x_0, double distance Relaxation zone for the sediment bed in x-direction
Within the given distance in m from $x = x_0$ in m, the bed is smoothly relaxed towards its initial level and the transport rates and bed shear stress are reduced, reaching zero at x_0. This method is preferred over S 71 and S 72 because of the smooth transition of the sediment bed. Several locations can be defined.

default: not set

● ○ ● ●

S 76 * double x_s, double x_e, double y_s, double y_e Fixed-bed zone with sediment supply
In the cells whose centre lies inside the rectangle $x_s \le x \le x_e$, $y_s \le y \le y_e$ (in m) the bed level is held (no bed change from the Exner equation, as the inflow guard S 78), but unlike S 74 the bedload transport is computed there, so the zone supplies sediment to the reach downstream. Several zones can be defined.

default: not set

● ○ ● ●

S 77 double x_s, double x_e Active region of the bed update algorithms in x-direction
Restricts the bed filter (S 100), the sandslide algorithm (S 90) and the relaxation S 73 to $x_s < x < x_e$ (in m). The bed level is transferred to the CFD topography inside this window and, outside it, wherever the bed has changed (e.g. sandslide next to the window, relaxation S 75).

default: $-1.0 \cdot 10^{20}$; $1.0 \cdot 10^{20}$ m (whole domain)

● ○ ● ●

S 78 int Inflow guard
When ON, the bed is kept fixed in the cells at the inflow boundary and at the patches of the patchBC (B 400s).

0 OFF

1 ON

default: 1

● ○ ● ●

S 80 int Critical shear stress reduction for a sloping bed
Reduces or increases the critical bed shear stress on sloping beds with the factor r, limited to $0.01 \leq r \leq 1.5$. Uses the angle of repose S 81/S 82.

0 OFF

1 Parker and Kovacs

2 Dey empirical

3 Dey analytical

4 Fredsøe

5 Fredsøe, with longitudinal and transverse parts weighted with the bed slopes along and across the flow

default: 0

● ○ ● ●

S 81 double Angle of repose
Mean angle of repose ϕ in degrees. The effective angle varies between $\phi \pm$ S 82 depending on the direction of the bed slope relative to the flow. Used by S 80 and the sandslide algorithm S 90.

default: 35.0 °

● ○ ● ●

S 82 double Variation of the angle of repose $\Delta\phi$
See S 81.

default: 5.0 °

● ○ ● ●

S 84 int Limiter for the critical bed shear stress reduction
Treatment of the reduction factor of S 80 where the bed is steeper than the angle of repose.

1 Fredsøe: reduction factor clipped to $0.01 \leq r \leq 1.5$

2 function based: $r = 0.1/|\gamma| + 0.1$ for slopes above the angle of repose

default: 1

● ○ ● ●

S 85 int Bedload direction correction
Corrects the bedload transport for the bed slope along the flow direction after Koch and Flokstra (1980): the transport capacity $q_{b,e}$ is multiplied by $1 - \min(1,\ 1.3\partial z_b/\partial s)$ with the bed slope $\partial z_b/\partial s$ in flow direction. Together with S 80 > 0 the slope effect is counted twice and a note is printed; with S 80 the option S 85 0 is recommended.

0 OFF

1 ON

default: 1

● ○ ● ●

S 86 int Transverse bed slope effect
Gravity turns the bed load down the transverse bed slope, $\mathbf{q}_b = q_b(\hat{s} - \nabla z_b / f(\theta))$ with $f(\theta) = A_{sh}\theta^{B_{sh}}(d/h)^{C_{sh}}$ (S 87 – S 89; Talmon, van Mierlo and Struiksma 1995). The part $-q_b/f\,\partial z_b/\partial n$ is a conservative, diffusive face flux of its own, closed at walls, cells without an erodible bed and cells held by the inflow guard S 78. Its explicit diffusion additionally limits the sediment time step. Requires S 31 1 or 2.

0 OFF

1 ON

default: 0

● ○ ● ●

S 87 double Slope factor A_{sh} of the transverse bed slope effect
Coefficient A_{sh} of $f(\theta)$ for S 86 1.

default: 9.0

● ○ ● ●

S 88 double Shields exponent B_{sh} of the transverse bed slope effect
Exponent of the Shields parameter θ in $f(\theta)$ for S 86 1.

default: 0.5

● ○ ● ●

S 89 double Exponent C_{sh} of d/h for the transverse bed slope effect
Exponent of the ratio of grain size d (S 20, or the diameter of the fraction with S 51) and water depth h in $f(\theta)$ for S 86 1.

default: 0.3

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S 90 int Sandslide algorithm
Corrects bed slopes that exceed the angle of repose (S 81, S 82) after each bed update. Iterations S 91, correction factor S 92, extra angle S 93, calling interval S 94.

0 OFF

1 explicit full correction

2 explicit partial correction

3 steepest descent

4 weighted multi-direction

5 diffusion

default: 0

● ○ ● ●

S 91 int Maximum number of sandslide iterations
The iteration stops earlier when no cell needs correction.

default: 1

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S 92 double Sandslide correction factor
Dimensionless factor on the bed level correction per sandslide iteration (S 90 1 – 4).

default: 1.0

● ○ ● ●

S 93 double Delta angle for the sandslide correction
Additional angle in degrees by which an over-steep slope is flattened below the angle of repose (S 90 1 and 2).

default: 0.0 °

● ○ ● ●

S 94 int Sandslide interval
The sandslide algorithm is called every S 94 sediment iterations.

default: 1

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S 95 int Spiral-flow correction of the bedload direction
For the depth-averaged SFLOW (A 10 2; NHFLOW and CFD resolve the secondary flow, other modules stop the run). The helical flow in a bend turns the near-bed flow towards the centre of curvature of the streamlines by the angle δ, $\tan\delta = \alpha_I\, h/R_s$ with $\alpha_I = E_{spir}\, 2/\kappa^2 (1 - \sqrt{g}/(\kappa C))$ (S 96; Engelund 1974, Struiksma et al. 1985), the signed curvature $1/R_s$ of the streamlines of the depth-averaged velocity and the Chezy coefficient $C = \sqrt{g}|U|/u_*$ from the bed shear stress. The transport direction of every face is turned by δ. Requires S 31 1 or 2.

0 OFF

1 equilibrium spiral flow

2 spiral flow with adaptation (Kalkwijk and Booij 1986): the intensity I relaxes towards $I_e = h|U|/R_s$ along the flow over the adaptation length L_a = S 97 $(m-1)\, h/(2\kappa^2)$, $m = \kappa C/\sqrt{g}$, so the spiral flow builds up after the entrance of a bend and decays after its exit; $\tan\delta = \alpha_I I/|U|$. Solved with upwind Gauss-Seidel sweeps (at most S 49); the intensity is added to the sediment output (ST_spiral).

default: 0

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S 96 double Spiral-flow intensity factor E_{spir}
Calibration factor of the spiral-flow coefficient α_I for S 95.

default: 1.0

● ○ ○ ○

S 97 double Factor of the spiral-flow adaptation length
Factor on the adaptation length L_a of the spiral flow for S 95 2; 0 gives the equilibrium spiral flow. Must be ≥ 0.

default: 1.0

● ○ ● ●

S 100 int Number of outer bed filter iterations
Spatial smoothing of the bed change rate before the bed update, restricted to sediment cells within S 77. Faces to cells without an erodible bed, to cells outside S 77 and to the domain boundary carry no flux, so the filter conserves the bed volume. Larger numbers increase the filter effect; 0 switches the filter off. The filter weight is S 102.

default: 3

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S 101 int Number of inner bed filter iterations

Corrector iterations of each outer filter iteration (Van Cittert sharpening). Larger numbers move the bed change closer to the unfiltered values; recommended range 1 – 5.

default: 5

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S 102 double Bed filter weight

Fraction of the centre value kept per filter pass. Each pass smooths in x- and then in y-direction, $f_i = h_i + \beta\,(h_{i-1} - 2h_i + h_{i+1})$ with $\beta = 0.25\,(1 - S102)$ (exchange weighted with the smaller cell area on non-uniform grids). With 0.0 a pass is the 1-2-1 filter, which removes the $2\Delta x$ wave and keeps the long waves (with S 100 3 and S 101 5: $3\Delta x$ 0.56, $4\Delta x$ 0.95). A value of 1.0 means no smoothing.

default: 0.0

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S 103 double Bed celerity Courant number for the sediment time step

Additionally limits the sediment time step with the celerity of bed forms, $\Delta t_{sed} \leq S103\,\Delta x / c$ with $c = S35/(1-n)\,\partial q_b/\partial z_b$. Only bed level differences larger than d_{50} are considered. 0.0 switches the limit off.

default: 0.0

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S 104 double Morphological acceleration limit for the sediment time step

Limits the sediment time step such that the bed forms move relative to the flow at most at S 104 times the near-bed flow speed, $c\,\Delta t_{sed}/\Delta t_{flow} \leq S104 \cdot U$, with the bed celerity c of S 103 (including S 35), the maximum near-bed flow speed U and the flow time Δt_{flow} since the previous bed update. The error of the morphological acceleration is of the order of this ratio. The effective morphological factor and the ratio c/U are printed with the sediment time step. 0.0 switches the limit off.

default: 0.0

6.16 T :: Turbulence

Valid for:
S F N C

○ ○ ○ ●

T 10 int Turbulence model

Turbulence model of REEF3D::CFD. The turbulence models of SFLOW and NHFLOW are selected with A 260 and A 560. The EARSM option T 10 12 has been removed; the run stops with a message, use T 10 2 instead. Any other value not listed below stops the run as well.

0 OFF

1 k-ϵ model

2 k-ω model

21 URANS with k-ϵ model

22 URANS with k-ω model

31 LES with Smagorinsky SGS model

33 LES with WALE model

default: 0

○ ○ ○ ●

T 12 int Convection discretization for the turbulence model

Other values stop the run with a message.

0 OFF

1 FOU

5 WENO HJ for non-uniform grids

55 same as 5 (the former uniform-grid WENO HJ has been removed)

default: 5

○ ○ ○ ●

T 21 int Type of LES filter

Used with T 10 31 and 33. The Smagorinsky or WALE eddy viscosity is computed from the strain rate of the filtered velocity $\mathbf{u}'$. Other values stop the run with a message.

0 box: no filtering, $\mathbf{u}' = \mathbf{u}$

1 first-order high-pass filter, $\mathbf{u}' = (I - G)\mathbf{u}$ with the discrete filter G = (1,2,1)/4 in each direction

2 second-order high-pass filter, $\mathbf{u}' = (I - G)^2\mathbf{u}$ (Stolz, Schlatter and Kleiser 2005), so that the eddy viscosity acts only on the smallest resolved scales

default: 0

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T 23 double C_3 coefficient for the URANS filter
Dimensionless coefficient of the URANS eddy viscosity filter of the k-ω model (T 10 22).

default: 1.0

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T 31 double Factor for the eddy viscosity limiter
Coefficient of the strain-based limiter $\nu_t \leq T31 \cdot k/|S|$. In CFD it is always applied with the k-ϵ model and activated with T 34 for the k-ω model; in NHFLOW it is activated with A 564 1 and is the realizability bound of the SST limiter A 564 2 (not with the EARSM A 559); in SFLOW it is applied with the depth-averaged k-ω model (A 260 2).

default: $1/\sqrt{3} \approx 0.577$

○ ○ ● ●

T 33 int Turbulent kinetic energy source term in the momentum equations
Adds the isotropic part of the Reynolds stresses, $-\frac{2}{3}\nabla k$, as source term to the momentum equations (two-equation models).

0 OFF

1 ON

default: 0

○ ○ ○ ●

T 34 int Eddy viscosity limiter
Strain-based eddy viscosity limiter of the k-ω model, with the coefficient T 31.

0 OFF

1 ON

default: 1

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T 36 int Free surface boundary condition for the turbulent dissipation
Damps the turbulence at the free surface with the turbulence length scale y' at the interface (Celik and Rodi 1984), for the k-ϵ and the k-ω model: the dissipation is limited from below by the surface value $\epsilon_s = c_\mu^{3/4} k^{3/2}/(\kappa y')$ (k-ϵ) or $\omega_s = k^{1/2}/(c_\mu^{1/4} \kappa y')$ (k-ω), weighted with $0.5(1+\cos(\pi\phi/\epsilon))$ within the interface band $|\phi| < \epsilon$ of width T 38. The damping only lowers the eddy viscosity. y' is at least half the cell height.

0 OFF

1 ON, y' = T 37

2 ON, including the distance d_w from the nearest wall, $1/y' = 1/T37 + 1/d_w$

3 ON, y' = T 37 $\cdot h$ with the local water depth h

default: 0

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T 37 double Virtual origin for the distance y' above the free surface
Used by the free surface dissipation boundary condition (T 36 in CFD, A 567 in NHFLOW); for T 36 3, A 567 3 and, with the NHFLOW k-ϵ model, A 567 1 it is a dimensionless factor on the water depth.

default: 0.07

○ ○ ○ ●

T 38 double Width factor for the free surface turbulence damping
Half-width ϵ of the free surface zone for T 36 and T 39, as a multiple of the average cell size.

default: 1.6

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T 39 int Minimum eddy viscosity at the free surface
Within the free surface zone (width T 38) the eddy viscosity of the k-ϵ and k-ω models is limited from below by a Smagorinsky subgrid-scale eddy viscosity ($C_s = 0.2$), weighted with $0.5(1 + \cos(\pi\phi/\epsilon))$.

0 OFF

1 ON

default: 0

○ ○ ○ ●

T 41 int k-ω stabilization
Stabilization of the eddy viscosity in nearly potential flow regions after Larsen and Fuhrman (2018), with the coefficient T 42. For NHFLOW use A 565.

0 OFF

1 ON

default: 0

○ ○ ● ●

T 42 double Coefficient λ_2 for the k-ω stabilization
Coefficient of the stabilization limiter (T 41 in CFD, A 565 in NHFLOW, not with the EARSM A 559); smaller values give a weaker limitation.

default: 0.05

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T 44 double Height factor for the wall velocity at the sediment bed
In the wall boundary condition of the k-ϵ and k-ω models at a sediment bed (S 10 > 0) the velocity for the bed shear stress is interpolated at the height T 44 $\cdot \Delta z$ above the bed, and the log law is evaluated at that height.

default: 1.0

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T 45 int Buoyancy term in the k-equation
Adds turbulence production or damping due to density stratification (k-ϵ and k-ω models). A sink due to stable stratification is treated implicitly, so that k stays positive.

0 OFF

1 ON

default: 0

6.17 W :: Water and Material Properties

Valid for:
S F N C

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W 1 double Density of phase 1 (water)

default: 998.2 kg/m^3

● ● ● ●

W 2 double Kinematic viscosity of phase 1 (water)

default: $1.004 \cdot 10^{-6}$ m^2/s

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W 3 double Density of phase 2 (air)
In SFLOW only used for the wind stress of the meteorological forcing (W 200). In FNPF and NHFLOW only used for the wind forcing (in NHFLOW also for W 200) and, in NHFLOW, for the parts of nets above the free surface.

default: 1.205 kg/m^3

○ ○ ○ ●

W 4 double Kinematic viscosity of phase 2 (air)

default: $1.41 \cdot 10^{-5}$ m^2/s

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W 5 double Surface tension coefficient
Surface tension between phase 1 and phase 2; 0.0 switches surface tension off.

default: 0.0 N/m

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W 6 double Density of phase 3
Density of the third fluid (e.g. oil) for the multiphase level set model (F 300).

default: 840.0 kg/m^3

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W 7 double Kinematic viscosity of phase 3
Viscosity of the third fluid for the multiphase level set model (F 300).

default: $3.0 \cdot 10^{-4}$ m^2/s

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W 10 double Discharge
Inflow discharge used by ioFlow (B 60), by the wave generation with current and by the river initialisation (F 65 2).

default: 0.0 m^3/s

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W 11 double *u*, double *v*, double *w* Inflow velocities on surface 1

Fixed velocities in m/s on surface 1 (requires an inflow boundary condition on surface 1 in DIVEMesh). Options W 11 – W 16 are made for benchmark and other simple test and validation cases. For more complex and real world cases, either B 60 1 (ioFlow) or the B 400s (patchBC) options are recommended.

default: 0.0 ; 0.0 ; 0.0 m/s

○ ○ ○ ●

W 12 double *u*, double *v*, double *w* Inflow velocities on surface 2

Requires an inflow boundary condition on surface 2 in DIVEMesh; see W 11.

default: 0.0 ; 0.0 ; 0.0 m/s

○ ○ ○ ●

W 13 double *u*, double *v*, double *w* Inflow velocities on surface 3

Requires an inflow boundary condition on surface 3 in DIVEMesh; see W 11.

default: 0.0 ; 0.0 ; 0.0 m/s

○ ○ ○ ●

W 14 double *u*, double *v*, double *w* Inflow velocities on surface 4

Requires an inflow boundary condition on surface 4 in DIVEMesh; see W 11.

default: 0.0 ; 0.0 ; 0.0 m/s

○ ○ ○ ●

W 15 double *u*, double *v*, double *w* Inflow velocities on surface 5

Requires an inflow boundary condition on surface 5 in DIVEMesh; see W 11.

default: 0.0 ; 0.0 ; 0.0 m/s

○ ○ ○ ●

W 16 double *u*, double *v*, double *w* Inflow velocities on surface 6

Requires an inflow boundary condition on surface 6 in DIVEMesh; see W 11.

default: 0.0 ; 0.0 ; 0.0 m/s

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W 20 double Acceleration due to gravity, x-component

In FNPF only used in the force calculation of floating bodies, in NHFLOW in the force calculation of floating bodies, for DEM particles (E 10) and for the rod trees (Z 20); SFLOW only uses W 22.

default: 0.0 m/s^2

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W 21 double Acceleration due to gravity, y-component

See W 20.

default: 0.0 m/s^2

● ● ● ●

W 22 double Acceleration due to gravity, z-component

Must be set for all free surface simulations, typically -9.81. Also used by REEF3D::SEASTATE (A 10 7, Section 6.1.4).

default: 0.0 m/s^2

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W 29 double x, double y, double z Additional constant pressure gradient

Constant source term added to the momentum equations like a body force, i.e. the pressure gradient divided by the density in m/s^2, e.g. to drive the flow with periodic boundary conditions.

default: 0.0 ; 0.0 ; 0.0 m/s^2

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W 30 int Compressibility of air

Calculates the density of phase 2 from the ideal gas law using the local pressure and the temperature W 31.

0 OFF

1 ON

default: 0

○ ○ ○ ●

W 31 double Air temperature for the compressibility model

Temperature in $°C$ used with W 30 1.

default: 20.0 $°C$

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W 32 int Isentropic air compressibility

Adds the compressibility of the air phase to the pressure Poisson equation, $\frac{1}{\gamma\, p_{abs}} \frac{Dp}{Dt} = -\nabla \cdot \vec{u}$ with the ratio of specific heats γ (W 33) and the absolute pressure p_{abs} = W 36 + p (at least 0.01 W 36), e.g. for trapped air pockets in wave impacts. The water stays incompressible and the density of the air is not changed. In an air band next to the interface (W 34) the compressibility is switched off; this is compensated in the adjacent layer (W 35), where the weight of the compressibility term rises to $1.5 + W34/W35$ and returns to 1 over a further W 35. Requires the projection method (D 30 1), the level set method (no VOF, F 80 0) and an SSP Runge-Kutta scheme (N 40 2 or 3; note that X 10 > 0 changes N 40 3 to 4); cannot be combined with W 30. Otherwise the run stops with a message. With porous media or floating bodies a warning is printed (not verified, the compressibility is not scaled with the porosity), and the CFD mesh refinement (G 1) is switched off with a message.

0 OFF

1 ON, local time derivative $\partial p/\partial t$

2 ON, material derivative $\partial p/\partial t + \vec{u} \cdot \nabla p$

default: 0

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W 33 double Ratio of specific heats γ of the air for W 32

default: 1.4

○ ○ ○ ●

W 34 double Interface band width for W 32

Distance in m from the interface into the air within which the air is treated as incompressible. A value ≤ 0 uses the local interface half width (F 45 cells) plus one cell.

default: -1.0 (automatic)

○ ○ ○ ●

W 35 double Compensation layer width for W 32

Width in m of the layer beyond the interface band W 34 in which the compressibility is weighted up (see W 32). A value ≤ 0 uses two local cell sizes.

default: -1.0 (automatic)

○ ○ ○ ●

W 36 double Reference pressure for W 32

Ambient absolute pressure in Pa that is added to the computed pressure to give the absolute pressure of the isentropic air compressibility.

default: 101325.0 Pa

○ ○ ○ ●

W 41 * double x_c, double y_c, double z_s, double z_e, double U, double β Velocity vertical line source for phase 1

Vertical line source at (x_c, y_c) in m between z_s and z_e in m with the velocity U in m/s. Here, β is the direction of the flow in degrees. Only applied without ioFlow and wave generation (B 60 0, B 90 0, B 180 0).

default: not set

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W 50 double Air inflow velocity

Velocity in m/s in x-direction imposed in the air phase at the inflow boundary, e.g. to model wind. Only active if the command is given.

default: 0.0 m/s (not set)

● ○ ○ ●

W 90 int Non-Newtonian rheology

Rheology of phase 1. In SFLOW only option 1 is available, as a Bingham-type bed friction; other values switch the rheology off with a message.

0 OFF

1 Herschel-Bulkley (yield stress W 96 or W 101, consistency W 97, exponent W 98)

2 three-phase mixture: Herschel-Bulkley slurry (W 98, W 105, W 106) and Coulomb viscoplastic gravel (W 106, W 107), weighted with W 108 (CFD only)

default: 0

○ ○ ○ ●

W 95 double Maximum kinematic viscosity ν_0

Upper limit of the rheological viscosity for W 90 > 0; it regularizes the viscosity at vanishing shear rates.

default: 0.001 m^2/s

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W 96 double Yield stress τ_0

Prescribed yield stress for W 101 0. It is also used in the first time step for all W 101 options.

default: 1.0 Pa

● ○ ○ ●

W 97 double Consistency index K

Consistency index of the Herschel-Bulkley model, $\tau = \tau_0 + K\dot{\gamma}^n$, in Pas^n.

default: $1.0 \cdot 10^{-5}$ Pas^n

○ ○ ○ ●

W 98 double Power law index n

$n < 1$: shear thinning, $n > 1$: shear thickening, $n = 1$: Bingham fluid.

default: 1.0

● ○ ○ ●

W 101 int Yield stress model

Pressure-dependent yield stress models use the angle of repose and cohesion W 102, the pressure selected with W 111 and the regularization W 103. Options $3-5$ use the buoyant part of the pressure (interstitial fluid density $1000\,kg/m^3$). SFLOW supports options 0 and 1; options $2-5$ use the Mohr-Coulomb yield stress of option 1 there.

0 prescribed yield stress W 96

1 Mohr-Coulomb, $\tau_0 = p\tan\phi + c$

2 Drucker-Prager type, $\tau_0 = (p\tan\phi + c)(1 - e^{-W103\dot{\gamma}})$

3 Herschel-Bulkley-Coulomb with buoyant pressure

4 as 3, with shear rate generated excess pore pressure, $p\,e^{-W104\dot{\gamma}}$

5 as 3, with linear shear rate coupling, $p - W104\dot{\gamma}$

default: 0

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W 102 double ϕ, double c Angle of repose and cohesion for the Mohr-Coulomb yield stress
Angle of repose ϕ in degrees and cohesion c in Pa, used with W 101 > 0.

default: 30.0 ; 0.0

○ ○ ○ ●

W 103 double Transition factor for the Mohr-Coulomb yield stress
Regularization time scale in s in the factor $(1 - e^{-W103\dot{\gamma}})$ of W 101 $2-5$; larger values give a sharper transition.

default: 1.0 s

○ ○ ○ ●

W 104 double Shear rate coupling factor for the pore pressure
Coefficient of the shear rate dependent reduction of the effective pressure: time scale in s for W 101 4, and coefficient in Pas for W 101 5.

default: 1.0

○ ○ ○ ●

W 105 double τ_{00}, double C, double C_{kc}, double C_i, double C_m Slurry yield stress parameters for W 90 2
Calibration yield stress τ_{00} in Pa (accounts for the grid dependency of the shear rate), total volumetric solid concentration C, and the volumetric concentrations of kaolinite and chlorite C_{kc}, illite C_i and montmorillonite C_m (all in m^3/m^3). The slurry yield stress is $\tau_y = \tau_0 C^2 e^{22CP_1}$, with $\tau_0 = \tau_{00}$ for $C < 0.47$ and $\tau_0 = \tau_{00} e^{5(C-0.47)}$ otherwise, and P_1 derived from $C_{kc} + 1.3C_i + 1.7C_m$.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ○ ●

W 106 double b, double m_y Slurry consistency factor and gravel regularization for W 90 2
The slurry consistency is $k = b\tau_y$. m_y in s is the regularization parameter of the Coulomb viscoplastic gravel viscosity, $\mu = \mu_{min} + p\sin\delta\,(1 - e^{-m_y\dot{\gamma}})/\dot{\gamma}$.

default: 0.3 ; 0.2

○ ○ ○ ●

W 107 double μ_{min}, double δ, double μ_{max} Gravel viscosity parameters for W 90 2

Minimum dynamic viscosity μ_{min} in Pas, internal friction angle δ in degrees and maximum dynamic viscosity μ_{max} in Pas of the Coulomb viscoplastic gravel phase. $\mu_{max} = 0$ (default) applies no upper limit. The pressure in the gravel viscosity is the gauge pressure, limited to non-negative values.

default: 0.0 ; 0.0 ; 0.0

○ ○ ○ ●

W 108 double Slurry volume fraction a_2 for W 90 2

The mixture viscosity is $(a_2\mu_{slurry} + (1 - a_2)\mu_{gravel})/\rho$, with ρ = W 1 as mean density of the mixture.

default: 0.3

○ ○ ○ ●

W 110 int Numerical treatment of the rheology

Determines whether the viscous (rheology) part and the yield stress part of the Herschel-Bulkley model enter as viscosity or as momentum source term.

1 viscosity for rheology and yield stress

2 viscosity for rheology, momentum source term for yield stress

3 momentum source term for rheology and yield stress

4 viscosity for rheology, yield stress gradient as momentum source term

5 rheology and yield stress gradient as momentum source term

6 viscosity for rheology, frictional source term from the pressure gradient, $\tan\phi\nabla p$

7 as 6

default: 1

○ ○ ○ ●

W 111 int Pressure for the Mohr-Coulomb yield stress

1 hydrostatic

2 total pressure (hydrostatic and dynamic)

3 hydrostatic pressure away from the interface and total pressure close to the interface (see W 112)

4 average of hydrostatic and total pressure

5 hydrostatic for the first 10 iterations, total pressure afterwards

default: 1

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W 112 double Threshold factor for W 111 3

The total pressure is used where the level set distance from the interface is smaller than W 112 $\cdot\Delta x$.

default: 2.1

6.17.1 Meteorological forcing

W 200 – W 227 drive REEF3D::SFLOW (A 10 2) and REEF3D::NHFLOW (A 10 5) with the wind and the air pressure of a meteorological field, e.g. for storm surge, and add the Coriolis force (W 211). In SFLOW, both act with the explicit (A 226 0) and the semi-implicit (A 226 1) free surface scheme. The same wind field is the wind input of REEF3D::SEASTATE with A 730 2 (stand-alone or coupled through A 750 1, Section 6.1.4); there the drag law W 207 is used with A 729 1. Stand-alone SEASTATE (A 10 7 with A 730 2) is the only case where W 200 is used outside SFLOW and NHFLOW; W 200 $>$ 0 with any other model stops the run with a message, as does W 211 $>$ 0 outside SFLOW and NHFLOW. In NHFLOW, W 200 cannot be combined with the wind forcing A 570, and the grids of `curvgrid.dat` (G 50 2) are not supported yet (the curvilinear grids of DIVEMesh are).

Equations: once per time step, at the model time of the start of the step, every cell takes the 10 m wind $\vec{U}_{10}$ and the air pressure p_a at mean sea level from the source (W 200). The surface stress and the momentum sources of the wet cells are

$$\vec{\tau}_s = r(t)\,\rho_a\,C_d(|\vec{U}_{10}|)\,|\vec{U}_{10}|\,\vec{U}_{10},$$

$$\frac{\partial(Uh)}{\partial t} = \ldots + \frac{\tau_{s,x}}{\rho_w} - \frac{h}{\rho_w}\,r(t)\frac{\partial p_a}{\partial x} + f\,Vh, \qquad \frac{\partial(Vh)}{\partial t} = \ldots + \frac{\tau_{s,y}}{\rho_w} - \frac{h}{\rho_w}\,r(t)\frac{\partial p_a}{\partial y} - f\,Uh,$$

with ρ_a = W 3, ρ_w = W 1, the drag coefficient C_d of W 207, the ramp $r(t)$ of W 204 and the Coriolis parameter f of W 211. The pressure gradient (W 208) is taken with central differences of the cell values. In NHFLOW the surface stress acts in the top σ-layer, the pressure gradient and the Coriolis force act in every layer; the friction velocity of the surface stress also enters the algebraic eddy viscosity of A 547. The Coriolis force is explicit in the Runge-Kutta stages (and in the Runge-Kutta predictor of the semi-implicit schemes). At the open boundaries (B 520 methods 3 – 5) and in the relaxation and beach zones, the target water level is raised by the inverse barometer $\eta_{ib} = -r(t)\,(p_a - p_{ref})/(\rho_w g)$ with p_{ref} = W 209 (W 210). Every grid of the run (refined patches, blocks) evaluates the forcing on its own cells.

Time and space: REEF3D model time 0 corresponds to the date-time W 203, or to the first record of the input file. The sources are interpolated bilinearly in space and linearly in time, and held at the first or last record outside the records. The geographic frame W 205, W 206 places the grid on the earth: a spherical transverse Mercator projection (R = 6371 km) about the longitude and latitude λ_0, φ_0, with the model x-axis rotated counter-clockwise from east by the given angle. The frame is needed for the vortex (W 200 3 and 4), which is given in longitude and latitude, and for the Coriolis parameter of every cell (W 211 2). Gridded met

files are already in model coordinates; the frame they were written in is stored in the file and compared with W 205, W 206 (a warning is printed if they differ).

Input files (in the simulation folder):

- `met-forcing.r3m` (W 200 2, background of W 200 4): binary file (little-endian) with u_{10}, v_{10} (m/s, along the model x- and y-axes) and optionally p_a (Pa; otherwise 101325 Pa) on a regular grid in model coordinates. A 144-byte header (`REEF3DMT`, version 1, `nx ny nvar moving nrec frame epsg`, `x0 y0 dx dy`, the frame and the averaging period of u_{10}) is followed by `nrec` records in increasing time, each with the time (s since 1970-01-01 00:00:00 UTC), the origin of the grid of the record and the `nx` × `ny` values of each variable. With `nx` = `ny` = 1 the file is a spatially uniform time series. Each MPI rank reads only the part of the two records around the current time that its cells need. The file is written by `tools/met_forcing.py`, e.g. from ERA5, NORA3 or BARRA-C2 NetCDF files, from a table `YYYYMMDD.HHMMSS U10 dir [pa]`, or from an old SEASTATE wind file `seastate-wind.dat`.
- `met-nest.r3m` (W 200 4 with W 220 1 x): a met file of the same format on a moving grid (each record has its own origin), e.g. a storm-following HAFS nest (`tools/met_forcing.py nest`).
- `met-vortex.trk` (W 200 3, W 200 4 with W 220 x 1 or W 227 1): ASCII track of a parametric tropical cyclone vortex in SI units. The first line (after comment lines starting with #) is `REEF3D-MET-TRACK 1 nrec` ρ_a, then each record has 64 numbers: `YYYYMMDD.HHMMSS lon lat ut vt pc pn vmaxg rmw B80 vms bs xn rn Bg0 phi0` followed by `R Rm Bg phi` for the quadrants NE, SE, SW, NW and the 34, 50 and 64 kt isotachs. The parameters are fitted from an ATCF best track or IBTrACS with `tools/met_track.py fit`. The air density ρ_a of the track is used in the vortex profiles, W 3 in the surface stress.

Output: with W 200 > 0, the free surface VTP files of SFLOW and NHFLOW contain `wind10` (vector, m/s), `air_pressure` (Pa) and `wind_stress` (vector, τ_s in N/m^2). The SFLOW maximum output (P 112) adds the maximum wind speed and the minimum air pressure. The settings and the model time 0 are printed at the start of the run.

Valid for:

S	F	N	C
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W 200 int Meteorological forcing

Source of the wind and the air pressure (see the introduction of this subsection). Also the wind field of SEASTATE with A 730 2.

0 OFF

1 uniform and constant wind (W 201) and air pressure (W 202)

2 met file `met-forcing.r3m`

3 parametric vortex from `met-vortex.trk` (W 215 – W 219), needs W 205

4 blend of the background `met-forcing.r3m`, the nest `met-nest.r3m` and the vortex `met-vortex.trk` (W 220 – W 227)

default: 0

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W 201 double U_{10}, double direction Uniform wind for W 200 1

Wind speed at 10 m height (m/s, not negative) and the direction the wind blows to (deg, counter-clockwise from the positive x-axis).

default: 0.0 ; 0.0

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W 202 double Uniform air pressure for W 200 1

A uniform pressure has no gradient; it only enters the inverse barometer (W 209, W 210).

default: 101325.0 *Pa*

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W 203 double Start date-time of the meteorological forcing

Date-time `YYYYMMDD.HHMMSS` (UTC) of model time 0. With 0, model time 0 is the first record of `met-forcing.r3m` (W 200 2 and 4) or of `met-vortex.trk` (W 200 3); W 200 1 then has no date. In SEASTATE runs it replaces A 780; if A 780 is given as well, both must agree, otherwise the run stops with a message.

default: 0.0

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W 204 double Ramp time of the meteorological forcing

The surface stress, the air pressure gradient and the inverse barometer are multiplied by $r(t) = 0.5(1 - \cos(\pi t/T))$ for $t < T$ = W 204 (s). Not applied to the Coriolis force and to the wind of SEASTATE. 0 switches the ramp off.

default: 0.0 *s*

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W 205 double λ_0, double φ_0, double rotation Geographic frame

Longitude and latitude (deg) of the centre of the transverse Mercator projection and the angle (deg) of the model x-axis, counter-clockwise from east. Required for the vortex (W 200 3, W 200 4 with W 220 x 1 or W 227 1) and for W 211 2; it gives the latitude for W 211 1 when W 212 is 0. Also used by SEASTATE with the vortex wind (A 730 2). The frame is the same as `-origin LON0 LAT0 -rotation ROT` of `tools/met_forcing.py` and `tools/seastate_forcing.py`.

default: not set

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W 206 double X_0, double Y_0 Model coordinates of the frame origin
Model coordinates (m) of the point (λ_0, φ_0) of W 205.

default: 0.0 ; 0.0 m

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W 207 int Drag law of the meteorological forcing
Drag coefficient $C_d(U)$ of the 10 m wind speed U for the surface stress, evaluated per cell. Options 1 – 4 are those of A 570. Also the drag law of the SEASTATE wind input with A 729 1.

0 automatic: 1 for W 200 1 and 2, 5 for W 200 3 and 4

1 Garratt (1977), $C_d = (0.75 + 0.067\,U) \cdot 10^{-3}$, $3.5 \cdot 10^{-3}$ above 41 m/s

2 Wu (1982), $C_d = 1.2875 \cdot 10^{-3}$ below 7.5 m/s, $(0.8 + 0.065\,U) \cdot 10^{-3}$ above

3 Smith and Banke (1975), $C_d = (0.63 + 0.066\,U) \cdot 10^{-3}$

4 Zijlema et al. (2012), $C_d = (0.55 + 2.97\,x - 1.49\,x^2) \cdot 10^{-3}$, $x = U/31.5$, not negative

5 Garratt, capped at W 213 (drag saturation in hurricane winds)

6 Zijlema et al., beyond its peak at 31.4 m/s not below W 214

default: 0

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W 208 int Air pressure gradient
Momentum source $-(h/\rho_w)\nabla p_a$ of the meteorological forcing.

0 OFF

1 ON

default: 1

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W 209 double Reference air pressure of the inverse barometer
Pressure p_{ref} at which the inverse barometer $\eta_{ib} = -r(t)\,(p_a - p_{ref})/(\rho_w g)$ of W 210 is zero.

default: 101325.0 Pa

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W 210 int Inverse barometer at the open boundaries

Raises the target water level of the open boundaries (B 520 methods 3 – 5, including the tidal edges with a background) and of the relaxation and beach zones of SFLOW and NHFLOW by the inverse barometer (W 209), so that a low pressure enters the domain with its static response.

0 OFF

1 ON

default: 1

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W 211 int Coriolis force

Adds $f\,V\,h$ to the x-momentum and $-f\,U\,h$ to the y-momentum of the wet cells (in NHFLOW in every layer), with $f = 2\Omega \sin\varphi$, $\Omega = 7.2921159 \cdot 10^{-5}$ 1/s. Independent of W 200. The force is explicit in the Runge-Kutta stages; SSP-RK3 is stable for $f\,\Delta t < 1.7$.

0 OFF

1 f-plane: constant f = W 212, or from the latitude φ_0 of W 205 if W 212 is 0

2 f of every cell from its latitude, needs W 205

default: 0

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W 212 double Coriolis parameter for W 211 1

Constant Coriolis parameter f (1/s, negative in the southern hemisphere). With 0, $f = 2\Omega \sin\varphi_0$ with the latitude of W 205.

default: 0.0 $1/s$

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W 213 double Maximum drag coefficient for W 207 5

Cap of the Garratt drag coefficient (saturation of the drag in hurricane winds, Powell et al. 2003, Donelan et al. 2004); the default is reached at 26.1 m/s. Must be positive. Also used by the SEASTATE wind input with A 729 1.

default: $2.5 \cdot 10^{-3}$

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W 214 double Minimum drag coefficient for W 207 6

Floor of the Zijlema et al. drag coefficient beyond its peak ($2.03{\cdot}10^{-3}$ at 31.4 m/s), which avoids the zero drag at 68 m/s. Must not be negative. Also used by the SEASTATE wind input with A 729 1.

default: $1.5 \cdot 10^{-3}$

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W 215 int Vortex model for W 200 3 and 4

Wind and pressure profile of the parametric tropical cyclone vortex of the track file `met-vortex.trk`, evaluated at the longitude and latitude of every cell with f at the storm centre (clockwise rotation in the southern hemisphere). The pressure is $p = p_c + \Delta p\, e^{-(R_m/r)^B}$ (model 2: $e^{-\phi(R_m/r)^{Bg}}$).

1 Holland (1980), symmetric gradient wind

2 GAHM, generalized asymmetric Holland model (Gao 2018, ADCIRC NWS 20): R_m, B_g, ϕ per quadrant, interpolated between the isotachs and in azimuth

3 Holland, Belanger and Fritz (2010), surface wind profile

default: 2

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W 216 double Reduction of the gradient wind to 10 m

Factor from the gradient wind to the 10 m wind for the vortex models W 215 1 and 2. Must be positive.

default: 0.9

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W 217 double Wind averaging factor of the vortex

Factor from 1-minute to 10-minute mean wind applied to the surface wind of the vortex. Must be positive.

default: 0.8928

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W 218 int Inflow angle of the vortex

Turns the surface wind of the vortex towards the centre.

0 none

1 as ADCIRC: $10\,r/R_m$ deg inside R_m, rising to 25 deg at 1.2 R_m, 25 deg beyond

default: 1

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W 219 int Storm translation in the vortex wind

Adds the translation velocity of the storm centre (`ut vt` of the track) to the surface wind of the vortex.

0 none

1 scaled by V_g/V_{max} (as ADCIRC)

2 in full

default: 1

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W 220 int nest, int vortex Layers of the blend W 200 4

Switches the nest layer `met-nest.r3m` and the vortex layer `met-vortex.trk` (0 off, 1 on) on top of the background `met-forcing.r3m`. Each layer replaces the field below with its weight w: $f = f_{below} + w\,(f_{layer} - f_{below})$, for u_{10}, v_{10} and p_a. Without p_a in the nest file, the pressure of the background is kept.

default: 0 ; 1

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W 221 double width, double T Nest taper for W 200 4

The weight of the nest rises from 0 at the edge of the nest grid to 1 at the distance width (m, positive) inside it, and in time over T (s) after the first and before the last nest record; outside the nest records it is 0.

default: 50000.0 ; 3600.0

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W 222 int basis, double r_1, double r_2 Vortex radii for W 200 4

The vortex has the weight 1 inside the distance r_1 from the storm centre and 0 beyond r_2 ($0 \le r_1 < r_2$). The basis gives the unit of r_1 and r_2: 1 multiples of the radius of maximum wind of the track, 2 multiples of the mean 34 kt radius (3 rmw for a record without 34 kt isotachs), 3 metres.

default: 1 ; 3.0 ; 6.0

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W 223 int Vortex pressure in the blend W 200 4

0 pressure of the vortex as fitted (outer pressure `pn` of the track)

1 the pressure deficit of the vortex is rescaled so that its mean on the ring r_2 (16 points) equals that of the background, $p'_v = p_c + (p_v - p_c)\max((\langle p_b \rangle - p_c)/(\langle p_v \rangle - p_c), 0)$; the wind is not changed

default: 1

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W 224 double Vortex taper in time for W 200 4

The weight of the vortex rises over W 224 (s) after the first record of the track and falls over W 224 before its last record; outside the track the vortex is off. 0 gives a step.

default: 21600.0 *s*

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W 225 int Taper shape for W 200 4

Shape of the weights of W 221, W 222 and W 224 over the normalized distance or time *s*.

1 cosine, $0.5\,(1 - \cos(\pi s))$

2 linear, *s* (as ADCIRC NWS 29/30)

default: 1

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W 226 int Environment of the vortex for W 200 4

0 OFF, the vortex alone inside r_1 (as ADCIRC NWS 29/30)

1 the large-scale flow of the background is added to the vortex: the mean of background minus vortex wind on the ring r_2 is added to the vortex wind, and the gradient of background minus vortex pressure (least-squares plane through the ring) to the vortex pressure

default: 0

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W 227 int Time interpolation of the nest for W 200 4

0 linear in time at fixed points

1 storm-relative: the two nest records around the current time are shifted with the storm centre of the track before the linear interpolation, so that the core of a moving storm is carried instead of cross-faded. Needs `met-vortex.trk` and W 205, also with the vortex layer off

default: 0

6.18 X :: 6DOF

The X commands control the rigid-body (6DOF) module for floating and moving bodies, including moorings, nets, membranes, prescribed-motion wavemakers and the ship module (X 350). The module is available for CFD and NHFLOW (two-way and one-way coupling), for FNPF (two-way coupling of one or several bodies, X 20; all body geometries X 110 – X 180 are triangulated; moorings, the power take-off X 500 – X 508 and the ship module X 350, no nets; free-surface treatment next to the body with X 17, X 18, X 27 and X 47, settings of the added mass and load solves with X 28 – X 37) and, in a simplified form, for SFLOW (one-way coupling and ship-induced pressure fields). A minimal free-floating body needs:

- X 10 to activate the module (X 10 1 for two-way coupling),
- X 11 to select the degrees of freedom (default: all six are solved),
- X 21 (density) or X 22 (mass) of the body; the centre of gravity (X 23) and moments of inertia (X 24) are computed from the geometry if they are not given,
- the body geometry: either primitives (X 110, X 131–X 133, X 153, X 163, X 164) or an ASCII STL file `floating.stl` (X 180, optionally scaled, moved and rotated with X 181–X 183).

By default (X 185 4), the surface triangulation of the body is remeshed to the local grid spacing at the start (X 184 – X 190); holes and gaps in an STL surface are closed. The initial position is the position of the geometry as defined; an initial rotation around the centre of gravity can be added with X 101 and initial velocities with X 102 and X 103.

Valid for:

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X 10 int Turn on 6DOF algorithm for floating bodies

Activates the rigid-body module. In FNPF (A 10 3) the bodies are resolved in the Laplace problem (Section 5.2); moorings and the power take-off (X 500) are supported there, nets are ignored. Whenever X 10 > 0, D 22 is set to 2, and the NHFLOW/CFD time scheme N 40 is changed from 3 or 23 to 4 (not with mesh refinement, G 1 > 0) and from 13 to 44. With REEF3D::CORAL (Z 40), the bodies can follow the dynamics of the CORAL model (`fluid_body` in `coral.txt`, Section 5.12). In the runtime coupling of FNPF with a CFD or NHFLOW model (B 92 62), the FNPF bodies can follow the 6DOF bodies of the receiving model (I 243).

0 OFF

1 Two-way coupling: body motion from the hydrodynamic forces (CFD, NHFLOW and FNPF)

2 One-way coupling: the body moves according to X 11 and the prescribed or initial velocities without hydrodynamic feedback (SFLOW and NHFLOW)

3 Ship-wave mode: the body acts as an external moving pressure field on the free surface (SFLOW and NHFLOW, requires X 400)

4 Prescribed wavemaker geometry from X 170 or X 172 driven by `wavemaker.dat` (NHFLOW)

default: 0

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X 11 int u, int v, int w, int p, int q, int r Degrees of freedom of the floating body
Sets the treatment of the three linear (u, v, w) and three angular (p, q, r) velocities individually. Prescribed degrees of freedom take their velocity from X 210/X 211 or from a motion file (X 240). In 2D simulations the sway velocity v and the rotations p and r are always switched off; fixed and prescribed degrees of freedom receive no hydrodynamic load and no damping (X 25, X 26).

0 fixed (OFF)

1 solve the 6DOF equation of motion

2 prescribed

default: 1 ; 1 ; 1 ; 1 ; 1 ; 1

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X 13 int Water level in the columns pierced by the body in NHFLOW
In the sigma grid of NHFLOW, every column reaches up to the free surface, also where a surface-piercing body occupies its upper part; this level enters the hydrostatic flux, the pressure boundary condition and the hydrostatic pressure on the hull. With 1, the level in the pierced columns (footprint of the body, ramped over half the direct forcing width A 526 at the waterline) is set after every water level update to the harmonic extension of the surrounding free surface (Laplace equation over the footprint), as in FNPF, so that the restoring forces come from the level of the surrounding water. Only for X 10 1 or 2 without porous body (X 16); otherwise a message is printed and X 13 is ignored. With nets, membranes or the ship module (X 320, X 330, X 350) the footprint of the previous ray cast is used. A log is written every X 19 time steps to `REEF3D_NHFLOW_6DOF/REEF3D_6DOF_pierced.dat`. Not available on curvilinear grids (G 50) and with the semi-implicit scheme (A 534 1); the run stops. With 2, the water columns under the hull end at the hull bottom instead (as in SWASH): the σ-layers of a covered column run from the bed to the keel, with no body cells and no direct forcing; the top of the column moves with the hull, the hydrostatic pressure terms take the piezometric head under the hull, which follows from the depth-integrated continuity of the footprint (a 2D Poisson equation over the footprint, solved by SOR), and the hull loads are the pressure on the keel and, at the hull sides, the pressure of the free neighbour columns. Only for X 10 1 or 2 with A 510 2 or 3, without porous body, on Cartesian grids (otherwise a message is printed and X 13 2 is ignored), and not with nets, membranes or the ship module (a message is printed). A log is written

every X 19 time steps to `REEF3D_NHFLOW_6DOF/REEF3D_6DOF_hull.dat`. Option 2 is a prototype: the added mass is close to potential theory, but the radiated waves and the damping are too large; the sharp hull X 15 8 is the alternative.

0 OFF

1 ON, level in the pierced columns from the surrounding free surface

2 water columns ending at the hull bottom (prototype)

default: 0

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X 14 int Preserve tangential velocities for forcing

Controls how the body velocity is imposed by the direct forcing in CFD. With 1, cells inside the body receive the full rigid-body velocity, while in the smoothing layer outside the body (X 41) each velocity component is forced only with the weight of the corresponding component of the surface normal, which preserves tangential velocities.

0 OFF (full forcing everywhere)

1 ON

default: 1

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X 15 int Type of direct forcing near the body surface for NHFLOW

Controls how the body velocity is imposed by the direct forcing in NHFLOW (also for the X 10 4 wavemakers). The forcing layer thickness is set with A 526.

0 full forcing of all velocity components (no-slip)

1 full forcing inside the body; outside, each component weighted with the magnitude of the corresponding normal component, with a smooth transition over a thin shell around the surface

2 full forcing inside the body; outside, only the normal velocity component is forced (free slip, tangential velocity preserved)

3 as 1, but only the cells inside the body are forced, and the momentum fluxes across the body surface are taken from the fluid side; for bodies moving through the grid at speed (towed or self-propelled ships), which otherwise carry the water of the forcing layer along

4 as 2 (normal projection outside the body), with the smooth transition over the shell of option 1; independent of the orientation of the hull to the grid

5 as 1, with a sharp switch at the body surface instead of the shell transition (for 2D cases and grid-aligned hulls)

6 as 1, and the faces between body and fluid cells act as reflective walls moving with the body in the HLL fluxes (mirror state of the fluid side about the body velocity, no continuity flux through these faces); for wave reflection and wave loads. On curvilinear grids (G 50) the fluxes of option 3 are used instead

7 as 6, with the forcing of option 3 (only the cells inside the body); for bodies at speed in waves

8 sharp hull: no direct forcing; the hull is a sharp internal boundary of the σ-grid, the staircase of the faces between the cells on the two sides of the body surface. The links across the surface are blocked (wall fluxes with the rigid-body velocity, blocked links in the pressure Poisson equation, as the sharp membranes of X 330), and the water inside the hull stays in the grid as enclosed water moving with the hull, so that the σ-layers run continuously across the hull side. The hull loads are the outer pressure on the blocked faces; the coupling is stabilised with an added mass of twice the body mass and inertia. The pressure scheme A 520 2 is changed to 1. With the mesh refinement (G 1), the zone around the hull (G 12) must keep the hull inside the finest level; there is no subcycling (as G 7 0), and with placed patches (G 40 1) the zone is kept whole on one rank. A log is written every X 19 time steps to `REEF3D_NHFLOW_6DOF/REEF3D_6DOF_hullbody.dat`. For one surface-piercing body; not for submerged or porous bodies (X 16) or curvilinear grids (G 50)

default: 1

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X 16 double n, double d_{50}, double α, double β Porous floating body

Treats the NHFLOW floating body (X 10 1) as a moving porous medium instead of a rigid body imposed by direct forcing. Given are the porosity n of the body, the characteristic grain or element size d_{50} in m and the linear (Darcy) and quadratic (Forchheimer) resistance coefficients α and β, with the same closure as the NHFLOW VRANS objects (B 201; the KC correction B 264 is applied as well). The porosity field $1-H(1-n)$ follows the body, and the Darcy-Forchheimer resistance acts on the fluid velocity relative to the body velocity; it is integrated implicitly over each Runge-Kutta stage, so that large coefficients remain stable. Its reaction is added to the body forces (written to the viscous force columns of the force file), the pressure load on the body surface is scaled with $(1-n)$, and no viscous wall forces are computed on the envelope. The forces are always integrated over the triangulated surface (as X 60 1). Can be combined with static VRANS structures (B 200 1). Giving X 16 activates the porous body.

default: not set (rigid body); 0.5 ; 0.01 ; 1000.0 ; 1.1

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X 17 double Relaxation of the free surface next to the body footprint in FNPF

For a surface-piercing body in FNPF, the free-surface elevation η in the columns directly adjacent to the footprint of the body is relaxed once per Runge-Kutta stage towards the mean of its neighbours, $\eta = (1-w)\,\eta + w\,\bar{\eta}$, with the weight w = X 17 (limited to 1). It damps a two-cell disturbance of η that can grow at the waterline of a hull moving at speed, where columns are uncovered every time step; the velocity potential at the free surface is not relaxed. 0.0 switches the relaxation off.

default: 0.0 (off)

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X 18 double Time constant of the running mean of χ at body-fixed points in FNPF

Time constant τ in s. In FNPF the hull pressure is computed with the time derivative of the potential following the body points, χ. With X 18 $>$ 0, the running mean of the free-surface data of χ is computed on a body-fixed grid around the body (horizontal extent of the body plus 5 cells, footprint excluded) with a second-order low-pass filter (two exponential stages with the time constant τ, updated once per time step) and removed from the free-surface data. This removes the mean residual that columns entering and leaving the footprint leave near the waterline of a moving surface-piercing hull (e.g. a spurious yaw moment in a steady turn), while oscillations with periods well below τ (waves, heave, pitch) are kept. Only used with a single body (X 20 1). 0.0 switches it off.

default: 0.0 (off)

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X 19 int Print out interval for 6DOF log files

The 6DOF log files (position, velocity and forces in the folder `REEF3D_*_6DOF`) are written every X 19 time steps. Also used for the FSI log files (Z 10) and the output of the ship module (X 350).

default: 1

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X 20 int Number of floating bodies

Number of 6DOF bodies in CFD and FNPF. Body 0 reads its geometry from `floating.stl`, body $n \geq 1$ from `floating-n.stl` (X 180 1). All other X settings are shared by all bodies. In FNPF the bodies are coupled through their added mass, including the cross terms; with several bodies the running mean X 18 is not used, and the FNPF mesh refinement with subcycling (G 7 1) stops the run. NHFLOW supports one body only (the run stops for X 20 $>$ 1), SFLOW always uses one body.

default: 1

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X 21 double Floating body homogeneous density ρ in [kg/m^3]

The mass is computed from this density and the body volume. The density is used by default; if X 22 is given, X 22 takes precedence. For a porous floating body (X 16), X 21 and X 22 define the mass of the solid skeleton, i.e. X 21 is the bulk density $\rho_s(1-n)$.

default: 900.0

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X 22 double Floating body mass m in [kg]
If given, the mass is prescribed and the density is computed from mass and body volume; X 21 is then ignored.

default: 0.0 (not set, mass from X 21)

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X 23 double x_G, double y_G, double z_G Floating body centre of gravity in [m]
If not given, it is calculated from the geometry. The centre of gravity is also the origin of the body-fixed (non-inertial) coordinate system and the centre of rotation.

default: calculated from the geometry

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X 24 double I_x, double I_y, double I_z Floating body moments of inertia in [kg m^2]
Principal moments of inertia about the centre of gravity. If given, the products of inertia are set to zero. If not given, the full inertia tensor is calculated from the geometry and the density, about the centroid of the geometry and, if X 23 is given, shifted to the prescribed centre of gravity (parallel-axis theorem).

default: calculated from the geometry

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X 25 double C_p, double C_q, double C_r Damping coefficients for rotational motion around the three axes
Linear damping moment $-C\,\omega$ added to the moment balance of the solved rotational degrees of freedom, in [N m s].

default: 0.0 ; 0.0 ; 0.0

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X 26 double C_u, double C_v, double C_w Damping coefficients for translational motion along the three axes
Linear damping force $-C\,u$ added to the force balance of the solved translational degrees of freedom, in [N s/m].

default: 0.0 ; 0.0 ; 0.0

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X 27 int Footprint of the body in the sigma layers in FNPF

A free-surface column belongs to the footprint of a surface-piercing body in FNPF if its free-surface node lies inside the body. With X 27 $n > 0$, a column also belongs to the footprint if a body node lies within n sigma layers below its free-surface node; the nodes above it become body nodes. This removes columns in which the free surface lies only one or two nodes above the hull, e.g. where the side of a hull passes almost through the column centres, and which can become unstable. 0 switches it off.

default: 0 (off)

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X 28 int Update interval of the added mass in FNPF

In FNPF, the added mass of the floating bodies (the unit-mode solves of the free degrees of freedom, with several bodies the cross terms) is computed at the first stage of every X 28-th time step; in between, the last added mass is used, while the pressure loads are new at every stage. 1 computes it every time step. See also X 29.

default: 1

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X 29 double Δx, double $\Delta\alpha$ Motion thresholds for the added mass update in FNPF

With X 28 > 1, the added mass is computed again before the X 28-th step as soon as a body has moved by more than Δx in [m] (centre of gravity) or turned by more than $\Delta\alpha$ in [deg] (any Euler angle) since it was last computed. 0.0 switches the respective threshold off.

default: 0.0 ; 0.0 (off)

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X 30 double Solver tolerance of the body load solves in FNPF

The Laplace solves of the body loads in FNPF (the pressure potential and the unit modes of the added mass) are solved to the tolerance X 30 $\times$ N 44 instead of the tolerance N 44 of the potential.

default: 1.0

○ ● ○ ○

X 35 int Initial guess of the body load solves in FNPF

By default, a Laplace solve of the body loads in FNPF starts from the last solution of its kind (the pressure potential of the previous stage, a unit mode of the previous added mass). With 1, it starts from the linear extrapolation in time of the last two solutions

of the same kind (the same Runge-Kutta stage of the previous two time steps, or the previous two added masses; extrapolation factor limited to 2). The results change within the solver tolerance.

0 OFF

1 ON

default: 0

○ ● ○ ○

X 36 int Added mass on the patches of the body in FNPF with mesh refinement

With the FNPF mesh refinement around the bodies (G 1 with the zones G 12, without subcycling G 7 1), the unit-mode solves of the added mass of a body are solved only on the patches that reach into its zone (window), with zero potential at their edge and on level 0, instead of on all grids. The cross added mass between bodies is then taken from the bodies inside the window only. Without the mesh refinement zones, or with G 7 1, a message is printed and the unit modes are solved on all grids.

0 OFF, unit modes on all grids

1 window solves with calibration: at the first added mass and after a new layout of the patches (and every X 37 updates), every body is also solved once on all grids, and the differences of the added mass and of the cross terms are added to the window results

2 window solves without calibration

default: 0

○ ● ○ ○

X 37 int Calibration interval of X 36 1

With X 36 1, the calibration on all grids is also repeated every X 37 added mass updates. 0: only at the first added mass and after a new layout of the patches.

default: 0

○ ○ ● ●

X 38 int Type of viscous force calculation on the body surface

Used with the STL force calculation (X 60 1). The velocity is sampled at the distance X 43 from the surface. On NHFLOW curvilinear grids (G 50) only option 0 is available; otherwise the run stops.

0 CFD: viscous stresses from the local velocity gradients with the molecular viscosity; NHFLOW: no viscous force

1 CFD: wall-gradient approximation with molecular and eddy viscosity; NHFLOW: skin friction with $C_f = 10^{-4}$

2 CFD: logarithmic wall function (roughness $k_s = 10^{-5}$ m)

default: 0

● ○ ● ●

X 41 double Thickness of the continuous direct forcing layer
Half-width of the smoothed Heaviside function around the body surface, in multiples of the mean cell size. In CFD it is used for floating bodies and for the direct forcing of solids and topography. In SFLOW and in the NHFLOW ship-wave mode (X 10 3) it sets the width of the pressure field transition; otherwise NHFLOW uses A 526.

default: 0.6

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X 42 double Distance for pressure force evaluation
Distance from the surface triangles along the outward normal at which the pressure is interpolated for the force integration (X 60 1), in multiples of the local cell size.

default: 0.0

○ ○ ● ●

X 43 double Distance for shear stress evaluation
Distance from the surface triangles along the outward normal at which the velocities are interpolated for the viscous forces (X 38), in multiples of the local cell size.

default: 1.0

○ ○ ○ ●

X 45 int Type of free surface level set convection discretization at the floating body
Modifies the stencil of the fifth-order WENO Hamilton-Jacobi scheme near the floating body. Only effective with F 35 5.

0 standard stencil, convection also inside the floating body

1 gradients set to zero for stencil cells closer than $0.6\,dx$ to the body surface

2 gradients weighted with the averaged body indicator function

3 gradients set to zero inside the body

4 gradients weighted with the upwind body indicator function

5 gradients set to zero inside solids and topography

6 gradients inside the body set to a large value (10^{20})

default: 0

○ ● ○ ○

X 47 double w, int n Sawtooth filter at the body footprint in FNPF

For surface-piercing bodies in FNPF, the free-surface elevation η and the free-surface potential in the free-surface columns within n columns of the footprint of a body are filtered once per Runge-Kutta stage with the 5-point filter of Longuet-Higgins and Cokelet (1976), $f = (1-w)\,f + w\,F(f)$ with $F(f)_i = (-f_{i-2} + 4f_{i-1} + 10f_i + 4f_{i+1} - f_{i+2})/16$, in x- and then in y-direction. The filter removes the two-cell mode that the waterline of a hull can feed in the neighbouring columns and changes a smooth field only at fourth order; unlike X 17, its damping does not decrease with the grid spacing. The weight w is limited to 1, the width n to $1-3$ columns. $w = 0.0$ switches the filter off.

default: 0.0 (off) ; 2

○ ○ ○ ●

X 48 int Level set extrapolation into direct forcing solids

If ON, the level set and the k-ω turbulence quantities are copied from the first fluid cell into the two adjacent cells inside direct-forcing bodies (floating bodies, solids, topography).

0 OFF

1 ON

default: 0

● ● ● ●

X 50 int Type of print out format for 6DOF structure

Since the VTP print out is binary, it is significantly faster than the STL print out. This becomes runtime relevant for STL files with a large number of triangles.

1 VTP

2 STL

default: 1

○ ○ ● ●

X 60 int Type of hydrodynamic force calculation

With 1 the pressure (X 42) and viscous stresses (X 38, X 43) are integrated over the surface triangles. With 2 the forces are computed from a triangulation reconstructed from the level set function; the surface refinement X 185 is then skipped. On NHFLOW curvilinear grids (G 50) only option 1 is available; otherwise the run stops.

1 STL

2 LSM

default: 1

● ● ● ●

X 101 double ϕ_0, double θ_0, double ψ_0 Initial Euler angles in [deg]
Rotates the body around its centre of gravity at the start. Positive angles rotate the body clockwise. Mooring end points on the body are rotated with it only if X 313 1.

default: 0.0 ; 0.0 ; 0.0

● ● ● ●

X 102 * double u_0, double v_0, double w_0 Initial linear floating body velocity in [m/s]
If given several times, the values are added.

default: 0.0 ; 0.0 ; 0.0

● ● ● ●

X 103 double p_0, double q_0, double r_0 Initial angular floating body velocity in [rad/s]
Angular velocities about the body-fixed axes; the initial angular momentum is computed with the inertia tensor of the body.

default: 0.0 ; 0.0 ; 0.0

● ● ● ●

X 110 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Rectangular box floating body in [m]
Several boxes can be combined into one body. The first box also defines length and width of the pressure field for X 400 2.

default: na

● ● ● ●

X 131 double r, double h, double x_c, double y_c, double z_c Cylinder in x-direction floating body
Radius r, length h and centre point (x_c, y_c, z_c) in [m]. The cylinder is triangulated with circumferential segments of about $0.75\,dx$ (at least 8).

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

● ● ● ●

X 132 double r, double h, double x_c, double y_c, double z_c Cylinder in y-direction floating body
Radius r, length h and centre point (x_c, y_c, z_c) in [m]. Triangulated as X 131.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

● ● ● ●

X 133 double r, double h, double x_c, double y_c, double z_c Cylinder in z-direction floating body
Radius r, height h and centre point (x_c, y_c, z_c) in [m]. Triangulated as X 131.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

● ● ● ●

X 153 double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Two-sided wedge in z-direction floating body in [m]

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

● ● ● ●

X 163 * double x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3, x_4, y_4, z_4, x_5, y_5, z_5, x_6, y_6, z_6 Wedge floating body
Each of the 6 corner points is given by its coordinates in [m], see the figure below.

default: na

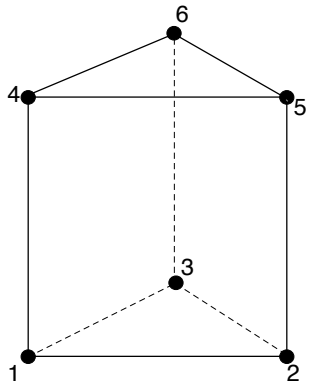

Figure 6.1: Definition of the wedge points for X 163

● ● ● ●

X 164 * double x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3, x_4, y_4, z_4, x_5, y_5, z_5, x_6, y_6, z_6, x_7, y_7, z_7, x_8, y_8, z_8
Hexahedron floating body
Each of the 8 corner points is given by its coordinates in [m], see the figure below.

default: na

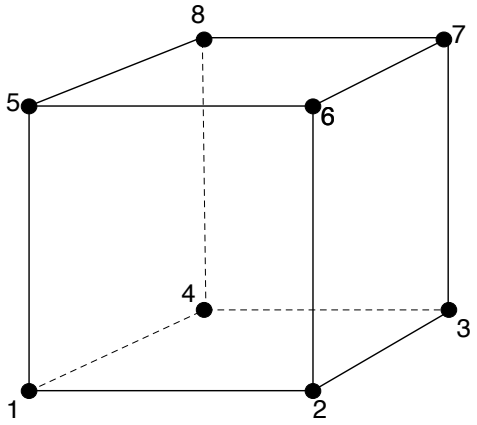

Figure 6.2: Definition of the hexahedron points for X 164

○ ○ ● ○

X 170 double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Piston wavemaker in [m]
Box-shaped piston moved with the displacement time series in `wavemaker.dat` (columns t [s], x [m]; lines with t smaller than B 117 are skipped, and the time is shifted by −B 117 and by X 241). The displacement and velocity are interpolated linearly to the Runge-Kutta stage times and scaled with B 118; before the first and after the last line the paddle rests. Requires X 10 4.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ● ○

X 171 double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{hinge}, double z_{end} Flap wavemaker in [m]
Intended for X 10 4; currently not implemented.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ● ○

X 172 double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{hinge1}, double z_{hinge2}, double z_{end} Double-hinged flap wavemaker in [m]
Requires X 10 4. The motion is read from `wavemaker.dat` (columns t [s], horizontal displacement of the lower and of the upper flap segment [m], or angles in [rad] with B 116 2); lines with t smaller than B 117 are skipped and the time is shifted by −B 117. The motion is interpolated to the Runge-Kutta stage times and scaled with B 118.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

● ● ● ●

X 180 int Read STL file `floating.stl` for floating body geometry

The file must be in ASCII STL format. With X 20 > 1, the further bodies are read from `floating-1.stl`, `floating-2.stl`, ...

0 OFF

1 ON

default: 0

● ● ● ●

X 181 double s_x, double s_y, double s_z Scale STL geometry in x-, y-, and z-direction

Scaling factors applied to the STL coordinates, before X 182 and X 183.

default: 1.0 ; 1.0 ; 1.0

● ● ● ●

X 182 double Δx, double Δy, double Δz 3D linear translation of STL object in [m]

Applied after the scaling X 181 and before the rotation X 183.

default: 0.0 ; 0.0 ; 0.0

● ● ● ●

X 183 double x_{origin}, double y_{origin}, double z_{origin}, double ϕ, double θ, double ψ 3D rotation of STL object

Rotation around the point $(x_{origin}, y_{origin}, z_{origin})$ in [m] by the angles in [deg], applied after X 181 and X 182. Positive angles rotate the body clockwise.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

● ● ● ●

X 184 int Hole filling for the surface remeshing

Treatment of open boundaries of the body surface with X 185 4. Gaps between nearly coincident edges are closed and holes are patched; if the surface was repaired, the volume and the mass (X 21) or density (X 22) are computed from the closed surface. With 0, a warning is printed if the surface is not closed.

0 OFF, holes are kept as open boundaries

1 close gaps and fill holes with a harmonic (flat, membrane-like) patch

2 close gaps and fill holes with a biharmonic (curvature-continuous) patch

default: 1

● ● ● ●

X 185 int Type of geometry refinement for triangulated surface mesh

Options 1 – 3 split surface triangles whose mean edge length exceeds the critical length (reference cell size times X 186) into four. Option 4 re-triangulates the whole surface with an adaptive isotropic remesher: the target edge length is X 186 times the local cell size of the fluid grid (X 190), sharp edges (X 187), corners and open boundaries are preserved, all vertices stay on the original surface and holes are closed (X 184). The remeshing is done on one process and broadcast, so the result does not depend on the domain decomposition. If the estimated number of triangles exceeds $2 \cdot 10^6$, the input triangles are kept (increase X 186). For sigma grids (NHFLOW, FNPF), the vertical cell size is taken from the still water column, and with mesh refinement around the body (G 1) the horizontal cell size is that of the finest refinement level; in CFD with mesh refinement (G 1, X 10 1) the cell size is divided by 2^{G1}. The refinement is skipped for X 60 2 and for the wavemakers X 170–X 172.

0 OFF

1 split, based on averaged dx

2 split, based on global minimum dx

3 split, based on local minimum dx

4 remeshing to the local grid spacing

default: 4

● ● ● ●

X 186 double Geometry refinement factor for triangulated surface mesh

Critical triangle size for X 185 1 – 3, or target edge length for X 185 4, relative to the reference cell size, i.e. in numbers of fluid cells.

default: 1.0

● ● ● ●

X 187 double Feature angle for the surface remeshing

Edges between surface triangles whose normals differ by more than this angle in degrees are treated as sharp features and preserved by the remeshing (X 185 4).

default: 30.0

○ ○ ○ ●

X 188 int Ray casting algorithm for the floating body level set

Method used to compute the distance function of the body from the surface triangles in CFD.

1 directional ray casting in x-, y- and z-direction

2 direct distance calculation

default: 1

● ● ● ●

X 189 int Number of iterations for the surface remeshing
Number of split, collapse, flip and relaxation cycles of the remeshing (X 185 4).

default: 10

● ● ● ●

X 190 int Cell size for the surface remeshing
Local cell size that is multiplied with X 186 to give the target edge length of the remeshing (X 185 4). Only the active grid directions are used (no y in 2D, no z for a single cell layer).

1 grid metric: the target length is set per direction, so that every edge spans about X 186 cells in each direction (triangles stretched like the cells)

2 geometric mean of the cell spacings dx, dy, dz (isotropic)

3 smallest cell spacing (isotropic)

default: 2

● ● ● ●

X 205 int Ramp up function type
Shape of the ramp functions X 206 and X 207.

1 linear

2 sine

default: 1

● ● ● ●

X 206 double t_s, double t_e External motion ramp up start time and end time in [s]
Ramps the prescribed velocities (X 210, X 211, X 240) from zero at t_s to full value at t_e; before t_s the prescribed velocities are zero.

default: not set (no ramp)

● ○ ● ○

X 207 double t_s, double t_e Draft ramp up start time and end time in [s]
Ramps the external pressure field of the ship-wave mode (X 10 3, X 400). Without X 207 the free surface is initialised with the depression of the full draft.

default: not set (no ramp)

● ● ● ●

X 210 double u, double v, double w Prescribed constant linear velocities of the floating body in [m/s]
Applied to the degrees of freedom set to 2 in X 11, ramped with X 206.

default: 0.0 ; 0.0 ; 0.0

● ● ● ●

X 211 double p, double q, double r Prescribed constant angular velocities of the floating body in [rad/s]
Applied to the degrees of freedom set to 2 in X 11, ramped with X 206.

default: 0.0 ; 0.0 ; 0.0

● ● ● ●

X 240 int Read motion file for 6DOF motions
The motion is read from `6DOF_motion.dat`; the velocities are computed from the differences between consecutive lines and applied to the degrees of freedom set to 2 in X 11 (ramped with X 206). CoG = course over ground (yaw angle) in [deg].

0 OFF

1 Translation format: t, x, y, z

11 Course over Ground format: t, x, y, CoG (acts on u, v and r)

21 Piston wavemaker format: t, x (acts on u)

default: 0

● ● ● ●

X 241 double Time shift for motion files in [s]
Added to the time column of `6DOF_motion.dat` (X 240) and of the piston `wavemaker.dat` (X 170).

default: 0.0

○ ● ● ●

X 310 int Turn on mooring modelling for floating bodies

The lines are defined with X 311 (options 1–3) or X 312 (option 4). Results are written to `REEF3D_*_6DOF_Mooring`. The state of the lines is included in the checkpoints (P 48, I 45), also without a floating body (X 10 0).

0 OFF

1 Static catenary

2 Quasi-static bar model

3 Dynamic model

4 Spring (using X 312)

default: 0

○ ● ● ●

X 311 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double w, double ρ_c, double EA, double d, double l, double H, double P, double Q Mooring line configuration

Each mooring line has to be specified separately with this command. $\vec{x}_{start}$ is the anchor point at the bottom and $\vec{x}_{end}$ the fairlead on the body, in [m]. w is the mass per unit length in air [kg/m], ρ_c the density of the line material [kg/m^3] (used for the submerged weight), EA the axial stiffness [N], d the line diameter [m] and l the unstretched length [m]. H is the number of elements for the numerical approaches. P and Q are read but currently not used. Do not combine with X 312.

default: na

○ ● ● ●

X 312 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double k, double T_0 Spring configuration

Each spring has to be specified separately with this command (X 310 4). $\vec{x}_{start}$ is the fixed point at the bottom and $\vec{x}_{end}$ the attachment point on the body, in [m]. k is the stiffness [N/m] and T_0 the pre-tension [N]. Do not combine with X 311.

default: na

○ ● ● ●

X 313 int Initial rotation of mooring end points with the floating body

If ON, the fairlead points $\vec{x}_{end}$ of X 311/X 312 are rotated with the initial body rotation X 101.

0 OFF

1 ON

default: 0

○ ● ● ●

X 314 double T_1, T_2, ... Breaking tension of the mooring lines in [N]
One value for each mooring line (in the order of X 311 or X 312), all on one line. A line breaks when its tension reaches the value. Set the value to zero to disable breaking for an individual line.

default: not set (no breaking)

○ ● ● ●

X 315 double t_1, t_2, ... Breaking time of the mooring lines in [s]
One value for each mooring line (in the order of X 311 or X 312), all on one line. A line breaks at the given simulation time. Set the value to zero to disable breaking for an individual line.

default: not set (no breaking)

○ ○ ● ●

X 320 * int Turn on net modelling
Each net has to be specified separately with this command, together with one X 321 and one X 322 per net. Nets are attached to the floating body, or fixed if X 10 is 0. Results are written to `REEF3D_*_6DOF_Net`. The state of the nets is included in the checkpoints (P 48, I 45). Not available on NHFLOW curvilinear grids (G 50); the run stops.

0 OFF

2 Static net cylinder

3 Static net wall

4 Static sheets following floating body (cylinder of diameter D and height L (X 322) with a bottom cone and bulkheads)

12 Dynamic net cylinder

13 Dynamic net wall

14 Dynamic net cone

default: 0

○ ○ ● ●

X 321 * double Sn, double d_t, double l_t, double d_k, double ρ, double N_d, double N_l Net material

Each net has to be specified separately with this command. Sn is the solidity, d_t the twine diameter [m], l_t the twine length [m], d_k the knot diameter [m] (0.0 if knotless) and ρ the density of the net material [kg/m^3]. The net is discretised with N_d meshes along D and N_l meshes along L (X 322).

default: na

○ ○ ● ●

X 322 * double D, double L, double x_0, double y_0, double z_0, double ϕ, double θ, double ψ Net dimensions

Each net has to be specified separately with this command; the lines are assigned to the nets in the order of the X 320 and X 321 lines, and their number must not exceed the number of X 321 lines (which sets the number of nets). D is the diameter or width of the net and L its height, in [m]. The structure is first rotated around the origin by the angles around the x-, y- and z-axis in [deg]. Then the centre point of the bottom of the structure is moved to the position (x_0, y_0, z_0) in [m].

default: na

○ ○ ● ●

X 323 double m_s, double d_s, double l_s Sinker for dynamic net models

Mass in air [kg], diameter [m] and length [m] of the sinkers, used for all dynamic nets (X 320 12–14).

default: 0.0 ; 0.0 ; 0.0

○ ○ ● ●

X 324 * double x, double y, double z Knot probes for dynamic net models

Location in [m]; the dynamic net model tracks the knot nearest to this point.

default: na

○ ○ ● ●

X 325 double Δt, double r_x, double r_y, double r_z Time step and relaxation for dynamic net models

Time step of the net solver in [s] and relaxation factors in x-, y- and z-direction. Set the time step to zero to use the time step of the fluid solver.

default: 0.001 ; 0.01 ; 0.01 ; 0.01

○ ○ ● ○

X 330 int Impermeable membranes (closed flexible fish cages)

Turns on impermeable membranes in NHFLOW, read from the file `membrane.dat` in the case directory (the program stops if it is missing or if a line cannot be read). The membrane is a triangulated surface that acts on the fluid as a thin anisotropic porous layer (porous jump) with a hydraulic resistance; the resistance is integrated implicitly and also enters the pressure Poisson equation, so that no flow passes the membrane through the projection. Use A 520 1 or 2; for moving membranes A 520 2 is changed to 1. The NHFLOW mesh refinement (G 1) is switched off with membranes; on curvilinear grids (G 50) and with the semi-implicit scheme (A 534 1) the run stops. Each membrane starts with a line `membrane box` x_0 x_1 y_0 y_1 z_{bottom} z_{top}, `membrane cylinder` x_c y_c R z_{bottom} z_{top} or `membrane cylcone` x_c y_c R z_{tip} z_{cone} z_{top} (cylinder on a conical floor) (in m), followed by optional parameter lines for this membrane (# starts a comment), e.g. `name`, `resistance` R_n [R_t] (normal and tangential resistance in m/s, default $R_n = 10^4$, R_t = 0 for a fixed and R_n for a moving membrane), `thickness` δ (half width of the layer in m, default $1.5\max(dx, dy, dz)$), `mesh` h (triangle edge length in m), `fill` Δh (initial inner water level above the outside level in m) or `filling` λ (filled fraction of the bag volume below the still water level, optionally pumped out over `drain` T in s), `mobility layer|link|sharp` (pressure coupling through the smeared porous layer (default), through the cell links across the membrane only, or with sharply blocked links and no layer forcing), `floorpressure`, `tau`, `projections`, `poisson`, `print` dt (VTP output interval in s, default P 20/P 30, 0 off), and `structure fixed|rigid|flexible`. A fixed membrane stays in place; a rigid one moves with the floating body (X 10); a flexible one is a mass-spring membrane (`mass` in kg/m^2, `density` in kg/m^3, `stiffness` Et in N/m, `damping` ratio, `compression` stiffness fraction, `bending` stiffness EI, `smoothing`, `sinker` weight in N/m along the floor edge, `attach` z) whose top edge follows the floating body like the nets, or stays in place without X 10. In 3D, the top edge of a flexible membrane without floating body can instead be a floating pipe ring, `collar` D m EA EI [C_d [C_a]], moored with linear springs `mooring` x_a y_a z_a k T_0 (one line per spring). The membrane loads act on the floating body as external forces, stabilised with an added mass (`bodyaddedmass`). The fluid-structure coupling of a flexible membrane is iterated in every Runge-Kutta stage by default (`coupling` iterated|staggered, with further `coupling...` keywords for the tolerances and the quasi-Newton acceleration; `addedmass` for the staggered coupling with `mobility sharp`). Time series per membrane (inner and outer water level, leakage discharge, forces, maximum tension) are written to `REEF3D_NHFLOW_Membrane`, the membrane geometry to `REEF3D_NHFLOW_Membrane_VTP`.

0 OFF

1 ON, membranes from `membrane.dat`

default: 0

○ ● ● ●

X 350 int Ship module

Adds semi-empirical hull loads, propulsion and steering to the two-way coupled floating body (X 10 1) in CFD, NHFLOW and FNPF. The parameters are read from the file `ship.dat` in the case directory (one keyword per line, # starts a comment, all keywords optional; without the file the defaults are used). The hull is the 6DOF surface triangulation; the ship frame is the body frame (x forward, y to port, z up at the start), headings are counter-clockwise from x, angles are given in degrees. The keywords are:

- hull: `lpp` L (length for the Reynolds number in m, default waterline length), `wetted_surface` S (m^2, default hull below the still water level), `form_factor` k, `friction` 0|1 (ITTC-1957 frictional resistance $-\frac{1}{2}\rho S(1+k)C_F|u|u$; default 1, 0 for CFD and with X 38 1), `viscosity` ν (default W 2), `roll_damping` B_{44} B_{44q} (linear and quadratic roll damping), `crossflow` C_d (cross-flow drag over `strips` n draft strips, default 40), `thrust` T [x z] (constant thrust along the ship x-axis in N), `current` U_x U_y (uniform current in m/s; the models use the velocity relative to it), `approach` U t_r t_f (the speed relative to the water is ramped smoothly from 0 to U over t_r and held by a controller until t_f in s, with sway and yaw held at zero and the rudder at 0; the ship is free afterwards).
- propeller (body-force actuator disk with the Hough-Ordway radial distribution of thrust and torque): `propeller` x y z D (disk centre relative to the centre of gravity in the ship frame and diameter in m), `propeller_hub` (hub/tip radius ratio, default 0.2), `propeller_kt` and `propeller_kq` (coefficients of the quadratic open-water curves $K_T(J)$ and $K_Q(J)$), `propeller_rps` n (1/s), `propeller_sense` ±1 (right- or left-handed), `propeller_thickness` (axial extent of the disk in m, default $\max(0.2D, 4\,dx)$), `propeller_inflow` `wake` w (advance velocity $(1-w)u$, default) or `sample` d (relative fluid velocity on rings d m ahead of the disk), `propeller_source` 0|1 (1: the thrust and torque act on the fluid as a momentum source, NHFLOW and CFD only, default there; 0: the thrust $(1-t)T$ acts on the hull only, with `thrust_deduction` t).
- rudder (MMG standard method): `rudder` x_R z_R A_R Λ (position, area, aspect ratio), `rudder_mmg` t_R a_H x_H ε κ l_R γ_R (interaction coefficients, x_H and l_R in units of L), `rudder_angle` `fixed` δ | `zigzag` δ ψ [t_0] | `autopilot` ψ K_p K_d [K_i], `rudder_rate` (default 15 deg/s), `rudder_max` (default 35 deg), `rudder_falpha`, `rudder_gamma`.
- MMG manoeuvring model (Yasukawa & Yoshimura 2015): `mmg_hull` (17 hull derivatives R_0, X_{vv}, ..., N_{rrr} about midship), `mmg_added_mass` m_x m_y J_z (solved implicitly with the body), `mmg_fluid` 0|1|2 (0: the surge, sway and yaw loads of the flow solver are replaced by the MMG model, 1: both are added (default), 2: hybrid with FNPF, the solver supplies the ideal-fluid loads and the Munk moment is taken out of the MMG hull moment), `hybrid_munk` $A_{22}-A_{11}$ (kg, the Munk moment taken out with `mmg_fluid` 2; default m_y-m_x of `mmg_added_mass`), `hybrid_surge` 0|1 (1: with `mmg_fluid` 2 the MMG surge velocity term $m_y v_m r - m_x v r$ is added, for grids that do not resolve the speed loss in a turn; default 0), `mmg_draft` d, `propeller_wake_mmg` C_1 C_{2+} C_{2-} x_P.

The loads are written every X 19 time steps to `REEF3D_*_6DOF/REEF3D_ship_<n>.dat`.

0 OFF

1 ON, parameters from `ship.dat`

default: 0

● ○ ● ○

X 400 int External moving pressure field for the ship-wave mode

Type of pressure field representing the body in X 10 3. The field follows the body motion (X 11, X 210, X 211, X 240) and is ramped with X 207.

0 OFF

2 Box: $p = -p_0\left(1 - c_l(x/L)^4\right)\left(1 - c_b(y/B)^2\right)\exp\left(-a(y/B)^2\right)$ inside the box, with length L and width B from the first X 110 and x, y relative to the centre of gravity

3 1D Gaussian: $p = -p_0\exp\left(-(x/a)^2\right)$ (same sign convention as options 2 and 10), with x relative to the centre of gravity and a in [m]

10 STL: $p = -\rho g \cdot draft$ from the body geometry (see X 410)

default: 0

● ○ ● ○

X 401 double p_0, double c_l, double c_b, double a Coefficients for the external moving pressure field

Pressure amplitude p_0 in [Pa] and shape coefficients for X 400 2 and 3 (see there).

default: 0.0 ; 2.0 ; 16.0 ; 16.0

● ○ ● ○

X 410 int Add local free surface for draft calculation

For X 400 10: adds the free surface elevation sampled just outside the body edge (offset X 41 cells) to the draft.

0 OFF

1 ON

default: 0

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X 500 int Power take-off (PTO)

Turns on a power take-off between the floating body and the ground, e.g. for a heaving point absorber, in FNPF and NHFLOW (in NHFLOW it acts on the motion only for X 10 1). The PTO is a prismatic joint (X 501) with the PTO coordinate q in m ($q = 0$ at the initial position) and its velocity $\dot{q}$, on which the elements X 502 – X 504 and the latch X 507 act; their forces are summed, and the resulting force F

acts along the joint axis at the attachment point (force and moment on the body). The generator part of the force (dampers and spring-dampers) can be limited (X 505) and controlled (X 506 – X 508); drive-train losses are added with X 510. The degrees of freedom of the body are set as usual with X 11. The time series of q, $\dot{q}$, the total PTO force, the absorbed power $P = -F\dot{q}$ of the generator, the absorbed energy, the generator gain (0 while declutched), the latch state, the saturation state of the limits (0 free, 1 force limit, 2 power limit), the force of each element and, with X 510, the dissipated power and energy of the drive-train losses are written to `REEF3D_FNPF_6DOF/REEF3D_6DOF_pto_<n>.dat` (NHFLOW: `REEF3D_NHFLOW_6DOF`). With the explicit coupling a warning is printed when the PTO damping or stiffness reaches the stability limit of the time step. When the body follows the dynamics of REEF3D::CORAL (Z 40), X 500 is not used (the PTO belongs to the CORAL model). The state of the PTO is included in the checkpoints (P 48, I 45), and its time series file is continued at a restart.

0 OFF

1 explicit coupling: the PTO force is added to the external forces like the mooring forces

2 linearly implicit coupling of the stiff elements (end-stops X 504, latch X 507) together with the added mass (FNPF; NHFLOW without added mass); dampers and springs stay explicit

default: 0

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X 501 double a_x, double a_y, double a_z, double x_a, double y_a, double z_a PTO joint axis and attachment point

Direction (a_x, a_y, a_z) of the joint axis, fixed in the inertial frame (it is normalised; a zero vector stops the program), and the attachment point (x_a, y_a, z_a) on the body in [m] at the start of the simulation. If X 501 is not given, the PTO acts in heave, $(0, 0, 1)$, at the centre of gravity.

default: 0.0 ; 0.0 ; 1.0 ; centre of gravity

○ ● ● ○

X 502 double B PTO linear damper

Adds a linear damper $F = -B\,\dot{q}$ with the damping coefficient B in [N s/m].

default: not set; 0.0

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X 503 double K, double B PTO spring-damper

Adds a linear spring-damper $F = -K\,q - B\,\dot{q}$ with the stiffness K in [N/m] and the damping coefficient B in [N s/m].

default: not set; 0.0 ; 0.0

○ ● ● ○

X 504 double q_{min}, double q_{max}, double K, double C PTO end-stops

Limits the stroke of the PTO to $q_{min} \le q \le q_{max}$ in [m]. Beyond a limit, a penalty force $F = -K\,(q - q_{lim}) - C\,\dot{q}$ with the end-stop stiffness K in [N/m] and damping C in [N s/m] acts; the force only pushes the body back into the stroke (it is set to zero when the damping part would pull). The end-stop force does not count as absorbed power. Stiff end-stops may require X 500 2.

default: not set; -10^{20} ; 10^{20} ; 0.0 ; 0.0

○ ● ● ○

X 505 double F_{max}, double P_{max} PTO generator force and power limits

Limits the generator force, i.e. the sum of the dampers and spring-dampers (X 502, X 503, X 506) after the declutching gain, to $|F| \le F_{max}$ in [N] and then to $|F\,\dot{q}| \le P_{max}$ in [W]. The end-stops, the latch and the drive-train losses (X 510) are not limited. A value of 0.0 applies no limit.

default: not set; 0.0 ; 0.0

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X 506 double T, double C PTO tuned passive damping

Sets the damping of the linear damper X 502 (or adds one) once at the start to the optimal passive damping of a single-degree-of-freedom absorber at the period T in [s], $B_{opt} = \sqrt{(B_{rad} + B_{loss})^2 + (\omega(m + A) - C/\omega)^2}$ with the loss damping B_{loss} of X 510 (0 without), $\omega = 2\pi/T$, the effective rigid-body mass m of the body along the joint and the hydrostatic stiffness C along the joint in [N/m] (e.g. $\rho g A_{wp}$ for a heave joint). The added mass A in [kg] and the radiation damping B_{rad} in [kg/s] along the joint are interpolated linearly in the table `pto_hydro.dat` in the case directory (columns ω [rad/s], A, B_{rad}; blank, tab or comma separated, lines not starting with a number are skipped, at least two lines; outside the table the end values are used). With $T \le 0$ the wave period of the wave input (B 93, or from B 91; peak period for irregular waves) is used. The run stops if no period is available, if $C \le 0$ or if `pto_hydro.dat` is missing.

default: not set; 0.0 ; 0.0

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X 507 double t_{hold}, double K, double C PTO latching

Latching control: at every zero crossing of the PTO velocity $\dot{q}$, the body is held at its position for the time t_{hold} in [s] by a stiff latch, $F = -K\,(q - q_0) - C\,\dot{q}$ with the latch stiffness K in [N/m] and damping C in [N s/m]. The time of the zero crossing is

interpolated within the time step. The latch force does not count as absorbed power. X 500 2 is recommended (a message is printed with X 500 1).

default: not set; 0.0 ; 0.0 ; 0.0

○ ● ● ○

X 508 double t_{off} PTO declutching

Declutching control: at every zero crossing of the PTO velocity $\dot{q}$, the generator (dampers and spring-dampers) is switched off for the time t_{off} in [s]; the end-stops stay active.

default: not set; 0.0

○ ● ● ○

X 510 double B, double F_c, double v_ε PTO drive-train losses

Adds the losses of the drive train (belt, bearings, seals) to the PTO (X 500): a linear loss damper $F = -B\,\dot{q}$ with B in [N s/m] and a regularised Coulomb friction $F = -F_c \tanh(\dot{q}/v_\varepsilon)$ with F_c in [N] and the regularisation velocity v_ε in [m/s] (≤ 0: 10^{-3} m/s). The losses act on the body in every state (also latched or declutched), are not limited by X 505, and their power is dissipated, not absorbed: it is written as the loss power and energy to the PTO time series. A value of 0.0 omits the element; negative B or F_c stop the run. X 506 tunes the damping with B in parallel to the radiation damping.

default: not set; 0.0 ; 0.0 ; 0.0

6.19 Z :: FSI

The Z commands control the coupling of deformable structures with the flow: flexible strips (Z 10 – Z 12, CFD), flexible rod trees (Z 20, Z 21, CFD and NHFLOW, input file `rodtree.dat`), finite element solids (Z 30, Z 31, CFD and NHFLOW) and the structural solver REEF3D::CORAL (Z 40, CFD, NHFLOW and FNPF, input file `coral.txt`, Section 5.12). The FEM module computes elastic, plastic and concrete solids (optionally reinforced) and rigid bodies on a voxel lattice of 8-node hexahedra with explicit time integration (subcycling inside the fluid time step), including cracking, element erosion, debris and contact with the ground and between fragments. It needs its own input file `fem.dat` in the case directory (one keyword per line, # starts a comment, SI units), with the main keywords `lattice` h_x h_y h_z [o_x o_y o_z] (element size; without it the element size is `resolution coarse|normal|fine` or a factor times the smallest fluid cell, default 1), `material` [id] followed by a preset (`concrete` C20–C50, `steel` S235–S460, `aluminium 6061`, `timber` C16|C24|C30|GL24h|GL28h, `rock granite|basalt|limestone|sandstone`, `masonry brick`, `rubber`) or by the numbers of the type (`elastic` ρ E ν | `plastic` ρ E ν σ_y H ε_{fail} | `concrete` ρ E ν f_t G_f f_c G_c [d_{erode}] | `rigid` ρ), optionally followed by key-value overrides and the flag `rigid`, `rebar` [preset]

`x|y|z` fraction (reinforcement of the last concrete material), `box` x_0 x_1 y_0 y_1 z_0 z_1 [id], `stl` file [id] [`move` dx dy dz] [`rotate` deg] [`scale` s], `remove` / `remove_stl` (openings), `fix` box [xyz] | `base` | `top` | `bed` (supports), `ground` z | `bed` [k_{fac} μ], `monitor` name x y z | `auto`, `loads` `hybrid|pressure|reaction`, `fragments` `rigid|deformable`, `check` (writes a check report of the model and stops) and `print` dt, and further numerical keywords (e.g. `element adaptive|full|reduced`, `cfl`, `threads`, `distribute auto|on|off` (solid split over the MPI ranks), `strong on|off` (strong coupling of rigid bodies with NHFLOW, default off), `damping`, `relax`, `bulk_viscosity`, `erode_J`, `contact`, `gravity`, `settle`, `snap`, `walls`, `shear`, `added_mass`, `points_per_face`, `forcing`, `debris_cd`, `debris_damping`). Gravity is taken from W 20 – W 22 and the fluid density from W 1; 2D runs are computed in plane strain with one voxel layer in y. The flow sees the solid through a direct forcing on its surface; the fluid loads are transferred to the solid nodes. The results are written to the folder `REEF3D_FEM` (VTU files of the deformed solid with displacement, velocity, load, von Mises stress, damage, plastic strain and utilisation, a log file and the monitor time series).

Valid for:

S	F	N	C
○	○	○	●

Z 10 int Turn on FSI algorithm for flexible bodies
Log files are written with the interval X 19.

0 OFF

1 Strips

default: 0

○	○	○	●

Z 11 * double x, double y, double z, double L, double W, double T, double ρ, double E, double I_x, double I_y, double I_z, double ν, double N Flexible strip definition

Each strip has to be specified separately with this command. (x, y, z) is the initial position of the central bottom point of the strip in [m]. L is the length, W the width and T the thickness of the strip in [m] ($L \gg W \gg T$ must hold). ρ is the density of the material [kg/m^3], E the Young modulus [N/m^2], (I_x, I_y, I_z) are the second moments of area [m^4] and ν is the Poisson ratio. N is the number of elements per strip.

default: na

○	○	○	●

Z 12 double $c_{d,X}$, double $c_{d,Y}$, double $c_{d,Z}$, double $c_{k,X}$, double $c_{k,Y}$, double $c_{k,Z}$ Structural damping of the strips

Velocity-related viscous damping in the local X, Y, Z directions of the beam, applied to all strips. $\vec{c}_d$ acts on the translational motions, $\vec{c}_k$ on the rotational motions.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

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Z 20 int Flexible rod trees

Branched flexible rods (soft corals, gorgonians, flexible vegetation) that are thinner than the grid cells, coupled to the flow in an unresolved way (actuator-line type): the fluid velocity is sampled at Lagrangian points along each rod element, Morison-type loads (normal and tangential drag, Froude-Krylov and added-mass inertia, buoyancy and gravity, scaled with the submerged fraction) act on the elements, and the reaction is spread back to the fluid with a 4-point Peskin kernel as a body force. The structure is a discrete Cosserat rod of rigid cylinder elements connected by elastic joints, with the roots clamped to the bed; it is advanced once per time step in the final Runge-Kutta stage. Independent of Z 10. The colonies are read from the file `rodtree.dat` in the case directory (the program stops if it is missing or invalid; # starts a comment). Global keywords: `integrator` implicit|explicit, `substeps` n, `mode` dynamic|quasistatic|rigid, `solver` tree|sparse, `reaction` full|drag|none. Each colony is a block `colony` [name] ... `end` with `node` id x y z r (m), `edge` parent child, `clamp` node, `material` E (Pa) ν ρ (kg/m^3), `hydro` C_{dn} C_{dt} C_a [C_m, default $1 + C_a$], `damping` β (s), `refine` n, and optionally `instance` Δx Δy Δz (copies of the colony), `represent` weight spacing (representative colonies for canopies), `polyps` h_p ϕ_p $C_{d,p}$ [k_t] and `polyp_response` U_r ΔU_r τ_p. Per colony, the tip displacement, the base force, the hydrodynamic force, the number of submerged elements and the polyp extension are written every time step to `REEF3D_RodTree/REEF3D_RodTree_<name>.dat`.

0 OFF

1 rod trees from `rodtree.dat`, unresolved coupling

default: 0

○ ○ ● ●

Z 21 double Print interval for the rod trees

Interval in s at which the rod-tree geometry is written as VTP files (`REEF3D_RodTree/REEF3D-RodTree-<n>.vtp`, Z 20 1). A value ≤ 0 writes no VTP files.

default: 0.0 (off)

○ ○ ● ●

Z 30 int FEM solid structures

Deformable, failing and collapsing solid structures and rigid debris computed with the finite element module and two-way coupled with CFD or NHFLOW, read from `fem.dat` (see the introduction of this section; the program stops if the file is missing). In every Runge-Kutta stage the surface velocity of the solid is imposed on the fluid by direct forcing (spread with a 3-point kernel, as for the strips), and in the final stage the fluid loads are transferred to the solid, which is then advanced over the fluid time step with subcycling. The coupling is staggered (in NHFLOW, `strong on` in `fem.dat` repeats the fluid step until the motion of the rigid bodies has converged); deformable materials lighter than about 1.1 times the fluid density can become unstable (a warning is printed),

floating debris should use a rigid material. In NHFLOW the cells inside the structure are solid and every water column reaches from the bed to the free surface, so a structure standing out of the water cannot be overtopped (use CFD for that); the NHFLOW mesh refinement (G 1) is switched off. Not available on NHFLOW curvilinear grids (G 50) and with the semi-implicit scheme (A 534 1); the run stops.

0 OFF

1 FEM solids from `fem.dat`, coupled with CFD or NHFLOW

default: 0

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Z 31 double Print interval for the FEM solids

Interval in s at which the deformed solid is written as VTU files (`REEF3D_FEM/REEF3D-FEM-<n>.vtu`, Z 30 1). A value ≤ 0 writes no VTU files. The keyword `print` in `fem.dat` overrides Z 31.

default: 0.0 (off)

○ ● ● ●

Z 40 int REEF3D::CORAL structural solver

Runs the CORAL model of the file `coral.txt` in the case directory together with the fluid solver (CFD, NHFLOW or FNPF), advanced in step with the fluid time; the program stops if the file is missing or invalid. CORAL holds rigid and flexible bodies, joints, mooring lines, power take-offs, controllers and engineering hydrodynamic and aerodynamic loads (Section 5.12). With `fluid_body` in `coral.txt`, the 6DOF bodies of REEF3D (X 10 1) follow the dynamics of the CORAL model, which receives their fluid loads; their moorings (X 310) and PTO (X 500) are then not used. With `one_way` or `fluid_member`, the loads of a fixed or prescribed 6DOF body act on a CORAL body or are distributed along a flexible member, and with `fluid_kinematics` (FNPF, NHFLOW) CORAL samples the fluid velocity and free surface (with `bathymetry fluid` also the bed under the dynamic mooring lines). The time series are written to `REEF3D_CORAL`. The state of the model is included in the checkpoints (P 48, I 45).

0 OFF

1 ON, CORAL model from `coral.txt`

default: 0

7. Compiling the Code

REEF3D is written in C++20 and has the following dependencies:

- a C++ compiler with C++20 support (e.g. GCC or Clang), used through the MPI compiler wrapper `mpicxx`,
- MPI for the parallelization (OpenMPI is recommended),
- optionally HYPRE, only for the HYPRE solvers N 10 10–39, which are intended for benchmarking. The default build has no HYPRE dependency; the pressure and Poisson equations are then solved with the built-in geometric multigrid solver REEFMG (N 10 1), and N 10 10–39 fall back to REEFMG,
- Eigen 5.0.0 (linear algebra, used by the 6DOF, mooring, net and beam modules). Eigen is a header-only library that is shipped with the REEF3D source code (folder `ThirdParty/eigen-5.0.0`), so it does not need to be installed separately.

DIVEMesh only requires a C++20 compiler (g++); it does not use MPI or HYPRE. Neither REEF3D nor DIVEMesh contain Fortran code, so a Fortran compiler (gfortran) is not needed for them. It is only used by the optional HYPRE library for its Fortran interface; HYPRE can be configured without it (Section 7.1.4).

The recommended route is:

1. install the dependencies, preferably with the package manager of the system (Section 7.1) or from source (Section 7.1.3; HYPRE, Section 7.1.4, only if the HYPRE solvers are needed),
2. download the source code (Section 7.2),
3. compile DIVEMesh and REEF3D with the Makefiles (Section 7.3) or with CMake (Section 7.4).

Windows users can work in the Windows Subsystem for Linux (Section 7.6). Alternatively, Docker can be used on all operating systems (Section 7.7). Developers can set up the CodeLite IDE as described in Section 7.5.

7.1 Install Required Packages

7.1.1 Ubuntu and Debian

Compilers, build tools and OpenMPI are installed with:
sudo apt update
sudo apt install build-essential git cmake openmpi-bin libopenmpi-dev

This is all that is needed for the default build. HYPRE is optional and only required for the HYPRE solvers N 10 10–39 (benchmarking, see Section 7.1.4). It is available as a distribution package as well:
sudo apt install libhypre-dev

For a build with HYPRE, both the Makefile and CMake expect the HYPRE header files in `$HYPRE_DIR/include` and the library in `$HYPRE_DIR/lib` (default `HYPRE_DIR=/usr/local/hypre`). If the distribution package installs HYPRE in a different layout, either adapt the include and library paths in the Makefile, or build HYPRE from source into `/usr/local/hypre` as described in Section 7.1.4. Building HYPRE from source is also what the Docker image does and is the most robust option.

7.1.2 macOS

- Install the Apple command line tools, which include the C and C++ compilers:
  xcode-select --install

  If problems occur in the compilation although the command line tools are installed, a possible solution is to remove and reinstall them. The command to remove the command line tools is:
  sudo rm -rf /Library/Developer/CommandLineTools

- With Homebrew (https://brew.sh), OpenMPI and CMake are installed with:
  brew install open-mpi cmake

  HYPRE is optional and only needed for the HYPRE solvers N 10 10–39 (Section 7.1.4); it is installed with **brew install hypre**. The HYPRE installation directory is then given by **brew --prefix hypre**; pass it to the build as shown in Sections 7.3 and 7.4.

- Confirm that the compiler wrapper works (in a new terminal):
  mpicxx --version

7.1.3 Installing OpenMPI from source

REEF3D depends on MPI for the parallelization of the code. MPI is very portable and the de-facto standard for the parallelization of high-performance computations; see https://www.mpi-forum.org/. We recommend OpenMPI for macOS and Linux. On HPC clusters, use the MPI installation provided by the cluster (usually loaded with `module load`). If no package is available, OpenMPI can be compiled from source:

- Download the source code from https://www.open-mpi.org/software/ and unpack the tar.gz archive.

- Open a terminal, change into the unpacked folder and type:
  ./configure --prefix=/usr/local/openmpi

- Compile:
  make -j n all (with n being the number of available cores on your computer)

- Install:
  sudo make install
- Add the OpenMPI executables to the PATH and the libraries to the library path, e.g. by adding the following lines to the file ∼/.bashrc (or ∼/.zshrc on macOS):
  export PATH=/usr/local/openmpi/bin:$PATH
  export LD_LIBRARY_PATH=/usr/local/openmpi/lib:$LD_LIBRARY_PATH
  The file .bashrc is a hidden file in your home folder; **ls -a** shows it.
- Open a new terminal and check the installation with **which mpicxx** and **mpirun --version**.

7.1.4 Installing HYPRE from source (optional)

HYPRE Cen [2015], Falgout et al. [2006] is a library for high-performance linear solvers and preconditioners. HYPRE is optional: it is only needed for the HYPRE solvers N 10 10–39, which are kept for benchmarking, and only when REEF3D is compiled with **make HYPRE=1** (Section 7.3) or -DREEF3D_USE_HYPRE=ON (Section 7.4). The default build does not use HYPRE, and this section can be skipped. HYPRE can be downloaded from the GitHub repository: https://github.com/hypre-space/hypre/releases. HYPRE has to be compiled with the same MPI installation that is used for REEF3D.

- Download the source code and unpack the archive.
- Open a terminal, change into the src folder of the HYPRE archive and type:
  ./configure --prefix=/usr/local/hypre
  If no Fortran compiler is installed, add --**disable-fortran**.
- Compile:
  make -j n all (with n the number of available cores)
- Install:
  sudo make install

The installation creates the folders /usr/local/hypre/include and /usr/local/hypre/lib, which is the default location used by the REEF3D Makefile and CMake for a build with HYPRE. If HYPRE is built as a shared library, add /usr/local/hypre/lib to LD_LIBRARY_PATH in the same way as for OpenMPI.

7.2 Getting the Source Code

The source code of REEF3D and DIVEMesh is downloaded from GitHub with:
git clone https://github.com/REEF3D/REEF3D.git
git clone https://github.com/REEF3D/DIVEMesh.git

Alternatively, the source code can be downloaded as archives from www.reef3d.com or the GitHub release pages. To update an existing clone, type **git pull** inside the folder and recompile.

7.3 User Compilation & Installation: Makefile

For users who are interested in running the code only, the easiest way is the compilation of REEF3D and DIVEMesh with the Makefiles given in the code repositories. Open the terminal and go to the main directory of each code.

- For DIVEMesh, type:
  make -j n (with n the number of available cores)
  The executable is `bin/DiveMESH`. With **make -j n OPENMP=1**, DIVEMesh is compiled with OpenMP.

- For REEF3D, type:
  make -j n
  The executable is `bin/REEF3D`; the object files are placed in the `build` folder.

The plain **make** builds REEF3D without HYPRE; the pressure and Poisson equations are solved with REEFMG (N 10 1), and the HYPRE solver options N 10 10–39 fall back to REEFMG. The HYPRE solvers are only intended for benchmarking. To compile them, HYPRE has to be installed (Section 7.1.4) and the build is started with:
make -j n HYPRE=1
In this case the object files are placed in the folder `build_hypre`, and the HYPRE library is expected in `/usr/local/hypre`. If HYPRE is installed elsewhere, the path is given on the command line, e.g. with Homebrew:
make -j n HYPRE=1 HYPRE_DIR=$(brew --prefix hypre)

The REEF3D Makefile offers the following targets:

release (default) optimized build with `-O3 -march=native -flto`. Because of `-march=native`, the executable is optimized for the processor of the computer on which it is compiled and may not run on older or different processor types. On clusters, compile on the type of node the simulations run on.

all optimized build (`-O3`) without processor-specific optimization.

dev optimized build with extensive compiler warnings.

debug build without optimization and with debug symbols (`-O0 -g`).

clean removes the `build` and `bin` folders.

For example, **make -j 8 debug**. Run **make clean** before switching between targets. In case of errors during the compilation, check that `mpicxx` is found and, for a build with `HYPRE=1`, that `HYPRE_DIR` is correct.

7.4 User Compilation & Installation: CMake

REEF3D can also be compiled with CMake (version 3.21 or newer). In the main directory of REEF3D, type:
cmake -S . -B build-cmake -DCMAKE_BUILD_TYPE=Release
cmake --build build-cmake -j n

By default, REEF3D is built without HYPRE (Poisson solves with REEFMG, N 10 1). To build the HYPRE solvers N 10 10–39 for benchmarking, add `-DREEF3D_USE_HYPRE=ON`, e.g.:
**cmake -S . -B build-cmake -DCMAKE_BUILD_TYPE=Release **
-DREEF3D_USE_HYPRE=ON -DHYPRE_DIR=/usr/local/hypre

The executable is written to `bin/REEF3D`, as with the Makefile. A separate build folder such as `build-cmake` avoids mixing the CMake files with the object files of the Makefile in `build`. The following options are available:

CMAKE_BUILD_TYPE `Release` (default), `Debug`, `Dev` or `All`; these correspond to the Makefile targets release, debug, dev and all.

REEF3D_USE_HYPRE `OFF` (default) or `ON`; builds the HYPRE solvers N 10 10–39 (benchmarking only). Without it, N 10 10–39 fall back to REEFMG (N 10 1).

HYPRE_DIR only used with `REEF3D_USE_HYPRE=ON`: HYPRE installation prefix (default `/usr/local/hypre`). If no HYPRE library is found in `$HYPRE_DIR/lib`, CMake downloads HYPRE 3.2.0 from GitHub and builds it as part of REEF3D. This requires git and internet access.

MPI is located automatically by CMake and has to be installed beforehand. Eigen is taken from `ThirdParty/eigen-5.0.0`; if that folder is missing, an installed Eigen 5.0.0 is used or it is downloaded.

7.5 Developer Compilation & Installation: CodeLite

REEF3D is written in highly modular C++ in order to provide efficient code development and maintenance. REEF3D is designed with cross-platform usage in mind. The source code can be compiled under macOS, Windows (WSL) and Linux. Since the user experience should be the same on all platforms when developing the code, CodeLite was chosen for the Integrated Development Environment (IDE). The reasons for this are:

- it is free under the GPL license
- it is cross-platform and works under macOS, Windows and Linux
- it works with a variety of compilers
- it works with the MPI library
- support for parallel builds

CodeLite can be downloaded from https://codelite.org. The REEF3D source contains a CodeLite project file (`src/REEF3D.project`), so all header and source files are sorted in a tree structure. This should make the orientation and navigation in the source code easier.

The following settings are needed in CodeLite to compile REEF3D. First, go to Settings > Build Settings and add a new compiler named mpicxx. The remaining settings are shown in the following screenshots. Check the location of your compilers with **which gcc**, **which g++** and **which mpicxx** to enter the appropriate paths. The compiler options have to include `-std=c++20`; compare with the flags in the Makefile if in doubt.

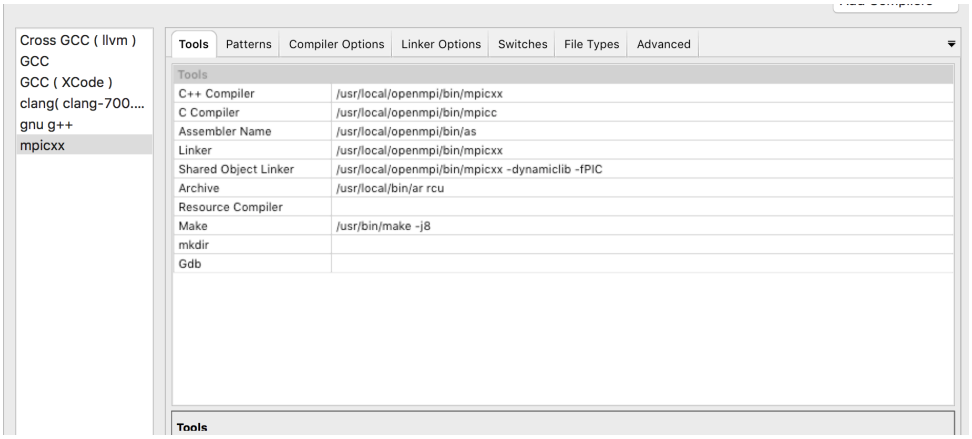


Figure 7.1: Setup of the mpicxx compiler in CodeLite.

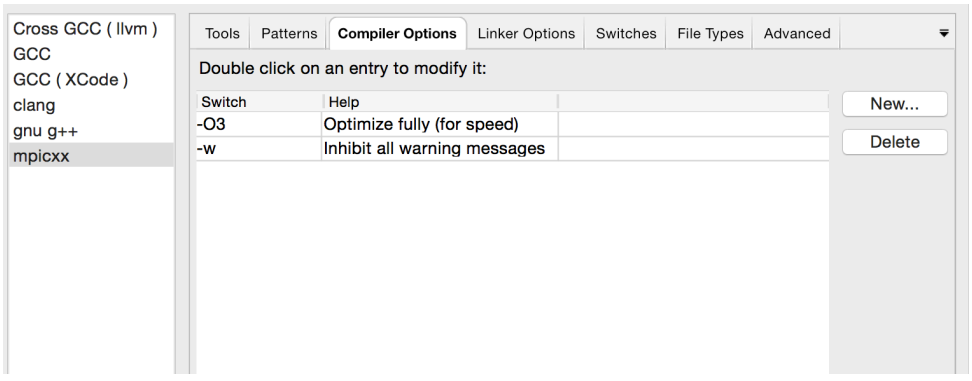


Figure 7.2: Compiler options for mpicxx.

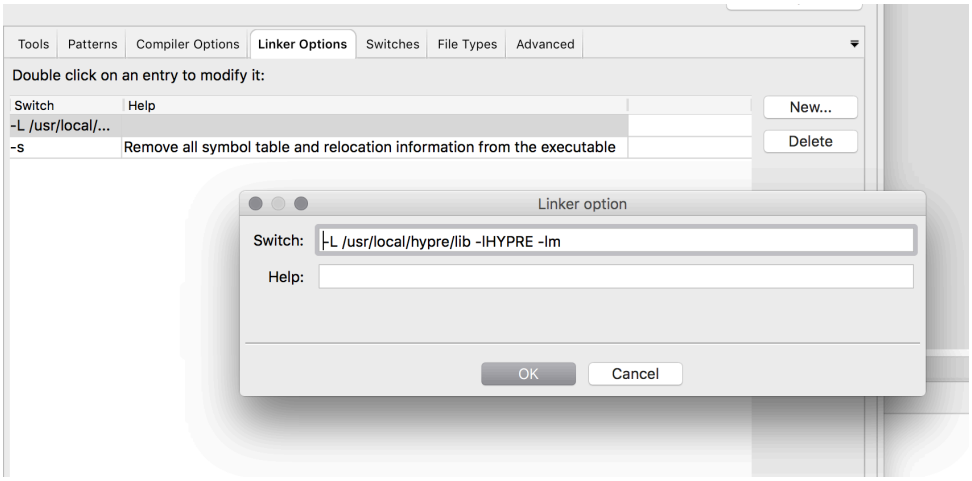


Figure 7.3: Linker options for mpicxx, link to HYPRE (only for a build with HYPRE).

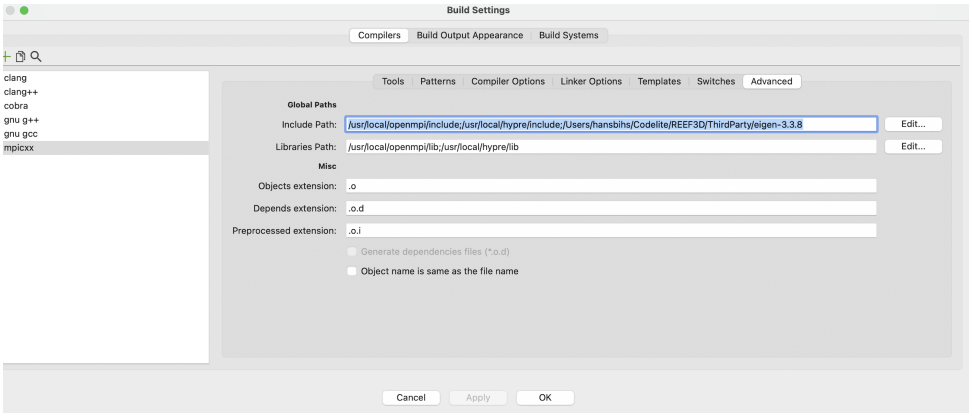


Figure 7.4: Add the include and library paths in the advanced tab of the mpicxx compiler: the HYPRE include and lib folders, and the Eigen headers shipped with REEF3D in `ThirdParty/eigen-5.0.0`. Adapt the paths to the file structure of the computer. The HYPRE folders are only needed for a build with HYPRE.

Afterwards, the code can be compiled by clicking Build > Build Project or pressing F7.

7.6 Installing REEF3D on Windows (WSL)

On Windows 10 (version 2004 or newer) and Windows 11, REEF3D is compiled and run in the Windows Subsystem for Linux (WSL 2), which provides a full Ubuntu installation.

- Open PowerShell as administrator and type:
  wsl --install
  This installs WSL 2 with Ubuntu as the default distribution. Restart the computer when prompted.
- Start Ubuntu from the Start menu. At the first start, you are asked for a user name and password for the Linux system; these are independent of your Windows credentials.

- Install the dependencies and compile DIVEMesh and REEF3D exactly as described for Ubuntu in Sections 7.1, 7.2 and 7.3.

- The Windows drives are available inside Ubuntu under `/mnt/c/`, `/mnt/d/`, ... For better performance, keep the source code and the simulation folders in the Linux file system (e.g. in your Ubuntu home folder) rather than under `/mnt/c/`.

- The Linux file system is accessible from the Windows Explorer under
  `\\wsl.localhost\Ubuntu\home\<user name>`
  (also shown as "Linux" in the navigation pane). ParaView installed on Windows can open the result files directly from there (Section 9.2).

7.7 Installing & Compiling REEF3D using Docker

Docker provides an alternative route for the compilation and installation of REEF3D and DIVEMesh that is independent of the operating system. The `Dockerfile` in the REEF3D repository creates an Ubuntu image in which OpenMPI and HYPRE are compiled from source, and the current versions of DIVEMesh and REEF3D are cloned from GitHub and compiled. The installation steps are as follows:

- Download and install Docker Desktop from:
  https://www.docker.com/products/docker-desktop/

  Adjust the number of CPU cores and the RAM available to Docker in the settings (optional), see Fig. 7.5.

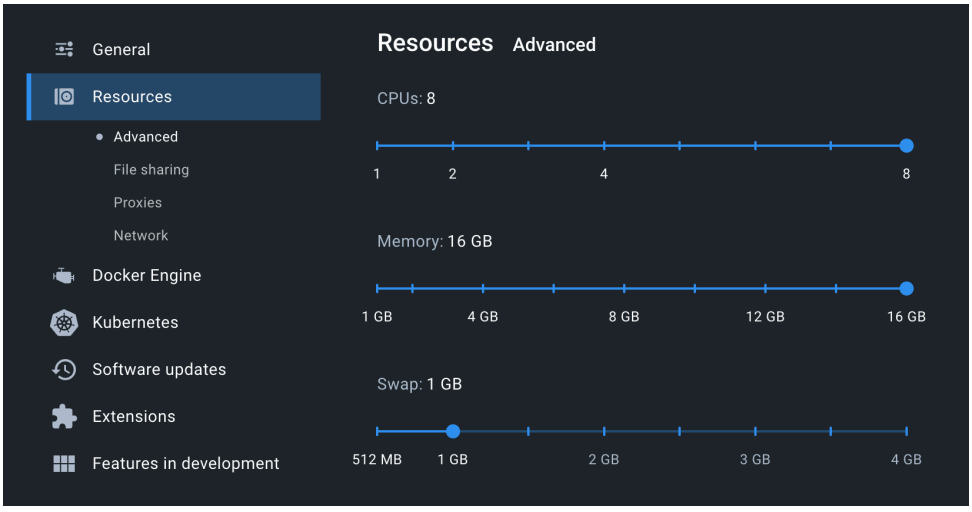


Figure 7.5: Docker resources

- Copy the file `Dockerfile` from the REEF3D repository into an empty folder and open a terminal in this folder. Only the Dockerfile is needed; the source code is downloaded during the build.

- Build the image, called reef3d, with:
  docker build -t reef3d .

  The process takes several minutes. Afterwards, all dependencies are installed and DIVEMesh and REEF3D are compiled. Because the Dockerfile always clones the latest source code, an update requires rebuilding the image without cache: **docker build --no-cache -t reef3d .**

- Create a folder called *simulations* on your computer. It is mounted into the container as `/simulations` and is the interface between the container and the local machine: everything written there is visible on both sides, everything else stays inside the container.

- Create and start the container. Replace `</local/path/simulations>` with the full path of your *simulations* folder.

  attached (the terminal is connected to the container directly):
  docker run -it --name reef3d -v </local/path/simulations>:/simulations reef3d

  detached (the container runs in the background):
  docker run -dit --name reef3d -v </local/path/simulations>:/simulations reef3d
  and connect to it with:
  docker attach reef3d

  In both cases, a bash shell opens in the folder `/simulations`. When the shell starts, the executables `DiveMESH` and `REEF3D` are copied into `/simulations`, which is on the PATH inside the container. Leaving the shell with **exit** stops the container; pressing Ctrl+P followed by Ctrl+Q detaches from it and leaves it running.

- Place each case in a subfolder of *simulations*, change into it inside the container and run DIVEMesh and REEF3D as described in Chapter 8, e.g. **DiveMESH** followed by **mpirun -n 4 REEF3D**.

- Other useful commands:
  start the stopped container: **docker start -ai reef3d**
  stop the container: **docker stop reef3d**
  restart the container: **docker restart reef3d**
  open an additional shell in a running container: **docker exec -it reef3d bash**

As in the Makefile build, the executables in the image are compiled with `-march=native` for the processor on which the image was built.

8. Running the Code

Every simulation is run in its own folder, the simulation folder. A simulation consists of two steps: DIVEMesh generates the grid and splits it into subdomains, and REEF3D then runs the simulation in parallel on these subdomains.

8.1 The Simulation Folder

Before the simulation, the folder contains the two input files and any further input files that the commands refer to:

```
my_case/
    control.txt        input for DIVEMesh (grid, geometry, M 10)
    ctrl.txt           input for REEF3D (see Chapter Functions)
    ...                optional input, e.g. STL files or bathymetry data
```

DIVEMesh adds the folder `DIVEMesh_Grid` with one grid file per subdomain (`grid-000001.dat`, `grid-000002.dat`, ...; with M 50 1 or 2 in `control.txt` also the global grid file `grid-global.dat`, from which REEF3D can decompose the grid itself, M 40) and further folders with log and visualization files (e.g. `DIVEMesh_Log`, `DIVEMesh_Paraview`). REEF3D reads the grid files and writes all results into folders starting with `REEF3D_` (see Section 8.6 and Chapter 9), and a record of every run into `REEF3D_Case`. Result folders from a previous run in the same folder are not deleted automatically: files with the same name are overwritten, others remain. Delete or move old results before a new run.

The executables `DiveMESH` and `REEF3D` (in the `bin` folders of the source code after compilation) are either copied into the simulation folder or, more conveniently, made available through the PATH, e.g. by adding the following line to ∼/`.bashrc`:
export PATH=/path/to/REEF3D/bin:/path/to/DIVEMesh/bin:$PATH
In the following, the executables are assumed to be on the PATH; if they have been copied into the simulation folder instead, use **./DiveMESH** and **./REEF3D**.

8.2 DIVEMesh

Open a terminal in the simulation folder and type:
DiveMESH
DIVEMesh reads `control.txt` (see the DIVEMesh User Guide), runs serially and generates the grid files. The number of subdomains is set with **M 10** in `control.txt`. If DIVEMesh stops with a message about a missing input parameter, check the commands in `control.txt` against the DIVEMesh User Guide. The DIVEMesh output in `DIVEMesh_Paraview` can be opened in ParaView to check the grid, the geometry and the domain decomposition before the simulation is started.

8.3 REEF3D

REEF3D is started with MPI from the simulation folder:
mpirun -n *number of processes* **REEF3D**
For example, for a grid decomposed into four subdomains:
mpirun -n 4 REEF3D
Alternatively, **mpiexec -n 4 REEF3D** can be used.

The number of MPI processes has to be the same as **M 10** in `control.txt` (the number of subdomains generated by DIVEMesh) and **M 10** in `ctrl.txt`. REEF3D checks this when reading the grid and stops with the message `Inconsistent M 10 parameter` if the three numbers differ. If DIVEMesh has not been run, REEF3D stops with `Could not open DIVEMesh grid file`. As a guideline, M 10 should not exceed the number of physical cores available.

At the start, REEF3D prints `read ctrl` while reading `ctrl.txt`. If a command has too few values, REEF3D stops with `missing input parameter in ctrl.txt`; unknown commands are ignored without a warning (see Chapter 6). During the simulation, the iteration number, the simulated time, the time step and solver information are printed to the terminal every P 12 iterations. The simulation ends when the maximum simulated time N 41 or the maximum number of iterations N 45 is reached. To keep the terminal output, redirect it to a file, e.g.:
mpirun -n 4 REEF3D > reef3d_output.txt 2>&1

If DIVEMesh has written the global grid file `DIVEMesh_Grid/grid-global.dat` (DIVEMesh M 50 1 or 2), REEF3D can decompose the grid itself for the number of processes of the run (M 40 1 in `ctrl.txt`; also when the grid files of the processes are missing). M 10 in `ctrl.txt` must then equal the number of MPI processes, but not M 10 of DIVEMesh, and with M 41 1 the weighted partition with the load map of the mesh refinement or of SEASTATE (G 45) is used (Section 6.11).

8.4 Restarting from Checkpoints

SFLOW, FNPF, NHFLOW, CFD and stand-alone SEASTATE (A 10 7) can write checkpoints and continue a run from them, also on another number of processes and another domain decomposition. With P 48 n, a checkpoint is written every n iterations into the folder `REEF3D_Checkpoint/<count>` (the iteration number as eight-digit folder name), and when it is complete, its name is written into `REEF3D_Checkpoint/latest`. Stationary SEASTATE runs (A 700 2) write it every n iterations within a time step into `REEF3D_Checkpoint/<count>_<iteration>`, and the restart continues the step at that iteration; SEASTATE keeps only the latest checkpoint of the run. To continue the run, add I 45 1 to `ctrl.txt` (and increase N 41 if needed) and start REEF3D again in the same folder: the run continues from the latest checkpoint with its simulated time, iteration number and output counters, and the time series files are appended to. Instead of 1, I 45 can name the iteration <count> of an older checkpoint. Optionally, P 48 n Δt also writes a checkpoint every Δt seconds of simulated time. SFLOW, FNPF, NHFLOW and CFD also write a checkpoint at the end of the run, and a run that receives the signal SIGTERM or SIGUSR1 (e.g. from the batch system before the end of the job) writes one at the end

of the current time step and stops. The folders are listed in the order they were written in `REEF3D_Checkpoint/history`; only the newest P 49 checkpoints are kept (default 2, SEASTATE 1), the latest never removed. The grid must be the same (a multi-block grid cut with G 55 also needs the same number of processes); for another number of processes, let REEF3D decompose the global grid file (M 40 1, see above), e.g.

```
P 48 1000       // first run: checkpoint every 1000 iterations
I 45 1          // restart: continue from REEF3D_Checkpoint/latest
M 40 1          // decompose DIVEMesh_Grid/grid-global.dat
M 10 16         // number of processes of the restart
```

and **mpirun -n 16 REEF3D**. Features that the checkpoints do not cover yet (SEASTATE coupled to a host model (A 750) and the SEASTATE surfbeat model (A 770 1), 6DOF wavemaker objects (X 10 4), FSI and FEM structures (Z 10, Z 30), and in NHFLOW Lagrangian particles (L 10) and membranes (X 330), see Section 6.9) are reported at the start: P 48 then writes no checkpoint, and I 45 stops with a message. The older state files (P 40) remain the input of the hydrodynamic coupling; the restart from them (I 40) is available for CFD only and on the same decomposition.

With G 46 on several processes, a run checks the load of the processes every G 47 iterations. If the load is too uneven, it writes a checkpoint, the load map and the marker file `REEF3D_Checkpoint/rebalance`, and stops with exit code 3. The next start in the same folder sees the marker and sets I 45, M 40 and M 41 to 1, i.e. it continues from the checkpoint on a new decomposition with the weighted partition of the load map; this needs the global grid file. Stand-alone SEASTATE runs measure the load as the time of the solves of their cells, check it every G 47 iterations within a time step when stationary, and stop only if the weighted partition lowers the load by at least 10 % by the load map and the restart pays off over the rest of the run (see G 46). The script `tools/reef3d_rebalance.sh` of the repository runs the start command given to it, e.g.

```
tools/reef3d_rebalance.sh mpirun -np 64 ./REEF3D
```

and repeats it as long as the marker file exists after the run (exit code 3, or 0 from an mpirun that does not pass the exit code on); it stops on any other exit code, and after at most `REEF3D_REBALANCE_MAX` restarts (environment variable, default 20). The restart removes the marker, also when it fails.

8.5 Coupled Runs: Several Models in One Job

With the runtime coupling (B 92 62), an FNPF model and one or more receiving models (CFD, NHFLOW or FNPF) run side by side in one MPI job. Each model keeps its own simulation folder with `control.txt`, `ctrl.txt`, its DIVEMesh grid and its output. DIVEMesh is run in each folder as usual, then all models are started with one **mpirun** command with one app context per model and the working directory of each:
mpirun -n 2 -wdir fnpf REEF3D : -n 8 -wdir cfd REEF3D
The order of the app contexts gives the app numbers 0, 1, 2, . . . , which are printed at the

start together with the role of each model. The number of processes of each app context has to match M 10 of its folder. For launchers without `-wdir` or without app contexts (e.g. `srun -multi-prog`), the command-line options `-dir <path>` and `-app <n>` of REEF3D give the folder and the app number of a process. A run with one model is unchanged.

- The FNPF model needs no coupling command. It sends to every model that names it as source.
- A receiver sets B 90 1 and B 92 62 and its usual wave generation and absorption (B 96, B 98, B 99, B 520). Its source is the lowest FNPF app that is not a receiver itself, or the app given with I 246.
- Both models start at the same time. I 241 maps the receiver time to the FNPF time; with I 30 1 the receiver starts from the FNPF state at its start time.
- With feedback (I 243, I 244), use linear time interpolation (I 242 1). At the end, both models report how long they waited for each other; I 247 sets the feedback lag.
- When the receiver ends, the FNPF model ends as well; when the FNPF model ends first, the receiver keeps its last state.
- An error in one model stops the whole job.
- Restart: give all models the same checkpoint times (P 48) and restart all of them with I 45 1.

FNPF is usually much cheaper per time step than the receiver: one FNPF process with a 2D FNPF model keeps up with about ten CFD processes. The tutorials in Chapter 14 cover the one-way coupling, oblique waves from 2D FNPF into 3D CFD, nesting, floating bodies, two-way coupling with CFD and NHFLOW, two-way coupling in 3D and a chain of three models.

8.6 Log Files

Besides the result files, REEF3D writes log files that are useful to monitor a simulation and to check it afterwards:

REEF3D_Log (CFD and NHFLOW) contains `REEF3D_Version.dat` (version and git branch of the executable), `REEF3D_mainlog.dat` (per iteration: time step, simulated time, computing time, pressure solver iterations and time, volumes, in- and outflow), `REEF3D_maxlog.dat` (maximum velocities, viscosity, turbulence quantities and pressure), `REEF3D_volumelog.dat` and `REEF3D_solverlog.dat` (iterations and time of the individual solvers). If activated, the sediment log and discharge gauges are written here as well.

REEF3D_SFLOW_Log (SFLOW) contains `REEF3D_SFLOW_mainlog.dat` with the same type of information.

REEF3D_Log-Probes (CFD and NHFLOW) contains the locations of the wave gauges as a `.vtu` file, which can be loaded in ParaView together with the results.

REEF3D_Log-Wave contains the theoretical time series of the generated waves at the locations given with P 58 (by default at $x = y = 0$).

REEF3D_SEASTATE_Log (SEASTATE) contains `REEF3D_SEASTATE_integral.dat` with one line per time step (total energy, maximum and mean H_s, solver iterations) and, for stationary runs, the convergence history; with G 45 or G 46 also the load map `REEF3D_SEASTATE_loadmap.dat`.

REEF3D_Case (all modules) contains the run log `REEF3D_<M>_run.jsonl` and a copy of the `ctrl.txt` of each run, `ctrl-<run id>.txt`. The run id is the start time (UTC) with a random suffix. Each run appends one JSON record per line to the run log: a run record (module, code version, start time, host, number of processes, the copy of `ctrl.txt` with its SHA-256 checksum, still water level and domain extent), one record per output stream (folder, file pattern and format, or the group in the LAGOON store), one record per written output (step, simulated time, iteration) and an end record with the status. The run log follows the run-log format of LAGOON and shows afterwards with which input and code version a result was computed.

FNPF does not write a main log file; its progress is followed through the terminal output. The version log is also helpful when reporting a problem, as it identifies the exact code version used.

8.7 Running on HPC Clusters

On HPC clusters, simulations are submitted to a queueing system instead of being started directly. The details (module names, partitions, accounts, whether `srun` or `mpirun` is used) differ between clusters; consult the documentation of your cluster. REEF3D should be compiled on the cluster with the MPI library provided there (Chapter 7). DIVEMesh runs serially and is usually fast, so it can be run on the login node or as the first step of the job. The following generic job script for the Slurm queueing system runs a case with 128 subdomains on two nodes with 64 cores each (`M 10 128` in both input files):

```
#!/bin/bash
#SBATCH --job-name=my_case
#SBATCH --nodes=2
#SBATCH --ntasks-per-node=64
#SBATCH --time=24:00:00
#SBATCH --account=<project>        # cluster specific

module load OpenMPI                # module names are cluster specific
                                   # (HYPRE only for a make HYPRE=1 build)

cd $SLURM_SUBMIT_DIR
DiveMESH                           # optional, if not run before
srun REEF3D > reef3d_output.txt 2>&1
                                   # or: mpirun -n $SLURM_NTASKS REEF3D
```

The total number of tasks (here $2 \times 64 = 128$) has to be equal to M 10. The job is submitted with **sbatch job.sh**, its status is checked with **squeue -u $USER**, and it is cancelled with **scancel** *job-id*. Large simulations write large amounts of output; use the work or scratch file system of the cluster for the simulation folder and choose the print intervals (P 20, P 30) accordingly.

9. Post-Processing

9.1 Output Files

REEF3D writes its results into folders in the simulation folder whose names start with `REEF3D_` followed by the module name. Field data are written in the VTK XML formats: every MPI process writes one file per output time (`.vtu` or `.vtp`), and a small parallel master file (`.pvtu` or `.pvtp`) that references the files of all processes. **Always open the .pvtu or .pvtp files in ParaView**, not the individual per-process files. File names contain an eight-digit output counter, e.g. `REEF3D-CFD-00000012.pvtu`, and the simulated time is stored in each file, so that ParaView loads the files of one folder as a time series.

The most important folders are:

- `REEF3D_CFD_VTU`, `REEF3D_NHFLOW_VTU`, `REEF3D_FNPF_VTU`: three-dimensional flow field (`.pvtu`), written with the print intervals P 20 (iterations) or P 30 (simulated time). The format is selected with P 10; with the non-default formats the folders end in `_VTS` (`.pvts`) or, for CFD, `_VTR` (`.pvtr`).
- `REEF3D_CFD_FSF`, `REEF3D_NHFLOW_VTP_FSF`, `REEF3D_FNPF_VTP_FSF`: free surface as a surface (`.pvtp`), activated with P 180.
- `REEF3D_SFLOW_VTP_FSF`, `REEF3D_SFLOW_VTP_BED`: SFLOW results. SFLOW is depth-averaged and writes its results as surfaces (`.pvtp`) for any P 10 $>$ 0; the print interval is set with P 181 or P 182. With the meteorological forcing (W 200, Section 6.17.1), the free surface files of SFLOW and NHFLOW also contain the 10 m wind `wind10`, the air pressure `air_pressure` and the surface stress `wind_stress`.
- `REEF3D_SFLOW_MAX`: maxima of an SFLOW run per cell (P 112), e.g. highest water level, largest depth and speed, flood duration and inundation class, and with W 200 the maximum wind and minimum air pressure (`.pvtp`, overwritten during the run).
- `REEF3D_NHFLOW_VTP_BED`, `REEF3D_FNPF_VTP_BED`, `REEF3D_CFD_Topo`: bed surface; the CFD sediment bed is activated with P 190.
- `REEF3D_*_6DOF_VTP`: surface of floating bodies (6DOF).
- `REEF3D_*_WSF`: time series of the wave gauges (P 51, P 50) as text files, see Section 9.5.
- `REEF3D_NHFLOW_Particles`, `REEF3D_CFD_Particles`: Lagrangian particles as point clouds (`.vtp`), CSV track file and log file, activated with L 10 (Section 6.10).
- `REEF3D_SFLOW_AMR`, `REEF3D_FNPF_AMR`, `REEF3D_NHFLOW_AMR`: all levels of the mesh refinement (G 1, Section 6.7.1) as `.vtm` multiblock files, with a log file and the wave gauges; `REEF3D_CFD_AMR`: the patches of the CFD mesh refinement (`.vtr` with a `.vtm` multiblock file) at the output times of P 10. The load statistics and load map of G 43 and G 45 are written into the same folders.

- `REEF3D_NHFLOW_Tidal`: maps of the long-run statistics of NHFLOW (P 160): harmonic analysis of the tidal constituents and time averages (`.vts` with a `.vtm` multiblock file).
- `REEF3D_CORAL`: time series of the structural solver REEF3D::CORAL (Z 40, Section 5.12).
- `REEF3D_Checkpoint`: checkpoints for the restart (P 48, I 45, Section 8.4), one folder per checkpoint named by its iteration, with `main.dat`, one `piece-<rank>.dat` per process, copies of `ctrl.txt` and `control.txt` and the tables `outputs-<rank>.txt` of the output file sizes; the files `latest` (the newest complete checkpoint) and `history` (the checkpoints in the order they were written, P 49).
- `REEF3D_FNPF_Breaking_Log`: breaking cells of FNPF and, with the breaking model A 350 4, the log of the breaking crests (P 59).
- `REEF3D_SEASTATE_VTP` (2D fields such as H_s, periods and mean direction, `.pvtp`, for any P 10 > 0), `REEF3D_SEASTATE_Log` (integral values and convergence), `REEF3D_SEASTATE_AMR` (refined patches, G 1) and `REEF3D_SEASTATE_Spectra` (handover spectra, A 760): results of REEF3D::SEASTATE, see Section 6.1.4.
- `REEF3D_<M>.lagoon`: LAGOON store (P 18), a Zarr v3 store read by LAGOON, the post-processing and visualisation tool of REEF3D. It holds the 3D fields, free surface and bed surfaces, mesh refinement grids, floating bodies and particles of the run. With P 18 1 it replaces these VTU/VTP files, with P 18 2 both are written. A store of an earlier run in the same folder is renamed to `REEF3D_<M>-<run id>.lagoon-earlier`.
- `REEF3D_*_ProbePoint`, `REEF3D_CFD_Force`, `REEF3D_*_Sediment`, ...: further time series from probes, force calculations, sediment transport and other options, as text files (`.dat`).

The log folders and the run log in `REEF3D_Case` are described in Section 8.6. Solid geometries and the bathymetry prepared by DIVEMesh can be visualized from the `.vtp` files that DIVEMesh writes.

9.2 Paraview

ParaView (www.paraview.org) is an open-source visualization tool. It is cross-platform and works on macOS, Windows and Linux. ParaView supports parallel output files as well as parallel rendering through distributed computing. Because it is very flexible, efficient and free, it is the preferred program for visualizing REEF3D's results.

9.3 Windows Users (WSL)

When REEF3D runs in WSL (Section 7.6), ParaView can be installed on Windows as usual. The result files in the Linux file system are opened directly in the Windows version of ParaView under `\\wsl.localhost\Ubuntu\home\<user name>\...`. For very large data sets, the ParaView server `pvserver` of a Linux ParaView installation can alternatively be started in Ubuntu and connected to from the Windows ParaView client with File > Connect; client and server must have the same ParaView version.

9.4 Visualising Two-Phase Flow Results in Paraview

REEF3D can write out a range of different result files, depending on the hydrodynamic module used. In this section, the procedure is presented how to visualize the free surface obtained from two-phase flow simulations with the CFD module. For this type of calculation, the level set method is employed, which defines the free surface implicitly: the free surface is the zero contour of the level set function (the variable phi in the ParaView drop-down menu). In the following, the procedure is presented on how to extract the zero level set contour from the results of the CFD module.

Open the .pvtu file series in the folder `REEF3D_CFD_VTU` in ParaView through File > Open. Click on Outline in the ParaView toolbar as shown in Fig. 9.1 to obtain the outline of the wave tank. Click on the Slice filter on the toolbar (circled in Fig. 9.1) and select "y-normal" in the properties sidebar to generate a 2D slice along the length of the numerical wave tank as shown in Fig. 9.2. The slice can be colored according to the various variables available in the drop-down menu. To obtain the free surface, select the slice in the pipeline browser, click on the Contour filter on the toolbar and select Contour by: phi in the properties sidebar with the value 0, as shown in Fig. 9.3. The geometry of the structure used in the wave tank is included by adding the .vtp file generated by DIVEMesh to obtain the result shown in Fig. 9.4.

To obtain a 3D visualization of the free surface, open the .pvtu file in ParaView and directly apply the Contour filter as shown in Fig. 9.5. Alternatively, the free surface can be written directly as a surface with P 180 (folder `REEF3D_CFD_FSF`).

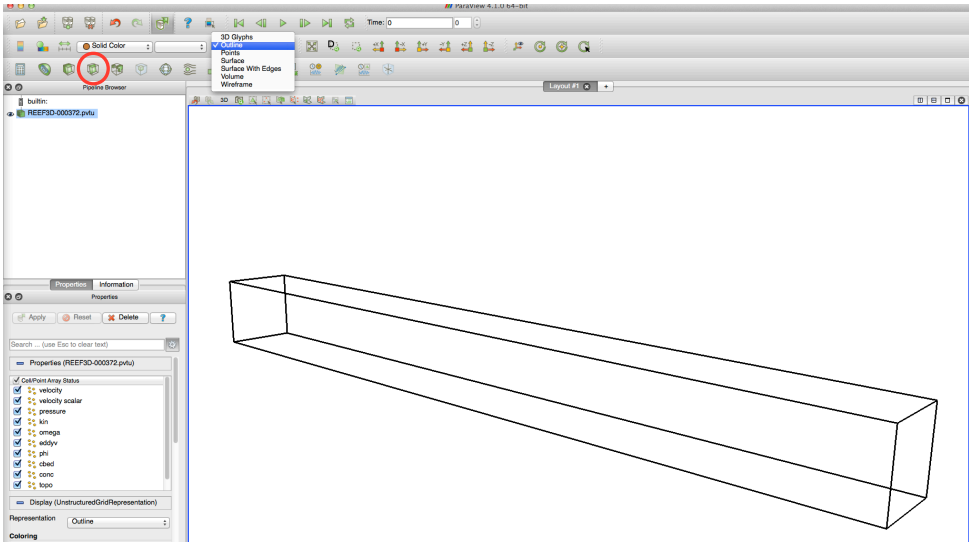


Figure 9.1: Outline of the numerical wave tank

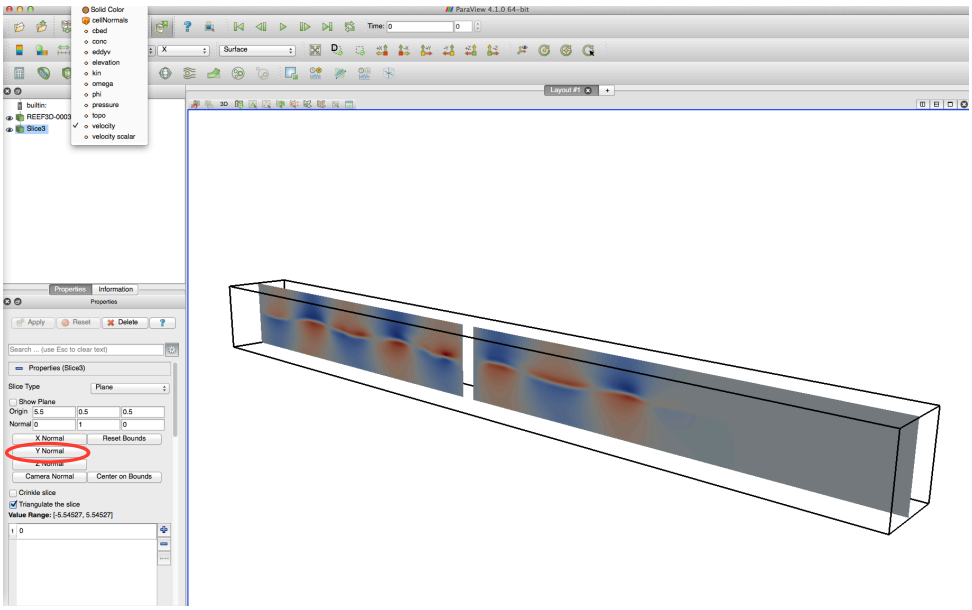


Figure 9.2: Creating a 2D slice

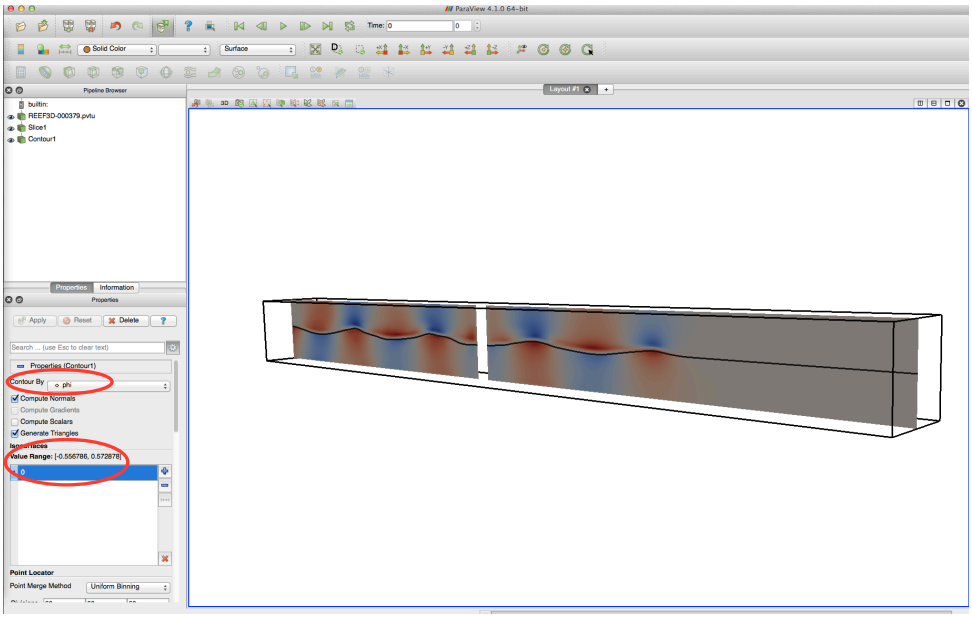


Figure 9.3: Generating the free surface

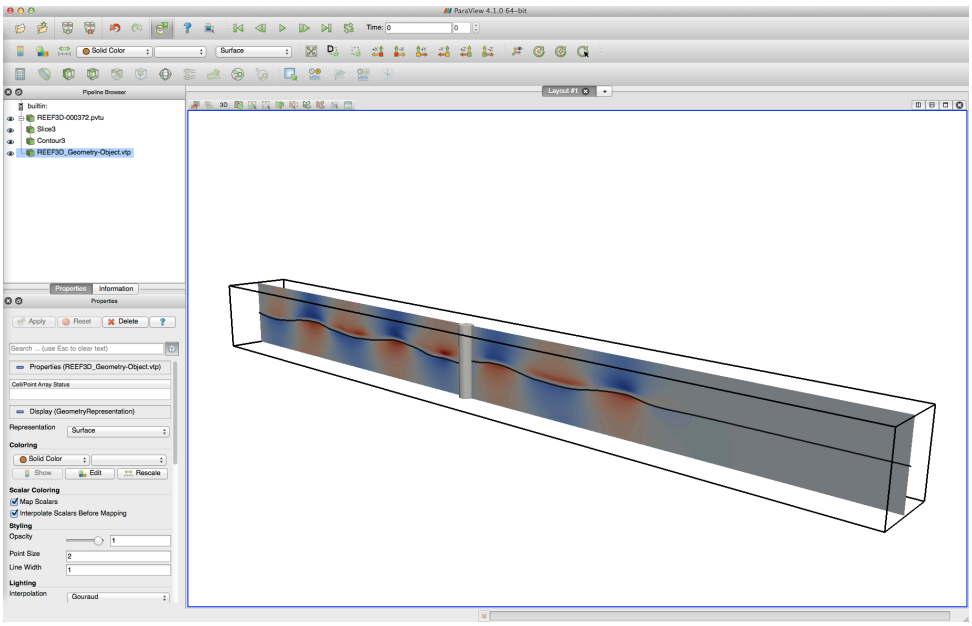


Figure 9.4: Including the geometry of the structure

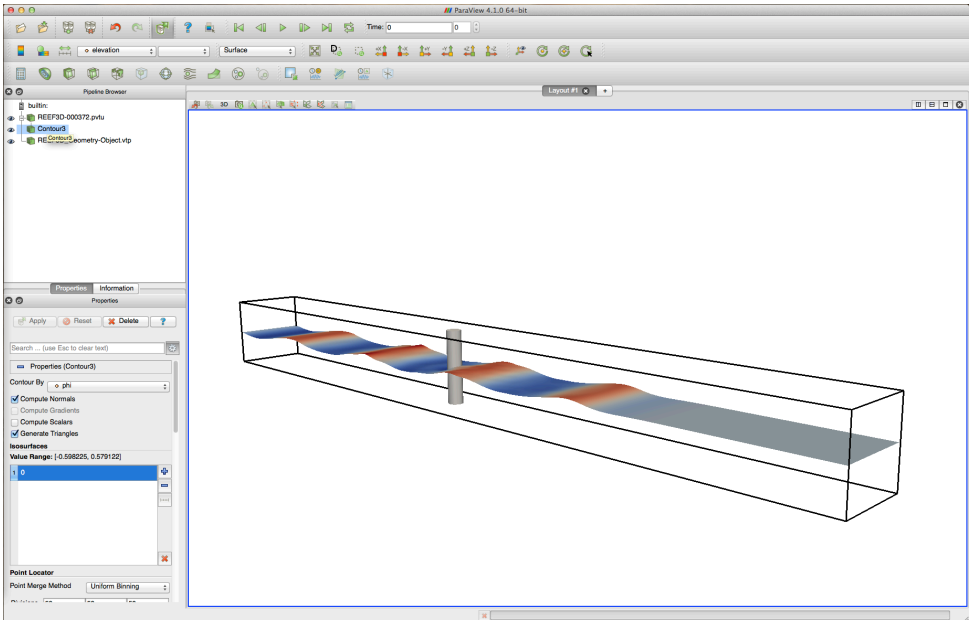


Figure 9.5: Visualizing the free surface in 3D

Generating screenshots and animations from Paraview

Use File > Save Screenshot from the drop-down menu, input the desired resolution, click OK and select the file type in the next window to save a screenshot from ParaView.

To generate an animation, use File > Save Animation from the drop-down menu. Enter the desired frame rate and the range of time steps to be included in the animation in Frame range. For an animation in real time, the frame rate is $1/\Delta t_{P30}$ frames per second, where Δt_{P30} is the print interval set with P 30 (e.g. 10 frames per second for P 30 0.1). Select the desired file type in the next window.

9.5 Time Series from Gauges and Probes

Wave gauges (P 51), probes and force calculations write plain text files that can be plotted with any tool, e.g. Python, Matlab or a spreadsheet program. As an example, the wave gauge file of the FNPF module `REEF3D_FNPF_WSF/REEF3D-FNPF-WSF-HG.dat` (and the corresponding files `REEF3D-CFD-WSF-HG.dat`, `REEF3D-NHFLOW-WSF-HG.dat` and `REEF3D-SFLOW-WSF-HG.dat` of the other modules) has the following structure:

```
number of gauges:  2

x_coord     y_coord
1    5.0    0.5
2    10.0   0.5


time    P1    P2

0.0      0.0      0.0
0.0125   0.0003  -0.0001
...
```

The header lists the number and the x- and y-coordinates of the gauges. It is followed by one line per time step with the simulated time in s and the free surface value of each gauge in m. Because of the adaptive time stepping, the time interval between the lines is in general not constant; interpolate the data to a constant time step before a spectral analysis. FNPF, NHFLOW and SFLOW write the free surface elevation η relative to the still water level. CFD writes the vertical coordinate z of the free surface (the zero level set), so the still water depth has to be subtracted to obtain η. A value of -10^{20} means that no free surface was found at the gauge location, e.g. because it lies outside the domain.

The file can be read and plotted with Python as follows:

```
import numpy as np
import matplotlib.pyplot as plt
```

```
fname = "REEF3D_FNPF_WSF/REEF3D-FNPF-WSF-HG.dat"
with open(fname) as f:
    lines = f.readlines()

# data start after the header line beginning with "time"
start = next(i for i, l in enumerate(lines) if l.startswith("time")) + 1
data = np.loadtxt(lines[start:], ndmin=2)
t, eta = data[:, 0], data[:, 1:]      # eta[:, n] is gauge n+1

plt.plot(t, eta[:, 0], label="gauge 1")
plt.xlabel("t [s]")
plt.ylabel("eta [m]")
plt.legend()
plt.show()
```

10. Tutorial | REEF3D::SFLOW

REEF3D::SFLOW solves the non-hydrostatic shallow water equations: all flow quantities are depth-averaged, so the model runs on a horizontal 2D grid and is very efficient for wave propagation in shallow and intermediate water depth and for open channel flow. The tutorials below are 2D wave flumes with a single cell in y-direction. For visualization, open the REEF3D_SFLOW_VTP_FSF folder (free surface: eta, velocity, pressure, ...) and the REEF3D_SFLOW_VTP_BED folder (bed) in ParaView; wave gauge time series are written to REEF3D_SFLOW_WSF. Each tutorial folder contains the DIVEMesh input 'control.txt' and the REEF3D input 'ctrl.txt'. Run DIVEMesh (**./DiveMESH**) in the folder first, then REEF3D with **mpirun -n** *M 10* **reef3d**.

Note that in SFLOW the ParaView output interval is set with P 182 (time) or P 181 (iterations), not with P 30. The ctrl.txt files in the repository contain P 30, which SFLOW does not read, so without further changes only the initial state is written to REEF3D_SFLOW_VTP_FSF and REEF3D_SFLOW_VTP_BED. Add e.g. P 182 0.1 to obtain a time series of .vtp files.

10.1 2nd-order Stokes waves

Repository folder: `Tutorials/REEF3D_SFLOW/8_1 Stokes Wave Propagation`

Regular 2nd-order Stokes waves propagate over a flat bed in a 28 m long 2D flume. The waves are generated in a relaxation zone at the left end and absorbed by a relaxation beach at the right end. The case demonstrates regular wave propagation with SFLOW and the comparison of the computed surface elevation with the wave theory at three gauges (P 50 and P 51 at the same locations).

- Wave: H = 0.05 m, L = 4.0 m (B 91), water depth d = 0.5 m (F 60)
- Domain: 28 m × 0.02 m, cell size dx = 0.02 m (1400 cells in x)
- Wave generation zone 4 m, numerical beach 8 m (B 96), simulated time 60 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

10.1.1 DIVEMesh: control.txt

```
C 11 6 // side 1 (x_min): wave generation
C 12 3 // side 2 (y_max): symmetry plane
C 13 3 // side 3 (y_min): symmetry plane
C 14 7 // side 4 (x_max): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.02 // uniform cell size dx [m]
B 10 0.0 28.0 0.0 0.02 0.0 1.0 // rectangular domain x_s, x_e, y_s, y_e, z_s, z_e [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)
```

10.1.2 REEF3D: ctrl.txt

A 10 2 // hydrodynamic model: REEF3D::SFLOW

B 90 1 // turn on ioWave (numerical wave tank)
B 92 4 // wave type: 2nd-order Stokes waves
B 91 0.05 4.0 // wave height H [m], wavelength L [m]
B 96 4.0 8.00 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1

F 60 0.5 // still water level [m]

N 41 60.0 // maximum simulated time [s]
N 47 0.2 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtp)
P 30 0.1 // no effect in SFLOW (the .vtp interval is set with P 182)

P 50 4.0 0.01 // theoretical wave gauge 1 at x, y [m]
P 51 4.0 0.01 // wave gauge 1 at x, y [m]

P 50 14.5 0.01 // theoretical wave gauge 2 at x, y [m]
P 51 14.5 0.01 // wave gauge 2 at x, y [m]

P 50 19.0 0.01 // theoretical wave gauge 3 at x, y [m]
P 51 19.0 0.01 // wave gauge 3 at x, y [m]

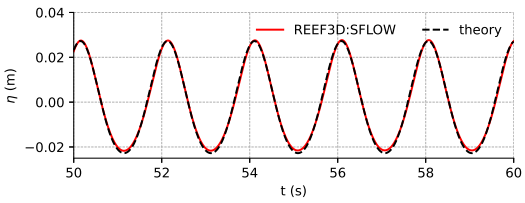
P 54 100 // wsfline interval in iterations; no effect here, no P 52 is given

W 10 0.0 // discharge [m^3/s], not used (ioFlow is off)
W 22 -9.81 // gravity in z-direction [m/s^2]

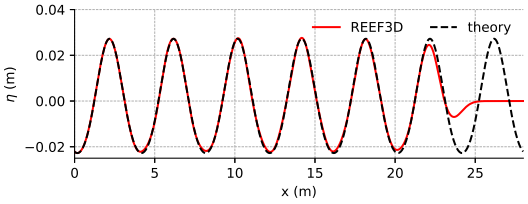
10.1.3 What to look at

Plot the time series in REEF3D_SFLOW_WSF: 'REEF3D-SFLOW-WSF-HG.dat' contains the computed surface elevation at the three P 51 gauges (x = 4.0, 14.5 and 19.0 m) and 'REEF3D-SFLOW-WSF-HG-THEORY.dat' the theoretical elevation at the same P 50 locations. For ParaView, open REEF3D_SFLOW_VTP_FSF and color by eta or velocity (add P 182 for more than the initial state, see the chapter introduction). The spatial profile shown below requires a free surface line: P 54 alone has no effect, add e.g. P 52 0.01 to write REEF3D_SFLOW_WSFLINE files.

10.1.4 Results



(a) Free surface elevation evolution over time at wave gauge 2 (x =14.5 m)



(b) Free surface elevation in space at t =60 s

Figure 10.1: Free surface elevations in the simulation of 2nd-order Stokes waves in intermediate water depth.

10.2 Cnoidal waves over constant water depth

Repository folder: `Tutorials/REEF3D_SFLOW/8_2 Cnoidal Wave Propagation`

Cnoidal waves propagate over a flat bed in the same 28 m flume as the Stokes wave case. In addition to the wave input, this ctrl.txt sets the SFLOW numerics explicitly with the A 2xx commands: 3rd-order Runge-Kutta time stepping, WENO flux reconstruction and the quadratic non-hydrostatic pressure. Note that A 243 0 and A 246 0 switch off wetting-drying and the breaking wave algorithm, which are both on by default.

- Wave: H = 0.21 m, L = 4.0 m (B 91), water depth d = 0.5 m (F 60)
- Domain: 28 m × 0.02 m, cell size dx = 0.02 m (1400 cells in x)
- Wave generation zone 4 m, numerical beach 8 m (B 96), simulated time 60 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

10.2.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.02 // uniform cell size dx [m]
B 10 0.0 28.0 0.0 0.02 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

10.2.2 REEF3D: ctrl.txt

A 10 2 // hydrodynamic model: REEF3D::SFLOW

A 210 3 // SFLOW time scheme: 3rd-order Runge-Kutta
A 211 4 // SFLOW convection: WENO flux reconstruction

A 220 2 // SFLOW quadratic non-hydrostatic pressure
A 223 0.5 // blending factor for current/previous η in the hydrostatic pressure gradient

A 240 1 // SFLOW free surface algorithm on
A 243 0 // wetting and drying off
A 246 0 // SFLOW breaking wave algorithm off

B 90 1 // turn on ioWave (numerical wave tank)
B 92 8 // wave type: 5th-order cnoidal waves
B 91 0.21 4.0 // wave height H [m], wavelength L [m]
B 96 4.0 8.00 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1

F 60 0.5 // still water level [m]

N 41 60.0 // maximum simulated time [s]
N 47 0.2 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtp)
P 30 0.5 // no effect in SFLOW (the .vtp interval is set with P 182)

P 50 4.0 0.01 // theoretical wave gauge 1 at x, y [m]
P 51 4.0 0.01 // wave gauge 1 at x, y [m]

P 50 14.5 0.01 // theoretical wave gauge 2 at x, y [m]
P 51 14.5 0.01 // wave gauge 2 at x, y [m]

P 50 19.0 0.01 // theoretical wave gauge 3 at x, y [m]
P 51 19.0 0.01 // wave gauge 3 at x, y [m]

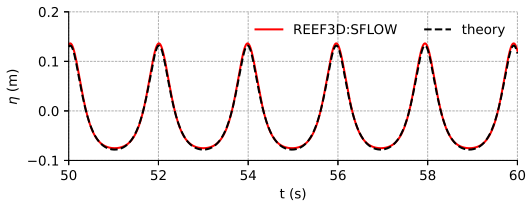
P 54 100 // wsfline interval in iterations; no effect here, no P 52 is given

W 22 -9.81 // gravity in z-direction [m/s^2]

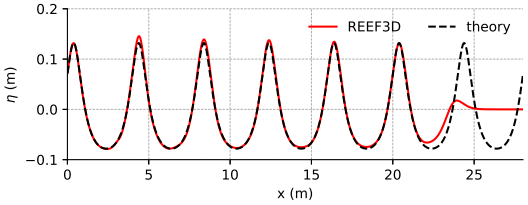
10.2.3 What to look at

Compare the columns of 'REEF3D-SFLOW-WSF-HG.dat' (P 51) and 'REEF3D-SFLOW-WSF-HG-THEORY.dat' (P 50) in REEF3D_SFLOW_WSF at x = 4.0, 14.5 and 19.0 m; the cnoidal wave shape with its sharp crests and flat troughs should be preserved along the flume. As in the previous case, add P 182 for a time series of REEF3D_SFLOW_VTP_FSF files and P 52 for the spatial profile.

10.2.4 Results



(a) Free surface elevation evolution over time at wave gauge 2 (x =14.5 m)



(b) Free surface elevation in space at t =60 s

Figure 10.2: Free surface elevations in the simulation of cnoidal waves in intermediate water depth.

10.3 Solitary wave over constant water depth

Repository folder: `Tutorials/REEF3D_SFLOW/8_3 Solitary Wave Propagation`

A single solitary wave is generated at the left end of the flume and propagates over a flat bed. The case shows that SFLOW keeps the shape of the solitary wave during propagation; the surface elevation at a gauge in the middle of the flume is compared with the theoretical solution. The numerical settings (A 2xx) are the same as in the cnoidal wave case.

- Wave: H = 0.05 m (B 91), water depth d = 0.5 m (F 60)
- Domain: 28 m × 0.02 m, cell size dx = 0.02 m (1400 cells in x)
- Wave generation zone 4 m, numerical beach 8 m (B 96), simulated time 20 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

10.3.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.02 // uniform cell size dx [m]
B 10 0.0 28.0 0.0 0.02 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

10.3.2 REEF3D: ctrl.txt

A 10 2 // hydrodynamic model: REEF3D::SFLOW

A 210 3 // SFLOW time scheme: 3rd-order Runge-Kutta
A 211 4 // SFLOW convection: WENO flux reconstruction

A 220 2 // SFLOW quadratic non-hydrostatic pressure
A 223 0.5 // blending factor for current/previous η in the hydrostatic pressure gradient

A 240 1 // SFLOW free surface algorithm on
A 243 0 // wetting and drying off
A 246 0 // SFLOW breaking wave algorithm off

B 90 1 // turn on ioWave (numerical wave tank)
B 92 9 // wave type: 1st-order solitary wave
B 91 0.05 4.0 // wave height H [m], wavelength L [m]
B 96 4.0 8.00 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1

F 60 0.5 // still water level [m]

N 41 20.0 // maximum simulated time [s]
N 47 0.2 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 30 0.5 // no effect in SFLOW (the .vtp interval is set with P 182)

P 50 14.5 0.01 // theoretical wave gauge 1 at x, y [m]
P 51 14.5 0.01 // wave gauge 1 at x, y [m]

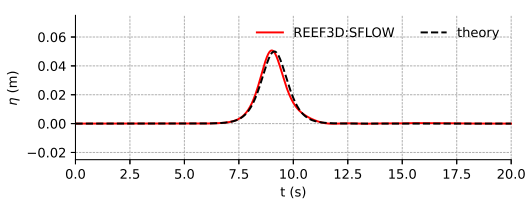
P 54 100 // wsfline interval in iterations; no effect here, no P 52 is given

W 22 -9.81 // gravity in z-direction [m/s^2]

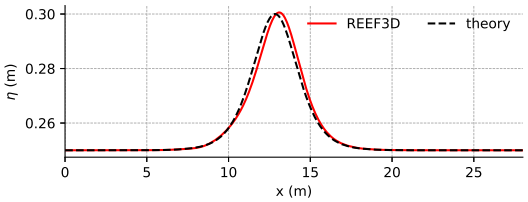
10.3.3 What to look at

Plot the single gauge at x = 14.5 m from 'REEF3D-SFLOW-WSF-HG.dat' (P 51) together with 'REEF3D-SFLOW-WSF-HG-THEORY.dat' (P 50) in REEF3D_SFLOW_WSF. Note that this ctrl.txt has no P 10 line; P 10 is 1 by default, so the initial .vtp files are still written. Add P 182 for a time series and P 52 for the spatial profile.

10.3.4 Results



(a) Free surface elevation evolution over time at the wave gauge (x =14.5 m)



(b) Free surface elevation in space at t =8.4155 s

Figure 10.3: Free surface elevations in the simulation of a solitary wave in intermediate water depth.

10.4 Wave propagation over a submerged bar

Repository folder: `Tutorials/REEF3D_SFLOW/8_4 Wave Propagation over a Submerged Bar`

Regular waves propagate over a submerged trapezoidal bar following the laboratory experiments of Beji and Battjes [1993]. The bar is built from solid objects in DIVEMesh: a front slope from x = 11 m to 17 m, a 2 m long crest and a back slope to x = 22 m, 0.3 m high. On the front slope the waves steepen and higher harmonics are generated, which are released behind the bar. The case tests the dispersion properties of SFLOW.

- Wave: 2nd-order Stokes, H = 0.021 m, T = 2.525 s (B 93), water depth d = 0.4 m, 0.1 m on the bar crest
- Domain: 38 m × 0.02 m, cell size dx = 0.02 m (1900 cells in x)
- Wave generation zone 5 m, numerical beach 9.5 m (B 96), simulated time 60 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

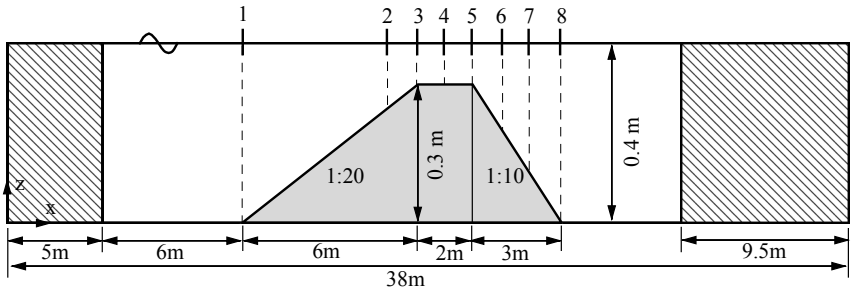


Figure 10.4: Experimental setup submerged bar

10.4.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane

C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.02 // uniform cell size dx [m]
B 10 0.0 38.0 0.0 0.02 0.0 0.8 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 61 11.0 17.0 0.0 0.02 0.0 0.3 // front slope of the bar, rising from z = 0 to 0.3 m
S 10 17.0 19.0 0.0 0.02 0.0 0.3 // bar crest, 0.3 m high
S 61 19.0 22.0 0.0 0.02 0.3 0.0 // back slope of the bar, falling from 0.3 m to 0

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

10.4.2 REEF3D: ctrl.txt

A 10 2 // hydrodynamic model: REEF3D::SFLOW

B 90 1 // turn on ioWave (numerical wave tank)
B 92 4 // wave type: 2nd-order Stokes waves
B 93 0.021 2.525 // wave height H [m], wave period T [s]
B 96 5.0 9.50 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1
b 120 0.0 // lower-case letter: this line is ignored (deactivated)

F 60 0.40 // still water level [m]

N 41 60.0 // maximum simulated time [s]
N 47 0.2 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 51 11.0 0.01 // wave gauge 1 at x, y [m]
P 51 16.0 0.01 // wave gauge 2 at x, y [m]
P 51 17.0 0.01 // wave gauge 3 at x, y [m]
P 51 18.0 0.01 // wave gauge 4 at x, y [m]
P 51 19.0 0.01 // wave gauge 5 at x, y [m]
P 51 20.0 0.01 // wave gauge 6 at x, y [m]
P 51 21.0 0.01 // wave gauge 7 at x, y [m]
P 51 22.0 0.01 // wave gauge 8 at x, y [m]
P 51 24.0 0.01 // wave gauge 9 at x, y [m]

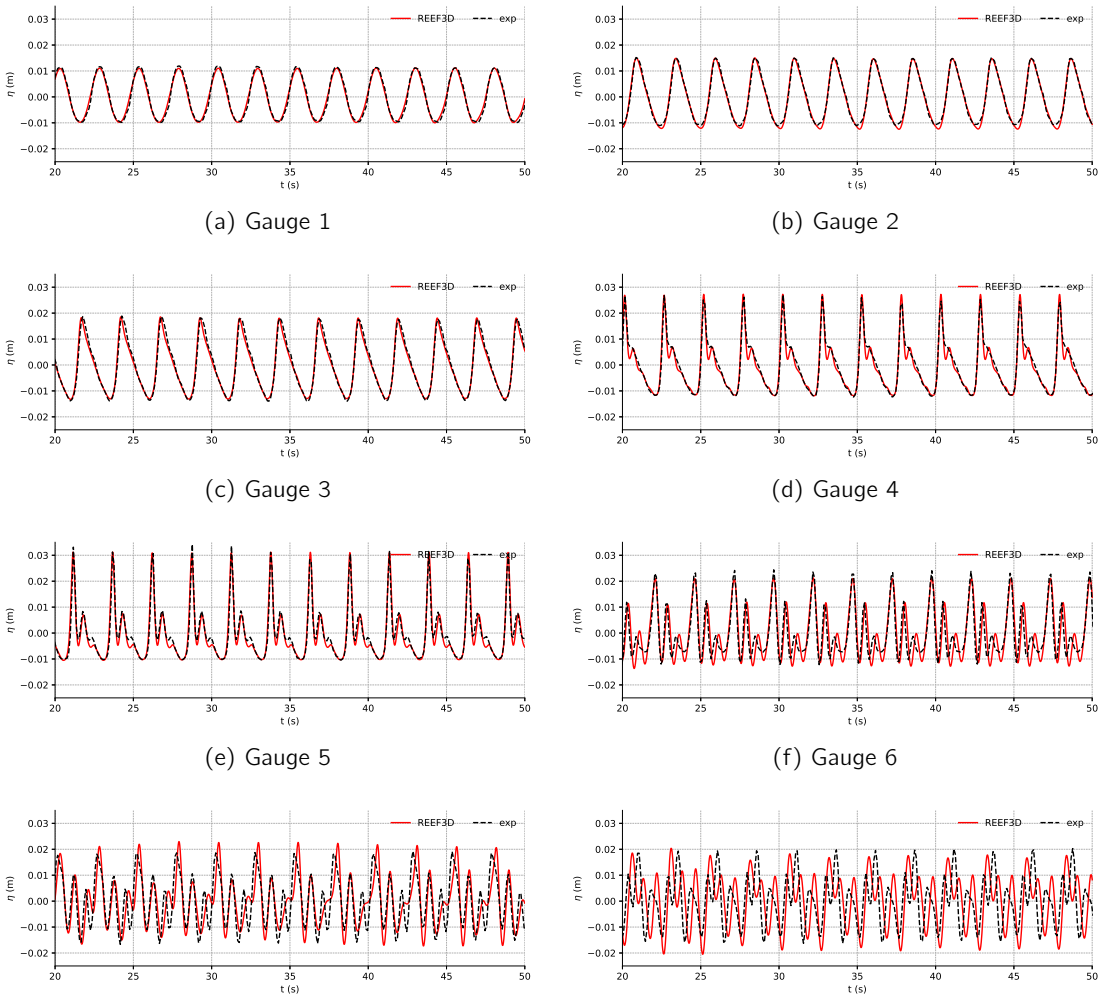
W 10 0.0 // discharge [m^3/s], not used (ioFlow is off)
W 22 -9.81 // gravity in z-direction [m/s^2]

10.4.3 What to look at

The nine P 51 gauges at x = 11, 16, 17, 18, 19, 20, 21, 22 and 24 m are written to 'REEF3D-SFLOW-WSF-HG.dat' in REEF3D_SFLOW_WSF, one column per gauge in the order given in ctrl.txt. Plot them over the last wave periods and compare with the experimental data of Beji and Battjes [1993]; the figures below show the first eight gauges. The bar geometry is visible in the bed elevation of the REEF3D_SFLOW_VTP_BED file.

10.4.4 Results

Note that the high-frequency bound waves released behind the top of the submerged bar are in relatively deep water, which explains the discrepancies with the experiment at wave gauges 6 and onwards. Be aware of the water depth condition when applying depth-averaged models.



(a) Gauge 1 (b) Gauge 2

(c) Gauge 3 (d) Gauge 4

(e) Gauge 5 (f) Gauge 6

(g) Gauge 7 (h) Gauge 8

Figure 10.5: Free surface elevations in the simulation of wave propagation over a submerged bar.

11. Tutorial | REEF3D::FNPF

REEF3D::FNPF is a fully nonlinear potential flow model: the Laplace equation for the velocity potential is solved on a sigma grid that follows the bed and the free surface. It is used for wave propagation from deep to shallow water over large domains and for long irregular sea states. For visualization, open the REEF3D_FNPF_VTU folder (.pvtu files with the velocity potential Fi, velocity and elevation) and, where P 180 is set, the REEF3D_FNPF_VTP_FSF folder (free surface: eta, Fifsf, breaking, ...); the bed is written to REEF3D_FNPF_VTP_BED. Wave gauge time series are written to REEF3D_FNPF_WSF. Each tutorial folder contains 'control.txt' for DIVEMesh and 'ctrl.txt' for REEF3D; run DIVEMesh (**./DiveMESH**) first and then **mpirun -n** *M 10* **reef3d**.

In FNPF the vertical extent given in B 10 of control.txt does not set the water depth: the vertical grid is normalized to a sigma coordinate between the bed and the free surface, whose still water level is given by F 60. Only the number of vertical cells (B 2) and their distribution (B 103, B 113, B 116) matter.

11.1 2nd-order Stokes wave in intermediate water depth

Repository folder: `Tutorials/REEF3D_FNPF/9_1 Regular Wave Propagation`

Regular 2nd-order Stokes waves propagate over a flat bed in a 40 m long 2D wave tank. The waves are generated in a relaxation zone at the left end and absorbed in a relaxation beach at the right end. The vertical grid is clustered towards the free surface. The computed surface elevation is compared with the wave theory both in time (gauges) and in space (free surface line).

- Wave: H = 0.04 m, L = 4.0 m (B 91), water depth d = 2.0 m (F 60)
- Domain: 40 m long, 400 × 1 × 10 cells (B 2), dx = 0.1 m
- Wave generation zone 4 m, numerical beach 8 m (B 96), simulated time 40 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

11.1.1 DIVEMesh: control.txt

```
C 11 6 // side 1 (x_min): wave generation
C 12 3 // side 2 (y_max): symmetry plane
C 13 3 // side 3 (y_min): symmetry plane
C 14 7 // side 4 (x_max): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.1 // uniform cell size dx [m]
B 2 400 1 10 // number of cells N_x, N_y, N_z (overrides B 1)
B 10 0.0 40.0 0.0 0.1 0.0 2.0 // rectangular domain x_s,x_e,y_s,y_e,z_s,z_e [m]
```

B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 2.5 // stretching factor in z-direction
B 116 2.0 // focus point of the z-stretching [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

11.1.2 REEF3D: ctrl.txt

A 10 3 // hydrodynamic model: REEF3D::FNPF
A 310 3 // FNPF time scheme: 3rd-order TVD Runge-Kutta
A 311 4 // FNPF free surface BCs: 5th-order WENO
A 343 0 // wetting and drying off

B 90 1 // turn on ioWave (numerical wave tank)
B 92 4 // wave type: 2nd-order Stokes waves
B 91 0.04 4.0 // wave height H [m], wavelength L [m]
B 96 4.0 8.0 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1

F 60 2.0 // still water level [m]

N 41 40.0 // maximum simulated time [s]
N 47 1.0 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

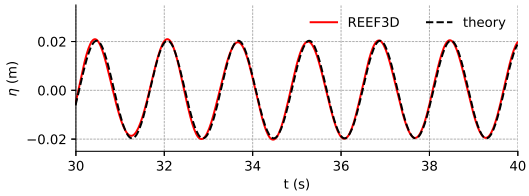
P 10 1 // print ParaView files (.vtu)
P 30 0.5 // print interval in simulated time [s]
P 50 15.0 0.005 // theoretical wave gauge 1 at x, y [m]
P 50 20.0 0.005 // theoretical wave gauge 2 at x, y [m]
P 50 25.0 0.005 // theoretical wave gauge 3 at x, y [m]
P 51 15.0 0.005 // wave gauge 1 at x, y [m]
P 51 20.0 0.005 // wave gauge 2 at x, y [m]
P 51 25.0 0.005 // wave gauge 3 at x, y [m]
P 52 0.005 // free surface line (wsfline) along x at this y [m]
P 53 1 // add the theoretical wave to the wsfline files
P 54 100 // wsfline print interval in iterations (needs P 52)

W 22 -9.81 // gravity in z-direction [m/s^2]

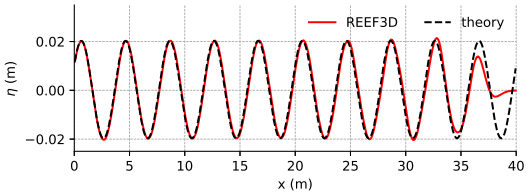
11.1.3 What to look at

Plot 'REEF3D-FNPF-WSF-HG.dat' (P 51) against 'REEF3D-FNPF-WSF-HG-THEORY.dat' (P 50) in REEF3D_FNPF_WSF for the gauges at x = 15, 20 and 25 m. The files in REEF3D_FNPF_WSFLINE contain the free surface along the tank at y = 0.005 m every 100 iterations, together with the theoretical profile (P 53 1). In ParaView, open REEF3D_FNPF_VTU (every 0.5 s) and color by Fi or velocity.

11.1.4 Results



(a) Free surface elevation evolution over time at wave gauge 2 (x =20 m)



(b) Free surface elevation in space at t =40 s

Figure 11.1: Free surface elevations in the simulation of 2nd-order Stokes waves in intermediate water depth.

11.2 Wave propagation over a submerged bar

Repository folder: `Tutorials/REEF3D_FNPF/9_2 Wave Propagation Over a Submerged Bar`

Regular waves propagate over a submerged trapezoidal bar following the laboratory experiments of Beji and Battjes [1993]. The bar is built from solid objects in DIVEMesh (front slope x = 11–17 m, crest x = 17–19 m, back slope x = 19–22 m, 0.3 m high), and the sigma grid follows the resulting bed. The generation of higher harmonics on the bar and their release behind it test the nonlinear and dispersive properties of FNPF.

- Wave: 2nd-order Stokes, H = 0.021 m, T = 2.525 s (B 93), water depth d = 0.4 m, 0.1 m on the bar crest
- Domain: 40 m long, 1000 × 1 × 10 cells (B 2), dx = 0.04 m
- Wave generation zone 5 m, numerical beach 10 m (B 96), simulated time 100 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

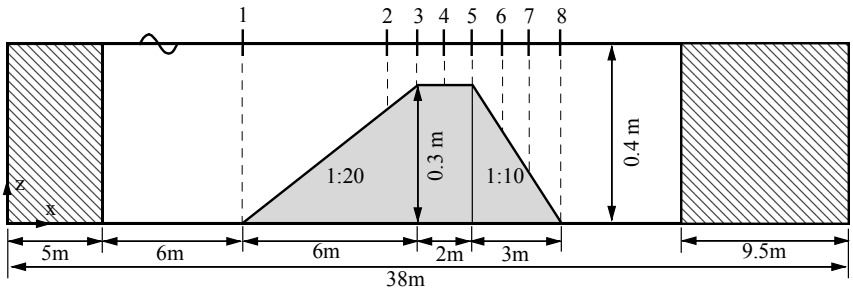


Figure 11.2: Experimental setup submerged bar

11.2.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane

```
C 13 3 // side 3 (y_min): symmetry plane
C 14 7 // side 4 (x_max): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.04 // uniform cell size dx [m]
B 2 1000 1 10 // number of cells N_x, N_y, N_z (overrides B 1)
B 10 0.0 40.0 0.0 0.04 0.0 1.0 // rectangular domain x_s, x_e, y_s, y_e, z_s, z_e [m]

S 61 11.0 17.0 0.0 0.04 0.0 0.3 // front slope of the bar, rising from z = 0 to 0.3 m
S 10 17.0 19.0 0.0 0.04 0.0 0.3 // bar crest, 0.3 m high
S 61 19.0 22.0 0.0 0.04 0.3 0.0 // back slope of the bar, falling from 0.3 m to 0

B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 3.0 // stretching factor in z-direction
B 116 1.0 // focus point of the z-stretching [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)
```

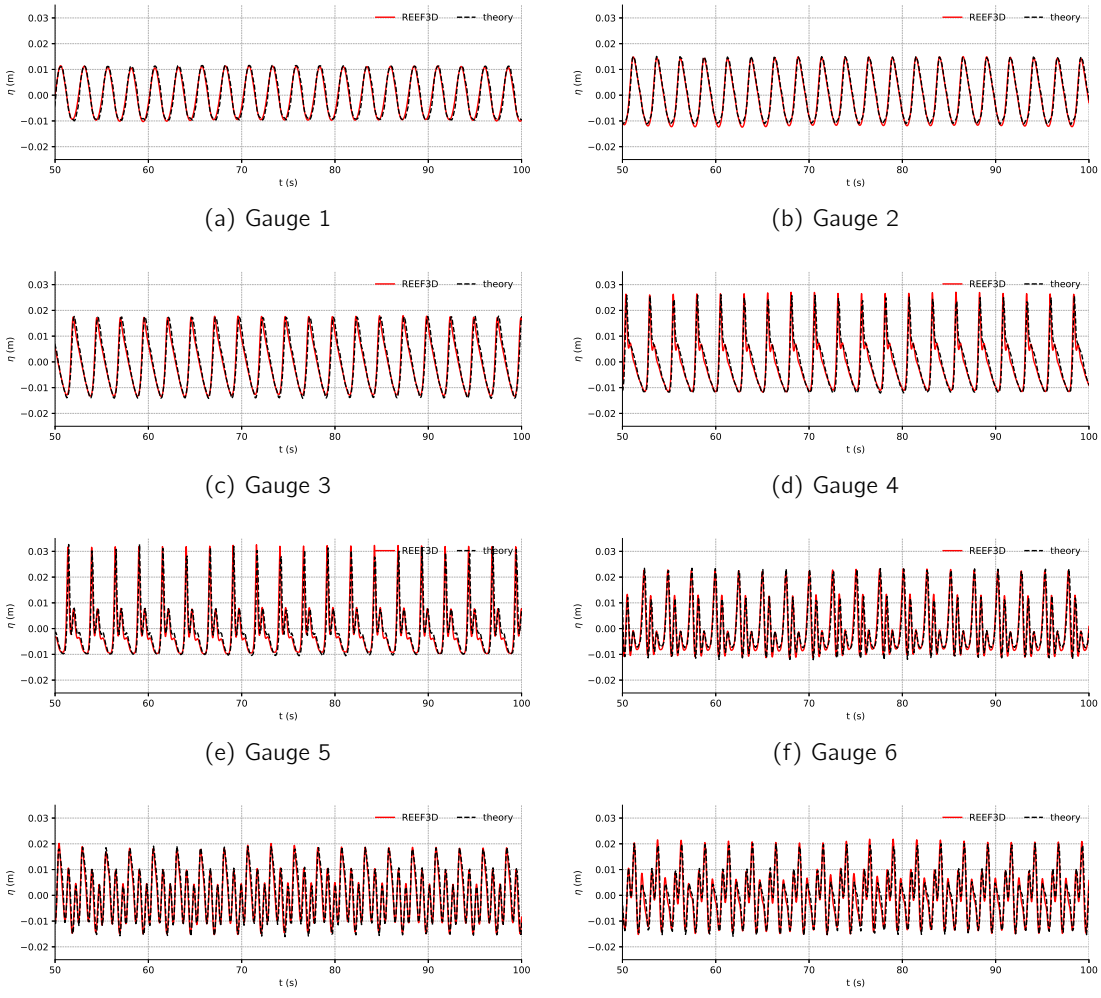
11.2.2 REEF3D: ctrl.txt

```
A 10 3 // hydrodynamic model: REEF3D::FNPF
A 310 3 // FNPF time scheme: 3rd-order TVD Runge-Kutta
A 311 4 // FNPF free surface BCs: 5th-order WENO
A 343 0 // wetting and drying off

B 90 1 // turn on ioWave (numerical wave tank)
B 92 4 // wave type: 2nd-order Stokes waves
B 93 0.021 2.525 // wave height H [m], wave period T [s]
B 96 5.0 10.0 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1

F 60 0.4 // still water level [m]

N 41 100.0 // maximum simulated time [s]
N 47 1.0 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.5 // print interval in simulated time [s]

P 51 11.0 0.005 // wave gauge 1 at x, y [m]
P 51 16.0 0.005 // wave gauge 2 at x, y [m]
P 51 17.0 0.005 // wave gauge 3 at x, y [m]
P 51 18.0 0.005 // wave gauge 4 at x, y [m]
P 51 19.0 0.005 // wave gauge 5 at x, y [m]
P 51 20.0 0.005 // wave gauge 6 at x, y [m]
P 51 21.0 0.005 // wave gauge 7 at x, y [m]
P 51 22.0 0.005 // wave gauge 8 at x, y [m]
P 51 24.0 0.005 // wave gauge 9 at x, y [m]

W 22 -9.81 // gravity in z-direction [m/s^2]
```

11.2.3 What to look at

'REEF3D-FNPF-WSF-HG.dat' in REEF3D_FNPF_WSF contains one column per P 51 gauge (x = 11, 16, 17, 18, 19, 20, 21, 22 and 24 m) in the order given in ctrl.txt; compare them with the measurements of Beji and Battjes [1993]. The figures below show the first eight gauges. In ParaView, REEF3D_FNPF_VTU (every 0.5 s) shows the wave field over the bar.

11.2.4 Results



(a) Gauge 1

(b) Gauge 2

(c) Gauge 3

(d) Gauge 4

(e) Gauge 5

(f) Gauge 6

(g) Gauge 7

(h) Gauge 8

Figure 11.3: Free surface elevations in the simulation of wave propagation over a submerged bar.

11.3 Wave breaking over a mild slope

Repository folder: `Tutorials/REEF3D_FNPF/9_3 Wave Breaking Over a Mild Slope`

Cnoidal waves shoal and break on a mild slope of about 1:35, similar to the plunging breaker experiment of Ting and Kirby [1995] that is also used in the CFD tutorials (shifted by 8 m in x). The waves are generated with the Dirichlet method (B 98 3), which is required in FNPF for wave theories that do not prescribe the velocity potential directly, such as cnoidal waves (see B 92). Breaking is modelled with the viscous damping algorithm (A 350 1) with breaking detection and filtering for shallow and deep water; wetting and drying (A 343 1) handles the shoreline on the slope.

- Wave: 5th-order cnoidal, H = 0.128 m, T = 5.0 s (B 93), water depth d = 0.4 m
- Domain: 30 m long, 600 × 1 × 10 cells (B 2), dx = 0.05 m, slope from x = 5.8 m
- Numerical beach 9 m (B 96), simulated time 160 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

11.3.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.05 // uniform cell size dx [m]
B 2 600 1 10 // number of cells N_x, N_y, N_z (overrides B 1)
B 10 0.0 30.0 0.0 0.05 0.0 0.748 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 1.0 // stretching factor in z-direction
B 116 0.4 // focus point of the z-stretching [m]

S 61 5.8 32.0 0.0 0.05 0.0 0.748 // bed slope (about 1:35) starting at x = 5.8 m

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

11.3.2 REEF3D: ctrl.txt

A 10 3 // hydrodynamic model: REEF3D::FNPF

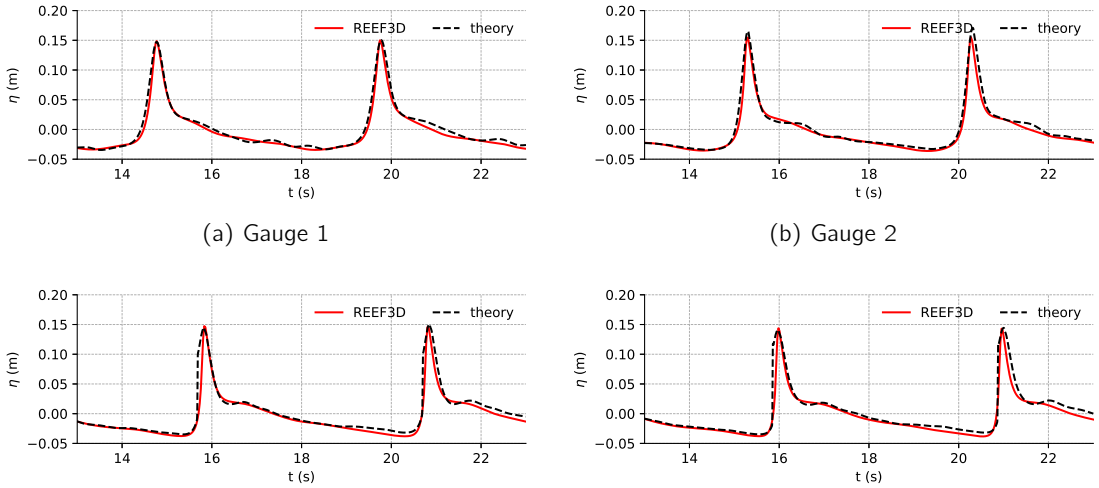
A 341 2.0 // coastline relaxation zone width, factor times dx
A 343 1 // wetting and drying on
A 350 1 // breaking wave algorithm: viscous damping (Baquet)
A 351 3 // breaking detection: shallow and deep water
A 352 3 // additional filtering for breaking in shallow and deep water
A 365 0.0025 // viscosity for breaking wave energy dissipation

```
B 90 1 // turn on ioWave (numerical wave tank)
B 92 8 // wave type: 5th-order cnoidal waves
B 93 0.128 5.0 // wave height H [m], wave period T [s]
B 96 9.5 9.0 // lengths of wave generation zone and numerical beach [m]
B 98 3 // wave generation: Dirichlet (DWG)
B 99 1 // numerical beach: relaxation method 1

F 60 0.4 // still water level [m]

N 41 160.0 // maximum simulated time [s]
N 47 1.0 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 23 1 // add the test array to the ParaView output
P 30 0.1 // print interval in simulated time [s]

P 51 11.8 0.0025 // wave gauge 1 at x, y [m]
P 51 12.8 0.0025 // wave gauge 2 at x, y [m]
P 51 13.8 0.0025 // wave gauge 3 at x, y [m]
P 51 14.1 0.0025 // wave gauge 4 at x, y [m]

W 22 -9.81 // gravity in z-direction [m/s^2]
```

11.3.3 What to look at

Plot the four P 51 gauges at x = 11.8, 12.8, 13.8 and 14.1 m from 'REEF3D-FNPF-WSF-HG.dat' in REEF3D_FNPF_WSF; they lie around the breaking point on the slope. In ParaView, open REEF3D_FNPF_VTU (every 0.1 s) and color the free surface by the breaking flag to see where the breaking algorithm is active; P 23 1 adds the test array to these files.

11.3.4 Results



(a) Gauge 1

(b) Gauge 2

(c) Gauge 3

(d) Gauge 4

Figure 11.4: Free surface elevations in the simulation of breaking waves over a mild slope.

11.4 Irregular wave sea state in 2D

Repository folder: `Tutorials/REEF3D_FNPF/9_4 Irregular Wave Propagation`

A long-crested irregular sea state based on a JONSWAP spectrum is simulated in a 6 km long 2D wave tank in deep water for 12800 s, i.e. more than the typical 3-hour design sea state. The spectrum is discretized into 2048 components with the peak enhance method (B 84 1). A fixed time step is used, and the breaking algorithm with deep water detection keeps steep waves stable. To keep the output small, the .vtu files are only written in the last 50 s (P 35).

- Wave: JONSWAP, H_s = 5.0 m, T_p = 12.0 s, γ = 3.0, ω = 0.32–1.4 rad/s, water depth d = 600 m
- Domain: 6000 m long, 6000 × 1 × 20 cells (B 2), dx = 1.0 m
- Wave generation zone 600 m, numerical beach 1200 m (B 96), fixed time step 0.03 s, simulated time 12800 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

11.4.1 DIVEMesh: control.txt

```
C 11 6 // side 1 (x_min): wave generation
C 12 3 // side 2 (y_max): symmetry plane
C 13 3 // side 3 (y_min): symmetry plane
C 14 7 // side 4 (x_max): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane
```

B 1 1.0 // uniform cell size dx [m]
B 2 6000 1 20 // number of cells N_x, N_y, N_z (overrides B 1)
B 10 0.0 6000.0 0.0 1.0 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 3.0 // stretching factor in z-direction
B 116 1.0 // focus point of the z-stretching [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

11.4.2 REEF3D: ctrl.txt

A 10 3 // hydrodynamic model: REEF3D::FNPF
A 310 3 // FNPF time scheme: 3rd-order TVD Runge-Kutta
A 311 4 // FNPF free surface BCs: 5th-order WENO
A 343 0 // wetting and drying off
A 350 1 // breaking wave algorithm: viscous damping (Baquet)
A 351 2 // breaking detection: deep water
A 352 0 // no additional filtering in breaking regions

B 84 1 // spectrum discretisation: peak enhance method
B 85 2 // JONSWAP spectrum
B 86 2048 // number of wave components
B 87 0.32 1.4 // lower and upper angular frequency ω_s, ω_e [rad/s]
B 88 3.0 // JONSWAP peak enhancement factor γ
B 90 1 // turn on ioWave (numerical wave tank)
B 92 31 // wave type: 1st-order irregular waves
B 93 5.00 12.00 // significant wave height H_s [m], peak period T_p [s]
B 96 600.0 1200.0 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1
b 139 16000 // lower-case letter: ignored (B 139 would set the random seed of the spectrum)

F 60 600.0 // still water level [m]

N 41 12800.0 // maximum simulated time [s]
n 47 1.0 // lower-case letter: ignored (not needed with the fixed time step)
N 48 0 // adaptive time stepping off
N 49 0.03 // fixed time step [s]

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 35 12750.0 12800.0 1.0 // print window: start [s], end [s], interval [s]

P 51 300.00 0.5 // wave gauge 1 at x, y [m]
P 51 1500.0 0.5 // wave gauge 2 at x, y [m]
P 51 3500.0 0.5 // wave gauge 3 at x, y [m]

W 22 -9.81 // gravity in z-direction [m/s^2]

11.4.3 What to look at

The main result is the surface elevation time series in 'REEF3D-FNPF-WSF-HG.dat' (REEF3D_FNPF_WSF) at x = 300, 1500 and 3500 m. Compute the spectrum of a gauge time series (discarding the initial ramp-up) and compare it with the input JONSWAP spectrum, see the figure below for gauge 2. REEF3D_FNPF_VTU contains snapshots every 1 s between 12750 s and 12800 s.

11.4.4 Results

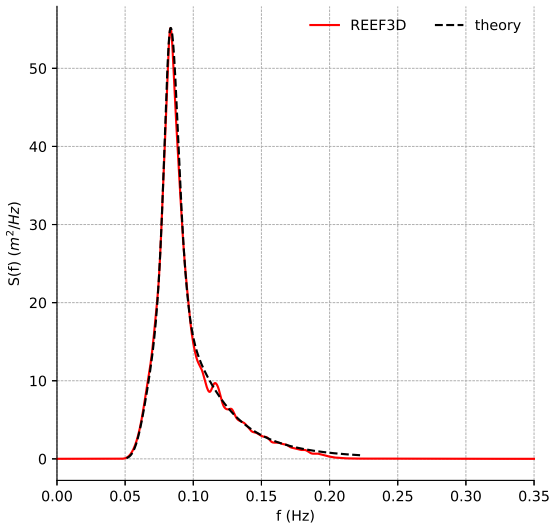


Figure 11.5: Wave spectrum in the simulation of a 3-hour irregular sea state at wave gauge 2.

11.5 Wave propagation over natural bathymetry

Repository folder: `Tutorials/REEF3D_FNPF/9_5 Wave Propagation Over Natural Bathymetry`

Regular waves propagate over a measured natural bathymetry in a 4 km × 4 km domain. DIVEMesh interpolates the bed from the point cloud in 'geo.dat' (about 10^6 x, y, z points, required in the case folder) with local inverse distance interpolation. Waves are generated along part of the left boundary (B 108), and relaxation beaches are placed along the remaining open boundaries (B 107). Coastlines are treated with wetting-drying and a coastline relaxation zone, and breaking with the viscous damping algorithm including filtering.

- Wave: 2nd-order Stokes, H = 2.0 m, T = 10.0 s (B 93), still water level z = 300 m (F 60), bed from geo.dat
- Domain: 4000 m × 4000 m, 800 × 800 × 10 cells (B 2), dx = 5.0 m
- Relaxation zones 200 m wide (B 107, B 108), simulated time 600 s
- Extra input file: geo.dat (read by DIVEMesh)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

11.5.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 5.0 // uniform cell size dx [m]
B 2 800 800 10 // number of cells N_x, N_y, N_z (overrides B 1)
B 10 0.0 4000.0 0.0 4000.0 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 2.5 // stretching factor in z-direction
B 116 1.0 // focus point of the z-stretching [m]

G 10 1 // turn on geodat interpolation of the bed from geo.dat
G 15 2 // local inverse distance interpolation
G 20 0 // automatic grid size from geo.dat extent: off
G 31 14 // number of outer smoothing filter iterations
G 41 1 // print SWAN bottom file from the interpolated bed

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

11.5.2 REEF3D: ctrl.txt

A 10 3 // hydrodynamic model: REEF3D::FNPF
A 310 3 // FNPF time scheme: 3rd-order TVD Runge-Kutta
A 311 5 // FNPF free surface BCs: 5th-order WENO

A 341 2.0 // coastline relaxation zone width, factor times dx
A 343 1 // wetting and drying on
A 344 0.1 // wetting-drying threshold water depth [m]
A 346 2.1 // damping viscosity in the coastline relaxation zone

A 350 1 // breaking wave algorithm: viscous damping (Baquet)
A 351 3 // breaking detection: shallow and deep water
A 352 3 // additional filtering for breaking in shallow and deep water
A 361 5 // breaking filter: outer iterations
A 362 2 // breaking filter: inner iterations
A 365 1.86 // viscosity for breaking wave energy dissipation

B 90 1 // turn on ioWave (numerical wave tank)
B 92 4 // wave type: 2nd-order Stokes waves
B 93 2.0 10.0 // wave height H [m], wave period T [s]
B 96 200.0 400.0 // lengths of wave generation zone and numerical beach [m]
B 107 0.0 4000.0 0.0 0.0 200.0 // beach zone along the side $y = 0$, 200 m wide

B 107 0.0 4000.0 4000.0 4000.0 200.0 // beach zone along the side y = 4000 m
B 107 4000.0 4000.0 0.0 4000.0 200.0 // beach zone along the end x = 4000 m
B 107 0.0 0.0 2900.0 3500.0 200.0 // beach zone on the inflow side x = 0 for y = 2900–3500 m
B 108 0.0 0.0 200.0 2900.0 200.0 // wave generation zone on x = 0 for y = 200–2900 m
B 98 2 // wave generation: relaxation method 2
B 99 2 // numerical beach: relaxation method 2

F 60 300.0 // still water level [m]

I 30 0 // full wave tank initialization off

N 41 600.0 // maximum simulated time [s]
N 47 1.0 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 10.0 // print interval in simulated time [s]

P 51 2000.0 2000.0 // wave gauge at the domain centre x = y = 2000 m

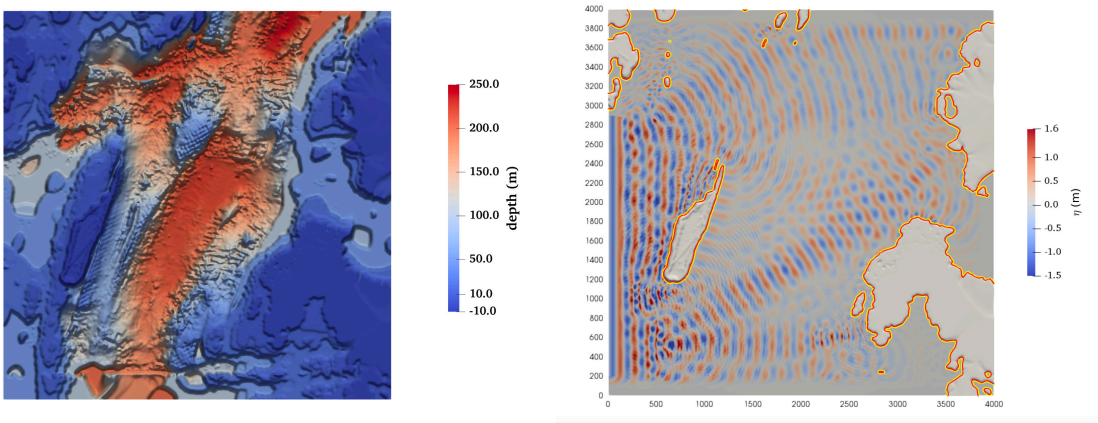
P 180 1 // print free surface .vtp files
P 185 0.0 120.0 0.1 // free surface .vtp print window: start [s], end [s], interval [s]
P 185 480.0 600.0 0.1 // free surface .vtp print window: start [s], end [s], interval [s]

W 22 -9.81 // gravity in z-direction [m/s^2]

11.5.3 What to look at

Open REEF3D_FNPF_VTP_FSF in ParaView: free surface files are written every 0.1 s in the windows 0–120 s and 480–600 s (P 185); color by eta to see refraction and shoaling over the bathymetry, and by breaking and coastline to see where breaking and wetting-drying are active. The bed is in REEF3D_FNPF_VTP_BED, and REEF3D_FNPF_VTU contains the full field every 10 s. The single P 51 gauge at the domain centre is written to REEF3D_FNPF_WSF.

11.5.4 Results



(a) Bathymetry

(b) Free surface elevation

Figure 11.6: Free surface elevations in the simulation of wave propagation over natural bathymetry

11.6 3D short-crested multi-directional irregular wave

Repository folder: `Tutorials/REEF3D_FNPF/9_6 Short Crested Multi-directional Irregular Wave`

A short-crested irregular sea state is generated in a 2 km × 2 km basin in deep water. The JONSWAP spectrum is discretized with the equal energy method (B 84 2) and spread over 16 directions with a Mitsuyasu spreading function around the principal direction 0° (B 130–B 136). An array of six wave gauges (one centre gauge and five gauges on a circle of about 28 m radius) is used to estimate the directional spectrum. I 30 1 initializes the whole tank with the wave field, which is useful for a quick check of the setup; for production runs I 30 0 is recommended.

- Wave: JONSWAP, H_s = 4.5 m, T_p = 12.0 s, γ = 3.3, ω = 0.39–1.01 rad/s, spreading parameter 10, water depth d = 600 m
- Domain: 2000 m × 2000 m, 1000 × 1000 × 10 cells (B 2), dx = 2.0 m
- Wave generation zone 400 m, numerical beach 800 m (B 96), fixed time step 0.08 s, simulated time 12800 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

11.6.1 DIVEMesh: control.txt

```
C 11 6 // side 1 (x_min): wave generation
C 12 3 // side 2 (y_max): symmetry plane
C 13 3 // side 3 (y_min): symmetry plane
C 14 7 // side 4 (x_max): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane
```

B 1 2.0 // uniform cell size dx [m]
B 2 1000 1000 10 // number of cells N_x, N_y, N_z (overrides B 1)
B 10 0.0 2000.0 0.0 2000.0 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 2.875 // stretching factor in z-direction
B 116 1.0 // focus point of the z-stretching [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

11.6.2 REEF3D: ctrl.txt

A 10 3 // hydrodynamic model: REEF3D::FNPF
A 310 3 // FNPF time scheme: 3rd-order TVD Runge-Kutta
A 311 4 // FNPF free surface BCs: 5th-order WENO
A 343 0 // wetting and drying off
A 350 1 // breaking wave algorithm: viscous damping (Baquet)
A 351 2 // breaking detection: deep water
A 352 0 // no additional filtering in breaking regions
A 365 1.86 // viscosity for breaking wave energy dissipation

B 84 2 // spectrum discretisation: equal energy method
B 85 2 // JONSWAP spectrum
B 86 1024 // number of wave components
B 87 0.39 1.01 // lower and upper angular frequency ω_s, ω_e [rad/s]
B 88 3.3 // JONSWAP peak enhancement factor γ
B 90 1 // turn on ioWave (numerical wave tank)
B 92 31 // wave type: 1st-order irregular waves
B 93 4.5 12.0 // significant wave height H_s [m], peak period T_p [s]
B 96 400.0 800.0 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1
B 130 2 // directional spreading: Mitsuyasu
B 131 0.0 // principal wave direction [deg]
B 132 -90.0 90.0 // start and end angle of the spreading [deg]
B 133 16 // number of directions
B 134 10.0 // spreading parameter of the spreading function
B 136 4 // multi-directional method: equal energy method (with B 84 2)

F 60 600.0 // still water level [m]

I 30 1 // full wave tank initialization on (quick setup check)

N 41 12800.0 // maximum simulated time [s]
N 48 0 // adaptive time stepping off
N 49 0.08 // fixed time step [s]

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 35 12700.0 12800.0 0.5 // print window: start [s], end [s], interval [s]

P 51 800.0 1000.0 // wave gauge array for the directional analysis: gauge 1
P 51 777.2464 983.4685 // gauge 2
P 51 808.6911 973.2515 // gauge 3
P 51 828.1250 1000.0000 // gauge 4
P 51 808.6911 1026.7485 // gauge 5
P 51 777.2464 1016.5315 // gauge 6

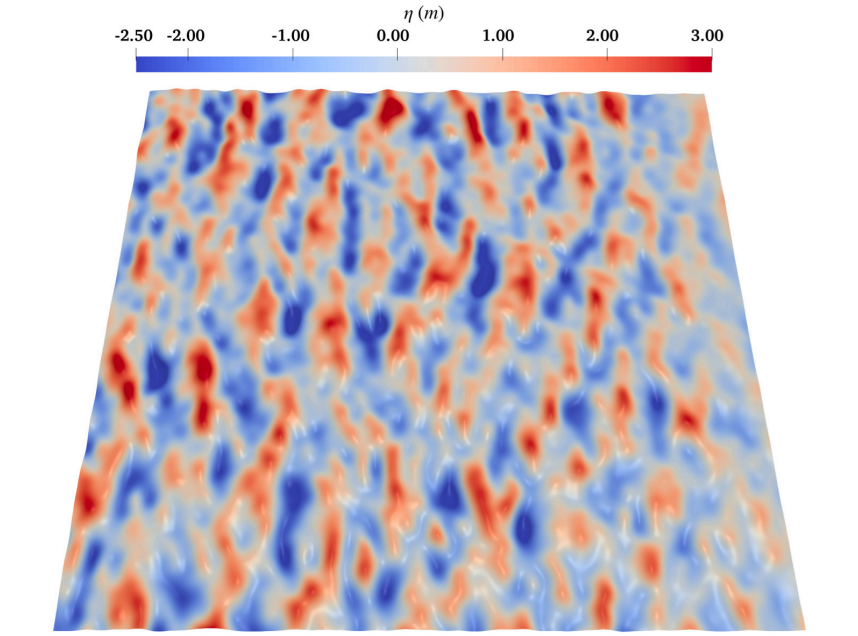
P 180 1 // print free surface .vtp files
P 185 0.0 300.0 0.5 // free surface .vtp print window: start [s], end [s], interval [s]
P 185 12500.0 12800.0 0.5 // free surface .vtp print window: start [s], end [s], interval [s]

W 22 -9.81 // gravity in z-direction [m/s^2]

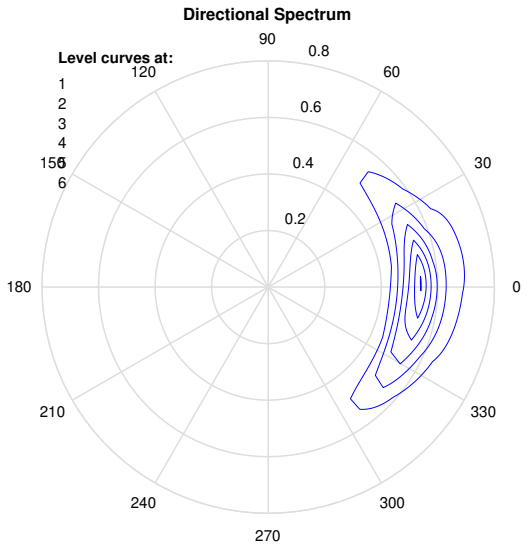
11.6.3 What to look at

Use the six columns of 'REEF3D-FNPF-WSF-HG.dat' in REEF3D_FNPF_WSF to estimate the directional spectrum and compare it with the input spreading. REEF3D_FNPF_VTP_FSF contains the free surface every 0.5 s for 0–300 s and 12500–12800 s (P 185), which shows the short-crested wave field; REEF3D_FNPF_VTU contains snapshots every 0.5 s between 12700 s and 12800 s (P 35).

11.6.4 Results



(a) Free surface elevation



(b) Directional spectrum

Figure 11.7: Free surface elevations and directional spectrum in the simulation of a short-crested irregular sea.

12. Tutorial | REEF3D::NHFLOW

REEF3D::NHFLOW is a three-dimensional non-hydrostatic wave model on a sigma grid that follows the bed and the free surface; the free surface is computed with a shock-capturing Godunov-type scheme and the dynamic pressure with a pressure correction. In contrast to the depth-averaged SFLOW, NHFLOW resolves the vertical flow structure, typically with only a few vertical cells. For visualization, open the REEF3D_NHFLOW_VTU folder (.pvtu files with velocity, pressure and elevation) and, where P 180 is set, the REEF3D_NHFLOW_VTP_FSF folder (free surface: eta, velocity, breaking, ...); wave gauge time series are written to REEF3D_NHFLOW_WSF. As in FNPF, the vertical extent of the DIVEMesh domain is normalized to the sigma coordinate: the water depth follows from F 60 and the bed, and B 2 sets the number of vertical cells. Run DIVEMesh (**./DiveMESH**) in the case folder first and then **mpirun -n** *M 10* **reef3d**.

12.1 5th-order Stokes waves

Repository folder: `Tutorials/REEF3D_NHFLOW/1_Stokes5`

Regular 5th-order Stokes waves propagate over a flat bed in a 200 m long 2D wave tank; the setup is the same as the CFD tutorial for 5th-order Stokes waves (`11_9`), which allows a direct comparison of the two models. The waves are generated in a relaxation zone of about one wavelength and absorbed in a relaxation beach of about two wavelengths. The vertical grid has only five cells, clustered towards the free surface (B 103, B 113, B 116). Computed and theoretical surface elevations are written both at gauges and along the tank.

- Wave: H = 1.0 m, T = 4.5 s (B 93), water depth d = 4.01 m (F 60), wavelength about 25 m
- Domain: 200 m long, 350 × 1 × 5 cells (B 2 overrides B 1), $dx \approx 0.57$ m
- Wave generation zone 25 m, numerical beach 50 m (B 96), simulated time 120 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

12.1.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.05 // uniform cell size dx [m]
B 2 350 1 5 // number of cells N_x, N_y, N_z (overrides B 1)

B 10 0.0 200.0 0.0 0.05 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]
B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 2.5 // stretching factor in z-direction
B 116 1.0 // focus point of the z-stretching [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

12.1.2 REEF3D: ctrl.txt

A 10 5 // hydrodynamic model: REEF3D::NHFLOW

B 90 1 // turn on ioWave (numerical wave tank)
B 92 5 // wave type: 5th-order Stokes waves (Fenton)
B 93 1.0 4.5 // wave height H [m], wave period T [s]
B 96 25.0 50.0 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1

F 60 4.01 // still water level [m]

I 12 1 // hydrostatic pressure initialization (CFD only, no effect in NHFLOW)

N 41 120.0 // maximum simulated time [s]
N 47 1.0 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.25 // print interval in simulated time [s]
P 50 15.0 0.005 // theoretical wave gauge 1 at x, y [m]
P 50 50.0 0.005 // theoretical wave gauge 2 at x, y [m]
P 50 75.0 0.005 // theoretical wave gauge 3 at x, y [m]
P 50 125.0 0.005 // theoretical wave gauge 4 at x, y [m]

P 51 15.0 0.005 // wave gauge 1 at x, y [m]
P 51 50.0 0.005 // wave gauge 2 at x, y [m]
P 51 75.0 0.005 // wave gauge 3 at x, y [m]
P 51 125.0 0.005 // wave gauge 4 at x, y [m]

P 52 0.005 // free surface line (wsfline) along x at this y [m]
P 53 1 // add the theoretical wave to the wsfline files
P 55 0.05 // wsfline and line probe print interval [s]

P 180 1 // print free surface .vtp files
P 182 0.05 // free surface .vtp print interval [s]

W 22 -9.81 // gravity in z-direction [m/s^2]

12.1.3 What to look at

Plot 'REEF3D-NHFLOW-WSF-HG.dat' (P 51) against 'REEF3D-NHFLOW-WSF-HG-THEORY.dat' (P 50) in REEF3D_NHFLOW_WSF at x = 15, 50, 75 and 125 m; compare crest height and phase with the theory along the tank. REEF3D_NHFLOW_WSFLINE contains the free surface along the tank with the theoretical profile every 0.05 s (P 52, P 53, P 55). In ParaView, open REEF3D_NHFLOW_VTP_FSF (every 0.05 s) for the free surface and REEF3D_NHFLOW_VTU (every 0.25 s) for the velocity field under the waves.

12.2 Wave propagation over a submerged bar

Repository folder: `Tutorials/REEF3D_NHFLOW/2_Submerged Bar`

Regular waves propagate over a submerged trapezoidal bar following the laboratory experiments of Beji and Battjes [1993], with the same geometry as the CFD tutorial (11_10): front slope x = 6–12 m, crest x = 12–14 m, back slope x = 14–17 m, 0.3 m high. The case tests how well NHFLOW with three vertical cells represents the generation of higher harmonics on the bar and their release behind it. Nine gauges are placed along the tank.

- Wave: linear, H = 0.02 m, T = 2.02 s (B 93), water depth d = 0.4 m, 0.1 m on the bar crest
- Domain: 35 m long, 1200 × 1 × 3 cells (B 2), $dx \approx 0.029$ m
- Wave generation zone 3.73 m, numerical beach 8.73 m (B 96), simulated time 60 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

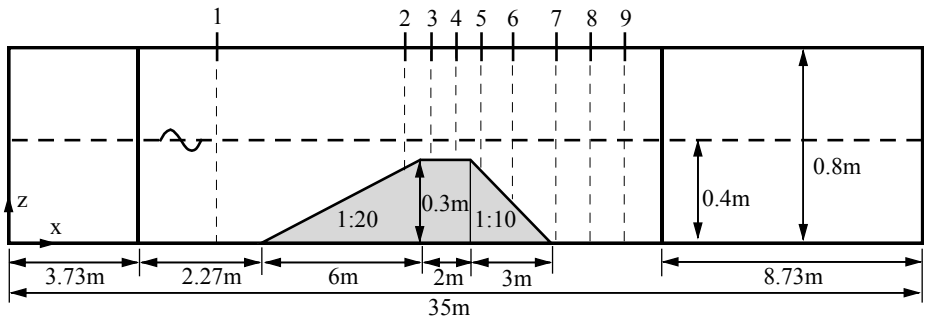


Figure 12.1: Experimental setup submerged bar

12.2.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach

C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 2 1200 1 3 // number of cells N_x, N_y, N_z (overrides B 1)
B 10 0.0 35.0 0.0 0.01 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 61 6.0 12.0 0.0 0.01 0.0 0.3 // front slope of the bar, rising from $z = 0$ to 0.3 m
S 10 12.0 14.0 0.0 0.01 0.0 0.3 // bar crest, 0.3 m high
S 61 14.0 17.0 0.0 0.01 0.3 0.0 // back slope of the bar, falling from 0.3 m to 0

B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 2.5 // stretching factor in z-direction
B 116 1.0 // focus point of the z-stretching [m]

M 10 6 // number of processes (partitions)

12.2.2 REEF3D: ctrl.txt

A 10 5 // hydrodynamic model: REEF3D::NHFLOW

B 90 1 // turn on ioWave (numerical wave tank)
B 92 2 // wave type: linear waves
B 93 0.02 2.02 // wave height H [m], wave period T [s]
B 96 3.73 8.73 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 2 // numerical beach: relaxation method 2

F 60 0.4 // still water level [m]

N 41 60.0 // maximum simulated time [s]
N 47 1.0 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.5 // print interval in simulated time [s]

W 22 -9.81 // gravity in z-direction [m/s^2]

P 51 4.0 0.005 // wave gauge 1 at x, y [m]
P 51 10.5 0.005 // wave gauge 2 at x, y [m]
P 51 12.5 0.005 // wave gauge 3 at x, y [m]
P 51 13.5 0.005 // wave gauge 4 at x, y [m]
P 51 14.5 0.005 // wave gauge 5 at x, y [m]
P 51 15.7 0.005 // wave gauge 6 at x, y [m]
P 51 17.3 0.005 // wave gauge 7 at x, y [m]
P 51 19.0 0.005 // wave gauge 8 at x, y [m]
P 51 21.0 0.005 // wave gauge 9 at x, y [m]

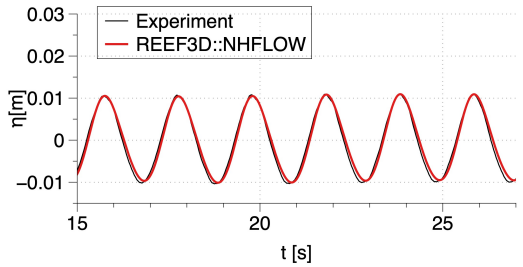
P 54 100 // wsfline interval in iterations; no effect here, no P 52 is given

12.2.3 What to look at

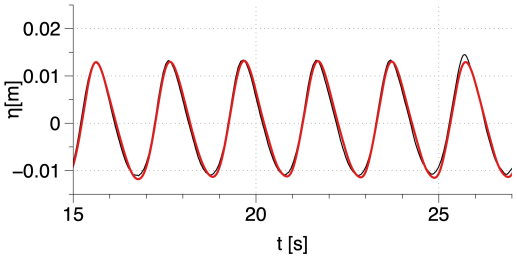
'REEF3D-NHFLOW-WSF-HG.dat' in REEF3D_NHFLOW_WSF contains one column per P 51 gauge (x = 4.0, 10.5, 12.5, 13.5, 14.5, 15.7, 17.3, 19.0 and 21.0 m) in the order given in ctrl.txt; compare them with the measurements of Beji and Battjes [1993], see the figures below. REEF3D_NHFLOW_VTU (every 0.5 s) shows the flow field over the bar. The P 54 line has no effect because no free surface line (P 52) is defined.

12.2.4 Results

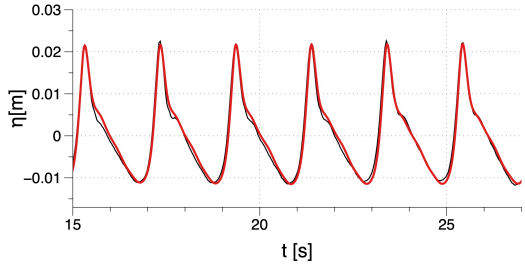
Behind the bar, the bound higher harmonics are released as free waves that are in relatively deep water. NHFLOW is a three-dimensional sigma-grid model, not a depth-averaged one, but with only three vertical cells (B 2) its dispersion accuracy for these short waves is limited, which explains the discrepancies with the experiment at wave gauges 6 and onwards. Be aware of the relative water depth of all relevant wave components when choosing the number of vertical cells.



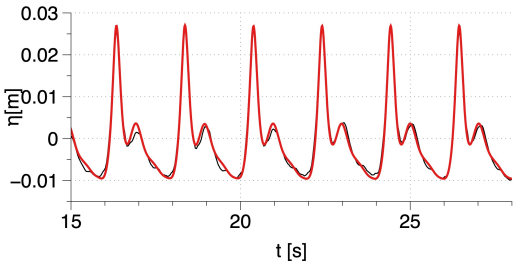
(a) Gauge 1



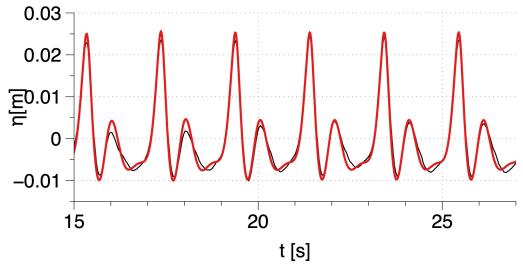
(b) Gauge 2



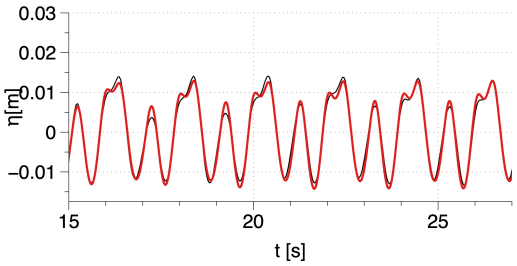
(c) Gauge 3



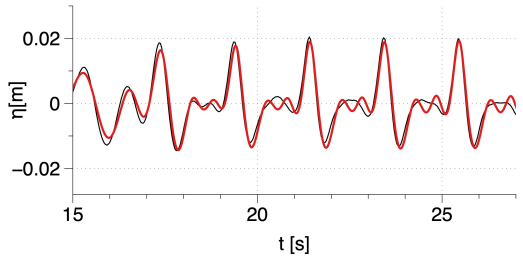
(d) Gauge 4



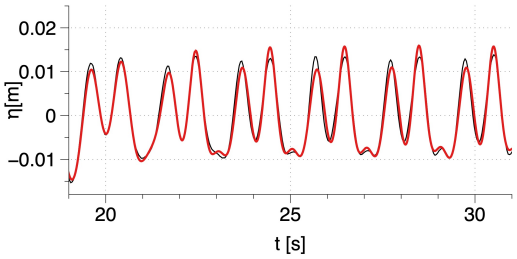
(e) Gauge 5



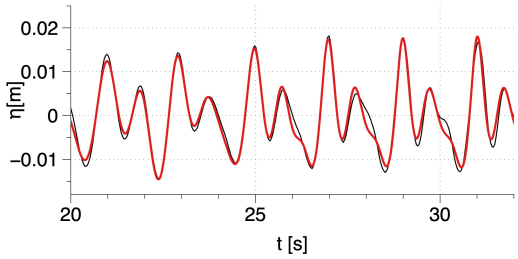
(f) Gauge 6



(g) Gauge 7



(h) Gauge 8



(i) Gauge 9

Figure 12.2: Free surface elevations in the simulation of wave propagation over a submerged bar.

13. Tutorial | REEF3D::CFD

REEF3D::CFD solves the three-dimensional incompressible Navier-Stokes equations on a staggered rectilinear grid, with the level set method for the free surface in two-phase flow. It is used for violent free surface flows, breaking waves, wave forces on structures, floating bodies and sediment transport. For visualization, open the REEF3D_CFD_VTU folder (.pvtu files with velocity, pressure, eddyv, phi, ...) in ParaView; the free surface is the zero contour of the level set function phi, see Chapter 9. Depending on the P commands, further results are written to REEF3D_CFD_WSF (wave gauges), REEF3D_CFD_WSFLINE (free surface lines), REEF3D_CFD_ProbePoint and REEF3D_CFD_ProbeLine (probes), REEF3D_CFD_Force (forces), REEF3D_CFD_FSF (free surface .vtp) and REEF3D_CFD_6DOF (floating bodies). Each tutorial folder contains 'control.txt' for DIVEMesh and 'ctrl.txt' for REEF3D; run DIVEMesh (**./DiveMESH**) first and then **mpirun -n** *M 10* **reef3d**. Commands that are not given in ctrl.txt take their default values (e.g. D 10 4, D 20 2, D 30 1, F 30 3, F 40 3, N 40 3, P 10 1).

13.1 2D Dam Break

Repository folder: `Tutorials/REEF3D_CFD/11_1 2D Dam Break`

A water column of 0.15 m width and 0.3 m height (F 54, F 56) collapses in a closed 2D box and hits the opposite wall. The case demonstrates the two-phase flow capabilities of REEF3D::CFD with the level set method for a strongly deforming free surface. All boundaries are walls, and no turbulence model is used. Comments to the control files are marked by blue fonts.

- Domain: 0.6 m × 0.005 m × 0.6 m, cell size dx = 0.005 m (120 × 1 × 120 cells)
- Water column: $x \leq 0.15$ m, $z \leq 0.3$ m; the run stops after 25000 iterations (N 45)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.1.1 DIVEMesh: control.txt

C 11 21 // side 1 (x_{min}): wall
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 21 // side 4 (x_{max}): wall
C 15 21 // side 5 (bottom): wall
C 16 21 // side 6 (top): wall

B 1 0.005 // uniform cell size dx [m]
B 10 0.0 0.6 0.0 0.005 0.0 0.6 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

M 10 6 // number of processes (partitions)

13.1.2 REEF3D: ctrl.txt

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 50 4 // level set not fixed at inflow or outflow
F 54 0.15 // water column ends at this x-coordinate [m]
F 56 0.3 // water column ends at this z-coordinate [m]

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 45 25000 // maximum number of iterations
N 47 0.1 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

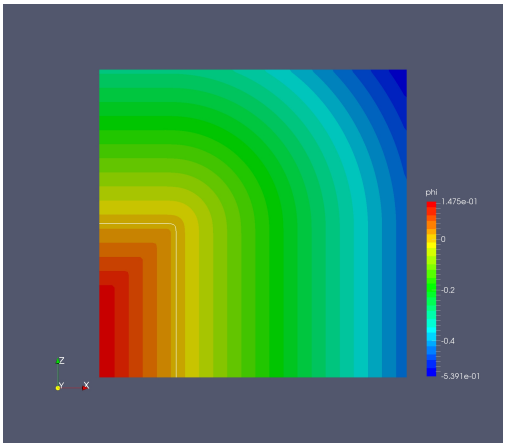
P 10 1 // print ParaView files (.vtu)
P 30 0.01 // print interval in simulated time [s]

W 22 -9.81 // gravity in z-direction [m/s^2]

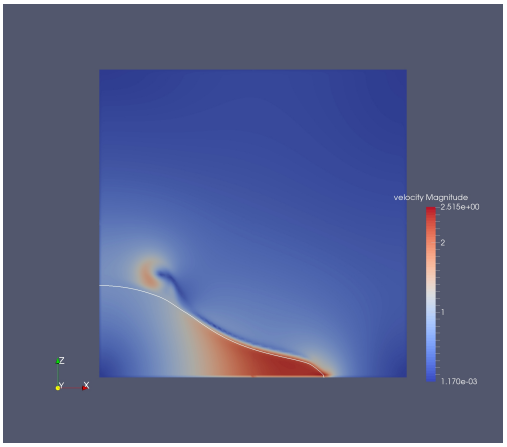
13.1.3 What to look at

Open REEF3D_CFD_VTU in ParaView (one file every 0.01 s), extract the free surface as the contour phi = 0 and color it or a slice by velocity, as in the figure below.

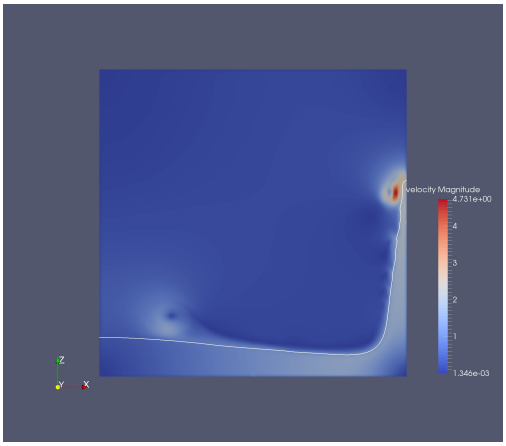
13.1.4 Results



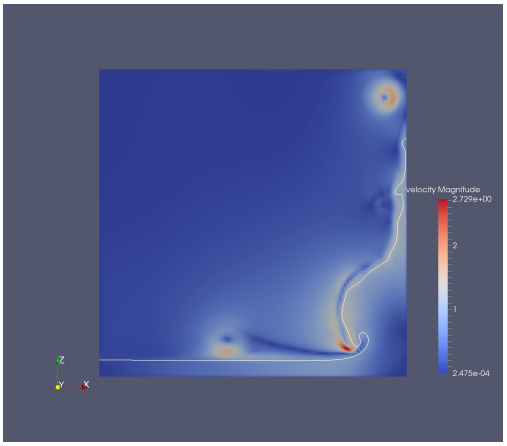
(a) Initial Condition with level set function



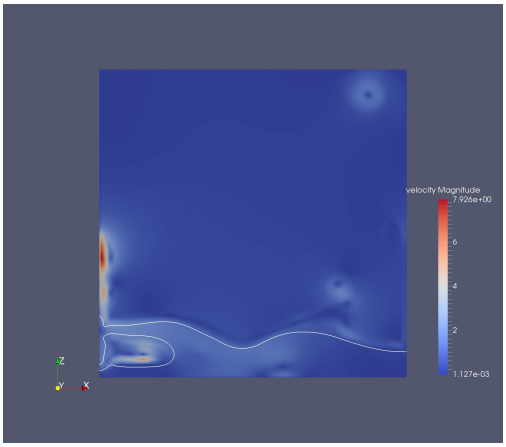
(b) after t = 0.2 sec with velocity magnitude



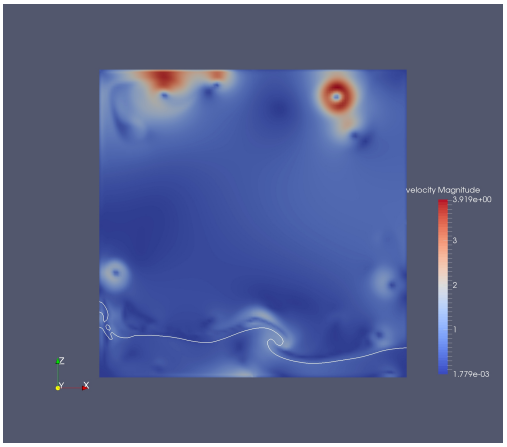
(c) after t = 0.4 sec with velocity magnitude



(d) after t = 0.7 sec with velocity magnitude



(e) after t = 1.0 sec with velocity magnitude



(f) after t = 1.5 sec with velocity magnitude

Figure 13.1: Dam Break

13.2 3D Dam Break with Obstacle

Repository folder: `Tutorials/REEF3D_CFD/11_2 3D Dam Break with Obstacle`

A water column over the full width of a closed 3D tank collapses and hits a rectangular obstacle. The flow splits around the obstacle and runs up on it, which gives a fully three-dimensional free surface. This case is optimized for fast processing. In order to get a more detailed free surface result, change B 1 to e.g. 0.0125.

- Domain: 2.0 m × 1.0 m × 1.0 m, cell size dx = 0.025 m (80 × 40 × 40 cells)
- Water column: $x \leq 0.5$ m, $z \leq 0.7$ m; obstacle x = 1.2–1.4 m, y = 0.4–0.6 m
- Simulated time 25 s (N 41), at most 50000 iterations (N 45)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.2.1 DIVEMesh: control.txt

C 11 21 // side 1 (x_{min}): wall
C 12 21 // side 2 (y_{max}): wall
C 13 21 // side 3 (y_{min}): wall
C 14 21 // side 4 (x_{max}): wall
C 15 21 // side 5 (bottom): wall
C 16 21 // side 6 (top): wall

B 1 0.025 // uniform cell size dx [m]
B 10 0.0 2.0 0.0 1.0 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 10 1.2 1.4 0.4 0.6 0.0 1.0 // rectangular obstacle (0.2 m × 0.2 m, full height)

M 10 6 // number of processes (partitions)

13.2.2 REEF3D: ctrl.txt

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 54 0.5 // water column ends at this x-coordinate [m]
F 56 0.7 // water column ends at this z-coordinate [m]

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 25.0 // maximum simulated time [s]
N 45 50000 // maximum number of iterations
N 47 0.2 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.01 // print interval in simulated time [s]

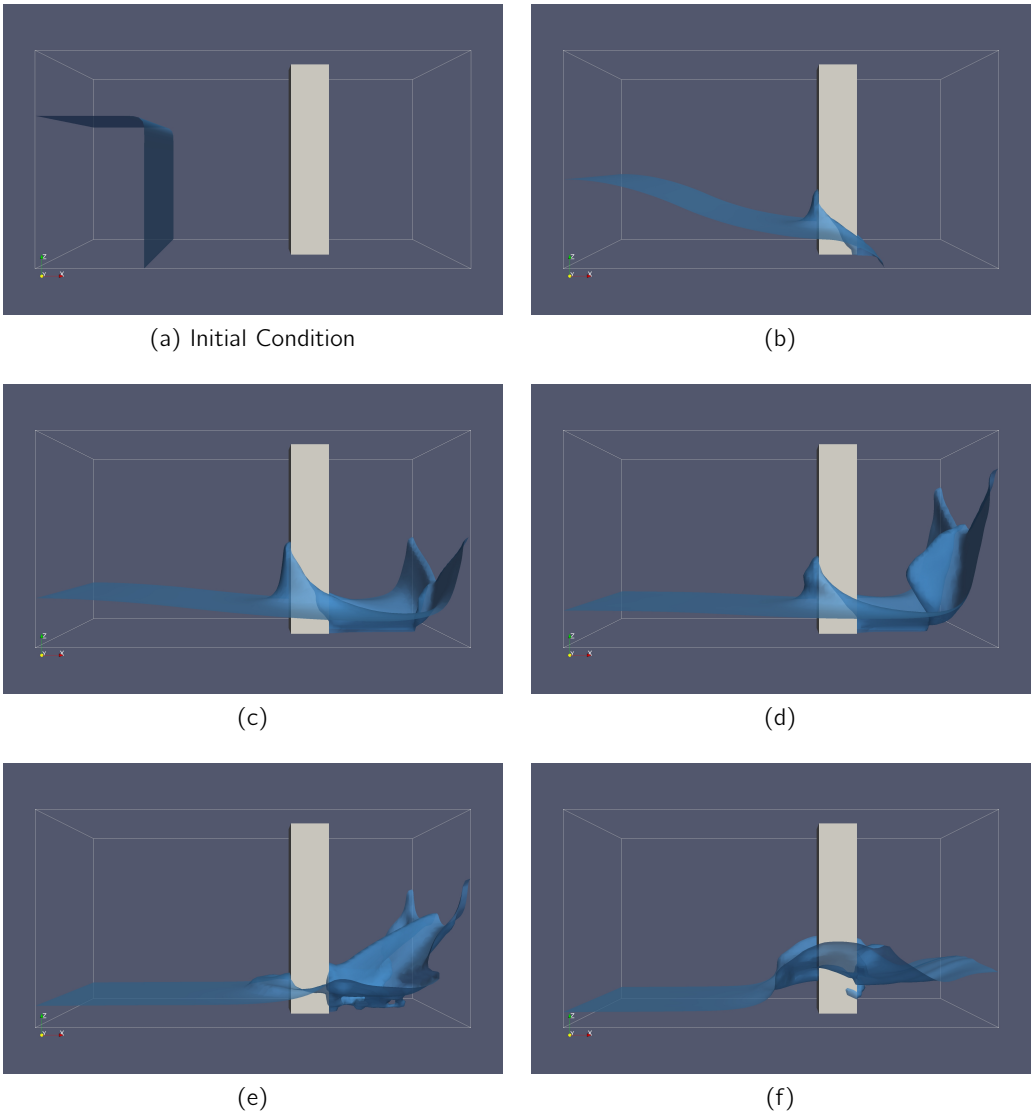
T 10 0 // no turbulence model

W 22 -9.81 // gravity in z-direction [m/s^2]

13.2.3 What to look at

Open REEF3D_CFD_VTU (one file every 0.01 s) and apply a contour filter for phi = 0 directly on the 3D data to see the free surface around the obstacle.

13.2.4 Results



(a) Initial Condition (b)

(c) (d)

(e) (f)

Figure 13.2: 3D Dam Break with Obstacle

13.3 2D Vortex Shedding

Repository folder: `Tutorials/REEF3D_CFD/11_3 2D Vortex Shedding`

Single-phase flow (F 30 0, F 40 0) passes a circular cylinder in a 2D channel with walls at the bottom and top. The inflow discharge is prescribed with ioFlow (B 60 1, W 10), and the velocities are initialized with the potential flow solver (I 11 1). With the mean velocity of $6 \cdot 10^{-4}$ m/s, the cylinder diameter of 0.2 m and the default water viscosity, the Reynolds number is about 120, so a laminar vortex street develops behind the cylinder; no turbulence model is used.

- Domain: 4.0 m × 0.01 m × 1.0 m, cell size dx = 0.01 m (400 × 1 × 100 cells)
- Cylinder: x = 1.0 m, z = 0.5 m, radius 0.1 m; discharge $6 \cdot 10^{-6}$ m^3/s
- 10000 iterations (N 45), output every 10 iterations (P 20)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.3.1 DIVEMesh: control.txt

C 11 1 // side 1 (x_{min}): inflow
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 2 // side 4 (x_{max}): outflow
C 15 21 // side 5 (bottom): wall
C 16 21 // side 6 (top): wall

B 1 0.01 // uniform cell size dx [m]
B 10 0.0 4.0 0.0 0.01 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 32 1.0 0.5 0.1 // cylinder in y-direction at x = 1.0 m, z = 0.5 m, radius 0.1 m

M 10 6 // number of processes (partitions)

13.3.2 REEF3D: ctrl.txt

B 10 0 // wall functions for the velocities off
B 50 0.0001 // wall roughness k_s [m]
B 60 1 // ioFlow: constant inflow discharge from W 10

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 1 // diffusion: explicit
D 30 1 // pressure: projection method

F 30 0 // level set transport off (single-phase flow)
F 40 0 // reinitialization off

I 11 1 // initialize velocities with the potential flow solver

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 45 10000 // maximum number of iterations
N 47 0.3 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 20 10 // print interval in iterations
P 62 0.0 0.0 0.005 0.005 0.0 1.0 // vertical line probe at the inlet x = 0 m

T 10 0 // no turbulence model

W 10 0.000006 // inflow discharge [m^3/s]

13.3.3 What to look at

Open REEF3D_CFD_VTU and color by velocity or pressure; streamlines or line integral convolution of the velocity show the vortex street as in the figure below. The vertical line probe at the inlet (P 62) is written to REEF3D_CFD_ProbeLine every 10 iterations (default P 54) and shows the inflow profile.

13.3.4 Results

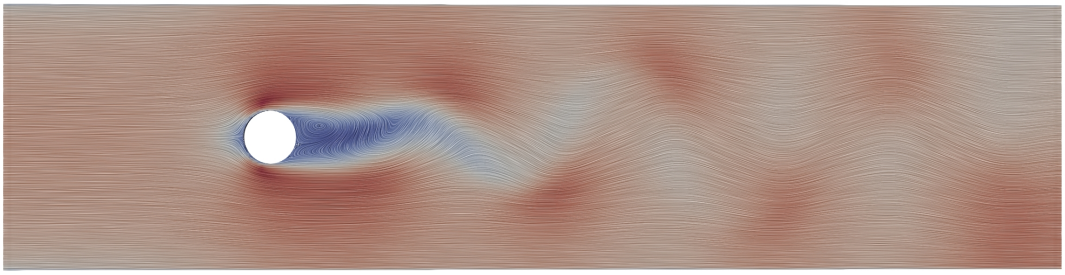


Figure 13.3: 2D vortex shedding with stream lines

13.4 Flow through a Narrow Contraction

Repository folder: `Tutorials/REEF3D_CFD/11_4 Narrow Contraction`

For the flow through a narrow contraction the open channel flow capabilities in ioFlow are used: a constant discharge enters at the inflow boundary and the water level is fixed at the outflow (F 50 2). The channel narrows from 0.6 m to 0.2 m width between x = 0.8 m and 1.1 m, stays narrow up to x = 1.4 m and widens again up to x = 1.7 m; the geometry is built from wedges (S 83) and boxes (S 10). The flow is initialized with the potential flow solver (I 10 1) and the k-ω model is used. Just for information: the grid is rather coarse for this flow situation. The grid can be easily refined by changing the B 1 parameter in the DIVEMesh 'control.txt' file, depending on the performance of your computer.

- Domain: 2.5 m × 0.6 m × 0.5 m, cell size dx = 0.025 m (100 × 24 × 20 cells)
- Water depth 0.15 m (F 60), discharge 0.05 m^3/s (W 10), i.e. about 0.56 m/s upstream
- 12000 iterations (N 45), output every 10 iterations (P 20)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.4.1 DIVEMesh: control.txt

C 11 1 // side 1 (x_{min}): inflow
C 12 21 // side 2 (y_{max}): wall
C 13 21 // side 3 (y_{min}): wall
C 14 2 // side 4 (x_{max}): outflow
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.025 // uniform cell size dx [m]

B 10 0.0 2.5 0.0 0.6 0.0 0.5 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 83 0.8 0.0 0.0 1.1 0.0 0.0 1.1 0.2 0.0 0.8 0.0 0.5 1.1 0.0 0.5 1.1 0.2 0.5 // wedge (six corner points): converging part of the contraction
S 10 1.1 1.4 0.0 0.2 0.0 0.5 // box: narrow section, side y = 0–0.2 m
S 83 1.4 0.0 0.0 1.7 0.0 0.0 1.4 0.2 0.0 1.4 0.0 0.5 1.7 0.0 0.5 1.4 0.2 0.5 // wedge (six corner points): diverging part of the contraction

S 83 0.8 0.6 0.0 1.1 0.6 0.0 1.1 0.4 0.0 0.8 0.6 0.5 1.1 0.6 0.5 1.1 0.4 0.5 // wedge (six corner points): converging part of the contraction
S 10 1.1 1.4 0.4 0.6 0.0 0.5 // box: narrow section, side y = 0.4–0.6 m
S 83 1.4 0.6 0.0 1.7 0.6 0.0 1.4 0.4 0.0 1.4 0.6 0.5 1.7 0.6 0.5 1.4 0.4 0.5 // wedge (six corner points): diverging part of the contraction

M 10 6 // number of processes (partitions)

13.4.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 11 1 // wall functions for the turbulence model on
B 50 0.0001 // wall roughness k_s [m]
B 60 1 // ioFlow: constant inflow discharge from W 10

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 0.5 // length for the initial reinitialization instead of the domain length [m]

F 50 2 // level set fixed at the outflow
F 60 0.15 // still water level [m]

I 10 1 // initialize everything (potential flow velocities, pressure, turbulence)

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 45 12000 // maximum number of iterations
N 47 0.3 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 20 10 // print interval in iterations

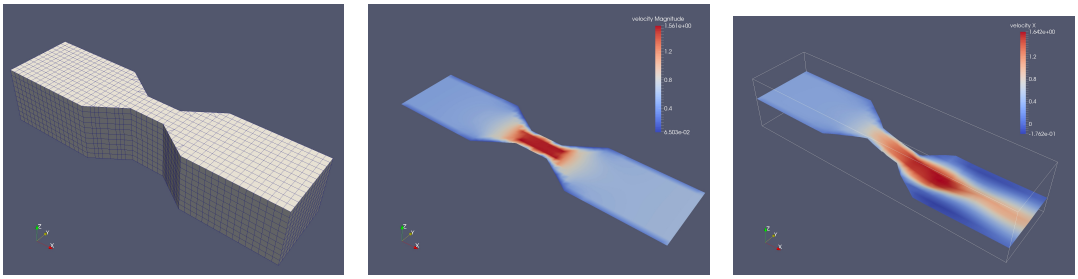
T 10 2 // k-ω turbulence model

W 10 0.05 // inflow discharge [m^3/s]
W 22 -9.81 // gravity in z-direction [m/s^2]

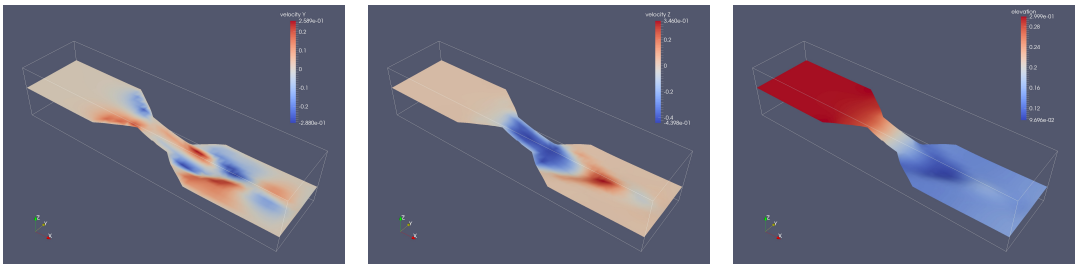
13.4.3 What to look at

Open REEF3D_CFD_VTU: the first file shows the initial potential flow velocities; later files show the acceleration through the contraction (velocity components U, V, W) and the drop of the free surface (contour phi = 0), as in the figure below.

13.4.4 Results



(a) Mesh (b) Initial velocity with potential flow solver (c) U



(d) V (e) W (f) Free surface elevation

Figure 13.4: Narrow Contraction

13.5 Flow around a Circular Pier

Repository folder: `Tutorials/REEF3D_CFD/11_5 Flow around a Circular Pier`

Open channel flow around a vertical circular cylinder. As in the previous case, ioFlow prescribes the discharge at the inflow and keeps the water level fixed at the outflow; the flow is initialized with the potential flow solver and the k-ω model is used. The case shows the free surface deformation and the three-dimensional flow field around the pier.

- Domain: 1.5 m $\times$ 1.0 m $\times$ 0.4 m, cell size dx = 0.025 m (60 $\times$ 40 $\times$ 16 cells)
- Pier at $x = y$ = 0.5 m, diameter 0.15 m; water depth 0.2 m, discharge 0.1 m^3/s (mean velocity 0.5 m/s)
- 100000 iterations (N 45), output every 10 iterations (P 20)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.5.1 DIVEMesh: control.txt

C 11 1 // side 1 (x_{min}): inflow
C 12 21 // side 2 (y_{max}): wall
C 13 21 // side 3 (y_{min}): wall
C 14 2 // side 4 (x_{max}): outflow
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.025 // uniform cell size dx [m]

B 10 0.0 1.5 0.0 1.0 0.0 0.4 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 33 0.5 0.5 0.075 // vertical cylinder at $x = y = 0.5$ m, radius 0.075 m

M 10 6 // number of processes (partitions)

13.5.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 50 0.0001 // wall roughness k_s [m]
B 60 1 // ioFlow: constant inflow discharge from W 10

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 50 2 // level set fixed at the outflow
F 60 0.2 // still water level [m]

I 10 1 // initialize everything (potential flow velocities, pressure, turbulence)

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 45 100000 // maximum number of iterations
N 47 0.3 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 20 10 // print interval in iterations

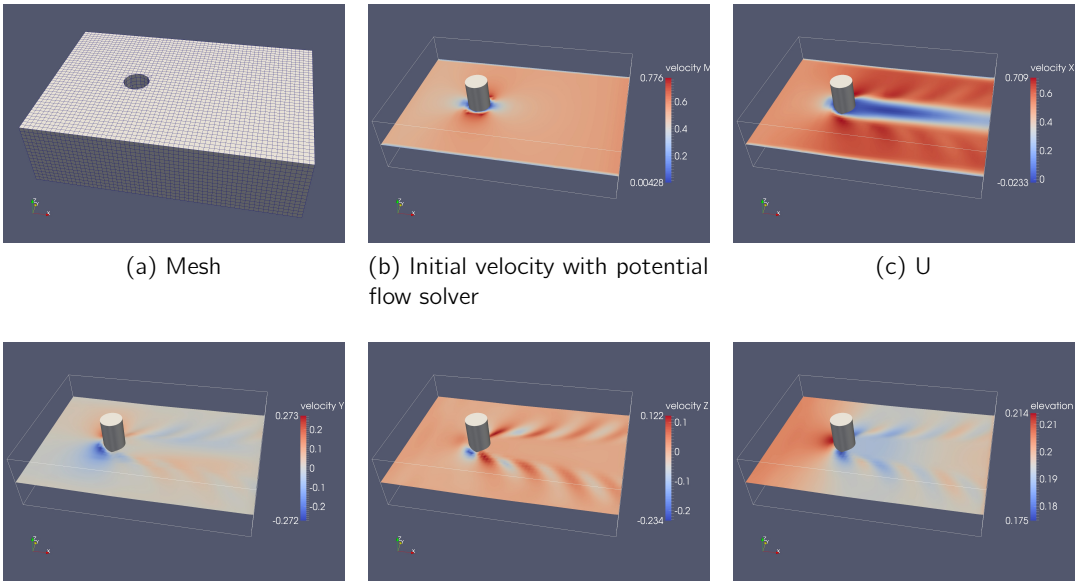
T 10 2 // k-ω turbulence model

W 10 0.1 // inflow discharge [m^3/s]
W 22 -9.81 // gravity in z-direction [m/s^2]

13.5.3 What to look at

Open REEF3D_CFD_VTU and use a horizontal slice below the free surface to see the velocity field around the pier, and the contour phi = 0 for the free surface with the bow wave in front of the pier.

13.5.4 Results



(a) Mesh

(b) Initial velocity with potential flow solver

(c) U

(d) V

(e) W

(f) Free surface elevation

Figure 13.5: Flow around Circular Cylinder

13.6 Rectangular Wave Tank

Repository folder: `Tutorials/REEF3D_CFD/11_6 Rectangular Wave Tank`

For the case of a simple rectangular wave tank without obstacles, the simulation can be run in 2D. Linear waves are generated with the relaxation method, and the propagating waves are compared with the theoretical solution throughout the wave tank. The idea is that if the wave propagation is represented well by the numerical model, the numerical results will match the theoretical free surface profile in the tank. In this example, the wave has a relatively low steepness, and good convergence can be reached for dx = 0.05 m already, which improves further on finer grids. For steeper waves, finer grids may be required.

- Wave: linear, H = 0.02 m, L = 4.0 m (B 91), water depth d = 0.5 m
- Domain: 30 m × 0.05 m × 1.0 m, cell size dx = 0.05 m (600 × 1 × 20 cells)
- Wave generation zone 4 m, numerical beach 8 m (B 96), simulated time 90 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.6.1 DIVEMesh: control.txt

```
C 11 6 // side 1 (x_min): wave generation
C 12 3 // side 2 (y_max): symmetry plane
C 13 3 // side 3 (y_min): symmetry plane
C 14 7 // side 4 (x_max): numerical beach
```

C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.05 // uniform cell size dx [m]
B 10 0.0 30.0 0.0 0.05 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

M 10 6 // number of processes (partitions)

13.6.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 2 // wave type: linear waves
B 91 0.02 4.0 // wave height H [m], wavelength L [m]
B 96 4.0 8.0 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 2 // numerical beach: relaxation method 2

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 1.0 // length for the initial reinitialization instead of the domain length [m]
F 60 0.5 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 90.0 // maximum simulated time [s]
N 47 0.25 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.05 // print interval in simulated time [s]

W 22 -9.81 // gravity in z-direction [m/s^2]

P 52 0.025 // free surface line (wsfline) along x at this y [m]
P 53 1 // add the theoretical wave to the wsfline files
P 54 500 // wsfline print interval in iterations (needs P 52)

13.6.3 What to look at

The files in REEF3D_CFD_WSFLINE contain the numerical free surface along the tank at y = 0.025 m every 500 iterations together with the theoretical profile (P 53 1); plot both to obtain the comparison in the last figure. In ParaView, REEF3D_CFD_VTU (every 0.05 s) shows the velocities U and W under the waves.

13.6.4 Results

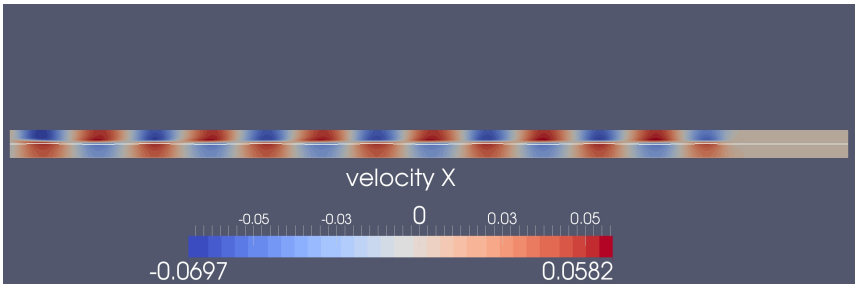


Figure 13.6: Wave tank with free surface and U

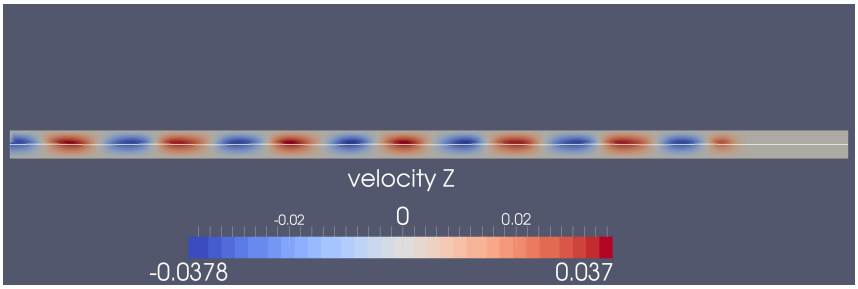


Figure 13.7: Wave tank with free surface and W

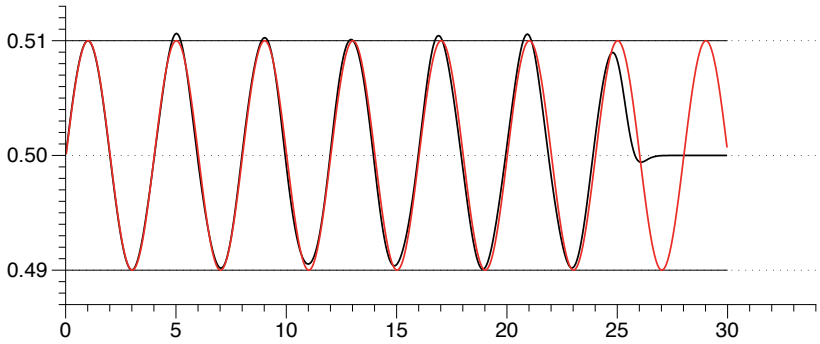


Figure 13.8: Free surface along the tank, numerical result (black line) vs theory (red line)

13.7 Rectangular Wave Tank with DWG and AWA

Repository folder: `Tutorials/REEF3D_CFD/11_7 Rectangular Wave Tank DWG-AWA`

Similar to the previous case, a two-dimensional rectangular wave tank is used. Now, different wave generation and absorption methods are used, namely Dirichlet wave generation (DWG) and active wave absorption (AWA). With DWG, the values for the horizontal and vertical velocities are prescribed at the inlet boundary without further modulation. The level set function is allowed to move freely, but due to the correct flow momentum it will follow the prescribed wave theory. AWA generates a wave opposite to the reflected one at the outlet boundary, thus cancelling out the unwanted reflections. AWA is based on shallow water theory, which also has consequences for the type of applications it should be used for, i.e. shallow to intermediate water with preferably longer waves. Overall, the DWG-AWA combination requires a smaller tank or basin compared to the relaxation methods (RM). AWA, DWG and RM can be used for generation and absorption in any combination.

- Wave: 5th-order Stokes, H = 0.1 m, L = 2.0 m (B 91), water depth d = 0.5 m
- Domain: 15 m × 0.025 m × 1.0 m, cell size dx = 0.025 m (600 × 1 × 40 cells)
- Simulated time 60 s; state files for a hot-start every 1 s (P 40, P 42; for a restart on any decomposition use the checkpoints P 48 and I 45, Section 8.4)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.7.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.025 // uniform cell size dx [m]

B 10 0.0 15.0 0.0 0.025 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

M 20 2 // decomposition method 2 (improved rectangular)

M 10 6 // number of processes (partitions)

13.7.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 5 // wave type: 5th-order Stokes waves (Fenton)
B 91 0.1 2.0 // wave height H [m], wavelength L [m]
B 96 2.0 4.0 // lengths of wave generation zone and numerical beach [m]
B 98 3 // wave generation: Dirichlet (DWG)
B 99 3 // numerical beach: active wave absorption (AWA)

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 1.0 // length for the initial reinitialization instead of the domain length [m]
F 60 0.50 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 60.0 // maximum simulated time [s]
N 47 0.25 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.1 // print interval in simulated time [s]
P 40 1 // print state files for hot-start
P 42 1.0 // state file interval [s]

T 10 0 // no turbulence model

W 22 -9.81 // gravity in z-direction [m/s^2]

P 52 0.0125 // free surface line (wsfline) along x at this y [m]
P 53 1 // add the theoretical wave to the wsfline files
P 55 0.50 // wsfline and line probe print interval [s]

P 51 2.50 0.0125 // wave gauge 1 at x, y [m]
P 51 5.00 0.0125 // wave gauge 2 at x, y [m]
P 51 15.0 0.0125 // wave gauge on the outlet boundary (x_{max} = 15 m)
P 51 20.0 0.0125 // wave gauge outside the domain (x_{max} = 15 m)

13.7.3 What to look at

Compare the numerical and theoretical free surface profiles in REEF3D_CFD_WSFLINE (every 0.5 s, P 55) to check that no reflections build up in the short tank, and plot the gauges in 'REEF3D-CFD-WSF-HG.dat' (REEF3D_CFD_WSF). Note that the last two P 51 gauges are located at x = 15 m and 20 m, i.e. on the outlet boundary and outside the 15 m long domain. REEF3D_CFD_VTU (every 0.1 s) shows the velocities U and W.

13.7.4 Results

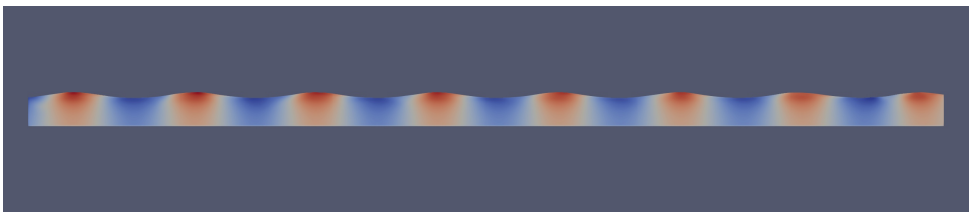


Figure 13.9: DWG-AWA Wave tank with free surface and U

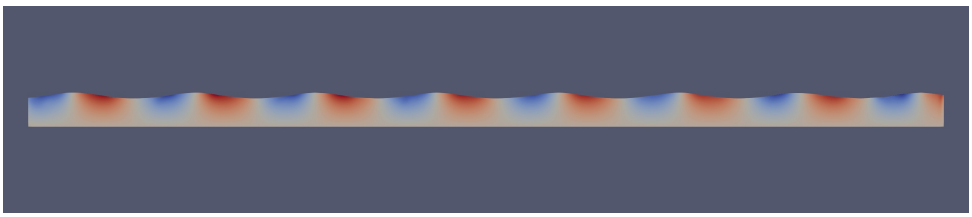


Figure 13.10: DWG-AWA Wave tank with free surface and W

13.8 Rectangular Wave Tank with Wavemaker

Repository folder: `Tutorials/REEF3D_CFD/11_8 Rectangular Wave Tank Wavemaker`

Following the previous NWT examples, now the waves are generated with wavemaker kinematics input, see e.g. Aggarwal et al. [2018a] for more information. The motion of a flap wavemaker is given as a time series of the flap angle in the file 'wavemaker.dat' (two columns: time t [s] and angle β [rad], B 116 2), which must be in the case folder. The flap kinematics are applied at the inlet boundary (B 98 3), and the waves are absorbed with active wave absorption at the outlet (B 99 3).

- Flap from z = -0.25 m to 1.46 m (B 111), input time shifted by 4.0 s (B 117), water depth d = 0.75 m
- Domain: 15 m × 0.025 m × 1.5 m, cell size dx = 0.025 m (600 × 1 × 60 cells)
- Simulated time 60 s; output every 10 iterations (P 20), state files every 1 s
- Extra input file: wavemaker.dat (read by REEF3D)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.8.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 8 // side 4 (x_{max}): numerical beach, type 8 (handled like type 7 in REEF3D)
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.025 // uniform cell size dx [m]
B 10 0.0 15.0 0.0 0.025 0.0 1.5 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

13.8.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 22 // wave type: flap wavemaker, kinematics from wavemaker.dat
B 98 3 // wave generation: Dirichlet (DWG)
B 99 3 // numerical beach: active wave absorption (AWA)
B 111 -0.25 1.46 // flap wavemaker from z_{start} = -0.25 m (below the bed) to z_{end} = 1.46 m
B 116 2 // flap wavemaker input as angle $\beta(t)$ [rad]

B 117 4.0 // time shift [s]: rows before t = 4.0 s in wavemaker.dat are skipped, t = 4.0 s becomes t = 0

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 1.5 // length for the initial reinitialization instead of the domain length [m]
F 60 0.75 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 60.0 // maximum simulated time [s]
N 47 0.1 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

T 10 0 // no turbulence model

W 22 -9.81 // gravity in z-direction [m/s^2]

P 10 1 // print ParaView files (.vtu)
P 20 10 // print interval in iterations
P 40 1 // print state files for hot-start
P 42 1.0 // state file interval [s]
P 52 0.015 // free surface line (wsfline) along x at this y [m]
P 55 0.01 // wsfline print interval [s]

P 51 5.0 0.015 // wave gauge 1 at x, y [m]
P 51 10.0 0.015 // wave gauge 2 at x, y [m]

13.8.3 REEF3D: wavemaker.dat

The 'wavemaker.dat' file contains the time dependent kinematics of the wavemaker.

13.8.4 What to look at

Plot the two gauges at x = 5 m and 10 m from 'REEF3D-CFD-WSF-HG.dat' in REEF3D_CFD_WSF, and the free surface along the tank in REEF3D_CFD_WSFLINE (every 0.01 s). REEF3D_CFD_VTU shows the generated waves with the velocities U and W, as in the figures below.

13.8.5 Results

Figure 13.11: Wavemaker generated waves with free surface and U

Figure 13.12: Wavemaker generated waves with free surface and W

13.9 5th-order Stokes Waves

Repository folder: `Tutorials/REEF3D_CFD/11_9 5th-order Stokes Waves`

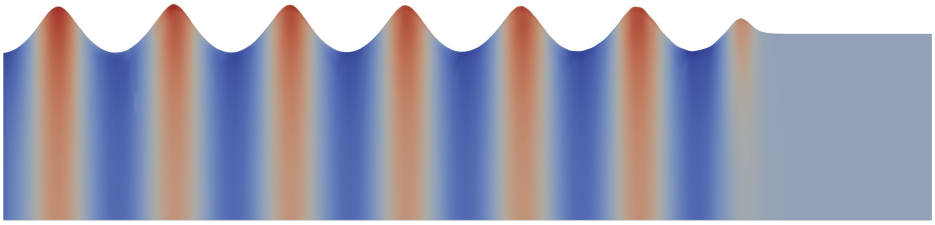


Figure 13.13: 5th-order Stokes Waves setup

This verification case shows the propagation of 5th-order Stokes waves in a 200 m long tank and highlights the use of velocity probes both from the model and the wave theory. The waves are generated in a relaxation zone of about one wavelength and absorbed in a relaxation beach of about two wavelengths. The same wave is used in the NHFLOW tutorial `1_Stokes5`. The numerical schemes not given in ctrl.txt take their default values.

- Wave: H = 1.0 m, T = 4.5 s (B 93), water depth d = 4.01 m (F 60), wavelength about 25 m
- Domain: 200 m × 0.025 m × 8.0 m, cell size dx = 0.025 m (8000 × 1 × 320 cells)
- Wave generation zone 25 m, numerical beach 50 m (B 96), simulated time 120 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.9.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane

C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.025 // uniform cell size dx [m]

B 10 0.0 200.0 0.0 0.025 0.0 8.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

13.9.2 REEF3D: ctrl.txt

B 90 1 // turn on ioWave (numerical wave tank)
B 92 5 // wave type: 5th-order Stokes waves (Fenton)
B 93 1.0 4.5 // wave height H [m], wave period T [s]
B 96 25.0 50.0 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 2 // numerical beach: relaxation method 2

D 30 2 // pressure: projection method (same solver as D 30 1 in the current code)

F 60 4.01 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 120.0 // maximum simulated time [s]
N 47 0.3 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.25 // print interval in simulated time [s]

P 50 15.0 0.005 // theoretical wave gauge 1 at x, y [m]
P 50 50.0 0.005 // theoretical wave gauge 2 at x, y [m]
P 50 75.0 0.005 // theoretical wave gauge 3 at x, y [m]
P 50 100.0 0.005 // theoretical wave gauge 4 at x, y [m]
P 50 125.0 0.005 // theoretical wave gauge 5 at x, y [m]

P 51 15.0 0.005 // wave gauge 1 at x, y [m]
P 51 50.0 0.005 // wave gauge 2 at x, y [m]
P 51 75.0 0.005 // wave gauge 3 at x, y [m]
P 51 100.0 0.005 // wave gauge 4 at x, y [m]
P 51 125.0 0.005 // wave gauge 5 at x, y [m]

P 52 0.005 // free surface line (wsfline) along x at this y [m]
P 53 1 // add the theoretical wave to the wsfline files
P 55 0.05 // wsfline and line probe print interval [s]

P 65 25.0 0.025 3.0 // velocity probe 1 at x, y, z [m]
P 66 25.0 0.025 3.0 // theoretical velocity probe 1 at x, y, z [m]

P 65 75.0 0.025 3.0 // velocity probe 2 at x, y, z [m]
P 66 75.0 0.025 3.0 // theoretical velocity probe 2 at x, y, z [m]

P 65 125.0 0.025 3.0 // velocity probe 3 at x, y, z [m]
P 66 125.0 0.025 3.0 // theoretical velocity probe 3 at x, y, z [m]

W 22 -9.81 // gravity in z-direction [m/s^2]

13.9.3 What to look at

Plot 'REEF3D-CFD-WSF-HG.dat' (P 51) against 'REEF3D-CFD-WSF-HG-THEORY.dat' (P 50) in REEF3D_CFD_WSF for the gauges at x = 15, 50, 75, 100 and 125 m. The velocity probes at z = 3.0 m write 'REEF3D-CFD-Vel-Probe-n.dat' (P 65) and the corresponding theoretical velocities 'REEF3D-CFD-Vel-Probe-Theory-n.dat' (P 66) to REEF3D_CFD_ProbePoint; compare the horizontal and vertical velocity at x = 25, 75 and 125 m. REEF3D_CFD_WSFLINE contains the numerical and theoretical free surface along the tank every 0.05 s. Note that the first gauge in the current ctrl.txt is at x = 15 m, while the first free surface figure below is labelled x = 25 m.

13.9.4 Free Surface Results

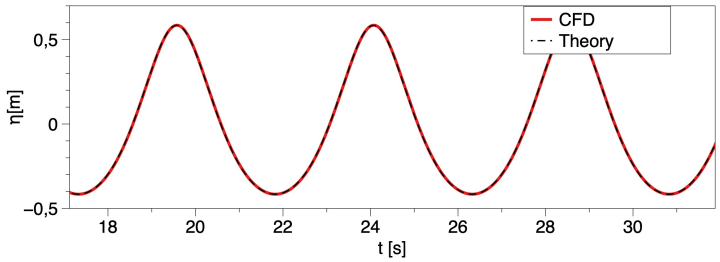


Figure 13.14: Wave gage 1, x = 25m

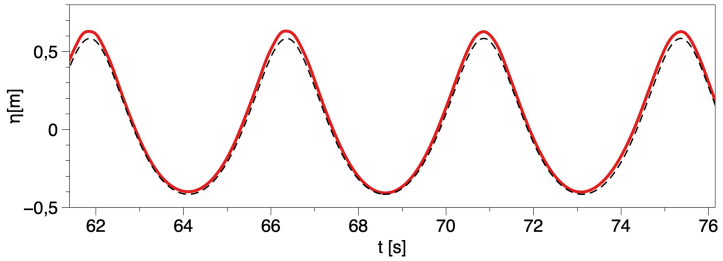


Figure 13.15: Wave gage 2, x = 50m

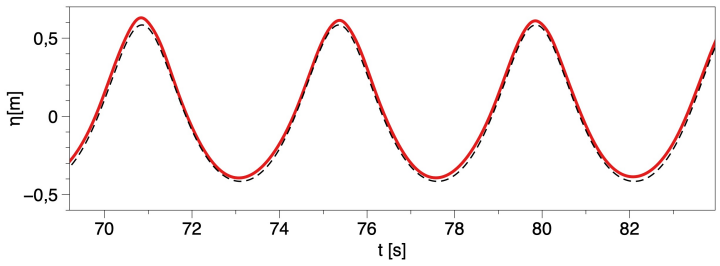


Figure 13.16: Wave gage 3, x = 75m

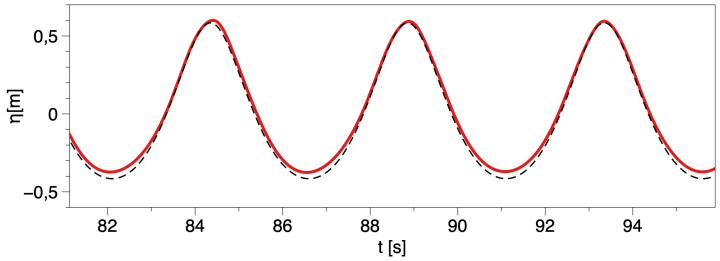


Figure 13.17: Wave gage 4, x = 100m

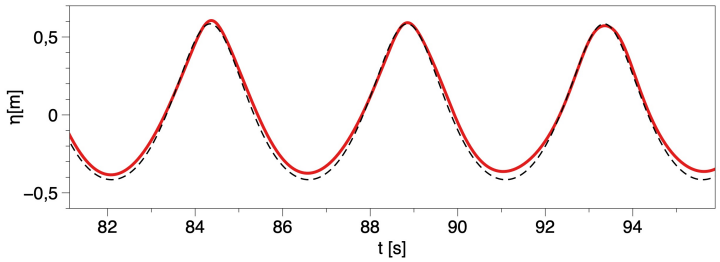


Figure 13.18: Wave gage 5, x = 125m

13.9.5 Velocity Results

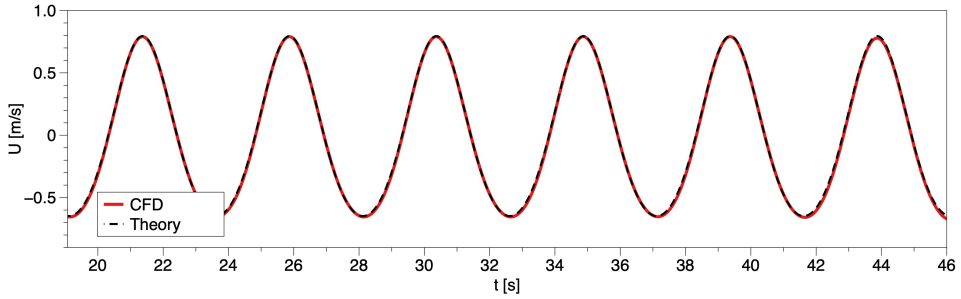


Figure 13.19: Horizontal Velocity, x = 25m, z=3m

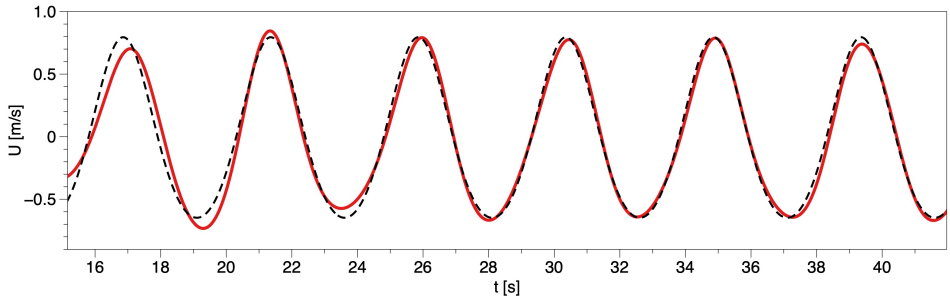


Figure 13.20: Horizontal Velocity, x = 75m, z=3m

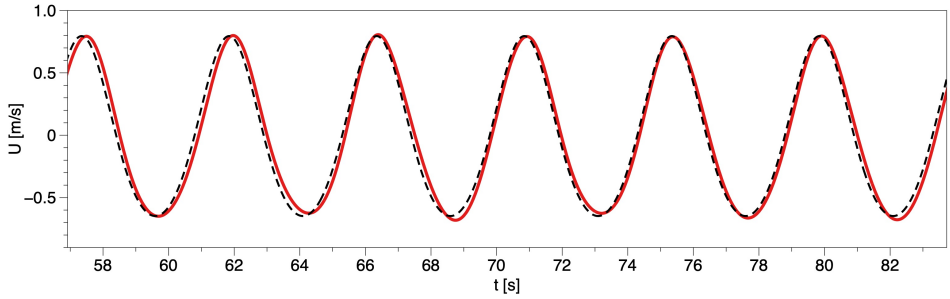


Figure 13.21: Horizontal Velocity, x = 125m, z=3m

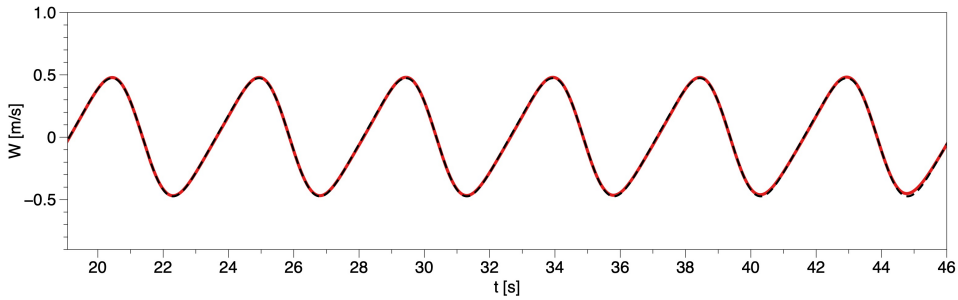


Figure 13.22: Vertical Velocity, x = 25m, z=3m

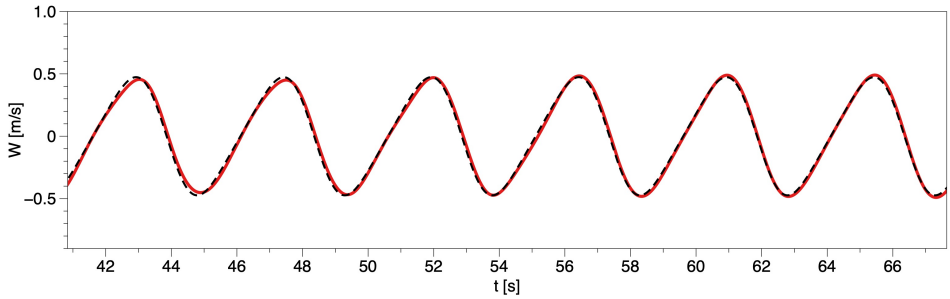


Figure 13.23: Vertical Velocity, x = 75m, z=3m

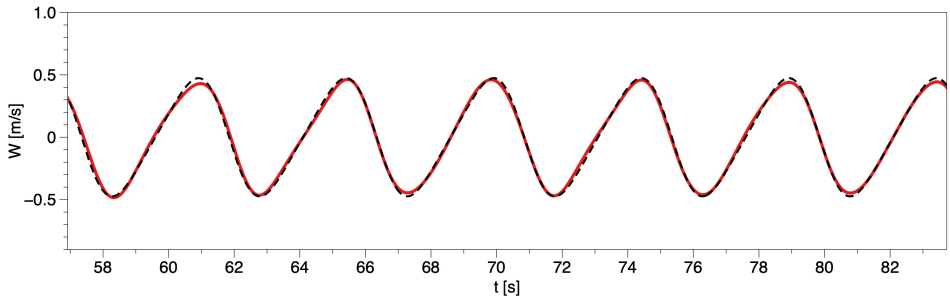


Figure 13.24: Vertical Velocity, x = 125m, z=3m

13.10 Wave Propagation over a Submerged Bar

Repository folder: `Tutorials/REEF3D_CFD/11_10 Submerged Bar Wave Propagation`

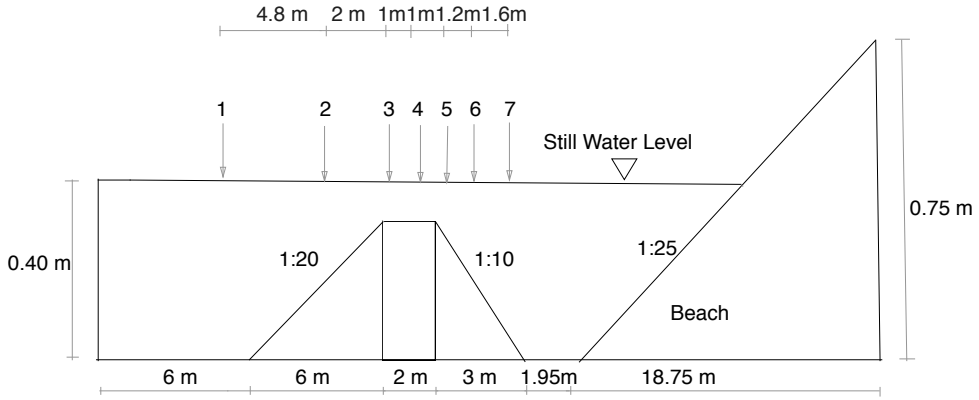


Figure 13.25: Experimental setup submerged bar

The well-known benchmark case for wave propagation over a submerged bar is presented here. The description of the experimental setup can be found in the original paper by Beji and Battjes [1993]. The bar consists of a front slope from x = 6 m to 12 m, a crest from 12 m to 14 m and a back slope to 17 m, 0.3 m high. The simulations are run in 2D with wave gauges at several locations along the wave tank.

- Wave: linear, H = 0.02 m, L = 3.73 m (B 91), water depth d = 0.4 m, 0.1 m on the bar crest
- Domain: 24 m × 0.01 m × 0.8 m, cell size dx = 0.01 m (2400 × 1 × 80 cells)
- Wave generation zone and numerical beach 3.73 m each (B 96), simulated time 60 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.10.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.01 // uniform cell size dx [m]
B 10 0.0 24.0 0.0 0.01 0.0 0.8 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 61 6.0 12.0 0.0 0.01 0.0 0.3 // front slope of the bar, rising from z = 0 to 0.3 m
S 10 12.0 14.0 0.0 0.01 0.0 0.3 // bar crest, 0.3 m high
S 61 14.0 17.0 0.0 0.01 0.3 0.0 // back slope of the bar, falling from 0.3 m to 0

M 10 6 // number of processes (partitions)

13.10.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 2 // wave type: linear waves
B 91 0.02 3.73 // wave height H [m], wavelength L [m]
B 96 3.73 3.73 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 2 // numerical beach: relaxation method 2

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 0.8 // length for the initial reinitialization instead of the domain length [m]
F 60 0.4 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 60.0 // maximum simulated time [s]
N 47 0.1 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.5 // print interval in simulated time [s]

W 22 -9.81 // gravity in z-direction [m/s^2]

P 52 0.005 // free surface line (wsfline) along x at this y [m]
P 53 1 // add the theoretical wave to the wsfline files
P 54 10 // wsfline print interval in iterations (needs P 52)

P 51 2.0 0.005 // wave gauge 1 at x, y [m]
P 51 4.0 0.005 // wave gauge 2 at x, y [m]
P 51 5.2 0.005 // wave gauge 3 at x, y [m]
P 51 10.5 0.005 // wave gauge 4 at x, y [m]
P 51 12.5 0.005 // wave gauge 5 at x, y [m]
P 51 13.5 0.005 // wave gauge 6 at x, y [m]
P 51 14.5 0.005 // wave gauge 7 at x, y [m]

13.10.3 What to look at

'REEF3D-CFD-WSF-HG.dat' in REEF3D_CFD_WSF contains one column per P 51 gauge (x = 2.0, 4.0, 5.2, 10.5, 12.5, 13.5 and 14.5 m), in the order given in ctrl.txt; compare them with the measurements of Beji and Battjes [1993] as in the figures below. REEF3D_CFD_WSFLINE contains the free surface along the tank every 10 iterations together with the theoretical wave (P 53 1).

13.10.4 Results

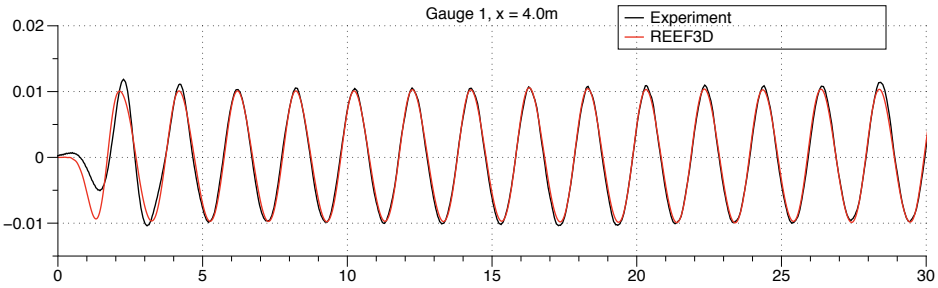


Figure 13.26: Wave gage 1

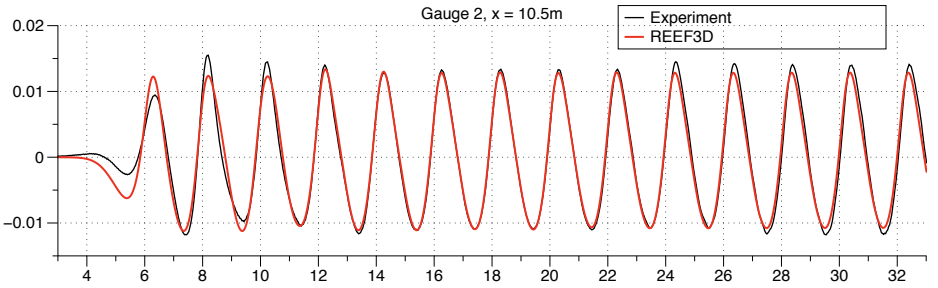


Figure 13.27: Wave gage 2

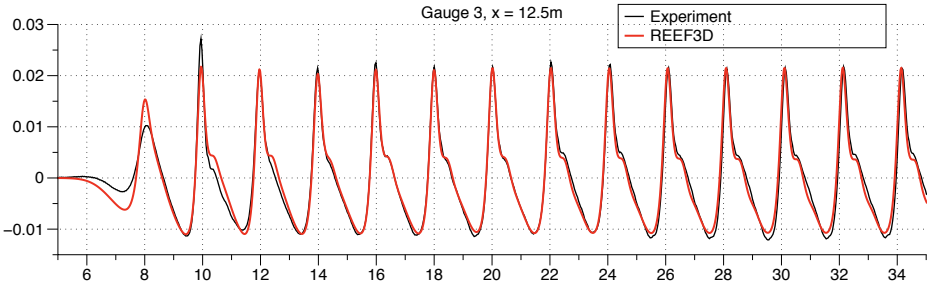


Figure 13.28: Wave gage 3

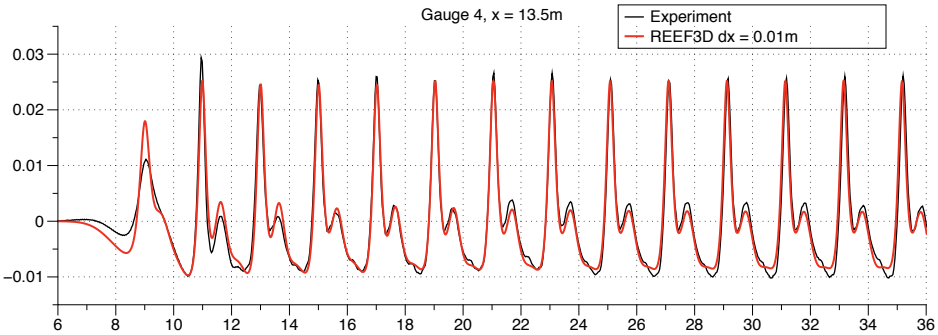


Figure 13.29: Wave gage 4

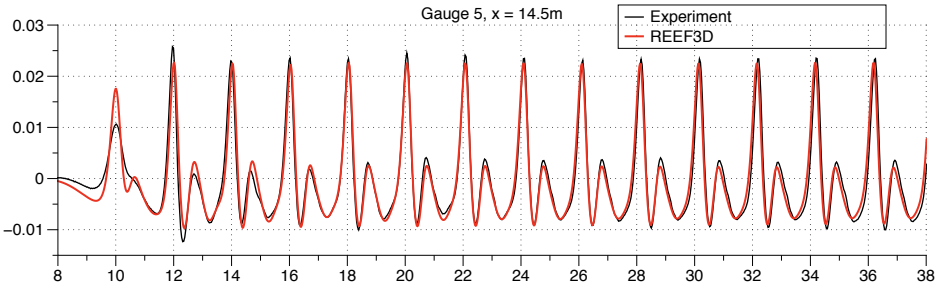


Figure 13.30: Wave gage 5

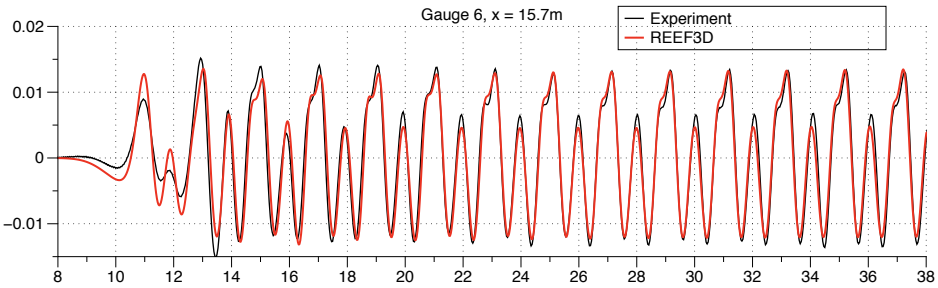


Figure 13.31: Wave gage 6

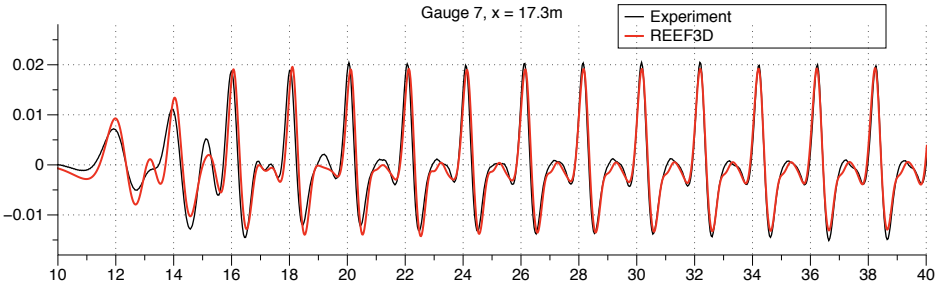


Figure 13.32: Wave gage 7

13.11 Plunging Breaking Waves over Slope

Repository folder: `Tutorials/REEF3D_CFD/11_11 Plunging Breaking Waves Slope`

The well-known benchmark case for plunging breaking waves over a slope by Ting and Kirby [1995] is shown here. Cnoidal waves are generated with the relaxation method and break on a 1:35 slope; there is no numerical beach (zero length in B 96), the wave energy is dissipated by breaking on the slope. The breaking waves require a relatively fine grid, thus 128 cores on NOTUR's supercomputer facilities were used to simulate the case. For more information on this case and breaking waves over slopes in general, please have a look at these in-depth studies: Alagan Chella et al. [2015b], Alagan Chella et al. [2015c] and Alagan Chella et al. [2015a].

- Wave: 5th-order cnoidal, H = 0.128 m, T = 5.0 s (B 93), water depth d = 0.4 m
- Domain: 40 m × 0.005 m × 1.0 m, cell size dx = 0.005 m (8000 × 1 × 200 cells), slope from x = 13.8 m

- Wave generation zone 9.8 m (B 96), simulated time 60 s, k-ω model with free surface damping (T 36 2)
- Run: **./DiveMESH**, then **mpirun -n 128 reef3d**

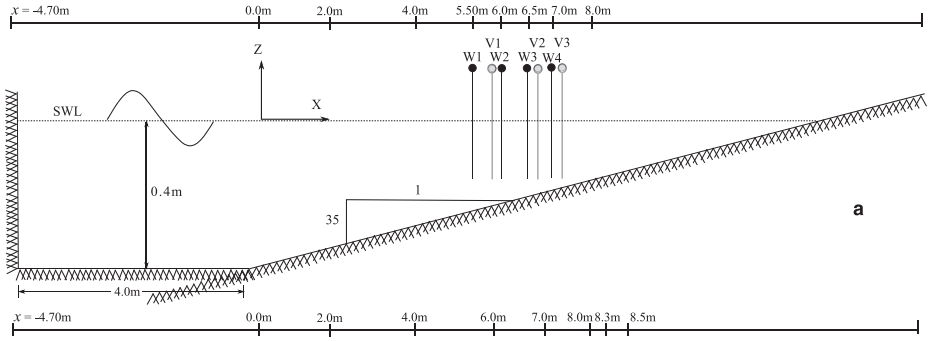


Figure 13.33: Experimental setup plunging breaking waves over slope

13.11.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.005 // uniform cell size dx [m]

B 10 0.0 40.0 0.0 0.005 0.0 1.00 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 61 13.8 40.0 0.0 0.005 0.0 0.748 // bed slope (about 1:35) starting at x = 13.8 m

M 10 128 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

13.11.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 11 1 // wall functions for the turbulence model on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 8 // wave type: 5th-order cnoidal waves
B 93 0.128 5.0 // wave height H [m], wave period T [s]
B 96 9.8 0.0 // wave generation zone 9.8 m, beach length 0 m
B 99 2 // numerical beach: relaxation method 2 (zero length in B 96)
B 98 2 // wave generation: relaxation method 2

```
D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 1.0 // length for the initial reinitialization instead of the domain length [m]
F 60 0.4 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 60.0 // maximum simulated time [s]
N 47 0.1 // CFL number for adaptive time stepping

M 10 128 // number of processes (same as in control.txt)

P 30 0.05 // print interval in simulated time [s]
P 40 1 // print state files for hot-start
P 42 0.50 // state file interval [s]

T 10 2 // k-ω turbulence model
T 36 2 // free surface turbulence damping on, including wall distance

W 10 0.0 // discharge [m^3/s]
W 22 -9.81 // gravity in z-direction [m/s^2]

P 51 19.8 0.0025 // wave gauge 1 at x, y [m]
P 51 20.8 0.0025 // wave gauge 2 at x, y [m]
P 51 21.8 0.0025 // wave gauge 3 at x, y [m]
P 51 22.1 0.0025 // wave gauge 4 at x, y [m]

P 52 0.0025 // free surface line (wsfline) along x at this y [m]

P 61 21.095 0.0025 0.35 // point probe 1 at x, y, z [m]
P 61 21.095 0.0025 0.25 // point probe 2 at x, y, z [m]
P 61 21.095 0.0025 0.4 // point probe 3 at x, y, z [m]

P 61 22.025 0.0025 0.35 // point probe 4 at x, y, z [m]
P 61 22.025 0.0025 0.26 // point probe 5 at x, y, z [m]
P 61 22.025 0.0025 0.4 // point probe 6 at x, y, z [m]
P 61 21.595 0.0025 0.4 // point probe 7 at x, y, z [m]

P 61 22.145 0.0025 0.35 // point probe 8 at x, y, z [m]
P 61 22.145 0.0025 0.28 // point probe 9 at x, y, z [m]
P 61 22.145 0.0025 0.4 // point probe 10 at x, y, z [m]
```

13.11.3 What to look at

The four P 51 gauges at x = 19.8, 20.8, 21.8 and 22.1 m are written to 'REEF3D-CFD-WSF-HG.dat' in REEF3D_CFD_WSF; compare them with the measurements of Ting and Kirby [1995]. The P 61 point probes in the breaking region write velocities and pressure to 'REEF3D-CFD-Probe-Point-n.dat' in REEF3D_CFD_ProbePoint. REEF3D_CFD_VTU (every 0.05 s; P 10 is 1 by default) shows the plunging jet; color by velocity as in the first figure below. State files are written every 0.5 s (P 40, P 42).

13.11.4 Results

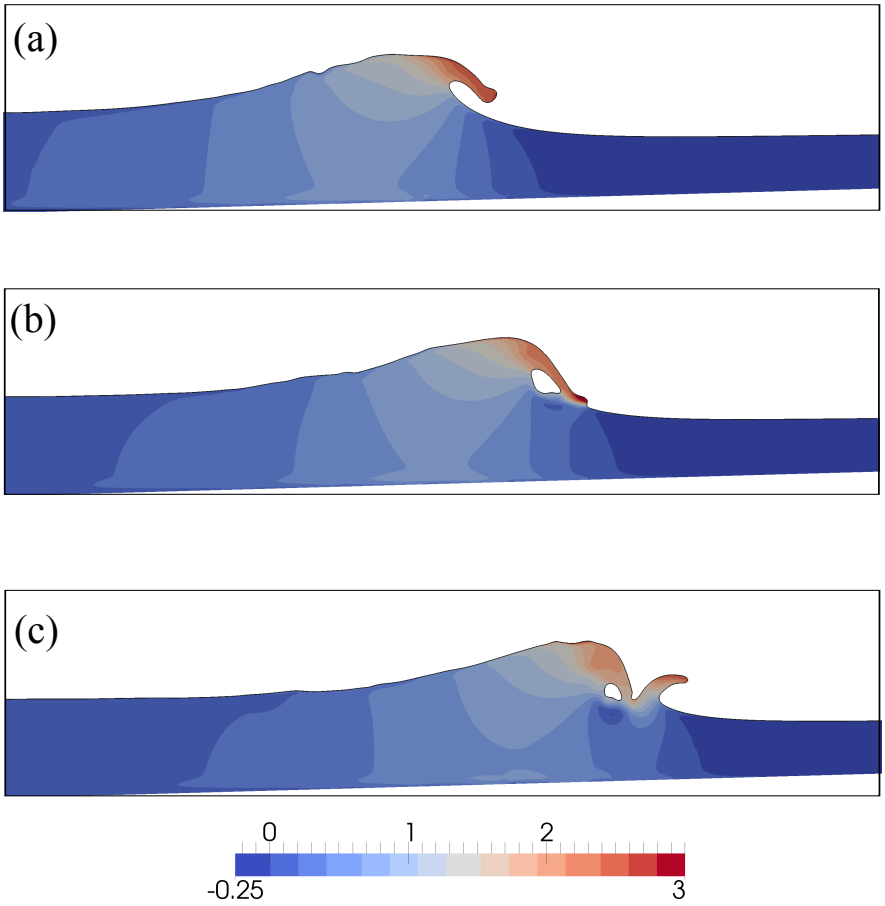


Figure 13.34: Variation of horizontal velocity, U (m/s) under the plunging breaker at t= 10.85s (a), 10.95s (b) and 11.05s (c)

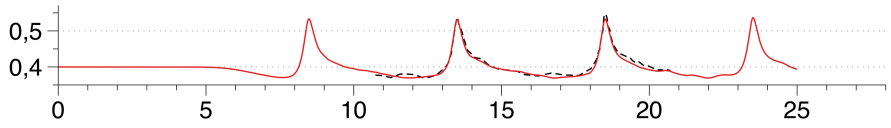


Figure 13.35: Wave gage 1, black experiment Ting and Kirby [1995], red REEF3D

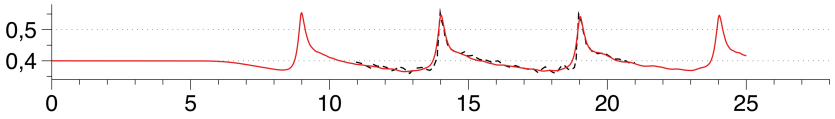


Figure 13.36: Wave gage 2, black experiment Ting and Kirby [1995], red REEF3D

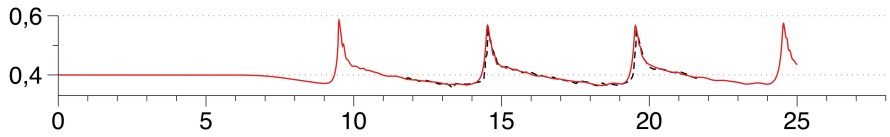


Figure 13.37: Wave gage 3, black experiment Ting and Kirby [1995], red REEF3D

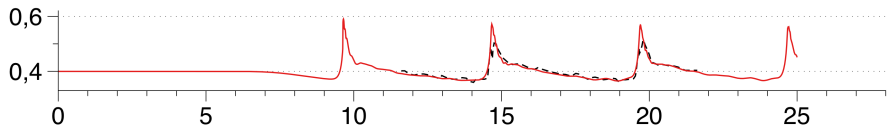


Figure 13.38: Wave gage 4, black experiment Ting and Kirby [1995], red REEF3D

13.12 Plunging Breaking Waves over Slope with Non-uniform Grid

Repository folder: `Tutorials/REEF3D_CFD/11_12 Plunging Breaking Waves Slope NUG`

The simulation of plunging breaking waves over a slope from the previous section by Ting and Kirby [1995] is shown here with a non-uniform grid (NUG) setup. The grid is refined in x-direction around the breaking region (x = 20 m, B 101 5) and in z-direction around the still water level (z = 0.4 m, B 103 5), and the tank is shortened to 30 m. This reduces the number of cells compared to the uniform grid, so the case runs with 6 processes. The ctrl.txt is identical to the one above, with the exception of the M 10 parameter.

- Domain: 30 m × 0.005 m × 1.0 m, 3000 × 1 × 80 cells (B 2) with sinh stretching (stretching factor 2.5) in x and z
- Wave and numerics as in the previous section
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.12.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 2 3000 1 80 // number of cells N_x, N_y, N_z (overrides B 1)

B 10 0.0 30.0 0.0 0.005 0.0 1.00 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

```
B 101 5 // x-direction stretching: point focus (sinh), uses B 111 and B 114
B 111 2.5 // stretching factor in x-direction
B 114 20.0 // focus point of the x-stretching [m]

B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 2.5 // stretching factor in z-direction
B 116 0.40 // focus point of the z-stretching [m]

S 61 13.8 30.0 0.0 0.005 0.0 0.463 // bed slope (about 1:35) starting at x = 13.8 m

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)
```

13.12.2 REEF3D: ctrl.txt

```
B 10 1 // wall functions for the velocities on
B 11 1 // wall functions for the turbulence model on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 8 // wave type: 5th-order cnoidal waves
B 93 0.128 5.0 // wave height H [m], wave period T [s]
B 96 9.8 0.0 // wave generation zone 9.8 m, beach length 0 m
B 99 2 // numerical beach: relaxation method 2 (zero length in B 96)
B 98 2 // wave generation: relaxation method 2

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 1.0 // length for the initial reinitialization instead of the domain length [m]
F 60 0.4 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 60.0 // maximum simulated time [s]
N 47 0.1 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 30 0.05 // print interval in simulated time [s]
P 40 1 // print state files for hot-start
P 42 0.50 // state file interval [s]

T 10 2 // k-ω turbulence model
T 36 2 // free surface turbulence damping on, including wall distance

W 10 0.0 // discharge [m^3/s]
W 22 -9.81 // gravity in z-direction [m/s^2]
```

P 51 19.8 0.0025 // wave gauge 1 at x, y [m]
P 51 20.8 0.0025 // wave gauge 2 at x, y [m]
P 51 21.8 0.0025 // wave gauge 3 at x, y [m]
P 51 22.1 0.0025 // wave gauge 4 at x, y [m]

P 52 0.0025 // free surface line (wsfline) along x at this y [m]

P 61 21.095 0.0025 0.35 // point probe 1 at x, y, z [m]
P 61 21.095 0.0025 0.25 // point probe 2 at x, y, z [m]
P 61 21.095 0.0025 0.4 // point probe 3 at x, y, z [m]

P 61 22.025 0.0025 0.35 // point probe 4 at x, y, z [m]
P 61 22.025 0.0025 0.26 // point probe 5 at x, y, z [m]
P 61 22.025 0.0025 0.4 // point probe 6 at x, y, z [m]
P 61 21.595 0.0025 0.4 // point probe 7 at x, y, z [m]

P 61 22.145 0.0025 0.35 // point probe 8 at x, y, z [m]
P 61 22.145 0.0025 0.28 // point probe 9 at x, y, z [m]
P 61 22.145 0.0025 0.4 // point probe 10 at x, y, z [m]

13.12.3 What to look at

The same output as in the previous section is written. Compare the gauges in REEF3D_CFD_WSF with those of the uniform grid case, and inspect the stretched grid in ParaView by showing the REEF3D_CFD_VTU data with the surface-with-edges representation.

13.13 Shoaling over Irregular Topography

Repository folder: `Tutorials/REEF3D_CFD/11_13 Shoaling over Irregular Bathymetry`

In this case, wave propagation over irregular topography is calculated using phase-accurate irregular waves based on Boers' experiment Boers [1996]. Here, the bed topography is generated using a 'geo.dat' file with x, y, z point data. These coordinates are interpolated using a local inverse distance algorithm in DIVEMesh. The waves are generated using wave reconstruction (B 92 51), a very convenient way to create phase-accurate irregular waves based on a single wave gauge Aggarwal et al. [2018b]; the wave components are read from 'waverecon.dat' and applied with Dirichlet wave generation. No numerical beach is used (B 99 0); the waves break on the bar and on the foreshore.

- Water depth d = 0.75 m at the wave maker, irregular waves from waverecon.dat
- Domain: 36.3 m × 0.01 m × 1.0 m, cell size dx = 0.01 m (3630 × 1 × 100 cells)
- Simulated time 200 s, k-ω model with free surface damping (T 36 2)
- Extra input files: geo.dat (read by DIVEMesh), waverecon.dat (read by REEF3D)
- Run: **./DiveMESH**, then **mpirun -n 128 reef3d**

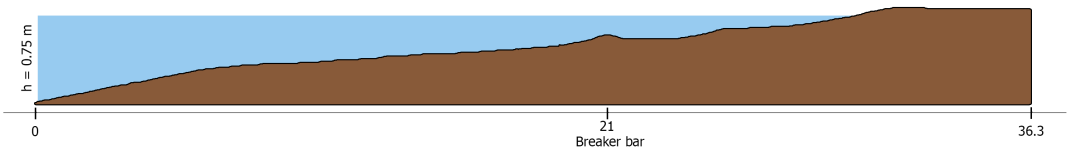


Figure 13.39: Experimental setup for shoaling over irregular bathymetry Elakel [2018]

13.13.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.01 // uniform cell size dx [m]

B 10 0.0 36.3 0.0 0.01 0.0 1.00 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

G 10 1 // turn on geodat interpolation of the bed from geo.dat

G 15 2 // local inverse distance interpolation
G 31 14 // number of outer smoothing filter iterations

M 10 128 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

13.13.2 DIVEMesh: geo.dat

In the 'geo.dat' file all bathymetry coordinates are given in ASCII format (three columns x, y, z) and are then used to interpolate the bed in DIVEMesh. The bed can be visualized with the zero contour of 'topo' in the .vtu files read by ParaView.

13.13.3 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 11 1 // wall functions for the turbulence model on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 51 // wave type: wave reconstruction, 1st-order irregular waves from waverecon.dat
B 98 3 // wave generation: Dirichlet (DWG)
B 99 0 // numerical beach off

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

```
F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 1.0 // length for the initial reinitialization instead of the domain length [m]
F 60 0.75 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 200.0 // maximum simulated time [s]
N 47 0.1 // CFL number for adaptive time stepping

M 10 128 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.05 // print interval in simulated time [s]
P 40 1 // print state files for hot-start
P 42 0.50 // state file interval [s]

T 10 2 // k-ω turbulence model
T 31 0.816 // eddy viscosity limiter factor in phase 1
T 36 2 // free surface turbulence damping on, including wall distance

W 22 -9.81 // gravity in z-direction [m/s^2]

P 51 0.0 0.005 // wave gauge 1 at x, y [m]
P 51 4.0 0.005 // wave gauge 2 at x, y [m]
P 51 8.0 0.005 // wave gauge 3 at x, y [m]
P 51 12.0 0.005 // wave gauge 4 at x, y [m]
P 51 16.0 0.005 // wave gauge 5 at x, y [m]
P 51 20.0 0.005 // wave gauge 6 at x, y [m]

P 61 14.37 0.005 0.74 // point probe 1 at x, y, z [m]
```

13.13.4 REEF3D: waverecon.dat

The 'waverecon.dat' file contains the individual wave components, with the amplitude, angular frequency and phase shift in three columns, in order to generate phase-accurate irregular waves.

13.13.5 What to look at

Compare the six P 51 gauges at x = 0, 4, 8, 12, 16 and 20 m in 'REEF3D-CFD-WSF-HG.dat' (REEF3D_CFD_WSF) with the measurements; the gauge at x = 0 m lies on the wave generation boundary. The point probe (P 61) is written to REEF3D_CFD_ProbePoint. In ParaView, REEF3D_CFD_VTU (every 0.05 s) shows the shoaling and breaking waves; the bed is the zero contour of topo.

13.13.6 Results

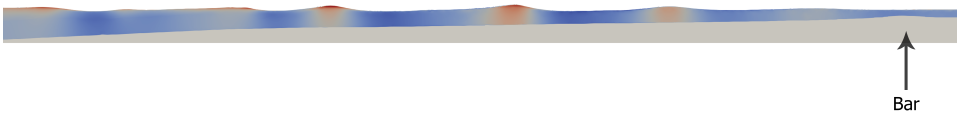


Figure 13.40: Shoaling of irregular waves for the Boers case Elakel [2018]

Figure 13.41: Breaking waves on the bar for the Boers case Elakel [2018]

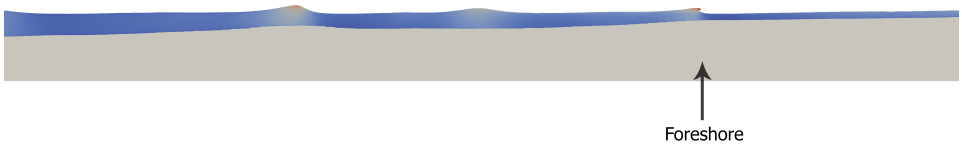


Figure 13.42: Breaking waves on the foreshore for the Boers case Elakel [2018]

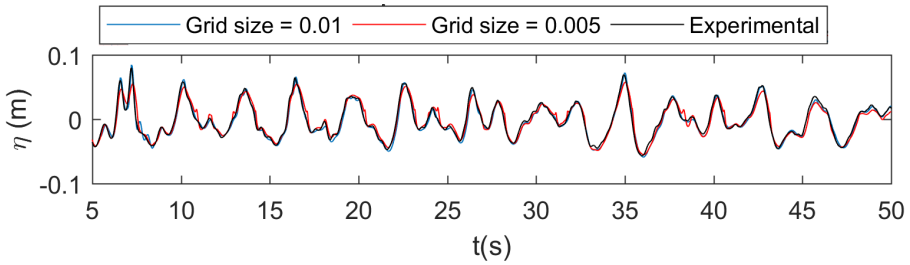


Figure 13.43: Wave gage 1

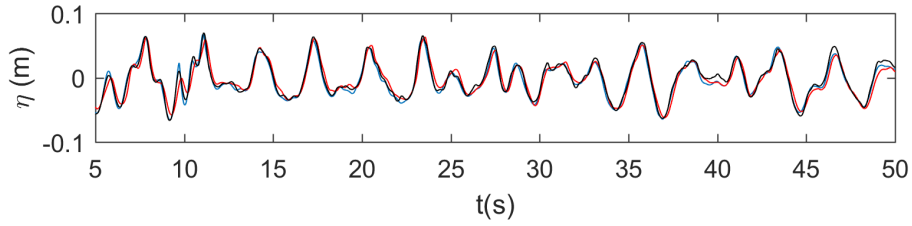


Figure 13.44: Wave gage 2

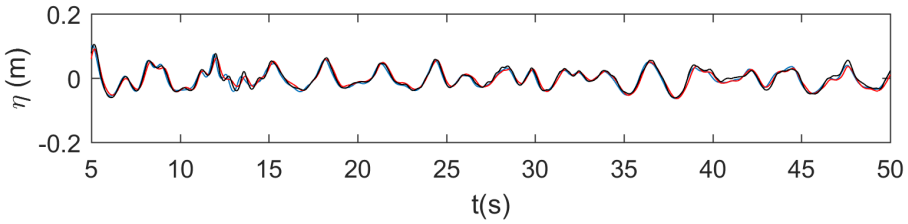


Figure 13.45: Wave gage 3

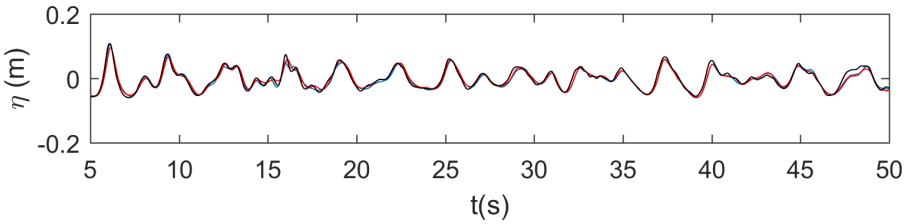


Figure 13.46: Wave gage 4

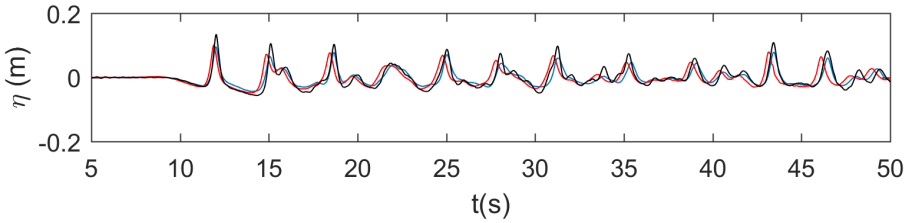


Figure 13.47: Wave gage 5

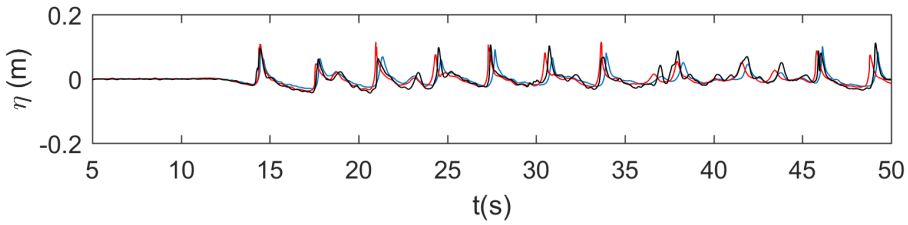


Figure 13.48: Wave gage 6

13.14 Non-Breaking Wave Forces

Repository folder: `Tutorials/REEF3D_CFD/11_14 Non-Breaking Wave Forces`

In many scenarios, wave forces need to be computed. Based on the experiments by Chen et al. [2014], numerical benchmark results were produced, see Bihs et al. [2016a] for more detail. In addition to the previous NWT examples, the use of the force box is shown in this example. The forces on a solid within the force box are calculated and printed out. It is important to give some margin around the solid in order to capture all surface facets in the

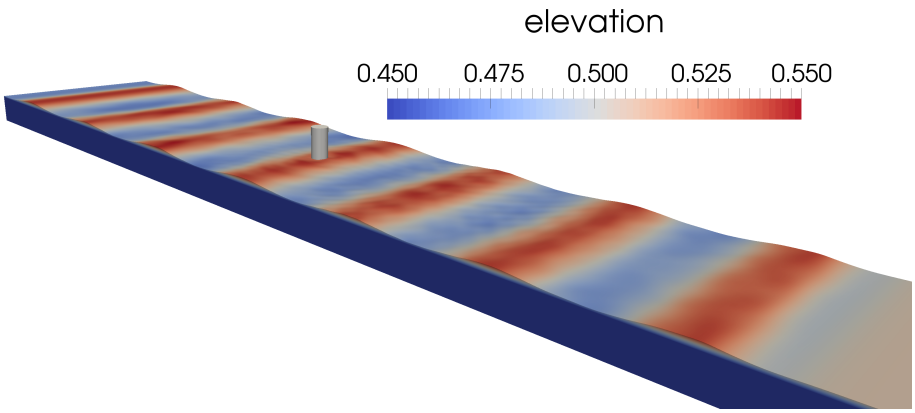


Figure 13.49: Non-breaking wave forces

pressure integration algorithm. More REEF3D related literature on non-breaking wave forces can be found here: Kamath et al. [2015b], Kamath et al. [2016b] and Kamath et al. [2015a].

- Wave: 2nd-order Stokes, H = 0.07 m, T = 1.22 s (B 93), water depth d = 0.505 m
- Domain: 18 m × 3 m × 1 m, cell size dx = 0.025 m (720 × 120 × 40 cells), side walls
- Cylinder at x = 7.5 m, y = 1.5 m, diameter 0.25 m; wave generation zone 2.11 m, numerical beach 4.22 m
- Simulated time 50 s; the folder also contains the measured data 'expt_data_wsf.txt' and 'expt_data_force.txt' (two columns: time, value)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.14.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 21 // side 2 (y_{max}): wall
C 13 21 // side 3 (y_{min}): wall
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.025 // uniform cell size dx [m]

B 10 0.0 18.0 0.0 3.0 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 33 7.50 1.50 0.125 // vertical cylinder at x = 7.5 m, y = 1.5 m, radius 0.125 m

M 20 2 // decomposition method 2 (improved rectangular)

M 10 6 // number of processes (partitions)

13.14.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 11 1 // wall functions for the turbulence model on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 4 // wave type: 2nd-order Stokes waves
B 93 0.07 1.22 // wave height H [m], wave period T [s]
B 96 2.11 4.22 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 2 // numerical beach: relaxation method 2

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 1.0 // length for the initial reinitialization instead of the domain length [m]
F 60 0.505 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 50.0 // maximum simulated time [s]
N 47 0.1 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 35 15.0 30.0 0.05 // print window: start [s], end [s], interval [s]
P 40 1 // print state files for hot-start
P 42 1.0 // state file interval [s]

T 10 2 // k-ω turbulence model
T 36 2 // free surface turbulence damping on, including wall distance

W 22 -9.81 // gravity in z-direction [m/s^2]

P 55 1.0 // wsfline/line probe interval [s]; no effect here, no P 52 or P 62 is given

P 51 5.00 1.50 // wave gauge 1 at x, y [m]

P 81 7.325 7.675 1.325 1.675 0.0 1.0 // force box around the cylinder

13.14.3 What to look at

The forces from the P 81 force box are written to 'REEF3D_CFD_Force-1.dat' in REEF3D_CFD_Force; plot the inline force F_x over time and compare it with 'expt_data_force.txt'. The gauge at x = 5 m (P 51) in 'REEF3D-CFD-WSF-HG.dat' (REEF3D_CFD_WSF) can be compared with 'expt_data_wsf.txt'. REEF3D_CFD_VTU contains the 3D field every 0.05 s between 15 s and 30 s (P 35).

13.14.4 Results

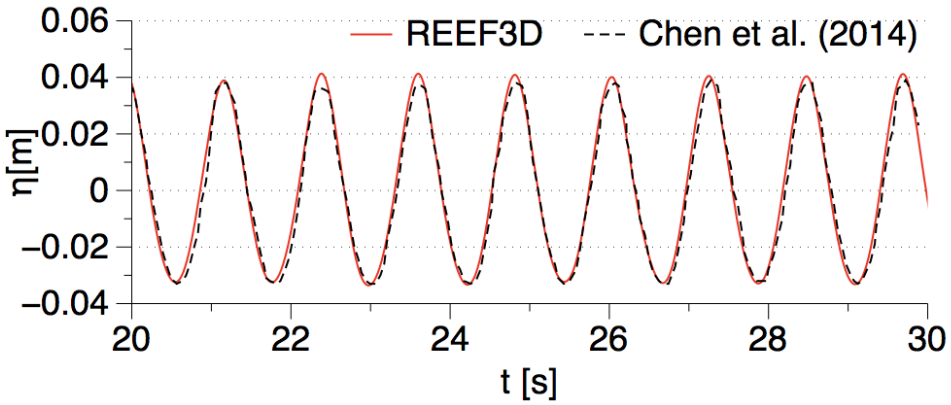


Figure 13.50: Free surface comparison between REEF3D and the experimental data Chen et al. [2014]

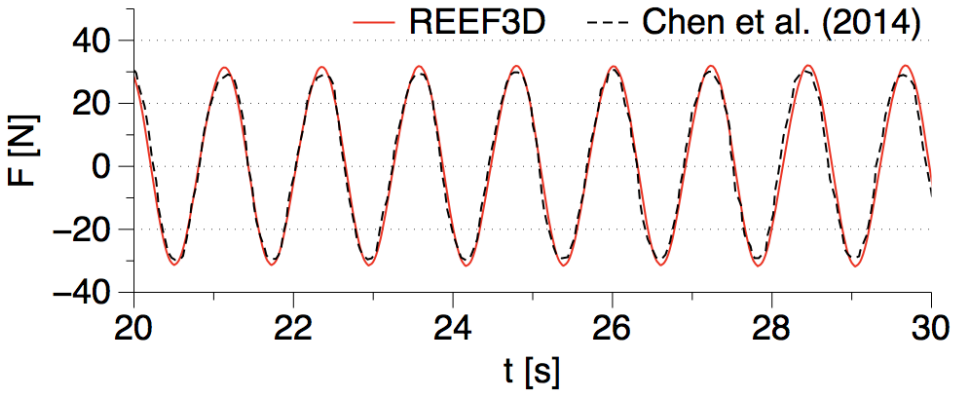


Figure 13.51: Wave Force comparison between REEF3D and the experimental data Chen et al. [2014]

13.15 Breaking Wave Forces

Repository folder: `Tutorials/REEF3D_CFD/11_15 Breaking Wave Forces`

Extending the previous scenario, breaking wave forces are simulated by using the case from Kamath et al. [2016a] based on an experimental study in the GWK in Hannover Irschik et al. [2002]. The waves are shoaling on a 1:10 slope and are breaking directly at the vertical cylinder. The slamming forces are calculated using the force box approach. REEF3D has been extensively validated and applied for breaking wave forces: Bihs et al. [2016b], Alagan Chella et al. [2019c], Alagan Chella et al. [2019b], Alagan Chella et al. [2019a].

- Wave: 5th-order Stokes, H = 1.3 m, T = 4.0 s (B 93), water depth d = 3.8 m, 1.5 m at the cylinder
- Domain: 54 m × 5 m × 7 m, cell size dx = 0.05 m (1080 × 100 × 140 cells), 1:10 slope from x = 21 m to 44 m

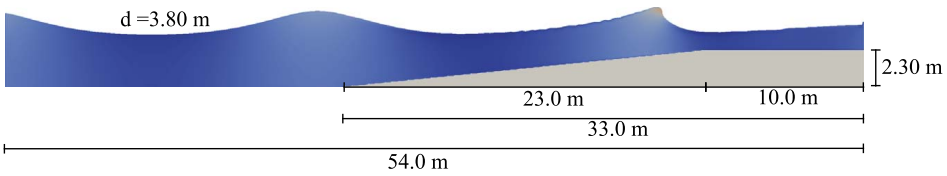


Figure 13.52: Breaking wave forces setup

- Cylinder at x = 44.0 m, y = 2.5 m, diameter 0.7 m; wave generation zone 21 m, active wave absorption at the outlet
- Simulated time 50 s (at most 200000 iterations); the folder also contains the measured data 'expt_data_wsf.txt' and 'expt_data_force.txt'
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.15.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 21 // side 2 (y_{max}): wall
C 13 21 // side 3 (y_{min}): wall
C 14 8 // side 4 (x_{max}): numerical beach, type 8 (handled like type 7 in REEF3D)
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.05 // uniform cell size dx [m]

B 10 0.0 54.0 0.0 5.0 0.0 7.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 33 44.0 2.50 0.35 // vertical cylinder at x = 44.0 m, y = 2.5 m, radius 0.35 m

S 61 21.0 44.0 0.0 5.0 0.0 2.30 // 1:10 slope from x = 21 m to 44 m
S 10 44.0 54.0 0.0 5.0 0.0 2.30 // horizontal bed 2.3 m high behind the slope

M 20 2 // decomposition method 2 (improved rectangular)

M 10 6 // number of processes (partitions)

13.15.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 5 // wave type: 5th-order Stokes waves (Fenton)
B 93 1.3 4.0 // wave height H [m], wave period T [s]
B 96 21.0 0.0 // lengths of wave generation zone and numerical beach [m]

B 98 2 // wave generation: relaxation method 2
B 99 3 // numerical beach: active wave absorption (AWA)

```
D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 7.0 // length for the initial reinitialization instead of the domain length [m]
F 60 3.80 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 50.0 // maximum simulated time [s]
N 45 200000 // maximum number of iterations
N 47 0.1 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 35 23.5 25.2 0.05 // print window: start [s], end [s], interval [s]
P 180 1 // print free surface .vtp files
P 182 0.1 // free surface .vtp print interval [s]

T 10 2 // k-ω turbulence model
T 36 2 // free surface turbulence damping on, including wall distance

W 22 -9.81 // gravity in z-direction [m/s^2]

P 51 22.0 2.50 // wave gauge 1 at x, y [m]
P 51 30.0 2.50 // wave gauge 2 at x, y [m]
P 51 35.0 2.50 // wave gauge 3 at x, y [m]
P 51 40.0 2.50 // wave gauge 4 at x, y [m]
P 51 43.65 4.80 // wave gauge 5 at x, y [m]
P 51 43.65 2.50 // wave gauge 6 at x, y [m]
P 51 44.5 2.505 // wave gauge 7 at x, y [m]
P 51 47.0 2.5 // wave gauge 8 at x, y [m]
P 51 50.5 2.5 // wave gauge 9 at x, y [m]
P 51 52.0 2.5 // wave gauge 10 at x, y [m]
P 51 53.0 2.5 // wave gauge 11 at x, y [m]

P 81 43.30 44.70 1.80 3.20 2.30 8.0 // force box around the cylinder
```

13.15.3 What to look at

Plot F_x from 'REEF3D_CFD_Force-1.dat' in REEF3D_CFD_Force together with 'expt_data_force.txt', and the 11 gauges in 'REEF3D-CFD-WSF-HG.dat' (REEF3D_CFD_WSF) together with 'expt_data_wsf.txt'. For the visualization of the breaker, REEF3D_CFD_FSF contains the free surface as .vtp every 0.1 s (P 180, P 182), and REEF3D_CFD_VTU the 3D field every 0.05 s between 23.5 s and 25.2 s (P 35).

13.15.4 Results

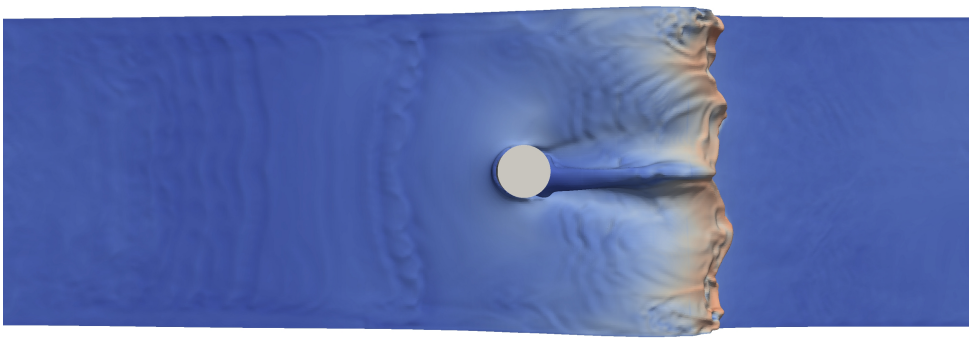


Figure 13.53: Breaking wave impact with REEF3D, plane view

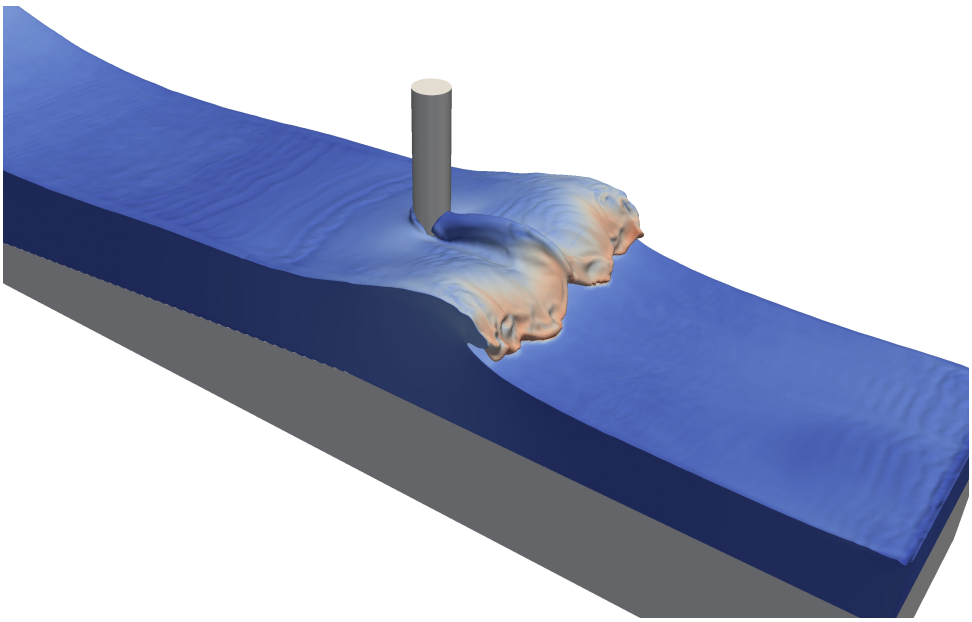


Figure 13.54: Breaking wave impact with REEF3D, side view

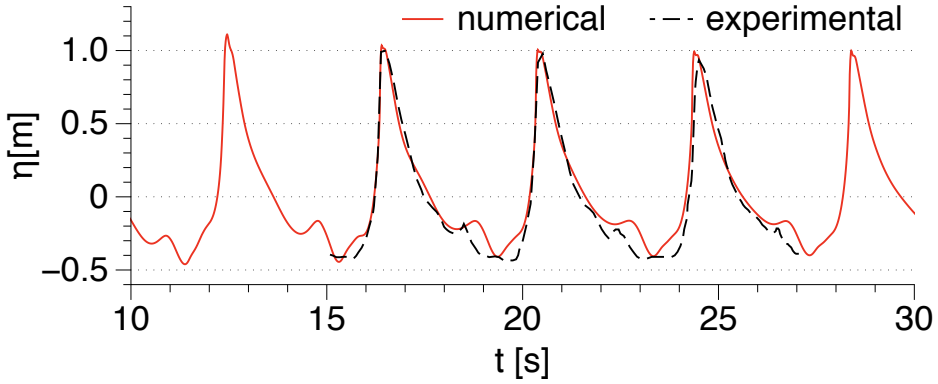


Figure 13.55: Free surface comparison between REEF3D and the experimental data Irschik et al. [2002]

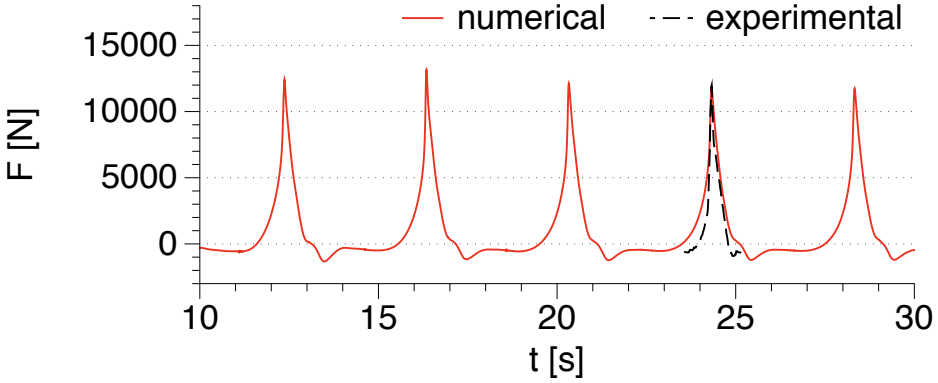


Figure 13.56: Wave Force comparison between REEF3D and the experimental data Irschik et al. [2002]

13.16 Heave Decay of a Sphere with Non-uniform Grid

Repository folder: `Tutorials/REEF3D_CFD/11_16 Heave Decay of Sphere`

Setup for simulations including floating bodies using the continuous direct forcing approach. As an example, the heave decay of a sphere can be simulated using the following input files together with the STL file 'floating.stl' provided in the tutorial folder. The sphere has a radius of 0.5 m and a density of 500 kg/m^3 (X 21), i.e. half the water density, so at rest its centre lies at the still water level z = 2.0 m. X 182 places the centre at z = 2.2 m, and the sphere is released from this position with only the translational degrees of freedom active (X 11). Relaxation zones along the four side walls (B 107) damp the radiated waves. It is recommended to use the cell-based stretching shown in this example to define a box with uniform cell sizes around the floating body.

- Domain: 6 m × 6 m × 4 m, cell-based stretching (B 101–B 103 11) with Δx_{min} = 0.05 m in a 1.5 m focus zone around $x = y$ = 3 m, z = 2 m and Δx_{max} = 0.3 m
- Still water level 2.0 m, initial heave offset 0.2 m, simulated time 10 s
- Extra input file: floating.stl (read by REEF3D)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.16.1 DIVEMesh: control.txt

C 11 21 // side 1 (x_{min}): wall
C 12 21 // side 2 (y_{max}): wall
C 13 21 // side 3 (y_{min}): wall
C 14 21 // side 4 (x_{max}): wall
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.05 // cell size [m], same as Δx_{min} of the cell-based stretching
B 10 0.0 6.0 0.0 6.0 0.0 4.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

B 101 11 // x-direction stretching: cell-size based, uses B 127
B 127 0.05 0.3 3.0 1.5 1.1 // x: Δx_{min}, Δx_{max}, focus point x_f, focus zone length δ_f, cell ratio r
B 102 11 // y-direction stretching: cell-size based, uses B 128
B 128 0.05 0.3 3.0 1.5 1.1 // y: Δy_{min}, Δy_{max}, focus point y_f, focus zone length δ_f, cell ratio r
B 103 11 // z-direction stretching: cell-size based, uses B 129
B 129 0.05 0.3 2.0 1.5 1.1 // z: Δz_{min}, Δz_{max}, focus point z_f, focus zone length δ_f, cell ratio r

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

13.16.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 50 0.00001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (used here for the damping zones only)
B 99 2 // numerical beach: relaxation method 2, zones from B 107
B 107 0.0 0.0 0.0 6.0 0.3 // damping zone along the wall x = 0, 0.3 m wide
B 107 6.0 6.0 0.0 6.0 0.3 // damping zone along the wall x = 6 m
B 107 0.0 6.0 0.0 0.0 0.3 // damping zone along the wall y = 0
B 107 0.0 6.0 6.0 6.0 0.3 // damping zone along the wall y = 6 m

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 35 5 // level set convection: 5th-order WENO (HJ)
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 60 2.0 // still water level [m]

I 10 1 // initialize everything (potential flow velocities, pressure, turbulence)

N 40 4 // momentum: 3rd-order low-storage Runge-Kutta (used with 6DOF)
N 41 10 // maximum simulated time [s]
N 47 0.3 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.1 // print interval in simulated time [s]

W 1 1000.0 // density of water [kg/m^3]
W 22 -9.81 // gravity in z-direction [m/s^2]

X 10 1 // turn on the 6DOF floating body algorithm
X 11 1 1 1 0 0 0 // degrees of freedom u, v, w, p, q, r: translations free, rotations fixed
X 21 500.0 // density of the floating body [kg/m^3]
X 180 1 // read the floating body geometry from floating.stl
X 182 3.0 3.0 2.2 // translate the STL geometry by $\Delta x, \Delta y, \Delta z$ [m]

13.16.3 What to look at

The motion of the body is written to REEF3D_CFD_6DOF: plot the vertical position over time from the 'REEF3D_6DOF_position' file to obtain the heave decay curve, and use the 'REEF3D_6DOF_forces' file for the forces on the body. The body surface is written as .vtp to REEF3D_CFD_6DOF_VTP, which can be loaded in ParaView together with REEF3D_CFD_VTU (every 0.1 s) to show the radiated waves (contour phi = 0).

13.17 Abutment Scour

Repository folder: `Tutorials/REEF3D_CFD/11_17 Abutment Scour`

Local scour at a vertical-wall abutment in an open channel flume. The abutment is a box of 0.1 m length in flow direction that protrudes 0.2 m from the side wall into the 0.6 m wide channel, standing on a sediment bed with its top at $z = 0$ (S 57). The discharge is prescribed with ioFlow; the bed evolves with the Engelund-Fredsøe bedload formula, critical shear stress reduction on slopes, bedload direction correction and the sandslide algorithm. Near the inlet the bed is kept at its initial level with a bed relaxation zone (S 73). The case demonstrates the coupling of the CFD flow solver with the sediment transport module.

- Domain: x = 0.5–4.0 m, y = 0–0.6 m, z = -0.3–0.2 m, cell size dx = 0.02 m (175 × 30 × 25 cells)
- Flow depth 0.095 m above the bed (F 60), discharge 0.02679 m^3/s (mean velocity about 0.47 m/s)
- Sediment: d_{50} = 1.9 mm, density 2570 kg/m^3, critical Shields parameter 0.03; sediment transport starts after 100 iterations and the bed is updated every 10 iterations
- The run ends after 7200 s of sediment time (S 19) or 500000 iterations (N 45)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

Note: this ctrl.txt contains S 17 0. With the default S 15 0 the sediment time step is S 17 times the computed value, i.e. zero, so in the current code the bed does not change and the run is only ended by N 45. Use the default S 17 1.0 to obtain a developing scour hole.

13.17.1 DIVEMesh: control.txt

C 11 1 // side 1 (x_{min}): inflow
C 12 21 // side 2 (y_{max}): wall
C 13 21 // side 3 (y_{min}): wall
C 14 2 // side 4 (x_{max}): outflow
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.02 // uniform cell size dx [m]
b 2 140 24 30 // lower-case letter: ignored, the grid follows B 1
B 10 0.5 4.0 0.0 0.6 -0.3 0.2 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 10 1.70 1.8 0.4 0.6 -0.3 0.2 // abutment: box attached to the side wall $y = 0.6$ m

B 103 0 // no vertical stretching (B 113 and B 116 are then not used)
B 113 6.0 // stretching factor in z-direction (not used with B 103 0)
B 116 0.312 // focus point of the z-stretching (not used with B 103 0)

M 10 6 // number of processes (partitions)

13.17.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 11 1 // wall functions for the turbulence model on
B 20 2 // no-slip velocity boundary condition
B 50 0.001 // wall roughness k_s [m]
B 60 1 // ioFlow: constant inflow discharge from W 10

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 46 0 // no Picard volume correction after reinitialization
F 60 0.095 // still water level [m]

I 10 1 // initialize everything (potential flow velocities, pressure, turbulence)

N 40 2 // momentum: 2nd-order TVD Runge-Kutta
N 45 500000 // maximum number of iterations
N 47 0.2 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 20 100 // print interval in iterations
P 23 1 // add the test array to the ParaView output
P 26 1 // add the sediment concentration (ST_conc) to the .vtu files
P 76 1 // add bedload q_b, q_{be} to the .vtu files

```
P 77 1 // add bed slope parameters to the .vtu files
P 78 1 // add sediment parameters (dh, bed change, reduction, ...) to the .vtu files
P 79 1 // add bed shear stress to the ParaView output

T 10 2 // k-ω turbulence model
T 36 1 // free surface turbulence damping on

W 10 0.02679 // inflow discharge [m^3/s]
W 22 -9.81 // gravity in z-direction [m/s^2]

S 10 1 // sediment transport on
S 11 3 // bedload formula: Engelund and Fredsøe
S 13 5.0 // maximum sediment time step [s]
S 14 0.5 // CFL factor for the sediment time step
S 16 4 // bed shear stress from turbulent kinetic energy
S 17 0 // decoupling factor for the sediment time step; 0 gives a zero sediment time step
(see note)
S 19 7200 // maximum sediment (morphological) time [s], ends the run
S 20 0.0019 // sediment d50 [m]
S 21 10.0 // factor on d50 for ks in the bed shear stress
S 22 2570.0 // sediment density [kg/m^3]
S 24 0.75 // porosity of the sediment layer
S 30 0.03 // critical Shields parameter

S 31 2 // Exner formulation, version 2

S 41 1 // sediment start criterion: iterations
S 42 1 // bed update interval criterion: iterations
S 43 100 // flow iterations before sediment transport starts
S 44 10 // flow iterations between bed updates
S 50 3 // bed level fixed at inflow and outflow
S 57 0.0 // initial sediment bed level (top of the sediment) [m]
S 73 0.00 0.75 0.0 0.5 0.0 // bed kept at z = 0 within 0.75 m of the inlet x = 0.5 m

S 80 4 // critical shear stress reduction on slopes: Fredsøe
S 81 30.0 // angle of repose [deg]
S 82 7.5 // deltaphi parameter [deg]
S 85 1 // bedload direction correction (Koch and Flokstra)
S 90 1 // sandslide algorithm: explicit full correction
S 91 5 // number of sandslide iterations
S 92 0.25 // sandslide correction factor
```

13.17.3 What to look at

With sediment transport active, the bed is written to the .vtu files as the variable topo: in ParaView open REEF3D_CFD_VTU (every 100 iterations, P 20) and show the contour topo = 0 colored by elevation to follow the development of the scour hole at the upstream corner of the abutment. P 76–P 79 add the sediment variables (prefix ST_, e.g. bedload, bed change and bed shear stress) and P 26 the sediment concentration to the same files; P 23 adds the test array. The free surface is the contour phi = 0.

13.18 Pier Scour

Repository folder: `Tutorials/REEF3D_CFD/11_18 Pier Scour`

Local scour around a vertical circular pier in an open channel flume. The pier stands in the middle of a 2 m wide channel on a sediment bed with its top at $z = 0$ (S 57). The discharge is prescribed with ioFlow with a logarithmic inflow profile (B 61 2), and the bed evolves with the Engelund-Fredsøe bedload formula, slope effects and the sandslide algorithm, with a bed update after every flow iteration after the first 100 iterations. The bed is kept at its initial level near the inlet (S 73); the second, lower-case s 73 line for the outlet is deactivated.

- Domain: x = 1.0–5.5 m, y = 0–2.0 m, z = -0.3–0.5 m, cell size dx = 0.02 m (225 × 100 × 40 cells)
- Pier at x = 2.5 m, y = 1.0 m, diameter 0.2 m; flow depth 0.3 m, discharge 0.18 m^3/s (mean velocity 0.3 m/s)
- Sediment: d_{50} = 0.97 mm, density 2570 kg/m^3, critical Shields parameter 0.035
- The run ends after 75600 s of sediment time (S 19); no N 41 or N 45 limit is set
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.18.1 DIVEMesh: control.txt

```
C 11 1 // side 1 (x_min): inflow
C 12 21 // side 2 (y_max): wall
C 13 21 // side 3 (y_min): wall
C 14 2 // side 4 (x_max): outflow
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.02 // uniform cell size dx [m]
B 10 1.0 5.5 0.0 2.0 -0.3 0.5 // rectangular domain x_s, x_e, y_s, y_e, z_s, z_e [m]

S 33 2.5 1.0 0.1 // pier: vertical cylinder at x = 2.5 m, y = 1.0 m, radius 0.1 m

M 10 6 // number of processes (partitions)
M 13 0 // no partitioning in z-direction
```

13.18.2 REEF3D: ctrl.txt

```
B 10 1 // wall functions for the velocities on
B 11 1 // wall functions for the turbulence model on
B 50 0.006 // wall roughness k_s [m]
B 60 1 // ioFlow: constant inflow discharge from W 10
B 61 2 // ioFlow inflow profile: logarithmic
```

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 60 0.3 // still water level [m]

I 10 1 // initialize everything (potential flow velocities, pressure, turbulence)

N 40 2 // momentum: 2nd-order TVD Runge-Kutta

N 47 0.3 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 20 100 // print interval in iterations
P 23 1 // add the test array to the ParaView output
P 26 1 // add the sediment concentration (ST_conc) to the .vtu files
P 79 1 // add bed shear stress to the ParaView output
P 122 1 // print maximum bed change

T 10 2 // k-ω turbulence model
T 36 1 // free surface turbulence damping on

W 10 0.18 // inflow discharge [m^3/s]
W 22 -9.81 // gravity in z-direction [m/s^2]

S 10 1 // sediment transport on
S 11 3 // bedload formula: Engelund and Fredsøe
S 12 0 // suspended load off
S 13 5.0 // maximum sediment time step [s]
S 14 0.3 // CFL factor for the sediment time step
S 16 4 // bed shear stress from turbulent kinetic energy
S 19 75600 // maximum sediment (morphological) time [s], ends the run
S 20 0.00097 // sediment d_{50} [m]
S 21 5.0 // factor on d_{50} for k_s in the bed shear stress
S 22 2570.0 // sediment density [kg/m^3]
S 24 0.75 // porosity of the sediment layer
S 30 0.035 // critical Shields parameter

S 41 1 // sediment start criterion: iterations
S 42 1 // bed update interval criterion: iterations
S 43 100 // flow iterations before sediment transport starts
S 44 1 // flow iterations between bed updates
S 50 1 // bed level fixed at the inflow
S 57 0.0 // initial sediment bed level (top of the sediment) [m]
S 73 0.00 0.5 0.0 1.0 0.0 // bed kept at $z = 0$ within 0.5 m of the inlet $x = 1.0$ m
s 73 0.00 0.25 0.0 5.5 0.0 // lower-case letter: ignored (would fix the bed near the outlet $x = 5.5$ m)
S 80 4 // critical shear stress reduction on slopes: Fredsøe
S 81 35.0 // angle of repose [deg]
S 82 10.0 // deltaphi parameter [deg]
S 90 1 // sandslide algorithm: explicit full correction
S 91 10 // number of sandslide iterations

13.18.3 What to look at

The maximum bed change is written to 'REEF3D-CFD-Sediment-Max.dat' in REEF3D_CFD_Sediment (P 122); plot it over time to follow the development of the scour depth. In ParaView, open REEF3D_CFD_VTU (every 100 iterations, P 20) and show the contour topo = 0 for the bed with the scour hole and the deposition downstream of the pier; P 79 adds the bed shear stress (ST_ variables), P 26 the sediment concentration and P 23 the test array.

14. Tutorial | Coupled Models

The runtime coupling (B 92 62, Section 8.5) runs an FNPF model and one or more receiving models in one MPI job. Each tutorial folder in `Tutorials/REEF3D_Coupling` contains one subfolder per model with its `control.txt` and `ctrl.txt`, and a script `run.sh` that runs DIVEMesh in every subfolder and starts all models with one **mpirun** command; the order of the app contexts in that command gives the app numbers. All cases use the same wave: 5th-order Stokes waves with H = 0.1 m and T = 1.6 s in d = 1.0 m (L = 3.76 m), generated in a 40 m long 2D FNPF tank (dx = 0.1 m, 10 σ-layers) with a 4 m generation zone and an 8 m beach. Gauges at the same positions in both models (P 51) show how closely the receiver follows the FNPF waves; FNPF writes the gauges of the state at $t + \Delta t$ with the label t, so its curves lead by one FNPF time step.

14.1 2D FNPF $\rightarrow$ 2D CFD

Folder: `1_FNPF_CFD_2D`. The CFD domain covers x = 6–18 m of the FNPF tank (dx = 0.025 m). It generates the waves in a 3 m relaxation zone from the FNPF solution (B 92 62, B 98 2) and absorbs them in a 4 m beach. The CFD coordinates equal the FNPF coordinates (B 105 not given, I 231–I 233 zero).

- Run: **mpirun -np 1 -wdir fnpf REEF3D : -np 1 -wdir cfd REEF3D**
- Compare REEF3D_CFD_WSF and REEF3D_FNPF_WSF at x = 8, 10, 12, 14 m; the CFD waves follow FNPF within a few mm.

14.2 2D FNPF $\rightarrow$ 3D CFD, oblique waves

Folder: `2_FNPF2D_CFD3D_oblique`. A 3D CFD box (8 m $\times$ 4 m, dx = 0.1 m, two processes) receives long-crested waves from the 2D FNPF model at 30° (B 105 30.0 0.0 0.0): the CFD point is rotated into the FNPF line, and the FNPF velocity is rotated back. Besides the generation zone at x_{min} (B 96), relaxation zones towards the FNPF field are set on the other three edges (B 520 / B 521, method 1), so that the oblique waves are not reflected by the side walls.

- The FNPF gauges are placed at the FNPF positions of the CFD gauges, $x_F = x\cos 30° + y\sin 30°$.

14.3 FNPF $\rightarrow$ FNPF nesting

Folder: `3_FNPF_FNPF_nesting`. A coarse FNPF model (dx = 0.2 m, app 0) drives a fine FNPF model (dx = 0.05 m, x = 6–18 m, app 1) with B 92 62. The source also sends the potential, and the fine model relaxes η and Φ_s in its generation zone towards it.

14.4 Floating body: CFD drives FNPF

Folder: `4_FNPF_CFD_floating_body`. A box of 1.0 m × 0.4 m with half the density of water floats at x = 12 m in both models (X 10 1, X 110). In CFD it moves freely in surge, heave and pitch (X 11 1 0 1 0 1 0); with I 243 1 its motion drives the FNPF body, which therefore radiates the waves of the CFD motion into the FNPF domain. Compare REEF3D_CFD_6DOF/REEF3D_6DOF_position_0.dat with REEF3D_FNPF_6DOF/REEF3D_6DOF_pos
In the 2D CFD domain the box spans exactly the domain width (y = 0–0.025 m).

14.5 Two-way coupling: reflection into FNPF

Folder: `5_FNPF_CFD_two-way`. The CFD domain (x = 6–18 m) ends in a vertical wall (C 12 21, no beach), which reflects the waves. With the overlap box I 244 10.0 16.5 -1.0 1.0 (ramp I 245 1.0, linear time interpolation I 242 1), FNPF is relaxed towards the CFD solution in the box, so that the reflected waves enter the FNPF domain. In front of the CFD domain, the FNPF gauges at x = 5.0–7.5 m show the standing wave with a node near x = 6 m, as in an FNPF tank that ends with a wall at x = 18 m. Without I 244, FNPF shows only the incident wave. The simulation needs 24 s, until the reflection has travelled back to the gauges.

14.6 Chain: regional FNPF → local FNPF → CFD

Folder: `6_FNPF_FNPF_CFD_chain`. Three models: a regional FNPF model (dx = 0.2 m, app 0), a local FNPF model (dx = 0.05 m, x = 4–20 m, app 1, B 92 62) and a CFD box (x = 8–18 m, app 2, B 92 62). The CFD model takes its waves from the local FNPF model with I 246 1; without I 246 it would take them from the regional model, the lowest FNPF app that is no receiver.

- Run: **mpirun -np 1 -wdir regional REEF3D : -np 1 -wdir local REEF3D : -np 1 -wdir cfd REEF3D**

14.7 Two-way coupling with NHFLOW

Folder: `7_FNPF_NHFLOW_two-way`. As tutorial 5, with an NHFLOW model (x = 6–18 m, dx = 0.05 m, 10 σ layers) in place of CFD. NHFLOW ends in a wall and feeds its free surface back with I 244 10.0 16.5 -1.0 1.0; the FNPF gauges at x = 5.0–7.5 m show the standing wave.

14.8 Two-way coupling in 3D

Folder: `8_FNPF3D_CFD3D_two-way`. A 3D FNPF tank (0–40 m × 2 m) with waves at 15° (B 105 15) and a 3D CFD box (x = 6–14 m, dx = 0.1 m) that ends in a wall. The overlap box I 244 9.5 13.0 -1.0 3.0 spans the whole width, so it is ramped only in x (I 245 0.8). The FNPF gauges at y = 0.5 m and 1.5 m show the standing pattern of the oblique waves in front of the wall, which differs across the tank.

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